Springer Texts in Business and Economics

Dmitry Ivanov
Alexander Tsipoulanidis
Jörn Schönberger

Global Supply Chain and Operations Management

A Decision-Oriented Introduction to the Creation of Value

Fourth Edition

Springer

Springer Texts in Business and Economics

Springer Texts in Business and Economics (STBE) delivers high-quality instructional content for undergraduates and graduates in all areas of Business/Management Science and Economics. The series is comprised of self-contained books with a broad and comprehensive coverage that are suitable for class as well as for individual self-study. All texts are authored by established experts in their fields and offer a solid methodological background, often accompanied by problems and exercises.

Dmitry Ivanov • Alexander Tsipoulanidis • Jörn Schönberger

Global Supply Chain and Operations Management

A Decision-Oriented Introduction to the Creation of Value

Fourth Edition

Dmitry Ivanov
Department of Business Administration
Berlin School of Economics and Law
Berlin, Germany

Alexander Tsipoulanidis
Department of Business Administration
Berlin School of Economics and Law
Berlin, Germany

Jörn Schönberger
Faculty of Transportation and Traffic
Science "Friedrich List"
Technical University of Dresden
Dresden, Germany

ISSN 2192-4333 ISSN 2192-4341 (electronic)
Springer Texts in Business and Economics
ISBN 978-3-031-95858-8 ISBN 978-3-031-95859-5 (eBook)
https://doi.org/10.1007/978-3-031-95859-5

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2017, 2019, 2021, 2025

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

If disposing of this product, please recycle the paper.

Dmitry Ivanov
To my parents, who inspired dreams and without whom this book would have never been completed
To my wife and children who make dreams come true, inspiring with love and smile, and without whom this book would have been completed much earlier
To all: Live with a clear conscience, a pure soul, and a strong spirit—and all will be well.

Alexander Tsipoulanidis
To my family:
Joanna, Marina, Irini, and Ursula—I love you all so much!
To my father Ioannis (1934–2002) and Reimar (1949–2023) —we miss you!

Jörn Schönberger
To my family:
Maybe this book explains what I'm doing in the lecture hall …

Preface

About This Book

Supply chain and operations management (SCOM) is one of the major areas in any company. Both a start-up and a global corporation will share a common set of problems to be solved and decisions to be made. Where to locate factories and warehouses? How to be resilient against disruption risk? Which suppliers do fit best? How much to produce and when? How much inventory to place and where? How to deliver products on time and in an environmentally friendly fashion?

Of course, these decisions could be taken by intuition. However, the complexity and uncertainty of supply chains (SC) and operations are too high for intuition-based decisions. SCOM helps to approach these problems from the method-based perspective and supports informed decision-making. Moreover, SCOM is not just a business concern. When SCs do not work properly, our society will not receive goods and services, and people will lose jobs. SCOM is important for business, economics, and governments alike.

In everyday life, we all make SCOM decisions. If you move to a new flat, location planning is necessary. Next, you need a plan for designing the overall process. This includes capacity planning, transportation planning, and human resource planning. You also need to replenish some items and do procurement planning. Finally, a detailed schedule for the day of the move is needed.

Similarly, building a new house involves many SCOM decisions. Again, it starts with location selection. If you decide to coordinate the overall process by yourself, it is necessary to coordinate the entire supply chain of different manufacturers and workers. In turn, they need the detailed data of your plans and forecasted data to plan their own process and sourcing activities. In order to avoid traffic jams at the building site, detailed coordination at the vehicle routing level is needed.

SCOM belongs to the most exciting management areas. These functionalities are tangible and in high demand in all industries and services. This study book intends to provide both the introduction to and advanced knowledge in the SCOM field. Providing readers with a working knowledge of SCOM, this textbook can be used in core, special, and advanced classes. Therefore, the book is targeted at a broad range of students and professionals involved in SCOM.

The special focus is on bridging theory and practice. Since managers use both quantitative and qualitative methods in making their decisions, the book follows these practical knowledge requirements. Decision-oriented and method-oriented perspectives determine the philosophy of the book. In addition, because of the extensive use of information technology and optimization techniques in SCOM, we pay particular attention to this aspect.

Distinctively, in the fourth edition we consider traditional SCOM paradigms such as profitability maximization along with resilience, sustainability, and digitalization perspectives. To this end, we deliver an understanding for both major SCOM determinants and future-oriented developments.

Next, a strong global focus with more than 80 up-to-date cases and practical examples from all over the world is a distinguishing feature of this textbook. (Disclaimer: The company names in case studies can be fictitious (unless we explicitly provide references to secondary sources and web sites of the companies), and any similarities are purely coincidental.) The case studies encompass different industries and services and consider examples of successful and failed SCOM practices in Europe, America, Asia, Africa, and Australia.

Finally, following the expectations of modern students and our teaching experiences in SCOM over the past 25 years, we divided this textbook into a hardback and an electronic supplement. In the hardback, basic theoretical concepts, case studies, applications, and numerical examples are explained. The e-supplement supports the hardback and provides students and teachers with additional case studies, numerical tasks, Excel files, and slides.

The *e-supplement* of this book can be accessed via the URL www.global-supply-chain-management.de or www.prof-ivanov.de without further registration. For course instructors, a special area is set up that contains further material. The e-supplement is updated with additional topics, exercises, and cases.

This second edition of our textbook consists of 18 chapters divided into four parts:

Part I Introduction to Supply Chain and Operations Management

- Chapter 1 Basics of Supply Chain and Operations Management
- Chapter 2 Examples from Different Industries, Services, and Continents
- Chapter 3 Processes, Systems, and Models

Part II Designing Operations and Supply Network: Strategic Perspective

- Chapter 4 Supply Chain Strategy
- Chapter 5 Sourcing Strategy
- Chapter 6 Production Strategy
- Chapter 7 Facility Location
- Chapter 8 Transportation and Distribution Network Design
- Chapter 9 Factory Planning and Process Design
- Chapter 10 Layout Planning

Part III Matching Demand and Supply: Tactical and Operative Planning

- Chapter 11 Demand Forecasting
- Chapter 12 Production and Material Requirements Planning
- Chapter 13 Inventory Management
- Chapter 14 Scheduling and Routing

Part IV Advanced Topics in Supply Chain and Operations Management

- Chapter 15 Supply Chain Risk Management and Resilience
- Chapter 16 Digital Supply Chain, Smart Operations, and Industry 4.0
- Chapter 17 Pricing and Revenue-Oriented Capacity Allocation
- Chapter 18 Yield Management

Each *chapter* contains the following elements:

- Introductory case study
- Learning objectives
- Theory with practical insights and case studies
- Tasks with example solutions
- Key points and outlook
- Additional tasks and case studies placed in e-supplement
- Further supplementary materials: online tutorial, Excel files, and videos.

Each chapter starts with an introductory case study. Subsequently, major decision areas and methods for decision support are handled. Finally, applications can be taught based on additional case studies and numerical tasks. The summary of key points and an outlook end each chapter. Throughout the book, practical insights are highlighted.

The fourth edition contains numerous extensions and updates in light of the insights learned for SCOM through the COVID-19 pandemic as well as new and rapid developments in SCOM digitalization and artificial intelligence. In particular, the fourth edition provides the following special extensions as compared to the previous edition:

– A multitude of new case studies related to resilience, sustainability, and digitalization in SCOM,
– Extensions to SCOM resilience and viability, and
– Extensions in the area of artificial intelligence (AI) in SCOM, e.g., new materials on using AI methods for demand forecasting and inventory management.

We have started to develop a set of satellite books to accompany our main textbook by an extended introduction to some specific topic. As the first satellite book, *Introduction to Supply Chain Resilience* written by Prof. Dmitry Ivanov appeared in Springer in 2021, followed by the second edition in 2025. In 2024, the second satellite book, *Introduction to Supply Chain Analytics* written by Prof. Dmitry Ivanov appeared in Springer. The third satellite book, *Introduction to Operations and Supply Chain Simulation with AnyLogic*, appears in 2025 authored by Dmitry Ivanov, Will Millhiser, and Phu Nguyen.

The existing chapters have been updated and new case studies have been included. The preface provides a guideline (Table 1) for instructors concerning how to select and structure the materials from this book for different courses in supply

Table 1 Recommendations for instructors on course structuring at different teaching levels using this textbook

Undergraduate core course in operations management			
1, 4.2, 5.2, 7.4, 7.5, 8.6, 9, 10, 12.2, 12.6, 13.2–13.4, 14.3, and 14.5			
Undergraduate major courses in supply chain and operations management			
Sourcing and Production Management	*Distribution and Transportation Management*	*Supply Chain Management*	*Selected Topics in Supply Chain and Operations Management*
1.2.1, 4.2, 4.3.1, 4.3.2, 5, 6, 9.3–9.5, 11, 12.2–12.5, 13.2, 13.5–13.7, 14.5	7, 8, 11, 14.2– 14.4, 18	1.2.2, 3.3, 4.2– 4.5, 5.3, 7.2, 7.4, 7.5, 8, 13.5–13.9	2, 3, 7.3, 16, 17, 18
Graduate courses in supply chain and operations management			
Advanced Operations Management	*Global Supply Chain Management*	*International Logistics Management*	*Supply Chain and Operations Analysis*
2, 9.2, 9.5, 9.6, 16, 17,18 AnyLogic Handbook in E-Supplement; Case Studies in E-Supplement	2, 4.3, 4.4, 5.3, 11.3, 13.5, 13.6, 15, 16 anyLogistix Handbook in E-Supplement; Case Studies in E-Supplement	2, 8, 10, 16 Case Studies in E-Supplement	3.4, 7.3 AnyLogic and anyLogistix Handbooks in E-Supplement; LP and CWLP Excel Spreadsheet Templates in E-Supplement

chain and operations management and with regard to different educational levels such as general undergraduate, specialized undergraduate, and graduate courses. The companion web site for this textbook www.global-supply-chain-management.de (linked to www.prof-ivanov.de) has been updated accordingly. In addition, the book is now supported by e-manuals for supply chain and operations simulation and optimization in AnyLogic and anyLogistix, which can be found at www.prof-ivanov.de.

In the e-supplement, different additional materials can be found, which are highlighted in each chapter. The advantage of using the e-supplement is that it offers the possibility of updating the case studies and adding additional materials more dynamically than producing new editions of the textbook. Another advantage is to be able to keep the hardback text short and concise.

Table 1 summarizes some recommendations for instructors concerning how to structure courses at different teaching levels using this textbook. This structure relies on the teaching concept in the bachelor's specialization "Supply Chain and Operations Management" and the master program "Global Supply Chain and Operations Management" at the Berlin School of Economics and Law.

The authors gratefully acknowledge all those who have helped us in bringing this book to publication. First and foremost, we have greatly benefited from the wealth of literature published on the subjects of SCOM and related topics. We thank Dr. Marina Ivanova for coauthoring Chaps. 4 and 6. We would like to thank all our colleagues from the Berlin School of Economics and Law and TU Dresden. The book has benefited immensely from their valuable insights, comments, and suggestions.

We thank companies the AnyLogic Company, Knorr-Bremse Berlin Systeme für Schienenfahrzeuge GmbH, OTLG, REWE, and SupplyOn for their permission to prepare new case studies and use company materials. We thank our students Tamara Erdenberger, Benjamin Bock, and Hiran Prathapage for technical assistance in artwork design and Alexander Reichardt, Katharina Schönhoff, Laura Seyfarth, Anna Putintseva, Nikolai Chadin, Daniil Demlanenko, and Mohiuddin Shojib who helped us to prepare case studies, numerical tasks, and technical support. We cordially thank Meghan Stewart for her thorough proofreading of the manuscript. We especially thank our former students in M.A. "Global Supply Chain and Operations Management" at the Berlin School of Economics and Law and now research associates at the Digital-AI Supply Chain Lab at the Berlin School of Economics and Law Vivian Samstag, Phu Nguyen, and Hiran Prathapage for great help in updating the chapters, preparing new case studies and numerical exercises, and setting up the slides for the book, which are available on the companion web site www.prof-ivanov.de.

Finally, we wish to thank Dr. Christian Rauscher, Editorial Director at Springer Nature, and Mrs. Jialin Yin, Editor at Springer Nature, and the entire Springer production team for their assistance and guidance in successfully completing this book.

Last but not least—we cordially thank our families who supported us enormously during our work on the book.

Berlin, Germany — Dmitry Ivanov
Berlin, Germany — Alexander Tsipoulanidis
Dresden, Germany — Jörn Schönberger
March 2025

Competing Interests The authors have no competing interests to declare that are relevant to the content of this manuscript.

Companion Web Site

This book is accompanied by free web sites www.prof-ivanov.de and www.global-supply-chain-management.de. On these web sites, you will find a lot of up-to-date complementary material such as slides for all chapters, case studies, Excel spreadsheet templates, tasks and answers, figures from the book, and simulation games. This area is updated continuously.

Contents

Part I Introduction to Supply Chain and Operations Management

1 Basics of Supply Chain and Operations Management 3
1.1 Supply Chains and Operations in Action 3
1.1.1 The Magic Supply Chain and the Best Operations Manager 4
1.1.2 Schneider Electric: The Best Supply Chain in 2023–2024 6
1.2 Basic Definitions and Decisions 9
1.2.1 The Transformation Process, Value Creation, and Operations Function 9
1.2.2 Supply Chain Management 12
1.2.3 Decisions in Supply Chain and Operations Management . . . 14
1.3 Careers and Future Challenges in Supply Chain and Operations Management 16
1.4 Key Points 20
Bibliography 22

2 Examples from Different Industries, Services, and Continents 25
2.1 Examples of Operations and Supply Chains in Manufacturing 25
2.1.1 Nike: Sourcing Strategy in the Integrated Supply Chain 25
2.1.2 AfriGlobal: Establishing a Sophisticated Supply Chain Management System in Africa 27
2.1.3 Tesla Gigafactory Berlin-Brandenburg 28
2.2 Examples of Operations and Supply Chains in Services 30
2.2.1 SCOM in Restaurants: Case Study Starbucks Corporation . . 30
2.2.2 Beyond Transit Management at Changi Airport in Singapore 31
2.2.3 Operations Issues in Car-Sharing 33
2.3 Examples of Digital Supply Chains and Smart Operations 35
2.3.1 Amazon Robots 35
2.3.2 Vestas Wind Systems: Revolutionizing Wind Turbine Supply Chain with 3D Printing 36

2.3.3 Energy-Efficient Car Designs with Machine Learning at Volkswagen . . . 37
2.3.4 DHL: Artificial Intelligence for Supply Chain Resilience . . . 38
2.4 Examples of Resilient Supply Chains and Operations . . . 39
2.4.1 Ford: Repurposing the Supply Chain in a Week—From Car Manufacturing to Saving Lives During the COVID-19 Pandemic . . . 39
2.4.2 Toyota: Supply Chain Disruption Management . . . 40
2.4.3 Time-Critical Supply Chains: Disaster Management and Humanitarian Logistics . . . 41
2.4.4 Resilient Sourcing Strategy Adoption at AGCO Corporation: Reacting to COVID-19 Virus Outbreak . . . 43
2.4.5 Capacity Disruption at BASF . . . 44
2.5 Examples of Sustainable Supply Chains and Operations . . . 45
2.5.1 Circular Supply Chain at Unilever . . . 45
2.5.2 Coca-Cola' Sustainable Supply Chain . . . 46
2.5.3 Circular Economy: Worth Buying Twice with Déjà Vu . . . 48
2.5.4 The COVID-19 Vaccine Supply Chain . . . 49
2.6 Examples of Artificial Intelligence in Supply Chains and Operations . . . 49
Bibliography . . . 50

3 Processes, Systems, and Models . . . 57
3.1 Introductory Case Study: AirSupply . . . 57
3.1.1 E-Procurement . . . 58
3.1.2 Vendor-Managed Inventory . . . 59
3.1.3 Implementation . . . 60
3.2 Business Process Management . . . 61
3.2.1 Process Optimization and Reengineering . . . 61
3.2.2 Business Process Modeling . . . 63
3.3 Management Information Systems . . . 65
3.3.1 Role of Information Technology in Supply Chain and Operations Management . . . 65
3.3.2 Types of Management Information Systems . . . 66
3.3.3 Management Information Systems and Organization . . . 67
3.3.4 ERP Systems . . . 68
3.3.5 APS Systems . . . 69
3.3.6 SCEM and RFID . . . 71
3.3.7 Business Analytics and E-Business . . . 72
3.4 Problem Solving Methods and Research Methodologies . . . 75
3.4.1 Problems, Systems, and Decision-Making . . . 75
3.4.2 Models and Modeling . . . 78
3.4.3 Model-Based Decision-Making . . . 79
3.4.4 Principles of Decision-Making in SCOM . . . 80
3.4.5 Quantitative Models and Operations Research . . . 82

3.4.6 Integrated and Digital Decision-Making Support 83
3.4.7 Research Methodologies 86
3.5 Integration of Business Analytics, Simulation, and Optimization . . . 87
3.6 Key Points 90
Bibliography 90

Part II Designing Operations and Supply Network: Strategic Perspective

4 Operations and Supply Chain Strategy 97
4.1 Introductory Case Study: "Quick and Affordable" Zara, UNIQLO & Primark 97
4.1.1 Zara's Three Success Factors: Speed, Speed, and Speed. . . . 97
4.1.2 UNIQLO: Basic, Casual Wear at Top Quality 99
4.1.3 Primark: It Is All About Money 100
4.2 Operations and Supply Chain Strategies 101
4.2.1 Value Added and Costs 101
4.2.2 Operations Strategies 101
4.2.3 Supply Chain Strategies and "Strategic Fit" 102
4.3 Supply Chain Coordination 104
4.3.1 Bullwhip Effect 104
4.3.2 Vendor-Managed Inventory 108
4.3.3 Collaborative Planning, Forecasting, and Replenishment . . . 111
4.3.4 Supply Chain Contracting 113
4.3.5 Vertical Integration vs. Horizontal Coordination. 113
4.4 Supply Chain Resilience and Sustainability 114
4.4.1 Supply Chain Sustainability and Closed-Loop Supply Chain 115
4.4.2 Supply Chain Risks, Resilience, and Ripple Effect. 118
4.5 Supply Chain Viability. 122
4.5.1 Viable Supply Chain and Value Network Ecosystem 122
4.5.2 Reconfigurable Supply Chain 125
4.5.3 Intertwined Supply Networks 127
4.6 Key Points 128
Bibliography 128

5 Sourcing Strategy 135
5.1 Introductory Case Study: Sustainable Supplier Management at Samsung. 135
5.2 Sourcing Process and Principles 137
5.2.1 Procurement, Purchasing, and Sourcing 137
5.2.2 Sourcing Process 139
5.2.3 Make-or-Buy and Outsourcing 140
5.2.4 Organization of Sourcing Processes 143
5.3 Sourcing Strategies 144
5.3.1 Single vs. Dual and Multiple Sourcing 144
5.3.2 Local vs. Global Sourcing 145

5.3.3 Just-in-Time . . . 147
5.4 Supplier Relationship Management . . . 148
5.4.1 Strategic Supplier Analysis . . . 149
5.4.2 Supplier Selection . . . 151
5.4.3 Supplier Integration and Development . . . 153
5.5 Key Points . . . 155
Bibliography . . . 155

6 Production Strategy . . . 159
6.1 Introductory Case Study: DELL vs. Lenovo . . . 159
6.2 Postponement and Modularization . . . 163
6.2.1 Problem: Mass Production or Product Customization . . . 163
6.2.2 Principles: Postponement and Modularization . . . 164
6.2.3 Examples of Postponement Strategies . . . 165
6.2.4 Repurposing . . . 167
6.3 Push–Pull Views and Order Penetration Point . . . 168
6.4 Selection of a Production Strategy . . . 170
6.4.1 Types of Production Strategies . . . 170
6.4.2 Method: Lost-Sales Analysis . . . 175
6.5 Key Points . . . 177
Bibliography . . . 178

7 Facility Location Planning and Network Design . . . 181
7.1 Introductory Case Study: Power Pong Sports . . . 182
7.2 Supply Chain Design Framework . . . 184
7.3 Global Supply Chain Design . . . 186
7.3.1 Warehouse Location Problem and Its Formalization . . . 186
7.3.2 A Spreadsheet Approach to the WLP . . . 189
7.3.3 Branch-&-Bound: How the Solver Add-In Works . . . 195
7.3.4 Capacitated WLP . . . 200
7.4 Regional Facility Location . . . 206
7.4.1 Management Problem Description . . . 208
7.4.2 A Mathematical Model of the Decision Situation . . . 208
7.4.3 Solving the Mathematical Model: Center-of-Gravity Approach . . . 209
7.5 Factor-Ranking Analysis . . . 215
7.5.1 Case-Study OTLG Germany . . . 215
7.5.2 Factor-Rating Method . . . 218
7.5.3 Utility Value Analysis . . . 220
7.6 Combining Optimization and Simulation in Supply Chain Design . . . 224
7.7 Resilient, Sustainable, and Viable Supply Chain Design . . . 226
7.8 Key Points . . . 228
Bibliography . . . 230

8 Distribution and Transportation Network Design 233
8.1 Introductory Case Study: Designing a Distribution Network 234
8.2 Generic Transport Network Structures 236
8.3 Realizing Economies of Scale in Transportation 238
8.3.1 Consolidation of Shipments 238
8.3.2 Postponement 240
8.3.3 Milk-Runs 241
8.3.4 Transshipment 243
8.4 Trade-Off-Based Transportation Network Design 248
8.5 Capacity Allocation in a Many-to-Many Network 251
8.5.1 The Transportation Problem 252
8.5.2 Decision Model 252
8.5.3 Finding the First Feasible Model Solution 253
8.5.4 Optimality Check 256
8.5.5 Solution Improvement 260
8.6 Distribution Network Design 263
8.6.1 Types of Distribution Networks 263
8.6.2 Case Study: Seven–Eleven Japan 267
8.6.3 Transportation Modes 269
8.7 Key Points 271
Bibliography 272

9 Factory Planning and Process Design 275
9.1 Introductory Case Study: Factory Planning at Tesla 275
9.2 Factory Planning 277
9.2.1 Role of Factory Planning in SCOM 277
9.2.2 Processes of Factory Planning 277
9.3 Capacity Planning 282
9.3.1 Queuing Theory 283
9.3.2 Little's Law 287
9.3.3 Bottleneck Analysis/Theory of Constraints 292
9.3.4 Drum, Buffer, Rope 292
9.3.5 Break-Even Analysis 293
9.3.6 Decision Trees 296
9.3.7 Simulation: Case-Study AnyLogic 297
9.4 Process Flow Structures 299
9.4.1 Job Shop 299
9.4.2 Batch Shop 300
9.4.3 Assembly Line 300
9.4.4 Continuous Flow 305
9.4.5 Product-Process Matrix 305
9.5 Lean Production Systems 306
9.5.1 Lean Thinking 306
9.5.2 Lean Production Principles 308
9.5.3 Lean Supply Chain 313

9.6 SCOM Excellence and Digitalization 315
9.6.1 Operational Excellence 315
9.6.2 From Operational Excellence to SCOM Excellence 316
9.6.3 Digitalization as New Driver in SCOM Excellence 316
9.7 Key Points and Discussion Questions 319
Bibliography 321

10 Layout Planning 325
10.1 Introductory Case Study: OTLG Ludwigsfelde 325
10.2 Layout Planning in Manufacturing 326
10.2.1 Fixed Position Layout 327
10.2.2 Process Flow Layout 328
10.2.3 Product Flow Layout 331
10.2.4 Cell-Based Layout 334
10.3 Layout Planning in Warehouses 337
10.3.1 Incoming Area 337
10.3.2 Storage Area 337
10.3.3 Outbound and Order Pick-Up 338
10.3.4 Layout Concepts 338
10.4 Methods of Layout Planning 340
10.4.1 REL-Charts 340
10.4.2 Quadratic Assignment Problem 342
10.4.3 Simulation: Modeling Operations at Pharmaceutical Distribution Warehouses with AnyLogic 343
10.5 Key Points 345
10.6 Discussion 346
Bibliography 346

Part III Matching Demand and Supply: Tactical and Operative Planning

11 Demand Forecasting 351
11.1 Introductory Case Study 351
11.2 Forecasting Process and Methods 354
11.2.1 The Forecasting Process and Time Horizons 354
11.2.2 Forecasting Methods 356
11.2.3 Forecasting Quality 358
11.3 Statistical Methods 359
11.3.1 Linear Regression 359
11.3.2 Moving Average 361
11.3.3 Simple Exponential Smoothing 362
11.3.4 Double Exponential Smoothing 364
11.4 Case study: Demand forecast using machine learning 365
11.5 Key Points and Outlook 372
Bibliography 373

12 Production and Material Requirements Planning 375
12.1 Introductory Case Study: Integrated Operations and Supply Chain Planning 375
12.2 Planning Horizons/MRP-II 377
12.3 Sales and Operations Planning. 379
12.3.1 Role of Sales and Operations Planning 379
12.3.2 Options for Aggregate Planning. 380
12.3.3 Methods for Aggregate Planning 382
12.4 Sales and Production Planning with Linear Programing 383
12.4.1 Problem Description 383
12.4.2 Method: Linear Programming 385
12.4.3 Graphical Solution 386
12.5 Master Production Schedule and Rolling Planning 389
12.5.1 Master Production Schedule 389
12.5.2 Rolling Planning 389
12.6 Material Requirements Planning 390
12.6.1 Bill-of-Materials 391
12.6.2 MRP Calculation 393
12.7 Key Points 397
Bibliography 398

13 Inventory Management 399
13.1 Introductory Case Study: Amazon, Volkswagen, and DELL. 399
13.2 Role, Functions, and Types of Inventory 400
13.3 Material Analysis 403
13.3.1 ABC Analysis 403
13.3.2 XYZ Analysis. 404
13.4 Deterministic Models. 408
13.4.1 EOQ Model 408
13.4.2 EOQ Model with Discounts. 412
13.4.3 EPQ Model. 413
13.4.4 Reorder Point 414
13.5 Stochastic Models 415
13.5.1 Service Level and Safety Stock 415
13.5.2 Single Period Systems (“Newsvendor Problem”) 421
13.5.3 Safety Stock and Transportation Strategy 423
13.6 Inventory Control Policies 424
13.6.1 Fixed Parameters 425
13.6.2 Dynamic View 429
13.7 Dynamic Lot-Sizing Models 430
13.7.1 Least Unit Cost Heuristic. 431
13.7.2 Silver-Meal Heuristic. 432
13.7.3 Wagner–Whitin Model. 433
13.8 Aggregating Inventory 435

13.9 ATP/CTP: Available-To-Promise / Capable-To-Promise 438
13.10 Measuring Inventory Performance 439
13.11 Key Points and Outlook 442
Bibliography 444

14 Routing and Scheduling 447
14.1 Introductory Case Study: RED SEA BUS TRAVEL 448
14.2 Shortest Paths in a Network 449
14.2.1 Outline of the Shortest Path Problem in a Network 449
14.2.2 Mathematical Graphs 451
14.2.3 The SPP as Graph-Based Optimization Model 452
14.2.4 Dijkstra's Algorithm for the Identification of a Shortest S-T-Path 453
14.3 Round Trip Planning/Traveling Salesman Problem 456
14.3.1 Traveling Salesman Problem 456
14.3.2 A Mixed-Integer Linear Program for TSP-Modeling 459
14.3.3 Heuristic Search for High-Quality Round Trips 461
14.4 Vehicle Routing 467
14.4.1 Case Study ORION: Vehicle Routing at UPS 468
14.4.2 Decision Situation Outline 469
14.4.3 Current Approach for the Route Compilation 471
14.4.4 Capacitated Vehicle Routing Problem 472
14.4.5 The Sweep Algorithm 476
14.5 Machine Scheduling 480
14.5.1 The Problem of Scheduling a Machine 480
14.5.2 Priority Rule-Based Scheduling 483
14.5.3 Scheduling Algorithm of Moore 485
14.5.4 Scheduling Two Machines in a Flow Shop 487
14.5.5 Further Challenges in Machine Scheduling 488
14.6 Key Points 490
Bibliography 491

Part IV Advanced Topics in Supply Chain and Operations Management

15 Supply Chain Risk Management and Resilience 497
15.1 Introductory Case-Study: Toyota's Resilience to the Global Semiconductor Shortage 497
15.2 Uncertainty and Risks 499
15.3 Risk Management in the Supply Chain 502
15.3.1 Risk Classification 502
15.3.2 General Framework of Risk Management in the Supply Chain 504
15.4 Operational and Disruption Risks 506
15.5 Ripple Effect in the Supply Chain 509
15.6 Supply Chain Resilience 511

15.6.1 Resilience Framework . . . 511
15.6.2 Costs of Resilience . . . 514
15.7 Measuring Supply Chain Risks, Resilience, and the Ripple Effect . . . 518
15.7.1 Operational Risks . . . 518
15.7.2 Disruption Risks . . . 519
15.7.3 Resilience Quantification . . . 520
15.7.4 Ripple Effect Quantification . . . 523
15.8 Super Disruptions and Supply Chain Viability . . . 526
15.8.1 Super Disruptions . . . 526
15.8.2 Supply Chain Viability . . . 527
15.9 Key Points and Discussion Questions . . . 529
Bibliography . . . 530

16 Digital Supply Chain, Smart Operations, Artificial Intelligence, and Industry 4.0 . . . 539
16.1 Introductory Case Study: SupplyOn . . . 539
16.2 Development of Technology in SCOM . . . 541
16.2.1 Three Industrial Revolutions and Industry 4.0 Formation . . . 541
16.2.2 Fourth Industrial Revolution: Industry 4.0 . . . 543
16.2.3 Internet of Things . . . 545
16.2.4 Cyber-Physical Systems . . . 546
16.2.5 Smart, Connected Products . . . 546
16.2.6 Smart Supply Chains and Smart Value-Adding Networks . . . 548
16.3 Digital SCOM Framework . . . 549
16.4 Digital Technology in the "Plan" Processes . . . 550
16.4.1 Big Data Analytics . . . 550
16.4.2 The Digital Twin . . . 551
16.5 Digital Technology in "Source" Processes . . . 554
16.5.1 eProcurement . . . 554
16.5.2 Supplier Collaboration Portals . . . 555
16.5.3 Digital Trends for Excellence in Sourcing . . . 556
16.5.4 Blockchain . . . 556
16.5.5 Robotic Process Automation and Artificial Intelligence in Procurement . . . 557
16.6 Digital Technology in "Make" Processes . . . 559
16.6.1 3D Printing and Additive Manufacturing . . . 559
16.6.2 Virtual Reality and Augmented Reality . . . 561
16.6.3 Robotics . . . 562
16.7 Digital Technology in the "Delivery" Processes . . . 565
16.7.1 Drones or Unmanned Aerial Vehicles . . . 566
16.7.2 Smart Driverless Transportation Systems . . . 566
16.7.3 Smart Forklifts, Pallet Movers, and Cranes . . . 567

16.8 Possible Obstacles and Limitations of Digital SCOM. 569
16.9 Artificial Intelligence in SCOM. 571
16.10 Case Study "Smart Usage of Big Data Along the Supply Chain: Big Data Analytics Between Companies, Value-Added and the Impact of Risk and Complexity Management" 573
16.11 Key Points and Discussion Questions 576
Bibliography 579

17 Pricing and Revenue-Oriented Capacity Allocation 587
17.1 Case Study: FRISIA COASTAL SHIPPING: The Story of Jördis and Tjark 589
17.2 Noncompetitive Pricing 593
17.3 Pricing with Scarce Capacities. 598
17.4 Setting Optimal Prices in Resource Networks. 603
17.5 Dynamic Pricing: Pricing in Reaction to Observed Market Developments. 608
17.5.1 Business Extension by Coastal Tours 609
17.5.2 Decision Situation Modeling 613
17.5.3 Pricing and Capacity Distribution Over the Sales Period 615
17.5.4 Summary of Dynamic Pricing 624
17.6 Pricing Lessons Learned and Open Issues. 628
Bibliography 629

18 Yield Management 631
18.1 Flying on Rail Tracks Across Europe 632
18.2 Sales Operations by FORTE-RailServices. 634
18.3 Yield Management: Requirements and Opportunities. 638
18.3.1 Requirements 638
18.3.2 Opportunities 638
18.4 Challenges Associated with Yield Management 639
18.4.1 Deciding Upon the Opening and Closure of a Booking Class. 640
18.4.2 Network-Based Products 641
18.4.3 Uncertainty of Demand 642
18.4.4 Dynamicity: Nesting and Protection 643
18.5 Static Decision Tasks in Yield Management 644
18.5.1 Single Resource Situation 644
18.5.2 Resource Networks 648
18.5.3 From Quotas to Booking Limits 652
18.6 Dynamic Yield Management 656
18.6.1 Core Procedure for Online Decisions About Acceptance and Rejection 656
18.6.2 Bid-Price-Oriented Demand Stream Handling 657
18.6.3 Indirect Control and Booking Limit Adaptation 658

18.7 Yield Management Lessons Learned and Open Issues 667
18.7.1 Summary of Lessons Learned 667
18.7.2 Further Aspects of Yield Management 667
Bibliography 668

Appendix: Case-Study "Re-designing the Material Flow in a Global Manufacturing Network" 669

Index 675

About the Authors

Dmitry Ivanov Prof. Dr. Dr. habil. Dmitry Ivanov teaches Supply Chain and Operations Management at Berlin School of Economics and Law (HWR Berlin) where he leads the Digital Supply Chain Lab and is academic director of M.A. Global Supply Chain and Operations Management.

Prof. Ivanov is an internationally renowned expert in supply chain and operations management, industrial and control engineering, and artificial intelligence. He has made fundamental and influential contributions, particularly exploring structural dynamics and control in complex networks with applications to supply chain resilience, Industry 4.0, supply chain simulation, risk analytics, and digital supply chain twins. He is co-author of structural dynamics control method. His research coined several seminal academic and practical directions such as the ripple effect in supply chains and supply chain viability. He has applied different methodologies such as optimization, simulation, control theory, and artificial intelligence to a variety of supply chain and operations management problems. Most of his research stems from real practical context and focuses on the interface of supply chain management, operations research, industrial engineering, and digital technology.

He has been teaching courses in operations management, supply chain management, logistics, management information systems, and strategic management at undergraduate, master's, PhD, and executive MBA levels at different universities worldwide for about 25 years. He has given guest lectures and webinars, presented scholarly papers and has been a visiting professor at numerous universities across the globe. Author of anyLogistix educational complex for supply chain optimization and simulation that is in use at more than 300 universities worldwide. He enjoys being part of students' learning experience and creating a safe and active learning environment. He sees his job as an educator to equip future industry leaders with working knowledge and skills in management and technology that would shape more resilient, adaptable, and sustainable supply chains and operations.

His multi-disciplinary academic background includes industrial engineering and management, operations research, and applied control theory. His educational background includes industrial engineering and management during five years of diploma studies along with supply chain management, artificial intelligence, information systems, simulation, and control theory during PhD and habilitation. He gained PhD (Dr.rer.pol.), Doctor of Science (ScD), and Habilitation (Dr. habil.) degrees. Prior to becoming an academic, he was mainly engaged in industry and consulting, especially for process optimization in manufacturing and logistics and ERP systems. His practical expertise includes numerous projects on the application of operations research and process optimization methods to operations design, logistics, scheduling, and supply chain management. Prior to joining the Berlin School of Economics and Law, he was professor and interim chair of Operations Management at University of Hamburg.

Professor Ivanov's research makes impact. His papers belong to the most cited worldwide in the areas of supply chain resilience and digital supply chain. For example, he has been the most cited author of premier journals International Journal of Production Research and Transportation Research: Part E in 2020-2024. His research record includes over 480 publications, with over 180 papers in prestigious academic journals, multiple editions of the leading textbooks "Global Supply

Chain and Operations Management," "Introduction to Supply Chain Resilience," "Introduction to Operations and Supply Chain Simulation with AnyLogic," and "Introduction to Supply Chain Analytics", and several research books such as "Digital Supply Chain", "Supply Networks Dynamics and Control", "Structural Dynamics and Resilience in Supply Chain Risk Management", "Scheduling in Industry 4.0 and Cloud Manufacturing", and "Handbook of Ripple Effects in the Supply Chain". Professor Ivanov's research has been published in various academic journals, including Annals of Operations Research, Annual Reviews in Control, Computers and Industrial Engineering, European Journal of Operational Research, Expert Systems with Applications, IEEE Transactions on Engineering Management, IISE Transactions, International Journal of Information Management, International Journal of Integrated Supply Management, International Journal of Inventory Research, International Journal of Physical Distribution & Logistics Management, International Journal of Production Research, International Journal of Production Economics, International Journal of Technology Management, International Journal of Systems Science, International Transactions in Operational Research, Journal of Intelligent Manufacturing, Journal of Purchasing and Supply Management, Journal of Operational Research Society, Journal of Scheduling, Omega, Production Planning and Control, Supply Chain Management: International Journal, Transportation Research: Part E, etc.

He is a recipient of IISE Transactions Best Paper Award 2021, Best Paper and Most Cited Paper Awards of International Journal of Production Research (2018-2025), OMEGA Best Paper Award 2022, Annual Reviewer Award of International Journal of Production Economics (2020), Clarivate Highly Cited Researcher Award (2021-2025). Ranked #1 in the "Top Scientists in Business and Management in Germany, Austria, and Switzerland" ranking in 2024 published in the Wirtschaftwoche. Ranked #1 in Operations Research in Standford-Elsevier 2% Top Scientists List (2022-2025). In 2024, ScholarGPS ranked Dr. Ivanov a Highly Ranked Scholar #1 in the prior 5-years and # 2 in the lifetime rankings in the Supply Chain area, and #1 in the 5-years ranking in Control (Management) area. Listed in Top 100 Scientists in Berlin in 2025.

His research projects have been supported by funding from European Commission (Horizon 2020), DFG (German Research Foundation), IFAF Berlin, and industry. Principal investigator in several research projects on resilience and digital twins including European projects ACCURATE and CERERE (2023-2027) and the DFG Collaborative Research Cluster on Resilience of Global Supply Chains at HWR Berlin (2024-2029).

Dr. Ivanov is profiled in supply chain and operations management, operations research, and industrial engineering. He is passionate about bridging the knowledge of different disciplines and applying it to solution of practically relevant problems. He delivered numerous invited plenary, keynote, and panel talks at the conferences and global webinars of INFORMS, POMS, IFAC, IFIP, IFPR, IEEE, CIE, DSI, and Academy of Management. He is leading working groups, tracks, and sessions on the Digital Supply Chain, Supply Chain Risk Management and Resilience in global research communities. He is Editor-in-Chief of International Journal of Integrated Supply Management, Editor of Annals of Operations Research, and an Associate Editor in International Journal of Production Research, International Transactions in Operational Research, International Journal of Systems Science, and OMEGA. He is an editorial board member, associate and guest-editor in different journals, including Annals of Operations Research, Annual Reviews in Control, IISE Transactions, International Journal of Production Economics, International Journal of Production Research, International Transactions in Operational Research, International Journal of Information Management, International Journal of Inventory Research, International Journal of Physical Distribution and Logistics Management, International Journal of Systems Science, and OMEGA. He is Chairman of IFAC CC5 "Cyber-Physical Manufacturing Enterprise", former Chairman of IFAC TC 5.2 "Manufacturing Modelling for Management and Control" (2017-2023), and Co-Chairman of the IFAC TC 5.2 Working group "Supply Network Engineering". He regularly presents his research results and has been Chairman, IPC and Advisory Board member of over 70 international conferences in supply chain and operations management, industrial engineering, control and information sciences where he has organized numerous tracks and sessions (including IFAC MIM, IFAC INCOM, EURO, INFORMS, IISE, IFORS, DSI, POMS, OR, IFAC World Congress, IFIP PRO-VE, LDIC, to name a few).

He has been General Conference Chair of IFAC MIM 2019—one of the worldwide largest conferences in manufacturing, industrial engineering, operations, and supply chain management (750 participants).

His personal homepages are blog.hwr-berlin.de/Ivanov and prof-ivanov.de

Jörn Schönberger is professor of business management at the Technical University of Dresden. He is the Chair of Transport Services and Logistics, as part of the Friedrich List Faculty of Transportation and Traffic Science. Prior to his current position, he was professor of operations and supply chain management at the Berlin School of Economics and Law (BSEL) and senior researcher at the University of Bremen. For several years, he was a guest professor at the University of Rennes 1 for IT and logistics. In 2010, he was a temporary professor (sine spe) of operations research and supply chain management at RWTH Aachen.

Jörn Schönberger was visiting professor at the University of Vienna and visiting researcher at Copenhagen Business School. In addition, he is/was involved in several academic programs on different study levels at the University of Bremen, Technical University of Berlin, RWTH Aachen, University of Vienna, University of Rennes 1, European Business School, FOM University of Applied Sciences, University of Applied Sciences Bremen, Hamburg Distance University of Applied Sciences, and University of Applied Sciences HIWL. Most of his courses address topics at the interface between engineering and business management.

His research interests comprise model-based optimization and control of complex logistics systems. Furthermore, he investigates the synchronization of information and material flows. Another research direction covers the dynamics of complex logistics systems. Based on methodologies from mathematics, operations research, artificial intelligence, and information systems, he investigates applications from the fields of logistics, transportation, and traffic, manufacturing, and sports management.

Jörn Schönberger received a diploma in mathematics in 2000 and a PhD (2004) in business management from the University of Bremen. His doctoral thesis was awarded by the German Operations Research Society. He gained his Habilitation degree in business management also from the University of Bremen in 2010.

He is author of two monographs and three textbooks as well as editor of several books. He (co) authored several journal papers and numerous other publications. He regularly presents his research results at international scientific meetings. His service to the scientific community comprises the organization of several conferences as well as his work as a regular reviewer for the Deutsche Forschungsgemeinschaft (German Research Foundation) and several scientific journals as well as conferences. Jörn Schönberger is member of several scientific communities like the German Academic Association for Business Research (VHB) and the German Operations Research Society (GOR). Currently, he is the chairman of the section "Logistics" of the German Academic Association for Business Research (VHB).

Besides his academic career, he regularly works as a freelance consultant. He is/was involved in projects related to information management, information processing, database development and application, and process optimization.

Alexander Tsipoulanidis, MBA, is professor of supply chain and operations management at the Berlin School of Economics and Law (HWR Berlin) and faculty director of the MA Global Supply Chain and Operations Management program at HWR Berlin. He is lecturing on supply chain and operations management in various programs leading to bachelor's, master's, or MBA degrees. Alexander is mechanical engineer and industrial engineer with a focus on factory, layout, and material flow planning. At the Engineering Faculty of the University of Bristol (UK), he proceeded with his doctorate (PhD) in which he explored the significant influence of lean engineering and product development as a part of the lean enterprise. He holds an MBA in strategic management from Anglia Ruskin University Cambridge (UK), where he concentrated on competitive strategies in times of global hypercompetition.

During a Federal Ministry of Education and Research (BMBF) project that was scientifically supported by the Institute for Machine Tools and Factory Management (IWF) of the TU Berlin and the Fraunhofer Institute for Material Flow and Logistics (IML) in Dortmund, Alexander contributed to the "Production 2000" project regarding assembly-controlled logistics and value-added network works (Entwicklung von montagegesteuerten, integrierten Logistik- und Wertschöpfungsketten; Rahmenkonzept Produktion 2000 des Bundesministeriums für Bildung und Forschung) from a materials logistics perspective. As such, he developed practical solutions for process synchronization and performance improvements within a network of small series producers.

Alexander Tsipoulanidis has been working approximately 20 years for international enterprises (e.g., ABB, DaimlerChrysler, and Bombardier Transportation). There, he had different roles at various production sites, but also had executive positions at the global headquarters. His responsibilities have been related to capacity management of a European production network, planning of new factories on four continents, or the redesign of shop-floor layouts. Globally, he was conducting "Lean Operations Assessments and Consulting" of production facilities, developed site continuous improvement plans, and implemented the related supply chain strategies. Furthermore, he defined lean and manufacturing-oriented product development processes, conducted global supply chain and operations consultancy projects, and was involved in a global restructuring initiative aiming to make the respective enterprise lean, efficient, attractive, and profitable.

His current research focuses on the fundamentals of lean thinking in combination with the usage of the digital technology in various value-added domains. Recently, he has analyzed the development of lean and sustainable supply chain strategies in strong connection with the above-mentioned technologies. Furthermore, the connection of lean thinking and Industry 4.0/5.0 has been researched, results were presented at scientific and practice-focused conferences, and respective publications have been issued. Thus, Alexander's primary research objective is to explore the generation of competitive advantage by utilizing the Internet of Things in order to sustain Supply Chain and Operational Excellence in times of digitalization. Since 2021, more than 20 scientific projects with a strong focus on Digital Business and Technology Management (e.g. focusing on digital maturity assessments, AI usage in workforce staffing, digital ticketing for train operators, digital patient journey in a hospital, renewable energy for the logistics fleet in middle-mile delivery in e-commerce…) have been run together with globally active and well-recognized companies and institutions.

Part I

Introduction to Supply Chain and Operations Management

Basics of Supply Chain and Operations Management

1

Key Questions for This Chapter

- What is the transformation process and value creation?
- What is operations and operations management?
- What is a supply chain and supply chain management?
- Which decisions are within the scope of supply chain and operations management?
- What are efficiency and resilience paradigms?
- Which qualifications should a future supply chain and operations manager obtain?
- Which career paths are possible for supply chain and operations managers?

1.1 Supply Chains and Operations in Action

The job of supply chain (SC) and operations managers is exciting! What are you doing in this profession? Well, the variety of tasks is as diverse as the supply chains are. Imagine that you have an excellent product idea with great market potential. How do you organize the production of your product? How to bring it to the customers? This is what SC and operations management (SCOM) is about: *to match demand and supply in the most efficient, resilient, and sustainable fashion.* And this is true for both a start-up and a global corporation.

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_1) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_1.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_1

Imagine that you might design a global SC network for your company and deploy it at scale. Where to locate your factories and warehouses? How to organize and manage the supplier base? Not only costs matter. Can you remember how, during the COVID-19 pandemic, shelves in the stores became empty? SCs and operations should be resilient to such shocks and global crises. The SCOM offers a variety of methods for developing a resilient portfolio for SCs and operations.

SCs impact the environment (e.g., carbon emissions) and society (e.g., jobs for workers). Have you thought about how global warming will impact future SCs and operations? Sustainability is one of the key topics in SCOM. Once a global footprint is in place, we should start defining production and logistics strategies and factory planning. Some companies produce on stock, while others customize their products. What is the optimal capacity of your factories and warehouses, and which process structures and layouts fit their requirements best? You will learn approaches and methods to answer these questions.

Matching demand and supply starts with forecasting. Operations managers predict future demand using statistical and artificial intelligence (AI) methods. Forecasts trigger production planning and material acquisition. How to utilize the available production capacity best? Which product mix is the most profitable? When to place the next order release at suppliers, and how much inventory to hold? And not to forget—all these decisions happen in highly dynamic environments. You might have planned your master production schedule and placed material orders at suppliers. However, your sales department can change its demand planning ad hoc, so you should replan your manufacturing and purchasing. And what if you are in charge of transportation? How do you find the shortest routes to all your customers to reduce the number of trucks and logistics costs? SCOM methods will help you with these and many other questions.

Last but not least, SCOM is at the interface of the organization, quantitative methods, and technology. You will learn management principles, such as lean production. Quantitative methods such as mathematical programming and discrete-event simulation will provide decision-making support. Digital technology and AI will help you make informed decisions. Ready for an SCOM journey? Let us start!

1.1.1 The Magic Supply Chain and the Best Operations Manager

Santa Claus is one of the world's best SC and operations managers. He achieves incredible *performance*: He always delivers the right products to the right place at the right time, despite highly *uncertain demand* and a very complex SC with more than two billion customers.

His *strategy* and *organization* are customer-centered and strives to provide maximal satisfaction to children. The organization of his supply chain and operations management (SCOM) is structured as follows. The customer department is responsible for processing all the letters from children all over the world. This *demand* data is then given to the supply department. The *supply* department is responsible for buying the desired items from suppliers worldwide. The core of the supply department is the global purchasing team, which is responsible for *coordinating* all the *global* purchasing activities.

Since many of the children's wishes are country-specific, the regional *purchasing* departments (so-called lead buyers) are distributed worldwide and build optimal *SC design*. In some cases, the desired items are so specific that no supplier can be found. For such cases, Santa Claus has established some globally located production facilities to minimize total *transportation* costs and ensure on-time delivery of all Christmas gifts.

The customer department regularly analyzes the children's wishes. They noticed that there are a lot of similar items which are requested each year. In order to reduce purchasing fixed costs and use *scale effects*, Santa Claus organized a network of *warehouses* worldwide. Standard items are purchased in large batches and stored. If the actual demand in the current year is lower than the *forecasted demand*, this is not a problem—these items can be used again in the following year. Since there are millions of different items in each warehouse, Santa Claus created optimal *layouts* and *pick-up* processes to find the necessary items quickly and efficiently.

The SC and operations planning happens as follows. In January, Santa Claus and the customer department start to analyze the previous year's demand. During the first 6 months of the year, they create a projection of future demand. The basis for such a forecast is statistical analysis of the past and identification of future trends (e.g., new books, films, toys, etc.). After that, the supply department replenishes the items and distributes them to different warehouses. The production department *schedules* the *manufacturing* processes. From October, the first letters from children start to arrive. The busy period begins. From October to December, Santa Claus needs many assistants and enlarges the workforce.

Operations and SC execution are now responsible for bringing all the desired items to the children. It comprises many activities: transportation, purchasing, and manufacturing. Children are waiting impatiently to start the incoming goods inspection. No wrong pick-ups and bundles are admissible, and no shortage is allowed. More and more, children's wishes are not about items but rather about events they want to happen (e.g., holidays, etc.). *Service operations* are also a competence of Santa Claus.

Santa's SC is very *resilient*. Sometimes, storms and hurricanes disrupt transportation and logistics flows. In this case, contingency plans are activated, and backup routes are used. Moreover, Santa's SC is increasingly *digitalized* as children write electronic letters. To react faster, information systems for SC visibility have been

established along with the use of 3D printers to produce toys quickly and reduce transportation emissions. In addition, Santa Claus has established the most *sustainable* SC in the world using *transportation* by sleds. Sometimes, letters with very unusual wishes come at the very last moment, but Santa Claus's SC is prepared for the unpredictable—lastly, it is a magic SC.

1.1.2 Schneider Electric: The Best Supply Chain in 2023–2024

Schneider Electric, ranked #1 in the Gartner Global Supply Chain TOP 25 ranking in 2023 and 2024, builds its supply chain through regionalization and localization, dual sourcing, building spare capacity in plants, and developing R&D capability in semiconductors and electronics to achieve efficiency, resilience, and sustainability. Schneider Electric is a global leader in energy management and industrial automation; it operates in 100 countries worldwide. In 2024, the company run 162 factories and 84 distribution centers; it has a complex SC, which makes efficiency a significant target for its business strategy. Global SCs have experienced an unprecedented increase in disruptions due to geopolitical risks, environmental concerns, and regulatory pressure. Schneider Electric has been attempting to improve sustainability and resilience through transforming its SC. To satisfy the growing demand for electrification and renewable energy, the company has simultaneously expanded its market share in the US.

Managing a complex SC requires significant agility and foresight approach from Schneider Electric's management. Setting up effective logistics coordination is crucial since the company's operations cover various regions around the world. Through lowering its carbon footprint, the business must further enhance sustainability. To increase resilience, Schneider Electric should improve their strategies to maintain same level of service while balancing cost efficiency. Acknowledging these difficulties, the business takes steps to improve the sustainability, resilience, and technical sophistication of its SC.

To enhance resilience, Schneider Electric is improving its SC network through advanced digital tools. For SC and inventory optimization, the organization has created end-to-end control tower capabilities. This real-time visibility improves execution dependability and enables well-informed decision-making. Schneider Electric uses SC modeling and data analytics to improve decision-making in terms of facility locations, inventory levels, and transportation routes. This enables the business to proactively create mitigation solutions and foresee possible disruptions. The business takes the most of cloud-based SC management tools, real-time tracking, and AI-driven predictive analytics. These technologies enable the company to optimize inventories, foresee possible disruptions, and rapidly adapt to changing market conditions. For instance, Schneider Electric can examine supplier performance patterns and modify procurement tactics in accordance with them by utilizing AI-driven predictive analytics. Schneider Electric uses real-time analytics to take proactive steps to reduce risks, resulting in a more efficient supply chain flow.

One of the core elements of Schneider Electric's strategy is regionalization. Schneider Electric ensures that 92% of its SC operates within the same region as its manufacturing facilities. Majority of raw materials sources are situated relatively close to production sites. This method decreases dependence on geopolitical risk factors and eliminates dependency on long-distance transportation. Regionalization strategy allows for improved inventory control, leading to optimized lead times and operational efficiency. The company's Smart Factory in Lexington, Kentucky, which uses smart manufacturing technologies to lower energy usage and boost operational efficiency, is a noteworthy example. Thanks to this regional strategy, the North American market is more resilient and self-sufficient. Through establishing a decentralized network, the business can swiftly adjust to changes in demand and be aware of possible bottlenecks that may arise from global SC dependencies.

A key priority for company is sustainability. Schneider Electric actively collaborates with suppliers to support circular economy efforts and lower emissions. The business is supporting industry-wide decarbonization initiatives by funding renewable energy projects. At the Global Sustainable Supply Chain Summit 2021, Schneider Electric was recognized as the Best Global Sustainable Supply Chain organization, demonstrating its dedication to running more equitable and environmentally friendly SC. To reduce supply shortages, Schneider Electric continually diversifies sources for essential components. By giving them information and tools to assist them cut emissions, the company actively participates in its suppliers' decarbonization initiatives. This cooperative strategy tackles Scope 3 emissions and supports Schneider Electric's net-zero objectives. Supplier risk management is also crucial. The company actively identifies components that rely on a single-source supplier and seeks additional sources. Strong partnerships with key suppliers further enhance operational stability and innovation.

To meet the rising demand for automation and renewable energy solutions, Schneider Electric continues to expand in North America, particularly in the United States. By establishing new production facilities and utilizing smart manufacturing technologies, the corporation is making investments in regional manufacturing. Clean energy is encouraged by government policy, which opens significant options. Schneider Electric is supporting modernization and the integration of renewable energy sources by utilizing its energy management expertise. The purchase of a 75% share in Motivair Corp., a U.S.-based business that specializes in liquid cooling for high-performance computers, by Schneider Electric is one example of such an expansion endeavor. This calculated action supports Schneider's growth in North America by improving its data center cooling system options. Schneider Electric also works with American businesses and governmental entities to promote green initiatives. The organization supports industry initiatives to attain environmental sustainability and ensures long-term growth in the area by offering energy-efficient solutions and developing cooperation agreements.

Schneider Electric's approach to SC transformation highlights the benefits of resilience, sustainability, regionalization, and digital technology development. By expanding digital tools, diversifying suppliers, supplier collaboration strategies and optimizing logistics, the company is building a resilient SC capable of coping with various disruptions. Its dedication to regionalization initiatives, supplier

participation in decarbonization, and smart factory development further solidifies its standing as a pioneer in resilient and sustainable SCs. Its North American expansion further reinforces its commitment to sustainable energy solutions. As global supply chains continue to face uncertainties, businesses that invest in digital innovation and sustainability will be better positioned for long-term success.

Discussion questions.

- What are the key risks Schneider Electric faces in its global SC?
- How does a regionalized SC strategy enhance resilience and efficiency?
- How can digitalization improve SC resilience?
- How can companies collaborate with suppliers in terms of sustainability?
- How does supplier diversification help in risks mitigation? Which challenges can arise from this approach?
- What are the potential trade-offs between cost reduction and resilience investment?
- What factors are driving Schneider Electric's expansion in the North American market?

Sources

Banker, S. (2024, October 30). Schneider Electric's supply chain design journey. Forbes. https://www.forbes.com/sites/stevebanker/2024/10/30/schneider-electrics-supply-chain-design-journey/

BloombergNEF (2024). Schneider Electric sees $2 trillion boost to its biggest market. BloombergNEF. https://about.bnef.com/blog/schneider-electric-sees-2-trillion-boost-to-its-biggest-market/

Schneider Electric (2024). Schneider Electric factsheet. Schneider Electric. https://www.se.com/ww/en/assets/564/document/373425/schneider-electric-factsheet.pdf

Reuters (2024). Schneider Electric to buy data centre cooling firm Motivair for $850 million. Retrieved from https://www.reuters.com/markets/deals/schneider-electric-buy-data-centre-cooling-firm-motivair-850-million-2024-10-17/

Schneider Electric (2021). Schneider Electric named as best global sustainable supply chain organization spearheading climate action throughout its ecosystem. Retrieved from https://www.se.com/ww/en/about-us/newsroom/news/press-releases/schneider-electric-named-as-best-global-sustainable-supply-chain-organization-spearheading-climate-action-throughout-its-ecosystem-60c8994c244643554f295a44/

Supply Chain Digital (2024). Schneider Electric unlocks supply chain efficiency with IoT. Retrieved from https://supplychaindigital.com/articles/schneider-electric-unlocks-supply-chain-efficiency-with-iot

Supply Chain Outlook (2024). How localising supply chains can be the key to increasing resiliency. Retrieved from https://www.supplychain-outlook.com/expert-eye/how-glocalising-supply-chains-can-be-the-key-to-increasing-resiliency/

Time (2024). Supply chain decarbonization. Retrieved from https://time.com/7086071/supply-chain-decarbonization/

1.2 Basic Definitions and Decisions

1.2.1 The Transformation Process, Value Creation, and Operations Function

Operations is a function or system that transforms inputs (e.g., materials and labor) into outputs of higher value (e.g., products or services); in other words, the operations function is responsible for *matching demand and supply* (see Fig. 1.1).

The transformation process is the traditional way of thinking about operations management in terms of *planning* activities. In practice, SC and operations managers spend at least half of their working time handling uncertainties and risks. That is why the *control* function becomes increasingly important for establishing feedback between the planned and real processes.

One of the basic elements in management is the creation of *value added*. Identifying the "*value*" of a product or service means understanding and specifying what the customer is expecting to receive. We usually understand as value added any activity in a process that is essential to deliver the service or product to the customer. Understanding what the customer wants and does not want (to pay) are key aspects in identifying value-added activities. The most effective process is achieved by performing the minimum required number of value-added steps and no wasted ones. The reduction of waste and focusing on value-added activities may increase the efficiency of input resource usage and enable the most effective output of a process. This ideal goal is almost impossible to achieve in practice because of the historical development of processes in firms and continuous changes in the firms and markets.

The *operations* function, along with marketing and finance, is a part of any organization (see Fig. 1.2).

Operations management is concerned with managing resources to produce and deliver products and services efficiently and effectively.

Operations management deals with the design and management of products, processes, and services and is comprised of four stages: sourcing, production, distribution, and after-sales.

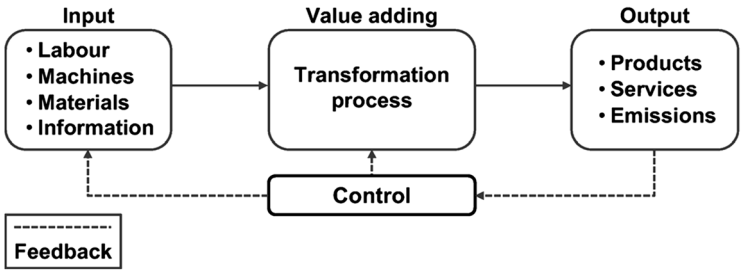


Fig. 1.1 Input-output view of the operations function

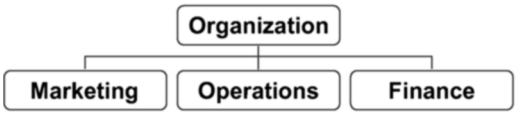


Fig. 1.2 Role of operations in an organization

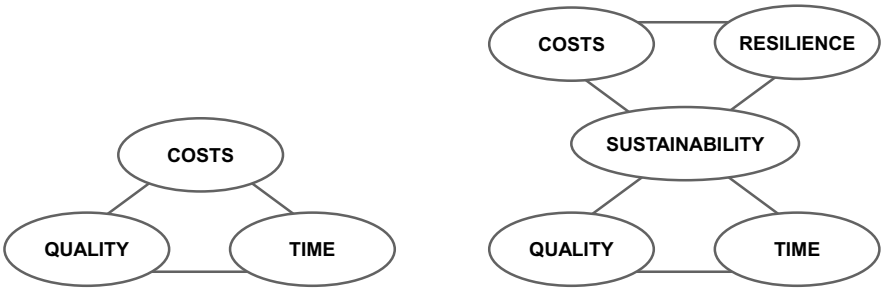


Fig. 1.3 Traditional and extended objectives triangle of operations and supply chain performance

The measurement of operations and SC *performance* is typically related to a traditional triangle of objectives, which is "costs-time-quality." In an extended framework, these three dimensions have been complemented by resilience and sustainability objectives (see Fig. 1.3).

Three major SCOM paradigms can be classified as follows:

Design-for-Efficiency: Efficient and responsive SCs and operations are coined by lean and agile principles. The key idea of such leagile operations and SC designs is to utilize the available resources (i.e., material, time, capital, technology, and workforce) at the highest possible degree of efficiency to avoid waste and maximize profitability.

Design-for-Resilience: Resilient SC and operations are designed to absorb unexpected, severe disruptions (e.g., natural disasters, fires at facilities, strikes, or pandemic outbreaks) and restore operations thereafter. Resilience helps mitigate the impact of disruptions using some redundancy (e.g., inventory, capacity buffers, or backup suppliers) and to recover to an original or even better performance later (Ivanov 2025).

Design-for-Sustainability: Sustainable supply chains and operations are designed to reduce the negative impacts of manufacturing and logistics on nature and society (e.g., CO2 emissions or unfair labor conditions). Sustainable supply chains and operations are built around the triple bottom line of "ecology–society–economics."

▶ **Practical Insights** In practice, the firms should target an integrated system that combines elements of efficiency, resilience, and sustainability. For example, PepsiCo has developed a global supply chain for beverages using coconut water. To utilize the advantages of

efficiency, PepsiCo has selected primary suppliers and manufacturers in Southeast Asia. To increase supply chain resilience in light of frequent natural disasters in this geographical region, PepsiCo established backup manufacturing capacities in other regions. Moreover, they utilized operations management models to use inventory to protect against disruptions efficiently. In terms of sustainability, PepsiCo established several initiatives around agriculture, recycling, and water usage in their sourcing regions.

The importance and domination of different objectives and paradigms are *dynamic*, i.e., they have changed over time since operations management has a long history. Over the last 60 years, a transition from a producers' market to a customers' markets has occurred. This transition began in the 1960s with the increasing role of marketing in conditions of *mass production* of similar products to an anonymous market. This period is known as the *economy of scale*. After filling the markets with products, quality problems came to the forefront. In the 1970s, total quality management (TQM) was established.

The increased quality triggered an *individualization* of customers' requirements in the 1980s. This was the launching point for the establishment of the economy of the customer. This period is characterized by efforts towards optimal inventory management and a reduction in production cycles.

In the 1980–1990s, handling high product variety challenged operations management. Another trend was the so-called *speed effect*. The speed of the reaction to market changes and cutting time-to-market became even more important. Consequently, the optimization of internal processes with external links to suppliers was simultaneously rooted in the concepts of lean production and just-in-time (JIT).

Throughout the 1990s, companies concentrated on development approaches to core competencies, outsourcing, innovations, and collaboration. These trends were caused by globalization, advancements in IT, and integration processes in the world economy. Particularly in the 1990s, a paradigm of supply chain management (SCM) was established that has shaped developments in SCOM into the twenty-first century.

From 2010 to 2025, trends such as digitalization, smart operations, artificial intelligence, predictive analytics, risk management, SC resilience and flexibility, intelligent information technologies, e-operations, leanness, and agility, the servitization of manufacturing, outsourcing, and globalization, additive manufacturing, and Industry 4.0 shaped the SCOM landscape in practice and research.

Moreover, novel frameworks of Viable Supply Chains and Reconfigurable Supply Chains have been coined (Dolgui et al. 2020; Ivanov 2022; Ivanov and Dolgui 2020). These new developments resulted in an extended SCOM framework that includes three traditional objectives (i.e., costs, time, and quality) and two new objectives, i.e., resilience and sustainability (cf. Fig. 1.3). In 2000–2015, SCs have adopted profitability-based practices with the goal of becoming lean and utilizing efficiency of globalized production and sourcing. However, such SCs became prone

to disruptions originating from natural/human-made disruptive events at different scales. Especially the COVID-19 pandemic has clearly shown the vulnerability of efficiency-oriented SCs. Numerous SC disruption examples clearly suggested the necessity of a new perspective on designing SC networks that are efficient in normal situations, yet resilient during crises. Traditional "Design-for-Efficiency" SCOM methods have been extended by "Design-for-Resilience" paradigm. We will discuss SC resilience and viability issues in detail in Chaps. 4 and 15. Moreover, SCs have very high impact on nature and society. Thus far, sustainability has become a major SCOM determinant.

1.2.2 Supply Chain Management

A *supply chain* is a network of organizations and processes wherein various enterprises (suppliers, manufacturers, distributors, and retailers) collaborate (cooperate and coordinate) along the entire value chain to acquire raw materials, convert these raw materials into specified final products, and deliver these final products to customers (see Fig. 1.4).

Supply chain management (SCM) is the integration and coordination of material, information, and financial flows across departments and enterprises to transform and use SC resources in the most rational way throughout the entire value chain, from raw material suppliers to customers. SCM is one of the key components of any organization and is responsible for balancing demand and supply along the entire value-adding chain (see Fig. 1.5).

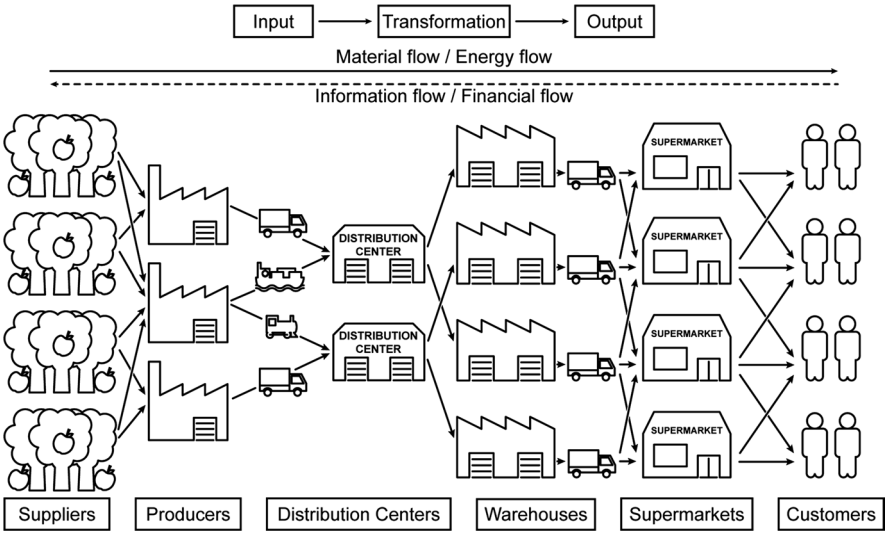


Fig. 1.4 Supply chain

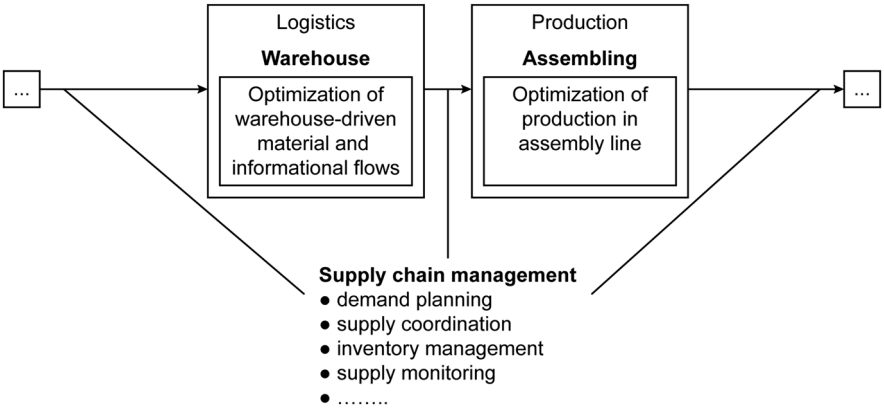


Fig. 1.5 Functions of logistics, production, and SCM in a value chain (from Ivanov and Sokolov 2010)

SCM integrates production and logistics processes. In practice, production, logistics, and SCM problems interact with each other and are tightly interlinked. Only two decades have passed since enterprise management and organizational structure have been considered from the functional perspective: marketing, research and development, procurement, warehousing, manufacturing, sales, and finance. The development of SCM was driven in the 1990s by three *main trends*: customer orientation, globalization of markets, and establishing an information society. These trends caused changes in the competitive strategies of enterprises and required new value chain management concepts.

The first use of the term "SCM" occurred in the article "SCM: Logistics Catches up with Strategy" by Oliver and Webber (1982). They set out to examine material flows from the raw material suppliers through the SC to the end consumers within an integrated framework that is now called SCM. The origins of SCM can be seen in early works on the postponement, system dynamics and the bullwhip effect (Forrester 1961), cooperation (Bowersox 1969), multi-echelon inventory management (Geoffrion and Graves 1974), JIT, and lean production. In the 1990s, SCM represented a managerial innovation due to its new way of integrated system thinking.

SCM, as the term implies, is primarily focused on the inter-organizational level. Another successful application of SCM depends on intra-organizational changes to a considerable extent. Even collaborative processes with extended information system applications are managed by people who work in different departments: marketing, procurement, sales, and production. The interests of these departments are usually in conflict with each other. Hence, not only outbound synchronizations but also internal organizational synchronizations are encompassed by the SC organization.

1.2.3 Decisions in Supply Chain and Operations Management

The main management task is making decisions. Decisions in SC and operations management (SCOM) are primarily directed towards matching demand and supply by designing and operating the transformation process in the most efficient way (Ivanov 2024). Put simply: SCOM is building a bridge between customers and suppliers (Fig. 1.6).

Consider an example. A *Chocolate SC* can be used as an illustration for depicting basic decisions in SCOM. What happens between the cocoa pod harvest and the placement of the chocolate on supermarket shelves? A complex SC is built to connect raw material suppliers and end customers (see Fig. 1.7).

To produce chocolate, cocoa pods are first harvested from cocoa trees, e.g., in Côte d'Ivoire. Cocoa pods are then moved to a processing station where they are packed into special carrier bags to avoid damage during transportation by container ship. At the harbor, the bags are packed into special containers and moved to a container ship that will bring them, e.g., to Hamburg.

Along with sugar and milk powder, vanilla is also one of the components needed for chocolate production. Sugar and milk powder are usually sourced locally to avoid long transportation. After unloading at the Hamburg harbor, through which up to 200,000 tons of cocoa pods are shipped each year, the transportation is continued by trucks. Simultaneously, container ships in Guatemala and Madagascar leave the harbor with cargoes full of vanilla.

The cocoa pods are delivered to a preliminary processing plant and separated into cocoa butter and cocoa mass, which are then moved in trucks by road to chocolate manufacturers. After getting all the ingredients, a multi-stage manufacturing process is started, the final result of which is chocolate. Chocolate is then packed and delivered to distribution centers in large batches on pallets. From here, small batches are finally delivered to the supermarket.

Fig. 1.6 SCOM as a bridge function for matching demand and supply

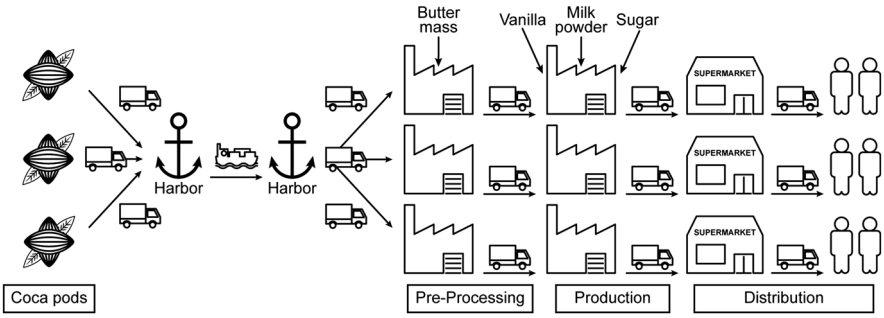


Fig. 1.7 Chocolate supply chain

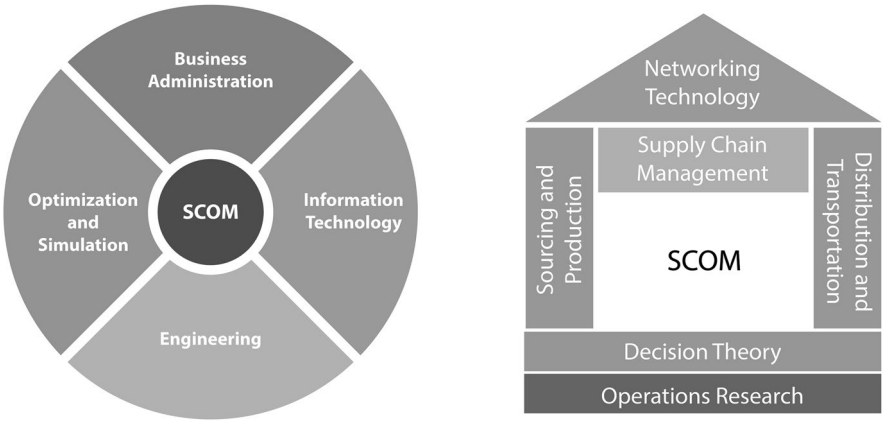


Fig. 1.8 House of SCOM

From the chocolate SC and Santa Claus's SC, it can be observed that the responsibilities of SCOM managers can be divided into sourcing, production, distribution, transportation, and SCM subject to business administration, information systems, optimization and simulation, and engineering (see Fig. 1.8).

The responsibilities of SCOM managers are multi-faceted (see Fig. 1.9). The decision-making areas in SCOM range from strategic to tactical and operative levels (Ivanov 2010). *Strategic* issues include, for example, determination of the size and location of manufacturing plants or distribution centers, decisions on the structure of service networks, factory planning, and design of the SC. *Tactical* issues include such decisions about production, transportation, and inventory planning. *Operative* issues involve production scheduling and control, inventory control, quality control, and inspection, vehicle routing, traffic and materials handling, and equipment maintenance policies.

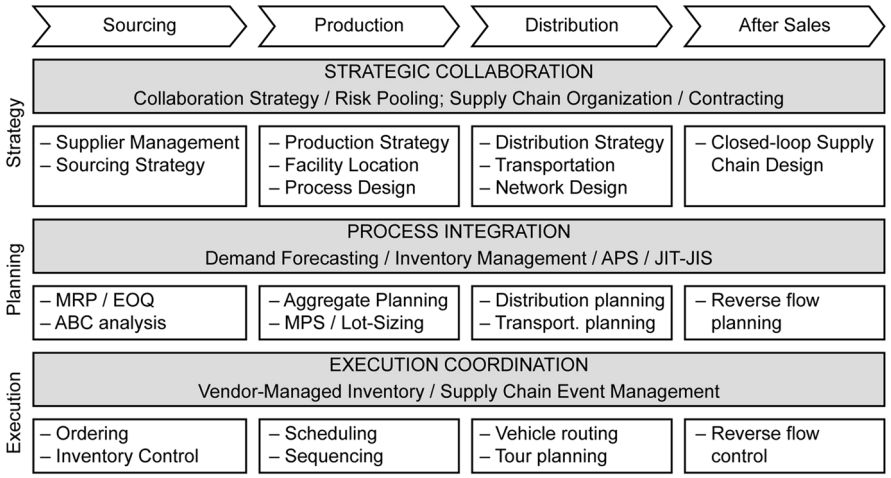


Fig. 1.9 Decision matrix in supply chain and operations management

This description holds true for many different organizations, including global brand manufacturers such as Apple or Toyota, major retailers such as Tesco or Wal-Mart, nonprofit organizations such as International Red Cross, or local petrol stations or hospitals. Purchasing, assembly, shipping, stocking, and even communicating are a few examples of the many different actions unfolding within these organizations, all united by a single purpose: to create value for a customer.

Moreover, SCOM methods are vital to designing and running firms and SCs resiliently (i.e., able to cope with disruptions and recover). One example is the COVID-19 pandemic, which required unprecedented efforts from SC and operations managers.

1.3 Careers and Future Challenges in Supply Chain and Operations Management

SCOM is everywhere! SCOM specialists should obtain different qualifications since SCOM is multidisciplinary in its nature. There are many *skills* and attributes that are important for SC and operations managers to succeed in the operations environment. These are:

- Systems and engineering knowledge
- Leadership and strong communication skills
- General multidisciplinary business knowledge
- Strong analytical problem-solving abilities and quantitative skills
- Negotiation and presentation skills.

To become an SC or operations manager, it is important to communicate well with people from all departments. The SCOM function is strongly related to other core and support functions of the organization, such as accounting and finance, product development, human resources, information systems, and marketing functions. Another important skill is time management.

Some of the *job responsibilities* common to SCOM are as follows:

- Global supply chain design
- Demand forecasting and capacity planning
- Production, pricing, sales, and distribution of products and services
- Managing the movement of goods into and out of production facilities
- Selecting suppliers and negotiations with them
- Managing inventory and collaborating with suppliers
- Planning warehouse and store layouts and designing production processes
- Process selection: design and implement the transformation processes that best meet the needs of the customers and firms
- Logistics: managing the movement of goods throughout the SC
- Risk manager: proactive SC design, risk monitoring, and real-time coordination in the event of disruptions
- Managing workforce, preparing schedules, and assigning specific duties
- Productivity and performance management
- Quality management

SC and operations managers have strategic responsibility but also control many of the everyday functions of a business or organization. They oversee and manage goods used at the facility, such as sales merchandise, inventory, or production materials. SC and operations managers also authorize and approve vendors and contract services at different locations worldwide. Starting positions for SCOM typically include operative responsibilities in procurement or sales departments and work as, for example, a consultant, purchasing manager, or SC analyst. With 5–10 years of practical experience, such positions as transportation manager, international logistics manager, warehouse operations manager, or SC manager can be achieved. With 10 or more years of experience, the production director or vice president of SCOM is a realistic position.

SCOM positions may include jobs in production planning, inventory control, materials control, work scheduling, quality control, and operations analysis. There are a wide variety of career options in the field of SCOM. Some key opportunity areas are SC manager, operations manager, business analyst, production planner, operations analyst, materials manager, quality control specialist, project manager, purchasing manager, industrial production manager, facility coordinator, logistics manager, and risk manager.

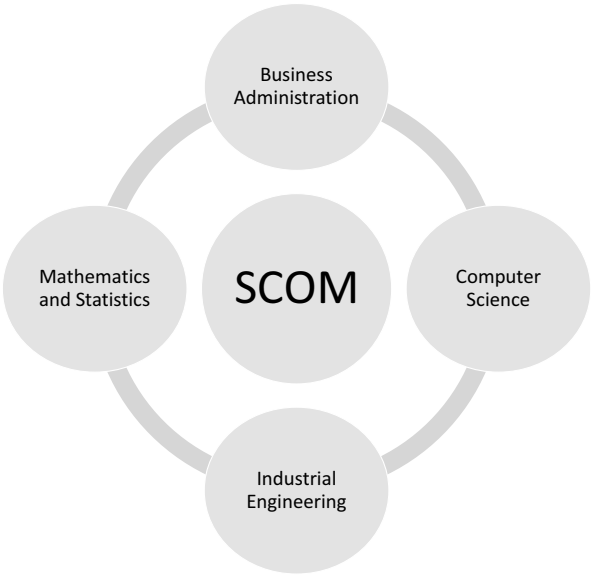


Fig. 1.10 Multidisciplinary perspective on SCOM education

SCOM education requires a multidisciplinary approach (Fig. 1.10). Managing SCs and operations in the twenty-first century is composed of knowledge and skills from business administration, industrial engineering, mathematics and statistics, and computer science. Through different educational stages (e.g., bachelor, master, industrial internship, and PhD), one should try obtain qualifications in these four areas and so develop a complete SCOM profile for a successful career. In addition, building of successful SCOM teams is also based on combining the four qualification areas through a multidisciplinary team composition and dynamic reallocation of the team leadership depending on the tasks.

Consider some of these and other possible *careers in SCOM*:

- *Plant managers* supervise and organize the daily operations of manufacturing plants. For this position, expertise in activities such as production planning, purchasing, and inventory management is needed.
- *Quality managers* aim to ensure that the product or service an organization provides is fit for purpose, consistent, and meets both external and internal requirements. This includes legal compliance and customer expectations. A quality manager, sometimes called a quality assurance manager, coordinates the activities required to meet quality standards. Statistical tools are required to monitor all aspects of services, timeliness, and workload management.
- *Process improvement consultants* take over activities, including designing and implementing such activities as lean production, six sigma, and cycle time reduction plans in both service and manufacturing processes.

- *Analysts* are key members of the operations team supporting data management, client reporting, trade processes, and problem resolution. They use analytical and quantitative methods to understand, predict, and improve SC processes.
- *Production managers* are involved in planning, coordinating, and controlling manufacturing processes. They ensure that goods and services are produced at the right cost and quality.
- *Service managers* plan and direct customer service teams to meet customers' needs and support company operations.
- *Sourcing managers* are involved with commercial and supplier aspects of product development and sourcing projects. They conduct supplier analysis, evaluate potential suppliers, and manage the overall supplier qualification process, develop, and create sourcing plans, manage requests for proposals and other sourcing documents, and evaluate and recommend purchasing and sourcing decisions.
- *International logistics managers* work closely with manufacturing, marketing, and purchasing to create efficient and effective global SC.
- *Transportation managers* are responsible for the execution, direction, and coordination of transportation. They ensure timely and cost-effective transportation of all incoming and outgoing shipments.
- *Warehouse managers* are responsible for managing inventory, avoiding stockouts, and ensuring material replenishment at minimal costs.
- *Risk managers* analyze possible risks, develop proactive operations and SC design, monitor risks, and coordinate activities for stabilization and recovery in the event of a disruption.

Since all organizations have operations and SC functions, there are many sectors for SCOM. Those sectors include, but are not limited to:

- Manufacturing companies
- Retail establishments
- Logistics
- Consulting companies
- IT companies
- Hospitals
- Banks and insurance companies
- Restaurants
- Airlines and airports
- Entertainment parks
- Building and construction companies
- Public transportation companies
- Government agencies
- Research corporations.

SC and operations managers work in an exciting and dynamic environment. This environment results from various challenging forces, from the globalization of world trade to the transfer of ideas, products, and money via the Internet. Some of the *challenges* are as follows:

- Digitalization and smart technologies
- Globalization and collaboration with suppliers and customers worldwide
- Risk management and resilience
- Data analytics in improving sales, promotions, and forecasting
- Shorter product lifecycles and fast-changing technology, materials, and processes (e.g., additive manufacturing, Internet of Things, and Industry 4.0)
- Sustainability and mass customization
- Higher requirements on multidisciplinary knowledge and competencies

▶ **Practical Insights** Excellence in Supply Chain and Operations Management has become a competitive advantage for companies. This requires a new kind of leader for managing complexity, risks, and diversity in global SCs and operations. This textbook provides the strategic, management, and analytical skills you need to launch your international career in global SCM, operations, and logistics. Three pillars build the textbook framework: practical orientation and creation of working knowledge, methodical focus, and personal development of students, the development of advanced communication, interaction, and organizational skills with the help of case studies and business simulation games.

A career in SCOM opens the door to many opportunities. The job itself is diverse, because of the variety of tasks completed by SC and operations managers. Another advantage is the high income expected, especially for experienced SC and operations managers. On the other hand, there may be long working hours and stress. Operations and SC managers generally have to work in a changing environment, so they have to be flexible. As can be observed in the real job offers, flexible managers with a willingness to travel a lot are always in demand.

1.4 Key Points

Operations is a function or system that transforms inputs into outputs of greater value. *Operations management* is involved with managing resources to produce and deliver products and services. It includes the stages of sourcing, production, distribution, and after sales. A *supply chain* is the network of organizations and processes along the entire value chain. SCM is a collaborative philosophy and a set of methods and tools for integrating and coordinating local logistics processes and their links with production processes from the perspective of the entire value chain and its total performance.

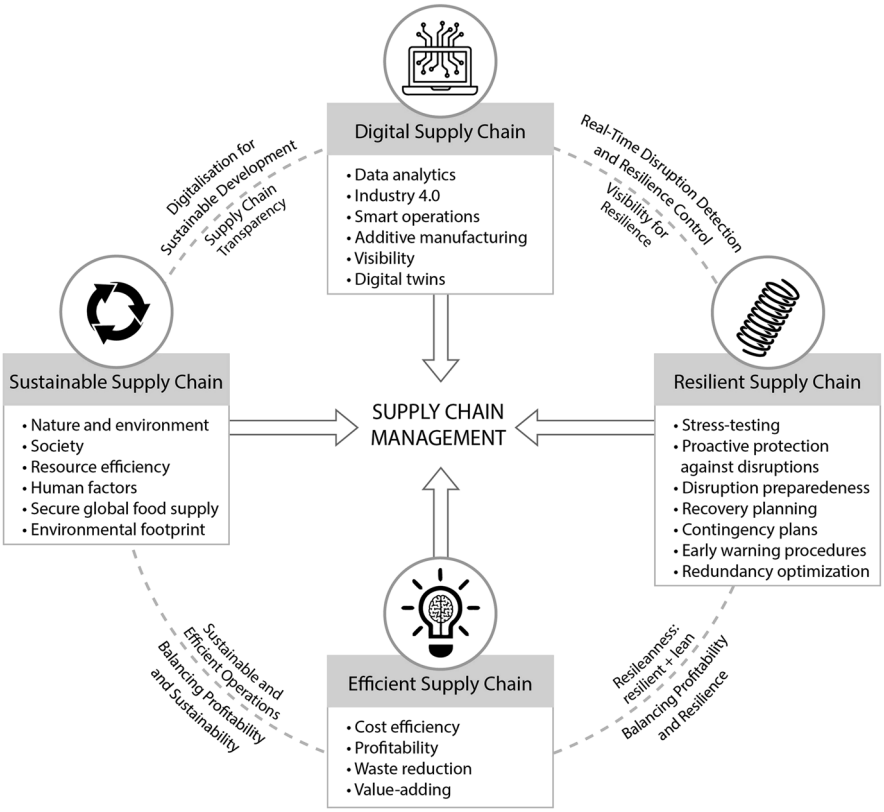


Fig. 1.11 Major paradigms of supply chain and operations management

SCOM is everywhere: in production, logistics, healthcare, airlines, entertainment parks, passenger transport, hotels, building and construction, etc. Key objectives of SCOM are costs, time, quality, and resilience. A career in SCOM is multi-faceted and requires multidisciplinary knowledge that is comprised of elements from business administration, optimization, engineering, and information systems. Examples include *logistics problem*s, warehouse management, transportation optimization, procurement quantity optimization, inventory management, cross-docking design, intermodal terminals design, etc. Accordingly, production management deals with optimizations in assembly lines, production cells, etc. *SCM problems* include SC design, demand planning, and supply coordination.

To summarize—SCOM is about how to use the available resources in the optimal way. The understanding of "optimal" differs across firms and management paradigms. "Optimal" can be understood in light of the profitability maximization (i.e., maximum effectiveness), cost-efficiency (i.e., minimum resource consumption), usage of resources in the most sustainable way (i.e., maximum sustainability), and resilience maximization. Thus far, SCOM spans perspectives of profitability, sustainability, digitalization, and resilience (Fig. 1.11).

Future developments of SCOM are related to the design and management of efficient, sustainable, digital, and resilient SC networks and operations in light of the 3P triad: Planet-People-Profit.

Bibliography

Bowersox DJ (1969) Physical distribution development, current status, and potential. J Mark 33:63–70
Bowersox DJ, Closs DJ, Copper MB (2023) Supply chain logistics management, 6th edn. McGraw-Hill, Boston, MA
Bozarth CB, Handfield RB (2019) Introduction to operations and supply chain management, 5th edn. Pearson, Harlow
Cachon C, Terwiesch C (2022) Operations management, 3rd edn. McGraw Hill, Boston, MA
Cachon GP, Girotra K, Netessine S (2020) Interesting, important, and impactful operations management. Manuf Serv Oper Manag 22(1):214–222
Choi TM (ed) (2016) Service supply chain systems. CRC Press, London
Chopra S (2019) Supply chain management: strategy, planning and operation, 7th edn. Pearson, Harlow
Christopher M (2022) Logistics and supply chain management. Creating value-adding networks, 6th edn. FT Press, New Jersey
Das A (2016) An introduction to operations management. CRC Press, Boca Raton, FL
Dolgui A, Proth J-M (2010) Supply chain engineering: useful methods and techniques. Springer, Berlin
Dolgui A, Ivanov D, Sokolov B (2020) Reconfigurable supply chain: the X-network. Int J Prod Res 58(13):4138–4163
Forrester JW (1961) Industrial dynamics. MIT Press, Cambridge
Foster ST, Sampso S, Wallin C, Webb S (2016) Managing supply chain and operations: an integrative approach. Pearson
Francis M, Fisher R, Thomas A, Rowland H (2014) The meaning of 'value' in purchasing, logistics and operations management. Int J Prod Res 52(22):6576–6589
Geoffrion AM, Graves GW (1974) Multicommodity distribution system design by benders decomposition. Manag Sci 29(5):822–844
Greasley A (2013) Operations management, 3rd edn. Wiley, New York
Heizer J, Render B, Munson C (2023) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow
http://careersinsupplychain.org
http://www.apics.org
Ivanov D (2010) A framework for aligning (re)planning decisions on supply chains strategy, design, tactics, and operations. Int J Prod Res 48(13):3999–4017
Ivanov D (2018) Structural dynamics and resilience in supply chain risk management. Springer, New York
Ivanov D (2022) Viable supply chain model: integrating agility, resilience and sustainability perspectives – lessons from and thinking beyond the COVID-19 pandemic. Ann Oper Res 319:1411–1431
Ivanov D, Dolgui A (2020) Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability. A position paper motivated by COVID-19 outbreak. Int J Prod Res 58(10):2904–2915
Ivanov D, Sokolov B (2010) Adaptive supply chain management. Springer, London
Ivanov D (2024) Introduction to supply chain analytics. Springer Nature, Cham
Ivanov D (2025) Introduction to Supply Chain Resilience and Viability, 2nd edn. Springer Nature, Cham

MacCarthy B, Ivanov D (eds) (2022) The digital supply chain. Elsevier, Amsterdam
Oliver RK, Webber MD (1982) Supply chain management: logistics catches up with strategy. In: Christopher M (ed) Logistics: the strategic issues. Chapman & Hall, London
Simchi-Levi D, Kaminsky P, Simchi-Levi E (2007) Designing and managing the supply chain: concepts, strategies and case studies, 3rd edn. McGraw-Hill, Boston, MA
Slack N, Brandon-Jones A, Burgess N (2022) Operations management, 10th edn. Pearson, Harlow
Stadtler H, Kilger C, Meyr H (2015) Supply chain management and advanced planning, 5th edn. Springer, Berlin

Examples from Different Industries, Services, and Continents

2

Key Items for This Chapter

- Examples of SCOM in manufacturing
- Examples of SCOM in services
- Examples of e-operations and supply chains
- Examples of resilient and sustainable supply chains and operations
- Examples of artificial intelligence in supply chains and operations

2.1 Examples of Operations and Supply Chains in Manufacturing

2.1.1 Nike: Sourcing Strategy in the Integrated Supply Chain

Since its establishment, Nike has become a global enterprise providing trainers and sports garments to customers worldwide. A *sourcing strategy* is essential for Nike since the company's production and logistics strategy is based on *outsourcing* (see Fig. 2.1).

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_2) contains supplementary material, which is available to authorized users.
Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de!

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_2

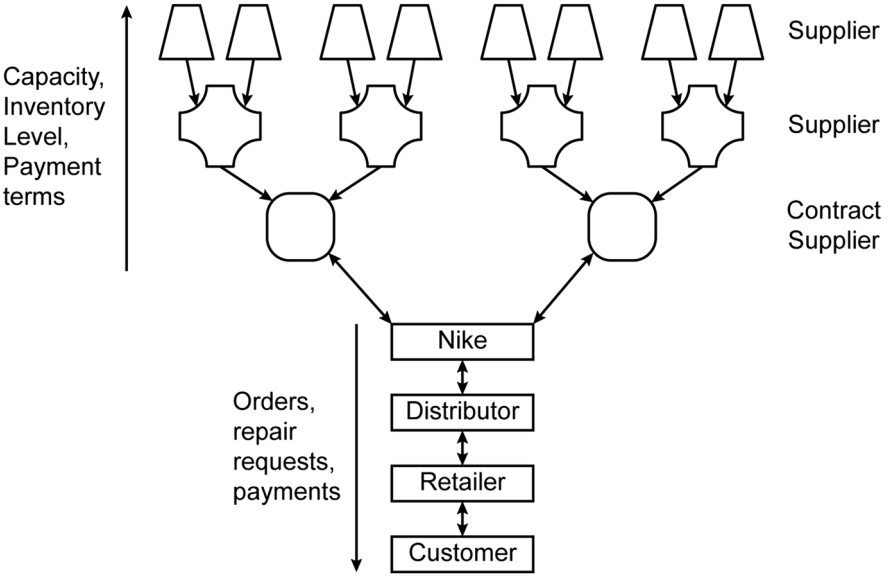


Fig. 2.1 Nike's supply chain

Nike executes a long-term sourcing consolidation strategy and is streamlining its SC operations. Nike assesses its contract manufacturing base in order to:

- Streamline the SC to focus on several contract manufacturing groups
- Build a strong and sustainable sourcing base for greater operational efficiencies and future growth
- Identify subcontractors able to deliver the best performance of products and innovation
- Align subcontractors in terms of Nike's corporate responsibility principles

Nike has been shifting from a risk-reduction focus—which devotes time and attention to the lowest factory performers—to a strategy that invests time and attention in strengthening relationships with the factories operating at the highest performance levels. In particular, a manufacturing index (MI) was implemented. It integrates scores from key performance areas into a single scoreboard rating that groups factories as Gold, Silver, Bronze, Yellow, or Red. Contract factories that are able to consistently exceed Nike's requirements in the areas equally weighted in quality, costs, delivery, and sustainability performance management and that show consistent performance leadership in the industry will achieve a silver rating in the MI. Contract factories that go beyond industry standards and demonstrate innovation and benchmark performance within the broader manufacturing landscape will achieve Gold. At a minimum, factories in Nike's SC will be expected to achieve and sustain a Bronze rating, indicating that the factory meets baseline standards and can self-govern through integrated systems and a lean approach.

The MI creates one overall score for each contract factory, enabling a consistent and comprehensive conversation about Nike's business with that factory. Nike develops incentives and sanctions based on the MI ratings. For example, Silver- and Gold-rated factories will be able to self-audit and calibrate with Nike staff and will have access to a range of Nike's technical assistance, leadership, and education resources, as well as possible innovation or community co-investment and priority consideration for orders.

Nike initiated several schemes to make its SC more sustainable and environmentally friendly. Nike has set up Sustainable Manufacturing & Sourcing-Sustainable—a new organizational structure within the company that has brought together labor compliance, health, safety, and environment, lean manufacturing, human resources management, climate and energy, and waste and water management.

Discussion (see Chap. 5)

- Which advantages and disadvantages do you see in the outsourcing strategy?
- Select and calculate at least two performance indicators to evaluate the inventory management performance at Nike!
- What do you think of the MI at Nike from the position of a contract manufacturer?

2.1.2 AfriGlobal: Establishing a Sophisticated Supply Chain Management System in Africa

The African multinational consumer goods company AfriGlobal launched 2024 an innovative supply chain management initiative in East Africa, focusing primarily on Kenya, Tanzania, and Uganda. The company implemented an up-to-date Transport Management System (TMS), which can optimize logistics operations across the entire region. This system allows for real-time tracking of shipments, as well as route optimization, load consolidation, and cutting transportation costs while improving delivery times.

2.1.2.1 Logistics and Distribution

AfriGlobal created a network that strategically located their distribution centers, which are stocked with highly advanced warehouse management systems that use automated storage and retrieval systems, RFID technology for inventory tracking, as well as energy-efficient lighting and climate control systems. To tackle the challenges of last-mile delivery in rural and urban areas of Africa, the company initiated a partnership with local logistics startups that use drone delivery for their distribution as an innovative method of delivery.

2.1.2.2 Sustainability Management

AfriGlobal initiated several sustainability measures in its supply chain. Local sourcing is one of these measures, which includes sourcing at least 60% of their raw

materials from local suppliers, which results in transportation emission reduction and support of regional economies. Furthermore, AfriGlobal installed solar panels at all distribution centers, which satisfy up to 70% of their energy requirements. For the reduction of plastic waste, AfriGlobal uses biodegradable packaging crafted from locally sourced materials. Additionally, the company uses water recycling systems in its manufacturing facilities to reduce its water consumption by 40%.

2.1.2.3 Technology Integration

AfriGlobal implemented a blockchain-based system to track products from their source to the customer to increase transparency and traceability. This system enabled the company to monitor product quality and safety in real-time, verify ethical sourcing practices, and enable accessible product information to enhance customer trust.

2.1.2.4 Community Engagement and Capacity Building

The company established special training centers in each country for local workers to enhance their skills in modern supply chain management techniques. Moreover, they partnered with local universities to implement supply chain management courses with African context. Lastly, AfriGlobal created a mentorship program that connects experienced supply chain experts with local businesses.

2.1.2.5 Impacts

After less than a year of AfriGlobals implementation strategy the company achieved significant results and impacts on the African supply chains. They reduced overall logistics costs by 25%, an improvement of 30% in on-time deliveries, decreased the carbon emissions from transportation by 40%, created over 5000 new jobs in the region, and increased local supplier participation in the supply chain by 15%. The AfriGlobal case study demonstrates that a focus on sustainability, investment in advanced technology, and engagement with local communities enables economic growth and environmental protection in these regions of Africa.

Discussion

- What problems might AfriGlobal experience in the future concerning its SC?
- What do you think about the introduction of direct shipments to customers?

2.1.3 Tesla Gigafactory Berlin-Brandenburg

The Tesla Gigafactory Berlin-Brandenburg is in Grünheide, Germany, which is approximately 30 km southeast of Berlin near the Berlin-Brandenburg Airport. The production site is designed to manufacture batteries, battery packs, powertrains, seats, and fully assembled Tesla Model Y vehicles. Its production capacity is planned to reach 500,000 cars per year with estimated costs of about 4 billion Euro. There are three major reasons why Tesla chose Grünheide as their first European

Gigafactory. These strategic factors include the proximity to Berlin, the access to transportation since it is located near the Berlin Brandenburg Airport and the A10 Autobahn as well as the available land that the industrial area provided.

The total area of the factory is 3,038,620 m2, including a cleared area of the first construction phase of 1,932,700 m2, a built-up area of 729,000 m2, and a paved traffic area of 301,000 m2. The factory was announced on November 12, 2019, and construction started in early 2020. After a delay caused by administrative issues, the factory opened officially on March 22, 2022.

Tesla's supply chain management strategy consists of five key elements: Vertical integration, local sourcing, direct sales model, strategic partnerships, and talent acquisition. The manufacturer uses vertical integration to lower the number of parts required for their chassis with proprietary battery technology and their own developed Gigapress. Furthermore, the Gigafactory serves as a European supply chain hub for facilitating economies of scale and reducing overall transportation costs. Moreover, the company uses a direct sales model that includes the operation of their own showrooms and direct vehicle sales through their website, which reduces distribution costs and controls customer experience. Additionally, Tesla collaborates with companies, including Panasonic, which enables them to leverage external expertise and resources in battery production. Lastly, Tesla focuses on acquiring talents from different industries to promote innovation and continuous improvement.

In addition to its supply chain management strategy, the manufacturer has implemented several sustainability initiatives. In August 2024, the company launched their own all-electric train service between Erkner train station and Tesla Süd, which is provided for free to employees and the public. This project estimates saving approximately 50 tons of CO2 emissions weekly. Another sustainability initiative is the renewable energy sources the factory uses, which they aim to be powered by soon and to contribute to the company's vision to increase the world's transition to sustainable energy.

In the future, Tesla Gigafactory Berlin plans to expand the factory and to increase production to 1 million vehicles per year. Overall, the company illustrates its devotion to vertical integration, sustainable manufacturing, and strategic supply chain management.

Discussion (see Chaps. 7 and 9)

- Why has Tesla started this large-scale project?
- What must be considered when choosing an appropriate location for such a project?
- What needs to be considered in terms of Facility Planning?
- Which challenges and potential problems in SCOM do exist for such a Gigafactory?

2.2 Examples of Operations and Supply Chains in Services

2.2.1 SCOM in Restaurants: Case Study Starbucks Corporation

The Starbucks Corporation, founded in 1971, is one of the world's largest coffee house chains, with more than 17,240 coffee shops in over 50 countries. Starbucks' product portfolio consists of food items, as well as coffee specialties, tea, and other refreshing drinks. Starbucks Corporation also offers roasted beans and several merchandise products. However, Starbucks' main product is coffee. Therefore, and because of Starbucks' large and complex global supply SC, this case focuses only on the coffee bean. In particular, the company's production and sourcing strategy and its transportation network are considered.

To decrease the lead time of coffee deliveries and to further decrease transportation costs, Starbucks aims to manufacture in the region where the coffee is sold. To regionalize coffee manufacturing, Starbucks owns five coffee roasting plants, four of which are located in the United States and the fifth in the Netherlands. In addition to its company-owned coffee roasting plants, Starbucks works with contract manufacturers in the United States, Canada, Europe, Asia, and Latin America.

Starbucks has spread its production across a vast territory. Nevertheless, transportation, logistics, and distribution are still the most significant parts of Starbucks' operating expenses. The existence of an efficient and effective single, global logistics system is essential for the company. Figure 2.2 shows the SC of the Starbucks Corporation.

The company has a multiple sourcing concept. The suppliers of the coffee beans are mainly located in Latin America, Asia, and Africa. To ensure ethical sourcing of high-quality coffee beans, Starbucks uses Coffee and Farmer Equity (CAFE) practices. CAFE is a set of guidelines that evaluates the social, economic, and environmental aspects of coffee production. This allows Starbucks to address sustainability issues, which have become very important in SCM.

Starbucks uses ships and trucks for transportation. The Starbucks Corporation normally delivers the unroasted beans in ocean containers to the United States and

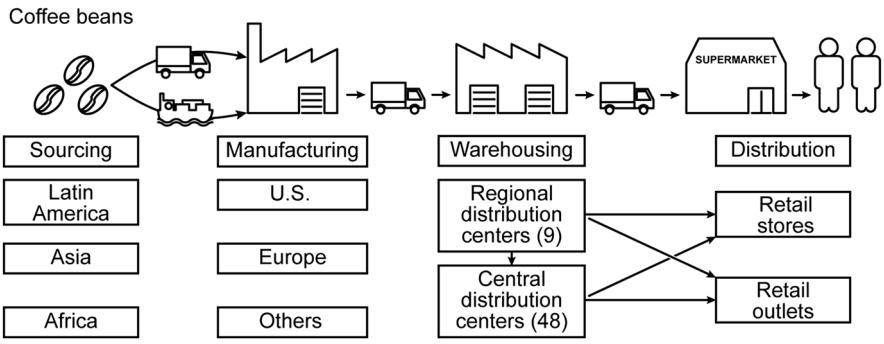


Fig. 2.2 Starbucks' SC

Europe. From the port of entry, the goods are trucked to a storage site close to one of the coffee roasting plants. Once the beans are roasted and packaged, trucks deliver the coffee to regional distribution centers. In total, Starbucks runs nine regional distribution centers, five in the United States, two in Europe, and two in Asia. Each of the distribution centers covers 200,000–300,000 square feet. Other goods needed for running a coffee shop are also stored there. From the distribution centers, the products are delivered either directly to the store or to central distribution centers, which are smaller warehouses. In total, the Starbucks Corporation has 48 central warehouses worldwide. From there the coffee beans and other products are frequently trucked to retail stores and retail outlets.

Discussion (see Chap. 8)

1. What kind of production strategy does the Starbucks Corporation use and why?
2. What kind of sourcing strategy is Starbucks using in terms of the number of suppliers and geography?
3. Which mode of transportation does Starbucks use?
4. What kind of transportation network does Starbucks use—direct shipping or via distribution centers?

2.2.2 Beyond Transit Management at Changi Airport in Singapore

Singapore Changi Airport is one of the busiest passenger hubs in Southeast Asia, widely acclaimed for its exceptional passenger experiences. Officially commenced its operations in 1981, the airport is adding unique features to its portfolio to give passengers a unique first impression about Singapore. Being constrained by essential resources (e.g., landmass, labor, water, etc.), every project of the airport comes with operational challenges. Today, Changi represents Singapore's globally renowned aviation facilities and management operations, and a distinct brand of Singaporean efficiency and reliability. The airport is recognized as the best airport in the world by many institutes according to their ranking systems and in fact, it is the most awarded airport in the world with 691 awards to date. In the year 2024, Changi managed to handle over 61.2 million passenger movements, 1.82 million airfreight movements and 333,000 commercial aircraft movements.

The following case describes how the airport has achieved effectiveness in its major operational areas.

Touchdown to Turnaround

Landing could be one of the few stops of the daily schedule for many aircraft. They should be ready to depart within a short time window. Typically, at Changi Airport, the time from touchdown to the next boarding is only around 60–90 min. The ground handlers, aerobridge operators, cleaners, in-flight catering staff, aircraft engineers, and technicians are on standby and ready to start their respective

operations once the plane is docked. The process begins with ground handlers chocking the wheels and signaling the aerobridge operator to proceed with docking, while allowing passengers to disembark. Simultaneously, specialized lifts and vehicles are used to unload the baggage and cargo. The maintenance and disposal services team carries out their tasks as the next step. In the meantime, after disembarkation, the cleaners move in to collect all the waste and replace all the used items. The technical operators remain busy throughout the entire process, addressing issues that were logged in during the previous flight.

Departing Flight Operations

The ground workers at Changi ensure the smooth take-off and landing operations 24/7 following an instructed checklist. To prepare each flight with speed and efficiency, special vehicles are used in almost every step.

Flight Catering Truck

According to the flight plans, the meals are first prepared in the kitchen. When the truck delivers the meals to the designated flight, a special lift efficiently raises them to the height of the plane's door. This allows the service trolleys to be easily wheeled into the aircraft's galley. catering trucks are arranged according to the height of all aircraft types.

Water Supply

Water is another essential supply for the crew and the passengers. Drinking water to the flight is ferried using portable water service vehicles according to the number of passengers and the crew. This should be calculated beforehand to avoid unnecessary weight which can increase the cost throughout the journey. Drinking water is pumped directly into the aircraft's freshwater system.

Refueling

The engineers constantly check the availability of the jet fuel stocks for flights to avoid unexpected delays. Jet fuels are stocked underground with safety reserves for emergency landing for fuels. A refueling truck should be scheduled for every flight to connect the underground tanks to the flight's fuel system. Vehicle itself does not carry any fuel to avoid accidents.

Tow Tug and Runway Maintenance

The row tug should be in place when the pilot receives the clearance from the control tower to push the plane out of the aircraft bay to its correct taxiing position. The runway is constantly examined and cleaned using a rubber removal machine for safe and efficient aircraft movements during landing and take-off.

Rescue Operations

The rescue operations at Changi airport are handled by the Airport Emergency Service (AES) team. Since the airports are located close to each other, the team is responsible not only for Changi Airport operations but also for Seletar Airport and

the RSAF airbases. Usual rescue operations include conducting exercises, executing aerodrome safety plans, and planning emergency response for large-scale events. To increase the success rate of the operations, AES maintains a sea rescue base with three types of vessels—Command Craft, Fast Craft and Landing Craft. Command crafts bring improved capabilities for managing command and control operations of the crash site. Fast crafts provide the speed and agility to rescue operations. Additionally, they can perform firefighting, surface rescue, and casualty conveyance with greater efficiency. Apart from firefighting and casualty management abilities, landing crafts can be beached with minimum effort.

Butterfly Garden in Terminal 3

Departing passengers and those in transit can enjoy a large, 330-square-meter butterfly garden with a waterfall. The garden is home to around 1000 butterflies from 25 to 45 species native to Southeast Asia. To maintain such a controlled environment, the horticulture team should manage their operations on time. The operations manager should ensure that 600 butterfly pupae are received from overseas every Wednesday to be released into the garden. Maintaining the correct moisture levels for successful hatching is crucial once the pupae are moved to the Emergence Cage. In the garden, potted nectar-producing plants should be replaced weekly to provide butterflies with a diverse diet of fresh, natural nectar. Additionally, the waterfall and misting systems should be maintained daily to keep the air humid for butterflies' comfort.

Discussion

- Which activities do comprise airport operations management?
- What are the challenges of airport operations management?
- Which role does information coordination play in airport operations management?

2.2.3 Operations Issues in Car-Sharing

The business model of car sharing is relatively new. However, its origins can be found in Zurich, Switzerland, in 1948. Similar to car sharing today, in a "Sefage" (Selbstfahrergemeinschaft = self-driving community) people could use one car together without having to buy one of their own. Changing preferences in passenger transport, particularly in bigger cities, have boosted the demand for car sharing. The advantages of car sharing are numerous, but high flexibility at a low cost compared to possessing a car is probably the main reason people increasingly choose car sharing. Business customers also benefit since they can use car sharing as an add-on for the company car fleet. Additionally, travel costs for employees can be reduced by using car sharing instead of taxis. But there is even more to car sharing. It decreases the total amount of cars in the city, which means less traffic volume and more parking spaces. Pollution has also decreased, especially since more and more car-sharing providers offer hybrid or completely electric-driven vehicles.

There are *two types of car sharing*: station-bound and free-floating. Station bound is the older version where the car is received and returned at the rental station. Most of the time, it is now possible to return the car to a different station if it belongs to the rental company. Free-floating is non-station-bound, meaning the car can be rented and left anywhere within a particular area. Charges are accrued either per minute and/or per kilometer. This option of car sharing is more appropriate for short distances.

Car-sharing providers face challenges in operations management concerning demand, capacity, location, and cost planning. Usually, car sharing is an additional business field for rental service providers or car manufacturers. For example, DriveNow is a subsidiary owned by BMW and SIXT. Car2Go was founded by Daimler and Europcar. In 2020 DriveNow and Car2Go have joined their forces and offered their services in a joint venture as ShareNow.[1] In 2022, ShareNow has been taken over by Stellantis and thus ShareNow belongs to the car-sharing brand Free2move, but the brand names ShareNow and Free2move co-exist.[2]

These companies already have expert knowledge in demand forecasting and capacity planning, which makes it easier to evaluate the number of cars needed at different locations. The challenge faced in this field is to meet demand at peak times and in different areas of the city.

For particular occasions, such as a big football game, it may happen that many vehicles are used to drive to one particular location. Accordingly, there may be a misallocation of cars within an area. It is easy for car-sharing providers to track these actions via GPS. To solve the problem, employees can be hired to spread the vehicles throughout the city. This option can be quite expensive, as employees need to be transported to the cars themselves, too. A cheaper option is to offer benefits for customers such as free miles, bonus points, or other discounts. Cost planning can also constitute an issue for operations management. Most car-sharing companies attempt to attract customers by offering free parking everywhere.

Discussion

- Which challenges of operations management are met by car sharing companies?
- What role does information technology play in car sharing?
- What do you think of sustainability issues regarding the car sharing concept?
- Compare station-bound and free-floating from an operations management perspective regarding the costs, time, quality, and management complexity.

[1] https://www.share-now.com/de/de/faq/about-share-now/#what-is-share-now (accessed 22.12.2020).

[2] https://www.free2move.com/de-DE/pressemitteilung/press-release-2022-07-18 (accessed 11.02.2025).

2.3 Examples of Digital Supply Chains and Smart Operations

2.3.1 Amazon Robots

The US mega-retailer Amazon has been consistently following a technological automation approach for the last several years. By increasingly using robots at their logistics centers, Amazon aims to achieve higher efficiency through cost decreases and faster process flows. However, the human workforce still plays an important role within most of the processes at Amazon's fulfillment centers (Singularity Hub 2017).

Amazon started introducing higher numbers of robots to their logistic processes in 2012, after acquiring the robotics company Kiva systems. In 2016, Amazon was using about 45,000 robots at its fulfillment centers globally, which represents a 50% increase compared to 30,000 robots used in 2015 (Quartz 2017). Along with an increase in robot use, the number of Amazon's employees rose by about 50% between 2015 and 2016. The greater need for human workforce is explained by the fact that the increased use of robots led to a decrease in shipping costs, which finally led to a higher demand for Amazon products (Singularity Hub 2017).

The robots working at Amazon's fulfillment centers can perform regular and predictable tasks in a controlled environment. These tasks include heavy lifting of packages, moving pallets between shelves, and shuttling goods from one end of the warehouse to the other. Tasks that include judgment, unpredictability, and fine motor skills still require a human workforce. Therefore, at Amazon's fulfillment centers, employees stock warehouse shelves according to how shelf space can be optimally used. Additionally, employees are responsible for product packaging and loading of trucks (Singularity Hub 2017).

Although many tasks are still completed by people, technology is consistently improving, which could lead to robotic takeovers in the near future. Amazon recently applied for a patent for a self-learning robot, including an automatic packaging system. This robot will be able to grab items ordered and package them appropriately, while secure data provides the opportunity to save the ideal packaging strategy of a product. Consequently, this task will no longer require a human workforce. However, according to Amazon, employees have no reason to be afraid of losing their jobs, as many tasks still require a human workforce, and additional working fields, such as machine maintenance, are evolving (Business Insider 2017).

In this case, Amazon clearly demonstrates its strong focus on robotic automation within the logistic processes, which will continue into the future. Besides the increased use of robots in fulfillment centers, drone delivery is one of the retailer's future automation ideas.

Discussion (see Chap. 16 for more details)

- Why is Amazon increasingly using robots at their fulfillment centers?
- Which challenges and potential problems exist?
- What are potential future opportunities and risks that come with increased automation in logistic processes?

2.3.2 Vestas Wind Systems: Revolutionizing Wind Turbine Supply Chain with 3D Printing

Vestas Wind Systems is the global leader of sustainable energy solutions which uses 3D printing for their supply chain transformation and operations management. The company faced several challenges before implementing its direct digital manufacturing (DDM) program in 2021 including 3D printing and Industry 4.0 technologies into their supply chain. These challenges included a long lead time for spare parts, high inventory costs, and logistical complications in global operations.

For digital inventory development, Vestas Wind Systems digitized over 2000 parts, which created a broad digital inventory. Furthermore, each part was properly 3D scanned and optimized for additive manufacturing. Lastly, the company developed a software platform for digital inventory management with easy access and version control. In addition, 10 strategic locations globally including sites in Denmark, Germany, Spain and the United States, were equipped with 3D printers. These 3D printers use a mix of three different technologies depending on the part. Fused Deposition Modeling (FDM) for larger and sturdy parts, Selective Laser Sintering (SLS) for complex geometries, and Continuous Fiber Reinforcement (CFR) for high-strength components.

This special 3D printing procedure requires also special material innovation. Therefore, Vestas collaborated with material scientists to develop custom composite materials and introduced a new carbon fiber-reinforced nylon material custom designed for their wind turbine components. Moreover, extensive testing of the newly developed material was required to ensure the quality of the 3D printed parts met or exceeded the previously used material that was not 3D printed. In addition to the material change, Vestas also used Industry 4.0 integration in their production sites. The company implemented a cloud-based Manufacturing Execution System (MES) which connects to all 3D printers and can manage their production flows. For the collection of real-time data on machine performance, material usage and printing quality, IoT sensors were deployed on the 3D printers. AI is used for reducing printers' downtime by predicting maintenance and blockchain based systems were developed to warrant traceability and authenticity of the parts.

To ensure the overall quality of the entire production process, Vestas implemented in-situ monitoring systems that can detect defects on the parts during the printing process. Moreover, computer vision and AI are used to automate post-print inspection, and a digital twin of each printed part ensures virtual testing and simulation before the actual physical production of the part.

Using this technology for its 3D printing process, Vestas Wind Systems was able to achieve several improvements in their operations. The average lead time for spare parts decreased from 4 to 6 weeks to a maximum of 48 h, while critical parts can be produced and shipped within 8 h. Inventory costs due to this project are $15 million yearly with a reduction of their physical inventory by 30% in the first year of implementation. Since 85% of all critical parts could now be produced on site the supply chain resilience increased, which reduced their dependency on single suppliers by 40%. Operational flexibility increased significantly since 95% of all service

employees were trained to use the digital inventory system, and the on-site capability of part production increased by 50% in all locations. Lastly, Vestas was able to reduce their carbon emission related to the spare parts logistics by 25% and cut their material waste by 15% due to optimized printing progress.

Discussion (see Chap. 16 for more details)

- How might Vestas' approach to implementing 3D printing and Industry 4.0 technologies be adapted for other industries?
- How does integrating multiple technologies (3D printing, IoT, AI, blockchain) contribute to the overall success of Vestas' supply chain transformation?
- How does distributed manufacturing through 3D printing enhance supply chain resilience? Are there any potential drawbacks?
- How does Vestas' implementation of 3D printing contribute to sustainability goals? Are there any potential adverse environmental impacts to consider?
- What regulatory challenges might companies face when implementing 3D printing for critical components in industries like renewable energy?

2.3.3 Energy-Efficient Car Designs with Machine Learning at Volkswagen

Volkswagen is known for designing beautiful, high-performance, and energy-efficient vehicle models. Energy efficiency and car design are interconnected aspects that should be analyzed not only in terms of energy but also comfortability, current market trends, safety, transportation rules and regulations, etc. The design process starts with designers going through multiple drafts, evaluating each, integrating feedback, and refining the design.

One of the most fundamental aspects of a car's design is its drag coefficient (e.g., its resistance to air). Operationally, estimating the drag coefficient can be expensive and time-consuming, according to current practices, which involve physical wind tunnels or computationally intensive simulations. As a solution, Volkswagen uses machine learning technology to estimate the drag coefficient much faster so they can introduce modern energy-efficient designs to the market on time.

Step one is to build 3D meshes of the car design. Car bodies are different in size and shape. In addition, a single vehicle can have large smooth surfaces and areas with delicately designed features. These requirements create a large number of 3D meshes for a single car. However, machine learning models require consistent representation to develop robust, generalized rules. As a solution, Volkswagen uses the "shrink wrapping" method to create 3D meshes. Starting with a general digital base mesh and deforming it according to the target mesh optimize the design process while providing a simplified representation for the machine learning model. The model then measures the drag coefficient according to the trained data. The model gives results with an average error of just 4% within a second. This is not as accurate as physical wind tunnel tests. However, this approach can help reduce a large pool

of design candidates to a smaller shortlist. In a nutshell, using machine learning for car designs:

- Reduce the time to process (fail or pass) the design test
- Minimize the cost
- Increase the accuracy rate (subject to model training level)
- Help to integrate with other systems to optimize the entire supply chain

With continuous training and integration with other AI methods, Volkswagen expects to achieve more accurate results in the near future.

2.3.4 DHL: Artificial Intelligence for Supply Chain Resilience

DHL is the global leader in logistics and faces increasingly more challenges through managing complex supply chain disruptions. Since the company's previously used methods of monitoring and addressing issues were becoming more outdated in a fast-changing global market. The company faced challenges that included delayed responses to disruptions, inefficient risk management, difficulty in processing of high amounts of global data, and inability to predict any supply chain disruptions proactively. For these reasons DHL integrated advanced AI technologies into their existing Resilience360 platform.

Furthermore, the company implemented a machine learning algorithm that is able to categorize and prioritize risks that was trained with historical data of past disruptions and their impacts. It includes automated risk scoring which is based on severe factors like severity, likelihood, and potential impact. Moreover, dynamic risk prioritizing is implemented, adapting to changing conditions and customized risk profiles for different industries or regions. AI powered incident monitoring with Natural Language Processing (NLP) and machine learning is used by DHL for real-time monitoring of global news media, social media and official reports with automated translation and analysis of the multi-language media sources, and intelligent filtering of media to reduce false information. DHL also used AI to predict any disruptions before they occur. With the analysis of historical and current data it can predict potential disruptions. The integration with weather data, geopolitical information, and economic indicators also helps predict disruptions, while scenario modeling is used to predict the potential impact of the disruption on the supply chain.

After the implementation of AI, DHL was able to significantly improve their early detection of potential risks by 60%, reduce their average lead time for risk identification from 3 days to 12 h, reduce their average response time to major disruptions by 30%, improved the successful risk mitigation by 25%, and a 50% reduction in false disruption predictions in the first year. In the future, DHL plans to integrate IoT data from shipments and warehouses for real-time visibility, develop more advanced AI models for long-term strategic planning, and expand their AI usage to sustainability measures in their supply chain management.

2.4 Examples of Resilient Supply Chains and Operations

2.4.1 Ford: Repurposing the Supply Chain in a Week—From Car Manufacturing to Saving Lives During the COVID-19 Pandemic

In the wake of the COVID-19 pandemic in the USA, Ford repurposed their production line to help the healthcare sector to cope with a shortage of PPE (personal protection equipment) within 2 weeks. Starting on March 20, 2020, designers and prototypers at Ford began to review and refine face shield concepts (Ford 2020). They identified suppliers and engaged engineering and program managers. In collaboration with the hospital managers, on March 21, face shields designs have been identified that could be built by a repurposed Ford's supply chain. On March 22, material planning and logistics started planning inbound and outbound deliveries for the newly established product. Suppliers started delivering materials to production sites. On the same day, the first prototype was completed, and an initial prototype production run was established. On March 23, new suppliers have been identified to resolve capacity problems with supply for some materials needed for the face shield production. On March 24, the first 5000 face shields arrived at the hospitals. Some problems needed to be operatively resolved. The supply of elastic bands was identified as not sufficient for full-speed production. Alternative product designs have been validated, and new suppliers have been identified and contacted. On March 25, production continued reaching 32,000 daily units on March 27. On March 28, a secondary design was approved, which removed the need for staples. Thus, in 1 week, Ford was able to go from idea to mass production. The production rate reached 225,000 shields daily, and the overall delivery of over 1 million units by April 5 has been realized.

Discussion

- What are the major decisions involved with SCOM when launching a new product? Which decisions have been done at Ford?
- What are the differences between developing a new product in an SC and repurposing the existing one?
- What is the role of time criticality in the case of Ford?
- Which major bottlenecks can you identify, which hindered to ramp-up the new SC for face shields? How they have been resolved at Ford?
- Discuss the importance of collaboration between commercial SCs, healthcare SCs, and governments.
- What steps could be taken in the future to quickly repurpose the SC? Do you think that some preparedness measures (in terms of production planning, supplier management, visibility, and digital technology) could help to react more effectively and efficiently?

2.4.2 Toyota: Supply Chain Disruption Management

A special focus of SCOM at Toyota is risk and disruption management. Many parts of Toyota's SC are located in areas that are likely to be hit by an earthquake. As such, the risk that Toyota's SC might suffer from those disasters is rising, and the damage could severely impact production and other activities. Given this context, it is essential to assume that Toyota, with its restricted resources, would suffer greatly from such a disaster and so should make preparations to affect early recovery. For these reasons, Toyota is reassessing its business continuity plan. The foremost premise of Toyota's business continuity plan is to work on preparedness before and recovery after a disaster happens.

As a part of the *preparedness*, Toyota addressed the difficulty of "energy, information and transport network fragmentation" once disasters occur, and developed a hybrid car for the Miyagi Prefectural Police, installed with external power provision systems. In addition to providing good fuel efficiency and environmental performance in normal operations, during disasters, these cars can be driven on fuel or electricity, and even have a power supply function that allows electricity to be drawn from the car.

Toyota's help in *disruption recovery* is characterized by sending workers to its disaster-affected production sites where they can be engaged in numerous activities, such as restoring facilities and distributing disaster relief supplies. Worker volunteers from the Toyota Corporation continue to help with restoration efforts for individuals in areas hit hardest by the latest disaster. Toyota additionally provides material support: trucks with relief supplies from the Toyota Corporation, which are gathered at two local production sites.

Learning from previous experiences, Toyota has prepared a nationwide framework that utilizes Japan's warehouses and logistics network for sending relief supplies to disaster-affected areas. In addition to stocking emergency supplies at the distributors nationwide, Toyota has also built a framework for sending relief supplies to the disaster-affected distributors. Taking into consideration of possible problems such as motor fuel shortages, this framework is important for delivering quick and reliable support to disaster-affected sites.

The risk management committee at Toyota organizes meetings twice a year to identify risks that may affect business activities and take preventative actions against their negative impacts. The committee members include the global chief risk officer (CRC), regional CRSs, and all senior managers and chief officers. They work to manage and prevent the major risks in the regions and report on any immediate and serious disruptions.

Toyota also has invested in new capabilities to improve SC resilience. Working with its partners, the company created a database to visualize supply networks for each component. If disaster strikes, Toyota can immediately identify the network at risk. The database identifies components that are supplied by only one manufacturer and are difficult to replace. The company then decreases dependence on solo providers by reducing the number of unique designs and sharing equipment specifications with parts production facilities and suppliers.

Discussion

- What are the main decisions involved with disruption management?
- Why is it risky to rely on single sourcing when considering the Tier-1 suppliers?
- How can we prepare an SC for maintaining business continuity in case of disruptions?
- What digital technologies can help improve resilience?

2.4.3 Time-Critical Supply Chains: Disaster Management and Humanitarian Logistics

Logistics is commonly associated with the business field. However, in recent years, particular efforts have been directed at investigating severe *SC disruptions*, which can be caused, for example, by natural disasters, political conflicts, terrorism, maritime piracy, economic crises, destruction of information systems, or transport infrastructure failures. In this setting, humanitarian logistics has become increasingly important. We can categorize natural disasters and human-made disasters, such as wars and terrorist attacks.

Disaster/humanitarian logistics is a relatively new field that has only recently received more attention (Dubey et al. 2019). This is due to increasing political conflicts along with the accelerating strength of natural disasters. The main challenge of humanitarian logistics is the coordination of activities in a destroyed environment. This can become extremely complex since almost all parameters are unknown up to the point the disaster starts and even after the event. Not knowing the type or amount of goods needed or the actual shipment location makes humanitarian logistics a particularly challenging field.

Each year more than 500 catastrophes of different kinds occur worldwide. The UN Humanitarian Response Depots (UNHRD), which is a UN organization under the head of the World Food Program, reacts quickly when disasters occur. Usually, the most necessary supplies are food and water. The organization has developed a concept whereby emergency supplies are stored in five hubs worldwide: in Ghana, UAE, Malaysia, Panama, and Italy. The hubs also contain medical kits, shelter items, and IT equipment. Through this network, the UNHRD is able to start supplying goods to every country in the world within 1–2 days.

The effectiveness of the emergency aid response relies on logistics speed and efficiency: logistics is key to humanitarian relief operations. Time is the most important factor as it influences how many lives can be saved. It is necessary to be prepared at all times. The disaster management cycle has four stages (Altay and Green 2006):

- *Mitigation* is a role played by the government, where the host government is responsible for putting protocols in place and taking action to reduce the probability of disasters.

- *Preparedness* refers to the various operations that occur before a disaster strikes. Here the physical network is designed, and information and communication technology (ICT) systems are developed together with bases of collaborators.
- *Response* refers to the various operations that are instantly implemented after a disaster occurs. First, all collaborators get involved; second, they try to restore basic services and deliver goods to the highest possible number of beneficiaries in the shortest time possible. Collaboration and coordination among the players are crucial.
- *Reconstruction* refers to the various operations in the aftermath of a disaster. It involves rehabilitation and has the aim of addressing problems from a long-term perspective, since the after-effects of the disaster can continue for decades.

When a disaster does occur, the local government needs to request the World Food Program's help. First, food supplies are sent out immediately. Following this, employees are sent to disaster areas to access information about demands for goods. They also evaluate the best means of transporting the goods to the affected area. Next, the emergency operation is planned and budgeted. To receive financial aid, the World Food Program requests funds from countries worldwide. Most funds are then received out of government expenditure. A team of logistics experts develops transport routes for all the required goods to the area where the disaster occurred using all possible transportation means. Reorganizing the ICT is of particular importance as it will help further operations through improved information flow. The World Food Program has an IT team that is specially trained to set up information infrastructure within 1 day.

In November 2013, one of the strongest typhoons recorded, "Haiyan," hit the Philippines. It killed thousands and destroyed such infrastructure as roads, ports, and hospitals. Many logistic problems were encountered from the beginning:

- Most of the roads, airport, and harbors were destroyed.
- Weather conditions after the typhoon were still destructive (rain, strong winds, etc.).
- No electricity was available due to a lack of fuel for generators, which also led to mobilization problems.
- Acute shortages of food and water and lack of medical supplies or proper hospital facilities were observed.

The World Food Program was working closely with the local government to provide as much help as possible. As soon as the typhoon hit, the organization supplied "high energy biscuits" and rice. Moreover, logistics and telecommunication support were also provided by the World Food Program. The goods financed by international donations were distributed via hubs in Cebu, Tacloban, Roxas, Ormoc, and Guiuan in the Philippines.

However, not only nonprofit organizations take responsibility for humanitarian aid. DHL, which is a German logistics company, also engaged in activities in the Philippines in 2013. In cooperation with the UN, the company built up a global

network of disaster response teams (DRT). These teams consisted of volunteers employed at DHL who were particularly trained for disaster situations. They are divided into three areas worldwide to be able to access all regions quickly. In the case of typhoon "Haiyan," the team supported the accessibility of the airport and increased throughput times so that goods could reach the victims faster. The DRT's greatest strength is its knowledge about logistics and how to solve crisis situations. Since building up in 2005, the teams have been successfully able to support victims of disasters more than 20 times.

For the future, the UN seeks to improve prevention. Naturally, it is impossible to reduce all risks to a minimum level, but improving prevention measures is still an option. A 10-year plan called the Hyogo Framework for Action (HFA) is designed to reduce the risks of natural disasters.

Discussion

- What is the difference between business and humanitarian logistics?
- Which four stages are included in the disaster management cycle?
- Where can you see the limitations of humanitarian logistics?
- What organizations are stakeholders in humanitarian logistics?

2.4.4 Resilient Sourcing Strategy Adoption at AGCO Corporation: Reacting to COVID-19 Virus Outbreak

AGCO Corporation is an agricultural equipment manufacturer. The company utilizes a multiple sourcing strategy combining local/domestic sources and international suppliers. The volume of local sourcing depends on the costs and some other factors that change dynamically. In the case of market-specific equipment, local sources are mostly used to increase responsiveness at times of disruptions. AGGO selects suppliers with consideration of both economic (costs) and risk criteria (Banker 2020). In addition, the company operates a digital sourcing platform to ensure an end-to-end communication with suppliers to identify disruptions in a timely manner. Risk management activities supported by the digital platform include the prediction of future scenarios and providing PPE (personal protection equipment) to suppliers to maintain manufacturing continuity.

Such preparedness and utilization of resilience practices at the "normal" times have allowed AGGO to survive through the COVID-19 pandemic. In the wake of the pandemic, various elements of resilience strategy to tackle SC disruptions by adopting multiple sourcing and supplier diversification strategies have been mobilized. Additional flexibility and adaptability were added by assemble-to-order and postponement production strategies entailing faster and customized deliveries to customers.

The COVID-19 action plan adopted by AGCO has been as follows. The supply side of AGCO had regular communication with Chinese suppliers which allowed it to respond quickly to the upcoming pandemic by searching for and utilizing

alternative sourcing channels. A task force was created comprised of plant managers, materials and logistics managers, purchasing, supplier quality, and finance managers. AGCO sourced/produced as many critical parts as possible in anticipation of the shutdowns of Chinese factories, and all the finished goods inventory was moved to European markets which have not been concerned with a pandemic yet. Even unconventional modes of transportation, e.g., railway shipment across Russia rather than using conventional, cheap sea shipping, have been activated. Even though these emergency transportation routes were 85% more expensive than sea freight, this recovery action allowed to avoid material shortages at European factories and so reducing the ripple effect.

In addition, shipment allocations have been centralized in decision-making processes to avoid disputes between different plants in the network regarding supply prioritization. Especially, factories producing equipment with higher gross margins and higher demand were prioritized for emergency supply allocations. AGCO also developed a special lockdown forecasting method, and so predicted South Korea's shutdown before it occurred. This helped to accelerate the impacted suppliers' shipments prior to the plant closures. Learning from this experience, they further improved the lockdown forecasting method and applied it to Europe, especially in Italy, where they predicted the closures 7 days prior to the official measures. AGCO used these 7 days of lead time to scale up deliveries from 240 suppliers before the lockdown announcement. The same policies have been adopted later in Brazil and North America.

We can see that both preparedness and reactive capabilities helped AGGO to achieve high resilience. They created flexible and adaptable SC network management at "normal" times and so were able to adapt efficiently and effectively in the wake of the COVID-19 pandemic that devastated SCs of many other companies.

Discussion

- What are the main resilience capabilities you can identify in this case study?
- What proactive and reactive measures can you see at AGGO?
- How can we prepare a supply chain to maintain business continuity in case of super-disruptions such as a global pandemic?
- Why is it important for resilience to design and manage SCs as flexible and adaptable systems at "normal" times?

2.4.5 Capacity Disruption at BASF

On October 17, 2016, as a result of an incorrect maintenance operation on a pipeline at BASF facility in Ludwigshafen (Germany), there was an explosion and subsequent fires at North Harbor, a terminal for the supply of raw materials such as naphtha, methanol, and compressed liquefied gases. More than 2.6 million tons of goods are handled there each year, and an average of seven ships a day moor at its docks. Two steam crackers, the starting point for producing basic chemicals, needed to be

stopped because they could no longer be supplied, and 22 were only partially working. The two steam crackers could have been restarted 2 days later, but only in May 2017 was the concept for reconstruction released whereby the reconstruction should be completed by September 2017. Restricted production output, a daily revenue decrease of 10–15% as compared to the previous year during the disruption period, impact on the basic chemicals division (about 21% of sales), delivery delays, limited access to key raw materials, exhausted product inventories, and a forecasted impact on 6% of BASF's annual earnings were some of the consequences of this incident. Logistics was temporarily shifted from ships and pipelines to trucks and trains. BASF was in close contact with its customers to keep them informed about the current availability of products to minimize the impact on customer deliveries. Because of BASF integrated "Verbundsystem" (networking system), comprised of various plants and delivery systems for feedstocks, the incident had an impact along the global supply chain (SC). This high and long-term impact is the so-called ripple effect.

BASF built a resilient SC; that is why the economic consequences of the aforementioned incident were considerably smaller than expected. BASF took process safety and risk prevention measures that included globally valid guidelines and requirements for buildings, etc., and practical security trainings for employees and support staff. Along with process safety and risk prevention measures, BASF has global emergency response management. This management consists of integrating worldwide group companies, joint ventures, partners, suppliers, and customers. Emergency phones and an integrated network of control centers (e.g., internal/external fire departments and rescue service) also enable this global emergency response management to work even more closely together. BASF was prepared for the incident in October 2016, but there is still long-term impact.

2.5 Examples of Sustainable Supply Chains and Operations

2.5.1 Circular Supply Chain at Unilever

Unilever is one of the major producers of consumer goods worldwide. With sustainable long-term growth forming the core of the company's business model, Unilever widely applies circular economy concepts as they pursue a zero-waste vision. In 2016, the company already achieved its goal of sending zero non-hazardous waste to landfills in more than 600 sites across 70 countries, including factories, warehouses, distribution centers, and offices. The company achieved this comprehensive waste reduction through the application of the 6 R approach: *refuse*, *reduce*, *reuse*, *recycle*, *recover,* and *rethink* as well as through supplying waste to different industries as a resource. To effectively implement its circular approaches across the entire SC, the company announced a partnership with the leading value-chain platform *2degrees,* which enables the company to share its lessons learned and experiences within a network of partners, aiming to further reduce waste for Unilever and all of its suppliers and business partners (Unilever 2019a; Grayson et al. 2018).

With only 9% of global plastic packaging waste being recycled—an estimated $80–120 billion dollar loss to the global economy each year—Unilever detected the need to apply and improve its circular approaches with an emphasis on its plastic packaging. In doing so, the company defined several milestones:

- 100% of the plastic packaging will be designed to be fully reusable, recyclable, or compostable by 2025.
- The content of recycled plastics in packaging will be increased by at least 25% by 2025.
- Help collect and process more plastic packaging than they sell.

In achieving these goals, the company explicitly embedded circular principles in its plastic packaging. By designing products for recyclability, areas such as modular packaging, design for disassembly and reassembly, more expansive use of refills, and recycling, are applied in innovative ways. Ultimately, by applying innovative, new business models, Unilever also involves the downstream part of its SC, namely the end customer, to take action. In France, for instance, Unilever is piloting a laundry detergent dispensing machine in supermarkets for the company's brands, so that customers can refill their empty laundry detergent bottles, thereby reducing their use of single-use plastics. Another example would be the company's participation in Loop™, which represents an innovative and waste-free shopping and delivery model for reusable packaging and refillable product formats, wherein refillable products are directly shipped to the customers and, after consumption, are sent back to be refilled (Unilever 2019b).

2.5.2 Coca-Cola' Sustainable Supply Chain

The Coca-Cola system created a value chain based on strong partnerships with suppliers, distributors, and retailers (Coca-Cola 2020). The Coca-Cola Company itself only manufactures the bases of the beverages, such as syrups and concentrates. It then distributes and sells these to its bottling partners (over 250 worldwide), located in every country where the products of Coca-Cola are sold (see Fig. 2.3).

The bottling partners manufacture packages and distribute the drinks as branded beverages to the points of sales (POS). The bottling partners also organize collaboration with suppliers. For example, Coca-Cola Beverages Vienna has had a long-term collaboration with SIG, a leading supplier of liquid packaging materials. Coca-Cola and SIG have recognized and embraced CPFR (collaborative planning, forecasting, and replenishment). SIG has electronic access to inventory levels and sales forecasts at Coca-Cola with no time delay (it was previously up to 15 days late). Coca-Cola receives exact delivery quantities and times. This has enabled Coca-Cola to reduce inventory levels for packaging material by 50%. This drastically reduces the processing time for orders and increases customer lock-in.

The bottling partners stay in close contact with grocery stores, supermarkets, convenience stores, cinemas, and all other customers providing drinks from Coca

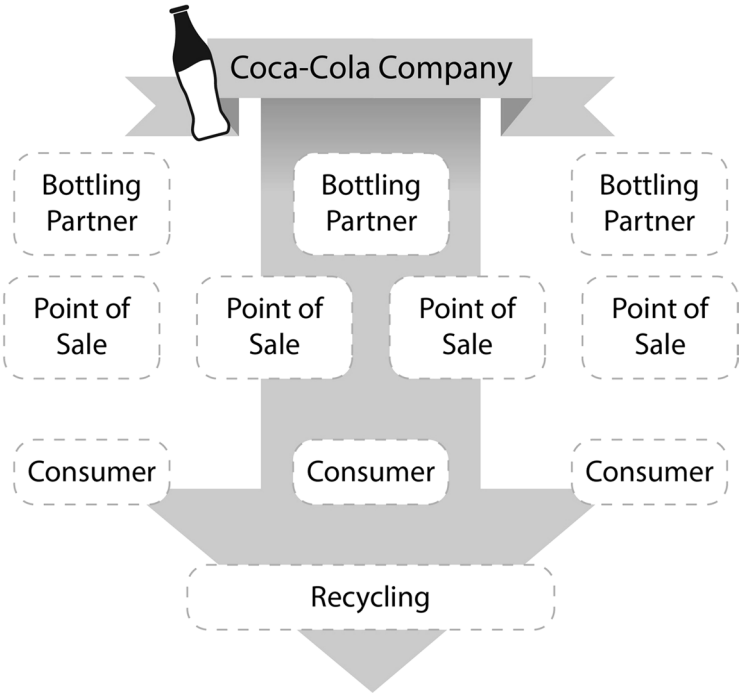


Fig. 2.3 Coca-Cola supply chain

Cola. After selling the beverages to customers, consumers are able to buy all Coca-Cola products from the different POS. Because every empty Coca-Cola bottle is recyclable it is up to consumers to take responsibility for recycling.

The whole value chain is based on the fact that the company and its suppliers do not go through a process from manufacturing to distributing the products to the consumer. However, consider a closed loop, from manufacturing to use, and recover material again for manufacturing. On the one hand, this minimizes production costs and, on the other hand, helps protect the environment. For instance, Coca-Cola contributed $2.0B (USD) to the planting of 25,000 acres of orange groves, thereby creating approximately 4100 jobs.

Because water is essential to the health of communities and communities are built by the people who buy drinks from Coca-Cola, the company has to find a way to replenish their water. To provide access to clean water and therefore create sustainable water provision for communities and, of course, the production of beverages, Coca-Cola contributed $18.4 M (USD) in 2012 to support 58 water initiatives worldwide.

In addition, supporting different projects on recycling and recovering used material and developing new technologies plays a crucial role in making the packaging process more sustainable. In 2009, the Coca-Cola system developed Plant Bottle packaging, a polyethylene terephthalate (PET) plastic bottle made partially from plants and completely recyclable.

Regarding new technologies and ecological concerns, Coca-Cola also introduced many projects to address climate change. Coca-Cola powers their trucks with a mix of efficient alternative fuels, for instance, biodiesel or natural gas. In North America, more than 750 trucks distributing products from Coca-Cola use about 30% less fuel than they would with conventional diesel.

2.5.3 Circular Economy: Worth Buying Twice with Déjà Vu

Déjà Vu was founded in Beijing in 2017 as a second-hand digital trading platform aligning with the *Five-Year Plan for Circular Economy Development* guidelines introduced by National Development and Reform Commission (NDRC) in China. Déjà Vu specializes in three primary product categories: books, clothing, and small electronics. The market value of these products is anticipated to experience substantial growth, surging from USD42 million in 2015 to approximately USD417.9 billion by 2025.

The consumer-to-consumer (C2C) market has been rising all over the world. However, particularly in China, second-hand consumers faced challenges such as counterfeit products, poor quality, and a lack of credibility, often resulting in financial losses and the acquisition of substandard items due to deceptive advertisements. As a solution, Déjà Vu operates its supply chain to improve quality and credibility, formulating a consumer-to-business-to-consumer (C2B2C) approach. Déjà Vu has 2 warehouses—in Tianjin or Kunshan. Once the first-hand customer feeds the article details into the system, the company accepts or rejects and sets the price according to the popularity and demand. A free courier service collects the product from the seller and transports it to the nearest warehouse. Each product is manually checked and repaired, cleaned, and refurbished as needed. When customers purchase a book, for instance, they can scan the ISBN code using Déjà Vu's WeChat mobile app to view the quality standards (i.e., "good condition," "medium quality," and "lower than average quality") and compare estimated prices with those of brand-new products.

Following successful experiments and pilot projects in Beijing and Shanghai, Déjà Vu expanded its supply chain network by opening its first-ever *Recycle Store* on Anfu Road in Shanghai, allowing customers to physically engage with its new product lines. Moreover, Déjà Vu is actively working to integrate circular design principles across its value chain. For example, by reusing courier boxes returned by customers, it has achieved a 3% monthly reduction in the demand for new courier boxes. These new expansions, combined with the growing popularity of circularity among the Chinese population, enabled Déjà Vu to attract 12 million users. By January 2024, the company had resold 36 million used books and 550,000 used clothing items through its platform, demonstrating its commitment to sustainability and circular economy practices.

2.5.4 The COVID-19 Vaccine Supply Chain

The COVID-19 pandemic created a massive challenge for healthcare systems worldwide since it required fast development, production, and distribution of vaccines, which has not been done before. The manufacturing of the vaccines was a critical part of the supply chain. For example, Pfizer (Biontech) used a network consisting of 20 facilities across Europe and the United States, which rapidly increased their previous production capacity from 1.3 billion doses of vaccine in 2021 to 4 billion doses in 2022. The competing company, Moderna, also expanded its production capacities by partnering with companies in Switzerland and Spain.

One of the biggest challenges for the vaccine supply chain was its cold storage that is needed. The Pfizer vaccine requires storing temperatures of −70 °C, while Moderna required storing at −20 °C. To fulfill these requirements, innovative solutions were developed that included Pfizer creating a special thermal shipper that was able to keep the vaccines in this container at −70 °C for 10 days without opening it. Equipped with temperature monitoring devices, it was ensured that the vaccine doses were kept at the correct temperature during the transport.

Another challenge was the vaccine distribution network, which involved both air and ground transportation. Airlines had to convert passenger planes to cargo flights in order to transport the vaccines, while ground transportation was handled by temperature-controlled cargo trucks. Main distribution centers were created in Kalamazoo (MI), Pleasant (WI), Puurs (Belgium), and Karlsruhe (Germany). To manage this complex network, UPS, for example, used RFID tags to track the vaccine shipments in real-time. For deployment of the vaccines worldwide, the COVAX Facility was created to ensure similar access to the vaccine for all countries, which aimed to distribute enough vaccines for 20% of the country's population.

2.6 Examples of Artificial Intelligence in Supply Chains and Operations

Artificial intelligence (AI) has significant potential to transform supply chain and operational management. Companies such as Amazon, Zalando, Hitachi, Schneider Electric, and Blue Yonder illustrate how AI can enhance operational activities (Porsche Consulting 2021). For instance, Blue Yonder's advanced AI analyzes 130,000 SKUs and 200 influencing variables to generate 150 million probability distributions, resulting in highly accurate demand forecasts (McKinsey 2016). Zalando utilizes neural networks to optimize picking routes for employees and picking carts, while Hitachi examines warehouse employees' problem-solving behaviors to develop an AI system that issues instructions, improving productivity by 8% (Porsche Consulting 2021).

These examples demonstrate AI's ability to enhance efficiency in supply chain operations and strengthen supply chain resilience. Companies like Resilinc and

Everstream have developed AI-driven solutions to proactively manage supply chain disruptions. Since 2010, Resilinc has offered a multi-tier visibility platform powered by AI and proprietary data. Two of their primary solutions are Multi-Tier Mapping and EventWatch for early alert systems and real-time insights. Their solutions are utilized by various organizations across 25 industries and 130 countries.

Resilinc combines autonomous AI mapping with supplier-validated data to comprehensively map the supply chain network. They utilize 12 years of supplier-validated data, including information on 800,000 suppliers and 1 million sites covering 4 million parts across 20 industries. This hybrid approach identifies vulnerable geographic hotspots and single points of failure, enabling proactive risk management. During a disruption, users can quickly identify available alternate sites and where to gain a head start.

EventWatchAI scans 104 million news sources for potential disruptions worldwide, ensuring real-time awareness of critical events that could impact the supply chain. The data in over 100 languages across more than 200 countries are processed. This predictive capability enables customers to outperform competitors when faced with unexpected disruptions. Customized alerts are classified into 50 event types and 400 risk categories, including extreme weather, cyberattacks, and labor issues. Real-time insights and "what-if" scenario planning empower organizations to strategize and maintain business continuity.

Efforts similar to these have been developed for decades. *DHL Resilience360* was launched in 2014 and later merged with Riskpulse to form Everstream in 2020. However, AI applications do have limitations. One major challenge is the understandability of results; users are more likely to utilize the tool when its results are comprehensible. Additionally, the supply network is constantly evolving, and current methods for mapping supply networks have several limitations (Pichler et al. 2023). Fake news can also negatively impact the quality of early alert systems. Lastly, we do not need to address all uncertainties, and more information does not necessarily lead to better decisions.

Bibliography

Bowersox DJ, Closs DJ, Copper MB (2023) Supply chain logistics management, 6th edn. McGraw Hill, Boston, MA

Chervenkova T, Ivanov D (2023) Adaptation strategies for building supply chain viability: A case study analysis of the global automotive industry re-purposing during the COVID-19 pandemic. Transp Res E 177:103249

Chopra S (2019) Supply chain management. Strategy, planning and operation, 7th edn. Pearson, Harlow

Das A, Gottlieb S, Ivanov D (2019) Managing disruptions and the ripple effect in digital supply chains: empirical case studies. In: Ivanov D et al (eds) Handbook of ripple effects in the supply chain. Springer, New York, pp 261–285

Dolgui A, Ivanov D, Sokolov B (2020) Reconfigurable supply chain: the X-network. Int J Prod Res 58(13):4138–4163

Dolgui A, Ivanov D (2023) Metaverse supply chain and operations management. Int J Prod Res 61(23):8179–8191

Fragapane G, Ivanov D, Peron M, Sgarbossa F, Strandhagen JO (2022) Increasing flexibility and productivity in industry 4.0 production networks with autonomous mobile robots and smart intralogistics. Ann Oper Res 308:125–143
Gleissner H, Moeller K (2011) Case studies in logistics. Gabler, Wiesbaden
Greasley A (2013) Operations management, 3rd edn. Wiley, New York
Heizer J, Render B, Munson C (2023) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow
Ivanov D (2020) Predicting the impact of epidemic outbreaks on global supply chains: a simulation-based analysis on the coronavirus outbreak (COVID-19/SARS-CoV-2) case. Transp Res E 136:101922
Ivanov D (2021) Supply chain viability and the COVID-19 pandemic: a conceptual and formal generalisation of four major adaptation strategies. Int J Prod Res 59(12):3535–3552
Ivanov D, Tang CS, Dolgui A, Battini D, Das A (2021) Researchers' perspectives on industry 4.0: multi-disciplinary analysis and opportunities for operations management. Int J Prod Res 59(7):2055–2078
Ivanov D, Dolgui A, Sokolov B (2022) Cloud supply chain: integrating Industry 4.0 and digital platforms in the "supply chain-as-a-service". Transp Res E Logist Transp Rev 160:102676
Ivanov D (2024) Introduction to supply chain analytics. Springer Nature, Cham
Ivanov D, Dolgui A (2025) Tariff shocks, ripple effect, and deep uncertainty in supply chains: we are entering a turbulence zone, please fasten your seatbelts. Int J Prod Res. https://doi.org/10.1080/00207543.2025.2520598
Ivanov D, Dolgui A, Sokolov B (eds) (2025) Handbook of ripple effects in the supply chain, 2nd edn. Springer, New York, ISBN 978-3-031-85507-8
Ivanov D (2025a) Introduction to supply chain resilience and viability, 2nd edn. Springer Nature, Cham
Ivanov, D. (2025b). Conceptual and formal models for design, adaptation, and control of digital twins in supply chain ecosystems. OMEGA 137:103356
MacCarthy B, Ivanov D (eds) (2022) The digital supply chain. Elsevier, Amsterdam
Meier O, Gruchmann T, Ivanov D (2023) Circular supply chain management with blockchain technology: a dynamic capabilities view. Transp Res E 176:103177
Panetto H, Iung B, Ivanov D, Weichhart G, Wang X (2019) Challenges for the cyber-physical manufacturing enterprises of the future. Annu Rev Control 47:200–213
Queiroz MM, Ivanov D, Dolgui A, Fosso WS (2022) Impacts of epidemic outbreaks on supply chains: mapping a research agenda amid the COVID-19 pandemic through a structured literature review. Ann Oper Res 319:1159–1196
Slack N, Brandon-Jones A, Burgess N (2022) Operations management, 10th edn. Pearson, Harlow
Sokolov B, Ivanov D, Dolgui A (eds) (2020) Scheduling in Industry 4.0 and cloud manufacturing. Springer, New York

References for Sect. 2.1.1

www.nike.com

References for Sect. 2.1.2

Achola VO (2024) Logist Res 17:6. https://doi.org/10.23773/2024_6
African sustainability matters (2024) The impact of sustainable supply chains on African economies. https://africasustainabilitymatters.com/the-impact-of-sustainable-supply-chains-on-african-economies/. Accessed 30 Dec 2024
https://afriglobal-logistics.com/

Kanyamuna V, Chawapiwa O, Bwanga C (2023) The effectiveness of service delivery in fast moving consumer goods supply value chain: a case study of Brands Africa Zambia Limited Company. Adv Soc Sci Res J 10(2):420–446. https://doi.org/10.14738/assrj.102.14177
Oyedijo A, Kusi-Sarpong S, Mubarik MS, Khan SA, Utulu K (2024) Multi-tier sustainable supply chain management: a case study of a global food retailer. Supply Chain Manag 29(1):68–97. https://doi.org/10.1108/SCM-05-2022-0205
Stroumpoulis A, Kopanaki E, Chountalas PT (2024) Enhancing sustainable supply chain management through digital transformation: a comparative case study analysis. Sustainability 16(16):6778. https://doi.org/10.3390/su16166778

References for Sect. 2.1.3

Logistik heute (2024) Nach Lieferproblemen: Tesla hat Produktion in Grünheide wiederaufgenommen. https://logistik-heute.de/news/nach-lieferproblemen-tesla-hat-produktion-gruenheide-wiederaufgenommen-85448.html. Accessed 30 Dec 2024
Sharma SK (2023) Defying conventional wisdom: A case study on Tesla's scalability and supply chain strategy. https://www.linkedin.com/pulse/defying-conventional-wisdom-case-study-teslas-supply-kumar-sharma. Accessed 29 Dec 2024
Supply Chain Digital (2024) Tesla: Transforming Supply Chain Sustainability. https://supplychaindigital.com/articles/tesla-transforming-supply-chain-sustainability Accessed 30 Dec 2024
Supply Chain Movement (2024) Tesla halts production in Berlin. https://www.supplychainmovement.com/tesla-halts-production-in-berlin/. Accessed 29 Dec 2024
Tesla (2021) Impact Report 2021. https://www.tesla.com/ns_videos/2021-tesla-impact-report.pdf. Accessed 30 Dec 2024

References for Sect. 2.2.1

Cooke JA (2012) From bean to cup: how starbucks transformed its supply chain. Supply Chain Quart 4(4):35–55
Starbuck Corporation (2014). www.starbucks.com. Accessed 01 Sept 2014

References for Sect. 2.2.2

Changi airport group (2024). https://www.changiairport.com/en.html. Accessed 29 Dec 2024

References for Sect. 2.2.3

Clark J (2014) What's next for car sharing? Europe Autonews. http://europe.autonews.com/article/20140207/ANE/140209903/whats-next-for-car-sharing. Accessed 21 Apr 2014
https://drive-now.com/
https://www.car2go.com

References for Sect. 2.3

Ayad A, Bohlke H, Menzel M (2022). https://cloud.google.com/blog/products/ai-machine-learning/volkswagen-uses-google-cloud-ai-for-more-efficient-cars. Accessed 29 Dec 2024
Bigrep (2024) Winds of change for Vestas: 3D printed tooling transforms wind turbines. https://bigrep.com/posts/vestas-wind-turbines-3d-printed-tooling/. Accessed 31 Dec 2024

Business Insider (2017) Mit dieser neuen Idee will Amazon die Packzentren völlig verändern. businessinsider.de/mit-dieser-neuen-idee-will-amazon-die-packzentren-voellig-veraendern-2017-2. Accessed 25 Feb 2017

Ciodive (2024) DHL Supply Chain experiments with generative AI. https://www.ciodive.com/news/dhl-supply-chain-experiments-with-generative-ai/732726/. Accessed 28 Dec 2024

Emerj (2024) AI for avoiding supply chain disruptions – two use cases. https://emerj.com/ai-for-avoiding-supply-chain-disruptions-two-use-cases/. Accessed 28 Dec 2024

Forbes (2023) From turbines to valves: 3D printing transforms energy manufacturers. https://www.forbes.com/sites/carolynschwaar/2023/12/03/from-turbines-to-valves-3d-printing-transforms-energy-manufacturers/. Accessed 31 Dec 2024

Helm B (2017) How this company makes $70 million selling random stuff on Amazon. https://www.inc.com/magazine/201603/burt-helm/pharmapacks-amazon-warehouse.html. Accessed 16 Oct 2017

https://www.dhl.com/global-en/delivered/innovation/ai-in-logistics.html

Johnson K, Lee ABH, Simchi-Levi D (2016) Analytics for an online retailer: demand forecasting and price optimization. Manuf Serv Oper Manag 18(1):69–85

Kimberling E (2024) The impact of artificial intelligence on supply chain management. https://www.linkedin.com/pulse/impact-artificial-intelligence-supply-chain-eric-kimberling-thwuc. Accessed 28 Dec 2024

Markforged (2023) 5 Beispiele für die Umgestaltung der Lieferkette durch 3D-Druck. https://markforged.com/de/resources/blog/3d-printing-supply-chain-5-examples. Accessed 31 Dec 2024

Osman M, Cui S, Tian Z, El-Said A (2023) From theory to practice: the impact of 3D printing on supply chain configurations and cost efficiency – a case study from Germany. In 4th Asia Pacific International Conference on Industrial Engineering and Operations Management. https://doi.org/10.46254/AP04.20230302

Quartz (2017) Amazon's massive fleet of robots hasn't slowed down its employment of humans. qz.com/885425/amazons-massive-fleet-of-robots-hasnt-slowed-down-its-employment-of-humans/. Accessed 25 Feb 2017

Simchi-Levi D, Wu MX (2018) Powering retailers' digitization through analytics and automation. Int J Prod Res 56(1–2):809–816

Singularity Hub (2017) How robots helped create 100,000 jobs at Amazon. singularityhub.com/2017/02/10/how-robots-helped-create-100000-jobs-at-amazon/. Accessed 25 Feb 2017

Tamir TS, Xiong G, Shen Z, Leng J, Fang Q, Yang Y, Jiang J, Lodhi E, Wang FY (2023) 3D printing in materials manufacturing industry: a realm of Industry 40. Heliyon 9(9):e19689. https://doi.org/10.1016/j.heliyon.2023.e19689

Vincent J (2018) Welcome to the automated warehouse of the future. https://www.theverge.com/2018/5/8/17331250/automated-warehouses-jobs-ocadoandover-amazon. Accessed 12 Jan 2019

References for Sect. 2.4.1

Ford (2020). https://www.at.ford.com/content/dam/atford/fna/images/articles/2020/04/face_shields/Face_Shield_Timeline.pdf. Accessed 2 Oct 2020

References for Sect. 2.4.2

http://www.toyota-global.com

Marsh P, Brown K, Anderlini J, Johnston T, Waldmeir P, Jung AS, Rickards J, Weitzman H, Ward A (2011) Japan crisis impact on the supply chain: global industries consider their options. Financial Times Europe, 17 Mar, p 16

References for Sect. 2.4.3

Altay N, Green WG (2006) OR/MS research in disaster operations management. Eur J Oper Res 175:475–493
DHL GmbH (2013) Disaster response teams leisten katastrophenhilfe. http://www.dpdhl.com/de/verantwortung/katastrophenmanagement/katastrophenhilfe_drt.html. Accessed 21 Mar 2014
Dubey R, Gunasekaran A, Papadopoulos T (2019) Disaster relief operations: past, present and future. Ann Oper Res 283(1–2):1–4
Ivanov D, Sokolov B, Dolgui A (2014) The ripple effect in supply chains: trade-off 'efficiency-flexibility-resilience' in disruption management. Int J Prod Res 52(7):2154–2172
Spens K, Kovács G (2009) Identifying challenges in humanitarian logistics. Int J Phys Distr Log Manag 39:506–528
Van Wassenhove L (2006) Humanitarian aid logistics: supply chain management in high gear. J Oper Res Soc 57:475–489
World Food Programme (2014) Responding to emergencies. http://www.wfp.org/emergencies. Accessed 20 Mar 2014

References for Sect. 2.4.4

Banker S (2020) A very agile supply chain: the inside story of AGCO's response to COVID-19. Logistics Viewpoints, 20 April 2020. [Online]. https://logisticsviewpoints.com/2020/04/20/a-very-agile-supply-chain-the-inside-story-of-agcos-response-to-covid-19/. Accessed 22 Jun 2020

References for Sect. 2.4.5

Dolgui A, Ivanov D, Sokolov B (2018) Ripple effect in the supply chain: an analysis and recent literature. Int J Prod Res 56(1–2):414–430

References for Sect. 2.5.1

Grayson D, Coulter C, Lee M (2018) All in. Why sustainability is the future of business. Routledge, Milton
Unilever (2019a) Going beyond zero waste to landfill. https://www.unilever.com/sustainable-living/reducing-environmental-impact/waste-and-packaging/going-beyond-zero-waste-to-landfill/
Unilever (2019b) Rethinking plastic packaging – towards a circular economy. https://www.unilever.com/sustainable-living/reducing-environmental-impact/waste-and-packaging/rethinking-plastic-packaging/

References for Sect. 2.5.2

Coca-Cola (2020). https://www.coca-colacompany.com/sustainable-business. Accessed 5 Oct 2020
Senger E (2003) Supply chain prototyp mit coca-cola beverages. Universität St Gallen, St Gallen

References for Sect. 2.5.3

Ellen MacArthur Foundation (2024). https://www.ellenmacarthurfoundation.org/circular-examples/enhancing-the-quality-of-resale-deja-vu. Accessed 29 Dec 2024

References for Sect. 2.5.4

Devex (2020) The unspoken COVID-19 vaccine challenges – distribution and corruption https://www.devex.com/news/opinion-the-unspoken-covid-19-vaccine-challenges-distribution-and-corruption-98437 Accessed 27 Dec 2024

DHL (2021) Vaccine distribution: putting our plans into action. https://www.dhl.com/global-en/delivered/global-trade/vaccine-distribution.html. Accessed 27 Dec 2024

https://www.capstonelogistics.com/blog/logistics-update-challenges-in-covid-19-vaccine-distribution/

https://www.iata.org/en/programs/cargo/pharma/vaccine-transport/

https://www.pfizer.com/science/coronavirus/vaccine/manufacturing-and-distribution

Kana BD, Arbuthnot P, Botwe BK, Choonara YE, Hassan F, Louzir H, Matsoso P, Moore PL, Muhairwe A, Naidoo K, Ndomondo-Sigonda M, Madhi SA (2023) Opportunities and challenges of leveraging COVID-19 vaccine innovation and technologies for developing sustainable vaccine manufacturing capabilities in Africa. Infect Dis 23(8):E288–E300

Peter LL, Schroeder L, Oliveira FN, Leiras A (2023) Logistics of Covid-19 vaccines: main challenges in theory and practice. Production 33:e20220036

Plumlogix (2023) Covid-19 vaccine management: challenges and solutions for success. https://plumlogix.com/covid-19-vaccine-management/. Accessed 27 Dec 2024

Wouters OJ, Shadlen KC, Salcher-Konrad M, Pollard AJ, Larson HJ, Teerawattananon Y, Jit M (2021) Challenges in ensuring global access to COVID-19 vaccines: production, affordability, allocation, and deployment. The Lancet 397(10278):P1023–P1034

References for Sect. 2.6

Everstream (2025). https://www.everstream.ai/. Accessed on 6 Jan 2025

McKinsey (2016). https://www.mckinsey.com/capabilities/operations/our-insights/big-data-and-the-supply-chain-the-big-supply-chain-analytics-landscape-part-1#/. Accessed 6 Jan 2025

Pichler A, Diem C, Brintrup A, Lafond F, Magerman G, Buiten G, Choi TY, Carvalho VM, Farmer JD, Thurner S (2023) Building an alliance to map global supply networks. Science 382(6668):270–272

Porsche Consulting (2021). https://www.porsche-consulting.com/sites/default/files/2023-04/boosting_ai_in_supply_chains_c_porsche_consulting_2021.pdf. Accessed 6 Jan 2025

Resilinc (2025). https://www.resilinc.com/. Accessed 6 Jan 2025

Processes, Systems, and Models

3

Learning Objectives for This Chapter

- Interrelations between business processes, quantitative models, and information systems
- Role of business process management in operations and supply chains
- Effects of management information systems in SCOM
- Management information technologies, e.g., ERP, APS, WMS, RFID
- Planning, problem, and decision
- Role of models and modeling in decision-making
- Quantitative methods of decision-making
- Modeling and data analytics

3.1 Introductory Case Study: AirSupply

Each time an aircraft is made, it is the result of assembling a multitude of parts representing a very large volume of orders. And what is true for the aircraft assembly line is true for the whole aerospace industry, where not only aircraft but also helicopters and satellites are built. Parts for manufacturing come from multiple suppliers from all over the world. Most of them are quite complex and need to fulfill the

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_3) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_3.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_3

highest quality standards. Each time, a delay can result in very high costs. It is essential that all suppliers involved in the manufacture of an aircraft have real-time visibility of demand and inventory to adapt to fluctuations and changes in customer requirements. As a result, it is also essential for customers and suppliers to have a common tool that enables them to better collaborate and gain visibility over demand as well as inventory (see Fig. 3.1).

One such tool is SupplyOn's AirSupply solution, which was developed in collaboration with Airbus/EADS and other major manufacturers and suppliers in the European aerospace industry. If AirSupply is mainly characterized by the networking of the whole Aerospace & Defense supply line, its main innovation lies in the fact that it provides all users with one single tool to view demand and inventory data. The solution combines major new technologies such as cloud computing, software as a service (SaaS), and e-procurement. It was developed in cooperation with SupplyOn, which has many years of experience with web-based SCM solutions in the automotive industry.

AirSupply offers services for a number of SC processes, covering the entire product cycle. The following main functions are provided (see Fig. 3.2):

- Forecast collaboration
- E-Procurement
- Vender-managed inventory (VMI)

3.1.1 E-Procurement

AirSupply is a solution for an integrated SC where e-procurement plays a crucial role. Other than traditional procurement using enterprise resource planning (ERP) and simple EDI (electronic data interchange) communication, the e-procurement

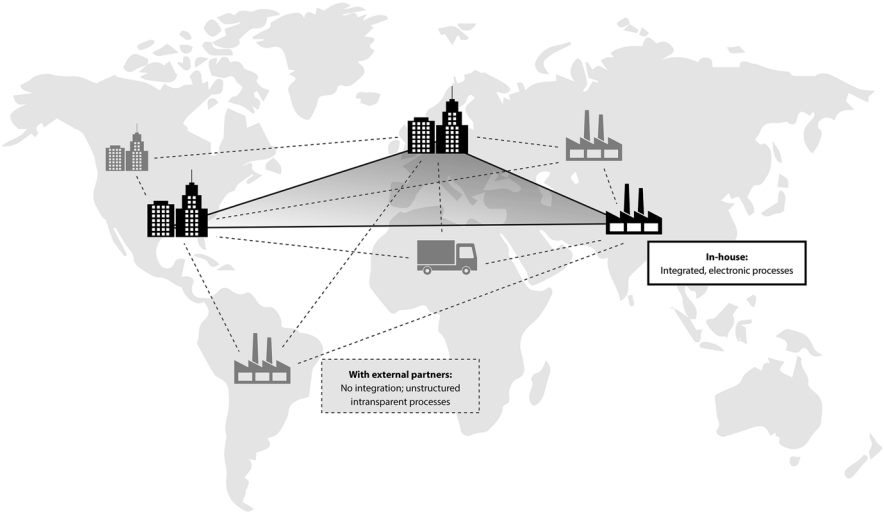


Fig. 3.1 Internal and external process integration in the supply chain

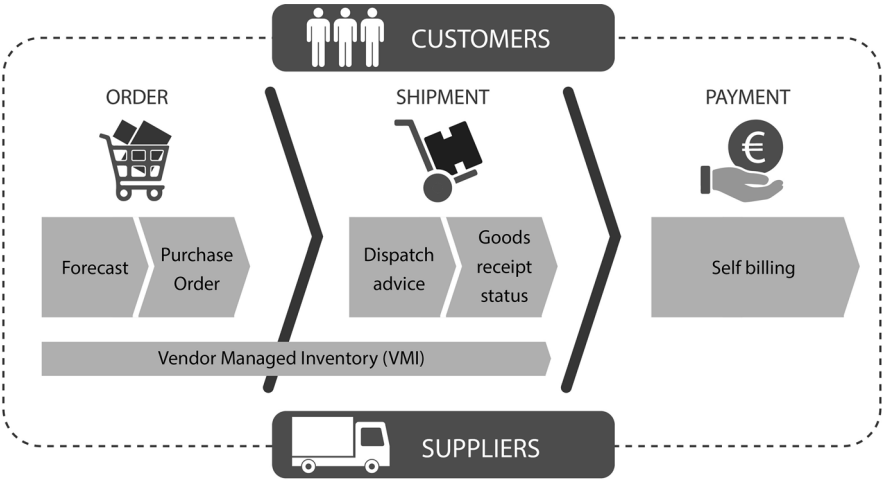


Fig. 3.2 Processes in supply chain management

solution provides real-time visibility of demand fluctuations and closer networking between suppliers and customers who all have the same view on demand.

Many of the aerospace industry's other characteristics, such as complex parts and high-quality standards, make SC visibility and collaboration crucial factors for maintaining a competitive edge. Each airplane is produced according to individual customer requirements, and several million parts are required at the right time to minimize inventories and prevent costly delays. Especially for suppliers using expensive parts from other suppliers, the financial risk can be high but can be mitigated through e-procurement.

The following description of the purchase order process shows how AirSupply facilitates SC processes. After the customer creates the purchase order (PO) in their ERP, three different scenarios can be supported by AirSupply:

- The supplier accepts the PO without changes.
- The supplier accepts the PO with changes. The customer either accepts or makes a new proposal.
- The supplier accepts the PO without changes, but the customer requests a modification. The supplier can accept and dispatch the goods or the customer can make a new proposal.

AirSupply automatically synchronizes the PO information from the cloud to the ERP system.

3.1.2 Vendor-Managed Inventory

The Vendor-Managed Inventory (VMI) process, in particular, shows a strong need for integrated SC information systems (see Fig. 3.3).

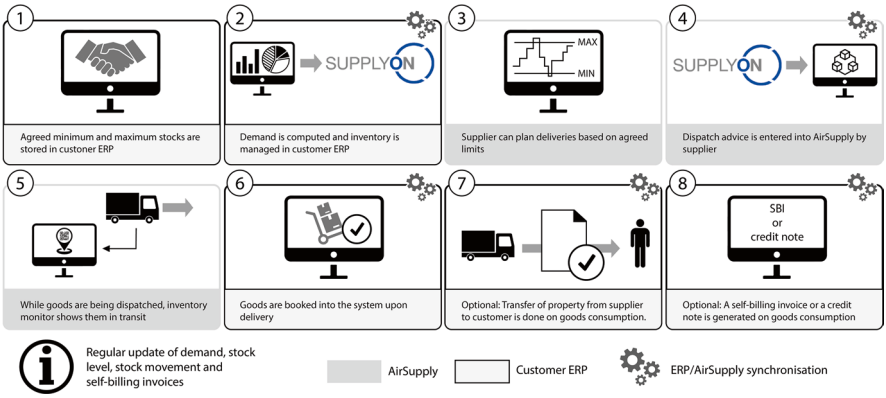


Fig. 3.3 Vendor-managed inventory process

In VMI, the supplier (or vendor) is responsible for managing the supply and usually agrees with the customer on a minimum and a maximum inventory level. This requires that the suppliers have timely and accurate information on the customer's inventory levels at all times. AirSupply strongly supports this SCM strategy by automatically advising suppliers when customers' inventory levels are critically low. Suppliers can also simulate deliveries to make adjustments prior to actually dispatching goods.

Closer collaboration has helped reduce safety stock and improve service level with the use of VMI. Manual work is also cut down as supplier data does not need to be entered, and disputes linked to reception, inspection, and billing can be prevented.

3.1.3 Implementation

AirSupply is based on a Hybrid Cloud with SaaS (software as a service) solution that allows the management of established internal processes. More than 1000 suppliers were integrated into AirSupply by the end of 2012, and the solution connects all major firms of the European aerospace industry and has become the dominant platform. The common cloud platform enables standardized and integrated procurement processes while at the same time offering customized services to each individual company. Since cloud computing helps companies keep their hardware and software investments lower and more flexible, AirSupply offers a huge savings potential. Furthermore, cloud computing providers can guarantee high data security and service reliability that a single company's IT, especially a smaller company, could hardly do. At first glance, cloud services are particularly attractive for small and medium-sized enterprises (SMEs) that want to access massive computing power and specialized software with low fixed costs. For larger companies, the importance of having control over their IT infrastructure to secure and protect crucial internal

processes could rather outweigh the benefits of cloud computing. However, the hybrid cloud solution can achieve the best combination of internal and external, fixed and variable IT systems for any company.

Discussion

- What benefits of cloud computing can you identify for SC integration?
- What benefits and drawbacks of e-procurement can you identify?
- How does the AirSupply platform encourage its users to increase collaboration and develop mutual trust to share information along the SC?
- What could prevent industry members from joining AirSupply?

3.2 Business Process Management

3.2.1 Process Optimization and Reengineering

A process is a content and logic sequence of functions or steps that are needed to create an object in a specified state. A *business process* is a network of activities for accomplishing a business function. Processes have input and output parameters and may be tied to one functional area or be cross-functional. Today companies are organized on the basis of processes. As said by W. Edwards Deming (Professor at Columbia University; 1900–1993), "if you can't describe what are you doing as a process, you don't know what are you doing."

Examples of processes include the following areas:

- Manufacturing and production, e.g., assembling the product
- Sourcing, e.g., selecting suppliers
- Human resources, e.g., hiring employees

As an example, consider the process "Order fulfillment" (Fig. 3.4):

It can be observed that fulfilling a customer order involves a complex set of steps that requires the close coordination of the sales, accounting, and manufacturing functions. The basic concept for managing processes in an organization is called *business process management (BPM)*. BPM contains a variety of tools, methodologies to analyze, design, and optimize processes. BPM is comprised of the following steps (Hammer and Champy 1993):

- Identify processes for change
- Analyze existing processes
- Design the new process
- Implement the new process
- Continuous measurement

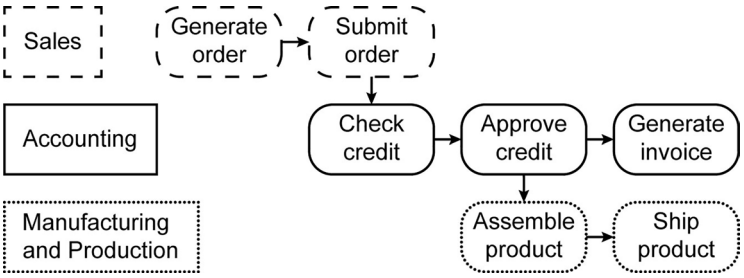


Fig. 3.4 Business process "Order fulfillment" [based on Laudon and Laudon 2021]

Processes can be optimized subject to finding their best state with regard to:

- Costs
- Capacity
- Time
- Quality
- Service level
- Reliability
- Flexibility
- Sustainability

The process analysis results may be characterized by *effectiveness* (the achievement of process goals) and *efficiency* (performing the process with minimum costs). Processes that are both effective and efficient are called *optimal*. Optimal processes are characterized by effectiveness, efficiency, controllability, stability, flexibility, analyzability, observability, reliability, documentability, and continuous improvement capabilities.

Most SCOM problems involve multiple objectives. The interrelation of objectives is called a "trade-off," which means that a manager has to find a compromise. For example, it is very difficult to achieve both low inventory and high customer service levels, or both high capacity utilization and short lead time.

▶ **Practical Insights** *Sustainability* has become more and more important in process optimization in production, logistics, and services. Consider two apples in a German supermarket: an apple from South Africa and an apple from Germany. Which apple is more sustainable? Different costs, such as transportation and warehousing costs, should be considered in the analysis. In autumn, a German apple would be more sustainable. In spring, an apple from South Africa can be more sustainable, despite transportation emissions, because of the high warehousing costs and the corresponding energy consumption and emissions in Germany.

Process optimization in organizations is known as *business process reengineering (BPR)*. BPR is an organization-wide philosophy of continuous process

improvement. It is comprised of different views of process management, such as operational, organizational, ethical, etc.

▶ **Practical Insights** Options for process improvement frequently lie in coordination. For example, in the automotive industry, the supply cycle between a customer order and car delivery is approximately 60 days. The car assembly itself accounts for only 2 of those days. This means that the value-adding processes comprise less than 5% of the total supply cycle. 95% are planning and coordination activities. Great potential for optimization can be found in these activities.

In SCs, the following options can be considered for *process improvement*:

- SC design and configuration (i.e., new market acquisition, lengthy time-to-market, non-resilient SC design)
- SC planning (weak flow capacities and lengthy supply cycle)
- SC operations (false priorities for customers' orders, imbalance of capacities and order volumes, too frequent disruptions, and high costs for recovery)
- SC performance evaluation (performance of different departments such as logistics, transport, and production is evaluated locally for each department without any general links to the larger SCM perspectives)
- SC execution (different levels of managers' qualifications, false or incomplete process documentation, weak consistency in process performance evaluation)

3.2.2 Business Process Modeling

In order to perform process optimization, as-is processes should be described. *Process models* describe SCOM activities from an information-processing perspective. Process modeling can be referred to as descriptive modeling and serves as an interface for developing information systems.

For *business process modeling*, a number of techniques and tools can be used. The most popular of these are as follows:

- SCOR (Supply Chain Operations Reference)
- ARIS (Architecture of Information Systems)
- UML (Unified Modeling Language)
- IDEF (Integration Definition for Function Modeling)

These approaches can also be used to model the *workflow* of decision-making processes. The process modeling serves to (1) describe processes and structures and (2) clearly illustrate those entities. For these purposes, different solutions have been developed, e.g., activity diagrams in UML and event-process chains (EPC) in ARIS.

The business process models typically have the following elements:

- Activities—Transform resources and information of one type into another type
- Decisions—A question that can be answered with Yes or No
- Roles—Sets of procedures
- Resources—People, or facilities, or computer programs assigned to roles
- Repository—Collection of business records

One of the well-known models for describing SC processes and measuring their performance is SCOR (Supply Chain Operations Reference Model) (see Fig. 3.5).

The SCOR model structure, which was presented in 1996 by the Supply Chain Council (later APICS), is not difficult to understand. Just imagine you are about to organize a party. What decisions are involved? First, we need to plan. Then we need to source food and drinks. The next step is to hold the party. Finally, your guests need to be delivered to their homes. The same principles are used in the SCOR model structure. It contains a description of planning, sourcing, making, delivering, and return processes in the SC at different levels of abstraction.

The main value of SCOR from the business process modeling point of view is that the standardized business process models are interlinked at three levels. In addition, a coherent system of performance indicators is correlated with the process models. Finally, the data origins used to calculate the performance indicators are explicitly provided. A *SCOR* project typically contains the following stages:

- Business process modeling
- Benchmarking analysis
- Business process reengineering
- Process reference model

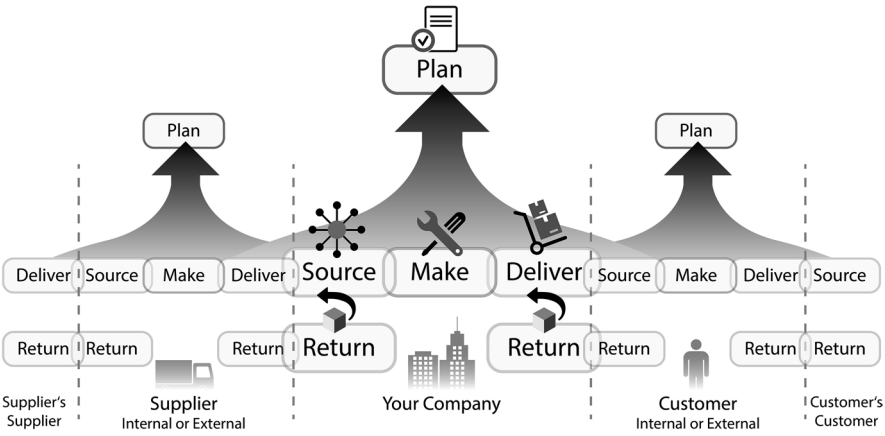


Fig. 3.5 SCOR model

The classical SCOR Model was continuously developed over time, and numerous improved versions were produced. In 2022, the Association for Supply Chain Management (ASCM) presented the new SCOR DS (Supply Chain Operations Reference Digital Standard), which is an adjusted and modernized version of the SCOR model.[1]

What are the differences or advancements of this new development? The classical SCOR Model primarily focuses on a linear SC. It deals with planning, sourcing, making, delivering, and return processes. The SCOR DS enhances the classical SCOR model by integrating SC resilience or sustainability. It is more dynamic and supports SC visibility and collaboration within the supply network.

In addition to the classical SCOR Model, the new processes are as follows:

- Orchestrate: To create an efficient SC connected to the business strategy and its rules and regulations, it considers state-of-the-art technology and human resource management.
- Order & Fulfill: This substitutes the former "Deliver" process. Now, customer orders ("Order") and the company's fulfillment activities ("Fulfill") are tackled separately as they require specific focus.
- Transform: This substitutes the former "Make" process to emphasize not only manufacturing companies but also to meet the needs of service or retail activities.

That means the revised processes according to SCOR DS are Plan, Order, Source, Transform, Fulfill, and Return. The enhancement from SCOR to SCOR DS should allow a broader range of organizations to apply these adjusted processes, use the respective metrics (KPIs), and develop the skills and teams to assess, measure, and improve performance.

3.3 Management Information Systems

3.3.1 Role of Information Technology in Supply Chain and Operations Management

SCOM managers say that where information is missing, material is also missing. A universal property of the management processes, irrespective of the problem domain, is that it has a notably *informational nature*, i.e., it is connected, first of all, with the collection, processing, analysis, and usage of data, information, and knowledge.

Management Information Systems (MIS) collect, process, store, and distribute information in order to support decision-making, coordination, and control. MIS use data, i.e., they are streams of raw facts. Information is data shaped into meaningful form. For companies, MIS is an instrument for *creating value*. Investments in the right information technology (IT) can result in superior returns in terms of

[1] https://www.ascm.org/corporate-solutions/standards-tools/scor-ds/ (accessed 12.01.2025).

productivity, revenue, and long-term strategic positioning. On the one hand, MIS automate steps that were done manually before. On the other hand, MIS enable entirely new processes by changing the flow of information, replacing sequential steps with parallel steps, eliminating delays in decision-making, and support new business models.

3.3.2 Types of Management Information Systems

The extensive development of IT for SCOM began in the 1980s. In the 1970s, first functions, such as accounting, were automated. In the 1980s, material requirements planning (MRP), sales and operations planning (S&OP), and MES (manufacturing execution systems) were developed. At the same time, additional functions, like product design and quality control, have been captured under the umbrella of CIM (computer integrated manufacturing).

Along with providing substantial advantages and help for managers, these systems were so called island solutions, without integration with each other. Since these systems used the same data to a great extent, in the 1990s, integrated enterprise planning systems (ERP) were developed.

In the twenty-first century, SCM sparked the development of APS (advanced planning and scheduling) systems and different IT for SC collaboration. At present, mobile technologies, Internet of Things, smart manufacturing and Industry 4.0, and cloud computing belong to the trends of IT development for SCOM.

The variety of the existing IT can be distinguished according to the following groups.

Planning at the Enterprise Level

- ERP (Enterprise Resource Planning)
- MES (Manufacturing Execution Systems)
- WMS (Warehouse Management Systems)

Planning and Control for SC Coordination

- APS (Advanced Planning Systems)
- TMS (Transportation Management Systems)
- SCEM (SC Event Management)

Real-Time Control

- RFID (Radio Frequency Identification)
- T&T (Trace and Tracking)
- ASN (Automated Shipping Notification)

Business Intelligence

- Big Data Analytics
- Machine Learning
- Cloud computing

SC Communication and Data Interchange

- EDI (Electronic Data Interchange)
- E-Commerce
- XML (Extensible Markup Language)
- Mobile technologies, Android

We will consider some of these further in the next chapter. Some advanced digital technologies are considered in Chap. 16.

3.3.3 Management Information Systems and Organization

IT provides a new level of coordination capabilities in SCs and have enabled a breakthrough in SC responsiveness and flexibility. Modern IT can potentially enable almost any integration and coordination concept. More important problems for efficient coordination lie in the organization sphere, collaboration culture, and trust.

First, the issues surrounding *investments in IT infrastructure* should be highlighted. Primarily, this concerns the amount a company should spend on IT. IT can be bought, rented, or used via cloud computing. IT can also be available via outsourcing. Whatever option is chosen, the *total cost of ownership (TCO)* model should be used to analyze direct and indirect costs. Note that hardware and software costs account for only about 20% of TCO. Other costs include installation, training, support, maintenance, infrastructure, downtime, space, and energy. TCO can be reduced by the use of cloud services, greater centralization, and standardization.

Second, *IT project management* is a critical issue. As in any project management, activities in an IT project include planning work, assessing risk, estimating resources required, organizing the work, assigning tasks, controlling project execution, reporting progress, and analyzing results. Five major variables for planning an IT project are scope, time, cost, quality, and risk.

In spite of sophisticated project management, empirical data shows that 30–40% of IT projects exceed schedule and/or budget or fail to perform as specified. Frequently, the reasons are as follows: failure to capture essential business requirements; failure to provide organizational benefits; complicated, poorly organized user interface; and inaccurate or inconsistent data.

A road map in an IT project should be elaborated at all the management levels and include the following parts:

- Purpose of the plan
- Strategic business plan rationale
- Current systems/situation
- New developments
- Management strategy
- Implementation plan
- Budget

Third, the issue of *global organizations* has become a critical challenge for many IT infrastructures, especially in global SCs. Unfortunately, many companies fail to build the right IT for their global SCs. As an example, consider a typical scenario of disorganization on a global scale. A multinational consumer goods company based in the United States and operating in Europe would like to expand into Asia. World headquarters and strategic management are located in the United States. The only centrally coordinated information system is financial controls and reporting. The company is divided into separate regional, national production, and marketing centers. Foreign divisions have separate IT systems, e-business systems of different divisions and centers are incompatible. In addition, each production facility uses different ERP systems, different hardware, and so on.

Recommendations on the right way of organizing IT in global companies include the following actions. First, it can be advised to share only core systems that support functionality critical to the company: systems that share some key elements can be partially coordinated. It is not advisable to have exactly the same systems across national boundaries: local variation is desirable since IT must be adapted to local requirements.

3.3.4 ERP Systems

Enterprise resource planning (ERP) systems are comprised of integrated software modules and a common central database. ERP modules include but are not limited to:

- Production Management
- Inventory Management
- Sales
- Human resources
- Marketing
- SCM

The modules are like a number of USB sticks that can be purchased all together or singularly. The working principle of ERP is based on the collection of data from different departments of a firm for use in nearly all of the firm's internal business activities. The advantage is that information entered in one process is immediately available for other processes. ERP uses:

- Master data (data not changed frequently, e.g., bill-of-materials)
- Process data (data changed operatively, e.g., current inventory level)

Advantages of ERP are as follows:

- Provides integration of the SC, production, finance, and marketing
- Creates a commonality of databases
- Can incorporate improved best processes
- Increases communication and collaboration between business units and sites
- Has an off-the-shelf software database
- May provide a strategic advantage and increase company assets

The *limitations* of ERP are as follows:

- Can be very expensive to purchase and even more so to customize
- Implementation may require major changes in the company and its processes
- Can be so complex that many companies cannot adjust to it
- Involves an ongoing, possibly never completed, process for implementation
- Expertise is limited with ongoing staffing problems

▶ **Practical Insights** An integrated ERP system is costly. Its implementation will change organizational processes and structure. The project will take into account many years. At the same time, there are many companies with "self-made" software for production planning, accounting, etc. The question is whether we should buy new ERP or develop and integrate self-made solutions? There is no absolute answer to this question. There are companies that develop their own solutions. In other cases, new ERP projects are successful. The key success factors depend much on the correspondence of the existing IT to the current and future needs of the company. It is also important how qualified and organized the IT department at the firm is.

3.3.5 APS Systems

At the beginning of the twenty-first century, SCM topics became more and more important. Some important modules such as "SC design" and "Distribution planning" were missing in ERP. In addition, ERP provided very limited support for mathematical optimization in scheduling. ERP also had other restrictions, such as planning against unlimited capacities. That is why a new generation of APS systems has been developed (Stadtler et al. 2015) (see Fig. 3.6).

An APS extracts data from ERP and uses it for optimization. Consider this example of scheduling in APS (see Fig. 3.7):

APS system:

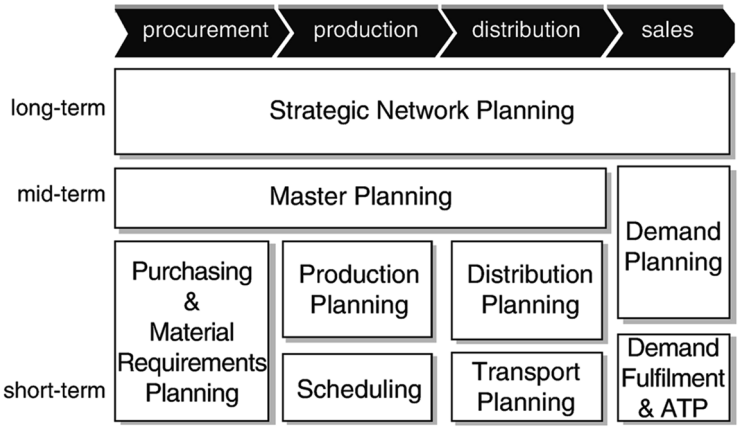


Fig. 3.6 Structure of APS system (Stadtler et al. 2015)

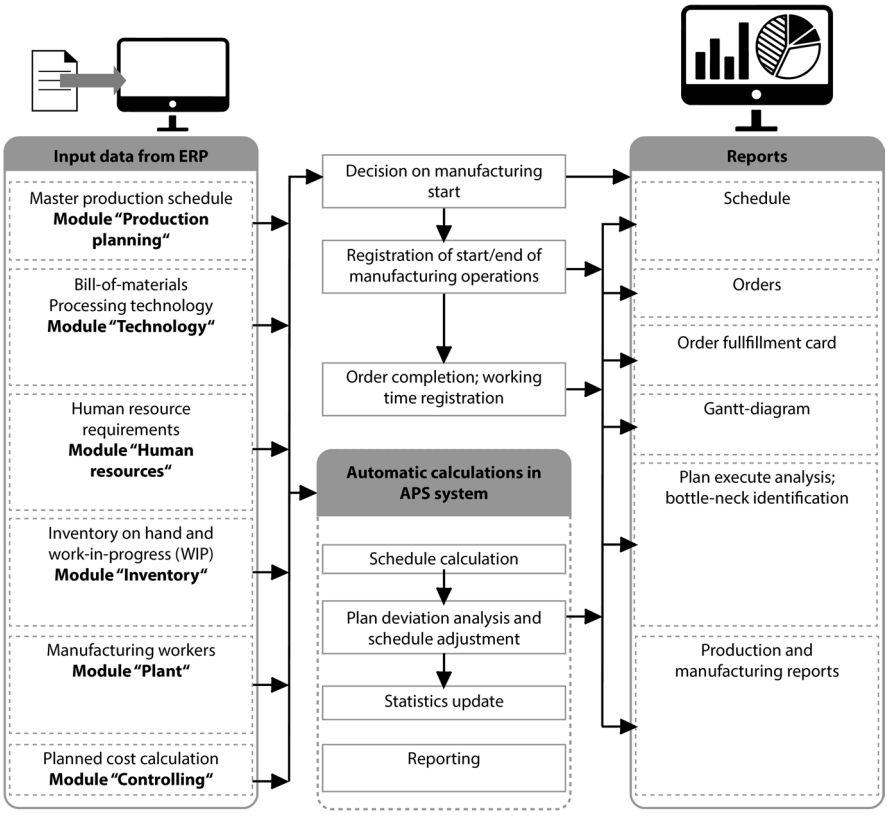


Fig. 3.7 Working principle of APS systems

- Extracts some input data from different ERP modules
- Performs automatic calculations on the basis of scientific optimization methodology
- Generates reports

APS modules are mainly dedicated to deterministic planning. However, uncertainties exist on both the inbound (unreliable suppliers, machine breakdowns) and outbound (unknown customer demand) sides. To hedge against uncertainty, buffers have to be installed—either in the form of safety stocks or safety times.

The practical application of APS is not easy. First, there is a discrepancy between the expectations of companies and the capabilities of the software. Second, optimization models can often not be solved within reasonable time limits, especially if interactive planning is desired. Third, the willingness to share information, e.g., about costs or capacities, is often rather limited. Problems also arise in using generic model formulations, which are often not sufficiently tailored to the needs of the specific application environment. APS appear to be most successful for intracompany SCs with centralized logistics control (Günther 2005).

3.3.6 SCEM and RFID

ERP and APS systems support activities planning. For *real-time control*, other systems are used. We consider two IT solutions for the *control* stage: SCEM and RFID. Supply Chain Event Management (SCEM) aims at timely identification of deviations in SCs. The basis for disruption *alerts* and disruption recovery is a tolerance area of execution parameters' admissible deviations. SCEM is composed of five main functions:

- Monitoring of processes
- Notification about an impermissible parameter deviation
- Simulating possible adjustment actions
- Selecting a control action to eliminate the deviation
- Measuring on the basis of performance indicators

SCEM is based on three *main drivers*. First, tracking and tracing (T&T) systems, RFID, and mobile devices are used to provide current information about process execution. Second, the method of management by exception is used to filter information and to compare actual parameter values with planned ones. Third, the method of event-oriented planning is used to identify sensitive adjustment actions in the case of negative events.

One of the basic technologies for facilitating feedback in SC execution is RFID. In recent years, applications and research literature on RFID have grown exponentially. RFID is an automatic identification technology composed of RFID tags and specialized RFID readers. *Tags* are also called RFID transponders (abbreviation of transmitters–responders). Tags are attached to or incorporated into any

kind of object (product, tool, animal, goods, human being, etc.) for the purpose of identification and tracking using radio waves. RFID *readers* read the information on tags and transfer it to a processing device. When applied to SCOM, RFID technology can provide several crucial advantages, but also has limitations (see Table 3.1).

The common target of RFID is to reduce costs created by manual operating, accelerate data receipt and transmission, and increase the preciseness and quality of data. RFID technology allows t*racking products throughout the SC* from supplier delivery to warehouses and points of sale. This contributes to reducing uncertainty concerning fluctuations in processes, improving the readability of the quantity and quality of items produced, improvements in safety (in particular, reduction of counterfeiting), and reduction of waste and theft.

RFID is also believed to provide crucial benefits in ensuring SC stability/continuity by means of improved information sharing and SC monitoring support through a faster exception management. SCs can potentially become more flexible, responsive, agile, and secure by applying RFID.

At the same time, the existing studies underline that RFID does neither explain nor solve the fluctuations of customer demand, transportation times, and inventory levels. It identifies and processes the data in the volume according to the tags, readers, and middleware functionalities, and at the places where they are installed. RFID also does not propose any control actions that should be taken to adapt an SC in the event of changes or disruptions at the execution stage.

3.3.7 Business Analytics and E-Business

During the last decade, new IT for SCOM has been developed. Details of managerial and technical implementation on these technologies differ, but most of them share attributes of intelligence. Since SCOM depends on the use of IT, these new tools will change the landscape of managerial concepts and decision support systems for SCOM in the coming years. Associating (and even embedding) IT and intelligence into SC planning and organizational structures has been discussed for over 10 years now. However, questions remain about the benefits and obstacles in allowing orders, deliveries, and products to be able to plan and control their own progress through an SC. With new IT, data management becomes a competitive advantage for companies. Trends in IT development for SCOM include:

Table 3.1 Advantages and limitations of RFID for SCOM

Advantages	Limitations
Up-to-date data	Costs
Contact-free	Technical issues
“Bulk” reading	Privacy issues
Protection/security	Heat resistance

- Business Intelligence (BI) and Big Data Analytics
- Cloud computing and SaaS (software as a service)
- E-Business, net marketplaces (e-hubs) and virtual enterprises
- IoS—Internet of Services/Internet of Things and Industry 4.0

BI comprises data and software tools for organizing and analyzing data. It is used to help managers and users make improved decisions. Presently, so-called Big Data Analytics (cf. Chap. 16) are becoming more and more important. Data mining belongs to the group of business analytics that is comprised of tools and techniques for analyzing data.

With the help of *cloud computing*, hardware/software can be used as a service without buying IT. This allows companies to minimize IT investments. The drawbacks are concerns about security and reliability.

E-business consists of the use of digital technology and the Internet to drive major business processes.

▶ **Practical Insights** In many e-commerce service SCs, especially in startups, the managers focus rather on the link between warehouses and customers and frequently undervalue the link between suppliers and warehouses. In manufacturing SCs the focus is rather on the link between suppliers and production, and the customer may play an underemphasized role in SCOM decisions. It is, therefore, important to consider integrating the SC from suppliers to customers for both service and manufacturing SCs.

E-commerce is a subset of e-business and consists of buying and selling goods and services through the Internet. Presently, e-commerce solutions are being developed to include mobile IT in SCOM. In order to implement e-business, *Electronic data interchange (EDI)* technology is used. EDI ensures computer-to-computer exchange of standard transactions such as invoices, or purchase orders. Major industries have EDI standards that define the structure and information fields of electronic documents. More companies are increasingly moving toward private networks that allow them to link to a wider variety of firms than EDI allows and share a wider range of information in a single system.

▶ **Practical Insights** Consider an example. Harley Davidson was facing problems with its suppliers, as the relationships were subpar and unsophisticated. Engineers focused on the suppliers' technical innovations and failed to consider whether they could produce the necessary volume, deliver on schedule, or meet cost targets. As a result, supplier components came in late, and production was often put at risk. To resolve this problem, the company developed an internet-based SCM strategy where a web portal was created which enabled Harley Davidson and its suppliers to conduct transactions, from placing purchase orders to invoicing, on the internet. Nearly 300 of Harley's 695 parts suppliers

logged into applications through the supplier portal. Suppliers can view production forecasts, account statuses, and are able to submit shipment notices and receive inventory-replenishment alerts. The web-portal has allowed Harley to achieve lower costs, consistent quality, and improved delivery performance.

Net marketplaces (e-hubs) are single markets for many buyers and sellers. They can be industry-owned or owned by an independent intermediary. E-hubs generate revenue from transaction fees and other services. The market principle uses prices established through negotiation, auction, or fixed prices.

In many industries, hierarchical SCs with predetermined suppliers' structures and long-term product programs evolve into *virtual enterprises and collaborative networks*. Their special feature is a customer-oriented networking of core competences and flexible, configurable SCs, conditioned by an enlargement of alternatives for searching for suitable partners with whom a cooperation enabled by Internet technologies is possible.

The main objective of a *virtual enterprise* is to allow a number of organizations to develop a common working environment or virtual breeding environment with the goal of maximizing flexibility and adaptability to environmental changes and developing a pool of competencies and resources. SCs in virtual enterprise are based on a customer-oriented networking of core competences through partner selection from a pool of available suppliers in a virtual environment according to customer requirements. Virtual enterprises focus on speed and flexibility. A virtual enterprise is enabled by building a united information space with extensive usage of web services.

Virtual enterprise structures are highly dynamic, and their life cycles can be very short. Remarkable are alternatives for SC configuration. It is a great advantage to react quickly to customers' requirements. A virtual enterprise also builds a structural–functional reserve for running an SC. Unfortunately, virtual enterprises are mostly considered from the information perspective and do not deal properly with managerial and organizational factors. Our practical experiences show that there are only a few (if any) organizations that have managed to apply the primary idea of a virtual enterprise. However, they often collaborate only a short time and then disperse, perhaps to form new networks with other enterprises. Two main obstacles cause this dispersal: trust and technical project documentation.

The development of the *Internet of Things* in a manufacturing context has been called *Industry 4.0* with reference to the three prior industrial revolutions: the steam machine, assembly line manufacturing, and computer integrated manufacturing (CIM). Even in our everyday life, WLAN-based communication between different devices such as refrigerators, kitchen stoves, and smartphones is possible. In Industry 4.0 manufacturing, both machines and materials are equipped with interconnected sensors in *cyber-physical systems*, take an active role in the process, and optimize it. The products and the machines communicate with each other. Materials will be able to communicate their purpose to the machines, making product lifecycles faster and more flexible. An important component of Industry 4.0 is *additive manufacturing* technology, especially 3D printing.

▶ **Practical Insights** Business Analytics, Artificial Intelligence (AI), and Industry 4.0 technology are path-breaking concepts for SCOM. All management activities are about data and information. By now, state-of-the-art IT allows storing and transmitting huge amounts of data. Managers must analyze all this data to extract what is necessary, and transfer it into information. In the future, managers will get the information they need at the right place, at the right time, and at the right quality using digital twins (see Chap. 16). The preliminary actions for sorting and combining data will be done automatically by Business Analytics, AI and Industry 4.0.

3.4 Problem Solving Methods and Research Methodologies

A baker bakes bread. A bus driver drives the bus. What do managers do? They make decisions. Decisions are made to solve problems. Decisions can be made regarding the planning and control of operations and SCs. In this paragraph, we learn the basic problem-solving methods in SCOM.

3.4.1 Problems, Systems, and Decision-Making

For a *problem* to exist there must be an individual (or a group of individuals). This person is referred to as the problem owner(s) (decision maker) who:

- Is dissatisfied with the current state of affairs within a real-life context, or has some unsatisfied present or future needs, i.e., who has some goals to be achieved or targets to be met
- Is capable of judging when these goals, objectives, or targets have been met to a satisfactory degree
- Has control over some aspects of the problem situation that affect the extent to which the goals, objectives, or targets can be achieved (Daellenbach and McNickle 2005)

Rather than assume that we have a well-structured problem with clearly defined objectives and alternative courses of action, it is better to:

- Represent the problem situation where issues are still vague or fuzzy
- Gain a comprehensive understanding of the various issues involved in decision-making
- Formulate the problem correctly to ensure an appropriate level of detail and provide insights into the problem solution
- Perform decision-making within the context of systems

Most problems exist in the system context. A *system* is a set of interrelated components, which make up a united whole. Each component interacts with or is related to at least one other component and they all serve a common objective. *Environment* is the aggregate of elements that do not belong to the system but influence it.

One basic system characteristic is structure. *Structure* is the characteristic of steady links and interaction modes of a system's elements; structure determines the system's integrity, organization, and functioning; structure is the framework of a system. In practice, many SC structures are disrupted (e.g., Toyota's SC was badly affected by the tsunami in March 2011), and this concerns the SC's *structure dynamics* (Ivanov et al. 2010).

Dynamics is a system's change and evolution in object and process states in space and time as driven by the perturbation and control influences resulting from both planned transition from a current state to a desired one, or adaptation of a system to a changed execution environment.

Decision is the selection of an activity (or a set of activities) from several alternatives. Decisions are subject to constraints that limit decision choices and objectives which make some decisions preferable to others. Managerial decisions affect system goal-orientation through output performance. Decisions shape system behavior with regard to a certain goal (or multiple goals). Objectives and criteria play the most important role in making decisions.

▶ **Practical Insights** Each management decision has two components. The first one is an *analytical* component and the second one is a *behavioral* component. The analytical component is supported by *quantitative analysis* business analytics methods, while the behavioral component is based on the intuition and leadership qualities of the decision maker as well as on external environment *behavior prediction and reaction* in regard to the decisions of suppliers, retailers, and customers.

Performance is a complex characteristic of the goal-oriented results of the system's functioning. Managers must consider potential and real performance, and take into account the conformity of the achieved results with the goals set by management at the planning stage. Performance is measured with certain metrics or indicators (e.g., customer service level, on-time delivery, costs, etc.)

Basic problems in decision-making are as follows:

- Optimality
- Multiple objectives
- Risk and uncertainty
- Complexity

3.4.1.1 Optimal Decisions and Multiple Objectives

The optimal decision is the best one. Optimal decisions are very "fragile" and presume certain problem dimensionality, fullness, and the certainty of the model. In addition, optimal solutions are usually very sensitive to deviations. Moreover,

decision-making is tightly interconnected with dynamics and should be considered as an adaptive tuning process and not as a "one-way" optimization.

▶ **Practical Insights** Consider an example. Determining the optimal order quantity for some purchased items is a typical optimization problem. This optimization is based on reducing ordering and inventory holding costs, and if so, the calculated order quantity is optimal only for the purchasing department. At the same time, this problem is tightly interconnected with transportation planning and customer service level improvement. So, the real problem situation is integrated inventory optimization.

The problems of applying optimization-based decision-making are interrelated with complexity, uncertainty, and multiple objectives. A particular feature of optimal decisions is *multi-objective* decision-making by managers with their own preferences that, in turn, are always changing. Hence, it becomes impossible to build any general selection function for multi-criteria decision-making. Finding optimal solutions is possible, but it can be very time-consuming. However, these optimal solutions can be used for benchmarking to estimate the quality of solutions obtained by heuristics or simulation models (see further in this chapter).

Multiple objectives conflict in the sense that they create a competition for common resources to achieve variously different performance objectives (financial, functional, and environmental). A multi-criteria decision-making (MCDM) strategy employs *trade-off* analysis techniques to identify compromise designs for which the competing criteria are mutually satisfied in a Pareto-optimal sense. For more detailed information on decision-making theory, we refer to the studies on operations research, decision theory, and quantitative analysis in management, e.g., Taha (2022), Render et al. (2023), Dolgui and Proth (2010), Yalaoui et al. (2012).

3.4.1.2 Risk and Uncertainty

Uncertainty is one of the most critical issues in making decisions. *Uncertainty* is a system property characterizing the incompleteness of our knowledge about the system and the conditions of its development. Uncertainty is a polysemic term (poly—many, sema—a sign). Historically, the first terms related to uncertainty were accident, probability, and possibility, which we relate to Aristotle. Until the twentieth century, the mathematical basics and factor description of uncertainty were founded on probability-frequency interpretation and were related to Pascal, Ferma, Bernoulli, and Laplace. Modern probability theory is based on the research of Kolmogorov, who introduced an axiomatic definition of probability as a measure related to a system of axioms of a so-called probability space.

Uncertainty begets risk. *Risk management* is a methodological approach for managing uncertain outcomes. The concept of risk is subject to various definitions. Knight (1921) classified "measurable" uncertainty as "risk." From the financial perspective of Markowitz (1952), risk is the variance of return. From the project management perspective, risk is a measure of the probability and consequence of not

achieving a defined project goal. According to March and Shapira (1987), risk is a product of the probability of occurrence of a negative event and the resulting amount of damage.

Generally, in decision theory, risk is a measure of the set of possible (negative) outcomes from a single rational decision and their probabilistic values. A particular feature of risk management in SCs (unlike in technical systems) is that people do not strive for a *100% guarantee* of the result: they consciously tend to take risks. There is a problem of contradiction between *objective risk* (determined by experts, applying quantitative scientific means) and *perceived risk* (perception of managers).

There are three types of uncertainty that affect SCOM (Klibi et al. 2010; Dolgui et al. 2018; de Lima et al. 2022):

- Random uncertainty (i.e., problem in demand and supply coordination)
- Hazard uncertainty (unusual events with high impact, i.e., ripple effect (Ivanov et al. 2014; Ivanov 2017b, 2018, 2020; Dolgui et al. 2018)
- Deep uncertainty (severe disruptions) no information (game theory and utility functions).

According to the certainty of data, the following cases can be classified:

- No information (treated in game theory and utility functions)
- Vague information (treated in fuzzy theory)
- Random and stochastic information (treated in probability theory)
- Determined information (treated in analytical models)

Uncertainty factors are usually divided into two groups:

- Stochastic factors
- Non-stochastic factors

The first group can be described via probability models. For the formal description of non-stochastic uncertainty, fuzzy description with known membership functions, subjective probabilities for the uncertainty factors, interval description, and a combined description of the uncertainty factors are used.

3.4.2 Models and Modeling

The concept of a model is widely applicable in natural human languages and is a general scientific term. It is characterized by polysemy, that is, widely expressed and reflecting different meanings depending on the applications and contexts. At present, there are numerous definitions of the concept of a model and modeling. To sum up the different definitions, the following views of models and modeling are presented.

A *model* is:

- A system whose investigation is a tool for obtaining information about another system
- A method of knowledge existence
- A multiple system map of the original object that, together with absolutely true content, contains conditionally true and false content, which reveals itself in the process of its creation and practical use

Modeling is one of the stages of cognitive activity of a subject, which involves the development (choice) of a model, use of it to conduct investigations, obtainment and analysis of the results, production of recommendations, and estimation of the quality of the model itself as applied to the solved problem and taking into account specific conditions (Fig. 3.8).

Because of the *finiteness* of the designed (applied) model (a limited number of elements and relations that describe objects belonging to infinitely diverse reality) and the *limited* resources (temporal, financial, and material) supplied for modeling, the model always reflects the original object in a simplified and approximate way. However, experience shows that these specific features of a model are admissible and do not oppose the solution of problems that are faced by subjects.

3.4.3 Model-Based Decision-Making

Decisions can be done by intuition, data-driven, and model-based. The model-based decision-making process is shown in Fig. 3.9.

We can observe that a *real management problem* is the initial point of the decision-making process. An example might be a facility location problem where we are given demand in some markets, possible locations and capacities of new facilities, fixed costs of having a facility in the SC, and transportation costs from each location to each market. We must decide where to locate the facilities and which quantities should be shipped from the facilities to the markets.

The next step is to transform the real problem into a *mathematical model*. For this transformation, we need to reduce the *complexity* of reality. This inevitably

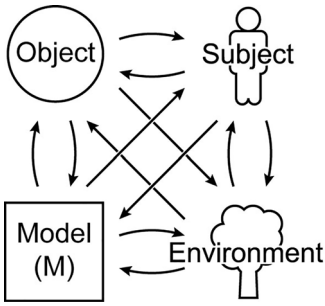


Fig. 3.8 Modeling stakeholders

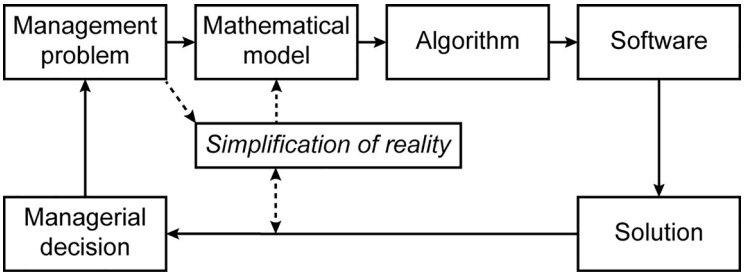


Fig. 3.9 Model-based decision-making process

results in simplification. For example, we would assume deterministic capacity in our facility location model instead of considering fluctuations in demand.

We simplify this representation of the management problem so that the model can be solved with existing *algorithms* within a reasonable time. In our example, we formulate the facility location problem as a mixed-integer linear programming model that can be solved with simplex and branch-bound algorithms.

For implementation, *software* is needed. Small instances can be solved with the help of Excel Solver, but for real data, a professional solver such as CPLEX, Lindo, AMPL, Matlab, GAMS, Gurobi, and XPRESS exist. The software calculates the *solution*. In our example, the solution suggests where to open facility locations and which product quantities should be shipped from each opened location to each of the markets so that total production and logistics costs are minimal.

The software calculates this solution. The most important question now is whether this solution is automatically our *decision*? NO! This is a solution to the mathematical problem. For the next stage, management expertise is needed to transfer this mathematical solution into managerial decisions. First of all, simplifications of reality should be reviewed. Second, so-called *soft facts* such as risks and flexibility should be included in the analysis. That is why we prefer talking about *decision-supporting quantitative methods*.

3.4.4 Principles of Decision-Making in SCOM

Any decision-support system in SCOM comprises the following elements: planning model, control algorithm, object (i.e., the supply chain), performance measurement, observation, and adaptation (Fig. 3.10).

Five main management tasks are as follows:

- Forecasting
- Planning
- Observation (monitoring)
- Performance analysis
- Adaptation (control)

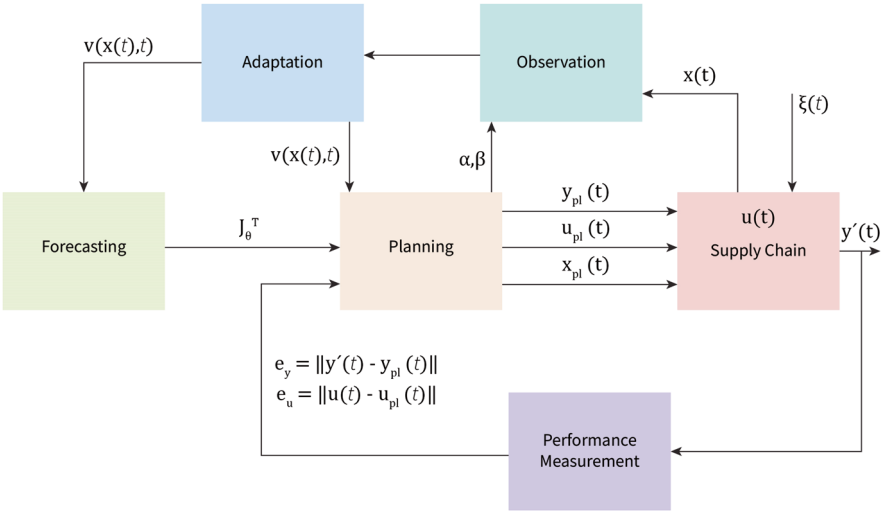


Fig. 3.10 Principles of decision-making in SCOM

The control mechanism for the supply chain can be expressed using control theory as a dynamic system, delineated by Eqs. (3.1), (3.2), and (3.3):

$$\dot{\mathbf{x}}(t) = \tilde{\phi}\left(\mathbf{x}(t), \mathbf{u}(t), \xi(t), \alpha, \beta, t\right) \tag{3.1}$$

$$\dot{\mathbf{y}}(t) = \tilde{\psi}\left(\mathbf{x}(t), \mathbf{u}(t), \xi(t), \alpha, \beta, t\right) \tag{3.2}$$

$$\mathbf{u}(t) = \mathbf{u}_{pl}^{\mathrm{T}}(t), \mathbf{v}^{\mathrm{T}}\left(\mathbf{x}(t), t\right)^{\mathrm{T}}; \ \mathbf{u}_{pl}(t) \in \mathbf{Q}\left(\mathbf{x}(t), t\right); \quad \mathbf{v}(t)\left(\mathbf{x}(t), t\right) \in \mathbf{V}\left(\mathbf{x}(t), t\right), \tag{3.3}$$

$$\xi(t) \in \Xi\left(\mathbf{x}(t), t\right); \quad \beta \in \mathrm{B}, \mathbf{x}(t) \in \mathbf{X}\left(\xi(t), t\right), \mathbf{x}(T_0) \in \mathbf{X}_0(\beta), \mathbf{x}(T_f) \in \mathbf{X}_f(\beta)$$

where B represents a domain of permissible parameter values (such as inventory levels) and $\mathbf{X}(\boldsymbol{\xi}(t), t)$ signifies a domain of allowable states.

Vectors defined in Eqs. (3.1) and (3.2) are subject to spatiotemporal constraints, as specified in set (3.3) at times $t = T_0$ and $t = T_f$.

Figure 3.10 delineates a system-cybernetic schema of decision-making support. Based on a forecasting, a certain level of performance is set, denoted as a multi-objective vector $\mathbf{J}_{\boldsymbol{\Theta}}^{\mathrm{T}}$ (e.g., annual revenues, production quantities, on-time delivery, and sales), where subscript $\boldsymbol{\Theta}$ corresponds to a set of models used in the dynamic system. A planning algorithm determines process plans for achieving the targets $\mathbf{J}_{\boldsymbol{\Theta}}^{\mathrm{T}}$ as control trajectories $\mathbf{u}_{pl}(t)$ (e.g., a production-distribution plan) aiming to transform available resources into the desired output with a specific performance level $\mathbf{y}(t)$ (e.g., monthly production quantities and costs). At each point of time, the supply chain is characterized by a state vector $\mathbf{x}(t)$ and performance vector $\mathbf{y}'(t)$.

$\mathbf{u}_{pl}(t)$ is subject real disruptions $\boldsymbol{\xi}$(t) which impact the supply chain. The state and performance vectors $\mathbf{x}(t)$ and $\mathbf{y}'(t)$ are continuously monitored and observed.

Performance deviations e from planned performance and process execution are recorded. If $e > 0$, $\boldsymbol{\alpha}(t)$ and $\boldsymbol{\beta}(t)$ vectors are formed representing structural and process elements performing with deviations from plan, respectively.

Finally, adaptation $\mathbf{v}(\mathbf{x}(t), t)$s are implemented through either re-planning using the control algorithm and the existing planning model or even through a model adaptation. $\mathbf{u}_{pl}(t)$ and $\mathbf{y}(t)$ are updated. The input data essential for supply chain adaptation are gathered in real-time subject to performance maximization (3.4):

$$J_G = J_G\left(\mathbf{x}(t), \mathbf{u}_{pl}(t), \mathbf{v}\left(\mathbf{x}(t), \xi\right), \xi\right) \to \max_{\mathbf{u} \in \Delta} \tag{3.4}$$

We note that J_G is a scalar form of $\mathbf{J}_{\Theta}^{\mathrm{T}}$. Through adaptation, a transition from the actual supply chain state trajectory $\mathbf{x}(t)$ to the planned (or a new one) $\mathbf{x}_{pl}(t)$ within the time interval $(t', t''] \in (T_0, T_f]$ or by the final time $\mathbf{x}(T_f) = \mathbf{x}_{pl}(T_f)$ as detailed in Eqs. (3.5) and (3.6):

$$\Delta\mathbf{x}(t) = \int_{t'}^{t''} \left(\mathbf{x}(\tau) - \mathbf{x}_{pl}(\tau)\right)^{\mathrm{T}} \left(\mathbf{x}(\tau) - \mathbf{x}_{pl}(\tau)\right) d\tau \tag{3.5}$$

$$\Delta\mathbf{x}(T_f) = \left(\mathbf{x}(T_f) - \mathbf{x}_{pl}(T_f)\right)^{\mathrm{T}} \left(\mathbf{x}(T_f) - \mathbf{x}_{pl}(T_f)\right) \tag{3.6}$$

By broadening the scope of the general case to include$\mathbf{v}(\mathbf{x}(t), t)$ Eqs. (3.1) through (3.3), we can obtain $\mathbf{u}_{pl}(t)$, $\boldsymbol{\nu}(\mathbf{x}(t), t)\boldsymbol{\alpha}(t)$ and $\boldsymbol{\beta}(t)$ so that the generalized function $J_G = J_G(\mathbf{J}(\mathbf{x}(t), \mathbf{u}(t), \boldsymbol{\nu}(x(t), t), \boldsymbol{\xi}(t)))$ achieves its extremal values (e.g., profit, service level, or sales).

3.4.5 Quantitative Models and Operations Research

Quantitative models for SCOM can be divided into three primary approaches. These are optimization, simulation, and heuristics. Hybrid models (e.g., optimization-based simulation models) also exist. *Optimization* is an analysis method that determines the best possible option of solving a particular operation or SC problem. Optimization has been a very visible and influential topic in the field of SCOM. The drawback of using optimization is the difficulty in developing a sufficiently detailed and accurate model representing the complexity and uncertainty of SCM while keeping the model simple enough to be solved.

Simulation imitates the behavior of one system with another. By making changes to the simulated SC, one can gain an understanding of the dynamics of the physical SC. Simulation is an ideal tool for further analyzing the performance of a proposed design derived from an optimization model (Sterman 2000; Ivanov 2016, 2017a, b).

Heuristics are intelligent rules that often lead to good solutions, not necessarily the best. Heuristic approaches typically are easier to implement and require less data. However, the quality of the solution is usually unknown. An option for estimating the quality of heuristic algorithms may be using optimization as a tool for "ideal" solutions to problems.

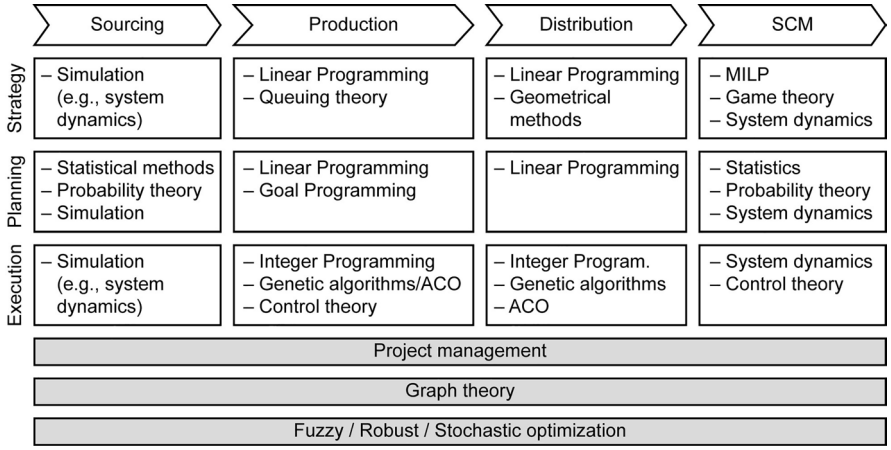


Fig. 3.11 OR methods for SCOM

Operations Research (OR) is the application of advanced analytical methods to help make better decisions (INFORMS definition). OR deals with problems that may be described with mathematical models to find optimal or good solutions. An overview of OR-based methods is presented in Fig. 3.11. Basic OR methods and their application to SCOM include but are not limited to:

- Linear programing (production planning; revenue management)
- Integer programing/Combinatorial Optimization (scheduling, routing)
- Mixed-integer linear programming (MILP) (SC design)
- Dynamic programing/Graph Theory (dynamic lot-sizing/SC contracting)
- Network theory (project management)
- Queuing theory (waiting lines: call centers, hotline services, process design, entertainment parks, services)
- Simulation (SC design, bullwhip-effect, etc.)
- Heuristics (genetic algorithms, ant colony optimization (ACO), application to scheduling and routing problems)
- Fuzzy/Robust/Stochastic optimization (data uncertainty)

The application of many of these methods to different SCOM problems will be considered in further chapters of this book.

3.4.6 Integrated and Digital Decision-Making Support

Having read thus far in this textbook, you already have an overview of:

- Decision makers in SCOM
- Business processes in SCOM

- Methods of decision-making support in SCOM
- IT in SCOM

Before starting to consider processes and decision-supporting methods in detail, let us summarize our knowledge according to the triangle "*Process→Model→IT*." Process modeling and reengineering are basic preliminaries for the successful application of quantitative models and IT. All three of these components are tightly interrelated with each other. The main blocks required for successful SCOM are presented in Fig. 3.12.

Many companies make a mistake by starting optimization directly with IT. However, before this, which data is necessary for which optimization models must be determined. Next, this data should be brought into correspondence with business processes and organizational structure.

In the block "Enterprise Supply Chain," typical functionality of ERP and APS systems can be used. In the block "Integrated SC," special EDI-driven software for SC coordination and collaboration is required. Finally, the block "Services and analytics" is responsible for reporting, data management, and KPI (key performance indicator) control. All of these models are based on business process models and mathematical models. Data management also comprises integration with some external IT like RFID, and payment security systems.

In Table 3.2, we summarize quantitative methods used in selected modules of the SCOM information system. In the next chapters of this book, we will learn many of these problems, models, and methods in detail.

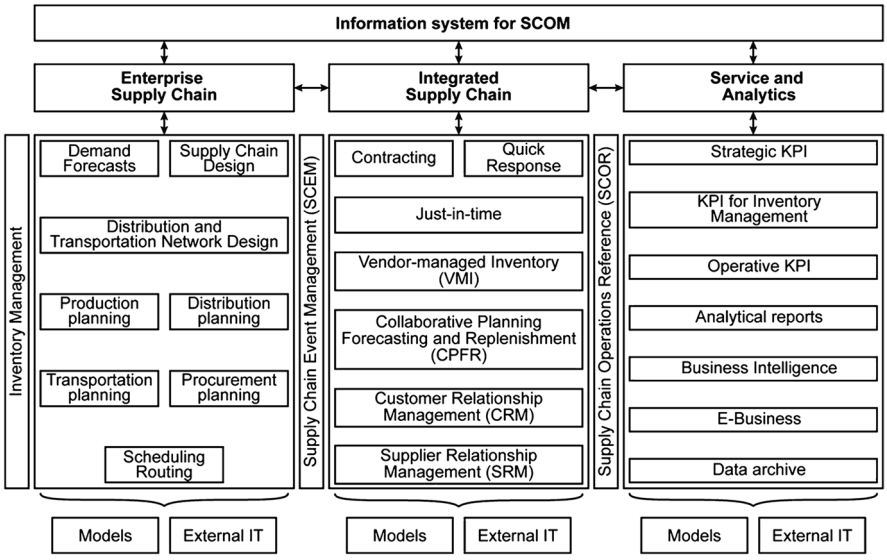


Fig. 3.12 Information system for SCOM

Table 3.2 Data, decisions, and methods in SCOM (based on Ivanov et al. 2010)

Input data	Performance indicators	Examples of solution methods	Result (output)
Demand forecasting			
Statistical sales data	Forecast error minimization	Exponential smoothing; Regression analysis	Demand forecast
Supply chain design			
Location data Demand forecasts; Capacity data; Transportation (time, costs, etc.)	Cost minimization	Center-of-gravity Warehouse location problem (WLP) CPLM (capacitated plant location model) Location-allocation model	How many production and distribution centers are needed, where, and of what capacity?
Distribution-transportation network design			
Customers and suppliers supply chain design demand forecasts; Transportation (time, costs, etc.)	Cost minimization; Service level maximization	Transportation problem and its multistage modifications	Which products and of what quantity should be shipped to which destinations?
Inventory management			
Demand; Lead-time Holding, ordering, and warehousing costs service level; Capacity data	Cost minimization; Service level maximization	Cycle inventory optimization (EOQ/EPQ); Safety inventory optimization; Multi-echelon inventory management; ATP/CTP	When and how much to order? What is the optimal service level? Where to place inventory in the SC?
Sales and operations planning			
Demand costs inventory capacity	Cost minimization Revenue maximization	Mathematical programming	Aggregate plan for 1–18 months
Distribution and transportation planning			
Sales and operations plan Costs inventory Transportation capacity	Cost minimization	Mathematical programming, heuristics	Aggregate plan for 1–18 months
Scheduling and routing			
Master production schedule; machine capacity; Due dates; costs	Minimize completion time Maximize utilization of facilities Minimize work-in-process (WIP) inventory Minimize flow time	Combinatorial optimization, heuristics	Schedules and delivery routes

Figure 3.13 extends our discussion about decision-support system toward their embedding into a digital environment (see more about digital twins of SC and operations in MacCarthy and Ivanov 2022 and Chap. 16).

The conceptual framework of planning and control in cyber-physical systems comprises four main elements, i.e., physical object, digital object, object model, and simulated model. Physical object is a real entity that evolves over time fulfilling some goal-oriented functions according to a plan and with some performance. Abstraction of the physical object is its model used for analyzing performance and behavior of and setting rules for a physical object based on information from optimization and simulation. A digital object is an observation of a physical object, i.e., a regularly updated, sensed footprint of a real object encapsulating data, information, and knowledge about a physical object and its dynamics. Based on modeling over the object model and using digital objects, a simulated object is created, representing the possible performance and behavior of a physical object based on information from simulations. A combination of four elements shown in Fig. 3.13 represents a decision-making environment for planning and control. This environment includes analyzing, predicting and improving performance and processes of and setting rules for a physical object based on its observation and information from simulations.

3.4.7 Research Methodologies

Quantitative methods, modeling, and optimization are popular ways of conducting research in SCOM. Other research methodologies are qualitative-oriented and

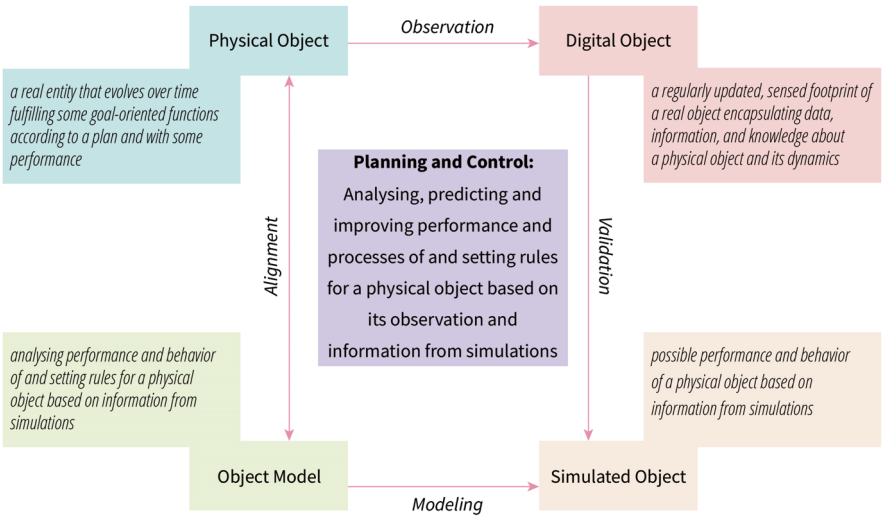


Fig. 3.13 Cyber-physical decision-making support

include case study research, action research, surveys, and interview-based research (Kotzab et al. 2005; Yin 2017).

Case study research is based on deductive analysis of existing practical examples described in the literature. A single case, several isolated cases, or even embedded cases can be analyzed to develop general recommendations for a problem.

Action research presumes the active participation of the investigator in the problem-solving. It can be, e.g., a traineeship in a company where students are involved in a project. Unlike in case study research, the results of the project are typically unknown at the beginning. In an ongoing project, the comparison of "as-is" and "to-be" states can be complicated. Benchmarking analysis can be successfully used here.

Surveys and interview-based research presumes data collection from different sources, e.g., interviews of experts in a particular area. The questionnaires are designed and evaluated with the help of statistical methods. Frequently, the structural equation method is used.

Finally, mixed methodologies can be used to combine quantitative and qualitative research within a certain research project (e.g., empirically grounded analytics).

3.5 Integration of Business Analytics, Simulation, and Optimization

Success in SC competition is highly dependent on analytics algorithms in combination with optimization and simulation modeling. Initially intended for process automation, business analytics techniques now disrupt markets and business models and have a significant impact on SCM development.

▶ **Practical Examples** Electronic retailers use their extensive customer transactional and behavioral data to offer new ways of trying, experiencing, and purchasing their products (e.g., Amazon with Alexa). PwC works with a large car company looking to introduce autonomous vehicles for the public (Wilkinson 2018). Part of this work employs deep reinforcement learning to develop rules. Together with simulation, deep reinforcement learning is used to determine "optimal" decision rules that allow the vehicles to maximize efficiency while also satisfying customer trip demand. The software environment for the project uses the extensible and practical environment of AnyLogic multi-method simulation software to lever the capabilities of Deep Learning for Java (DL4J) for the deep learning environment. Autonomous cars are becoming more common, and the features are already present in many consumer cars. Artificial intelligence (AI) becomes more pervasive in the real world with every project, and necessarily it must be part of simulation. This means AI will not only be part of simulations, but the simulation will also help to develop the AI.

In the past decades, simulation and optimization have played significant roles in solving complex problems in SCOM. Successful examples include production planning and scheduling, supply chain design, and routing optimization, to name a few. However, many problems remain challenging because of their complexity and large scales, and/or uncertainty and stochastic nature. On the other hand, the rapid rise of business analytics provides exciting opportunities for Operations Research and the reexamination of these hard optimization problems, as well as newly emerging problems in SCOM (Fig. 3.14).

Examples of SC and operations analytics applications include logistics and SC control with real-time data, inventory control and management using sensing data, dynamic resource allocation in Industry 4.0 customized assembly systems, improving forecasting models using Big Data, machine learning techniques for process control, SC visibility and risk control, optimizing systems based on predictive information (e.g., predictive maintenance), combining optimization and machine learning algorithms, and simulation-based modeling and optimization for stochastic systems (Ivanov et al. 2019).

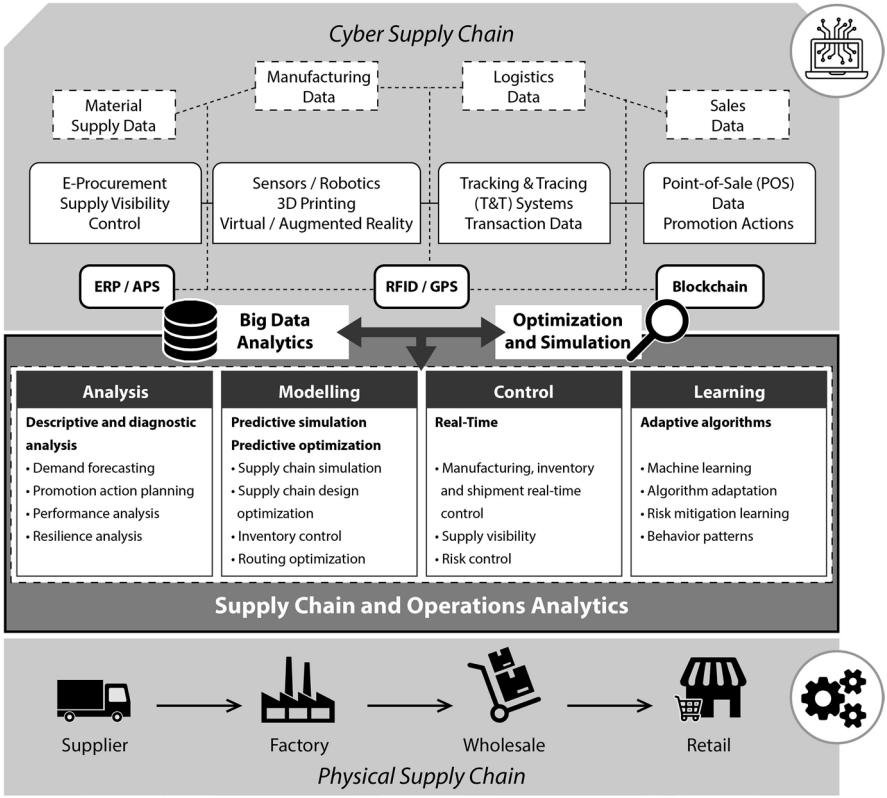


Fig. 3.14 Supply chain and operations analytics

Table 3.3 Data analytics applications to SCOM

	Descriptive analytics	Predictive analytics	Prescriptive analytics
Key question	What happened in the past?	What can happen in the future?	What is our course of action?
Methods	Statistical data analysis	Regression/machine learning	Neural networks/deep learning
Example	On-time delivery (OTD) of 90% last year	Why did we have 90% OTD last year? What OTD can we expect next year?	What to do to increase OTD from 90% to 95%?

The application areas of Supply Chain and Operations Analytics can be classified into four areas:

- Descriptive and diagnostic analysis
- Predictive simulation and prescriptive optimization
- Real-time control
- Adaptive learning

Table 3.3 provides some examples of data analytics applications to SCOM.

Table 3.3 seeks to illustrate some practical applications of data analytics to SCOM using a simple example of an on-time-delivery (OTD) performance indicator (i.e., the percentage of on-time fulfilled orders). *Descriptive analytics* can help observe retrospective dynamics of some parameters. For example, a descriptive analytics method can be used to compute that say 90% of our orders in the past arrived on-time, and 10% of our orders were late. However, we do not know why 10% of our orders were late and what OTD we can expect next year. To answer these questions, *predictive analytics* can be used. They make it possible to analyze the reasons for some events in the past and predict future developments based on past data. However, the predictive analytics do not tell us what should we do to improve our OTD and reach 95%? At this stage, *prescriptive analytics* need to be considered, which helps to reveal the course of future actions.

Finally, we note that sourcing, manufacturing, logistics, and sales data are distributed among very different systems, such as ERP, RFID, sensors, and blockchain. Big Data Analytics integrates this data to information used by artificial intelligence algorithms in the cyber SC and managers in the physical SC (cf. Fig. 3.14; see also Chap. 16). As such, a new generation of simulation and optimization models has arisen. The pervasive adoption of analytics and its integration with Operations Research shows that simulation and optimization are key, not only in modeling physical SC systems but also in modeling cyber SC systems and learning them. We refer to Ivanov (2024a), which is dedicated to SCOM analytics methods.

3.6 Key Points

Organizations are structured by processes. Any organization or business strategy is influenced by technology. A business process is a network of activities for accomplishing a business function. The basic concept of managing processes in an organization is called business process management and includes a variety of tools, methodologies to analyze, design, and optimize processes. The analysis of process results may be characterized by effectiveness (achieving process goals) and efficiency (performing the process with minimum costs). Process models describe SCOM activities from an information processing perspective.

At the enterprise level, such MIS as ERP, WMS, and TPS systems are used. At the SC level, APS and SCEM systems can be applied. During the last decade, new IT for operations and SC integration have been developed. Examples include data mining, cloud computing, Physical Internet, digital twins, pattern recognition, knowledge discovery, and early warning systems, to name a few. Trends in IT development for SCOM include Business Intelligence and data mining, cloud computing and SaaS, E-Business, and Industry 4.0.

For each decision, managers use both quantitative and qualitative models. Quantitative methods are based on Operations Research techniques. Qualitative research methodologies include case study, action, and survey research.

Acknowledgment The introductory case study is based on the materials of SupplyOn and is written with the permission of SupplyOn. We thank Ms. Cornelia Staib, Mr. Arvid Holzwarth (MBA), and Mr. Jonas Dahl for their help in preparing this case study.

Bibliography

Aldrighetti R, Calzavara M, Martignago M, Zennaro I, Battini D, Ivanov D (2024) A methodological framework for the design of efficient resilience in supply networks. Int J Prod Res 62(1–2):271–290

Bertrand JWM, Fransoo JC (2002) Modelling and simulation: operations management research methodologies using quantitative modeling. Int J Oper Prod Manag 22(2):241–264

Borshev A (2013) The big book of simulation modeling. AnyLogic North America, Chicago, IL

Brandenburg M, Rebs T (2015) Sustainable supply chain management: a modeling perspective. Ann Oper Res 229:213–252

Brintrup A, Wang Y, Tiwari A (2015) Supply networks as complex systems: a network science-based characterization. IEEE Syst J 99:1–12

Cavalcantea IM, Frazzon EM, Forcellinia FA, Ivanov D (2019) A supervised machine learning approach to data-driven simulation of resilient supplier selection in digital manufacturing. Int J Inf Manag 49:86–97

Chandrasekaran A, Linderman K, Sting FJ (2018) Avoiding epistemological silos and empirical elephants in OM: how to combine empirical and simulation methods. J Oper Manag 63(Suppl l):1–5

Check J, Schutt RK (2012) Survey research. In: Research methods in education. Sage, Thousand Oaks, CA, pp 159–185

Choi TM, Wallace SW, Wang Y (2018) Big data analytics in operations management. Prod Oper Manag 27(10):1868–1883

Coughlan P, Coghlan D (2002) Action research for operations management. Int J Oper Prod Manag 22(2):220–240
Daellenbach HG, McNickle DC (2005) Management science. Decision making through systems thinking. Palgrave Macmillan, Basingstoke
De Lima FA, Seuring S, Sauer PC (2022) A systematic literature review exploring uncertainty management and sustainability outcomes in circular supply chains. Int J Prod Res 60(19):6013–6064
Demirel G, MacCarthy BL, Ritterskamp D, Champneys A, Gross T (2019) Identifying dynamical instabilities in supply networks using generalized modeling. J Oper Manag 65(2):136–159
Dolgui A, Proth J-M (2010) Supply chain engineering: useful methods and techniques. Springer, Berlin
Dolgui A, Ivanov D, Sokolov B (2018) Ripple effect in the supply chain: an analysis and recent literature. Int J Prod Res 56(1–2):414–430
Dolgui A, Gusikhin O, Ivanov D, Li X, Stecke K (2024) A network-of-networks adaptation for cross-industry manufacturing repurposing. IISE Trans 56(6):666–682
Dubey R, Gunasekaran A, Childe SJ, Wamba SF, Roubaud D, Foropon C (2021) Empirical investigation of data analytics capability and organizational flexibility as complements to supply chain resilience. Int J Prod Res 59(1):110–128
Eisenhardt KM (1989) Building theory from case study research. Acad Manag Rev 14(4):532–550
Ellram LM (1996) The use of the case study method in logistics research. J Bus Logist 17(2):93–138
Fahimnia B, Tang CS, Davarzani H, Sarkis J (2015) Quantitative models for managing supply chain risks: a review. Eur J Oper Res 247:1–15
Gunasekaran A, Tiwari MK, Dubey R, Wamba SF (2016) Big data and predictive analytics applications in supply chain management. Comput Ind Eng 101:525–527
Günther H-O (2005) Supply chain management and advanced planning systems: a tutorial. In: Günther H-O, Mattfeld DC, Suhl L (eds) Supply chain management und logistik. Physica-Verlag, Heidelberg, pp 3–40
Hammer M, Champy J (1993) Reengineering the corporation: a manifesto for business revolution. Harper Collins, New York
Hancock DR, Algozzine R (2006) Doing case study research: a practical guide for beginning researchers. Teachers College Press, New York
Heizer J, Render B, Munson C (2023) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow
Hosseini S, Ivanov D, Dolgui A (2019) Review of quantitative methods for supply chain resilience analysis. Transp Res E 125:285–307
Ijioui R, Emmerich H, Ceyp M (2007) Supply chain event management: konzepte, prozesse, erfolgsfaktoren und praxisbeispiele. Physica-Verlag, Heidelberg
Ivanov D (2024a) Introduction to supply chain analytics. Springer Nature, Cham
Ivanov D (2024b) Exiting the COVID-19 pandemic: after-shock risks and avoidance of disruption tails in supply chains. Ann Oper Res 335:1627–1644
Ivanov D (2024c) Two views of supply chain resilience. Int J Prod Res 62(11):4031–4045
Ivanov D (2024d) Conceptualisation of a 7-element digital twin framework in supply chain and operations management. Int J Prod Res 62(6):2220–2232
Ivanov D (2021) Supply chain viability and the COVID-19 pandemic: a conceptual and formal generalisation of four major adaptation strategies. Int J Prod Res 59(12):3535–3552
Ivanov D (2010) A framework for aligning (re)planning decisions on supply chains strategy, design, tactics, and operations. Int J Prod Res 48(13):3999–4017
Ivanov D (2016) Operations and supply chain simulation with AnyLogic 7.2. www.anylogic.com/books
Ivanov D (2017a) Simulation-based single vs dual sourcing analysis in the supply chain with consideration of capacity disruptions, big data and demand patterns. Int J Integr Supply Manag 11(1):24–43
Ivanov D (2017b) Simulation-based ripple effect modelling in the supply chain. Int J Prod Res 55(7):2083–2101

Ivanov D (2018) Structural dynamics and resilience in supply chain risk management. Springer, New York
Ivanov D (2020) Supply chain simulation and optimization with anyLogistix. https://www.anylogistix.com/
Ivanov D (2023) Intelligent Digital Twin (iDT) for supply chain stress-testing, resilience, and viability. Int J Prod Econ 263:108938
Ivanov D, Dolgui A (2019) Low-certainty-need (LCN) supply chains: a new perspective in managing disruption risks and resilience. Int J Prod Res 57(15–16):5119–5136
Ivanov D, Dolgui A (2021a) A digital supply chain twin for managing the disruptions risks and resilience in the era of Industry 4.0. Prod Plan Control 32(9):775–788
Ivanov D, Dolgui A (2021b) OR-Methods for coping with the ripple effect in supply chains during COVID-19 pandemic: managerial insights and research implications. Int J Prod Econ 232:107921
Ivanov D, Sokolov B, Kaeschel J (2010) A multi-structural framework for adaptive supply chain planning and operations control with structure dynamics considerations. Eur J Oper Res 200(2):409–420
Ivanov D, Sokolov B, Dolgui A (2014) The ripple effect in supply chains: trade-off 'efficiency-flexibility-resilience' in disruption management. Int J Prod Res 52(7):2154–2172
Ivanov D, Dolgui A, Sokolov B (2019) The impact of digital technology and industry 4.0 on the ripple effect and supply chain risk analytics. Int J Prod Res 57(3):829–846
Ivanov D (2025) Conceptual and formal models for design, adaptation, and control of digital twins in supply chain ecosystems. OMEGA 137:103356
Ivanov D, Millhiser W, Nguyen P (2025) Introduction to operations and supply chain simulation with AnyLogic. Springer Nature, Cham, ISBN 978-3-031-73663-6
Jahani H, Jain R, Ivanov D (2023) Data science and big data analytics: a systematic review of methodologies used in the supply chain and logistics research. Ann Oper Res. https://doi.org/10.1007/s10479-023-05390-7
Karlsson C (2016) Research methods for operations management. Routledge, London
Klibi W, Martel A, Guitouni A (2010) The design of robust value-creating supply chain networks: a critical review. Eur J Oper Res 203(2):283–293
Knight F (1921) Risk, uncertainty, and profit, vol 31. Hart, Schaffner, and Marx prize essays. Houghton Mifflin, Boston, MA
Kohli R, Grover V (2008) Business value of IT: an essay on expanding research directions to keep up with the times. J Assoc Inf Syst 9(1):23–39
Kotzab H, Seuring S, Müller M, Reiner G (eds) (2005) Research methodologies in supply chain management. Physica-Verlag, Heidelberg
Kroenke DM (2014) MIS essentials, 4th edn. Pearson, Harlow
Kumar S, Mookerjee V, Shubham A (2018) Research in operations management and in-formation systems interface. Prod Oper Manag 27(11):1893–1900
Laudon KC, Laudon JP (2021) Management information systems, 17th edn. Pearson, Harlow
MacCarthy B, Ivanov D (eds) (2022) The digital supply chain. Elsevier, Amsterdam
March JG, Shapira Z (1987) Managerial perspectives on risk and risk taking. Manag Sci 33:1404–1418
Markowitz H (1952) Portfolio selection. J Financ VII(1):77–91
Meredith JR, Raturi A, Amoako-Gyampah K, Kaplan B (1998) Alternative research paradigms in operations. J Oper Manag 8(4):297–326
Motiwalla L, Thompson J (2012) Enterprise systems for management, 2nd edn. Pearson, Harlow
Oldenburg F, Hoberg K, Leopold H (2024) Process mining in supply chain management: state-of-the-art, use cases and research outlook. Int J Prod Res:1–16. https://doi.org/10.1080/00207543.2024.2412285
Panetto H, Iung B, Ivanov D, Weichhart G, Wang X (2019) Challenges for the cyber-physical manufacturing enterprises of the future. Annu Rev Control 47:200–213

Papadopoulos T, Gunasekaran A, Dubey R, Altay N, Childe SJ, Wamba SF (2017) The role of big data in explaining disaster resilience in supply chains for sustainability. J Clean Prod 142(2):1108–1118

Render B, Stair RM, Hanna ME, Hale TS (2023) Quantitative analysis for management, 14th edn. Pearson, Upper Saddle River, NJ

Rolf B, Jackson I, Müller M, Lang S, Reggelin T, Ivanov D (2023) A review on reinforcement learning algorithms and applications in supply chain management. Int J Prod Res 61(20):7151–7179

Rocco TS, Bliss LA, Gallagher S, Perez-Prado A (2003) Taking the next step: mixed methods research in organizational systems. Inf Technol Learn Perform J 21(1):19–29

Sawik T (2020) Supply chain disruption management, 2nd edn. Springer, New York

Schönberger J (2011) Model-based control of logistics processes in volatile environments: decision support for operations planning in supply consortia. Springer, New York

Seuring SA (2008) Assessing the rigor of case study research in supply chain management. Int J Supply Chain Manag 13(2):128–137

Stadtler H, Kilger C, Meyr H (2015) Supply chain management and advanced planning, 5th edn. Springer, Berlin

Sterman J (2000) Business dynamics: systems thinking and modeling for a complex world. McGraw-Hill, London

Taha HA (2022) Operations research: an introduction, 11th edn. Pearson, Harlow

Voss C, Tsikriktsis N, Frohlich M (2002) Case research in operations management. Int J Oper Prod Manag 22(2):195–219

Wilkinson G (2018) Integrating artificial intelligence with simulation modeling. https://www.anylogic.com/blog. Accessed 4 Feb 2018

Wu Z, Choi TY (2005) Supplier–supplier relationships in the buyer–supplier triad: building theories from eight case studies. J Oper Manag 24:27–52

Yalaoui A, Chehade H, Yalaoui F, Amodeo L (2012) Optimization of logistics. Wiley, New York

Yin RK (2017) Case study research: design and methods (applied social research methods), 6th edn. Sage, London

Part II

Designing Operations and Supply Network: Strategic Perspective

Operations and Supply Chain Strategy

4

Learning Objectives for This Chapter

- Operations strategies and "strategic fit"
- Efficient and effective supply chain strategies
- Bullwhip effect
- Ripple effect
- Vendor-Managed Inventory (VMI)
- Collaborative Planning, Forecasting, and Replenishment (CPFR)
- Supply chain resilience and sustainability
- Supply chain viability
- Reconfigurable supply chain and intertwined supply networks

4.1 Introductory Case Study: "Quick and Affordable" Zara, UNIQLO & Primark

4.1.1 Zara's Three Success Factors: Speed, Speed, and Speed

Zara is a global fashion retailer which sells its goods around the world. The retailer's international footprint proves that national borders are no hindrance to a shared fashion culture. Founded in 1975, the Spanish retailer Zara has stores strategically

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_4) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the E-Supplement to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_4.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_4

located in leading cities across the continents (https://www.inditex.com/brands/zara), launches over 10,000 new designs each year, and is recognized as one of the world's principal fashion retailers. It belongs to the INDITEX Group, which also holds common brand names like Pull&Bear, Bershka, and Stradivarius.

Zara is well known for its up-to-the-minute styles and products. The main key to their corporate strategy is their SC strategy. The following statement by INDITEX gives the first hint of this exclusive strategy: "Our approach to fashion—creativity, quality design and rapid turnaround to adjust to changing market demands—has allowed us to expand internationally at a fast pace and has generated an excellent public response to our retailers' collections" (Inditex 2014).

While many retailers design and produce around 80% of their season's inventory, Zara keeps back 50% to be produced in the middle of the season. This way, they can react if, for example, the hot trend of the current season is skinny instead of boot-cut jeans. All research and design activities are steered from Zara's headquarters in La Coruña in Spain. For production, Zara has its own high-tech factories and a number of subcontractors. However, Zara keeps outsourcing to a minimum and produces primarily in developed countries to provide better quality control. For high responsiveness, many garments are kept in a generic unprinted stage. This approach is called *the postponement* strategy since products can be modified from generic to finished according to customer demand.

Procurement offices in the UK, China, and the Netherlands deal with all purchasing activities. In manufacturing, Zara links the two sides of SC strategies. First, they import 40% of their products as finished goods from the best-cost countries. This strategy is used for "basics," or products that are unchanged by fashion trends such as white and black T-shirts. Second, all other materials are bought from Mauritius, New Zealand, Australia, Morocco, China, India, Turkey, Korea, Italy, and Germany. Zara tries to source locally instead of globally, thus reducing transportation costs.

Short lead times are another key factor for Zara. Where other fashion companies supply every 3 months, Zara replenishes its stores twice weekly. The company provides the necessary fundamentals for subcontractors to meet their agreed-upon delivery times.

To implement its SC strategy, ZARA redesigned its inventory management. Previously, goods were shipped from two central warehouses to each of the stores based on requests from individual store managers. These local decisions, when assessed on a global scale, inevitably led to inefficient warehousing, shipping, and logistics operations. Production overruns, inefficient SCs, and an ever-changing marketplace (to say the least) were some of the problems that Zara had to overcome.

A variety of SCOM models was used in redesigning and implementing an entirely new inventory management system. The new centralized decision-making system replaced all store-level inventory decisions, thus providing results that were more globally optimal. Since implementation, having the right products in the right places at the right time for customers led to an increase in sales from 3% to 4%.

Inventory at Zara is kept a smidge under the estimated sales levels. For example, if demand increases immensely for clutch purses, they can assign emergency orders. In this way, the company is able to deal with sudden demand changes. However,

when the opposite is the case, Zara's management determined that the potentially lost customers would be less costly than slow-moving or last season's products. Speed is Zara's key strategy. All products are to leave the warehouse after a maximum of 3 days. At the end of each season, Zara reduces the price of its products up to 30% to sell remaining stocks.

All items arrive at the stores ironed, on hangers, and with price tags, saving valuable time for staff. Zara's pricing is based not on the classical cost plus margin principle but on setting prices according to comparable items in the local market. Because of this, it is possible that the same product has a different price in each European country. To make price tagging easier, Zara previously attached a tag on its goods showing all prices and goods could easily be shifted from one country to another with reduced complexity. Today, bar codes are used to tag the products, and these can be read via a scanner, showing the local price in the local store. This also enables Zara to keep up-to-date information about when and where goods are sold. All information is analyzed at headquarters to adapt particular strategies if necessary.

Marketing is minimized, as Zara sees all promotion activities as distracting for the customer. The company has managed to present itself as a fashion retailer with ever-changing and up-to-date styles with good quality at affordable price levels. Customers value exactly these assets—making all additional marketing efforts unnecessary.

4.1.2 UNIQLO: Basic, Casual Wear at Top Quality

UNIQLO does the exact opposite of Zara but is no less successful. UNIQLO is one of the largest and fastest-growing Japanese companies and ranks third among fashion retailers worldwide following Gap and Marks & Spencer. The mother company, FAST RETAILING CO. LTD, which also owns brand names GU and Theory, was founded in 1963 in Japan, where it still has its head office in Yamaguchi-City. High-quality and affordable products are valued more highly than chasing the newest trend.

Products are rather casual and basic, making them occasion- and age-less, but still stylish, using various colors and cuts. As a result, UNIQLO meets customer demand for clothes without presenting them on the catwalk.

UNIQLO has multiple production and purchasing offices in Asia that look after more than 100 suppliers each week. This allows the company to maintain good quality control over its outsourced partners. If quality issues arise, they are immediately taken to the production units, where means for improvement are found. Currently, UNIQLO is seeking to expand its purchasing and production facilities to meet demand in the United Kingdom and the United States.

UNIQLO controls its inventory carefully to maintain optimum inventory levels for each week. At the end of each season, products are sold for 20–30% less than the initial price to get rid of inventory.

Additionally, the fashion retailer has found a new place for selling clothes: railway stations. The concept works fabulously for UNIQLO. Shinjuku Station is Tokyo's biggest rail station and the UNIQLO store there ranks 6th out of 770

UNIQLO stores. The popularity of so-called one-day packs with basics like socks and underwear has also risen.

UNIQLO follows the SPA model (specialty store retailer of private label apparel), which is a specialized model where fashion retailers track all business activities from the point of origin up to the point of sale. This approach enables UNIQLO to improve its business processes quickly and constantly, giving the company an advantage over its competitors. Through long-term relationships with suppliers, UNIQLO is always seeking ways to improve its SC. This affects selling prices. By optimizing its SC, the company has been able to lower prices up to one-third compared to its competitors.

UNIQLO promotes its products on posters, flyers, or TV, hinting which items will be available for sale the following week.

4.1.3 Primark: It Is All About Money

A lot for a little is what customers value when visiting Primark stores, leaving with huge bags full of new fashionable clothes which cost not even half of what competitors offer. Primark belongs to Associated British Foods, a company selling food and clothes. Since the first store opened in Dublin, Primark has quickly expanded and today has more than 250 stores all over Europe. In the future, the company plans to open even more stores in more countries.

The product is based on simple designs with fashionable prints. To save costs, Primark focuses on selling only the most popular sizes.

Primark's customers demand catwalk fashion items at supermarket price levels. That is why it is all about money for Primark. Primark's products are sourced from more than 600 suppliers, mainly from the best-cost countries. Buying and selling in high volumes and using economies of scale allows Primark to cut down selling prices for the customer. The company uses the off-season to produce in advance and uses cheaper production periods. Primark uses a typical and efficient make-to-stock strategy where the focus is to (guess what?) cut costs.

In addition, Primark has little markup on its products and avoids marketing efforts, trusting in mouth-to-mouth advertising. As a result, prices are comparatively low.

Criticism has arisen concerning the sustainability of low-cost retailers, as many workers in low-cost countries suffer poor working conditions. This has also caught the attention of some customers, which is why retailers such as Primark try to counter a negative ethical image, e.g., by disclosing their suppliers.

Discussion Questions

1. What is the danger of mixing elements ("hybrid strategy") from both agile and lean SC strategies?
2. Contrast and compare Zara, UNIQLO, and Primark concerning their primary goals, product designs, manufacturing, inventory, suppliers, lead times, and pricing strategies.

3. What factors are critical for each SC strategy?
4. What do you think of UNIQLO's decision to open new facilities in the UK and the US?
5. Which roles does information technology play in SC coordination?
6. How do the SC strategies of Zara, UNIQLO, and Primark support their competitive strategies to achieve "strategic fit"?
7. Are you concerned about sustainability issues in apparel SCs?

4.2 Operations and Supply Chain Strategies

4.2.1 Value Added and Costs

Value added and costs are two basic, dominant factors in SCOM. Value creation can be achieved in different ways, e.g., through innovation, unique service, or quality. Any process has value-added steps, but these require resources that increase the costs. Of course, good processes have to be designed with a high degree of value-adding time and short periods of non-value-added time. This increases both *effectiveness* and *efficiency*.

However, in practice, the opposite situation often occurs. In many cases, the value-added steps hardly comprise more than 20% of the total time, and more realistically estimations are about 5–7% (Vrijhoef and Koskela 2000; Christopher 2022). This relation between value-added time and costs can be depicted as shown in Fig. 4.1.

It can be observed from the car manufacturing example above that value-added time is short compared to the increase in costs. The general recommendation is, therefore, to condense the cost-creation steps to reduce the lead time and increase the service level. A nice illustration of this idea is the following: a doctor has to remove a tick, and the act itself takes only 10 s, but the patient will spend at least 10 min in the room since 9 min and 50 s will be required to fill in the numerous forms and medical insurance documents. In this case, the value-added time is only 1.7%.

4.2.2 Operations Strategies

Since different ways exist to create value and reduce costs, different operations strategies can be classified as follows:

- Innovation strategy
- Efficiency strategy
- Quality strategy
- Service strategy

The main driver in *innovation strategy* is excellent product or service innovation. Creating value through unique product or service properties may significantly

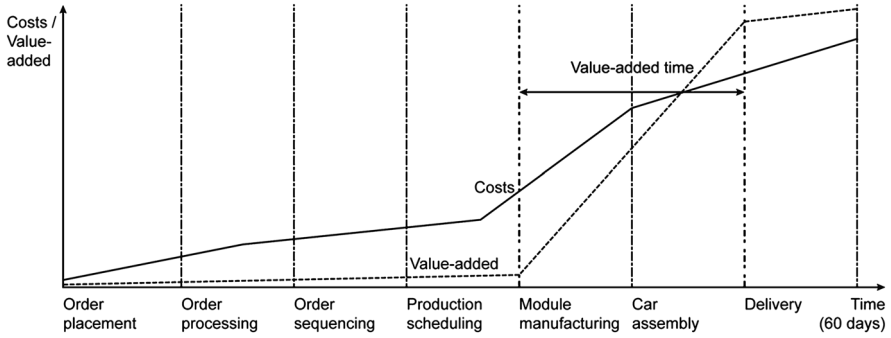


Fig. 4.1 Value added and costs

increase a company's margins while maintaining cost efficiencies. An example of this is Apple's transition from one iPhone model to the next generation. While the manufacturing costs were only slightly increased, the sales price grew steadily in most cases (Statista 2021).

In an *efficiency strategy*, price is the main competitive advantage. IKEA is well known for its highly efficient logistics and manufacturing. All the steps in the value chain support the efficiency strategy. Starting with a consideration of manufacturing and logistics at the product design stage, the operations at IKEA follow this philosophy in all other areas, such as production, sourcing, and shipments (Heizer et al. 2023).

A *quality strategy* ensures a competitive advantage based on superior product properties. "Made in Germany" has been associated with high-quality standards for many years. For such products, costs may be higher since in all operational areas decisions are taken in favor of quality. Quality control mechanisms are crucial for ensuring high quality in the SCs.

A service or *responsiveness strategy* means a high priority for customer preferences in operations decisions. Zara is known for its highly responsive operations and SCs. This is possible through sophisticated facility location planning and production in small batches. Both create additional costs, but these are covered by higher value-added activities (Chopra and Meindl 2015).

4.2.3 Supply Chain Strategies and "Strategic Fit"

An SC strategy should correspond to the competitive and operations strategies. Matching enterprise competitive strategy and SC strategy is called "strategic fit" (Chopra and Meindl 2015). *Strategic fit* presumes the alignment of objectives in different departments with the overall SC objectives. For example, typical decisions in marketing relate to product mix and pricing. This should be brought into correspondence with logistics decisions such as transportation, packaging, and transshipment.

▶ **Practical Insights** Consider an example. In an automotive company, the marketing and sales departments target increasing revenues. The supply chain department is responsible for meeting production targets at the lowest possible costs. When an automotive company decides to increase the individualization degree of its cars following marketing recommendations, the supply chain managers can compute the costs of this flexibility. Thus, company management can compare the benefits and costs of an increased car individualization strategy.

▶ **Practical Insights** Consider an example. Walmart has a marketplace strategy that mainly focuses on the high availability of a variety of reasonable quality products at a lower pricing level. Its products are not differentiated by technological advancement or unique design elements and can be purchased at other retailers as well. The competitive strength of Walmart lies in high availability and low prices, which Walmart's customers consequently expect from the mega-retailer. In order to be in line with the marketing focus introduced, Walmart has adapted its SC appropriately. For example, Walmart has its own SC infrastructure and distribution networks. In addition, they buy from low-cost producers only. This SC strategy fully supports Walmart's business focus of providing low prices and an everyday guarantee for product availability. Thus, Walmart represents a successful example of strategic fit (Chopra and Meindl 2015).

SC strategies and SC contributions to competitive and operations strategies are shown in Table 4.1. An SC strategy does not often target only one factor. Only decreasing prices to the lowest level may be an option. At the same time, an SC strategy needs to be matched to company and customer requirements. Consider efficient and responsive SC strategies in detail (see Table 4.2). Characteristics of agile SCs are a fast response to customer demand, low inventory, and flexible suppliers. Also, lead times are reduced to enable a swift reaction to demand fluctuations. Margins are comparatively high to fulfill the company's financial needs suitably. Responsive SCs are the best fit when market requirements are unpredictable and changeable.

When looking at lean SCs, almost always the opposite is the case. The market situation is stable in demand and price. The variety of products is also comparatively low. Efficient SCs concentrate mainly on lowering costs via high capacity utilization, minimizing inventory, and contracting suppliers from low-cost countries. Accordingly, margins are kept at a low level. Lean SC strategies also try to reduce lead time, but not at the expense of higher costs.

While choosing a strategy, it is sometimes advisable to go for a mixed approach. A poor strategy might endanger the company, perhaps by not positioning itself prominently enough in the market, with the additional risk of encountering other

Table 4.1 SCM strategies (based on Cohen and Roussel 2013)

Strategy	Advantages	Competitive basis	SCM contribution
Innovation	Unique brand or technology	Innovative and high demanded products	Quick introduction of new products and innovation
Costs	Cost-efficient organization	Low prices in product segment	Efficient, lean SC organization
Service/ response	Flexibility	Customer orientation	Agile SC is developed from a customer point of view
Quality	High-quality products	Unique product properties	Quality control along the SC

Table 4.2 Efficient and responsive SC strategies (based on Fisher 1997)

Criterion	Efficient supply chains	Responsive supply chains
Primary goal	Supply demand at the lowest cost	Respond quickly to demand
Product design strategy	Maximize performance at minimum product cost	Create modularity to allow postponement of product differentiation
Pricing strategy	Lower margins because price is a prime customer driver	Higher margins because price is not a prime customer driver
Manufacturing strategy	Lower costs through high utilization	Maintain capacity flexibility to buffer against demand/supply uncertainty
Inventory strategy	Minimize inventory to lower cost	Maintain buffer inventory to deal with demand/supply uncertainty
Lead time strategy	Reduce, but not at the expense of costs	Reduce aggressively, even if the costs are significant
Supplier strategy	Select based on cost and quality	Select based on speed, flexibility, reliability, and quality

negative factors. Some situations require a clear position in the lean or agile SC. However, many companies have a wide product range. For some products, market conditions may be stable; for others, however, they may be volatile and unpredictable. For these companies, a hybrid strategy of applying elements of responsiveness and efficiency for their particular products is advisable.

As a part of SC strategy, the definition of preassembled components, modules, or systems is paramount. These modules or systems should be pretested and should be ready to install shipped directly from the supplier to the point of usage on the shop floor of the producer. Furthermore, early cooperation between the supplier of the modules and its customer will also help to increase the usage of standard parts in higher volumes, which will also result in reduced costs due to economies of scale.

4.3 Supply Chain Coordination

4.3.1 Bullwhip Effect

The *bullwhip effect* is not a new phenomenon in the industrial world (Forrester 1961). The effect can be explained as the magnification of variability in orders in the

SC. In other words, irregular orders in the downstream part of the SC become more distinct upstream in the SC. This variance can interrupt the smoothness of SC processes as each link in the SC will over- or under-estimate product demand, resulting in exaggerated fluctuations (see Fig. 4.2).

Many retailers, each with little variability in their orders, can cause great variability for a smaller number of wholesalers, and even greater variability for a single manufacturer. The main reasons for the bullwhip effect can be divided into behavioral and operational areas (Lee et al. 1997; Sterman 2000):

Behavioral Causes

- Misuse of base-stock policies
- Misperceptions of feedback and time delays
- Panic ordering reactions after unmet demand
- Perceived risk of other players' bounded rationality

Operational Causes

- Dependent demand processing (demand is non-transparent and causes distortions in information in the SC).
- Lead time variability
- Lot-sizing/order synchronization
- Quantity discount
- Trade promotion and forward buying
- Anticipation of shortages

Negative consequences, higher safety stocks, inefficient production (surplus or shortage), and low or peak utilization of distribution channels can be recognized. As countermeasures to the bullwhip effect, the following actions can be identified (Table 4.3).

Consider how demand non-transparency leads to the bullwhip effect.

▶ **Practical Insights** In order to reduce the bullwhip effect, it is recommended to avoid panic-ordering reactions in the event of unexpected demand fluctuations. It is advisable to take into account time lags between orders and deliveries throughout the SC. It is important to understand where the demand is, where the inventory is, and where the costs are.

Task 4.1 Bullwhip Effect: Focus on Demand Non-transparency
A fast-moving consumer goods (FMCG) company is facing slight demand variation which has led to considerable variations in stocks on the supplier side. The company delivers its product to consumers through the manufacturer and three suppliers. Because of the relatively low cost of changing the production rate

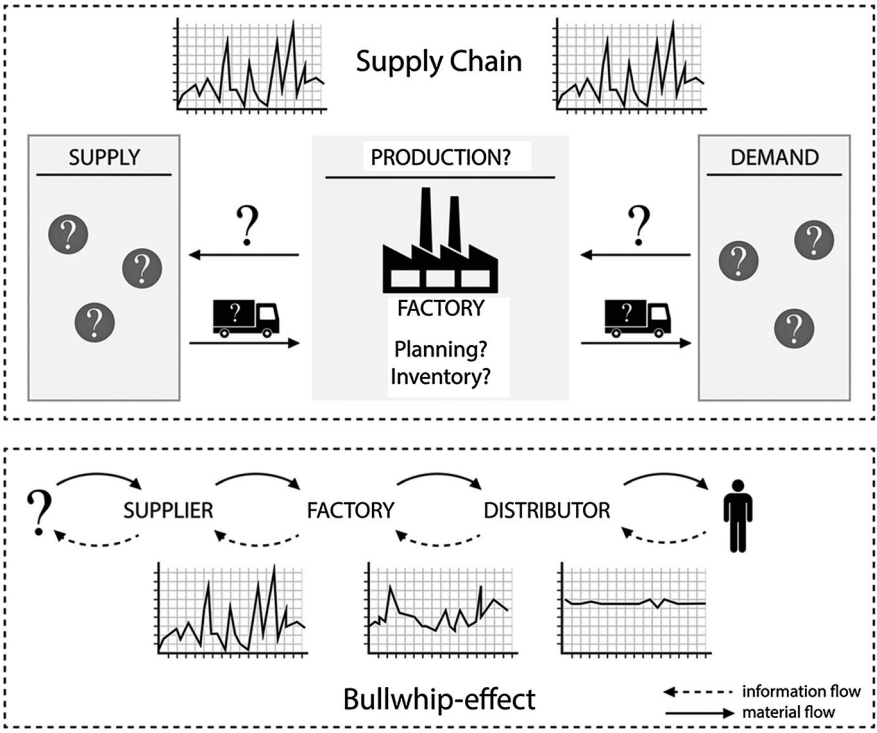


Fig. 4.2 Bullwhip effect in the SC

Table 4.3 Elimination of bullwhip effect

Reason for bullwhip effect	Countermeasures
Demand non-transparency	Information coordination
Neglecting to order in an attempt to reduce inventory	Automated ordering and monitoring of inventory in order to avoid overstock or shortage
Order batching	Coordinated and accurate lot size definition
Promotions	Use of everyday low prices (EDLP) instead of promotions
Shortage gaming	Validation of customer demand through historical data of customer ordering
Product returns	Policies to control returns or canceled orders.

compared to the cost of carrying inventory, the company has decided to change its production rate in order to reduce the capital commitment in the form of a finished product. The task shows the effect on the manufacturer of a 10% decrease in demand from customers.

$$\text{New order / Production rate} = \text{Demand new} - \text{Safety stock old} + \text{Safety stock new} \tag{4.1}$$

The initial data for analysis is presented in Table 4.4.

Table 4.4 Demand data

	Demand old	Order old	Safety stock old
Customer	1200	1200	300
Manufacturer	1200	1200	300
Supplier #1	1200	1200	300
Supplier #2	1200	1200	300
Supplier #3	1200	1200	300

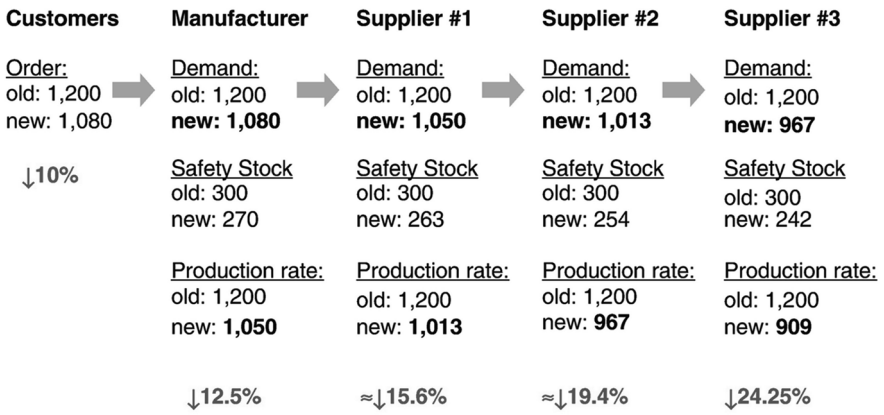


Fig. 4.3 Implications of the demand fluctuations on production rates

Now, demand has decreased by 10%. Safety stock is 25% of demand and is, therefore, able to cover demand of 1 week. New demand correlates with the new production rate of the predecessor SC member (see Fig. 4.3).

Each player in the SC assumes that the demand forecast (or orders) for the next period is the same as in the current period. Following this assumption, each supplier will be the same as in the current period, and each supplier will be planning their production rate to cover the demand/order for the next period, which will be equal to the demand/order for the current period (e.g., the new demand of supplier #3 orients itself by the new production rate of supplier #2). The problem is that only the manufacturer can see the changes in demand on the customer side. Other players in the SC cannot see the changes in demand because of nontransparency. For that reason, the players will change their production rate and safety stocks because the predecessor changed his order without informing other SC partners.

We can observe that a lack of demand transparency affects the shortage. For example, supplier #3 will not produce the right amount of pieces for customers' orders to satisfy their demand.

It can be concluded that changes on the customer side increase order quantity through the SC if demand is nontransparent. In general, communication, validation of demand, information sharing, computer-aided ordering, and better pricing strategies can help reduce the bullwhip effect in this situation.

4.3.2 Vendor-Managed Inventory

Inventory typically exists to manage the uncertainty of supply and demand (safety stock) and to take advantage of economy of scale (cycle inventory). At the same time, modern markets require more flexibility from SCs. Customer orders and demand change frequently. In building up high inventory, companies can increase their flexibility on the one hand. On the other hand, if demand changes, these inventory mountains will lead to losses. Consider a short example: HP and Canon worked jointly on the LaserJet printer production. Canon produced engines for different types of printers. However, HP was able to indicate changes in demand only 3 months in advance. Canon needed 6 months to change its production plans. As demand for LaserJet III drastically decreased, many engines for this printer type were in inventory and had to be written off (the Laserjet-mountain).

In order to face this challenge, SC coordination strategies extensively use the Internet and new IT. The ideas of these concepts are as follows (Heizer et al. 2023):

- Coordination instead of uncertainty
- "Replace inventory with information"
- Integrated SC
- Transparency of demand and inventory

At the plant level, inventory is insurance against the unknown; in the SC, this "unknown" became "known" through collaboration. Inventories in SCM are leverage for SC reliability and flexibility rather than buffers against blind uncertainty.

One of the strategies for SC coordination is vendor-managed inventory (VMI). With VMI, the vendor controls inventory on the buyer side. The buyer provides information on inventory and sales (see Fig. 4.4).

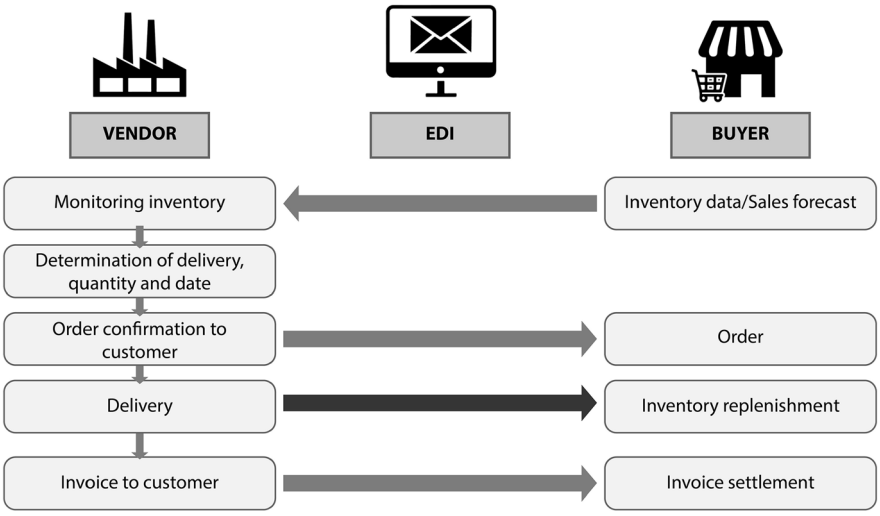


Fig. 4.4 Vendor-managed inventory strategy

VMI is a special concept of restocking and replenishing inventory that originated in the retail business, where stock availability is a significant, if not crucial, factor of a company's success. The buyer provides certain information to a supplier, who takes full responsibility for maintaining an agreed-upon inventory level, usually at the buyer's consumption location (usually a store). A third-party logistics provider (3PL) might be involved to control the required level of inventory by adjusting the demand and supply gaps.

The main advantage of this concept is that the real-time inventory information is made available to the supplier (manufacturer or a wholesaler), and a customer (distributor or retailer) relinquishes control of inventory to them. The vendor reviews every customer's item and is responsible for the inventory plan. Efficiency increases in process activities are based on Electronic Data Interchange (EDI) and collaborative SC organization (see Fig. 4.5).

The supplier receives an insight into stock level and demand forecast and ensures that the stock is within agreed-upon limits. Consider an example of the VMI process in SupplyOn (Fig. 4.6).

The following steps are included in the VMI process at SupplyOn (Fig. 4.6):

- Agree upon minimum and maximum stock limits
- Demand and inventory is transmitted from the internal ERP system to SupplyOn.
- Demand and inventory is shown in the Inventory Monitor; thus the supplier can plan deliveries based on the agreed limits.
- ASN (Advanced Shipment Notification) is entered by the supplier and transmitted to the internal ERP system.
- While the goods are being dispatched, the Inventory Monitor shows them as in-transit quantities.

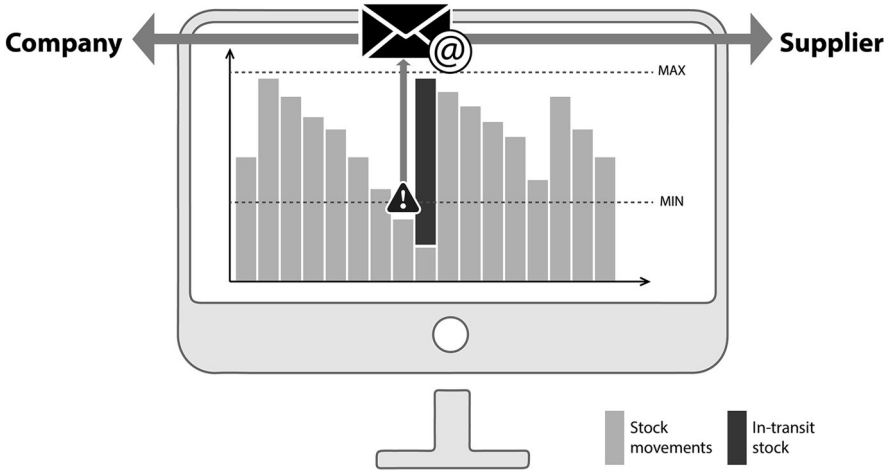


Fig. 4.5 Alerting concept in VMI (based on SupplyOn, used with permission)

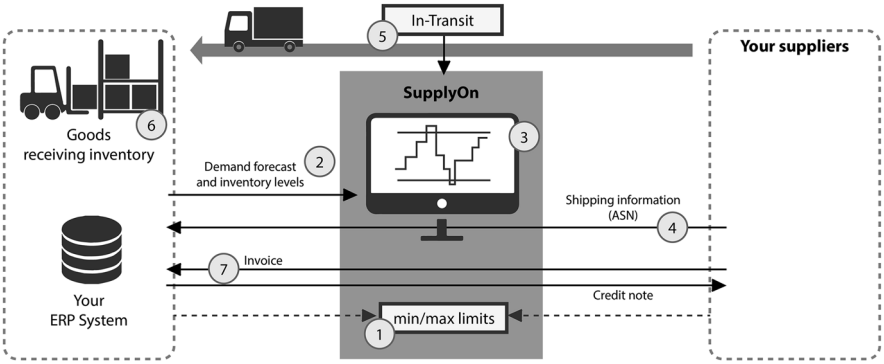


Fig. 4.6 VMI in SupplyOn (based on SupplyOn, used with permission)

- Goods are booked into the system upon delivery and become visible in the Stock Monitor.
- An invoice or credit note is generated as soon as the goods have been booked.

The *advantages of VMI* are as follows:

For the Vendor

- Early recognition of fluctuation in demand
- Optimization of production planning; increased volume
- Enforcement of discipline: measurements and communication
- Better planning and resource use via visibility
- Improved market analysis and elimination of non-value-added activities
- Closer customer ties and preferred status

For the Buyer

- Increase of inventory availability
- Reduction of procurement activities
- Fewer stock-outs with higher inventory turnover
- Optimal product mix
- Lower operating, purchasing, and administrative costs
- SC relationship strategic strength
- Greater customer satisfaction and increased sales

For Overall SC

- Optimization of inventory management and cost reduction
- Decrease of fixed capital (stocks)

- Improvement of financial planning
- Supports long-term collaboration

For End-User

- Increased service level and reduced stock-outs

However, it is proven that a sustainable VMI requires trust from both parties and willingness to invest (e.g., switching costs, EDI, etc.). A closer relationship implies more interdependency and cooperation. So, a customer needs to be ready to communicate promotions and other marketing campaigns with the supplier so that additional items can be available on time. Among other limitations of VMI are unforeseeable risks, as employee strikes in the event of stock-outs and the risk of customer loss.

The *limitations of VMI* can be stated as follows:

- Trust in the SC should be high.
- High costs of implementation and investments in IT; dependence on EDI
- Customer loss risk
- Non-foreseeable risks: employee strikes
- VMI mostly benefits the end-user and seller, while the vendor does most of the work.
- Additional processing activities for the vendor (costs)
- Supplier-dependent buyer

The VMI concept has been proven to be efficient in the following cases:

- Items with a high consumption amount
- Items with a high consumption value (compare with ABC analysis)
- Traditional procurement activities should be changed.

Lastly, VMI is suggested for lines of business with multiple outlets and the potential for long-term partnerships between vendor and customer. Simply put, a major retail store chain is more likely to benefit from VMI and recover the capital invested in switching to VMI than a small family-owned convenience store. VMI is recommended specifically for industries like retail, pharmaceutical, automotive, high-tech, and other industries where inventory expectation is a significant factor in gaining a competitive advantage.

4.3.3 Collaborative Planning, Forecasting, and Replenishment

Collaborative Planning, Forecasting, and Replenishment (CPFR) aims to integrate and coordinate actors and processes in the SC regarding the planning and

fulfillment of customer demand, production, and inventory (see Fig. 4.7). CPFR is the cooperative management of demand and supply through joint visibility and replenishment of products in the SC. CPFR is based on information exchange between suppliers and retailers, which helps in planning and satisfying customer demands through a joint system of shared information. CPFR includes three basic stages.

In the planning phase, the definition of collaboration areas, responsibilities, and a description of the collaboration mission, objectives, and framework takes place. Trading partners share the business plans and organizational information. Category information, such as the definition of roles, strategies, and tactics, needs to be explored. Seasonal events and frameworks for all events should be shared at this stage.

In the forecasting phase, sales and order forecasts need to be performed and shared. Moreover, exceptions for these forecasts should be classified. Events, promotion plans for products, new product information, individual forecast, and forecast constraints related to the sales should be shared.

In the replenishment area, lead time, logistics data, location changes, current inventory levels, inventory in transit, and POS (point-of-sale) data should be shared. In the last phase, a committed order from order forecasts should be generated.

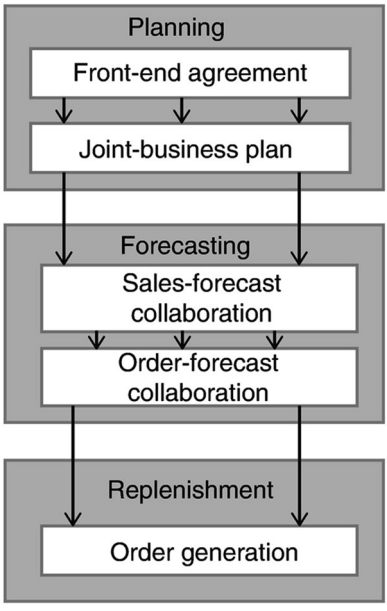


Fig. 4.7 CPFR strategy

4.3.4 Supply Chain Contracting

A supply contract specifies parameters governing the buyer–supplier relationship (Chopra 2019). *Contracts* have a significant impact on the behavior and performance of all stages in an SC. Two groups of contracts can be distinguished:

- Contracts for product availability and SC profit
- Contracts to coordinate SC costs

In the first group, the supplier designs a contract that encourages the buyer to purchase more items and increase the level of product availability. In other words, the supplier shares a part of the buyer's demand uncertainty. In this group, the following contracts are included:

- Buyback or returns contracts (a salvage price is used at which a retailer can return unsold inventory for a certain amount).
- Revenue-sharing contracts (a share of the retailer's revenue is paid to the supplier).
- Quantity flexibility contracts (the manufacturer allows the retailer to change the quantity ordered after observing demand).

Contracts to coordinate SC costs are based on quantity discount contracts. A *quantity discount contract* decreases overall costs but leads by tendency to higher lot sizes. This, in turn, facilitates higher levels of inventory in the SC. It is typically reasonable for commodity products for which the supplier has high fixed costs per lot. We refer to the textbook by Das (2016) regarding quantitative analysis of SC contracts.

4.3.5 Vertical Integration vs. Horizontal Coordination

An SC can be organized as a *vertical integration* or *horizontal coordination*. A vertical integration strategy means that the firm itself manages the whole SC of a firm. All (or almost all) value-adding activities are designed, planned, and controlled centrally by the focal company. Vertical integration should be considered a corporate strategy that is pursued in almost all areas of the company.

▶ **Practical Insights** Consider the example of Tesla. Tesla focuses on vertical integration, which is unique within the automotive industry. Most other manufacturers rely on the prevailing concept of horizontal coordination and use outsourcing to reduce the vertical range of manufacturing to save costs. Traditional car manufacturers focus on the assembly of final productions as their key competence, which accounts for about 20–25% of their value-adding activities. They develop and produce only a small number of their own parts, such as different body

> parts. Tesla concentrates a high amount of value-adding in-house. It designs and manufactures a wide range of its own components, such as traction batteries, electric powertrains, charging systems, and software chips in-house. In addition to the development and production of its own parts, Tesla also uses vertical integration in the after-sales market. Vertical integration for Tesla, therefore, is more than an approach to production and logistics planning. It is a corporate strategy that concentrates all relevant value-adding steps in-house to acquire know-how about e-mobility.

Vertical integration provides several advantages typical of centralized planning. It allows for high information availability, SC visibility, planning instead of negotiating, monitoring of the contracting process, and control. However, a vertical integration strategy may result in lower flexibility and agility compared with horizontal coordination.

Horizontal coordination is the most popular form of strategic SC organization. It is based on the networking of different firms into a value chain using contracting and negotiation mechanisms. For example, the focal companies in the automotive industry are networked with thousands of suppliers and retailers, which deliver parts and sell cars. The focal company itself is responsible for the design and assembly of the final products. In other industries, e.g., in the apparel and fashion sectors, the focal firms serve mostly as brands, whereas the actual manufacturing and logistics activities are outsourced. A horizontal coordination strategy can offer high flexibility and agility; nevertheless, it may lack visibility and control, making it necessary to utilize information coordination technology extensively. Horizontal coordination may also be challenged by changes in bargaining power and high SC volatility based on the different dynamics in economies and environments.

4.4 Supply Chain Resilience and Sustainability

SCs influence and are influenced by the environment. In light of ecological problems, natural disasters, and social development challenges, the necessity for new viewpoints on SCM has become even more obvious. The former paradigm of overall and unlimited customer satisfaction has naturally failed because of the limited resources available for this satisfaction.

In these settings, the duality of the main goals of SCM—maximizing service level and minimizing costs—will be enhanced by the third component—maintaining SC *resilience* and *sustainability* (Ivanov 2017). This "triangle" goal framework will build the *new SCM paradigm* that can be formulated as the maintenance of resilience and the harmonization of value chains with possibly full customer satisfaction and cost-efficient resource consumption to ensure the performance of production-ecological systems at an infinite time horizon (see Fig. 4.8).

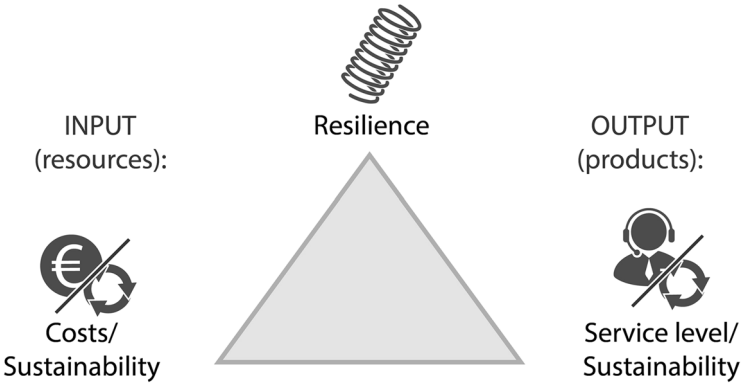


Fig. 4.8 Supply chain management resilience and sustainability

4.4.1 Supply Chain Sustainability and Closed-Loop Supply Chain

SC sustainability is based on the triple-bottom-line: "economy–ecology–society." Joint consideration of all these elements is crucial for SCM in the long-term perspective. The main objective of sustainable SCOM is to extend the scope from a mere optimization of economic performance to sustainable developments that meet the needs of the present generation without compromising the ability of future generations to meet their own needs (Seuring and Müller 2008; Tang and Zhou 2012; Brandenburg et al. 2019). Thus, sustainability in SCOM decision-making requires a balance between multiple economic, environmental, and social factors as well as exploiting the complementarity of different elements in the triple-bottomline. In SCOM such concepts as "Closed-Loop Supply Chain" and "Reverse Logistics" have been developed. They are based on the idea that an SC does not end at the point of sale but can include the after-sale area. We highlight SC sustainability issues based on two case studies of Coca-Cola and Mercadona.

Case Study Mercadona

The *objective* of this case study is to analyze how a company can efficiently implement a sustainable SC without having a negative impact on turnover. Coordination of the value chain from positions of intermodal logistics and a closed-loop SC makes it possible to decrease costs, improve on-time delivery and company image, and achieve higher customer satisfaction. All these factors have a positive effect on turnover.

Mercadona is a family business founded in Valencia and is the largest Spanish supermarket chain in terms of a turnover of 19.08 million euros in 2012. In 1981, the company had only eight shops in Valencia. Today this has risen to the considerable count of 1411 stores and 74,000 employees.

Mercadona is called the "Toyota of retail" because of its continuous improvement of *lean processes*. For example, the company packs fruit loosely instead of

selling it in containers to save packaging material, sells only a limited number of goods, so it can reduce cost due to the decreased variability, and has redesigned packaging to optimize transportation. Mercadona says profit is not the primary goal, but rather an investment in the long-term development of, for example, their employees. Being lean and always one step ahead of their competitors is a part of Mercadona's strategy. The whole SC supports this *strategy* of leanness and innovation to remain both efficient and effective in modern volatile markets. The target of recent years was to find a way to become more *sustainable* generally. The Mercadona Board of Management focused mainly on its SC.

Mercadona initially organized the *transportation* of its products by trucks and collaborated with the logistics company Acotral, which increasingly suffered delays due to road traffic. Since many goods go from the supplier in Sevilla in southern Spain to Valencia in the east, usage of rail services for the longest part of the distance was discussed as an option.

Renfe, a former state-owned train company, was suffering immensely from market liberalization. To secure its financial position, it sought a new strategy. Offering transporting solutions to supermarket chains became an attractive option.

Hence, Mercadona, Arcotral, and Renfe arrived at a collaborative agreement to involve rail as a transportation mode. The new *intermodal logistics* process was designed as follows. From the supplier in Seville, the products were transported by truck to the rail terminal in Seville. Here, the goods were *transhipped*, with Renfe taking the goods on two trains twice a week to Valencia. Of course, this concept only worked for non-fresh goods such as sugar and non-food goods such as batteries. From the Valencia rail terminal, Arcotral picked up the goods and brought them to Mercadona's distribution center to supply the regional supermarkets.

The crucial element of the redesigned SC was *coordination* between the companies. Renfe had to customize its services to be able to transport such a large amount of goods to Mercadona. Moreover, collaborative agreements can be successful only if based on trust and a *win-win* situation. As previously stated, Renfe customized their services. Further changes on their side included a *communication* tool that enabled the *tracking* of the product at each stage of the transportation process. Mercadona then incorporated this communication tool into its own system, which made it possible to have *visibility* throughout the whole SC. By offering this transportation solution, Renfe developed a new business concept. With such key customers as Mercadona, they resolved their own financial situation and secured a market position. In its turn, Mercadona became less vulnerable to road traffic, and deliveries were rarely delayed. This also saved costs in terms of maintaining safety stocks. While using rail, there were less noise emissions, fewer traffic jams and accidents, and 12,000 tons less CO_2 emissions each year. Mercadona demonstrated a best-practice example of how a company can, when implementing it correctly, increase sustainability and have positive effects on their business processes and turnover.

Discussion Questions

- The case study shows positive, real-life examples which can be applied to other businesses. Even though Mercadona is a first-class example of how to implement sustainability in SCM, where can you see possible limitations of this concept?
- Consider the new SC redesigned within the collaborative agreement between Mercadona, Renfe, and Arcotral. At which point could a reverse flow/closed-loop SC be implemented?
- As stated in the case study, collaborative agreements only work if they are based on "trust and a win-win situation for all." What challenges can you see that could become critical for the SC?

Finally, we consider the *Closed-Loop Supply Chain* concept. Closed-Loop SC management considers planning, implementing, and controlling of material and information flows from the point of consumption to the point of origin to recapture value or ensure proper disposal (Guide and Van Wassenhove 2009; Govindan et al. 2015; Gianesello et al. 2017). The purpose of the Closed-Loop SC is to increase product life by collecting the used products from the customers, performing an inspection, and enacting some recovering activities that depend on the type and quality of the returned products and, finally, reintroducing the products in the forward SC (Fig. 4.9).

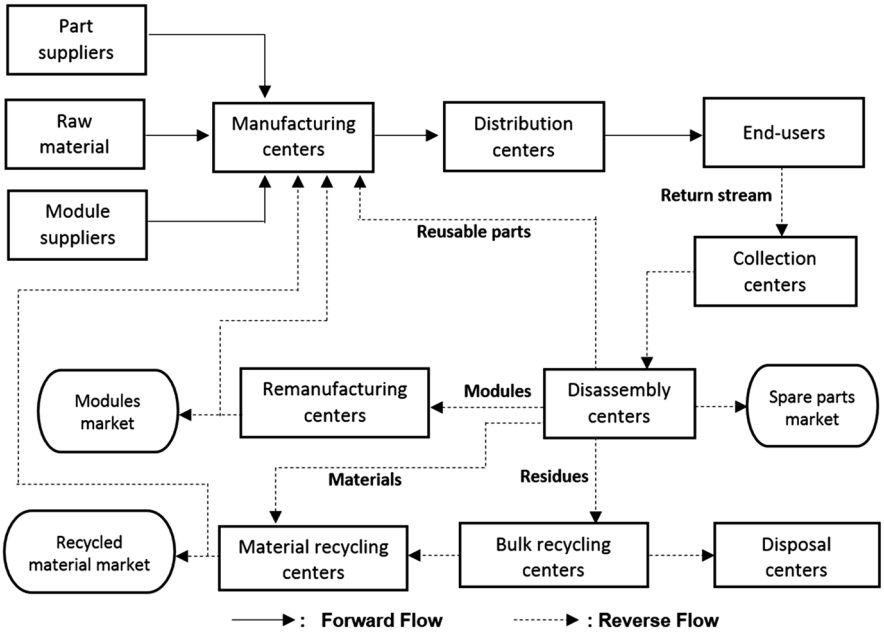


Fig. 4.9 General structure of a Closed-Loop Supply Chain (Jeihoonian et al. 2017)

The recovered products may be reintroduced in the forward SC as "good as new products" and used to satisfy the customers of the primary market. In other cases, the products are reintroduced to supply a secondary market, like used cars. In these cases, the recovered products are characterized by a lower level of quality and price compared with the brand-new products. For example, the closed-loop SC principles are vital for the electric car industry (Gianesello et al. 2017).

4.4.2 Supply Chain Risks, Resilience, and Ripple Effect

The design of an efficient and resilient SC capable of operations and demand fulfillment continuity despite disruptions is imperative and has been highlighted in literature and practice alike for the last two decades. However, the COVID-19 pandemic has unveiled the lack of resilience in many SCs, as complex networks failed from disruptions at local nodes, their propagation (i.e., the ripple effect), and the resulting missing connectivity. This subsection briefly introduces the SC risk and resilience area. In Chap. 15, we will elaborate on different aspects of SC risks, resilience, and the ripple effect in more detail.

In general, SC risks can be classified into demand, supply, process, and structure (see Fig. 4.10). Risks of demand and supply uncertainty are related to random uncertainty and a business-as-usual situation. Such risks are also known as *recurrent* or *operational risks*. SC managers achieved significant improvements in managing global SCs and mitigating recurrent SC risks through improved planning and execution (Chopra and Sodhi 2014). From 2010 to 2014, SC disruptions occurred at a greater frequency and intensity and thus had more significant consequences. Such *disruptive risks* now represent a new challenge for SC managers (Simchi-Levi et al. 2015; Ivanov et al. 2017a, b). Let us concentrate on the disruption side.

First, globalization and outsourcing trends make SCs more complex and less observable and controllable. According to complexity theory, such systems become more sensitive to disruptions. Special focus in this area is directed toward disruptions in transportation channels. Second, the efficiency paradigms of lean processes, single sourcing, etc. have failed in disruption situations. As a consequence, SCs have become more vulnerable, even to minor perturbations. Any disruption in a global SC, especially in its supply base, immediately affects

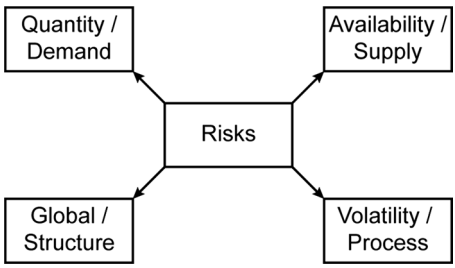


Fig. 4.10 Supply chain risks (Ivanov et al. 2014a)

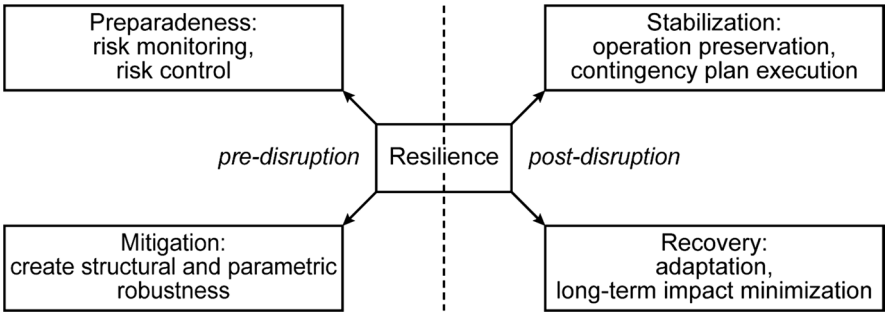


Fig. 4.11 Supply chain resilience (Ivanov et al. 2014a)

the entire SC. Third, with increased specialization and geographical concentration of manufacturing, disruptions in one or several nodes affect almost all nodes and links in the SC. Fourth, IT became a crucial element of global SCs since disruptions in IT may have a significant impact on disruptions in material flows.

Resilience is the ability to maintain, execute, and recover (adapt) planned execution along with achieving the planned (or adapted, but still acceptable) performance (Ivanov and Sokolov 2013). Building a resilient SC is based on mitigating risks, preparedness for disruptions, stabilization, and recovery (see Fig. 4.11).

SC resilience is close to the role of immune systems and can be understood better in their example (Ivanov 2025). A strong immune system and acting in line with nature and society help human beings achieving high performances. Similarly, the SC performance depends on resilience and sustainability. A weak immune system may result in performance degradation. Low SC resilience, if a disruption is experienced, also results in profitability reductions, mismatches of demand and supply, and destabilization of normal operations. Continuing the immune system analogy, we can understand the difference between risks and resilience. The immune system is an inherent property of any human being to absorb negative events, e.g., a virus. It also helps to recover after an infection. In SCOM terms, risks are negative events (i.e., a kind of virus that can cause an infection), and resilience is an inherent SC property responsible for absorbing the negative events and restoring normal operations (i.e., recovering).

Basic areas of resilience include system, process, and product resilience. To the basic elements of increasing SC resilience belong the following (Blackhurst et al. 2011; Pettit et al. 2013; Chopra and Sodhi 2014):

- Backup suppliers
- Excessive inventory
- Excessive capacity
- Information coordination and data transparency
- Process flexibility
- Product flexibility

In Table 4.5, the resilient SC is presented in comparison with the efficient SC strategy.

It can be observed in Table 4.5 that the concepts of efficiency and resilience are rather considered in contradictory settings with direct assumptions of efficiency reduction with resilience increases or vice versa (Aldrighetti et al. 2021). The recently developed *LCN (low-certainty-need) SC* framework suggests approaching SC resilience from another perspective. Rather than opposing efficiency and resilience, the LCN framework suggests considering their mutual intersections to enhance each other based on synergetic effects in terms of SC *lean resilience* (Ivanov and Dolgui 2019).

Coordination and sourcing strategies in the SC are also typical in practice. Many companies also invest in structural redundancy (e.g., Toyota extends its SC subject to multiple-sourcing and building new facilities on the supply side). All these four elements of resilience can be seen as strategies for mitigating at the *pre-disruption* stage and reacting at the *post-disruption* stage in the resilience framework.

For practical implementation, it is important to understand that risk and resilience analysis means little unless they are correlated with their impact on cost efficiency. The trade-off "efficiency–flexibility–resilience" becomes increasingly important in SCOM and has recently been considered a *ripple effect* in SCs (Ivanov et al. 2014a, b).

While the bullwhip effect describes high-probability-low-impact risks, the ripple effect focuses on low-frequency-high-impact disruptions (Dolgui et al. 2018; Ivanov and Dolgui 2021). Ripple effect in the SC results from disruption propagation of an initial disruption toward other SC stages in the supply, production, and distribution networks (Fig. 4.12).

Table 4.5 Efficient and resilient supply chain design (Ivanov and Dolgui 2019)

Criteria	Efficient supply chains	Resilient supply chain
Primary goal	Supply demand at the lowest cost	Ensure demand fulfillment in the presence of disruptions
Network organization	Centralized, global	Decentralization, diversification, localization, segmentation, fortification
Product design strategy	Standardization, maximize performance at minimum product cost	Postponement to ensure product flexibility, product substitution, capacity pooling
Pricing strategy	Lower margins because price is a prime customer driver	Higher prices caused by the costs of resilience
Manufacturing strategy	Lower costs through high utilization	Capacity reservations
Inventory strategy	Minimize inventory to lower cost	Risk mitigation inventory
Lead time strategy	Reduce, but not at the expense of costs	Lead time reservations
Sourcing strategy	Select suppliers based on cost and quality; single sourcing	Supplier risk exposure analysis; backup suppliers and dual sourcing

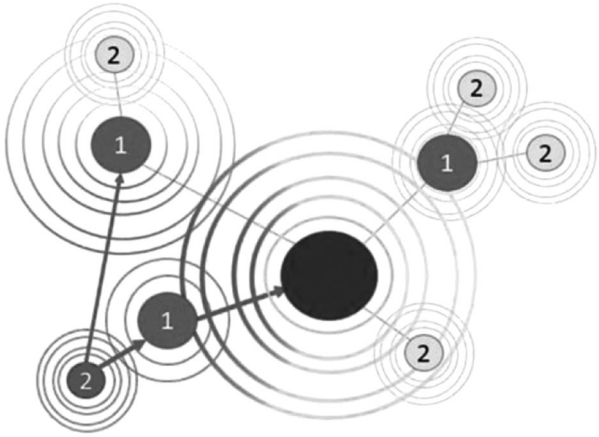


Fig. 4.12 Disruption propagation in the supply chain

The ripple effect is not an infrequent occurrence. In many cases, SC disruptions go beyond the disrupted stage; i.e., the original disruption causes disruption propagation in the SC, at times provoking still higher consequences (Ivanov 2025a, b). Ripple effect is a phenomenon of disruption propagations in the SC and their impact on output SC performance (e.g., sales, on-time delivery, and total profit). This effect may have more serious consequences than short-term performance decreases. It can result in market share losses (e.g., Toyota lost its market leader position after the tsunami in 2011 and needed to redesign its SC coordination mechanism or lose company value). Hendricks and Singhal (2005) quantified the negative effects of SC disruption through empirical analysis and found 33–40% lower stock returns relative to their benchmarks over a 3-year time period that started 1 year before and ended 2 years after a disruption: large negative effects on profitability, a 107% drop in operating income, 7% lower sales growth and an 11% growth in costs, 2 years with a lower performance level after a disruption.

The ripple effect is also known as the "*domino effect*" or "*snowball effect*." The reasons for the ripple effect are not difficult to identify (Dolgui and Ivanov 2021; Li et al. 2021). With increasing SC complexity and consequent pressure on increasing speed and efficiency, ever larger numbers of industries come to be distributed worldwide and concentrated in industrial districts. In addition, globalized SCs depend heavily on permanent transportation infrastructure availability. Mitigation and recovery strategies for the ripple effect are considered in the SC resilience framework (Fig. 4.11).

Table 4.6 summarizes the reasons and countermeasures for the ripple effect (Ivanov and Rozhkov 2020; Dolgui et al. 2018; Ivanov 2018a, b).

Literature provides evidence that disruption duration and propagation impact SC performance. Proactive strategies such as backup facilities and inventory have positive impacts concerning both performance and prevention of disruption propagation. Speed of recovery plays an important role in mitigating the performance

Table 4.6 Ripple effect reasons and countermeasures (Dolgui et al. 2018)

Reason	SCM impact	Ripple effect impact	Countermeasures
Leanness	Single sourcing	In the non-disrupted scenario, it is irrational to avoid lean practices. At the same time, a capacity disruption may result in a ripple effect and performance decreases. Recommendation to use capacity buffers or a backup facility as additional capacity reserves.	Multiple/dual sourcing/backup suppliers
	Low inventory		Risk mitigation inventory
	Inflexible capacity		Postponement
Complexity	Globalization	Without a coordinated contingency policy, disruption recovery and performance impact estimation can be very long-lasting and expensive. Coordinated control algorithms are needed to monitor SC behavior, identify disruptions, and adjust order allocation rules using a coordinated contingency policy.	Geographical sourcing diversification
	Decentralization		Global SC contingency plans
	Multistage SCs		Supplier segmentation according to disruption risks

impact of disruptions. An increase in SC resilience may imply cost increases in the SC. However, a missing resilience can have devastating consequences for the SCs as demonstrated through the COVID-19 pandemic (Ivanov 2020; Ivanov and Das 2020; Ivanov and Dolgui 2020; Queiroz et al. 2022).

4.5 Supply Chain Viability

To recapitulate, SC resilience is considered the SC's ability to bounce back from a severe disruption and recover to an "old normal" (Pettit et al. 2013, Hosseini et al. 2019). Notably, resilience theory has been developed for managing disruptions, which are usually defined as events. As a new super-disruption, the COVID-19 pandemic raised novel questions and decision-making contexts, which frequently exceeded the resilience scope. In this section, we explain one of the major SC strategic issues to survive during the super-disruption, i.e., the *viability*.

4.5.1 Viable Supply Chain and Value Network Ecosystem

Viability is an overarched resilience perspective in SCs. The *Viable Supply Chain* (VSC) model is based on adaptable structural SC designs for supply–demand allocations and, most importantly, establishing and controlling adaptive mechanisms for transitions between the structural designs. Ivanov (2022b) defined VSC as follows: "*Viable Supply Chain (VSC) is a dynamically adaptable and structurally changeable value-adding network able to (i) react agilely to positive changes, (ii) be resilient to absorb negative events and recover after the disruptions, and (iii)*

survive at the times of long-term, global disruptions by adjusting capacities, utilizations, and their allocations to demands in response to internal and external changes in line with the sustainable developments to secure the provision of society and markets with good and services in a long-term perspective."

The VSC framework thus takes the *SC ecosystem* perspective. The interactions in these SC ecosystems are very complex and are triggered by mutual interrelations and feedbacks between SCs, nature, society, and the economy. Viability can be considered a central lens to design and manage SCs in the post-pandemic world (Fig. 4.13).

Initially developed in terms of leanness and agility (and their combination as leagility), SC research has been extended by the perspectives of resilience and sustainability, followed by the advanced utilization of digital technologies and Industry 4.0. The current state-of-the-art SC management results from a number of remarkable transformations. In Fig. 4.13, these transformations are framed in a historical perspective. Being lean, responsive, and globalized in structural designs, SCs have also learned a great deal about how to act in line with nature and societal interests (i.e., becoming sustainable), how to strengthen their resilience during disruptions triggered by severe natural or human-made disasters, how to recover and manage the ripple effects, and how to utilize the advantages of digital technologies in SC management. However, in 2020, the leagility, resilience, and sustainability of SCs have been put to the test. SCs worldwide have experienced unprecedented shocks caused by the COVID-19 virus outbreak and global pandemic, which have been a new instigator of SC disruptions, unlike any seen in recent times. The COVID-19 outbreak and global pandemic have immensely affected all areas of the economy and society, raising a series of completely novel decision-making settings for SC researchers and practitioners.

The *viable value network ecosystem* is comprised of the following components (Fig. 4.14):

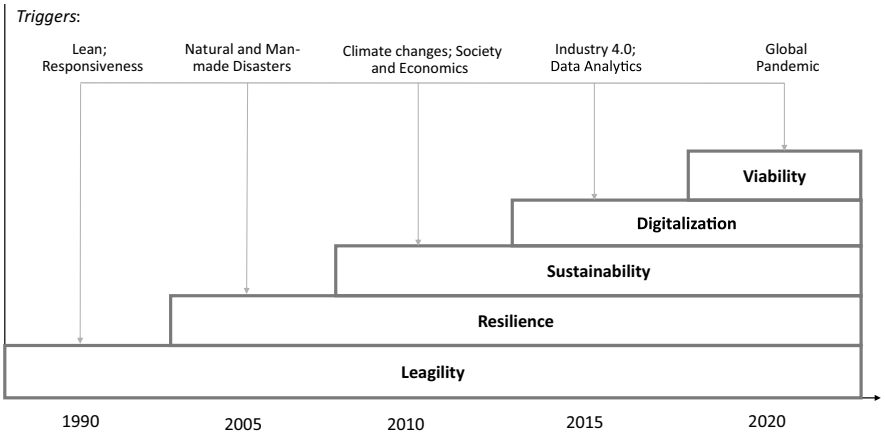


Fig. 4.13 Transformation of SCOM research over time (Ivanov 2022b)

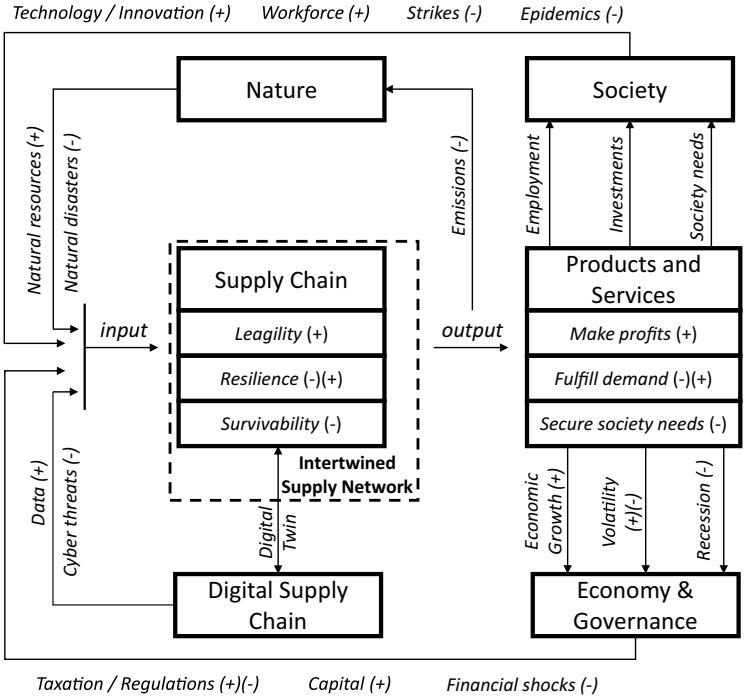


Fig. 4.14 Viable value network ecosystem (Ivanov 2022b)

- The SC itself
- The intertwined supply network (ISN) is an "*entirety of interconnected supply chains which, in their integrity, secure the provision of society and markets with goods and services*" (Ivanov and Dolgui 2020)
- Society
- Nature
- Economy and governance
- Digital SCs, which, in combination with the physical SC, represent a cyber physical system framing a digital SC twin

An SC can be considered viable if it can maintain an ecosystem balance (i.e., achieve homeostasis) within several feedback cycles (Ruel et al. 2024; Ivanov 2021). The positive cycle is concerned with profitability, developments, investments, efficiency, agility, and responsiveness. This is the time to use the advantages of technological developments, innovations, and market growth to react to positive changes. The volatile cycle is concerned with sustaining and recovering from the disruptions to fulfill the demand. The cycle of survivability, meanwhile, is intended to fulfill the minimum needs of the economy and society by securing the SC's products and services as provisions.

Within each of the three feedback cycles in the SC ecosystem, there is internal positive and negative feedback. For example, the interactions of the SC and nature are concerned with a positive cycle of using natural resources and a negative cycle of emissions as potential contributors to climate change. The interaction with society results in positive feedback, such as technological innovations and workforce development, and negative feedback in terms of possible labor strikes (disruptions at the SC resilience level) or global pandemics (disruptions at the SC survivability level).

In summary, the VSC framework integrates the perspectives of sustainability and resilience, extends them using survivability, and offers a VSC model of interactions between SCs and their ecosystems at the levels of profitability, resilience, and survivability.

The VSC ecosystem framework builds a foundation for the VSC model. With the development of this model, we argue in favor of multiple structural SC designs for matching supply and demand according to the three feedback cycles of the viable SC ecosystem framework and, most importantly, the establishment and control of adaptive mechanisms for transitions between the structural designs (Ivanov 2021). The rationale behind several structural designs for matching supply and demand stems from the following three feedback cycles:

- Leagility-oriented cycle
- Resilience-oriented cycle
- Survival-oriented cycle

4.5.2 Reconfigurable Supply Chain

The Reconfigurable SC framework (RSC) has been developed by Dolgui et al. (2020b). The RSC framework is a part of the realization of the VSC framework. As summarized in Table 4.7, RSC is comprehensively framed into the coordinates of Costs, Responsiveness, Adaptation, Flexibility, Data, and Sustainability. Such integration is not available in the individual SC concepts.

It can be observed from Table 4.7 that the RSC comprehensively combines elements driven from lean, agile, resilient, sustainable, and digital SCs. One objective of such a combination is to find ways to resolve the existing trade-offs. For example, efficiency and flexibility or efficiency and resilience can conflict with each other at times. If we build an excessive manufacturing capacity or hold more inventory (thereby increasing flexibility), we naturally incur more costs. The RSC aims at uncovering the potentials of resolving such conflicts through creating an integral view of the RSC. Examples include using order penetration points in SCs by both reducing inventory and increasing flexibility, multistage hub logistics systems, which increase flexibility, reducing transportation costs and increasing resilience (i.e., so-called LCN, or low-certainty-need, SCs), or using digital technologies to build supplier collaboration portals and use them for early warning signals, advanced

Table 4.7 The RSC combines features of leagile, resilient, digital, and sustainable SCs (Dolgui et al. 2020a, b)

	Cost-efficiency	Response to demand uncertainty	Structural adaptation	Process flexibility	Response to supply uncertainty	Data-driven	Sustainable development
Lean SC	x						
Agile SC		x		x			
Resilient SC			x		x		
Sustainable SC							x
Digital SC		x		x		x	
Reconfigurable SC	**x**	**x**	**x**	**x**	**x**	**x**	**x**

selling notices, and supply monitoring to increase resilience, inventory costs, and responsiveness. In this perspective, the RSC is a macro-framework emphasizing reconfigurability as the central issue of *viability-oriented value creation*.

▶ **Practical Insights** For example, luxury goods manufacturers have wholly transformed their operations to manufacture urgently needed items during the COVID-19 virus outbreak in 2020. LVMH, L'Oreal, and Coty have all repurposed their perfume and hair gel factories to produce hand sanitizer. Giorgio Armani, Burberry, Gucci, and Prada have also altered their designer clothing factories in Italy to produce masks, gloves, and nonsurgical gowns. Similarly, many automotive giants like Ford, Tesla, Mahindra Group, and Suzuki have shifted their production from automobiles to highly proprietary ventilators and hospital beds by collaborating with local manufacturers (Schatteman et al. 2020). Thus, reconfigurable supply chains play a critical role in rapid raw material sourcing, product design, development, testing, and distribution. Furthermore, some companies have resolved parts shortages for life-saving ventilators and masks by using 3D printers. This has triggered the role of new production technologies for additive manufacturing in spare parts supply chains.

4.5.3 Intertwined Supply Networks

An SC intertwining can be observed when firms are involved with different SCs at the same time. For example, Apple and Samsung play the role of both suppliers and competing focal firms in two different but intersecting SCs. Such an intertwining can also be encountered in industrial symbiosis as well as in circular and sharing economies. For example, a symbiosis of commercial and humanitarian logistics can exist when several business and humanitarian SCs share warehouse facilities. As seen during the COVID-19 pandemic, suppliers in the automotive sector can simultaneously become producers of valves for respirators as well (Ivanov and Dolgui 2020).

Ivanov and Dolgui (2020) introduced the new term, "*Intertwined Supply Network*." *An intertwined supply network* (ISN) is an "entirety of interconnected supply chains (SC), which, in their integrity, secure the provision of society and markets with goods and services" (Fig. 4.15).

In contrast to linearly directed SCs or supply networks (cf. Fig. 4.15) with static structures, the firms in the ISNs may exhibit multiple behaviors in buyer–supplier relations (i.e., behavioral dynamics) by acting as buyers and suppliers in interconnected or even competing SCs simultaneously. From the position of viability, the ISNs as a whole provide services to society (e.g., food service, mobility service, or communication service), which are required to ensure long-term survival. The analysis of survivability at the level of the ISN requires consideration at the same large

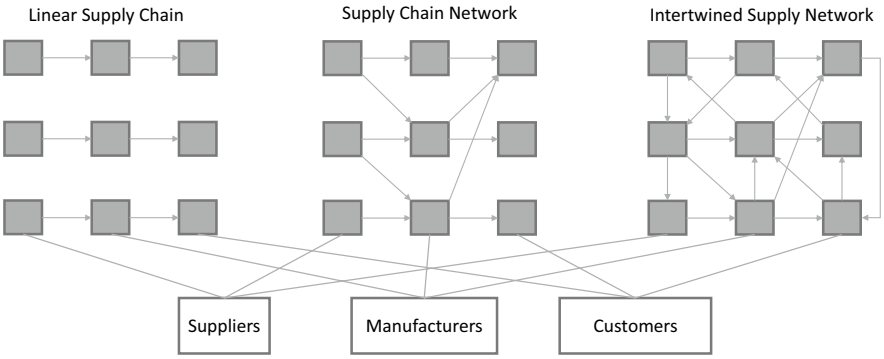


Fig. 4.15 Different types of supply networks (based on Ivanov and Dolgui 2020)

scale as the resilience of individual SCs. The example of the COVID-19 outbreak clearly shows the necessity of this new perspective.

4.6 Key Points

An SC strategy is one of the most important strategies in an enterprise. It should correspond to competitive and operations strategy. Such an alignment is called "strategic fit." SC strategies can be based on costs, agility, innovation, or quality. Frequently, hybrid strategies are very successful.

One of the crucial issues in SCM is SC coordination. Failed coordination in the SC can lead to the bullwhip effect. It can be prevented with the help of information coordination, sharing data about sales, demand forecasts, and inventory. VMI and CPFR are examples of coordination strategies. Sophisticated SC contracting can also help to ensure better SC coordination.

Finally, sustainability and resilience have become more and more important in SCM. The bottom-line "economy–ecology–society" has a direct impact on the future development of SCM. Resilient SCs are expected to continue their execution even if severe disruptions occur and parts of the SC are destroyed. An extended resilience perspective is viability which is based on adaptation of SCs to super disruptions. Viable SC model and Reconfigurable SC frameworks as well as the concept of the Intertwined Supply Network complement the resilience theory.

Bibliography

Aldrighetti R, Battini D, Ivanov D, Zennaro I (2021) Costs of resilience and disruptions in supply chain network design models: a review and future research directions. Int J Prod Econ 235:108103

Allaoui H, Guo YN, Sarkis J (2019) Decision support for collaboration planning in sustainable supply chains. J Clean Prod 229:761–774

Altay N, Gunasekaran A, Dubey R, Childe SJ (2018) Agility and resilience as antecedents of supply chain performance under moderating effects of organizational culture within humanitarian setting: a dynamic capability view. Prod Plan Control 29(14):1158–1174
Basole RC, Bellamy MA (2014) Supply network structure, visibility, and risk diffusion: a computational approach. Decis Sci 45(4):1–49
Behzadi G, O'Sullivan MJ, Olsen TL (2020) On metrics for supply chain resilience. Eur J Oper Res 287(1):145–158
Blackhurst J, Dunn KS, Craighead CW (2011) An empirically derived framework of global supply resiliency. J Bus Logist 32(4):374–391
Brandenburg M, Rebs T (2015) Sustainable supply chain management: a modeling perspective. Ann Oper Res 229:213–252
Brandenburg M, Gruchmann T, Oelze N (2019) Sustainable supply chain management—a conceptual framework and future research perspectives. Sustainability 11(24):7239
Brandon-Jones E, Squire B, Autry CW, Petersen KJ (2014) A contingent resource-based perspective of supply chain resilience and robustness. J Supply Chain Manag 50(3):55–73
Burke GJ, Carrillo JE, Vakharia AJ (2007) Single versus multiple supplier sourcing strategies. Eur J Oper Res 182(1):95–112
Carter C, Rogers D (2008) A framework of sustainable supply chain management: moving toward new theory. Int J Phys Distrib Logist Manag 38:360–387
Chen HY, Das A, Ivanov D (2019) Building resilience and managing post-disruption supply chain recovery: lessons from the information and communication technology industry. Int J Inf Manag 49:330–342
Chopra S (2019) Supply chain management. Strategy, planning and operation, 7th edn. Pearson, Harlow
Chopra S, Sodhi MS (2014) Reducing the risk of supply chain disruptions. MIT Sloan Manag Rev 55(3):73–80
Christopher M (2000a) The agile supply chain: competing in volatile markets. Ind Mark Manag 29(1):37–44
Christopher M (2022) Logistics and supply chain management. Creating value-adding networks, 6th edn. FT Press, New Jersey
Cohen S, Roussel J (2013) Strategic supply chain management: the five core disciplines for top performance, 2nd edn. McGraw-Hill, Boston, MA
Costantino N, Dotoli M, Falagario M, Fanti MP, Mangini AM (2012) A model for supply chain management of agile manufacturing supply chains. Int J Prod Econ 135(1):451–457
Das A (2016) An introduction to operations management. CRC Press, Boca Raton, FL
Dekker R, Bloemhof J, Mallidis I (2012) Operations research for green logistics – an overview of aspects, issues, contributions and challenges. Eur J Oper Res 219:671–679
Demirel G, MacCarthy BL, Ritterskamp D, Champneys A, Gross T (2019) Identifying dynamical instabilities in supply networks using generalized modeling. J Oper Manag 65(2):133–159
Dolgui A, Ivanov D, Sokolov B (2018) Ripple effect in the supply chain: an analysis and recent literature. Int J Prod Res 56(1–2):414–430
Dolgui A, Ivanov D, Rozhkov M (2020a) Does the ripple effect influence the bullwhip effect? An integrated analysis of structural and operational dynamics in the supply chain. Int J Prod Res 58(5):1285–1301
Dolgui A, Ivanov D, Sokolov B (2020b) Reconfigurable supply chain: the X-network. Int J Prod Res 58(13):4138–4163
Dolgui A, Ivanov D (2021) Ripple effect and supply chain disruption management: new trends and research directions. Int J Prod Res 59(1):102–109
Dubey R, Altay N, Gunasekaran A, Blome C, Papadopoulos T, Childe SJ (2018) Supply chain agility, adaptability and alignment: empirical evidence from the Indian auto components industry. Int J Oper Prod Manag 38(1):129–148
Dubey R, Gunasekaran A, Childe SJ, Fosso Wamba S, Roubaud D, Foropon C (2021) Empirical investigation of data analytics capability and organizational flexibility as complements to supply chain resilience. Int J Prod Res 59(1):110–128

Dolgui, A., Ivanov, D., & Simchi-Levi, D. (2025). Stress tests for supply chains: towards resilience and Int J Prod Res 63(9): 3254–3258
Dubey R, Gunasekaran A, Childe SJ, Papadopoulos T, Blome C, Luo Z (2019) Antecedents of resilient supply chains: an empirical study. IEEE Trans Eng Manag 66(1):8–19
Eckstein D, Goellner M, Blome C, Henke M (2015) The performance impact of supply chain agility and supply chain adaptability: the moderating effect of product complexity. Int J Prod Res 53(10):3028–3046
Ernst R, Kamrad B (2000) Evaluation of supply chain structures through modularization and postponement. Eur J Oper Res 124(3):495–510
Fahimnia B, Tang CS, Davarzani H, Sarkis J (2015) Quantitative models for managing supply chain risks: a review. Eur J Oper Res 247(1):1–15
Fahimnia B, Jabarzadeh A (2016) Marrying supply chain sustainability and resilience: a match made in heaven. Trans Res E 91:306–324
Fahimnia B, Jabarzadeh A, Sarkis J (2018) Greening versus resilience: a supply chain design perspective. Transp Res E 119:129–148
Fahimnia B, Sarkis J, Eshragh A (2014) A tradeoff model for green supply chain planning: a leanness-versus-greenness analysis. Omega 54:173–190
Fisher M (1997) What is the right supply chain for your product? Harv Bus Rev 75:105–116
Forrester JW (1961) Industrial dynamics. MIT Press, Cambridge
Ghadge A, Weiß M, Caldwell ND, Wilding R (2019) Managing cyber risk in supply chains: a review and research agenda. Supply Chain Manag Int J 25(2):223–240
Gianesello P, Ivanov D, Battini D (2017) Closed-loop supply chain simulation with disruption considerations: a case-study on tesla. Int J Inventory Res 4(4):257–280
Goldsby TJ, Griffis SE, Roath AS (2006) Modeling lean, agile, and leagile supply chain strategies. J Bus Logist 27(1):57–80
Govindan K, Soleimani H, Kannan D (2015) Reverse logistics and closed-loop supply chain: a comprehensive review to explore the future. Eur J Oper Res 240(3):603–626
Guide VDR Jr, Van Wassenhove LN (2009) OR FORUM – the evolution of closed-loop supply chain research. Oper Res 57(1):10–18
Guide VDR, Wassenhove LNV (2009) The evolution of closed-loop supply chain research. Oper Res 57(1):10–18
Gunasekaran A, Laib K-H, Cheng TCE (2008a) Responsive supply chain: a competitive strategy in a networked economy. Omega 36(4):549–564
Gupta V, Ivanov D, Choi T-M (2021) Competitive pricing of substitute products under supply disruption. Omega 101:102279
Haasis H-D, Kreowski H-J, Scholz-Reiter B (eds) (2008) Dynamics in logistics. Springer, Berlin
He J, Alavifard F, Ivanov D, Jahani H (2019) A real-option approach to mitigate disruption risk in the supply chain. Omega 88:133–149
Heizer J, Render B, Munson C (2023) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow
Hendricks KB, Singhal VR (2005) Association between supply chain glitches and operating performance. Manag Sci 51(5):695–711
Ho W, Zheng T, Yildiz H, Talluri S (2015) Supply chain risk management: a literature review. Int J Prod Res 53(16):5031–5069
Hosseini S, Ivanov D, Dolgui A (2019) Review of quantitative methods for supply chain resilience analysis. Transp Res E 125:285–307
Hosseini S, Ivanov D, Dolgui A (2020) Ripple effect modeling of supplier disruption: integrated Markov chain and dynamic Bayesian network approach. Int J Prod Res 58(11):3284–3303
Hosseini S, Ivanov D, Blackhurst J (2022) Conceptualization and measurement of supply chain resilience in an open-system context. IEEE Trans Eng Manag 69(6):3111–3126
Ivanov D (2025) Introduction to Supply Chain Resilience and Viability, 2nd edn. Springer Nature, Cham
Ivanov D (2022a) Lean resilience: AURA (Active usage of resilience assets) Framework for post-COVID-19 supply chain management. Int J Logis Manag 33(4):1196–1217

Ivanov D (2021) Supply chain viability and the COVID-19 pandemic: a conceptual and formal generalisation of four major adaptation strategies. Int J Prod Res 59(12):3535–3552
Ivanov D (2018a) Structural dynamics and resilience in supply chain risk management. Springer, New York
Ivanov D, Dolgui A, Sokolov B, Ivanova M (2017b) Literature review on disruption recovery in the supply chain. Int J Prod Res 55(20):6158–6174
Ivanov D, Pavlov A, Pavlov D, Sokolov B (2017a) Minimization of disruption-related return flows in the supply chain. Int J Prod Econ 183:503–513
Ivanov D, Rozhkov M (2020) Coordination of production and ordering policies under capacity disruption and product write-off risk: an analytical study with real-data based simulations of a fast moving consumer goods company. Ann Oper Res 291(1–2):387–407
Ivanov D, Sokolov B (2010) Adaptive supply chain management. Springer, London
Ivanov D, Sokolov B (2013) Control and system-theoretic identification of the supply chain dynamics domain for planning, analysis, and adaptation of performance under uncertainty. Eur J Oper Res 224(2):313–323
Ivanov D, Sokolov B, Dolgui A (2014a) The ripple effect in supply chains: trade-off 'efficiency-flexibility-resilience' in disruption management. Int J Prod Res 52(7):2154–2172
Ivanov D, Sokolov B, Pavlov A (2014b) Optimal distribution (re)planning in a centralized multistage network under conditions of ripple effect and structure dynamics. Eur J Oper Res 237(2):758–770
Ivanov D (2017) Simulation-based ripple effect modelling in the supply chain. Int J Prod Res 55(7):2083–2101
Ivanov D (2018b) Revealing interfaces of supply chain resilience and sustainability: a simulation study. Int J Prod Res 56(10):3507–3523
Ivanov D (2019) Disruption tails and revival policies: a simulation analysis of supply chain design and production-ordering systems in the recovery and post-disruption periods. Comput Ind Eng 127:558–570
Ivanov D (2020) Predicting the impact of epidemic outbreaks on the global supply chains: a simulation-based analysis on the example of coronavirus (COVID-19 / SARS-CoV-2) case. Trans Res E 136:101922
Ivanov D, Dolgui A (2021) OR-methods for coping with the ripple effect in supply chains during COVID-19 pandemic: managerial insights and research implications. Int J Prod Econ 232:107921
Ivanov D, Dolgui A (2020) Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability. A position paper motivated by COVID-19 outbreak. Int J Prod Res 58(10):2904–2915
Ivanov D, Dolgui A (2019) Low-certainty-need (LCN) supply chains: a new perspective in managing disruption risks and resilience. Int J Prod Res 57(15–16):5119–5136
Ivanov D (2022b) Viable supply chain model: integrating agility, resilience and sustainability perspectives. Lessons from and thinking beyond the COVID-19 pandemic. Ann Oper Res 319:1411–1431
Ivanov D, Das A (2020) Coronavirus (COVID-19 / SARS-CoV-2) and supply chain resilience: a research note. Int J Integr Supply Manag 13(1):90–102
Ivanov D (2025a) When is the supply chain resilient? Customer and operational perspectives. Int J Prod Res:1–16. https://doi.org/10.1080/00207543.2025.2454331
Ivanov D (2025b) Comparative analysis of product and network supply chain resilience. Int Trans Oper Res. https://doi.org/10.1111/itor.13612
Ivanov D, Dolgui A, Sokolov B (eds) (2025). Handbook of ripple effects in the supply chain, 2nd Edn. Springer, New York
Ivanov D, Dolgui A (2022) The shortage economy and its implications for supply chain and operations management. Int J Prod Res 60(24):7141–7154
Ivanov D (2024) Two views of supply chain resilience. Int J Prod Res 62(11):4031–4045

Ivanov D, Dolgui A, Blackhurst J, Choi TM (2023) Toward supply chain viability theory: from lessons learned through COVID-19 pandemic to viable ecosystems. Int J Prod Res 61(8):2402–2415

Ivanov D (2023) The Industry 5.0 framework: viability-based integration of the resilience, sustainability, and human-centricity perspectives. Int J Prod Res 61(5):1683–1695

Ivanov D, Dolgui A (2025) Tariff shocks, ripple effect, and deep uncertainty in supply chains: we are entering a turbulence zone, please fasten your seatbelts. Int J Prod Res. https://doi.org/10.1080/00207543.2025.2520598

Ivanov D (2025) Supply chain stress testing for tariff shocks and trade conflicts: methods, models, and reciprocal influence of operations and economics. IEEE Transactions on Engineering Management. https://doi.org/10.1109/TEM.2025.3599711

Ivanov D (2025c) No risk, no fun? A bioinspired adaptation-based framework for supply chain resilience in Industry 5.0. Int J Prod Res. https://doi.org/10.1080/00207543.2025.2496962

Jeihoonian M, Zanjani MK, Gendreau M (2017) Closed-loop supply chain network design under uncertain quality status: case of durable products. Int J Prod Econ 183B:470–486

Kleindorfer PR, Singhal K, van Wassenhove LN (2005) Sustainable operations management. Prod Oper Manag 14(4):482–492

Knemeyer AM, Zinn W, Eroglu C (2009) Proactive planning for catastrophic events in supply chains. J Oper Manag 27(2):141–153

Lee HL, Padmanabhan V, Whang S (1997) Information distortion in a supply chain: the bullwhip effect. Manag Sci 43(4):546–558

Li Y, Zobel CW (2020) Exploring supply chain network resilience in the presence of the ripple effect. Int J Prod Econ 228:107693

Li Y, Chen K, Collignon S, Ivanov D (2021) Ripple effect in the supply chain network: forward and backward disruption propagation, network health and firm vulnerability. Eur J Oper 291(3):1117–1131

Liberatore F, Scaparra MP, Daskin MS (2012) Hedging against disruptions with ripple effects in location analysis. Omega 40(2012):21–30

Lim MK, Bassamboo A, Chopra S, Daskin MS (2013) Facility location decisions with random disruptions and imperfect estimation. Manuf Serv Oper Manage 15(2):239–249

Losada C, Scaparra MP, O'Hanley JR (2012) Optimizing system resilience: a facility protection model with recovery time. Eur J Oper Res 217:519–530

MacCarthy BL, Blome C, Olhager J, Srai JS, Zhao X (2016) Supply chain evolution – theory, concepts and science. Int J Oper Prod Manag 36(12):1696–1718

Mehmood S, Nazir S, Fan J, Nazir Z (2024) Achieving supply chain sustainability: enhancing supply chain resilience, organizational performance, innovation and information sharing: empirical evidence from Chinese SMEs, Modern Supply Chain Research and Applications. https://doi.org/10.1108/MSCRA-01-2024-0002

Meisel F, Bierwirth C (2014) The design of make-to-order supply networks under uncertainties using simulation and optimization. Int J Prod Res 52(22):6590–6607

Panetto H, Iung B, Ivanov D, Weichhart G, Wang X (2019) Challenges for the cyber-physical manufacturing enterprises of the future. Annu Rev Control 47:200–213

Papadopoulos T, Gunasekaran A, Dubey R, Altay N, Childe SJ, Wamba SF (2017) The role of big data in explaining disaster resilience in supply chains for sustainability. J Clean Prod 142(2):1108–1118

Paul SK, Sarker R, Essam D (2017) A quantitative model for disruption mitigation in a supply chain. Eur J Oper Res 257(3):881–895

Pavlov A, Ivanov D, Pavlov D, Slinko A (2019) Optimization of network redundancy and contingency planning in sustainable and resilient supply chain resource management under conditions of structural dynamics. Ann Oper Res:1–30. https://doi.org/10.1007/s10479-019-03182-6

Pavlov A, Ivanov D, Werner F, Dolgui A, Sokolov B (2022) Integrated detection of disruption scenarios, the ripple effect dispersal and recovery paths in supply chains. Ann Oper Res 319:609–631

Pettit TL, Croxton KL, Fiksel J (2013) Ensuring supply chain resilience: development and implementation of an assessment tool. J Bus Logist 34(1):46–76
Pournader M, Kach A, Talluri S (2020) A review of the existing and emerging topics in supply chain risk management literature. Decis Sci 51(4):867–919
Queiroz MM, Ivanov D, Dolgui A, Fosso WS (2022) Impacts of epidemic outbreaks on supply chains: mapping a research agenda amid the COVID-19 pandemic through a structured literature review. Ann Oper Res 319:1159–1196
Ruel S, El Baz J, Ivanov D, Das A (2024) Supply chain viability: conceptualization, measurement, and nomological validation. Ann Oper Res 335:1107–1136
Sawik T (2016) On the risk-averse optimization of service level in a supply chain under disruption risks. Int J Prod Res 54(1):98–113
Sawik T (2020) Supply chain disruption management, 2nd edn. Springer, New York
Schmitt AJ, Singh M (2012) A quantitative analysis of disruption risk in a multi-echelon supply chain. Int J Prod Econ 139(1):23–32
Schmitt AJ, Sun SA, Snyder LV, Shen ZJM (2015) Centralization versus decentralization: risk pooling, risk diversification, and supply chain disruptions. Omega 52:201–212
Seuring S, Müller M (2008) From a literature review to a conceptual framework for sustainable supply chain management. J Clean Prod 16:1699–1710
Simchi-Levi D, Schmidt W, Wei Y (2014) From superstorms to factory fires: managing unpredictable supply chain disruptions. Harv Bus Rev 92(1–2):96
Snyder LV, Zümbül A, Peng P, Ying R, Schmitt AJ, Sinsoysal B (2016) OR/MS models for supply chain disruptions: a review. IIE Trans 48(2):89–109
Sokolov B, Ivanov D, Dolgui A, Pavlov A (2016) Structural analysis of the ripple effect in the supply chain. Int J Prod Res 54(1):152–169
Sterman JD (2000) Business dynamics: systems thinking and modeling for complex world. McGraw-Hill, Boston
Tang CS (2006) Perspectives in supply chain risk management. Int J Prod Econ 103:451–488
Tang CS (2007) Robust strategies for mitigating supply chain disruptions. Int J Log Res Appl 9(1):33–45
Tang C, Zhou S (2012) Research advances in environmentally and socially sustainable operations. Eur J Oper Res 223:585–594
Vrijhoef R, Koskela L (2000) The four roles of supply chain management in construction. Eur J Purch Supply Manage 6(3–4):169–178
Wilson MC (2007) The impact of transportation disruptions on supply chain performance. Transp Res E Logist Transp Rev 43:295–320
Zhang B, Mohammad J (2024) The effects of sustainability innovation and supply chain resilience on sustainability performance: Evidence from China's cold chain logistics industry. Cogent Bus Manag 11(1). https://doi.org/10.1080/23311975.2024.2353222

References for Sect. 4.1

Chopra S, Meindl P (2015) Supply chain management: strategy, planning, and operations, 6th edn. Pearson, London
Christopher M (2000b) The agile supply chain. Ind Mark Manag 29:37–44
Fast Retailing (2013) UNIQLO business strategy. http://www.fastretailing.com/eng/group/strategy/tactics.html. Accessed 2 Mar 2014
Ferdows K (2003) New world manufacturing order. Ind Eng 35(2):29–31
Gunasekaran A, Lai K, Cheng TCE (2008b) Responsive supply chain: a competitive strategy in a networked economy. Omega 36:549–564
Heizer J, Render B (2011) Operations management, 10th edn. Pearson, London
Inditex (2014) Who we are. http://www.inditex.com/en/who_we_are/our_group. Accessed 20 Feb 2014

Makioka H, Biragnet JP, Booker M (2009) Learning from the recession, the Japanese way. Business Week Online 16:18
Primark (2014) Über uns. http://www.primark.de/uber_uns/. Accessed 3 Mar 2014
Slack N, Brandon-Jones A, Johnston R (2011) Operations management, 1st edn. Pearson Education, London

References for Sect. 4.2

Statista (2021). https://www.statista.com/chart/11067/how-the-iphones-price-developed/. Accessed 11 Jan 2021

References for Sect. 4.4.1

European Logistics Association (2010). http://www.elabestlog.org
Grüttner A (2010) Revolution im Einzelhandel – wenn der Kunde zum "Boss" wird. Handelsblatt. www.handelsblatt.com
Mercadona (2013). https://www.mercadona.es
Renfe (2013). http://www.renfe.com
The Economist (2011) Spanish aisles. The Economist, June. http://www.economist.com

References for Sect. 4.4.2

Simchi-Levi D, Schmidt W, Wei Y, Zhang PY, Combs K, Ge Y, Gusikhin O, Sanders M, Zhang D (2015) Identifying risks and mitigating disruptions in the automotive supply chain. Interfaces 45(5):375–390

References for Sect. 4.5.3

Schatteman O, Woodhouse D, Terino J (2020) Amid crisis and disruption, supply networks designed for low-cost and minimal inventory pose a major risk, 27 April. [Online]. https://www.bain.com/insights/supply-chain-lessons-from-covid-19/. Accessed 18 June 2020

Sourcing Strategy

5

Learning Objectives for This Chapter

- Role of purchasing, procurement, and sourcing in SCOM
- Basic elements of the sourcing process
- Make-or-buy vs. outsourcing
- Organizational issues in sourcing
- Sourcing strategies according to the number of suppliers, geographical supplier distribution, and sourcing principles
- Methods of spend analysis and supplier selection
- Elements of supplier relationship management (SRM)

5.1 Introductory Case Study: Sustainable Supplier Management at Samsung

Based in South Korea, Samsung is a diversified multinational corporation. The company's largest division is Samsung Electronics Co., which primarily focuses on developing and creating electronic devices and products. Samsung Electronics operates worldwide in North and South America, Europe, Asia, Middle East Asia, and Africa. Moreover, the company primarily offers consumer electronics products,

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_5) contains supplementary material, which is available to authorized users.
Find additional case studies, Excel spreadsheet templates, and video streams in the E-Supplement to this book on www.global-supply-chain-management.de!
Supplementary InformationThe online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_5.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_5

such as smartphones, tablets, televisions, headphones, laptops, printers, and household appliances, including refrigerators, ovens, microwaves, and dishwashers.

Offering a wide variety of products and services, Samsung operates one of the largest supply chains in the world. However, managing this vast and *complex network of suppliers* increases the overall *coordination complexity*.

By shifting the focus to the supplier base, Samsung was able to implement a highly reliable and flexible supply chain, which increased both supplier and customer satisfaction significantly. By developing long-term supplier relationships that are built on trust, fairness, and transparency, the company and its suppliers are able to gain competitive advantages and reduce risks (Samsung 2019). To develop a supplier base that complies with continuously increasing expectations, requirements, and standards regarding sustainable aspects, Samsung vets its suppliers thoroughly using a strict *selection strategy*. The company evaluates a broad tier of suppliers based on environmental and human factors, such as health, safety, and labor. After classifying a suitable number of suppliers that fulfill the requirements, Samsung conducts on-site inspections, emphasizing regulatory and ethical compliance issues and operational aspects. Accordingly, Samsung is able to identify an appropriate and outstanding supplier base that not only saves money in the further development process but also helps the company gain a competitive advantage from the start. Another important tool for the supplier selection strategy is the company's International Procurement Center (IPC), which identifies excellent suppliers in strategically important regions.

After the selection process, the company focuses on developing *long-term relationships* by implementing the first pillar of sustainable supplier development, which is Supplier Operations. Samsung supports both new and existing suppliers in assessing and improving their performances regularly. Moreover, Samsung aims to increase its transparency by implementing a shared procurement portal called secbuy.com that enables suppliers and involved organizations to suggest ideas and proposals anytime. Due to Samsung's broad and complex supplier network, reaching and assessing each supplier can be difficult. However, by implementing shared and connected systems, such as secbuy.com, suppliers are linked directly to the whole network and can not only share problems but also suggest new ideas and concepts.

Supplier selection, as well as *supplier development and management*, can be a significant competitive advantage in the market. Samsung is focusing not only on its primary suppliers, which are especially important for the company's operations, but also on smaller second- and third-tier suppliers. As such, the whole supplier base benefits from Samsung's customer base, expertise, global network, and leadership. Since supply chains have become increasingly global, organizations also need to consider human rights aspects when planning their supply chain strategies. Samsung pursues a sustainable Supplier Work Environment committed to international human rights agreements and follows the code of conduct set by the Responsible Business Alliance (RBA). In particular, Samsung implemented its own Supplier Code of Conduct, which is customized to the company's specific operations and supplier base. Furthermore, Samsung monitors its suppliers regularly to ensure

compliance with all requirements and standards from the Electronic Industry Citizenship Coalition (EICC) code of conduct. The importance of sustainable and ethical business practices is essential for the efficiency of an organization, quality of its products and services, and its social image.

Samsung also established a so-called Eco-partner certification, which follows a specific set of rules and regulations regarding the substances used in products and suppliers' environmental quality management. As a result, the company and its suppliers are able to continuously improve operations and provide customers with sustainable, high-quality products. This improves business operations while also creating a *sustainable global image* that has long-term positive impacts on the company's performance.

Discussion

- Why is sustainable sourcing important?
- How can a firm achieve a competitive advantage through sustainable sourcing?
- What are the sustainability elements that go beyond merely economic criteria?

5.2 Sourcing Process and Principles

5.2.1 Procurement, Purchasing, and Sourcing

This section deals with the definition of procurement, purchasing, and sourcing, as there are various explanations and definitions of these terms in use. As Mangan et al. (2008, p. 76) pointed out, "the term sourcing is often used interchangeably with the terms purchasing and procurement, but to be more specific, sourcing relates to identifying and working with appropriate suppliers, while purchasing relates to the specific functions associated with the actual buying of goods and services from such suppliers." Sourcing needs to be understood as the entire "set of business processes required to purchase goods and services (Chopra and Meindl 2013, p. 66)." In line with Jacobs and Chase (2021, p. 449), sourcing is "a process, suitable for procuring products that are strategically important to the firm" and strategic sourcing is related to "the development and management of supplier relationships to acquire goods and services in a way that aids in achieving the needs of a business."

According to Lysons and Farrington (2012, p. 6), purchasing is "the process of procuring the proper requirement, at the time needed, for the lowest possible cost from a reliable source." Purchasing deals mostly with operational activities and is related to transactional ordering processes. Grant (2012, p. 270) elaborates that purchasing is related to "the functions associated with the actual buying of goods and services required by a company."

"Procurement is the process of acquiring goods, works and services, covering both acquisitions from third parties and from in-house providers. The process spans the whole lifecycle from identification of needs, through to the end of the useful life of an asset. It involves options appraisal and the critical "make or buy" decision"

(Lysons and Farrington 2012, p. 6). The CIPS (Chartered Institute of Procurement and Supply[1]) defines procurement as "the act of obtaining something, whether tangible or intangible, such as a product or service" and that purchasing is "the processes concerned with acquiring goods and services, including payment of invoices—it is part of the wider procurement process."

Summarizing the above definitions, one could consider procurement more as an organizational unit, in which the strategic processes related to sourcing activities are conducted as well as the operational tasks of purchasing are performed. Note that the above-mentioned terms are often used interchangeably or in different ways. The term "supply management" is also applied to strategic processes. In line with the SCOR Model, the term "sourcing" will be used predominantly within this book, although there are elaborated, specific differences. Table 5.1 summarizes the focus areas of sourcing, purchasing, and procurement.

The sourcing activities range from selecting suppliers to drawing up contracts, involvement in product design and collaboration, and evaluation of supplier performance. Broadly speaking, sourcing is the process of establishing and managing supplier relationships in the SC. In the narrow sense, it is related to the activities and

Table 5.1 Procurement, sourcing, and purchasing

Procurement might be considered as the organizational unit within a firm. It is the function that performs the activities to make the goods, materials, articles, products, or services available inside the company. To simplify it, one could conclude that procurement is the organizational body, in which strategic thinking (sourcing) and active work (purchasing) takes place.	
Sourcing occurs before the products are bought (i.e., purchased). That means it has a long-term, more strategic orientation and deals with identifying the best source (i.e., supplier or partner). The strategic evaluation also involves comparing some sourcing-related criteria. Besides the classical criteria of cost (e.g., price negotiations, discounts, or terms and conditions), quality, delivery time, flexibility, capacity buffers, and sustainability factors (e.g., ethics, fair payment, or environmental criteria) will be considered to support the strategic sourcing decision. The purpose is to identify a supplier for a high-quality product or service at a reasonable price while considering sustainability aspects.	*Purchasing* follows strategic and commercial negotiations and covers the operational processes of making parts, products, or services available for the company or requesting institution. In that sense, purchasing means buy the item or article from the approved source (the approved supplier/partner). Purchasing covers the active process elements, such as placing orders, contacting suppliers for replenishment, and communicating with them (e.g., regarding back orders or delays) until the products or services have been received. Checking the invoices against the initially placed orders and—once they have been verified—submitting the invoices to the accounts payable department also belong to purchasing.
Mainly strategic processes	Mainly operational processes
Often considered as "Source-to-Contract" = S2C	Often considered as "Purchase-to-Pay" = P2P

[1] https://www.cips.org/knowledge/glossary-of-terms/ (accessed 21 Dec 2020).

processes that provide the enterprise with materials, services, capital equipment, means of production, tools, and supplies for work from external suppliers or partners. In practice, sourcing integrates the strategic and operative decision-making levels and coordinates supply strategy and purchasing activities. It is a fundamental element of SCOM as it is the linking process that provides the organization with the inputs required to create their products or services. Increasingly, operational and repetitive purchasing activities are supported by software solutions, referred to as robotic process automation (RPA, see Chap. 16), with the purpose of increasing efficiency and reducing errors. A practical approach on how to achieve operational excellence in the purchase-to-pay (P2P) process has been described by Knotte et al. (2020).

Sourcing is an essential activity in the SC. The purchased parts and materials can account for over 60% of the cost of finished goods; for retail companies within the SC, this can be as high as 90% (see Stevenson 2018, p. 662). The quality of purchased material, costs of goods bought, delivery of goods or services on time, supplier management, and supplier relationships are some of the factors that significantly impact SC performance and place particular demands on procurement managers. As sourcing is a vital and broad subject, this chapter focuses on the principles, processes, and organizational factors around sourcing strategies in SCOM and related activities.

5.2.2 Sourcing Process

In general, the *sourcing process* connects the supply and demand sides, so the so-called sourcing objects will be made available to the requestor. This is illustrated in a simplified way in Fig. 5.1, which depicts the high-level linkage between the requesting and the supplying side. In order to better understand the details behind these processes, it is important to develop a clear understanding of the content and objectives of sourcing activities, which include the following elements:

- *Determination of Material Requirements*: The type, composition, configuration, quality, volume, location, and timing for the delivery of the sourcing objects.
- *Order Management*: Determination of order volumes, frequencies, times, and specification of logistical conditions. Also, the supervision of accurate deliveries, goods reception, invoice control, and approval can belong to the generic sourcing process.
- *Supplier Base Research, Observation, and Analysis*: Analysis and evaluation of the supplier base, assessment of potential new partnerships, elaboration of (strategic sourcing) recommendations, preparation of negotiations, and contracts.
- *Make or Buy Decisions*: Identification of internal/external value-adding scope depending on the core competencies, but also the comparison of capacities, lead times, and costs.
- *Supplier Management*: Auditing suppliers, running performance management assessments, e.g., regarding on-time deliveries, quality, reliability, and flexibility.

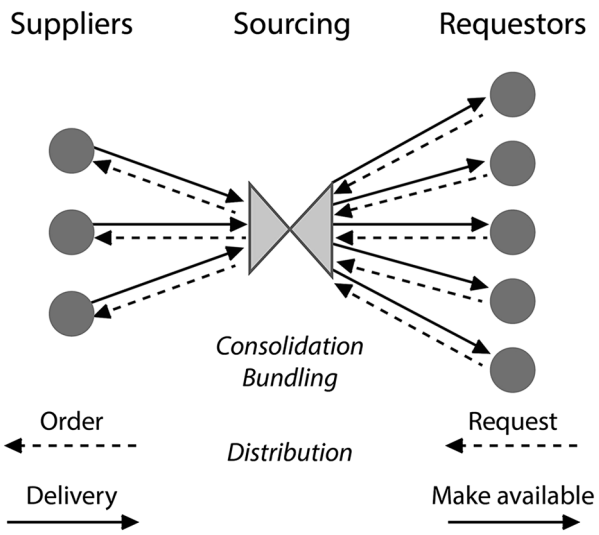


Fig. 5.1 Generic sourcing process (based on Kummer et al. 2018)

This is also related to supplier development or the initiation of a substitution of suppliers.

Sourcing objects can be product and non-product related. For example, in a restaurant, there are SCs both for food (i.e., product-related sourcing) and for kitchenware (i.e., non-product-related sourcing). Table 5.2 shows a selection of related examples, but of course, the table could be extended by additional items.

The examples in Table 5.2 also indicate that a company deals with a large number of different suppliers on a daily basis and is also in regular contact with potential new suppliers. To make this a bit more specific, we will show an example of companies from the heavy investment goods industry (like turbines, planes, and trains), in which approximately 35,000 suppliers are documented in the SRM System.

It is important to mention that not all of these suppliers are equally important. The company will deal with some suppliers on a very regular basis and with others more occasionally. Out of these 35,000 suppliers, for example, perhaps 500–600 form the core supplier base.

5.2.3 Make-or-Buy and Outsourcing

"*Make or buy*" is a strategic decision determining whether the sourcing objects are internally or externally sourced. Such strategic decisions might also be related to the question of core competencies. Companies might even have their in-house suppliers, which prepare certain modules or systems and present them to the assembly line once they are needed (this could be an exclusive steering wheel that is handmade for one car customer and which is presented to the assembly line). It could also be that

Table 5.2 Examples of sourcing objects

Examples of product-related sourcing objects	Examples of non-product-related sourcing objects
Raw material (e.g., metal plates, wood, stones) Auxiliary supplies (e.g., screws, nuts, bolts, glue) Consumables/operating supplies (e.g., oil/grease, gasoline) Purchased parts (e.g., sparks, valves, muffler parts) Purchased assembled modules (e.g., navigation system, bumper with grill and fog lights) Purchased developed systems (e.g., development and pre-assembly of the complete HVAC System fitted to the front module) Trade goods (e.g., bag for a notebook, DVDs for the camcorder)	Capital-intensive equipment (e.g., cranes, milling machines, warehouse shelves, and equipment) Spare parts for production or maintenance (e.g., drillers, cutting disks, machine spare parts, lightbulbs) Office supplies (e.g., paper, pens, staplers) Personnel (e.g., via recruiting firms, hiring consultants, or temporary staff) Information and business-related services (e.g., competitor analysis, planning and conduction of exhibitions, events, business travel bookings) Capital or financial services (obtaining loans, external financial support)

propulsion components for railway manufacturers are pre-assembled by in-house suppliers, or internal wing components are pre-assembled by an aircraft producer.

▶ **Practical Insights** For example, a car manufacturer keeps the creation of the tools they use to produce metal components such as fenders, doors, hoods, etc., in-house in order to avoid characteristics regarding the outward appearance of the car being handed over to competitors in the very early stages of the design phase. This decision to "make" is mainly driven by the desire to avoid imitation of the car design.

Make-or-buy decisions impact the overall level of in-house value creation; respectively, they are closely linked to decisions around production strategy. When we talk about make-or-buy, the sourcing team will need to decide jointly with manufacturing experts whether external purchasing ("buy") might be the economically preferable solution or if the respective part should be produced in-house ("make"). As detailed earlier, a detailed analysis of the TCO (*total costs of ownership*) is highly recommended for make-or-buy decisions. Such a decision will also need to consider competitive factors regarding core competencies or the fear of providing external parties with the knowledge that would offer the opportunity for imitations. In such cases, the in-house "make" will be the preferred choice, even when external sourcing might be considered the economically more attractive alternative.

On the other hand, if an organizational unit's own people have historically performed the relevant value-adding processes and the decision has been made to externalize certain processes (i.e., to buy), so-called outsourcing takes place. This means that *outsourcing* is a result of a make-or-buy decision. Some of the common issues in outsourcing analysis are as follows:

- Lower production cost
- Better use of available resources
- Focus on core competencies
- Cost restructuring
- Reduction of time-to-market
- Risk sharing
- Know-how sharing
- Quality issues
- Flexibility
- Tax benefits

▶ **Practical Insights** False outsourcing decisions may ruin a company. A supplier for a shipbuilding yard decided to outsource a welding process that was placed precisely in the middle of its technological process. However, the order quantity was relatively low (this was the reason for outsourcing), and the new subcontractor always scheduled these orders as unimportant since it was not interested in maintaining a high service level for such an unimportant customer. As a result, the lead time for deliveries at the shipbuilding yards increased from 10 days to 35–40 days. The procurement manager needed to contact the subcontractor frequently and ask them to speed up the orders. In a few months, the shipbuilding yard canceled the agreement. In the case of low machine utilization at a process step, it can be more sensible to in-sourcing additional orders for this machine rather than outsourcing a part of technological processes, especially if it is in the middle of the process.

Make-or-buy is not only connected to pure manufacturing processes. For example, raw materials or single loose parts can be evaluated; it can also be applied to modules or services, such as IT services, transportation logistics, or facility management (e.g., cleaning, maintenance, or security services). Furthermore, mixed solutions between buying and making are also possible to match market demands.

The following selection provides a summary based on Kummer et al. (2018), reflecting on possible parameters that might be considered when a company is deciding whether to "make" or "buy:"

- *Core Competence/Strategic Fit*
  What are the unique capabilities that are very difficult to imitate?
  What are the strategic objectives the company needs to meet?
  Does the focus on lean production require an adaptation of the production depth?
  Is it necessary to buy more in new markets because of local content requirements?
- *Variety and/or Stability on the Demand Side*
  What is the level of demand we can stably meet with our existing capacity, and when is it recommended to increase peak capacity needs by temporary supplementary external sourcing?
- *Production Capacity on the Sourcing Side*

Are there existing contracts (e.g., purchasing agreements or sales orders) that need to be fulfilled and which will impact the make-or-buy decision?

- *Autonomy*
  Do we jeopardize our competitive position by providing external parties with confidential drawings, intellectual property rights, and new research results?
  Is there a risk that we might even create a future competitor by going for a buy decision that will lead to a know-how transfer?
- *Dependency on Suppliers*
  By increasing the level of external purchases, dependency on suppliers and their quality, reliability, and punctuality can significantly influence the organization's reputation: Do we have a strong or weak market position, and how does it look for suppliers?
- *Evaluation of the Cost Structure*
  In the case of "make" decisions, the corresponding costs are linked to, e.g., material and labor costs, salaries, or the depreciation cost of the equipment in use. In the case of "buy," the decision will be related to purchasing, transportation, handling, storage, or transportation costs.
- *Financial Shortage or Need for Capital*
  Especially during financial shortages, the decision to externally source parts, modules, or services is an important factor as it relieves cost pressures.
  In practice, decisions will, in addition to the qualitative criteria, be combined with quantitative methods to substantiate the selection process further.

5.2.4 Organization of Sourcing Processes

The last dimension in managing the sourcing process is related to *organizing sourcing activities.* This dimension is focused on the organizational set-up of the buying party or parties. A company can buy goods individually, but it could also be that a few companies jointly specify the goods that should be sourced from one supplier. This means that these few companies are organizationally creating a sourcing alliance.

In terms of the appropriate terminology, we could summarize that, in contrast to the individual sourcing process, collective sourcing might be a reasonable strategy. Heizer and Render (2013) give as examples small manufacturers of motorboats that formed such alliances in order to aggregate their demand for motorboat engines; or hospitals that ordered clinical supplies in a cooperative, thus benefiting from the effects of larger ordering volume.

In the case of a global player in the telecommunications industry, a company might be in a supplier–customer relationship with approximately 20,000 suppliers in 80 countries. Many different suppliers and different parts exhibit different characteristics: It is important to stress that a company needs to define a sourcing strategy that fits its individual purpose. Sourcing strategies will be considered in the next section.

5.3 Sourcing Strategies

Sourcing strategies can be classified according to three basic features:

- Number of suppliers (single, dual, and multiple sourcing)
- Geographical supplier distribution (local, national, international, and global sourcing)
- Sourcing principles (sourcing on-stock, JIT, particular sourcing)

Let us consider these strategies in detail.

5.3.1 Single vs. Dual and Multiple Sourcing

In this area, the task is to manage the supplier base with the objective of determining the right number of suppliers. By reducing the supplier base, larger volumes can be ordered from one supplier (*single sourcing strategy*) to generate volume bundling (scale) effects. However, dependence on one supplier might be too high of a risk.

▶ **Practical Insights** Focusing on single sourcing provides many efficiency advantages. However, a number of recent disruptions have forced SC managers to rethink this lean sourcing strategy since cost savings can be overwhelmed by disruption impacts (Chopra and Sodhi 2014; Ivanov et al. 2014). Companies that used single-sourcing with suppliers in Japan or Thailand were drastically affected by the tsunami and floods in 2011. Many production factories worldwide were stopped for several months. Intel claimed to have lost $1.0 billion in sales during the fourth quarter of 2011. In 2010, the eruption of a volcano in Iceland interrupted many time-sensitive air shipments in global SCs. A fire in the Phillips Semiconductor plant in Albuquerque, New Mexico, in 2000 caused its major customer, Ericsson, to lose $200 million in potential revenue. Nokia was also affected by this disruption but found alternative supply sources within 3 days. However, this was very costly. Another risk of single sourcing is related to product standardization. If the sole supplier produces items that are used in many different models of a product, the impacts of a disaster can ripple very fast through the SC, as was the case for Toyota in 2011. This is why many companies like Samsung tend to have at least two suppliers, even if the second one provides only 20% of the volume (Sodhi and Lee 2007).

Thus, it might also be a reasonable strategy to cooperate with a second or third source for a part or module. This supplier strategy is in contrast to a single sourcing strategy and is referred to as *dual sourcing* or, with more than two, a *multiple sourcing strategy* which both better balance the global flows of material and provide risk reduction. For example, ZARA produces their trendiest items in Europe, close to highly dynamic and changing demand. Slow-moving items are produced in Turkey

and Asia, since lead time is not crucial and cost reduction can be achieved in manufacturing. Li & Fung Ltd., a contract manufacturing company from Hong Kong, has a variety of supply plans that enable them to shift production among suppliers flexibly.

The discussion above allows us to formulate some critical issues when deciding on single vs. dual or multiple sourcing. They include:

- Volume
- Product variety
- Demand uncertainty
- Lead time importance
- Disruption and other risks
- Transportation costs
- Manufacturing complexity
- Coordination complexity
- Post-sales issues

Some of the *advantages of single sourcing* are as follows:

- Long-term agreements
- Price stability
- Suppliers included in the product development process at a very early stage
- Low transactional costs
- Scale effects

Shortcomings of a single sourcing strategy include:

- High dependency
- Lead time, quality, and service issues
- Lack of collaboration with many suppliers
- Resilience risks due to high dependency in case of bankruptcy, insufficient capacity, strikes, fires, etc.

5.3.2 Local vs. Global Sourcing

Looking at the location of suppliers is another aspect of a sourcing strategy. This is related to the *geographic sourcing area strategy.* There are two extremes, starting from *local sourcing* (limited number of suppliers, but same norms, language, currency, shortest distance, and thus fastest reaction time in case of changes) vs. global sourcing (offering the broadest supplier base, but suffering from, e.g., long distances, different currencies, and norms standards). In between these, there is also national or continental sourcing.

In order to avoid bad decisions that might be made based only on the purchasing price, practice shows that a stringent focus on the TCO (total cost of ownership) approach is highly recommended. Due to shrinking time-to-market and the need for agile management of the SC, the pure advantage of a low purchasing price can

easily be eliminated by the need to order large lot sizes; these may travel around the globe for weeks, resulting in higher safety buffers, inventory, and transportation costs.

▶ **Practical Insights** Of the 17 Louis Vuitton (LV) manufacturing shops for leather goods, 12 are located in France, 3 in Spain, and 2 in the United States. All manufacturing processes and development are handled at a central workshop in Italy. LV uses external manufacturers only to supplement its manufacturing segment and achieve production flexibility in terms of volume. It purchases its materials from suppliers located around the world, with which LV has established partnership relationships. The supplier strategy implemented over the last few years has enabled requirements to be fulfilled in terms of volume, quality, and innovation (Heizer and Render 2013).

For *global sourcing*, items of high volume, steady demand, and low transportation costs are most preferable. However, different *chances and risks* in regard to costs, service, quality, and sustainability should be included in the analysis.

- Costs: labor, taxes, transportation, insurance, transshipment, duties, and transactions
- Quality: bill of materials, quality control, after-sales service, certifications, norms, standards
- Service: on-time delivery, responsiveness, flexibility, technical equipment, image, reliability
- Sustainability: ecological (e.g., transportation distance), political, economic, and social problems

Problems regarding global sourcing decision analysis are presented in Table 5.3. Global sourcing offers access to the broadest available range of suppliers (in contrast to local or national sourcing), and this provides many advantages. At the same time, greater efforts and certain language skills are required to establish relationships with global vendors or partners.

Global sourcing also leads to longer lead times and increased exposure to the ripple effect (Ivanov 2017). Factors such as currency risks or political instability, along with different cultures, norms, or standards, can be very influential and of very high importance.

Table 5.3 shows some advantages and disadvantages of the two extreme geographic sourcing strategies: local vs. global sourcing. Between these two extremes, there are also opportunities for following a national sourcing strategy; this would be related to expanding the sourcing radius, i.e., increasing the number of potential vendors, but continuing to keep risks at a low level. A continental sourcing strategy would further enhance the accessible supplier base but still limit the risk of very long transportation times and large volumes.

Table 5.3 Analysis of global sourcing decisions

	Local sourcing	Global sourcing
Advantages	• Same norms/standards • Easy to reach/short distances • Same culture, same currency, same political climate • Good basis for JIT deliveries • Lower disruption risks for overall SC	• Broadest variety of available vendors • Largest portfolio of products or services • Best opportunities to compare and negotiate with suppliers due to broadest supplier base
Disadvantages	• (Very) limited supplier base or no supplier base • Possibly limited bargaining power of buyer because of limitations on the supplier side	• Longer travel and transportation times, sustainability aspects • Longer response times in the event of changes • Possibly larger lot sizes • Potentially different norms/standards • Different cultures, currencies, and political uncertainty • Ripple effect risks

▶ **Practical Insights** In practice, per commodity and part, a reasonable strategy has to be determined by evaluating the corresponding opportunities and risks. In many cases, a mixed strategy with a combination of local and global sourcing elements can be a good choice. For example, Amazon prefers holding fast-moving items in distribution centers while slow-moving items tend to be stored centrally. Apple localized production in China, but the distribution network is global. Such SC segmentation also helps to reduce disruption risk implications (Chopra and Sodhi 2014).

5.3.3 Just-in-Time

Let us first have a look at the definition of Just-in-Time (JIT) provided by the American Production and Inventory Control Society (Lysons and Farrington 2012, p. 334–5):

> JIT is "a philosophy of manufacturing based on planned elimination of all waste and continuous improvement of productivity. It encompasses the successful execution of all manufacturing activities required to produce a final product from design engineering to delivery and including all stages of conversion from raw material onward. The primary elements include having only the required inventory when needed; to improve quality to zero defects; to reduce lead time by reducing set-up times, queue length and lot sizes; to incrementally revise the operations themselves; and to accomplish these things at minimum cost."

Making what the customer needs, when it is needed, and in quantity needed using the minimum resources of people, materials, and machinery—this is the core idea of JIT. JIT was developed in the 1950s by Toyota in Japan. Along with the development of the Toyota Production System (TPS), JIT became a core element of the lean production concept. It is focused on meeting exact demands following a production-synchronous replenishment mode, which is activated by a customer order and is linked to all predecessing value-adding steps. JIT principle closely connects the functional disciplines of production, procurement/sourcing, logistics, and sales. Furthermore, JIT requires the successful close cooperation of internally and externally involved parties since it is based on the pull principle.

JIT allows the reduction of inventory at the production site, cutting of lead times, and the increasing of productivity and responsiveness. High-value materials with good demand predictability and steady demand can be especially recommended for JIT.

5.4 Supplier Relationship Management

When selecting a sourcing strategy, the following aspects of SRM should be considered in practice (Geissbauer and D'heur 2010) (see Table 5.4).

SRM can be divided into four areas:

- Supplier analysis
- Supplier selection
- Supplier integration
- Supplier development (Fig. 5.2).

Let us consider these areas in detail.

Table 5.4 Sourcing strategy and SRM

Elements of the sourcing strategy	Corresponding sourcing activities
Supplier selection	Composition of an optimized supplier portfolio
Supplier evaluation	Systematic assessment of supplier capabilities and performance
Supplier development and integration	Increase supplier potential and capabilities
Supplier risk management	Prevention, reduction, or elimination of supplier risks
Supplier phase-out	Optimization of supplier portfolio
Supplier relationship management	Sustainable activities to improve supplier performance
Supplier information management	Information systems for administration of supplier data
Supplier audit	Identification of performance and technology level

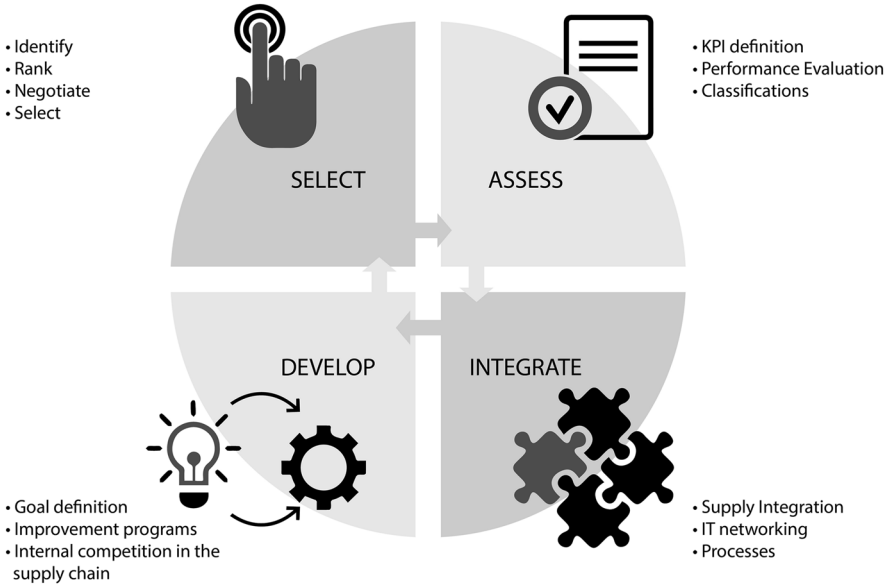


Fig. 5.2 Supplier relationship management functions

5.4.1 Strategic Supplier Analysis

First, the sourcing department is obliged to perform a *spend analysis* in order to analyze which supplier for which sales volume is covered for what type of part or commodity. This means the number of purchased items requires careful analysis to generate a basis for developing improvement scenarios. Such scenarios could be related to adjusting (increasing or reducing) the number of suppliers or to transferring additional responsibilities to the supplier. In general, *the opportunities are assessed* in the first step of the strategic analysis phase.

The *second strategic step* focuses on creating profiles for categories (e.g., paper, metal, rubber or seats, wheels, lamps, etc.) and clustering materials or services in order to identify repetitive sourcing patterns.

In order to develop a sourcing strategy in *step three*, the TCO has to be identified. This is important for developing comparable decision papers that will also be used to support make-or-buy decisions. The result is a condensed overview highlighting which items will be externally sourced compared to the parts or components produced internally.

In *strategic step four*, potential suppliers need to be identified and screened. It might well be that new suppliers are added after they have submitted a promising offer. Based on defined supplier selection criteria, the supplier base will need to be critically evaluated. A set of pre-defined selection criteria captures and clarifies distinct attributes. These help to facilitate and speed up the assessment process (Baily et al. 2021).

When business experts talk about a *spend analysis*, it means understanding how many suppliers there are and the monetary exchange per supplier. Bozarth and Handfield (2013, p. 214) state that a "s*pend analysis is the application of quantitative techniques to purchasing data in an effort to understand spending patterns better and identify opportunities for improvement.*" The spend analysis can be used to answer the following questions:

- What categories of products or services significantly influence company spending?
- How much are we spending with various suppliers?
- What are our spending patterns in different locations?

The main idea behind this analysis is to obtain a good understanding, based on data analysis, about the spending behavior of a company with its suppliers or partners. Depending on the question that needs to be answered, a corresponding path needs to be taken for review of the existing quantitative information. For example, practice shows that a company will spend a lot of their sourcing costs with only a few supply or service partners.

The Pareto analysis is a good solution for analyzing such a case. In other cases, where a large number of data requires careful attention, a regression analysis is also a good approach. The teams elaborating such analyses require good skills and competencies to run such data segregation.

Along with the spend analysis, some other methods are used in practice:

1. *Supplier industry analysis*—Review of the suppliers, their market shares, financial strengths, and geographic coverage to obtain a structured overview of the suppliers for certain goods and services, including an understanding of the role these suppliers play in the SC (Is it a local raw material provider or a globally acting partner who provides sub-assemblies?).
2. *Cost and performance analysis*—Examination of the cost drivers for the sourced goods and services as well as developing a solid key performance indicator (KPI) model. It is recommended that the value-adding steps be analyzed on the supplier side and then that cost drivers and performance drivers/influencers be mapped against the various transformation processes.
3. *Supplier role analysis*—Identification of the roles suppliers play in the design of the SC: which are the suppliers that will play an active role or coordinate parts of the SC vs. vendors that only deliver their goods or services (e.g., diesel fuel refilling) in passive mode.
4. *Business process analysis*—Assessment of the business processes between the purchasing and selling parties. In practice, this analysis will help answer the question of who from the customer side will exchange with whom, how often, and how with the suppliers of the corresponding commodity. This relates to the frequency of data exchange, processes of cooperation, documents to be transferred, and also the tools (e. g. EDI, WebEDI, or supplier portals) that need to be used to facilitate a structured and harmonized exchange of information, so that the business process becomes practically alive.

5. *Business benefit analysis*—Quantification of the saving potential per commodity and usage of these identified benefits for transparent target setting purposes. Furthermore, such a substantiated business benefit analysis will also help senior management to make decisions.
6. *Commodity plan implementation and execution*—Elaboration of a structured plan showing for what type of commodity with which supplier what type of strategy and tool should be implemented by whom from the organization and by when.
7. This means project management principles play a significant role in converting the strategy into a plan with defined tasks and work packages.

Since the commodity sourcing strategy has a huge impact on the entire design of the SC, it is recommended that, in addition to the sourcing specialists, representatives from, e.g., manufacturing, logistics/SC, quality, engineering, and finance should be part of a cross-functional commodity sourcing strategy team. Of course, the individual composition of the team needs to be identified case by case depending on the enterprise, its global footprint, and industry, but overall, a jointly developed strategy that considers the inputs and contributions of multiple relevant functions should be the objective.

5.4.2 Supplier Selection

In this phase, the sourcing team will *create a supplier shortlist* based on the identified supplier selection criteria. Different approaches are possible, but in principle, the task is to assess to which degree the capabilities of the suppliers meet the customer's selection criteria. Either they are completely fulfilled, or to a certain degree fulfilled, or they will not be fulfilled by the supplier. This means in order to identify the most appropriate vendor, commonly developed and accepted vendor selection attributes and processes will need to be defined and applied inside the organization.

The selection of vendors is a multi-criteria problem that includes qualitative and quantitative measures. According to Amid et al. (2011), the relative importance of the criteria, sub-criteria, and, in many cases, the weighting factors are determined by top management and purchasing managers. It is important to emphasize that careful and proper selection of suppliers is a crucial and important element in sourcing activities, because the performance of the overall SC significantly depends on every single party involved.

Once the shortlist is created, detailed contractual discussions and *negotiations* will be initiated. In this *step*, commercial conditions are especially on the agenda, but logistical requirements (e.g., packaging instructions, lot sizes, transportation devices, item identification tags, or delivery notes) are also part of the agreements. When the contractual negotiations are finalized, and the contracts are signed, these specified agreements will become part of the daily operational processes.

In practice, *weighted models* are frequently used to respect priorities or focus on more important criteria. For example, this might be quality, cost, or sustainability. The factors of cost, quality, and delivery performance, as well as capacity, experience, service, technical capability, and financial status, are used in practice. In addition, supplier risk exposure became an important evaluation factor in recent years (Simchi-Levi et al. 2015; Ivanov et al. 2017; Ivanov 2018; Dolgui et al. 2018; Yoon et al. 2018). The consideration of these factors facilitates long-term relationships with possible vendors or partners.

Task 5.1 Supplier Selection

To better understand the principle behind such systematic supplier selection, let us assume we are employees of a company, and it is our task to evaluate two alternative suppliers against the list of supplier selection criteria for our company. In our example, we should have the following supplier selection criteria with the corresponding importance weighting next to it (see Table 5.5). The sourcing department will have created a standard containing the relevant selection criteria. This list of criteria will be used for all suppliers in order to ensure comparability of the supplier selection that has been made in the past and for the selection of future suppliers (Table 5.5).

Table 5.5 Scoring analysis for supplier selection

Criteria	Weight (%)	Scores for supplier A	Supplier A rating	Scores for supplier B[a]	Supplier B rating
Cost of purchased items	20	4	0.80	7	1.40
Quality of purchased items	20	4	0.80	6	1.20
On-time delivery (OTD) performance	15	7	1.05	4	0.60
Sustainability standards compliance	25	6	1.50	8	2.00
Reputation of supplier in the market	10	3	0.30	4	0.40
References from other customers	5	8	0.40	4	0.20
Global footprint of supplier	5	9	0.45	6	0.30
Total	100		5.30		6.10

[a]The scale ranges from 0 points (very poor) up to 10 points (excellent)

Next to the company specific selection criteria are the individual weighting percentages. In total, we will always have a 100% split over the selection criteria. In addition, standard weighting factors will be used in practice to ensure the consistency and reproducibility of supplier selection.

In practice, you will find that the relevant sourcing department has agreed on requirements to assign points for each criterion. This will be multiplied line by line to calculate the overall supplier rating value. In our example, supplier A achieves in total a rating of 5.3 points, and for supplier B the total sum is 6.1 points. We would therefore select supplier B.

As the ratings in the table above are quite close to each other, one might also think about maintaining good relationships with supplier A, possibly for a future evaluation. Of course, it is not one supplier alone that is evaluated by the sourcing department: there are multiple companies that are assessed.

Finally, since the usage of the factor-ranking method for supplier selection is very subjective, a *sensitivity analysis* should be performed in the event of similar evaluations of some suppliers. Such an analysis helps to identify the impact of changes in weights and scores on the overall supplier evaluation.

In practice, supplier evaluation and selection is a complex multi-criteria decision-making (MCDM) problem that provides a framework for comparing and selecting suppliers. The aim of this process is to evaluate suppliers based on some predetermined quantitative and qualitative criteria related with the requirements of the company. Supplier evaluation and selection can be supported by different methods such as Analytical Hierarchy Process (AHP), Data Envelopment Analysis (DEA), Technique for Order Performance by Similarity to Ideal Solution (TOPSIS), Mathematical and Stochastic Programming, and Fuzzy Logic (Talluri et al. 2013; Yoon et al. 2018; Sawik 2020).

5.4.3 Supplier Integration and Development

In principle, the repetitive process of sourcing execution and SC collaboration starts now. The practice has shown that traditional purchase orders tend to be considered inflexible because quantities and delivery dates are fixed (simplified view). In contrast to purchase orders, *scheduling agreements* (flexible agreements) are preferred in business practice, because the exact demand date and quantity will be communicated from the ordering party to the supplier at a later stage. In this context concerning how to conduct sourcing execution and SC collaboration more practically, the usage of IT should also be mentioned. *Supplier collaboration portals* (e.g., SupplyOn, see Chap. 3) enable communication between partnering companies and thus represent an intelligent platform to exchange the necessary information to initiate the corresponding material flows.

Supplier development is focused on improving the supplier's ability to implement production improvements, e.g., by practical problem solving or by establishing continuous improvement processes. On the other hand, it strives to improve the supplier's cost, quality, and delivery by having supplier development teams at the

vendor's facility. As the details indicate, process- and results-oriented approaches are complementary.

Supplier development is related to active consulting support and providing the necessary guidance to identify problems, analyze them, and solve them practically. This might include the temporary provision of specialists or even be linked to financial support. Overall, the purpose is to secure, develop, and maintain a solid and reliable supplier base.

Lysons and Farrington (2012) suggested a typical nine-step approach to systematically organizing structure and following the supplier development process. These steps can be represented as follows:

1. *Identify the Critical Products*
   (Mainly strategic and bottleneck products).
2. *Identify the Critical Suppliers*
   (Assess their current and future capabilities; does it make sense to develop them?).
3. *Appraise the Performance of the Suppliers*
   (Usage of a standardized KPI framework, questionnaires, checklists, or company specific assessment methodology).
4. *Determine the Gap Between Current and Desired Supplier Performance*
   (Identify the gaps from a supply point of view and from a demand point of view, or combined gaps, such as the level of collaboration).
5. *Form a Cross-Functional Supplier Development Team*
   (Who run the assessments, identify the gaps, and develop and negotiate a possible solution approach).
6. *Meet with Supplier's Top Management Team*
   (To identify the required degree of collaborative relationship, this helps both sides to understand improvement needs and develop a relationship built on trust).
7. *Agree how the Identified Gaps Can Be Bridged*
   (E.g., to implement an IT solution, to form joint improvement teams, to execute a quality certification. . . develop a corresponding plan).
8. *Set Deadlines for the Achievement of the Improvements*
   (Having an agreed upon action plan with defined actions, action owners, deliverables, and due dates).
9. *Monitor Improvements*
   (During the implementation of the supplier development actions, but also afterwards to ensure corrective measures are sustained).

In practice, such a development of a supplier is related to the establishment of a long-lasting relationship that is based on mutual trust. Furthermore, the employees need to cooperate successfully together to align organizations and cultures. This means that besides the nine theoretical steps summarized earlier, soft human and cultural factors play an important role in achieving the objective of jointly improving supplier performance.

5.5 Key Points

In essence, in sourcing activities, the following questions need to be assessed and answers have to be developed in principle for the articles of the commodity categories:

1. *What should be sourced?*—These are the questions regarding the *sourcing objects* (e.g., single loose parts, modules, systems, services, or non-product-related objects?) It also refers to questions about make-or-buy, which accounts for outsourcing decisions.
2. *How should items be sourced?*—What is the *sourcing tool or process* that needs to be applied (e.g., do we consider manual sourcing or do we use IT tools such as portals or EDI? How well are business processes aligned between the supplying and buying parties?) What is the appropriate *sourcing organization*, i.e., whether to run the purchasing *individually* or to establish an alliance and thus follow the idea of *collaborative sourcing?*
3. *From whom to source?*—With *how many suppliers* or partners do we cooperate (do we focus on a *sole supplier*, do we prefer double sourcing, or is our strategy to purchase the goods or services from multiple vendors?) Which supplier demonstrates good performance or has further potential—who should be developed and who should be substituted?
4. *From where to source?*—Do we focus on *cooperation with local* suppliers, is our scope connected to national or continental vendors, or is the strategy to aim for *global sourcing*?
5. *When to source?*—How should sourcing be scheduled regarding early supplier involvement at the stage of product development in order to jointly develop a part or module and to agree on the most suitable delivery strategy *(*e.g., *JIT)*, including corresponding containers or stock keeping units (SKU)? Alternatively, the strategies of *stock sourcing or demand-tailored sourcing* might be applied.

Parallel to these five dimensions are other strategic factors concerning long-term relationships with suppliers, measurement of their performance, and steady analysis of spend patterns, which need to be conducted to ensure continuous evaluation and the improvement of sourcing activities. Of course, the effort involved will differ from company to company and industry to industry; thus, this chapter is intended to provide some guidance on fundamental sourcing strategy considerations, the related processes, and how relationships with suppliers are connected to the SCOM.

Bibliography

Amid A, Ghodsypour SH, O'Brien C (2011) A weighted max-min model for fuzzy multi-objective supplier selection in a supply chain. Int J Prod Econ 131(1):139–145

Baily P, Farmer D, Crocker B, Jessop D (2021) Procurement principles and management, 12th edn. Pearson, Harlow

Bozarth C, Handfield RB (2013) Introduction to operations and supply chain management, 3rd edn. Pearson, Harlow

Carter DE (2024) Influence of supplier relationship management on product quality. Int J Supply Chain Manag 9(5):66–77
Chopra S, Meindl P (2013) Supply chain management. Strategy, planning and operation, 5th edn. Pearson, Harlow
Chopra S, Sodhi MS (2014) Reducing the risk of supply chain disruptions. MIT Sloan Manag Rev 55(3):73–80
Cohen S, Roussel J (2013) Strategic supply chain management: the five core disciplines for top performance, 2nd edn. McGraw-Hill, Boston, MA
Dolgui A, Ivanov D, Sokolov B (2018) Ripple effect in the supply chain: an analysis and recent literature. Int J Prod Res 56(1–2):414–430
Echefaj K, Charkaoui A, Cherrafi A, Ivanov D (2024) Design of resilient and viable sourcing strategies in intertwined circular supply networks. Ann Oper Res. https://doi.org/10.1007/s10479-024-05873-1
Geissbauer R, D'heur M (2010) 2010–2012 Global supply chain trends – are our supply chains able to support the recovery? Lessons learned from the global recession. http://www.imperiallogistics.co.za/documents/Global-Supply-Chain-Trends-2010-2012.pdf. Accessed 15 Apr 2014
Grant D (2012) Logistics management. Person, Harlow
Gupta V, Ivanov D (2020) Dual sourcing under supply disruption with risk-averse suppliers in the sharing economy. Int J Prod Res 58(1):291–307
Heizer J, Render B (2013) Operations management: sustainability and supply chain management, 11th edn. Pearson, Harlow
Hosseini S, Morshedlou N, Ivanov D, Sarder MD, Barker K, Al Khaled A (2019) Resilient supplier selection and optimal order allocation under disruption risks. Int J Prod Econ 213:124–137
Ivanov D, Sokolov B, Dolgui A (2014) The ripple effect in supply chains: trade-off 'efficiency-flexibility-resilience' in disruption management. Int J Prod Res 52(7):2154–2172
Ivanov D, Dolgui A, Sokolov B, Ivanova M (2017) Literature review on disruption recovery in the supply chain. Int J Prod Res 55(20):6158–6174
Ivanov D (2018) Structural dynamics and resilience in supply chain risk management. Springer, New York
Ivanov D (2017) Simulation-based single vs dual sourcing analysis in the supply chain with consideration of capacity disruptions, Big Data and demand patterns. Int J Integr Supply Manag 11(1):24–43
Jacobs F, Chase R (2021) Operations and supply chain management, 16th edn. McGraw Hill, New York
Knotte D, Senger E, Tsipoulanidis A (2020) Operational excellence in Ihrem purchase-to-pay prozess – Ein Guide zur Realisierung von Potenzialen im operativen Einkauf. Invent, Capgemini
Kravchenko K, Gruchmann T, Ivanova M, Ivanov D (2024) Responding to the ripple effect from systemic disruptions: empirical evidence from the semiconductor shortage during COVID19. Modern Supply Chain Res Appl 6(4):354–375
Kummer S, Jammernegg W, Grün O (2018) Grundzüge der Beschaffung, Produktion und Logistik, 4th edn. Pearson, Harlow
Lysons K, Farrington B (2012) Purchasing and supply chain management, 8th edn. Pearson, Harlow
Maihami R, Kannan D, Fattahi M, Bai C (2024) Supply disruptions and the problem of pricing, advertising, and sourcing strategies in a retail duopoly market. Ann Oper Res 344:217–266
Mangan J, Lalwani C, Butcher T (2008) Global logistics and supply chain management. Wiley, New York
Nunes M (2024) Insights of the strategic approach of global sourcing and the modes of international sourcing strategies. Int Bus Res 9(9):134–141
Olle W, Ivanov D (2009) Just-in-time and just-in-sequence: example of Volkswagen Saxony. In: Ivanov D (ed) Supply chain management. St Petersburg Polytechnic University, St Petersburg. (in Russian)
Sawik T (2020) Supply chain disruption management, 2nd edn. Springer, New York

Simchi-Levi D, Schmidt W, Wei Y, Zhang PY, Combs K, Ge Y, Gusikhin O, Sander M, Zhang D (2015) Identifying risks and mitigating disruptions in the automotive supply chain. Interfaces 45(5):375–390

Sodhi MS, Lee S (2007) An analysis of sources of risk in the consumer electronics industry. J Oper Res Soc 58(11):1430–1439

Stevenson W (2018) Operations management, 13th edn. McGraw Hill, New York

Talluri S, Kull TJ, Yildiz H, Yoon J (2013) Assessing the efficiency of risk mitigation strategies in supply chains. J Bus Logist 34(4):253–269

Yoon J, Talluri S, Yildiz H, Ho W (2018) Models for supplier selection and risk mitigation: a holistic approach. Int J Prod Res 56(10):3636–3661

Zheng G, Ivanov D, Brintrup A (2025) An adaptive federated learning system for information sharing in supply chains. Int J Prod Res. https://doi.org/10.1080/00207543.2024.2432469

References for Sect. 5.1

Lawrence M (2019) How Samsung drives sustainability across its supply chain. https://www.csomagazine.com/supply-chain/how-samsung-drives-sustainability-across-its-supply-chain. Accessed 5 Jan 2020

Samsung (2019) Responsible management of supply chain. https://www.samsung.com/us/about-samsung/sustainability/supply-chain/. Accessed 11 Jan 2020

Production Strategy

6

Learning Objectives for This Chapter

- Push and pull views of the supply chain
- Mass customization and modularization
- Order penetration point and postponement
- Basic production strategies in the supply chain
- Analysis of order penetration point location
- Production repurposing strategy

6.1 Introductory Case Study: DELL vs. Lenovo

Dell

Dell was founded in 1984 by Michael Dell, who was 19 at the time. Starting off with only $1000 in start-up capital borrowed from Dell's father, the company opened its first subsidiary in the UK only 3 years later. In 1995, Dell became a global company and was named the No. 1 computer systems provider in 2001. What is it about Dell that makes it so special?

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_6) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_6.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_6

Initially, Dell was well known for its direct-sales model and its ATO or "assemble-to-order" approach. This means manufacturing and delivering individual PCs configured to particular customer specifications. Instead of pre-assembling a complete PC, Michael Dell created an SC where inventories of components were held in a few central locations, and as customers placed their orders, a Dell computer was assembled exactly to the customer's requirements. Distribution and shipping were made directly to the customer. Standard components were offered, and customers chose the features they wanted to add to the technical device. Strong customer focus is an important part of the direct model. The online channel has proven very effective for selling computer hardware, and by 2009 represented about half of the sales in this category. In 2007, Dell expanded its sales channels and started to sell PCs through retail stores.

Most of Dell's suppliers came from Asian countries, such as Japan (RAM chips), Taiwan (monitors), and Singapore (disk devices). All components were then delivered to manufacturing plants, for example, in Austin, Texas (Fig. 6.1).

Using the ATO strategy, cycle times were reduced, and inventory expenses were kept at a low level, as not all varieties of products needed to be stored, but only the standardized versions. In addition, assembly was easy because the features that were to be chosen were independent of the outer appearance of the device. So Dell was able to offer high-quality products at comparatively low prices.

The customer was able to choose between what type of processor, software, or storage they wanted in their device. However, the touch and feel experience of the device was lacking, and Dell had little presence in retail stores. Customers might have missed the presence of a sales expert during the purchasing process. Dell tried to overcome this obstacle by offering hotlines and live chat with sales representatives on their website.

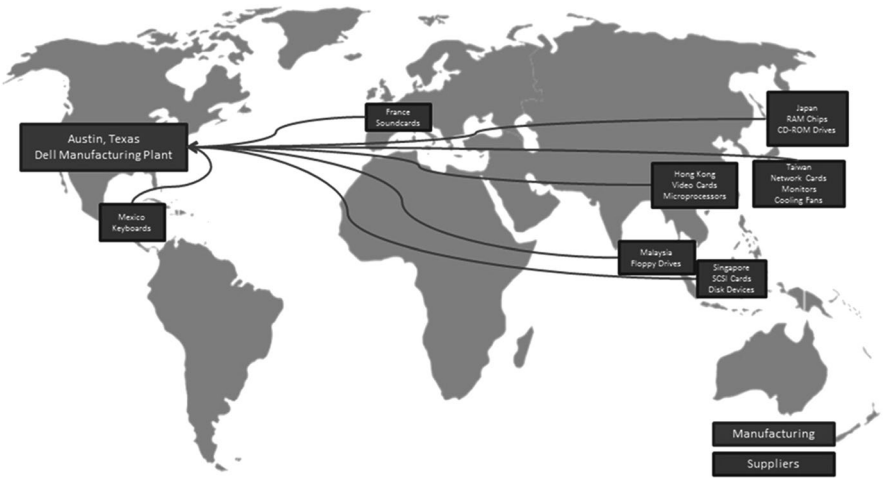


Fig. 6.1 DELL's supply chain

Such a system allowed Dell to create a more flexible SC and ensure efficiency. The lead time was initially 4–6 weeks. With the ATO model, a PC could be fully assembled in 1–2 days, and the customer would receive the order by the end of the week. This significantly reduced inventory holding costs incurred during shipping and improved customer satisfaction. With this strategy, Dell usually held inventory for less than 4 days. By using a postponement assembly strategy, Dell was able to generate cash much faster and reinvest this cash into improving their SC model.

Even though Dell was a so-called rule-maker when they introduced the direct model, market changes forced the company to rethink its strategy. The model was mainly meant to mass-produce personal computers for business customers. However, today, most sales are generated by notebooks for end customers. Here, DELL uses a combination of pre-assembled (e.g., Make-to-Stock) and customized (e.g., Make-to-Order) features.

Lenovo

The Lenovo brand appeared in 2004, but the company has a much longer history, starting in Beijing in 1984 as the New Technology Development Company (NTD Co.) of the Institute of Computing Technology (ICT) of the Chinese Academy of Sciences (CAS). In 1988, the company was expanded into Hong Kong and grew into the largest PC company in China. In 2004, it changed its name to Lenovo, and in 2005 it acquired the former IBM Personal Computing Division. Currently, Lenovo is one of the world's largest PC manufacturers. It has more than 77,000 employees in over 60 countries and operates in more than 180 countries worldwide.

A global SC at Lenovo was established in accordance with functional, geographic, and "internal customers" dimensions. In order to increase customer service in major markets, Lenovo established "strongholds" in Europe, America, Singapore, and Shenzhen, China, to provide customers with business and financial services (see Fig. 6.2).

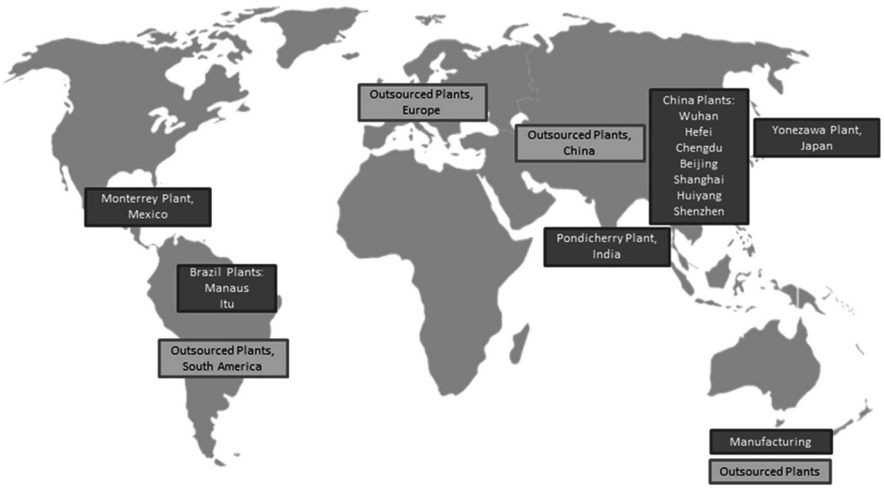


Fig. 6.2 Lenovo's supply chain

Lenovo's SC includes raw material procurement, production in three locations (Beijing, Shanghai, Huiyang), product distribution, transportation, warehousing (outsourcing), and payment. Lenovo's inventory cycle is now within 10 days, and the cash cycle can be achieved within 10–20 days. These performance indicators contribute to relatively high profits for Lenovo China.

In October 2005, Lenovo announced a new organizational structure composed of production, marketing, and logistics. Lenovo bases product development and marketing in Hong Kong. This enables Lenovo's technical staff to receive timely market and technical information and understand the progress of the market and technical factors. This also shortens the company's product development cycle, and company products can keep up with international trends.

Production is mainly organized in Shenzhen and other production areas in mainland China. The labor costs and price of real estate in mainland China are much lower than in Hong Kong. Locating the production processes on the Chinese mainland helps reduce manufacturing costs. In addition to Lenovo's own investment in the construction of production factories in Shenzhen and other places, Lenovo also develops cooperative relations with other domestic manufacturers.

In procurement and inventory management, Lenovo integrates procurement, production, distribution, and logistics into a unified system with over 300 suppliers. Purchase of raw materials such as monitors, hard drives, video cards, and even chassies and other parts are typically organized in collaboration with suppliers in Taiwan, which is a world supply centers for IT components.

In production and inventory management, Lenovo takes a combination of MTS with a large margin of safety and MTO production with a relatively low safety stock of finished goods to cover demand for only a couple of days. In mainland China, Lenovo's suppliers operate warehouses near Shanghai, Beijing, and Huiyang so that these three factories can receive deliveries faster. Recently, Lenovo opened production facilities in the United States to improve its relationship with its business customers and to be able to offer more flexible support.

Lenovo also uses a VMI (Vendor-Managed Inventory) purchasing pattern. By Lenovo's VMI project, its inventory manager, BAX Global Freight Logistics Ltd., as third-party logistics, is responsible for inventory control. The third-party logistics company is responsible for replenishing the production line at the right time with the right items in the right quantity. VMI relies on sharing information between suppliers, third-party logistics, and Lenovo and ensures timely delivery of materials needed for production. VMI brings considerable benefits to production and inventory management at Lenovo. First, its internal business processes are streamlined. Second, it places the inventory closer to production and enhances the elasticity of supply, and better responds to changes in market demand. Third, it improves inventory liquidity. Fourth, through visual inventory management, Lenovo can monitor suppliers' delivery capabilities online. Eventually, it will accelerate Lenovo's logistics, shorten lead times, and ensure timely delivery of the materials required for production: since the Lenovo SC is simplified, costs are reduced, and flexibility is enhanced.

In the distribution phase, Lenovo and its partners form a closer relationship, with timely information sharing and precise and efficient integration of operations. Lenovo's distribution channels are distributors, agents, distributors of contracts, and distributors of non-contracts. Lenovo subdivides functions of different channels and uses two-channel systems, i.e., one for home computers and one for business computers. Due to their different target customers, distributors can establish good relationships with their customers, which is helpful in opening up the market while avoiding strong competition among themselves. Lenovo adheres to the secondary distribution system, namely a maximum of two levels between Lenovo and consumers, because excessive levels would reduce the efficiency of operations and increase Lenovo's management costs. Lenovo has adopted a selective distribution strategy. On the one hand, Lenovo can reach markets through the distributors.

On the other hand, by selecting and training core agents and distributors, Lenovo is able to provide users with better solutions. Through the development of secondary channels, Lenovo flattens the structure of its distribution channel. Direct shipments are organized for MTO production. As with Dell products, customers can order Lenovo's products online, but without any adaptation tailored to customers.

To manage the complexity of the SC, Lenovo uses an ERP system from SAP. ERP helps Lenovo to run effective SC execution. By monitoring huge amounts of SC data in real time, Lenovo has access to more accurate information, helping to enable fast decision making, facilitating a responsive SC, and improving logistical efficiency. It also provides end-to-end visibility of the entire global SC and increases transparency and efficiency with real-time access across the SC.

Discussion

- Identify the basic stages in the value chain described in this case study.
- Explain the interconnections between the production and distribution strategies at Dell and Lenovo.
- Compare MTO and MTS strategies.
- Describe the advantages and possible limitations of VMI at Lenovo.

6.2 Postponement and Modularization

6.2.1 Problem: Mass Production or Product Customization

Henry Ford said: "A customer can have a car painted any color that he wants so long as it is black." Until the 1970s, mass production dominated the industries and markets. The advantages were low unit costs because of rare machine setups, large batch material ordering, and standard technologies and materials. Since the 1980s, customers have played a more and more important role in the SC. Products become individual and customized. This flexibility creates competitive advantages but is also costly. Machines should be set up frequently, lot-sizes in manufacturing and procurement become smaller, and the variety of technologies and materials rise.

In this context, the problem is to determine the right *production strategy* for the SC to combine the advantages of mass production and customization.

6.2.2 Principles: Postponement and Modularization

In an era of customer-oriented product individualization and global competition, companies are struggling to find ways to improve responsiveness without holding huge finished goods inventories. Postponement and modularization are useful tools in this regard.

The *postponement* concept was first introduced in literature by Alderson (1950), where it was observed that products tend to become differentiated as they approach the point of purchase; later, this concept was further developed by Bucklin (1965). In addition to postponed forward shipment of goods (time postponement) and maintaining goods at a central location in the channel (place postponement), certain manufacturing activities can also be postponed (Van Hoek 2001). Although this differentiation improves marketability, the manufacturability of the products becomes more complex.

Van Hoek (2001) defines *postponement* as "an organizational concept whereby some of the activities in the SC are not performed until customer orders are received." Postponement is especially useful for:

- Significant number of variants of an end-product with an uncertain split of demand on variants.
- Delivery time requested by customers must allow value-adding steps after receipt of customer orders (or reliable demand forecast).

Differentiation of a generic product into a specific end-product is shifted *closer to the consumer* by postponing identity changes, such as assembly or packaging, until the last possible SC location. This allows *safety stock* to be held as one generic product instead of multiple specific end-products. Especially in cases where the split of demand into specific end-products is uncertain, postponement with its *risk pooling* effect leads to a lower safety stock requirement and a lower risk of obsolescence of end-products. Furthermore, as less value has been added to the generic product than to the specific end product, *less capital* is bound in each stocked unit.

Modularization implies a product design approach whereby the product is assembled from a set of standardized constituent units. It provides opportunities for exploiting economies of scope and scale from a product design perspective. The key issue here is to design efficient linkage mechanisms in the constituent units so that any required combination can be conveniently assembled.

Baldwin and Clark (1997) describe modularity as "building a complex product or process from smaller subsystems that can be designed independently yet function together as a whole." The ideas of the integration of the product and process modularity have been extensively investigated in the *mass customization* approach.

Ernst and Kamrad (2000) introduced a conceptual framework for evaluating different SC structures in the context of modularization and postponement. In the analysis, modularization is linked to the postponement. The paper introduces taxonomy and develops a corresponding framework for characterizing four SC structures, defined according to the combined levels of modularization and postponement: rigid, postponed, modularized, and flexible.

6.2.3 Examples of Postponement Strategies

Asian Paints (Shah 2009)
Asian Paints is an Indian paint manufacturing firm that has employed a postponement strategy in its emulsions product category. Asian Paints offers four emulsion brands. In turn, each brand offers 150–250 shades. Offering a wide variety of color shades is essential in the emulsions market. An emulsion comprises a "base" and a combination of "stainers." The base provides the functional aspects, while the stainers provide the required shade. The base accounts for 99% of the final emulsion volume. A wide range of shades is developed using just ten stainers. At Asian Paints, the mixing of the base and the stainers, also known as "tinting," is carried out at the various sales points (SPs) distributed across the country. On average, there is one SP for every 400 retailers. The customer chooses from a range of 150–250 shades at the retailer point.

The retailer will immediately forward this order to the SP. The effective time for tinting is about 10 min. The customer collects the shade of choice within 1–2 days of placing the order. Except for certain fast-moving shades, the inventory at the SP is mainly bases and stainers. The SPs order these periodically from their designated regional distribution centers (RDCs). The factory warehouses replenish the RDC inventories periodically. The periodicity of ordering depends on the demand volumes. The RDC lead times (factory to RDC) are 2 weeks to 1 month, while the SP lead times (RDC to SP) are 1–2 days. Thus, the delivery period would be close to a month if the tinting operation took place at the factory itself. Tinting is a low-technology operation due to low capital expenditure and simplicity. The delayed differentiation due to the postponement of the tinting operation has reduced the inventory levels drastically. Customer service is high because of the reduced delivery period. Forecasting errors are also considerably reduced. There is no loss of scale economies as a result of the postponement. Being a simple operation, product quality is not diluted because operations were transferred from the factory to the SPs.

Asian Paints revolutionized the postponement concept in the Indian paint industry. Asian Paints has been offering substantially higher numbers of shades and maintains its finished goods inventories at about 60% of the industry average. This has helped them to maintain profitability consistently higher than the industry average and has probably contributed to increased market share over a period of time.

Automotive Industry (Appelfeller and Buchholz 2011; Kagermann et al. 2011)
Modular design is an important method for controlling and reducing complexity. Using identical components and modules significantly reduces diversity and

complexity. Limited, accurate, and defined components and modules, which are assembled to create individualized final products, result in outward diversity and inward standardization. In the automotive industry, this approach has become known as a *platform strategy* that can be used within or across product lines or brands.

From the customer's perspective, postponement means being able to specify requirements as late as possible in the production process. Vehicle manufacturers are particularly well known for this practice. For example, BMW customers are able to modify their personalized vehicle up to 5 days before assembly begins. In turn, a gearbox or seat supplier has to defer production until the precise requirements are known.

While the implementation costs are significant, postponement in combination with JIT has many benefits:

- Lower inventory costs for finished goods, but also for raw materials and work in process (WIP).
- Lower risk of product obsolescence.
- Simpler forecasting and greater responsiveness.
- Customized products can deliver competitive advantages, such as time to market and command a price premium.

Suppose a company shifts final assembly to a time and location close to the customer in time or space. In that case, it can supply more personalized solutions and avoid make-to-stock production, while still realizing economies of scale.

A good example of successful implementation of the postponement strategy is the automobile manufacturer Toyota (Belson 2011). Building brands such as Toyota, Lexus, and Scion in its own factories in Japan, Toyota afterward ships the cars with the aim of selling them internationally. Toyota employs port facilities, e.g., a 98-acre operation at Newark's port is a scaled-down assembly plant. Cars, destined for sale from Virginia to Maine, are modified at the Newark port according to a customer's special request at very short notice.

A range of so-called port-installed options can result in 21 different models. The average length of time spent at Toyota's facility is 8 days for the up to 12,500 vehicles arriving by ship at any one time. Between 2 and 6 days are required for the necessary adjustments, and it takes on average another 1.9 days for a car, once completed, to leave the facility.

By completing the modifications within 8 days and allowing the customer the possibility to adjust his/her order in regard to the car's configuration—including satellite radios, tires, rims, bumper protectors, body moldings, and a variety of exterior items—until 2 days before the car arrives in Newark, Toyota achieves better customer service and higher customer satisfaction. In addition, having established a close relationship with suppliers and only ordering daily the accessories needed for the next 2 days, the facility has reduced its just-in-time inventory by 70% within the last 4 years.

IKEA (Cheng et al. 2010; Kagermann et al. 2011)
With production in 41 different units worldwide, IKEA controls an SC for approximately 9500 finished products. The "Range Strategy and Product Development Supply Chain group is in charge of handling the SC as smoothly and efficiently as possible. IKEA developed the concept of 'design for logistics" (DFL). DFL focuses on the design of the product not only according to customer requirements but also to ensure efficient handling and transportation. This results in more efficient stocking of items and the so-called flatpack approach, in which most items are packaged in flat, easy-to-handle packages. IKEA ultimately aims for a 75% fill rate for its trucks. Due to vertical integration, IKEA produces "make-to-stock," which means they follow a push strategy. This strategy comes with many advantages on the one hand, but also with some risks on the other. For one, the "make-to-stock" approach requires accurate forecasting of demand to manage the trade-offs between costs (transportation cost/inventory cost) and service level (availability of items). Therefore, IKEA is engaging in risk when pooling activities such as:

- Postponement (the last step of the SC is the product assembly).
- Material standardization (uniform screws, planks, colors, techniques).
- Inventory pooling (at the warehouse or in-store).
- Product pooling (similar kinds of products).

IKEA has taken the concept of manufacturing postponement to an extremely high level: The final assembly of many furniture items occurs in people's houses. The company works closely with its suppliers to source and design attractive furniture that can be location-neutral and flat-packed whenever possible, making its products acceptable in many countries while keeping transportation and warehousing costs low. All products in IKEA retail stores are kept semi-finished (flat packs) and assembled by customers or couriers after home delivery. In this way, truckload capacities can be better utilized, and configurations can be easily made at customer locations.

6.2.4 Repurposing

Repurposing is a concept of utilizing the existing manufacturing and logistics capacities to produce a new, untypical product in an SC. The concept of repurposing has been used extensively during the COVID-19 pandemic in 2020. Healthcare SCs for PPE (personal protection equipment) items have not been able to cope with increased demand due to the COVID-19 outbreak. To address the PPE shortages, which have been hampering the healthcare sector worldwide, governments appealed to commercial companies, creating ad-hoc SCs to combat this supply deficit. Numerous cases have been observed of production systems and SCs being repurposed during the COVID-19 pandemic to meet the demand for such critical items (Ivanov and Dolgui 2020; Ivanov 2022). Utilizing an existing manufacturing line for a purpose other than its intended use is called "Repurposing." This is typically a

short-term strategy to overcome the shortage of critical resources, such as PPEs, diagnostic equipment, and clinical care equipment.

▶ **Practical Insights** Large companies can rapidly develop the production of necessary supplies by utilizing their extensive resources: engineering talent, manufacturing capabilities, purchasing, and logistics networks. Additionally, there is an extensive network of small manufacturers, including facilities with 3D printing equipment, that they can draw on as well. In contrast to large companies, many small manufacturers can not address the problem alone but may set up production as a coalition. To efficiently leverage small businesses' network, local administrations need to integrate and coordinate diverse production capabilities, design and engineering services, and logistics. For example, luxury goods manufacturers in Italy and France have re-designed their operations to produce urgently needed items during the COVID-19 virus outbreak. Within 72 h of the French government's call for business to pitch in, LVMH's perfume factories were able to produce hand sanitizer. Giorgio Armani, Gucci, and Prada repurposed their designer clothing factories in Italy to churn out medical overalls, and Burberry harnessed a trench coat plant to make face masks and nonsurgical gowns (Schatteman et al. 2020).

6.3 Push–Pull Views and Order Penetration Point

The push/pull view of the SC divides SC processes into two categories based on whether they are executed in response to a customer order (downstream SC) or in anticipation of customer orders (make-to-stock, upstream SC). Pull processes are initiated in response to a customer order. The pull advantages are responsiveness and a high degree of customer-oriented product individualization. Push processes are initiated and performed in anticipation of customer orders. The advantages of this are economy of scale (low manufacturing and transportation costs), flexibility (high level of inventory), and short supply times.

This view is very useful when considering strategic decisions related to SC design, because it forces a more global consideration of SC processes as they relate to the customer.

The main idea behind the postponement is to delay product differentiation at a point closer to the customer. This point is also known as the order penetration point (OPP) (see Fig. 6.3).

In the case of a customer inquiry or order, the goods can be delivered from a general inventory holding unit (trajectory α) or customized according to the inquiry or order (trajectory β). Upstream of the OPP, the processes are designed to be lean. Downstream the processes are designed to be agile.

Three categories are primarily relevant to the OPP location:

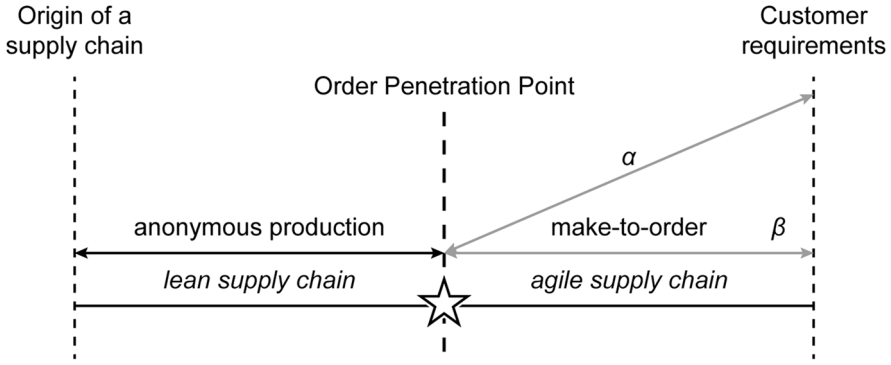


Fig. 6.3 Order penetration point

- Delay of product differentiation—postponement.
- Design and developing standard or generic configurable products that can be customized quickly—product modularization.
- Implementation of specific collaboration strategies, inventory strategies, buffers, etc. to fulfill service level objectives—agility.

In practice, a selected postponement strategy determines the OPP location. The further upstream the OPP is located in the SC, the more manufacturing and logistics activities are postponed. In general, the aim is to postpone manufacturing and logistics as much as possible and maintain high customer service standards.

The OPP location is dependent on several factors.

Factor 1. Costs of inventory holding. This factor has been highlighted in the literature as one of the most important for analyzing postponement strategies.
Factor 2. Costs of modularization. High costs of modularization are the main drawback of postponements.
Factor 3. Costs of information systems installation and running. However, the introduction of new IT costs a lot of money.
Factor 4. Costs of replenishment. Practical case studies, i.e., of DELL, point to the conclusion that SC responsiveness highly depends on the speed of replenishment of customer-individualized product units.

OPP location depends on products.

Factor 1. How many variants of a product are possible?
Factor 2. What is customized (product, batch sizes, and package)?
Factor 3. Which parts of the products can be customized from a technological point of view?
Factor 4. Which parts of the products can be modularized from the technological point of view?
Factor 5. Price of products.

Factor 6. What is the concurrent situation in the industry?
Factor 7. SC cycle for a product.
Factor 8. Customization cycle of a product.
Factor 9. Where is strategic inventory held?

OPP location depends on customers.

Factor 1. Who launches orders (individual consumer, organization, retail, or wholesaler)?
Factor 2. Who launches the customization (individual consumer or organization)?
Factor 3. Who is the end customer (individual consumer or organization)?
Factor 4. Number of customers.
Factor 5. Which contracts are used?

OPP location depends on suppliers.

Factor 1. How many alternative suppliers are available?
Factor 2. Possibility of strategic alliances under and with the suppliers.
Factor 3. What contracts are used?

6.4 Selection of a Production Strategy

We already know the principles of postponement, push/pull views, and OPP. Now we can integrate this knowledge and consider several production strategies.

6.4.1 Types of Production Strategies

With pull processes, execution is initiated in response to a customer order. With push processes, execution is initiated in anticipation of customer orders. Therefore, at the time of execution of a pull process, customer demand is known with certainty, whereas at the time of execution of a push process, demand is not known and must be forecasted (see Fig. 6.4).

The push/pull boundary is the place of the OPP that separates push processes from pull processes. A product is kept in a generic state for as long as possible. Differentiation of the generic product into a specific end-product is shifted closer to the consumer by postponing identity changes, such as assembly or packaging, until the last possible SC location. According to the placement of the OPP, the following *production strategies* can be determined:

- Make-to-stock (MTS)
- Distribute-to-order (DTO)/Configure-to-order (CTO)
- Assemble-to-order (ATO)
- Make-to-order (MTO)
- Engineer-to-order (ETO)

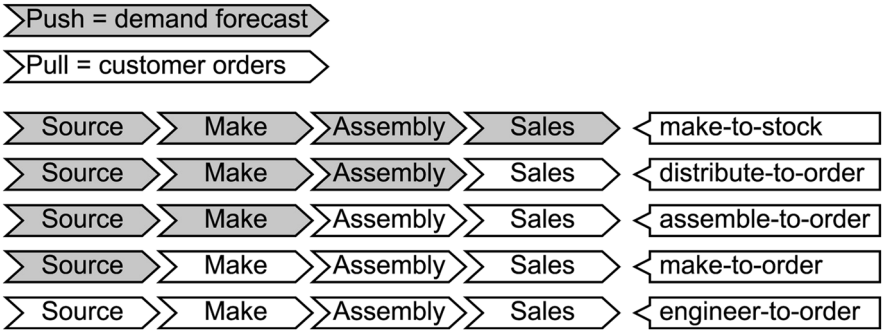


Fig. 6.4 Production strategies in the supply chain

A *make-to-stock strategy* is typical for the mass production of standard products like sugar and socks. MTS's advantages are low unit production costs and higher capacity utilization because of the scale effect and quantity flexibility regarding demand fluctuations resulting from product standardization. The disadvantages of MTS are that it has a higher inventory and lower production flexibility.

Distribute-to-order (or *configure-to-order*) strategies allow a small degree of customer individualization. It might have customer-specific packaging (as in the pharmaceutical industry) or some individual items added to a standard manufactured product. Advantages of DTO are higher flexibility regarding product structure and lower inventory.

Assemble-to-order presumes a higher degree of product individualization, which happens directly within the manufacturing process. A typical example of ATO is car manufacturing or DELL in the electronics industry. Assembly is performed individually for each customer from a set of standard modules. The advantages and shortcomings of ATO are similar to those of DTO/CTO, but with higher investments in the process, product flexibility, and lower inventory.

A *make-to-order* strategy is similar to ATO but also presumes a possibility of customer individualization at the module/component level. This means that not only the final assembly from some standard modules can be individualized, but also that the modules and components can be manufactured individually for each particular customer. MTO can be recommended for customer-specific products with fluctuating demand. The advantages and shortcomings of MTO are similar to ATO, but with even higher investments in flexibility and lower inventory.

Engineer-to-order has the highest degree of customer individualization. An example is the production of an aircraft, railway vehicle, or a house building where the customer is involved in the engineering and production processes right from the beginning and can select not only individual components, but also materials and designs. ETO is reasonable for complex products of high value where customer individualization is crucial.

In practice, hybrid production strategies, e.g., MTO-ETO, are frequently encountered (Jordan and Graves 1995; Alp and Tan 2008; Zschorn et al. 2017).

Case-Study Airbus: Production Strategy and Strategic Supplier Collaboration with the Help of AirSupply

Taking aircraft manufacturing at Airbus as an example, the following case study describes how Airbus implements the ETO production strategy and collaborates with its suppliers using AirSupply. Each time an Airbus aircraft is made, it is the result of assembling a multitude of parts representing a very large volume of orders. And what is valid for the Airbus assembly line is true for the whole aerospace industry where not only aircraft, but also helicopters and satellites are built. Parts for manufacturing come from multiple suppliers from all over the world. Most of them are quite complex and need to fulfill the highest quality standards. Every time delay can cause very high costs.

It is essential that all suppliers involved in the manufacture of aircraft have real-time visibility of the progress of demand and inventory to adapt to fluctuations and changes in customer requirements. In such a case, it is essential for customers and suppliers to have a common tool that will enable them to better collaborate and to gain visibility over demand as well as inventory.

In January 2009, EADS decided to initiate the development of a specific Aerospace and Defense industry solution to replace the current company's individual SCM solution. EADS is one of the leading Aerospace and Defense companies worldwide. The range of products includes aircraft from Airbus, helicopters from Eurocopter, and satellites from the Astrium division. Each aircraft is practically one of a kind. One of the reasons why commercial aircrafts are all configured differently is that the ordering airlines want to differentiate themselves from their competitors. The easiest way for passengers to spot these differences is by looking at the seats and entertainment systems.

With 75%, the value added by external suppliers in aircraft manufacturing is relatively high. A single long-range aircraft is made up of around 4 million individual parts, supplied by companies from many countries. Even at the final assembly line, workers put together components originating from at least four different countries: fuselage sections and the vertical stabilizers come from Germany, the cockpit and central fuselage section from France, the wings from England, and the horizontal stabilizers from Spain (see Fig. 6.5).

Where the engines come from depends on the customer's preference. The task is made even more complex by the generous length of time Airbus gives its customers to submit their final wishes. Including lead time, manufacturing takes around 2 years, and Airbus accepts customer change requests up to as late as 6 months before delivery (see Fig. 6.6).

Collaboration with suppliers begins only a few months after the order has been received. This is normal for the aerospace industry and is very different from the automotive industry, for example. Delivery dates are coordinated with suppliers at weekly intervals, on both the capacity planning and order levels. The production of components begins before the date on which the customer is able to request the last modifications. Final assembly takes around 1 month. This intensive process of coordination with suppliers is managed via AirSupply.

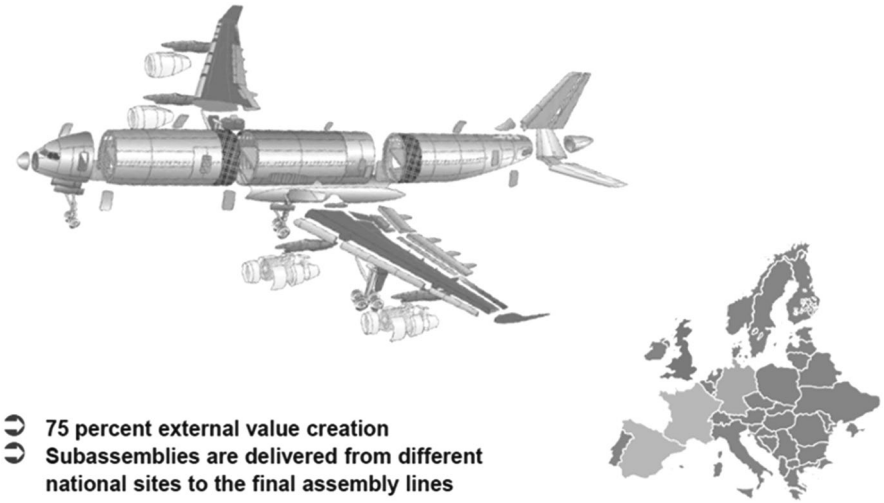


Fig. 6.5 Modular production strategy (based on SupplyOn)

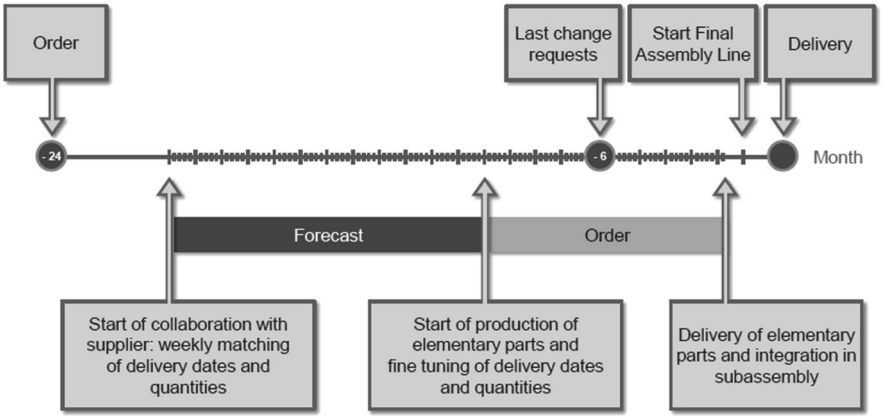


Fig. 6.6 Intense collaboration with suppliers throughout the whole production process of an aircraft

How does the collaboration look like in concrete terms? The first step is calculating future demands via the Airbus ERP system. This always takes place on the weekend. The following Monday, the Airbus Order Officer checks the generated demand planning (see Fig. 6.7).

The next step is the approval of the requirements planning and transmission to the suppliers via AirSupply. The supplier now checks whether he can provide what is needed by Airbus. If the answer is yes, he sends his confirmation. AirSupply can be used to propose alternative quantities and delivery dates if the answer is no. The Order Officer at Airbus then checks the supplier's proposal and accepts it. The

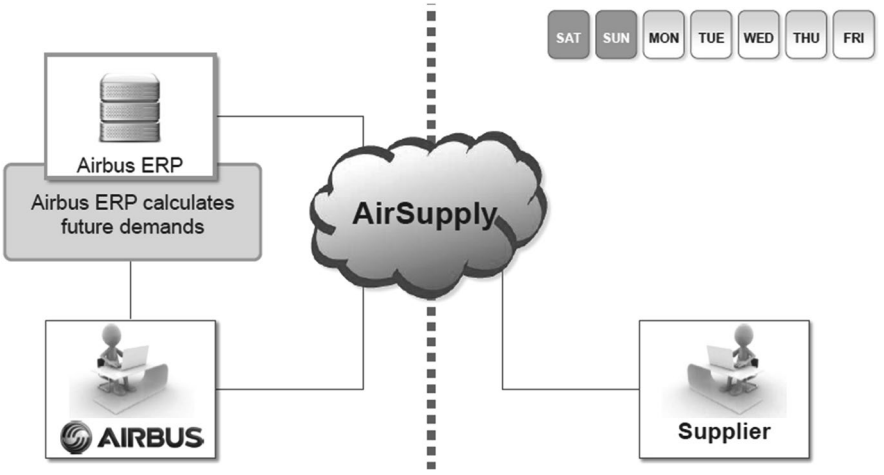


Fig. 6.7 Start of supplier collaboration

collaboration process has now been successfully completed, and the data is transferred back to the ERP system at Airbus.

It is evident that it would not have been possible to conduct this process by exchanging messages in a conventional EDI format. The rules required to logically represent this collaboration are not provided by classical ERP systems. This is why the entire set of logical rules governing collaboration has been moved to the cloud. The ERP systems simply process the results of the collaboration. From the system architecture perspective, the solution is embedded in SupplyOn's cloud infrastructure. This is made up of central components that control the registration of companies and users. At the current time, around 8000 companies from over 70 countries are using SupplyOn's solutions.

One of the most remarkable aspects of this project was that it created a solution for the entire European Aerospace and Defence industry and not only for the EADS group. To make this possible, the French aerospace industries association GIFAS formed the BoostAeroSpace initiative. SupplyOn designed the solution together with the lead partner EADS and the companies Dassault Aviation, Safran, and Thales. Up to 150 people were involved in the project in its most intensive phases, with a total of around 10,000 working days. After the completion of the design, development, and pilot phase, the solution went live at the end of 2011. Shortly after, the integration of the suppliers began.

The strategic goal of EADS and Airbus is to maintain the competitiveness of the European Aerospace and Defence industry in the long term. They are convinced that this can best be achieved by creating a common industry solution together with further leading companies in this sector. It was EADS's ambitious goal to build the best Aerospace and Defense industry SCM solution worldwide. This applies to the functionality of the solution as well as to the underlying business model. EADS and BoostAeroSpace opted for SupplyOn as their implementation and operating partner

due to its successful track record with cloud-based solutions for the automotive industry.

Discussion

1. What benefits of AirSupply for supplier collaboration can you identify?
2. Describe the advantages and risks of an ETO (engineer-to-order) production strategy.

In summary, the basic trade-off in selecting the right production strategy is between investments in flexibility and reduction of costs due to the scale effects or inventory reduction. Total flexibility, inventory, and unit costs should be balanced with customer satisfaction and service level. One possible approach to perform such an analysis is presented in the next section.

6.4.2 Method: Lost-Sales Analysis

This task investigates the issue of how to determine what the right production strategy and OPP location is in the SC. In the proposed model, we consider two strategies:

- MTS and delivering from a general inventory holding unit and.
- Introducing the agile MTO part downstream from the general inventory holding unit.

We assume that introducing the agile part of the SC downstream from the OPP leads to an increase in flexibility and costs. As such, we propose a "lost sales" based treatment of the OPP location determination: the OPP location can be determined by a comparison of the financial results of the two strategies (with and without OPP). With regard to this treatment, a quantitative estimation of the OPP location can be found by relating the above-mentioned strategies to each other, i.e., an OPP location index can be calculated at different stages of the SC according to the model (6.1) (Ivanova et al. 2015):

$$D = \frac{R_{\mathrm{a}} \cdot k_{\mathrm{a}} - \left(C_{\mathrm{u}}^{a} + C_{\mathrm{d}}^{a} + P^{a} + L^{a}\right)}{R - \left(C + k \cdot P + k \cdot L\right)} \rightarrow \max\left[t_0; T\right] \tag{6.1}$$

where

D is the index to characterize the efficiency of the OPP location,
R is the revenue in case of make-to-stock,
R_{a} is the revenue when agility is introduced downstream from the general inventory holding unit,
C are the SC costs in case of make-to-stock,
$C_{\mathrm{u}}{}^{a}$ are the SC costs upstream from the OPP when agility is introduced,

C_d^a are the SC costs downstream from the OPP when agility is introduced,
P are penalties of not-fulfilled contracts in case of make-to-stock,
P^a are penalties of not-fulfilled contracts when agility is introduced,
L are losses of rejected customer's inquiries in case of make-to-stock,
L^a are losses of rejected customer's inquiries when agility is introduced,
k, k_a are correcting coefficients to take into account future increases in sales due to an increase in responsiveness when agility is introduced, and,
t is instants of time within the whole SC cycle $[t_0; T]$.

Different OPP locations can be investigated, and the best one with the maximum value $D > 1$ is selected on the basis of the formula (6.1).

Task 6.1 Selection of an OPP Location

We consider an SC in which four potential OPP locations correspond to the following strategies: MTO—Make-to-Order, ATO—Assembly-to-Order, CTO—Configure-to-Order, and MTS—Make-to-Stock (Fig. 6.8).

The following data is considered (Table 6.1):

Calculate the indexes D and propose the OPP location. $D_{max} = 6.65$ and corresponds to the OPP at $t = 4$.

Recommend a manufacturing strategy! We recommend selecting the MTS strategy, as there is no need for any OPP in this SC.

Explain the push/pull view of the processes within an SC.

The push/pull view of the SC divides SC processes into two categories based on whether they are executed in response to a customer order or in anticipation of customer orders. Pull processes are initiated in response to a customer order. Push processes are initiated and performed in anticipation of customer orders. The push/pull boundary separates push processes from pull processes.

Explain the basic trade-off between responsiveness and efficiency of SCs in terms of inventory. The fundamental trade-off when making inventory decisions is between responsiveness and efficiency. Increasing inventory will generally make the SC more responsive to the customer. This choice, however, comes at a cost as added inventory decreases efficiency. Therefore, an SC manager can use inventory as a driver for reaching a targeted level of responsiveness and efficiency according to their competitive strategy.

Explain the meaning of the coefficients k! k; ka is correcting coefficients that account for future increases in sales due to the increase in responsiveness when agility is introduced.

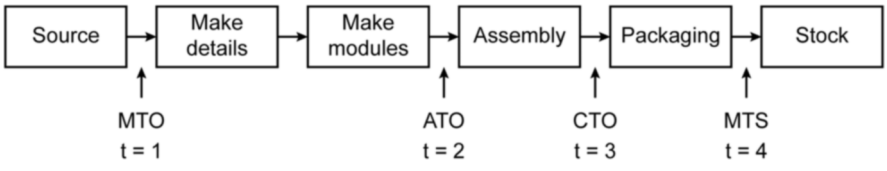


Fig. 6.8 Alternative OPP locations

Table 6.1 Comparison of OPP locations in the SC

T	1	2	3	4
R^a	160	180	200	220
R	220	220	220	220
$C_u{}^a$	40	40	40	40
$C_d{}^a$	160	160	160	160
C	192	192	192	192
P^a	3.5	3.5	3.5	3.5
P	7	7	7	7
L^a	2.5	2.5	2.5	2.5
L	11.2	11.2	11.2	11.2
k_a, k	1.15	1.15	1.15	1.15
D	–3.111739745	0.141442716	3.394625177	**6.647807638**

6.5 Key Points

Customers play more and more important roles in the SC. Products have become individual and customized. This flexibility creates competitive advantages, but is also costly. The right production strategy in the SC combines the advantages of efficient mass production and responsive product customization. The push/pull view of the SC divides SC processes into downstream (production in response to customer order) and upstream (production in anticipation of customer orders). Pull processes are initiated in response to a customer order. The advantages of the downstream are responsiveness and a high degree of customer-oriented product individualization. Push processes are initiated and performed in anticipation of customer orders. The advantages of the upstream are economy of scale and flexibility (high level of inventory and short supply times).

The first element of a production strategy from the push–pull point of view is postponement, which is an organizational concept whereby some of the activities in the SC are not performed until customer orders are received. Postponement is especially useful when there are many variants of an end-product with an uncertain split of demand on variants. In addition, delivery time requested by customers must allow value-adding steps after receipt of customer orders (or forecast).

The second element of a production strategy is OPP, which separates the push and pull parts of the SC. In practice, the selected postponement strategy determines the OPP location. The further upstream the OPP is located in the SC, the more manufacturing and logistics activities are postponed. In general, the aim is to postpone manufacturing and logistics as much as possible while maintaining high customer service standards.

According to the placement of the OPP, the following production strategies can be determined:

- Make-to-stock (MTS)
- Distribute-to-order (DTO)/Configure-to-order (CTO)
- Assemble-to-order (ATO)

- Make-to-order (MTO)
- Engineer-to-order (ETO)

Utilizing the existing manufacturing and logistics capacities to produce a new, untypical product in an SC is called "Repurposing." Typically, this is a short-term strategy to overcome the shortage of critical resources such as PPEs, diagnostic equipment, and clinical care equipment (such as disinfection liquids).

Acknowledgment The case study is written with permission of SupplyOn and uses the materials and pictures from SupplyOn. We thank Mr. Arvid Holzwarth (MBA) for his help in preparing this case study.

Bibliography

Alderson W (1950) Marketing efficiency and the principle of postponement. Cost Profit Outlook 3(4):15–18

Alp O, Tan T (2008) Tactical capacity management under capacity flexibility. IIE Trans 40:221–237

Appelfeller W, Buchholz W (2011) Supplier relationship management: strategie, organisation und IT des modernen Beschaffungsmanagements, 2nd edn. Gabler, Wiesbaden

Baldwin CY, Clark KB (1997) Managing in the age of modularity. Harv Bus Rev 75:84–93

Belson K (2011) Far from the factory, adding final touches. The New York Times, September 23. http://www.nytimes.com/2011/09/25/automobiles/far-from-the-factory-adding-final-touches.html?pagewanted=all&_r=0. Accessed 10 May 2014

Bucklin LP (1965) Postponement, speculation, and the structure of distribution channels. J Mark Res 2:26–31

Cheng TCE, Li J, Wan CLJ, Wang S (2010) Postponement strategies in supply chain management. International series in operations research and management science, vol 143. Springer, New York

Dolgui A, Gusikhin O, Ivanov D, Li X, Stecke K (2024) A network-of-networks adaptation for cross-industry manufacturing repurposing. IISE Trans 56(6):666–682

Ernst R, Kamrad B (2000) Evaluation of supply chain structures through modularization and postponement. Eur J Oper Res 124(3):495–510

Gunasekaran A, Laib K-H, Cheng TCE (2008) Responsive supply chain: a competitive strategy in a networked economy. Omega 36(4):549–564

Ivanov D (2022) Viable supply chain model: integrating agility, resilience and sustainability perspectives – lessons from and thinking beyond the COVID-19 pandemic. Ann Oper Res 319:1411–1431

Ivanov D, Dolgui A (2020) Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability. A position paper motivated by COVID-19 outbreak. Int J Prod Res 58(10):2904–2915

Ivanova M, Kaeschel J, Ivanov D (2015) Analysis of the order recovery point location in the supply chain. Int J Integr Supply Manag 9(4):329–342

Ivanov D (2023) The Industry 5.0 framework: viability-based integration of the resilience, sustainability, and human-centricity perspectives. Int J Prod Res 61(5):1683–1695

Jordan WC, Graves SC (1995) On the principles of the benefits of manufacturing process flexibility. Manag Sci 41:577–594

Kagermann H, Osterle H, Jordan JM (2011) IT-driven business models: global case studies in transformation. Wiley, Hoboken

Mosayebi M, Fathi M, Hedayati MK, Ivanov D (2024) Time-to-Adapt (TTA). Int J Prod Econ 278:109432

Olhager J (2003) Strategic positioning of the order penetration point. Int J Prod Econ 85(3):319–329
Purohit Y, Parkhi S, Karande K (2024) Mass customization in welding and manufacturing through standard work hour modeling. In: Mansour N, Bujosa Vadell LM (eds) Finance and law in the metaverse world. Contributions to finance and accounting. Springer, Cham. https://doi.org/10.1007/978-3-031-67547-8_3
Shah J (2009) Supply chain management: text and cases. Pearson, Chennai
van Hoek RI (2001) The rediscovery of postponement a literature review and directions for research. J Oper Manag 19(2):161–184
Wachter C, Beckschulte S, Hinrichs M, Sohnius F, Schmitt R (2024) Strategies for resilient manufacturing: a systematic literature review of failure management in production. Procedia CIRP 130:1393–1402
Zschorn L, Müller S, Ivanov D (2017) Capacity planning on key work stations in a hybrid MTO-ETO production system: a case-study on Siemens AG. Int J Invent Res 4(2/3):214–232

Reference for Sect. 6.1

Chopra S, Meindl P (2012) Supply chain management: strategy, planning, and operations, 5th edn. Pearson, London
CNN (2013). http://edition.cnn.com/video/data/2.0/video/business/2013/12/13/world-biz%2D%2Ddell-brian-gladden.cnn.html
Dell (2014). http://www.dell.com
Global Integrated Supply Chain System (2006) Successful cases of SCM in China. http://tinyurl.com/p8rhuh8
Hirsh L (2007). http://www.ecommercetimes.com/story/18779.html
http://www.it-times.de/news/nachricht/datum/2013/06/07/lenovo-eroeffnet-werkshallen-in-den-usa/
San Juan R (2013). http://www.gaebler.com/Dell-Direct-Model-to-Success.htm
Slack N, Brandon-Jones A, Johnston R (2011) Essentials of operations management. Pearson, London
Supply Chain Council (2012) Inside Lenovo's hybrid supply chain strategy. http://supply-chain.org/f/Zapko-Inside_Lenovos_Hybrid_Supply_Chain_Strategy.pdf
www.lenovo.com

Reference for Sect. 6.2

Schatteman O, Woodhouse D, Terino J (2020) Amid crisis and disruption, supply networks designed for low-cost and minimal inventory pose a major risk, 27 April. [Online]. https://www.bain.com/insights/supply-chain-lessons-from-covid-19/. Accessed 18 June 2020

Facility Location Planning and Network Design

7

We have already discussed the performance, success, and failures of SCs. Now, it is time to discuss approaches for creating such a network. In particular, it is necessary to clarify important decisions that must be made in order to set up an SC. Such a network setup comprises both decisions about the usage and installation of network nodes (represented by factories and/or warehouses or any other immobile equipment such as supply regions or markets) as well as transportation links connecting different network nodes to enable the flow of materials as well as information between these nodes. In order to set up a supply network, we have to cope with the challenge that long-lasting SC design (SCD) decisions (with several years of validity) have to be coordinated with mid-term network configuration decisions (commitments for some weeks or months) as well as with operational deployment decisions.

This chapter starts with an outline of a location planning problem in Sect. 7.1. Next, in Sect. 7.2, we analyze different decision tasks in a typical location planning scenario and reveal the interactions between individual decision situations of the three aforementioned categories (long-, mid-, as well as short-term planning). The generic decision task regarding which regions should be incorporated into a supply network is addressed in Sect. 7.3. Section 7.4 is dedicated to the identification of explicit location proposals for those regions that contribute to SC cost efficiency.

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_7) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_7.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_7

The multiple-factor search for the right location for a facility in a region is addressed in Sect. 7.5. Finally, in Sects. 7.6 and 7.7, we extend our discussion beyond merely economic objectives and consider sustainable, resilient, and viable SC network designs.

Learning Objectives for This Chapter

- Understand the importance of selecting the right facility locations
- Describe the main phases of location-related decision-making processes
- Apply quantitative analysis techniques to solve SCD problems
- Compute solutions to different settings of the warehouse location problem
- Compute facility location with the help of Steiner–Weber model
- Use center-of-gravity methods and the Miehle algorithm
- Understand the role of multiple-factor analysis in locating facilities
- Apply factor-ranking method to facility location decisions
- Understand the differences between efficient, resilient, sustainable, and viable SC designs

7.1 Introductory Case Study: Power Pong Sports

In 1856 Alexander Parks produced the first celluloid, a plastic material that can be easily melted and made into different shapes. Originally, Parks was looking for a material similar to ivory to make the production of billiard balls easier and cheaper. In the middle of the twentieth century celluloid was mainly used for film carriers and table tennis balls. However, it has also been an important ingredient in several explosive materials as well as weapons. It is a hazardous material since it is harmful and is spontaneously inflammable.

One of the most commonly used products based on celluloid and sold in high quantities worldwide is table tennis balls. Most of the annual overall production quantity originates in factories in China and Japan. However, table tennis balls are needed all over the world, so significant quantities are exported. For example, in Germany, more than 20 million table tennis balls are sold annually.

Transporting large quantities of products containing significant amounts of celluloid falls into the category of transporting dangerous goods. In 2001, a container loaded with 300,000 table tennis balls is reported to have exploded after being heated by the sun. Due to the aforementioned hazards, more and more countries have banned celluloid in all forms, from production to trading and handling. For these reasons, more and more products are being replaced by celluloid-free substitutes.

Referring to the ongoing worldwide ban on celluloid, Mr. Adham Sharara, chairman of the International Table Tennis Federation (ITTF), announced in 2011 that the ITTF was going to change the international table tennis rules to allow the use of table tennis balls made without any celluloid. After this announcement, some manufacturers of table tennis balls started to redesign their corresponding production

processes in their SCs. It turns out that any waiver of celluloid requires a significant redesign of production processes, and investment in new production facilities and technologies becomes necessary.

One of the major table tennis ball manufacturers is Power Pong Sport (PPS), which is headquartered in China. The most important markets for PPS in Europe (measured in goods sold) are France (FRA), Germany (GER), and the United Kingdom (UK). All other European countries are grouped into the markets of South-Eastern Europe (SEE), South-Western Europe (SWE), and Northern Europe (NEU). Because of the importance of the first three markets, PPS has long-lasting exclusive import contracts with one retailer of sports equipment in France (TriColor Sportive), Germany (TT Profi), and the UK (Competitive Fitness).

PPS has decided to build a new factory in Shanghai exclusively for the new table tennis balls manufactured without celluloid. Mr. Xu Chen is head of the logistics department at PPS and is in charge of new contracts with resellers in Europe to supply the European markets with the new celluloid-free table tennis balls. Mr. Chen has already made two significant decisions, subject to PPS's SC strategy:

- A contract made with a European retailer will extend for 1 year to ensure that significant quantities can be sold.
- Only the most important European markets will receive deliveries directly from PPS. This contributes to the realization of significant economies of scale for the shipment of table tennis balls from Shanghai to Marseille (TriColor Sportive), Bremerhaven (TT Profi), and Felixstowe (Competitive Fitness) by maritime container transportation in completely filled 20-foot containers.

Having monitored these settings, Mr. Chen is now going to identify reasonable ways to ensure that, in addition to the three major markets, all European countries will have access to the new PPS balls. The basic idea is that PPS will extend the contracts made with TriColor Sportive, TT Profi, and Competitive Fitness and distribute rights among these three resellers to deliver PPS balls in SEE, SWE, and NEU countries.

Mr. Chen proposes his idea to the PPS board of managers and gets a "go ahead" since his idea keeps the SC simple. However, Mr. Chen is instructed to keep the annual distribution costs as low as possible. At the same time, his distribution strategy is to ensure that the PPS balls are spread throughout the whole European market.

As a starting point for cost minimization, Mr. Chen contacts three resellers, invites their representatives to his office, and explains his proposal. All three representatives are pretty excited. In a few days, they get the OK from their management. Contract negotiations are established.

Mr. Chen asks the local representatives of the three resellers to estimate their annual transportation costs from the corresponding warehouse to six markets. Since PPS wants to be the first to distribute celluloid-free table tennis balls in Europe, it will cover all transportation costs for the first-year contract. Table 7.1 contains aggregate costs per 1000 € for the first year. The forwarding costs from the selected European resellers to the rest of Europe are independently assessed.

Table 7.1 Annual transportation costs between European resellers and markets as well as fixed annual payments to the European resellers (1000 €)

		All markets					
Reseller	Fixed annual costs	$j = 1$ FRA	$j = 2$ GER	$j = 3$ UK	$j = 4$ SEE	$j = 5$ SWE	$j = 6$ NEU
TriColor Sportive (FRA)	95	4	24	12	23	16	19
TT Profi (GER)	90	24	1	19	11	14	13
Competitive fitness (UK)	70	30	16	3	24	21	17

In order to make cooperation attractive for the three resellers, PPS pays an annual fixed amount to a reseller used to redistribute balls into other European countries, as shown in Table 7.1. Furthermore, PPS does not allow conflicts of interest between the three indirect markets SEE, SEW, and NEU. It guarantees that each of the six aforementioned markets is exclusively assigned to precisely one reseller.

Mr. Chen has a general agreement with the conditions of the three resellers. His next task is to decide about the way(s) in which the PPS balls can reach the six European markets.

Discussion

- Which resellers should PPS use to distribute the celluloid-free table tennis balls in Europe?
- Which reseller(s) should be assigned to SEE, SWE, and NEU, respectively?
- Is it necessary to use more than one reseller to serve a market?
- What happens if one of the resellers is unable to handle the expected demand from SEE, SWE, or NEU?
- Assume that a reseller has to open a new facility to handle the additional demand. Where should this facility be located?
- How can the annual costs be kept as low as possible?

7.2 Supply Chain Design Framework

The core decision to be made in the PPS case was the *selection of markets* (country) in which a base for the PPS operations should be installed. Previous choices influenced this decision, but also impact subsequent decisions. Strategic decisions made by the PPS board regarding the strategy for serving European markets through only a few entry gates were an implication of the strategic position of the PPS SC—to keep distribution costs as low and as simple as possible. These board decisions had to be considered by Mr. Chen during the selection of the gateway country to be used to deliver PPS balls into the six European markets.

Mr. Chen has to consider these guidelines in order to ensure that the *strategic fit* of the PPS distribution system is achieved and preserved. On the other hand, as soon as the gateway markets have been determined, these decisions are binding and must be considered when making subsequent decisions. For example, suppose Germany

was selected as the unique gateway country to serve the European markets. In that case, the planning of all transport operations must include consideration of the available infrastructure, regulations, and laws. Selection of the regions (countries) involved in SC operations is, therefore, an individual decision that is positioned in a sequence of other decision tasks that must be solved in order to set up and use the distribution network as part of the PPS SC.

In Fig. 7.1, a modified SCD framework from (Chopra and Meindl 2012) is presented to show the arrangement of all decision tasks required to set up and deploy the SC.

In this framework, the SC strategy is understood as the first decision phase. We refer to Chap. 4, Supply Chain Strategy, regarding this phase. Phase I addresses the compilation of those sourcing or supply regions/markets that are considered subject to the long-term fulfillment of market demand. Such a region might cover a continent, a country, an area of cultivation, or a fishing ground. Also, the segmentation of the total (global) market varies from local areas to continents. Phase I provides a list of geographic regions that could be promising for the setup of the SCD. In addition, suggestions for the installation of transportation links between supply and demand regions are provided by phase I.

In phase II, all regions are processed individually. In each region, specific locations (expressed by coordinates) are identified at which a facility can be installed or from which supply quantities can infiltrate the SC. The output of phase II is a shortlist of candidates for locations (with existing facilities or with prospective facilities) that have been selected for cost efficiency.

In phase III, the final selection of the locations to be considered in the SCD is carried out. Multiple criteria are important at this stage, and the analysis goes far

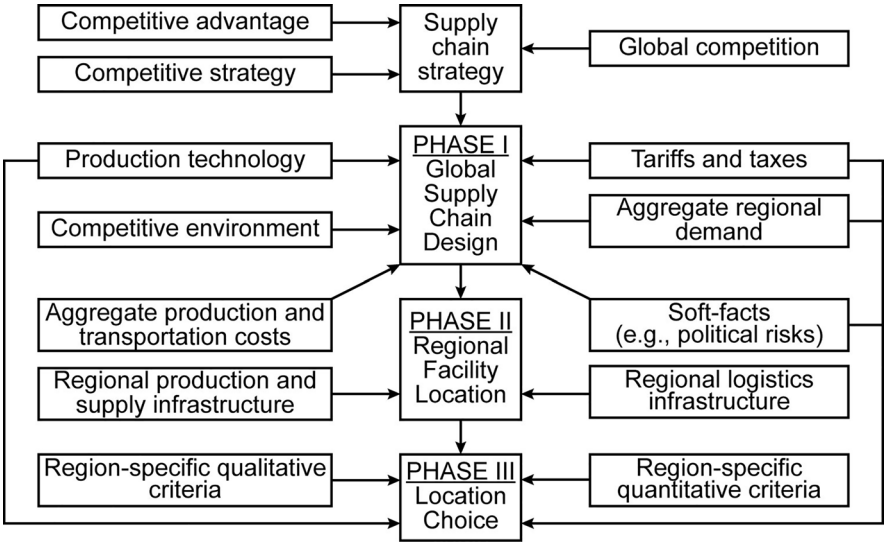


Fig. 7.1 Supply chain design framework (based on Chopra and Meindl 2012)

beyond cost minimization. Furthermore, capacities are allocated at all locations that are considered to contribute to fulfilling demand from the markets to be served.

Decisions associated with phases I–III fall into the category of strategic (long-term) SCD decisions. A revision of these decisions is hardly possible or implies significant costs. This chapter addresses these three decision phases of the location planning framework. We discuss and investigate tools to support SC managers in identifying the best decision alternatives. In Sect. 7.3, we introduce the *warehouse location problem* as a representative example of a core decision task in the selection of contributing regions (phase II). Tools for supporting the derivation of adequate location selection decisions are the focus of Sect. 7.4 (phase III). Decisions associated with phase III are considered in Sect. 7.5.

7.3 Global Supply Chain Design

Phase I of the location analysis framework addresses the selection of source regions that are incorporated into the overall SC. From the sourcing regions selected, markets are served such that the overall supply from the different sources covers the demand of each individual market. The PPS case is a typical representation of a phase I decision situation. By means of the PPS case presented in Sect. 7.1, we develop an appropriate mathematical optimization model (Sect. 7.3.1) as the starting point of rational decision-making that incorporates computational resources to derive optimal location decisions.

For solving this so-called (uncapacitated) *warehouse location problem* (WLP), we explain the usage of a spreadsheet approach (Sect. 7.3.2) as well as the configuration of a general purpose decision support algorithm called *branch-&-bound* (Sect. 7.3.3). Finally, we investigate the consequences of scarceness, i.e., the limited availability of the quantities offered in source regions, leading to the definition of the *capacitated warehouse location problem* (CWLP). For the CWLP, we also propose an appropriate mathematical optimization model and a decision support technique (Sect. 7.3.4).

7.3.1 Warehouse Location Problem and Its Formalization

The case of PPS is a representative example of a frequent decision situation related to SCD. First, a decision about regions (markets, countries) hosting a facility (factory, warehouse, partner) to be used as a network node is required ("location decision"). Second, a decision must also be made as to how those markets are supplied from a regional facility so that the demand of each market is covered ("supply decision"). Such a problem requiring both location and supply decisions is called a *warehouse location problem* or WLP (Daskin 1995; Drezner 1995; Melo et al. 2009; Askin et al. 2014). The overall goal to be achieved in solving a WLP is to keep the total costs of supplying all regions as low as possible. Therefore, for each region, it has to be decided if it is more beneficial to open a facility (creating fixed costs for

running the facility) or use transportation links to supply a region (creating transportation costs).

Let S denote the set of all regions in which a facility/warehouse can be installed or used (e.g., S = {GER; FRA; UK}) and let M be the set of all markets (e.g., M = {GER; FRA; UK; SEE; SWE; NEU}). The set $T := S \times M$ contains all possible transportation links between buyers and suppliers. If a facility $s \in S$ is opened then the annual costs rise by the amount f_s. The corresponding decision is coded again by the binary decision variables y_s ($s \in S$), which equals 1 if a facility is open, and 0—otherwise. The decision to use the transportation link $(s, m) \in T$ between the facility $s \in S$ and the market $m \in M$ increases the annual costs by the additional amount c_{sm}. Using the aforementioned sets, we are able to formally present the WLP as follows.

First, the *objective function* (7.1) is formulated:

$$Z = \sum_{s \in S} f_s \cdot y_s + \sum_{s \in S} \sum_{m \in M} C_{sm} \cdot x_{sm} \tag{7.1}$$

The sum of (annual) costs expressed in Eq. (7.1) has to be minimized by varying the values of the *decision variables* y_s and x_{sm}. The family y_s of binary decision variables represents the facility opening decisions. All these decision variables are allowed to be set to either 1 ("use this facility") or 0 ("do not use this facility"). Similarly, x_{sm} code the decisions about whether to use the transportation links in T between warehouses and markets. Although the two decision categories introduced to address different managerial decisions, they fall into the same type of decisions: exactly one of two options must be selected (binary decisions). Therefore, the WLP turns out to be a collection of interdependent binary decisions about opening the locations.

Each market has to be served from exactly one facility, as is the case in the PPS example. In order to ensure this condition when fixing the values for the x-decision variables, it is necessary to ensure that constraint (7.2) is respected.

$$\sum_{s \in S} x_{sm} = 1, \forall m \in M \tag{7.2}$$

If (7.2) remains unfilled, then at least one market in M remains unserved. Since the overall sum of costs for supplying all markets must be minimized: every solution in which a market $m \in M$ is connected with two or more facilities implies higher costs, and selecting one of these facilities for serving the markets may reduce costs. In summary, in a cost optimal solution of the WLP each market is supplied from exactly one facility $s \in S$.

Obviously, it is useless to install a transport link between market m and facility s if s is not opened, e.g., if we set $x_{sm} = 1$ if, and at the same time, $y_s = 0$, then we would end up with a useless and unrealizable solution for the WLP. In order to avoid such a failure, we introduce the constraints (7.3) and (7.4) that couple facility installation with transport link installation decisions and ensure that we install a transport link only if it has been decided that the origin facility should also be installed.

$$x_{sm} \leq y_s, \forall s \in S, \forall m \in M \tag{7.3}$$

$$y_s \in \{0;1\} \forall s \in S, x_{sm} \in \{0;1\} \forall (s,m) \in T \tag{7.4}$$

Using the mathematical model (7.1)–(7.4), we are now ready to state precisely the WLP problem as follows:

It is necessary to minimize the total costs for the installation of facilities and transportation links subject to Eq. (7.1), so that each market is served by exactly one facility (7.2). If we use a facility for supplying a market, then this facility must be open (7.3). Each available facility is either opened or closed and each available transportation link is either used or not (7.4).

A pure, formalized problem formulation is as follows: "minimize (7.1) while taking into account (7.2)–(7.4). The collection of mathematical expressions (7.1)–(7.4) is a mathematical model for the WLP. This model represents the underlying decision problem formally. A solution to this model is comprised of a selection of values for each of the *y*-decision variables as well as each of the *x*-decision variables. Such a solution is called *feasible*, if and only if, all constraints (7.2)–(7.4) are fulfilled, e.g., if the implementation of the selected values for the decision variables leads to logically true statements. Every feasible solution of the proposed WLP model that leads to a non-dominated objective function value is called an optimal solution of the WLP model. Such an optimal solution can be used to derive an optimal solution to the underlying real-world WLP.

If we want to use the WLP model to represent Mr. Chen's problem in the PPS case, we first have to collect all relevant planning data. The set S of potential regions hosting a facility is compiled as S = {GER; FRA; UK} and the set of markets M equals {GER; FRA; UK; SEE; SWE; NEU}. Consequently, the set of transportation links is formed as T = {(GER;GER); (GER;FRA); (GER;UK); (GER;SEE); (GER;SWE); (GER;NEU);(FRA;GER); (FRA;FRA); (FRA;UK); (FRA;SEE); (FRA;SWE); (FRA;NEU); (UK;GER); (UK;FRA); (UK;UK); (UK;SEE); (UK; SWE); (UK;NEU)}. The cost coefficients representing the fixed annual costs for an opened facility are as follows (cf. Table 7.1): f_{GER} = 95, f_{FRA} = 90 and f_{UK} = 70. Finally, the cost coefficients for the annual distribution costs in Europe are $c_{GER;GER}$ = 1; $c_{GER;FRA}$ = 24; $c_{GER;UK}$ = 19; $c_{GER;SEE}$ = 11; $c_{GER;SWE}$ = 14; $c_{GER;NEU}$ = 13;$c_{FRA;GER}$ = 24; $c_{FRA;FRA}$ = 4; $c_{FRA;UK}$ = 12; $c_{FRA;SEE}$ = 23; $c_{FRA;SWE}$ = 16; $c_{FRA;NEU}$ = 19; $c_{UK;GER}$ = 16; $c_{UK;FRA}$ = 30; $c_{UK;UK}$ = 3; $c_{UK;SEE}$ = 24; $c_{UK;SWE}$ = 21; $c_{UK;NEU}$ = 17.

Mr. Chen puts all the aforementioned data into the general WLP model and obtains the following mathematical optimization model that represents his SCD problem in Europe as mixed integer linear programming (MILP) model by (7.5)–(7.13):

$$\begin{aligned} \textit{Minimize } Z = \; & 90y_{GER} + 95y_{FRA} + 70y_{UK} + 1x_{GER;GER} \\ & +24x_{GER;FRA} + 19x_{GER;UK} + 11x_{GER;SEE} + 14x_{GER;SWE} \\ & +13x_{GER;NEU} + 24x_{FRA;GER} + 4x_{FRA;FRA} + 12x_{FRA;UK} \\ & +23x_{FRA;SEE} + 16x_{FRA;SWE} + 19x_{FRA;NEU} + 16x_{UK;GER} \\ & +30x_{UK;FRA} + 3x_{UK;UK} + 24x_{UK;SEE} + 21x_{UK;SWE} + 17x_{UK;NEU} \end{aligned} \tag{7.5}$$

So that

$$x_{GER;GER} + x_{FRA;GER} + X_{UK;GER} = 1 \tag{7.6}$$

$$x_{GER;FRA} + x_{FRA;FRA} + X_{UK;FRA} = 1 \tag{7.7}$$

$$x_{GER;UK} + x_{FRA;UK} + X_{UK;UK} = 1 \tag{7.8}$$

$$x_{GER;SEE} + x_{FRA;SEE} + X_{UK;SEE} = 1 \tag{7.9}$$

$$x_{GER;SWE} + x_{FRA;SWE} + X_{UK;SWE} = 1 \tag{7.10}$$

$$x_{GER;NEU} + x_{FRA;NEU} + X_{UK;NEU} = 1 \tag{7.11}$$

$$\begin{aligned}
&x_{GER;GER} \text{ " } y_{GER}, x_{GER;FRA} \text{ " } y_{GER}, x_{GER;FRA} \text{ " } y_{GER},\\
&x_{GER;UK} \text{ " } y_{GER}, x_{GER;SEE} \text{ " } y_{GER}, x_{GER;SWE} \text{ " } y_{GER},\\
&x_{GER;NEU} \text{ " } y_{GER}, x_{FRA;GER} \text{ " } y_{FRA}, x_{FRA;FRA} \text{ " } y_{FRA},\\
&x_{FRA;UK} \text{ " } y_{FRA}, x_{FRA;SEE} \text{ " } y_{FRA}, x_{FRA;SWE} \text{ " } y_{FRA},\\
&x_{FRA;NEU} \text{ " } y_{FRA}, x_{UK;GER} \text{ " } y_{UK}, x_{UK;FRA} \text{ " } y_{UK}, x_{UK;UK} \text{ " } y_{UK},\\
&x_{UK;SEE} \text{ " } y_{UK}, x_{UK;SWE} \text{ " } y_{UK}, x_{UK;NEU} \text{ " } y_{UK}
\end{aligned} \tag{7.12}$$

$$\begin{aligned}
&y_{GER} \in \{0;1\}, y_{FRA} \in \{0;1\}, y_{UK} \in \{0;1\}, x_{GER;GER} \in \{0;1\},\\
&x_{GER;FRA} \in \{0;1\}, x_{GER;UK} \in \{0;1\}, x_{GER;SEE} \in \{0;1\}\\
&x_{GER;SWE} \in \{0;1\}, x_{GER;NEU} \in \{0;1\}, x_{FRA;GER} \in \{0;1\},\\
&x_{FRA;FRA} \in \{0;1\}, x_{FRA;UK} \in \{0;1\}, x_{FRA;SEE} \in \{0;1\}, x_{FRA;SWE} \in \{0;1\},\\
&x_{FRA;NEU} \in \{0;1\}, x_{UK;GER} \in \{0;1\}, x_{UK;FRA} \in \{0;1\}, x_{UK;UK} \in \{0;1\}, x_{UK;SEE} \in \{0;1\},\\
&x_{UK;SWE} \in \{0;1\}, x_{UK;NEU} \in \{0;1\}
\end{aligned} \tag{7.13}$$

The fairly small SCD problem in the PPS case is represented by the mathematical model (7.5)–(7.13). Although only three potential facility sites and six markets are involved in the problem, the PPS case is quite a complex decision situation, and the proposed model is so complex that it is impossible to solve it manually. We need to utilize the support of a computer system.

7.3.2 A Spreadsheet Approach to the WLP

Even for a relatively small WLP case such as the PPS scenario, it is hardly possible to identify a minimal cost instantiation of the y_i- and the x_{ij}-values. A spreadsheet calculation schema as shown in Fig. 7.2 is a first step to managing the problem's inherent complexity and determining the costs of different decisions.

The upper part of this spreadsheet (rows 5–10) contains all relevant problem data, which are the annual fixed costs for each reseller (column C) and the costs for installing a transportation link from a reseller into a market (columns D–I).

In the middle part of the calculation scheme (rows 14–19), setup decisions can be typed into the gray shaded cells. The area covering cells C17–C19 is used to code

	A	B	C	D	E	F	G	H	I
1									
2									
3	PowerPong Sports – Distribution Costs								
4									
5				All markets					
6	i	re-sellers	annual fixed costs	j=1	j=2	j=3	j=4	j=5	j=6
7				FRA	GER	UK	SEE	SWE	NEU
8	1 (FRA)	TriColor Sportive (FRA)	95	4	24	12	23	16	19
9	2 (GER)	TT Profi (GER)	90	24	1	19	11	14	13
10	3 (UK)	Competitive Fitness (UK)	70	30	16	3	24	21	17
11									
12	Decisions about incorporated re-sellers and established delivery links from re-sellers into markets								
13									
14				delivery links (x_{ij})					
15	i	re-sellers	y_i	j=1	j=2	j=3	j=4	j=5	j=6
16				FRA	GER	UK	SEE	SWE	NEU
17	1 (FRA)	TriColor Sportive (FRA)							
18	2 (GER)	TT Profi (GER)							
19	3 (UK)	Competitive Fitness (UK)							
20	number of used sources			0	0	0	0	0	0
21									
22	Cost calculation (all values in thousand EUR)								
23									
24	annual fixed costs from selected re-sellers					0			
25	re-distribution costs					0			
26	total costs					0			

Fig. 7.2 Cost calculation sheet for the PPS-case

the decisions associated with incorporating a reseller. When "1" is inserted, then the corresponding reseller is considered, but if a "0" is typed in, then this reseller remains unconsidered. For example, if TriColor Sportive is considered in the PPS European distribution system, then cell C17 is filled with "1." Similarly, "1" and "0" are typed into the cell area D17–I19 to code the decisions on the installed transportation links from the resellers into the six markets. For each market, the number of selected resellers is calculated in row 20.

The costs for a coded collection of decisions are calculated in the lower part of the spreadsheet (rows 24–26). The corresponding formulas are given in Table 7.2.

Theoretically, one might use the spreadsheet to test every possible combination of "0" and "1" values in the gray-shaded cells. Unfortunately, there are $2^{21} = 2{,}097{,}152$ different combinations. First, each combination has to be checked for feasibility concerning the constraints (7.6)–(7.13). Second, for each feasible combination, the

Table 7.2 Payments to the European resellers (1000 €)

Cell	Formulas	
F24	=SUMPRODUCT(C17:C19;C8:C10)	Sum of annual fixed costs
F25	=SUMPRODUCT(D17:I19;D8:I10)	Annual link costs
F26	=F24 + F25	Total annual costs

associated costs have to be calculated using Eq. (7.5). If we assume that these two steps can be executed within 1 s, then it takes 2.5 days to find the best feasible solution for the PPS-WLP setting. Therefore, testing all combinations manually is impractical. Fortunately, spreadsheet calculation tools (e.g., Microsoft Excel, Open Office) provide special add-ins that assist us in testing different combinations. One add-in is called "*Solver.*"

The concept of the solver tool is as follows. The user specifies those cells that are variable (here: the gray-shaded cells) and the cell in which the objective function value is contained (here: cell F26), and then the solver proposes a first set of values and inserts these values tentatively into the variable cells. After this, the solver reads the value in the objective function cell. Using optimization algorithms, the solver can now decide if the proposed values form an optimal solution. In the first case, the solver returns the current solution proposal by definitively inserting the current values into the variable cells. In the latter case, the solver iterates the recent proposal and re-evaluates this new proposal by reading the updated objective function cell. This iteration is repeated by the solver until an optimal combination of values for the decision variables is found or if it decides that no better solution is available.

Note: An Excel file pre-configured with the PPS data and a suitable solver configuration can be found in the E-Supplement. The Excel solver is an add-in that comes with each Excel installation. In order to use this add-in, it is necessary to activate it before first use. Open Office also includes a solver tool

Before you can apply the solver add-in to the PPS case, you must execute a careful configuration to provide the solver with comprehensive problem information, e.g., input the model parameter and decision variables. Figure 7.3 exhibits the necessary parameter settings.

First, the cell that contains the objective function value (or in which the objective function value is calculated) is inserted into the "set objective" input box. Here, cell $F$26 contains the target value that is the subject of optimization.

Second, the optimization goal is specified, e.g., whether the objective function value is going to be minimized or maximized, or if a certain target value must be achieved is established. In the PPS case, the totals calculated in cell $F$26 are minimized.

Third, those fields that should be modified by the solver ("variable cells") must be specified. Here, all cells in the area $C$17:$I$19 can be varied by the solver. The area $C$17:$C$19 represents the decision variable vector (y_{FRA}, y_{GER}, y_{UK}), whereas the area $D$17 to $I$19 corresponds to the matrix of the x_{ij}-decision variables.

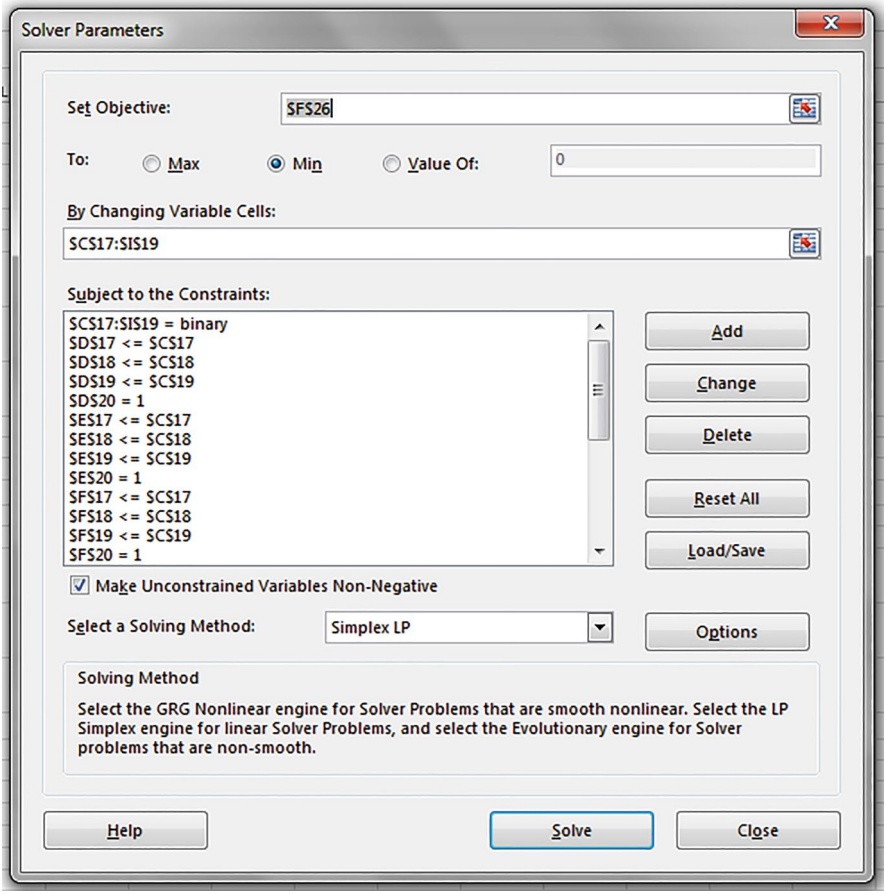


Fig. 7.3 Parameter settings for the solver

Fourth, you need to specify the constraints that must be considered/that belong to the model. Constraints are typed in one after another by clicking the add button in the solver parameter window. Figure 7.4 shows an example of how the constraint $x_{\text{GER; NEU}} \leq y_{\text{GER}}$ from Eq. (7.12) is specified.

For the input of the constraints (7.2), the solver refers to cells D20–I20 where the sum on the left side of Eq. (7.2) is calculated (Fig. 7.5).

If necessary, the decision variable domains must be restricted as shown in Fig. 7.6 to limit the values of y_i and x_{ij} to either 0 or 1.

Finally, it is necessary to specify the optimization algorithm that the solver invokes to solve the specified mathematical model. For the PPS model, we select the option simplex algorithm since the model (7.1)–(7.4) consists of a linear objective function and constraints. The solver detects that the decision variables may only be instantiated with the values 0 or 1.

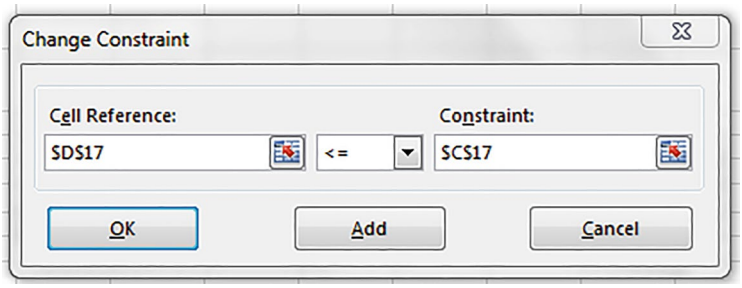


Fig. 7.4 Specification of constraints

Fig. 7.5 Compilation of constraints

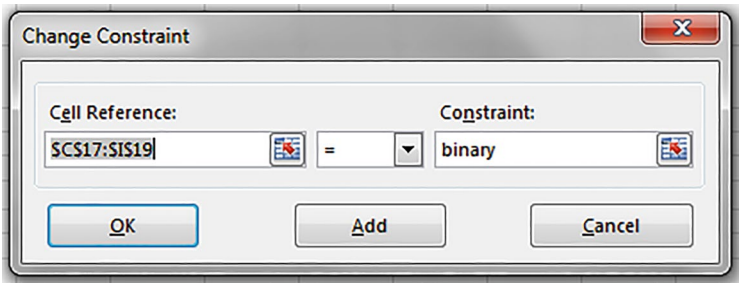


Fig. 7.6 Specification of the decision variable's domains

When the five steps mentioned above have been executed, the solving process is initiated by clicking on the "Solve" button. The optimization routines then start the search process and write the final solution proposal into the variable cells.

In Fig. 7.7 the values returned for the PPS case have been written into the variable cells.

The optimal design of the PPS-network in Europe is to agree to the contract with the reseller in Germany ($y_2 = 1$) and to forward balls from Germany to the five remaining markets ($x_{2j} = 1$ for all $j = 1, \ldots, 6$). The annual costs for this design account to 172 TEUR (1000 euro) p.a. and this sum includes the amount of 80

	A	B	C	D	E	F	G	H	I
1									
2									
3	PowerPong Sports – Distribution Costs								
4									
5			annual	All markets					
6	i	re-sellers	fixed	j=1	j=2	j=3	j=4	j=5	j=6
7			costs	FRA	GER	UK	SEE	SWE	NEU
8	1 (FRA)	TriColor Sportive (FRA)	95	4	24	12	23	16	19
9	2 (GER)	TT Profi (GER)	90	24	1	19	11	14	13
10	3 (UK)	Competitive Fitness (UK)	70	30	16	3	24	21	17
11									
12	Decisions about incorporated re-sellers and established delivery links from re-sellers into markets								
13									
14				delivery links (x_{ij})					
15	i	re-sellers	y_i	j=1	j=2	j=3	j=4	j=5	j=6
16				FRA	GER	UK	SEE	SWE	NEU
17	1 (FRA)	TriColor Sportive (FRA)	0	0	0	0	0	0	0
18	2 (GER)	TT Profi (GER)	1	1	1	1	1	1	1
19	3 (UK)	Competitive Fitness (UK)	0	0	0	0	0	0	0
20	number of used sources			1	1	1	1	1	1
21									
22	Cost calculation (all values in thousand EUR)								
23									
24	annual fixed costs from selected re-sellers					90			
25	re-distribution costs					82			
26	total costs					172			
27									

Fig. 7.7 Returned optimal solution for the PPS case

TEUR paid to TT Profi as a fixed annual amount to cover all the reseller's expenses for handling the intra-European orders, as well as the costs for the transportation links, which account for 92 TEUR p.a. This proposal reduces the total annual costs so that Mr. Chen will go on to sign the contract with TT Profi, but not the contracts with TriColor Sportive in France. In addition, no contract with Competitive Fitness in the UK will be signed.

From the methodological perspective, we conducted three steps consecutively to solve the real-world decision situation:

- The real-world decision situation is coded into a mathematical decision model ("modeling step").
- The formalized decision representation (the mathematical decision model) is processed by an optimization algorithm that returns an optimal solution of the model ("model solving step").

- The model solution is interpreted as a decision proposal for the real-world problem ("model solution implementation step").

The problem-solving approach reported here is a typical example of a *model-based decision approach*. The major challenge was representing the WLP using a mathematical model. Due to the *linear* structure of the objective function and the constraints, we were able to apply a black-box model-solving tool to derive the model solution.

Exercise
PPS's management board of directors has decided to redesign the market segmentation in Europe. The former market region SWE has been split into South-Western Europe I (SWE-I) and South-Western Europe II (SWE-II). SWE-I now demands one-third of the former SWE market. The annual costs for installing a transportation link are correspondingly split between SWE-I and SEW-II. Is it necessary to redesign the network in Europe?

Assume that SWE remains complete. However, due to the recent success of the Portuguese national table tennis teams, the Portuguese market has grown rapidly. Situated in this emergent market, an additional potential reseller offers PPS the following conditions for cooperation: Annual costs are fixed at 60,000 €, and for each of the six markets, equal forwarding costs of 20 € per packaging unit are possible. Is it beneficial for PPS to consider this fourth reseller candidate for its European distribution network?

7.3.3 Branch-&-Bound: How the Solver Add-In Works

So far we have learned how to code a WLP into a mathematical optimization model. Furthermore, we have learned how we can deploy a black-box tool such as Excel Solver to derive a high-quality (optimal) solution for the model.

Obviously, the knowledge of an appropriate decision model type is sufficient to solve the real-world challenge of the WLP. However, so far, we do not know how the black-box model solver processes the specified model, nor do we have a chance to judge the optimality of the proposed model solution. In this section, we will learn the basic principles of a model-solving algorithm that serves as the base for many decision model solvers. This technique is called *branch-&-bound* (b&b).

As the name b&b suggests, there are two interdependent activities for processing the model. One activity is to split the problem into smaller problems (*branching*), and the second activity is to estimate the best objective function value that may be achieved by solving a model of such a specific "branch" of the overall problem (*bounding*).

In this section, we explain an efficient branching strategy for a WLP model and an efficient bounding scheme to determine the lower bound of the objective function associated with a model for each splitting up of a WLP sub-problem. Although the idea of b&b is applied to a fairly comprehensive collection of decision models for a

large number of decision problems, both the branching strategy and the bounding scheme have to be adjusted for any specific model.

Let us begin the motivation and explanation of b&b algorithms with a simple observation that is valid for the WLP model of PPS. Independently from the decisions on the reseller selection, we can determine the least required transportation linkage costs in the following way. First, select a minimal value in each column associated with a target market in Table 7.1. Second, calculate the sum $Z(---)$ of the six selected values, e.g., $Z(---) = 4 + 1 + 3 + 11 + 14 + 13 = 46$ TEUR. This value is needed to maintain the distribution network independently from the incorporated resellers. The fact that no reseller is selected is expressed by the three dashes "---" where the first dash represents the selection to incorporate the first reseller ("1") or the decision to refrain from cooperating with the first reseller ("0").

We start the configuration of a b&b algorithm for the WLP with the derivation of an adequate branching strategy. This strategy recursively splits the model $M(---)$ of the given WLP model into two sub-models $M(1--)$ and $M(0--)$.

- In the model $M(1--)$, we add the constraint that the first supplier must be incorporated ($y_1 = 1$).
- In the model $M(0--)$, we postulate the constraint that the first supplier may not be considered ($y_1 = 0$).

Note that no solution of the WLP model remains unconsidered if we handle $M(1--)$ and $M(0--)$ instead of the initial WLP model, since y_1 equals either 1 or 0. However, one might now ask what the benefit of dealing with two models instead of one is. In order to justify this model split, we analyze the consequences of the model replacement.

The recently generated models $M(1--)$ and $M(0--)$ are "smaller" than the original model $M(---)$, since one of three requested decisions has already been made and only two decisions (about the second as well as the third reseller) must be solved.

The cost estimation value $Z(1--)$ associated with $M(1--)$ as well as the cost estimation value $Z(0--)$ associated with $M(0--)$ can be refined compared to the cost estimation $Z(---)$ associated with $M(---)$.

In the sub-problem represented by $M(1--)$, the first reseller is incorporated, and therefore the annual fixed cost of $f_1 = 95$ TEUR definitely has to be paid in addition to the least transportation costs $Z(---) = 46$ TEUR so that we get $Z(1--) = 46$ TEUR +95 TEUR = 141 TEUR: each decision alternative in which the first reseller is incorporated ($y_1 = y_{FRA} = 1$) creates annual costs of at least 141 TEUR.

In the sub-problem represented by $M(0--)$, the first reseller is not incorporated. Therefore, no transportation link can originate from this reseller ($x_{ij} = x_{FRAj} = 1, \forall j \in$ {FRA; GER; UK; SEE; SWE; NEU}). In particular, the link FRA to FRA cannot be used. The market FRA must be served from another reseller. Because of this, we cannot consider these transportation links for calculation of the least transport costs among the resellers and the markets. From the remaining links that can be used to serve FRA, we select the cheapest link, which is (GER;FRA). Since the linkage

costs $c_{FRA;FRA} = 4$ but $c_{GER;FRA} = 24$, the least transportation cost sum increases by 20 TEUR p.a. so that we have $Z(0\text{--}) = 46 - 4 + 24 = 46 + 20 = 66$ TEUR p.a.

Having generated the two models $M(1\text{--})$ and $M(0\text{--})$ by deciding about the incorporation of reseller 1, we continue with the decision about consideration of reseller 2. Both sub-problems represented by $M(1\text{--})$ as well as $M(0\text{--})$ are therefore split into two smaller problems $M(11\text{-})$ and $M(10\text{-})$, respectively M(01-) and M(00-). Each of these four sub-problems is then split again to incorporate reseller 3 into the cost evaluation (Fig. 7.8).

The tree structure of all possible sub-problems in the PPS case is shown in Fig. 7.8. Each node in the tree represents one sub-problem determined by combinations of involved/rejected/so far undecided resellers. A node represents an incomplete solution of the WLP where at least one reseller remains undecided. A node represents a solution of the WLP if there is a decision made for each reseller. The nodes in the first three upper levels of the tree in Fig. 7.8 represent incomplete solutions, but the nodes in the lower level represent the different solutions of the WLP. The solution of the model $M(000)$ is infeasible because no reseller is used, and this is impossible. The remaining seven solutions are feasible, and the solution of M (010), with a total cost of 172, is the cheapest solution available. Excel Solver has returned the same solution.

Overall, it was necessary to set up and evaluate 15 sub-problems to solve the PPS case. In general, WLPs come along with a significantly larger number of potential locations corresponding to resellers (or suppliers). Let N be the number of supply sources so that the number of sub-problems equals $1 + 2 + 2^2 + 2^3 + \ldots + 2^N$. For example, let $N = 5$ which requires the specification and evaluation (cost estimation) of $1 + 2 + 4 + 8 + 16 + 32 = 63$ sub-problems and if $N = 10$ then the number of sub-problems climbs up to $1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + 512 + 1024 = 2047$. The time needed to solve all these sub-problems is prohibitively high. However, we can use the objective function value of the first solution found during a b&b execution to reduce the number of sub-problems that must be set up and

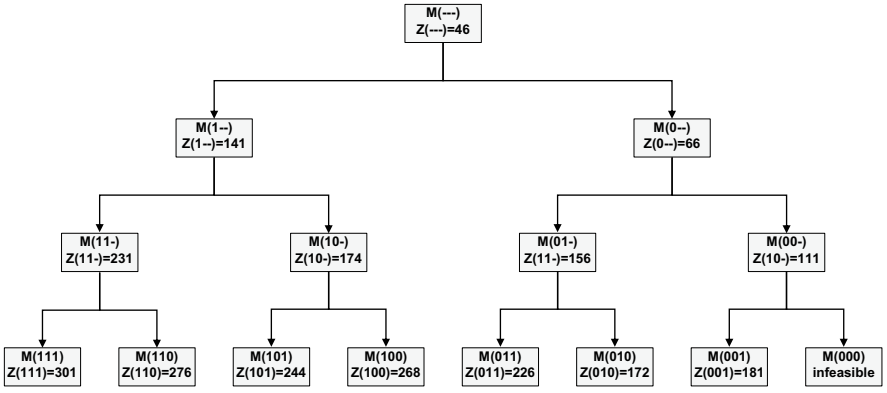


Fig. 7.8 Tree of sub-problems and cost estimations as well as solutions for the PPS case

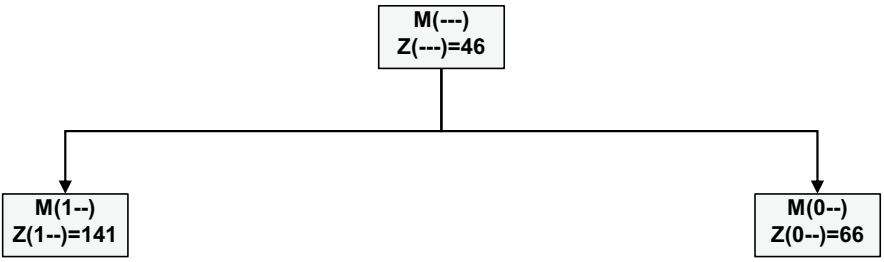


Fig. 7.9 Tree after the first branching step

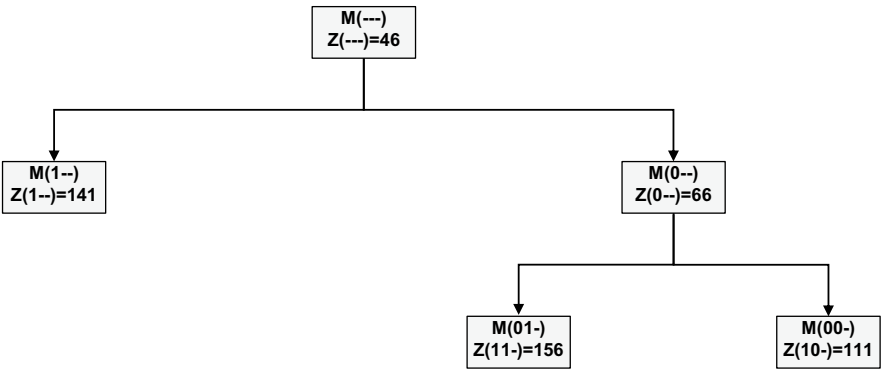


Fig. 7.10 Tree after the second branching step

evaluated. For this reason, we analyze the sequence in which the sub-problems are generated (Figs. 7.9, 7.10, 7.11, 7.12, and 7.13).

After the first branching step on the decision about the incorporation of reseller 1, we have two sub-problems. We now select the sub-problem with the lower cost estimation and assume that the first reseller remains unconsidered (Fig. 7.10).

The right-hand sub-problem represented by $M(0\text{--})$ has a lower cost estimation value and will be split up next. For this reason, we generate the sub-problems 01- and 00- and determine the lower bound of the costs (Fig. 7.11).

Figure 7.11 contains the current stage of the decision tree. The right-hand sub-problem represented by $M(000)$ is infeasible, but the left-hand sub-problem corresponds to the first complete and feasible solution with costs of 181. We set the current least cost to Z^{least}:=181 TEUR. Since both sub-problems 001 and 000 cannot be split up further, it is necessary to go back to the previous level and investigate the open sub-problem represented by $M(01\text{-})$. Such a step back into a higher level is called a backtracking step (Fig. 7.12).

After backtracking has been done, we expand the sub-problem 01- and obtain two new solutions for the WLP. The costs from solution 011 are 226 TEUR and do not improve the least cost Z^{least} of the best solution found so far. Solution 010 creates costs of 172 TEUR, which improves Z^{least} so that we may update Z^{least} and set Z^{least}:=

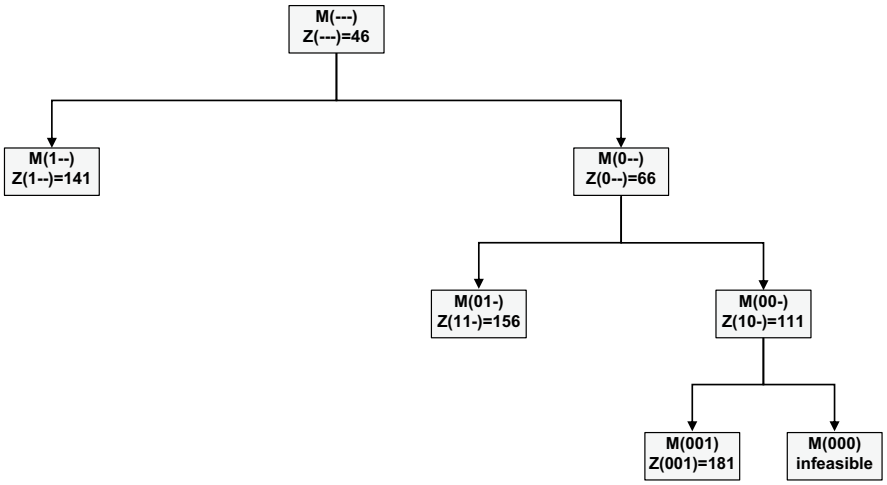


Fig. 7.11 Tree after the third branching step

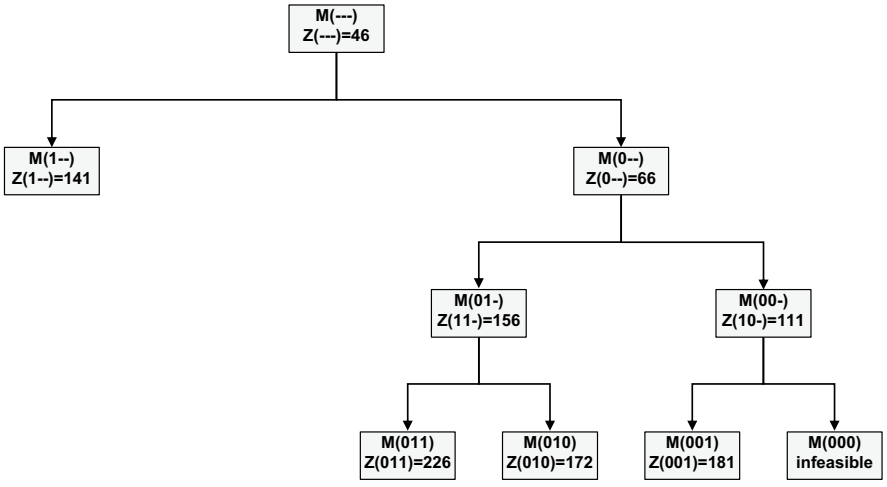


Fig. 7.12 Tree after backtracking and fourth branching step

172 TEUR, tracking back to level three. Since there are no open sub-problems left in level three which have not been processed, we track back into level 2 and process the open sub-problem 1-- (Fig. 7.13).

We generate sub-problem 11- as well as sub-problem 10- and determine the lower bound of the costs (Fig. 7.13). For both sub-problems, the lower bounds 231 and 174 exceed the costs 172 of the least cost solution found so far. Therefore, it is not necessary to analyze these two sub-problems any further since it is impossible to find an improved solution for the WLP model.

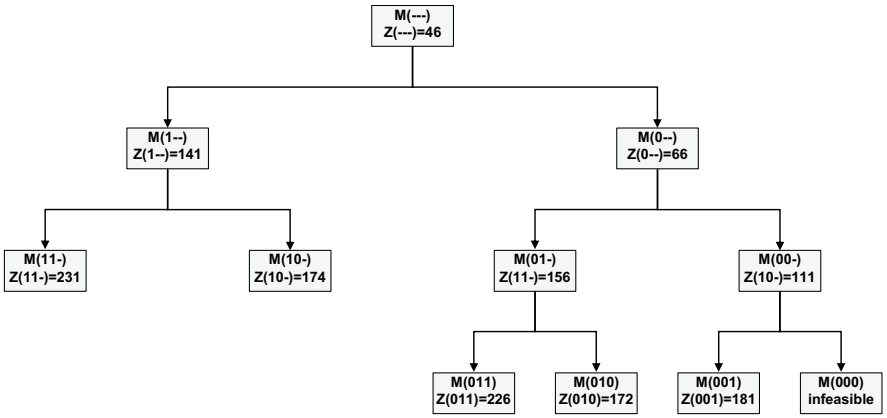


Fig. 7.13 Tree after double backtracking and fifth branching step

As we see from the final tree in Fig. 7.13, it is not necessary to generate all possible sub-problems if we save the best-found solution so far.

▶ **Practical Insights** For small instances, b&b is a suitable method. However, the solution to real-life problems involves a higher complexity. Thus, we have to solve the WLP using different heuristic methods (offering no guarantee of finding the best available solution) rather than exact algorithms. The reason for this is the high number of integer variables. However, heuristic procedures offer no guarantee of finding an optimal solution.

7.3.4 Capacitated WLP

We now investigate the impacts of limited capacities on SCD. Therefore, it is necessary that we know the quantities q_s ($s \in S$) that can be handled by each reseller per year. Furthermore, in order to assign a sufficient quantity to each reseller, we need to have the demand quantity d_m of each market ($m \in M$). For the PPS case, the maximal quantities that can be handled by a reseller are summarized in Table 7.3. Table 7.4 shows the expected annual demand from the six markets. All values are expressed in a number of packing units (parcels).

Of course, there is no reseller solely able to serve the total demand of 23,300 packing units. Therefore, at least two resellers should be incorporated. Consequently, at least a second annual fixed handling amount has to be paid. In contrast to the uncapacitated WLP, an optimal solution to the case investigated here called the CWLP or Capacitated Plant Location Model (CPLM), differs from a WLP solution because more than one reseller is considered in an optimal solution.

Again, we have to decide on the incorporation of resellers, so the corresponding decision is coded again by the binary decision variables y_s ($s \in S$). In the WLP we

Table 7.3 Annual handling quantities (packing units) of the reseller/market

s	q_s
TriColor sportive (FRA)	20,000
TT Profi (GER)	20,000
Competitive fitness (UK)	15,000

Table 7.4 Annual demand of six markets packing units

m	d_m
FRA	5000
GER	8500
UK	3200
SEE	2800
SWE	2100
NEU	1700

use the binary decision variable family x_{sm} to represent decisions about installing a transportation link connection supplier (reseller) s with market m. In the context of the CWLP, it is also necessary to determine the quantity shipped along a transportation link. Therefore, we use the family of non-negative decision variables z_{sm} to represent decisions about the installation and usage of transportation links. If z_{sm} equals 0, no link is installed to connect supplier s with market m. If z_{sm} is larger than 0, then the value z_{sm} is interpreted as the quantity shipped along the transportation link originating from supplier s and terminating in market m.

We have used the annual fixed cost value c_{sm} to determine the annual costs for incorporating the transportation link between supply location s and market m. This parameter determines the annual costs for moving the whole demand of the target market m from the supply source s. In the CWLP, we aim to distribute the total demand of a market among different transportation links; it is necessary to know the costs c'_{sm} for moving one package unit along the transportation link, connecting supply source s with market m. We can calculate the values for c'_{sm} from the values contained in Table 7.1 and explain this as an example of the transportation link between the UK reseller and the market SEE. Here, the total costs for delivering the market's demand of 2800 package units are 24,000 € per year, so each package unit shipped between the supplier in the UK and the market SEE has a cost of 24,000 €/2800 package units ≈ 8.57 €. Table 7.5 summarizes the shipment costs c'_{sm} per package unit.

Using the transportation cost coefficients per package unit as well as the decision variables z_{sm} for the shipped quantities, we can determine the overall distribution costs Z^{CWLP} per year according to Eq. (7.14).

$$Z^{CWLP} = \sum_{s \in S} f_s \cdot y_s + \sum_{s \in S} \sum_{m \in M} c'_{sm} \cdot z_{sm} \tag{7.14}$$

Using the flow quantity variables z_{sm}, we can re-formulate the demand covering constraint (7.2) so that a market supply from more than one supplier (reseller) also becomes possible [Eq. (7.15)].

Table 7.5 Shipment costs c'_{sm} between European resellers and markets per package unit

Reseller	All markets					
$i = 1,.. ..., 3$	$j = 1$ FRA	$j = 2$ GER	$j = 3$ UK	$j = 4$ SEE	$j = 5$ SWE	$j = 6$ NEU
TriColor sportive (FRA)	0.80	2.82	3.75	8.21	7.62	11.18
TT Profi (GER)	4.80	0.12	5.94	3.93	6.67	7.65
Competitive fitness (UK)	6.00	1.88	0.94	8.57	10.00	10.00

$$\sum_{s \in S} z_{sm} \geq d_m, \forall m \in M \tag{7.15}$$

Similarly, we have to ensure that no supplier (reseller) intends to deliver more quantities than are available to all markets together. Such a constraint is not contained in the model of the uncapacitated WLP. The left side of constraint (7.16) calculates the actual supply quantities delivered from supplier s to all other markets, and this quantity must not exceed the available quantity q_s.

$$\sum_{s \in S} z_{sm} \geq d_m, \forall m \in M \tag{7.16}$$

Again, it is necessary to ensure that a supplier (reseller) s sends out quantities only if this reseller s is closed, i.e., it is necessary to code the implication $z_{sm} > 0 \Rightarrow y_s = 1$ in a linear constraint. We can adjust the corresponding constraint (7.3) and obtain constraint (7.17). In this constraint, K represents a "sufficiently large" number, e.g., K can be set to 1,000,000 in the PSS-case scenario. If $z_{sm} > 0$ for a certain market m and if Eq. (7.17) should be valid, then it is necessary that $K \cdot y_s > 0$, which is equivalent to the fact that $y_s > 0$ and this means that supplier s is considered. This technique to model logical implications between non-binary and binary decision variables is called the "big-K-method" or "big-M-method." It is necessary to select an appropriate value for K for each modeled scenario.

$$z_{sm} \leq K \cdot y_s, \forall s \in S, \forall m \in M \tag{7.17}$$

Finally, the domains of the incorporated decision variables are declared by constraint (7.18).

$$y_s \in \{0;1\} \forall s \in S, z_{sm} \geq 0 \forall (s,m) \in T \tag{7.18}$$

Using the aforementioned mathematical expressions, we are now ready to state the CWLP challenge precisely:

It is necessary to minimize the total costs for the installation of facilities and the shipment of packages as expressed in (7.14) so that each market's demand is covered (7.15), but no supplier distributes more than its local stock quantity (7.16). If we use a facility for supplying a market, then this facility must be open (7.17). Each available facility is either opened or closed, and the number of packages to be shipped between each pair of suppliers and markets must be determined (7.18).

A pure formalized problem formulation of the CWLP is the following: minimize (7.14) while the constraints (7.15)–(7.18) are respected. Figure 7.14 represents the

CWLP problem in the form of an Excel spreadsheet. The necessity of considering the limited availabilities of supply quantities at the different suppliers (resellers) makes the CWLP solving process more challenging compared to the solving of the WLP. In particular, it becomes necessary to redefine the components of the b&b algorithm used for the WLP. A major difficulty is to decide whether a sub-problem has a feasible solution, i.e., to find out if suppliers already considered and still untreated can provide enough capacity to cover the demand from all markets. Evaluating a single sub-problem becomes complicated and requires huge computational effort. Therefore, we refer the reader to the scientific literature (Daskin 1995; Drezner 1995; Melo et al. 2009; Benyoucef et al. 2013; Askin et al. 2014) and report only a spreadsheet approach for the CWLP.

Figure 7.14 depicts a spreadsheet model that is used to provide all problem data to the Excel Solver. Special attention is paid to the preparation of the handling of the quantity constraints on the maximal supply quantities (7.15) as well as of the least provided quantities to cover the demand of the individual markets (7.16).

The upper part of the spreadsheet from row 1 to row 12 contains the problem parameter now including the locally available stock (column C) as well as the demand expressed from the individual markets (row 12). The cost matrix (E9:J11) contains the costs for shipping a single package unit instead of the annual costs for

	A	B	C	D	E	F	G	H	I	J	K	L
2												
3	PowerPong Sports - Distribution Costs, Demand and Availability											
4												
5	Distribution costs per loading unit in EUR											
6			max.		All markets							
7	i	re-sellers	handling	annual fixed costs	j=1	j=2	j=3	j=4	j=5	j=6		
8			quantities S_i		FRA	GER	UK	SEE	SWE	NEU		
9	1 (FRA)	TriColor Sportive (FRA)	20000	95000	0,80	2,82	3,75	8,21	7,62	11,18		
10	2 (GER)	TT Profi (GER)	20000	90000	4,80	0,12	5,94	3,93	6,67	7,65		
11	3 (UK)	Competitive Fitness (UK)	15000	70000	6,00	1,88	0,94	8,57	10,00	10,00		
12	demanded packages D_j		55000		5000	8500	3200	2800	2100	1700	23300	
13												
14	Decisions about incorporated re-sellers and established delivery links from re-sellers into markets											
15												
16					delivery quantities (z_{ij})							
17	i	re-sellers		y_i	j=1	j=2	j=3	j=4	j=5	j=6	deliverred	excess
18					FRA	GER	UK	SEE	SWE	NEU	quantities	quantities
19	1 (FRA)	TriColor Sportive (FRA)									0	-20000
20	2 (GER)	TT Profi (GER)									0	-20000
21	3 (UK)	Competitive Fitness (UK)									0	-15000
22	fulfilled demand				0	0	0	0	0	0		
23	uncovered deman				5000	8500	3200	2800	2100	1700		
24												
25	Cost calculation (all values in EUR)											
26												
27	annual fixed costs from selected re-sellers						0					
28	re-distribution costs						0					
29	total costs						0					

Fig. 7.14 A spreadsheet model of the CWLP

serving a complete market (the annual costs for installing a transportation link in the WLP).

The middle part of the spreadsheet from row 14 to row 23 contains the variable cells (D19:J21). In addition, we calculate the fulfilled demand (row 22) as well as the open (uncovered) demand for each market (row 23). The last-mentioned values are used later in formulating the constraints propagated to the solver. Furthermore, we can calculate the total sum of deliveries from each supplier (column K) and the excess of local stock (column L). We are going to incorporate the stock excess values into the constraints. The lower part of the spreadsheet contains the costs calculated from the parameters and the variable cells (rows 27–29).

In Figs. 7.15, 7.16, 7.17, and 7.18, we describe the building of the constraint system [Eqs. (7.15)–(7.18)] for CWLP.

According to the constraint shown in Fig. 7.15, we enforce the solver add-in to avoid uncovered demand. According to the constraint shown in Fig. 7.16, the solver add-in is instructed to prevent any excess of stock at the suppliers. The Big-*K*-method represents the logical dependencies between the usage of a supplier and the delivery quantities coded in constraint (7.17), which is shown in Fig. 7.17, where $K = 1{,}000{,}000{,}000$. Finally, Fig. 7.18 shows the domain specification for the shipped quantities represented by the decision variable family z_{sm}. In Fig. 7.19, the solution to CWLP is presented.

In order to test this model, we first use it to solve the WLP scenario. We assumed that each supplier (reseller) is able to handle the overall demand from all markets. Sufficiently high quantities of stock have been specified in the cells C9, C10, and C11. The returned optimal solution is the same as shown in Fig. 7.7. Only reseller #2 is used, and the total costs equal 172,000 €.

In the second experiment, we limit the stock size (max. Handling quantity for each reseller) according to the values compiled in Table 7.3. The optimal solution is shown in Fig. 7.20.

In this case, it is necessary to consider a second reseller since TT Profi is unable to handle the total annual demand for the six markets. In the new solution, the model suggests also considering reseller #3. It contributes 100 packages for delivery to market FRA and fulfills the completed UK market demand.

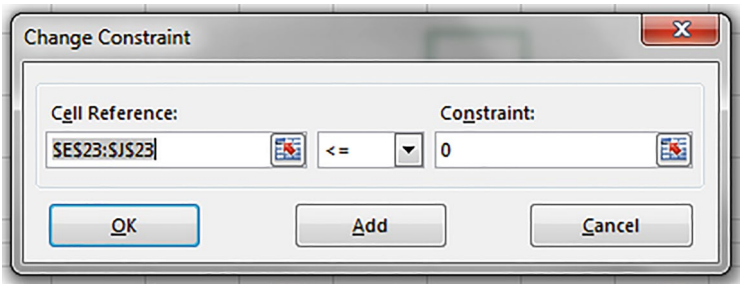


Fig. 7.15 Representation of the demand covering constraint (7.15) in the Excel Solver add-in

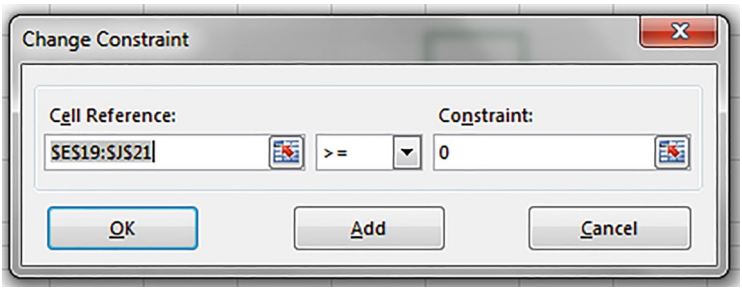


Fig. 7.16 Representation of the limited stock constraint (7.16) in the Excel Solver add-in

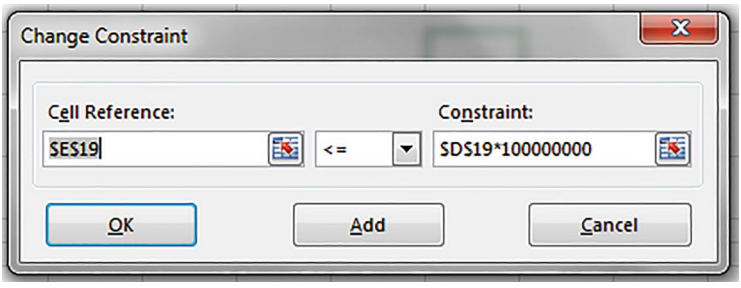


Fig. 7.17 Representation of the constraint family (7.17) in the Excel Solver add-in

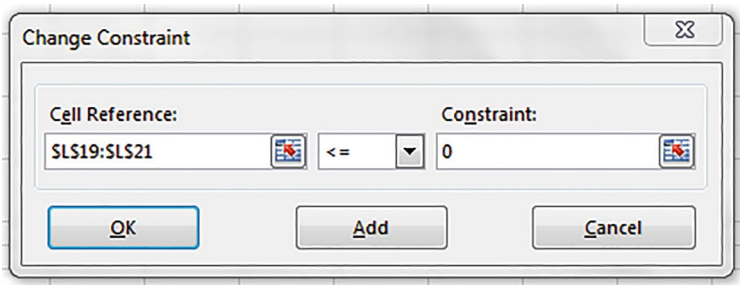


Fig. 7.18 Representation of the constraint family (7.18) in the Excel Solver add-in

The total costs have increased from 172,000 € to 226,120 €. This is an increase of 54,120 €. The consideration of the second reseller incurs 70,000 € in costs per year. Furthermore, the limited handling quantity requires the inclusion of the UK to save the GER market at increased transportation costs of 120 €, but we can profit from cheaper transportation costs into the UK, which saves 16,000 €. Therefore, we obtain a total saving of 16,000 €–120 € = 15,880 € in shipment costs, but we have to pay an additional 70,000 € in annual fixed charges to the second reseller.

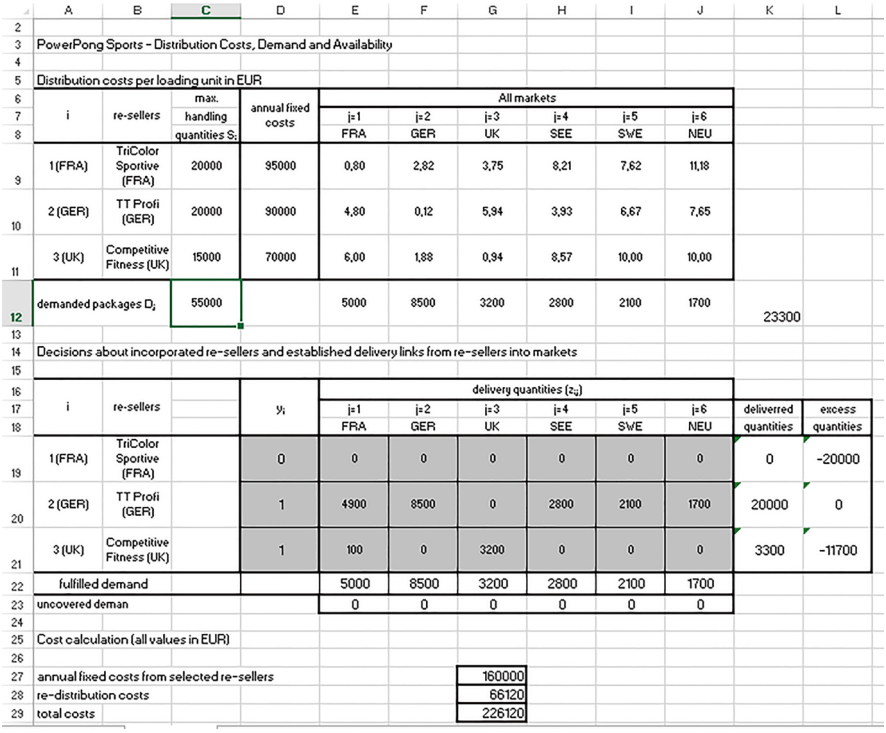
	A	B	C	D	E	F	G	H	I	J	K	L
2												
3	PowerPong Sports – Distribution Costs, Demand and Availability											
4												
5	Distribution costs per loading unit in EUR											
6			max.		All markets							
7	i	re-sellers	handling	annual fixed costs	j=1	j=2	j=3	j=4	j=5	j=6		
8			quantities S_i		FRA	GER	UK	SEE	SWE	NEU		
9	1 (FRA)	TriColor Sportive (FRA)	25000	95000	0,80	2,82	3,75	8,21	7,62	11,18		
10	2 (GER)	TT Profi (GER)	25000	90000	4,80	0,12	5,94	3,93	6,67	7,65		
11	3 (UK)	Competitive Fitness (UK)	25000	70000	6,00	1,88	0,94	8,57	10,00	10,00		
12	demanded packages D_j		75000		5000	8500	3200	2800	2100	1700	23300	
13												
14	Decisions about incorporated re-sellers and established delivery links from re-sellers into markets											
15												
16					delivery quantities (z_{ij})							
17	i	re-sellers		y_i	j=1	j=2	j=3	j=4	j=5	j=6	deliverred	excess
18					FRA	GER	UK	SEE	SWE	NEU	quantities	quantities
19	1 (FRA)	TriColor Sportive (FRA)		0	0	0	0	0	0	0	0	-25000
20	2 (GER)	TT Profi (GER)		1	5000	8500	3200	2800	2100	1700	23300	-1700
21	3 (UK)	Competitive Fitness (UK)		0	0	0	0	0	0	0	0	-25000
22	fulfilled demand				5000	8500	3200	2800	2100	1700		
23	uncovered deman				0	0	0	0	0	0		
24												
25	Cost calculation (all values in EUR)											
26												
27	annual fixed costs from selected re-sellers						90000					
28	re-distribution costs						82000					
29	total costs						172000					

Fig. 7.19 Returned optimal solution to the CWLP scenario in the PPS case

7.4 Regional Facility Location

The WLP-based approach supports the identification of regions that should be considered for setting up an SC. These regions might be continents, countries, states, or even farms, growing areas, or plantations supplying or consuming products. According to the location planning scheme outlined in Sect. 7.2, the outcome of phase I (the regions to be considered in the prospective SCD) is forwarded into phase II, where one or several locations have to be identified for each region as a network node.

In the context of the PPS case, the regions are the countries that host the resellers or the markets. A typical phase II decision is now to select a warehouse to which the packages are shipped from China and from where customers in the market receive their deliveries. We can now assume that the regional reseller's existing facilities are too small to handle the additional freight flow associated with the celluloid-free balls, and the reseller is looking for a new location in the associated country to build or rent a new warehouse. This section addresses the problem of identifying promising candidates for establishing a new facility that can handle incoming and outgoing materials as part of the global SC. In particular, we will



PowerPong Sports - Distribution Costs, Demand and Availability

Distribution costs per loading unit in EUR

i	re-sellers	max. handling quantities S_i	annual fixed costs	All markets j=1 FRA	j=2 GER	j=3 UK	j=4 SEE	j=5 SWE	j=6 NEU		
1 (FRA)	TriColor Sportive (FRA)	20000	95000	0,80	2,82	3,75	8,21	7,62	11,18		
2 (GER)	TT Profi (GER)	20000	90000	4,80	0,12	5,94	3,93	6,67	7,65		
3 (UK)	Competitive Fitness (UK)	15000	70000	6,00	1,88	0,94	8,57	10,00	10,00		
demanded packages D_j		55000		5000	8500	3200	2800	2100	1700	23300	

Decisions about incorporated re-sellers and established delivery links from re-sellers into markets

i	re-sellers	y_i	delivery quantities (z_{ij}) j=1 FRA	j=2 GER	j=3 UK	j=4 SEE	j=5 SWE	j=6 NEU	deliverred quantities	excess quantities
1 (FRA)	TriColor Sportive (FRA)	0	0	0	0	0	0	0	0	-20000
2 (GER)	TT Profi (GER)	1	4900	8500	0	2800	2100	1700	20000	0
3 (UK)	Competitive Fitness (UK)	1	100	0	3200	0	0	0	3300	-11700
fulfilled demand			5000	8500	3200	2800	2100	1700		
uncovered deman			0	0	0	0	0	0		

Cost calculation (all values in EUR)

annual fixed costs from selected re-sellers	160000
re-distribution costs	66120
total costs	226120

Fig. 7.20 Returned optimal solution for the CWLP scenario in the PPS case

- Become familiar with typical phase II decision problems related to location planning.
- Introduce a simple model of *center-of-gravity* representing a phase II decision situation.
- Develop simple mathematical calculations for the identification and determination of the coordinates of an optimal location.
- Consider available demand knowledge while deriving such an optimal location by solving the Steiner–Weber model.
- Learn how the *Miehle algorithm* can be used to solve the *Steiner–Weber model*.

We start in Sect. 7.4.1 with the verbal description of the typical phase II decision problem. Next, in Sect. 7.4.2, we propose a modeling approach for a phase II decision problem. In Sect. 7.4.3, we discuss a mathematical calculus for the derivation of the optimal solution of the location model.

7.4.1 Management Problem Description

We consider the phase II decision situation of TT Profi, the German reseller selected by PPS, as part of its global SC. TT Profi is looking for a new warehouse in Germany to receive the inbound material flow from the PPS factory in Shanghai. From this facility, the five major local retail partners in Germany, shown in Table 7.6, receive their deliveries.

Each delivery is executed by a small forwarding company which is paid according to the distance between the pickup point and the delivery point of a shipment. In order to keep the distribution costs as low as possible, TT Profi wants to open its new warehouse at a location that leads to the smallest annual transportation distance to be bridged. The chairman of TT Profi wants to know which coordinates the new warehouse should be placed in to minimize total transportation costs from the warehouse to all customers.

7.4.2 A Mathematical Model of the Decision Situation

The relevant issues in this decision problem are customer locations, distances from the warehouse to customers, and customer demands. Each customer location is represented by the ordered pair of $(x;y)$-coordinates. The $(x;y)$-coordinates of the five locations are given in the third as well as fourth column of Table 7.6. These data cannot be modified; they are input data or problem *parameters*. The coordinates of customer i are named $(x_i;y_i)$, e.g., $(x_1;y_1) = (10;-80)$ and so on. Demand for each customer is also given and denoted as $D(x_i;y_i)$.

In contrast, the $(x;y)$-coordinates $(p_x;p_y)$ of the new warehouse are variable and have to be determined. Consequently, p_x as well as p_y are the *decision variables* in the investigated decision scenario.

We *assume* that the total transportation cost sum is proportional to the distance and the transportation volume (i.e., the demand). This leads us to the formulation of the *objective function*, as shown in Eq. (7.19):

$$Z\left(p_x;p_y\right) = \sum_{i=1}^{N} d\left(\left(p_x,p_y\right);\left(x_i,y_i\right)\right) \cdot D\left(x_i,y_i\right) \rightarrow \min \tag{7.19}$$

Table 7.6 German customers of TT Profi

		Coordinates		Annual demand
i	Name	x_i	y_i	$D(x_i;y_i)$
1	Sport 1–2–3 KG	10	−80	3 t
2	TT direct	−45	−30	1 t
3	Sports and fun	60	50	1.5 t
4	Leisure outlet	45	−75	3.5 t
5	Raquets and more	−75	80	2 t

We can observe that the total transportation costs depend on the coordinates p_x and p_y of the prospective warehouses and distances. We assume that the total transportation cost sum from the prospective warehouse location $(p_x;p_y)$ to a customer location $(x_i;y_i)$ is more or less equivalent to the distance and demand. Therefore, the distance $d((p_x;p_y); (x_i;y_i))$ between the ith customer location and the warehouse should be determined to calculate transportation costs.

To minimize the payments to the forwarding company, we should vary p_x as well as p_y as long as $Z(p_x;p_y)$ becomes minimal. The minimization of (7.19) represents the decision situation of identifying the lowest cost facility for TT Profi, so that it can be understood as a model for the decision situation outlined in Sect. 7.4.1.

7.4.3 Solving the Mathematical Model: Center-of-Gravity Approach

We call each pair $(p_x;p_y)$ that minimizes (7.19) an optimal solution of the model (7.19). Since there are no further limitations to be considered that restrict the selection of values of p_x or p_y, the decision problem represented by (7.19) falls into the category of *global optimization* problems. All possible values of p_x and p_y represent a feasible solution of (7.19) and are, therefore, candidates to become the optimal solution.

There are two options for calculating the distances. The first option is to use *direct line distance* (Euclidean distance), and the second one is the *orthographic (square)* distance calculation. According to the first option, distance equals the Euclidean distance between these two points in the plane. This value can be determined according to Pythagoras' Theorem leading to Eq. (7.20).

$$d\left(\left(p_x;p_y\right);\left(x_i;y_i\right)\right)=\sqrt{\left(x_i-p_x\right)^2+\left(y_i-p_y\right)^2} \tag{7.20}$$

Figure 7.21 depicts the total distance from the warehouse to all customers in dependence of the warehouse coordinates $(p_x;p_y)$ (*without demand considerations*). We observe that the function Z shows exactly one minimum; this means there is exactly one pair $(p_x;p_y)$ that minimizes Z. Reading the exact values of this optimal pair is hardly possible. Furthermore, if the number of customers N is quite high, then drawing a similar picture requires huge computational efforts. For these reasons, a quicker and more reliable method for the determination of the pair of optimal warehouse coordinates is required.

The function Z is continuous and differentiable, and the decision variables are unrestricted. Hence, we can determine the optimal point of Z by differential calculus. The following consecutive steps have to be executed in the given order.

1. The first derivative Z' of Z is determined and.
2. The zero of Z' is determined.

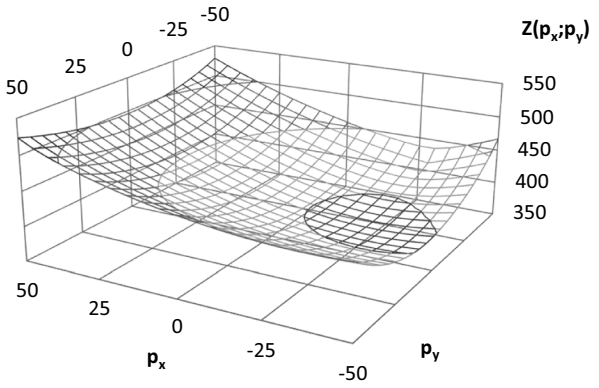


Fig. 7.21 Total distance to all customers (objective function value) in dependence of the values of p_x as well as p_y (all values given in kilometers)

$$\frac{\mathrm{d}Z}{\mathrm{d}p_x} = \sum_{i=1}^{N} \frac{p_x - x_i}{\sqrt{\left(x_i - p_x\right)^2 + \left(y_i - p_y\right)^2}} \cdot D\left(x_i; y_i\right) \tag{7.21}$$

$$\frac{\mathrm{d}Z}{\mathrm{d}p_y} = \sum_{i=1}^{N} \frac{p_y - y_i}{\sqrt{\left(x_i - p_x\right)^2 + \left(y_i - p_y\right)^2}} \cdot D\left(x_i; y_i\right) \tag{7.22}$$

We start with the situation in which we do not consider individual demand, i.e., $D(x_i;y_i) = 1$ for all i. Since Z is a function of the two decision variables p_x and p_y, the derivative Z' consists of the two partial derivatives (7.21) and (7.22). They can be determined by considering p_y and p_x respectively as invariant.

$$p_x = \frac{\sum_{i=1}^{N} x_i}{N} \tag{7.23}$$

$$p_y = \frac{\sum_{i=1}^{N} y_i}{N} \tag{7.24}$$

Setting the right-hand part of Eq. (7.21) equal to 0 leads to the formula (7.23), and setting the right-hand side of Eq. (7.22) equal to 0 results in the formula (7.24). These two formulas (7.23) and (7.24) can be used to determine the optimal values of the coordinates of the warehouse.

The application of the developed formulas determines the optimal location of the warehouse for TT Profi. We get the optimal coordinates $p_x = \frac{10 - 45 + 60 + 45 - 75}{5} = -1$ $p_y = \frac{-80 - 80 + 50 - 75 + 80}{5} = -11$. Using these two formulas enables us to derive an optimal solution of the model (7.19) (without demand considerations) from the problem parameters provided.

We now incorporate the *demand* associated with each customer subject in the model (7.19). This step is required since a forwarding company is typically paid according to the bridged distance and quantity moved. The product of distance (measured in km) and weight (measured in tons) is called transport performance. It is expressed in ton kilometers (tkm).

The model (7.19) is called the *center-of-gravity model* of location analysis (Chopra and Meindl 2012) or the *Steiner–Weber model* (Domschke and Drexl 1985). Using demand data, formulas (7.25) and (7.26) are used to calculate optimal coordinates.

$$p_x = \frac{\sum_{j=1}^{N} \frac{D(x_j;y_j) \cdot x_j}{\sqrt{(p_x - x_j)^2 + (p_y - y_j)^2}}}{\Sigma_{j=1}^{N} \frac{D(x_j;y_j)}{\sqrt{(p_x - x_j)^2 + (p_y - y_j)^2}}} \tag{7.25}$$

$$p_y = \frac{\Sigma_{j=1}^{N} \frac{D(x_j;y_j) \cdot y_j}{\sqrt{(p_x - x_j)^2 + (p_y - y_j)^2}}}{\Sigma_{j=1}^{N} \frac{D(x_j;y_j)}{\sqrt{(p_x - x_j)^2 + (p_y - y_j)^2}}} \tag{7.26}$$

The determination of an optimal pair of coordinates for the warehouse again requires the determination of the directional derivatives. These two functions are then set equal to 0, and we get the expressions (7.25) and (7.26), respectively, to express p_x and p_y. Unfortunately, these characterizations of p_x and p_y are recursive, meaning we need p_x (on the right-hand side of the equation) to calculate p_x. The same problem is observed for the determination of p_y. It is impossible to transform the two equations so that the recursion is avoided. We cannot determine the values of $(p_x;p_y)$ as we did in the case without demand data. Another approach is required.

In order to break the circulation in the formulas (7.25) and (7.26), Miehle (1958) proposes an approximation approach. The basic idea of the *Miehle algorithm* comprises the consecutive calculation of a sequence of solutions $(a_0;b_0)$, $(a_1;b_1)$, $(a_2;b_2)$, … of (7.19). The values of $(a_{i-1};b_{i-1})$ are used as p_x- and p_y-values in the right-hand part of the Eqs. (7.25) and (7.26), and they are used to calculate the next solution $(a_i;b_i)$. This approach represents an example of a recursive calculation. If we calculate the coordinate pairs in this way, then the sequence of objective function values $Z(a_0;b_0)$, $Z(a_1;b_1)$, $Z(a_2;b_2)$, … decreases and converges towards the minimal possible performance value. Therefore, the sequence of calculated coordinates approximates the optimal pair of coordinates of the new warehouse with respect to the minimization of the overall required transport performance.

We demonstrate the application of the Miehle algorithm and apply it to the example data given in Table 7.6. Ignoring customer demand, we can determine a first

coordinate pair $(a_0;b_0)$ by applying the formulas (7.23)–(7.24), so that we start with $(a_0;b_0) = (—1; 11)$. The required performance is $Z(—1;11) = 89{,}738$ km.

If we set $p_x = a_0 = -1$ and $p_y = b_0 = -11$ in the right-hand sides of (7.25) and (7.26), then we get the updated solution $(a_1;b_1) = (8.78; 36.16)$ with a reduced required performance value $Z(8.78, 36.16) = 80{,}837$ tkm. The next iterations are similarly repeated and an existing solution is replaced by an updated one, re-applying the formulas (7.25) and (7.26) (see Table 7.7).

Unfortunately, there is no guarantee that the iterative process terminates after a particular number of repetitions. For this reason, we stop the solution update as soon as we observe that the original solution $(a_i;b_i)$ and its update $(a_{i+1};b_{i+1})$ have become similar. Here, we call two solutions similar if the distance between the two represented points is less than 0.05 km. In the TT Profi example, we stop the Miehle algorithm after ten iterations since the tenth solution proposal (20.50; 66.10) and its update (20.49;–66.14) are less than 0.05 km away from each other (column 6 in Table 7.7).

Table 7.7 Iterations of the Miehle algorithm applied to the TT Profi data

Iteration	Old solution		Updated solution			Costs
i	a_i	b_i	a_{i+1}	b_{i+1}	Distance	$Z(a_i;b_i)$
0	−1.00	−11.00	8.78	−36.16	26.99	897.38
1	8.78	−36.16	14.69	−50.10	15.14	808.37
2	14.69	−50.10	18.05	−58.14	8.71	771.93
3	18.05	−58.14	19.64	−62.48	4.62	757.29
4	19.64	−62.48	20.28	−64.57	2.18	752.67
5	20.28	−64.57	20.49	−65.48	0.94	751.58
6	20.49	−65.48	20.53	−65.87	0.39	751.37
7	20.53	−65.87	20.52	−66.03	0.16	751.33
8	20.52	−66.03	20.50	−66.10	0.07	751.32
9	20.50	−66.10	20.49	−66.14	0.04	751.32

We have seen in the example application of the Miehle algorithm that the Z-objective function value decreases after each executed update and the original point and its update become more and more similar. Now, we summarize the steps of the Miehle approach:

Start: determine an initial pair of coordinates $(a_0;b_0)$ by applying (7.21) to determine a_0 and (7.22) for determining b_0.

Iteration: as long as the distance between the point $(a_i;b_i)$ and its update $(a_{i+1};b_{i+1})$ exceeds a given threshold, then start another update cycle. Otherwise, return $(a_{i+1};b_{i+1})$ as approximation of the (unknown) optimal solution of the center-of-gravity model (7.19).

The evolution of the intermediate solutions during the execution of the Miehle algorithm is shown in Fig. 7.22. We can observe that the original proposal (the dark triangle) is far away from the final proposal (the diamond). Since Leisure Outlet requests the largest quantity from TT Profi, it is reasonable to position the warehouse close to this important customer.

If the customer Rackets and More increases its annual demand quantity to 8 tons, then it is beneficial to place the new warehouse in the upper left-hand area of the operations field. It shows that the selection of the right warehouse location is responsive to the demand data. It is, therefore very important to take trustworthy and stable demand estimations to select the right warehouse (see Fig. 7.23).

Note: An Excel file containing a spreadsheet model of the Miehle approach to the TT Profi setting can be found in the E-Supplement

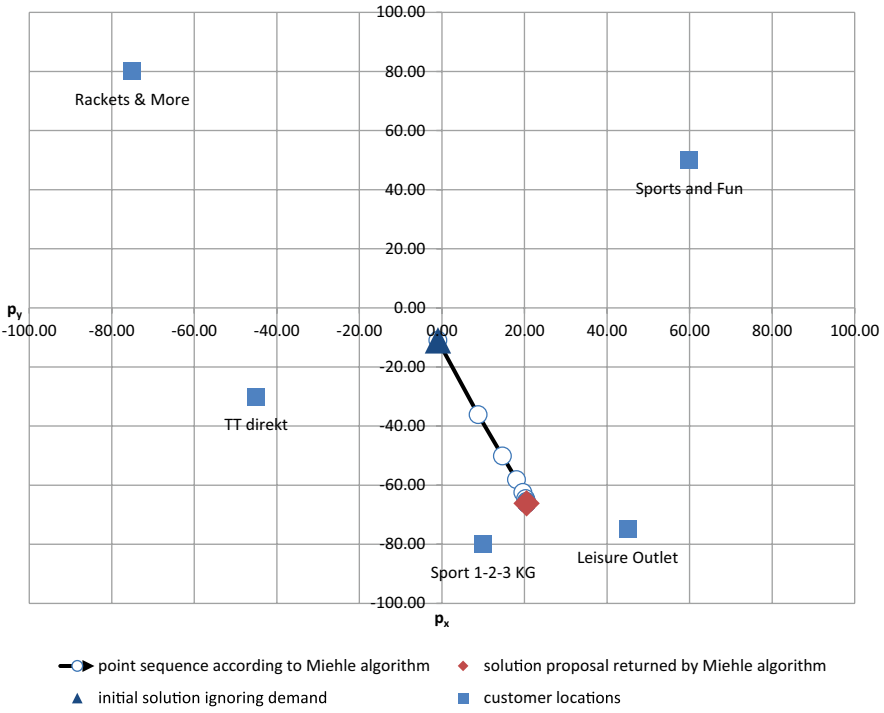


Fig. 7.22 Graphic representation of the progress of the Miehle algorithm

Letting the coordinates for the warehouse be (20.49; −66.14), the total transportation cost is 751.32 €, subject to (7.19).

In a second option for distance calculation, we use the rectangular method as shown in formula (7.27), followed by (7.28)–(7.29) to determine p_x; p_y:

$$d\left(\left(p_x;p_y\right);\left(x_i;y_i\right)\right)=\left|x_i-p_x\right|+\left|y_i-p_y\right| \tag{7.27}$$

$$p_x=\frac{\sum_{i=1}^{N}D_i\cdot x_i}{\sum_{i=1}^{N}D_i} \tag{7.28}$$

$$p_y=\frac{\sum_{i=1}^{N}D_i\cdot y_i}{\sum_{i=1}^{N}D_i} \tag{7.29}$$

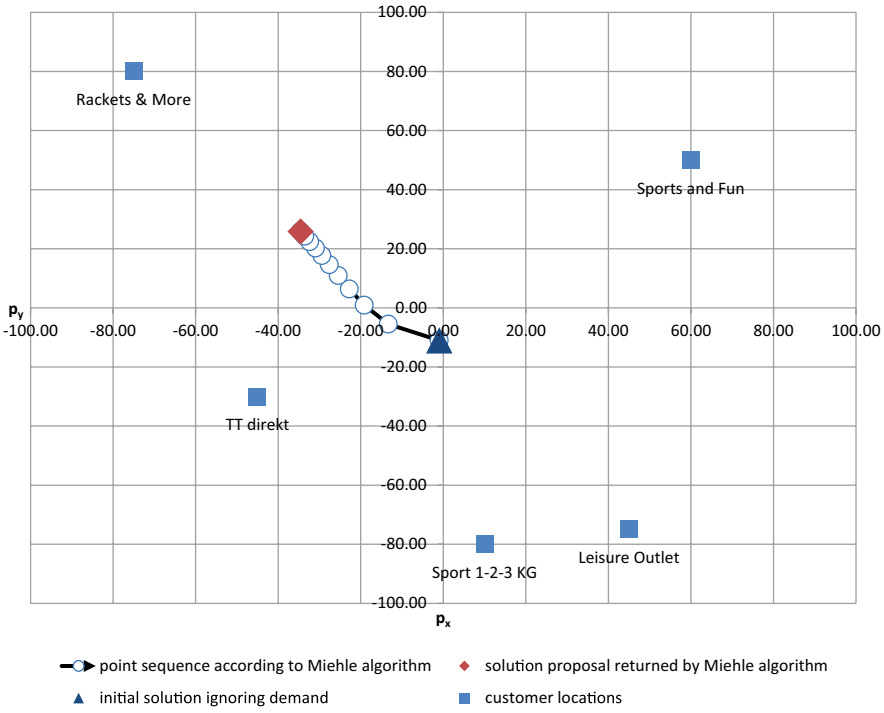


Fig. 7.23 Graphic representation of the progress of the Miehle algorithm if Rackets and More should demand 8 tons

Using data from Table 7.6, we get $p_x = \frac{30-45+90+157.5+150}{11} = 7.5$ and $p_x = \frac{240-30+75-262.5+160}{11} = -27$.

The calculation of distances is shown in Table 7.8.

Subject to Eq. (7.19), the total cost is calculated as follows, assuming that costs per kilometer are \$1.00:

$$Z = 55.5 \cdot 3 + 55.5 \cdot 1 + 129.5 \cdot 1.5 + 85.5 \cdot 3.5 + 189.5 \cdot 2 = 1094.5$$

We can observe that total costs in the second case are higher than the result of the Miehle algorithm. This can first be explained by the larger distances in the square method compared to the Euclidian distances. Second, we used a simplified procedure for coordinate calculations without improving them with the Miehle algorithm. Using the Excel file in the E-Supplement of this book, you can prove that letting initial coordinates from Table 7.7 be (7.5;–27), then the Miehle algorithm would provide the same coordinate solution as for the initial case with coordinates (−1;–11). Setting these optimal coordinates (20.49; −66.14) in (7.19) and using square distance metrics, calculate total transportation costs and compare this to the solution with Euclidian distances!

Table 7.8 German customers of TT Profi

		Coordinates		Distances from (7.5; 27)	Demand
i	Name	x_i	y_i	$d(x_i;y_i)$	D_i
1	Sport 1-2-3 KG	10	−80	55.5	3 t
2	TT direct	−45	−30	55.5	1 t
3	Sports and fun	60	50	129.5	1.5 t
4	Leisure outlet	45	−75	85.5	3.5 t
5	Racquets and more	−75	80	189.5	2 t

Note: In case of different costs per kilometer in different directions, Eqs. (7.25), (7.26), (7.28), and (7.29) need to be extended. More specifically, the numerators and denominators of the given equations need to be multiplied by the costs for each individual customer, i.e., in each individual component of the sum functions.

7.5 Factor-Ranking Analysis

7.5.1 Case-Study OTLG Germany

Volkswagen Original Teile Logistik GmbH & Co. KG (OTLG) is a service partner for spare parts of Volkswagen, Volkswagen Nutzfahrzeuge, Audi, Seat, and Škoda. OTLG is part of the wholesale level in the SC and is responsible for sales and marketing in Germany. OTLG operates at seven locations in Germany.

The company's annual revenue is almost 2.5 billion euros. It serves almost 5000 partners in Germany and occupies a warehouse space of more than 300,000 m^2 (equivalent to 47 football pitches). Annually 45,000,000 materials call-off, pickup, and delivery actions take place, corresponding to approximately 200,000 daily operations.

During the logistics optimization project DNO D (distribution network optimization), OTLG's distribution network was redesigned. Eleven distribution centers (DC) were consolidated into eight, as shown in Fig. 7.24.

Among them, a new DC VZ Brandenburg in Ludwigsfelde was established (see Fig. 7.25). The first step was to select a location for the new DC. Initially, many potential locations and selection criteria were identified, as shown in Fig. 7.26. Next, a shortlist of preferable locations was established (Fig. 7.27).

After evaluation, the location Preußenpark was selected because it had the best scores for the listed criteria. In Fig. 7.28, the new DC is shown.

First, excellent rail and road connections made the selected location preferable. These criteria are crucial for OTLG because of highly dynamic logistics. Daily, 250 trucks are used for regular supplies, and 200 small vans are used for day-to-day supplies. This ensures that 98% of orders are fulfilled within 24 h. The total SKU number reaches almost 100,000 positions with an inventory value of 43.2 million euros. Daily, in excess of 30,000 positions are delivered, making it possible to supply 402 VW/Audi and 476 SEAT/Škoda customers with original spare parts.

Fig. 7.24 OTLG footprint in Germany (© OTLG GmbH & Co. KG)

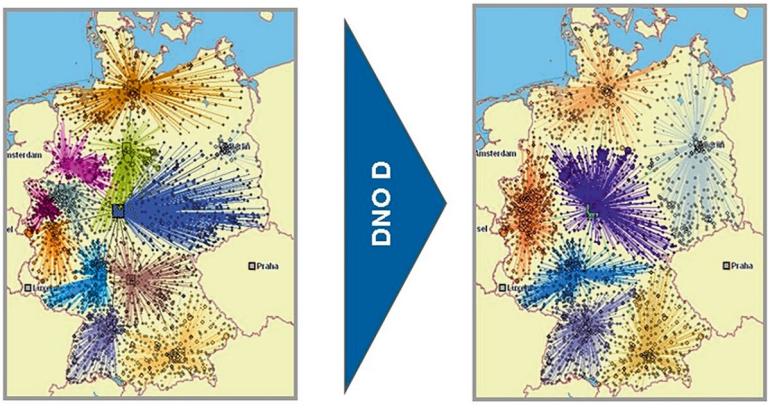


Fig. 7.25 Distribution network redesign at OTLG (© OTLG GmbH & Co. KG)

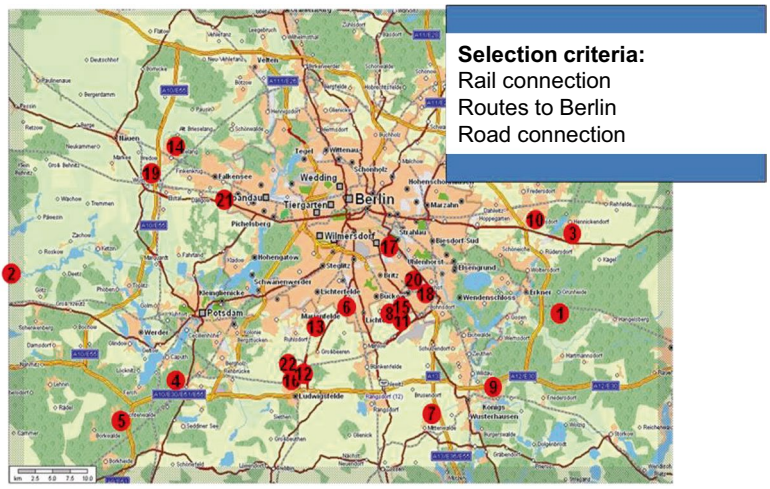


Fig. 7.26 Facility location planning at OTLG (© OTLG GmbH & Co. KG)

Fig. 7.27 Detailed stage of the facility location planning at OTLG (© OTLG GmbH & Co. KG)

Discussion

1. Analyze the redesign of the OTLG distribution network. Which type of logistics network has been implemented? Why look for a new location in Brandenburg?
2. Based on this case study, describe the basic steps in facility selection decisions! Why were the logistics criteria so important for OTLG? Which criteria would be important, e.g., for the case of a new hotel?

Fig. 7.28 New distribution center in Ludwigsfelde (© OTLG GmbH & Co. KG)

7.5.2 Factor-Rating Method

Having seen how OTLG GmbH & Co. KG conducted their analysis to identify the right location for their distribution center, we need to look deeper at the related decision criteria and decision-making procedure. The OTLG case study demonstrated that the region around Berlin and its excellent infrastructure (road/rail connection) highly influenced their decision. In other words, the availability of infrastructure was the factor with the highest priority for OTLG GmbH & Co. KG.

As you can imagine, not all companies will rate the criterion "infrastructure" as the most important in selecting a new facility location. This means that different companies will select different location criteria or location factors that have different priorities. Therefore, we need to take a closer look at the generic list of possible *plant location criteria*.

Location criteria can be divided into *quantitative* and *qualitative* areas, as summarized in Table 7.9.

▶ **Practical Insights** In practice, the list of location selection criteria is the first step in facility location planning. On the one hand, this list can become long very quickly. This is why building main and sub-groups of criteria is common practice to facilitate a clear understanding of the list

Table 7.9 Facility location selection criteria (selected examples)

Quantitative criteria	Qualitative criteria
Transportation costs	Infrastructure
Building and construction costs	Quality of labor
Rental costs	Transportation development
Labor costs	Purchasing power
Material costs	Options for financing (free trade zones, etc.)
Taxes	Suppliers
Financial support from local governments	Political risks
	Natural disaster risks
	Proximity to customers and suppliers
	Business climate
	Environmental regulations
	Competitive advantage
	Government and trading barriers

structure. On the other hand, many experts from different departments are involved in setting up this list. It may take quite a long time to find a compromise on which criteria should be included.

The next step is to evaluate different potential locations according to the defined factors. The *factor-rating method* is easy to understand. The team in charge of identifying the new location selects the relevant criteria. As we will see later, this depends on the industry specifics, i.e., each company in a certain industry will compose an individual company-centered list of site location factors.

To make the process very tangible, let us assume that we are consultants and we need to support another company like OTLG GmbH & Co. KG in their decision-making process. Imagine that in our consulting team, we have compiled the following list with six criteria, as shown in Table 7.10. According to importance, the team decides the maximum amount of points that can be obtained per location factor. For the criterion "Infrastructure" this might hypothetically be 300 points, maximal. Then the team assigns the points for each of the options and per factor.

▶ **Practical Insights** It might be a reasonable to let the team members make their rating individually and then calculate the averages per factor and option. The individual score sheets are archived so that the pathway of decision-making is fully transparent and reproducible for later reviews.

For example, if the operations management team consists of five members and each individual scores the first factor of the first option, the resulting scores might look like this:

Table 7.10 Factor-rating model

Location factor	Range min-max	Location A	Location B	Location C
Infrastructure	0–300 pts	213	232	204
Proximity to suppliers	0–200 pts	170	182	186
Proximity to customers	0–200 pts	180	171	192
Other facilities	0–150 pts	113	156	78
Quality of labor	0–100 pts	72	65	85
Cost of energy	0–50 pts	42	48	32
Sum	0–1000 pts	790	854	777

Member 1: scores 210
Member 2: scores 230
Member 3: scores 220
Member 4: scores 200
Member 5: scores 205

On average, this will result in 1065/5 = 213 points for the first factor "Infrastructure" for option A. For all other criteria, a similar exercise should be done in practice so that a completely transparent rating can be presented to senior management.

The option that receives the highest score is the one that is suggested as the most suitable new location. In our example, this is location B. If the scores of different options are very close to each other, a *sensitivity analysis* is mandatory.

Shortcomings of the factor-rating method include the fact that the weighting percentages assigned per factor are not clearly visible. A useful method for overcoming this is the so-called *utility value analysis*.

7.5.3 Utility Value Analysis

We will elaborate on the principle approach followed by the *utility value analysis* (see Günther and Tempelmeier 2009) in connection with a theoretical case in which a global player developed a strategy to produce and distribute its goods in Asia.

Just imagine that the top management of a globally functioning firm has asked the SCOM team to look for a suitable location to create a new factory in Asia. The sales department of that hypothetical global firm might have assessed that the market for the firm's products is growing significantly in, for example, India. As a consequence, the SCOM team will run an assessment of potential locations there.

The following list provides an overview for finding a location for a new production facility depending on the different levels of the location decision problem.

Step 1 Selection of the Economic Region First of all, the respective continent for the economic region or trade-zone where the search for an appropriate location should be conducted must be specified.

In the first step, the following criteria are extremely important:

- Attractiveness of the economic region
- Expected sales potential and/or market development
- Political stability of the region
- Legal requirements regarding the establishment of a production facility, etc.

Step 2 Selection of a Country Within the Economic Region Our theoretical case also shows that the options need to be further streamlined, i.e., the search for a site has to be narrowed down (here, it is focused on India). This means in the continent of Asia (Step 1); the sub-continent India has been identified in Step 2 for the execution of the location planning and location analysis. Besides the expected market development, there are also such location factors as:

- Labor availability
- Quality of human resources, for example, education levels
- Salary and/or wage levels
- Local/regional support for the location of a new facility
- Local/regional support regarding business development
- Availability of appropriate suppliers
- Availability of transport technology and/or infrastructure, which play an important role

Step 3 Selection of a Region Within the Country in the Economic Region In the third step, the sub-continent or maybe the country identified has to be further assessed by the SCOM team. This means a certain kind of county or district will be identified for the placement of the new production facility. In reality, there will most probably be multiple options available, and it will be the task of the experts to identify the best location for the company. Thus, further criteria have to be taken into consideration; these are be presented next.

When we refer back to our example, the decision-making operations team will need to evaluate different alternatives to identify the new facility in India. This means a long list of alternative sites is drawn up, and different options must be critically assessed. This critical assessment has to be reproducible and needs to be compiled in such a way that the top management can trace back the different steps that led to the identification of the preferred location(s). In practice, a table will be compiled that shows the different alternatives, which are then mapped and rated against a defined list of location factors (criteria). This is the fundamental principle of the utility value analysis. These criteria can be:

- Connection to available infrastructure (e.g., roads, rail, ports, airports)
- Availability of utilities (such as electricity or water for cooling)

- Availability of raw-material (e.g., iron for steel production)
- Availability of green-field properties or existing objects/facilities
- Subsidies or taxation benefits, etc.

Step 4 Selection of the Facility Location In Step 4, the final decision for the selected location has to be substantiated in case there are multiple possible properties available on the same piece of land. Imagine, for example, that there might be a newly created industrial park where the home community offers different properties for sale. Therefore, the above-mentioned location factor table will provide evidence for the identification of the most suitable plot of land for the new facility. Final decision factors could, therefore be:

- Geographic dimensioning of the land (either more longitudinal or rectangular or maybe triangular shape of the property)
- Topographic suitability or existing constraints (e.g., hilly grounds or existing electrical pylons)
- Cost of the property and later options for enlargement/expansion
- Opportunities for suppliers to locate nearby
- Environmental constraints (e.g., regarding emissions)
- Connection to existing infrastructure (accessibility for inbound flows and also shipment of finished goods outbound)

The principle *sequence of the activities* that need to be performed by the SCOM team to conduct such a utility value analysis is as follows:

1. Identify location options.
2. Determine decision criteria (location factors) and their measurement.
3. Determine weighting totaling 100% for the different criteria.
4. Evaluate every location option on a normalized scale to achieve "partial utility values" with a scale from 1 to 10 points (in which 10 = best).
5. Calculate the "total utility value" of a location option by multiplying "partial utility values" with weights and adding these values.
6. Choose the option with the highest "total utility value."

Table 7.11 depicts a simplified example of the potential application of the utility value analysis.

The possible location factors presented above are respective indications of inspiration for the SCOM team. In practice, the location criteria will be different depending on the respective industry and will have different priorities. For example, for a chemical company, ecological restrictions are extremely important, and for heavy industry, the availability of required resources will most probably have a significant influence on the decision-making process. Also, factors such as the company's other existing facilities in the given region might influence the decision about a new site.

Table 7.11 Utility value analysis

Location factor	Weight (%)	Option *X*		Option *Y*		Option *Z*	
		Points	Partial value	Points	Partial value	Points	Partial value
Infrastructure	*35*	7	2.45	8	2.80	6	2.10
Proximity to suppliers	*20*	6	1.20	7	1.40	8	1.60
Proximity to existing sites	*20*	8	1.60	4	0.80	9	1.80
Expansion potential	*10*	6	0.60	7	0.70	7	0.70
Quality of labor	*10*	8	0.80	3	0.30	8	0.80
Topography	*5*	10	0.50	8	0.40	9	0.45
Sum	100						
Total utility value			7.15		6.40		7.45

▶ **Practical Insights** Facility location planning decisions imply the usage of both quantitative and qualitative methods. Despite the rigor and technical power, simulation and optimization methods are not the dominant techniques in decision-making on facility location planning and SCD. In practice, these decisions are typically driven by corporate policies and are analyzed with the help of business cases and empirical data.

Case Study "Niedersachsen Park" (Based on www.niedersachsenpark.de)
The objective of this case study is to find out why the Adidas Group selected the Niedersachsenpark as a location for its new distribution center. The Adidas Group is the world's second-biggest sports goods manufacturer and is headquartered in Herzogenaurach, Germany. In Germany, the Adidas Group has three distribution centers. Two of them are in the area of the group's headquarter in Bavaria and the third one, which opened in 2013, is located in Niedersachsenpark, the largest industrial and commercial area in Lower Saxony. This newly established distribution center is the group's largest in terms of throughput. In 2015, the Adidas Group expected a throughput of 100 million pieces per year at this distribution center. In comparison, the two distribution centers in Bavaria (in Uffenheim and Scheinfeld) together achieve a total throughput of 80 million pieces.

Besides the expected throughput, the new distribution center in the Niedersachsenpark has further advantages, such as its location. In contrast to the locations in Bavaria, this one is directly located on highway A1, and the main harbors are easily accessible from this location. This is extremely important for the Adidas Group as the main components of its products arrive in Germany by sea (Hamburg, Bremen, Bremerhaven). Therefore, the group can decrease transportation costs from the harbors to the distribution centers. Furthermore, since Niedersachsenpark is a newly established industrial and commercial park, it was highly flexible in terms of construction and development plans for fulfilling the Adidas Group's requirements.

Additionally, the construction rights for the whole area are regulated. There was no possibility of public disputes that might cause long waiting times or potential reputation damages before they started operating at this location. Since there are no residential areas in the immediate vicinity of Niedersachsenpark, the distribution center is allowed to operate 24/7, as opposed to 17 h per day in Uffenheim and Scheinfeld. This allows for highly efficient processes at Niedersachsenpark. Another significant advantage of Niedersachsenpark was the chance to reserve areas for potential future expansion. Companies can reserve these areas in case they do not want to exclude possible future expansion, but as yet, do not have any concrete plans. This advantage creates a valuable degree of flexibility for the organizations located in Niedersachsenpark.

To sum up, the advantages outlined indicate that Niedersachsenpark was a favorable location for the Adidas Group's new distribution center. In many aspects, it surpasses the distribution centers in Bavaria. The case study reveals that Niedersachsenpark is the optimal location for the Adidas Group's new distribution center. Do you agree? Apply the factor-rating and utility value analysis methods to this case (N-Park = Niedersachsenpark; Bavaria = Uffenheim&Scheinfeld) (use Table 7.12).

7.6 Combining Optimization and Simulation in Supply Chain Design

Consider a combination of simulation and optimization that seeks to find optimal locations for facilities and allocate customers to those locations subject to supply chain profit maximization (i.e., we consider location-allocation problems). Figure 7.29 depicts major interdependencies between the parameters in supply chain design.

Network optimization can be used for a number of supply chain design problems, such as:

- *Incapacitated and capacitated plant location problem*
- *Distribution network design*

Table 7.12 Factor-rating and utility value analysis

Key success factor	Weight	Scores (out of 100)		Weighted scores	
		N-Park	Bavaria	N-Park	Bavaria
Total scores					

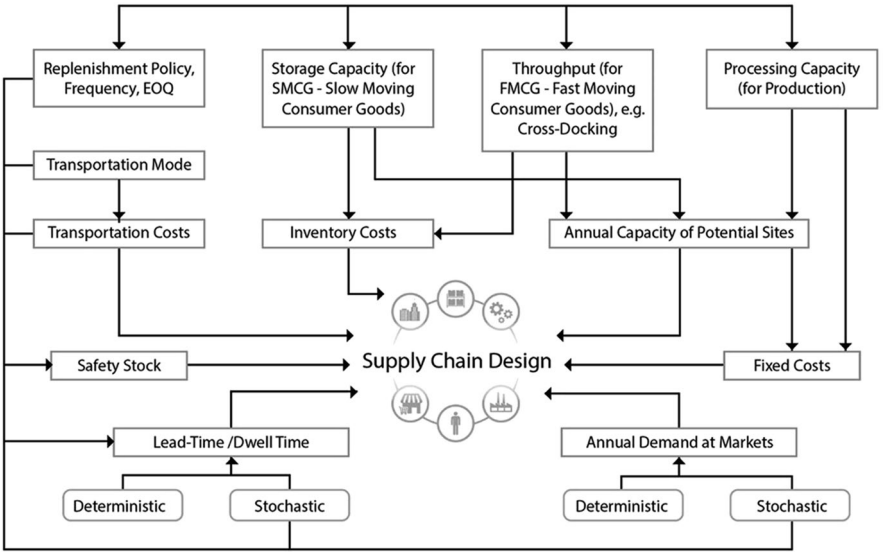


Fig. 7.29 Supply chain design analysis framework

- *Distribution network design with inventory, lead time, and transportation mode selection*
- *Production-distribution network design*
- *Hub location problem*
- *Supply network design with operational risks*
- *Supply network design with disruption risks*

In a generalized form, supply chain design using network optimization considers such parameters as

- Alternative facility locations
- Customers (markets)
- Production, inventory processing, and transportation costs
- Fixed facility costs and inventory holding costs
- Minimum and maximum throughputs and capacities in production, transportation, and storage
- Demand in the markets
- Number of periods and products
- Bill of materials

The variables to be optimized are facilities to be included in the supply chain design and quantities (flows) to be delivered from sources to destinations in the supply chain. The solutions are usually constrained by maximum/minimum demand in the markets and minimum and maximum throughputs and capacities in production, transportation, and storage. The objective function minimizes total costs.

Even though network optimization can lead to useful insights, some dynamic issues, such as inventory, sourcing, and shipment control policies are not considered within this framework of analysis. As such, simulation can be a valuable extension of network optimization because it enables consideration of time-dependent uncertainties, such as demand and lead-time fluctuations (i.e., operational risks) and facility breakdowns (i.e., disruption risks). Moreover, simulation can be used to validate optimization results in dynamic and uncertain environments (cf. Chap. 3).

7.7 Resilient, Sustainable, and Viable Supply Chain Design

So far, we have considered network optimization models that utilize the design-for-efficiency perspective. Our optimization was done with a single objective in mind: to minimize total SC costs. While efficiency and profitability are important and, indeed, central concerns in SCOM, network optimization models need to be extended in many cases. Three important extensions are resilience, sustainability**,** and viability.

Adopting carbon regulation mechanisms facilitates an evolution toward green and sustainable SCs, followed by the increased complexity of network optimization. Increasing concerns about the environmental impacts of SCs and international and government regulations have resulted in developments of *green SC design* problems that reach beyond economic aspects alone. In a green SC, the environmental impacts from SCs need to be minimized, complementing total cost minimization. In a *sustainable SC design*, the financial, environmental, and social impacts of the SC are considered simultaneously. The sustainability issues are usually addressed by defining a multi-objective optimization model for an SC. Solution techniques for multi-criteria decision-making (MCDM) problems are used in this case based on Pareto-optimization principles, weighting different objectives, and convolution (Homayoonia et al. 2023).

Resilient SC design adds aspects of disruption risk management to network designs. Due to natural disasters, strikes, or fires, some suppliers, factories, or warehouses can be disrupted and become unavailable for a more extended period, which might severely impact SC operations and performance. The key idea of a resilient SC design is to consider these uncertainties proactively (Ivanov 2021; Aldrighetti et al. 2021). The resilient SC design models can help identify an SC design's proneness to different disruptions by measuring disrupted and normal SC performance deviations. Moreover, resilient SC design models can be used to determine an optimal network design capable of withstanding disruptions. A stress-testing of SC designs can further be performed for some scenarios anticipating different upstream and downstream disruptions. Technically, the resilient SC design models are based on some disruption probability estimations. They are usually formulated either as a single-objective model with some resilience-related constraints or as a multi-criteria model with a trade-off between costs and resilience in an objective function. Such models are frequently non-linear and are formulated as stochastic or robust optimization models blended with heuristic solution methods.

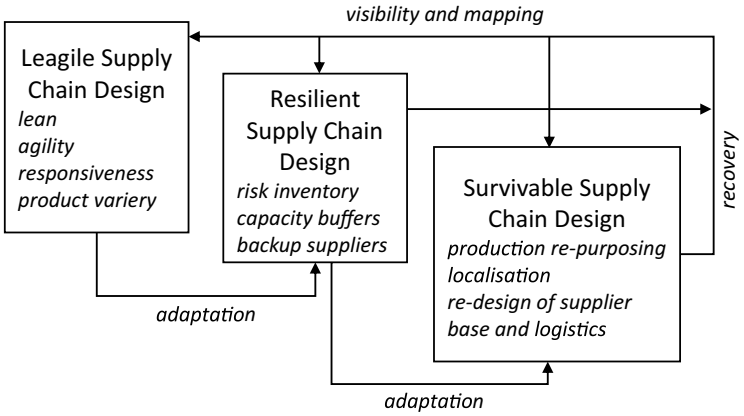


Fig. 7.30 Adaptive supply chain design model

Viable SC design extends the resilient SC design by the notion of survivability, e.g., during super disruptions such as the COVID-19 pandemic. The viable SC design capabilities include manufacturing repurposing, localization of SCs, and redesign of supplier base and logistics infrastructure to cope with radical and dynamically changing operational conditions. Most importantly, SCs need established and manageable adaptive mechanisms for fast transitions between efficient, resilient, and survivable network designs (Fig. 7.30).

The adaptive SC design model is presented in Fig. 7.30. The structural view represents three SC structures that are activated through adaptation and recovery actions. We illustrate using an example. Consider a car manufacturer with a global SC. This SC can have one structural design at times of economic stability and growth with full utilization of global sourcing, leanness, and agility advantages, and offering a wide variety of products to satisfy individual customer needs. The second structural design relates to disruptions. Its focus is on maintaining resilience in case of singular, local events such as natural disasters, strikes, or fires. This kind of design, i.e., the resilient SC, builds around proactive and reactive capabilities, such as risk mitigation inventory, capacity flexibilities, and back-up suppliers. Finally, in cases of long-term, global disruptions, such as the COVID-19 pandemic, the third SC design is adapted, which might be characterized by production changeovers (e.g., production of ventilators or masks instead of cars), reduced product variety, radical changes in supplier base and logistics, and production localization.

▶ **Practical Insights** For example, Ford utilizes three levels of the Viable Supply Chain model. Its major SC design is based on cost efficiency and profitability by utilizing the advantages of lean production such as Just-in-Time, agility (i.e., locating factories close to the markets), and global sourcing. To prepare for disruptions and transition to a resilient SC design, Ford evaluates the supplier risks, identifies critical suppliers,

analyzes the SC resilience in terms of Time-to-Survive and Time-to-Recover, and utilizes visibility technology to identify disruptions (Simchi-Levi et al. 2015; Ivanov 2025a, b) . In case of a severe crisis, such as the COVID-19 pandemic, Ford has launched the survivable SC design wherein the production systems and supplier base have been repurposed to produce face shields and ventilators instead of cars by using the available equipment and technology (Dolgui et al. 2024).

We note that it might be a very challenging task to operate and control three SC designs simultaneously, both in terms of efficiency and complexity. In addition, predicting all possible future scenarios and respective SC designs for matching supply and demand is nearly impossible. As such, the main role in the viable SC belongs to adaptation and recovery mechanisms, their design, establishment, training, and implementation. It might be instructive for firms to "virtually" design and simulate the SC structures for resilience and survivability and focus on the adaptation training to practice the SC changeability. We refer to further reading on the viable SC in Chaps. 4 and 15, as well as to the articles by Ivanov (2022), Ivanov and Dolgui (2020), and Ruel et al. (2024).

7.8 Key Points

This chapter introduced the problems, models, and techniques for managing location decisions in the context of a supply network setup. We received insights into location analysis processes and understand the importance of selecting the right location for facilities during the formation of a supply network. There are several different important decision tasks to be solved before the network can be set up. We have introduced an SCD framework for aligning these decisions. Finally, we have become familiar with the tools for supporting SCD decisions. The selection of an approach mainly depends on the problem data we want to consider for obtaining optimal decisions.

Section 7.3 provided an introduction to an important decision problem of location planning. The WLP addresses the challenge of researching markets and suppliers and forging transportation links between markets and suppliers from a longer term perspective. The goal of warehouse location planning is the minimization of the total costs for SCD. With the help of the PPS example, we analyzed a typical WLP setting and the relevant planning data.

We investigated a model-based approach to identify the best SCD. First, we proposed a mathematical optimization model for the WLP. Second, we configured and applied a spreadsheet decision support tool using the Excel Solver add-in. Third, we learned the technique of branch-&-bound as a general approach for deriving optimal design options for the WLP model. Finally, we analyzed the impacts of limited product availability at certain suppliers for the SCD and introduced the CWLP model for which we configured a spreadsheet-based tool to derive optimal solutions.

It can be concluded that mathematical programming techniques are useful and powerful tools for improving decisions for SCD. They allow for consideration of

demand, capacity, and costs while determining global SCD. At the same time, the application of these tools in real life should be considered subject to numerous limitations, such as assumptions on linear functions, deterministic parameter values, and computational complexity. A solution to the mathematical model is not automatically a managerial decision! Up-to-date research trends are involved in consideration of non-deterministic parameters (Lim et al. 2013), reliable SCD (Snyder and Daskin 2005; Klibi et al. 2010; Li et al. 2013), and the ripple effect in SCs (Ivanov et al. 2014).

Section 7.4 is considered the next stage in SCD, where all regions are processed individually. In each region, specific locations (expressed by coordinates) are identified at which a facility can be installed or from which supply quantities can infiltrate the SC. The output of this phase is suggestions for locations for opening facilities with regard to cost efficiency. We learned how to apply the center-of-gravity method to facility location decisions and how to determine optimal location, coordinated with the help of a simple average method and the Miehle algorithm. The center-of-gravity method can help determine the location of one warehouse subject to minimal transportation costs. It is a simple method that is easy to implement. The shortcomings are linear assumptions, the impossibility of considering multiple locations, deterministic data, and poor consideration of geographic and real road infrastructure.

In Sect. 7.5, the multiple criteria analysis was considered reaching beyond cost minimization. We learned factor-rating and utility value analysis methods and their advantages and limitations in practice. Factor-rating and utility value analysis methods allow for consideration of different qualitative and quantitative factors for location decisions which are easily understood by managers. Some shortcomings are the subjectivity of these methods regarding the selection of factors and their scoring and weighting. Sensitivity analysis is important for the practical application of factor-rating and utility value analysis methods.

Finally, in Sects. 7.6 and 7.7, we introduced a generalized model of SC design as well as notions of sustainable, resilient, and viable SC designs which extend the efficiency-oriented network modeling perspective toward multi-objective formulations. Such multi-objective SC design models can help in considering sustainability, resilience, and survivability factors when designing SC networks.

Recall Chap. 4, Supply Chain Strategy, and analyze the PPS case about the following:

- Discuss the SC strategy of PPS with respect to Europe and the celluloid-free table tennis balls. What is the market to be served? Is a reactive strategy preferred, or is a push strategy intended?
- What are the limitations given to Mr. Chen for the setup of the SCD in Europe?
- Assume that PPS wants to offer a 72 h delivery time to all customers in Europe when they order a packaging unit via a retail website. Discuss whether the SCD as intended by Mr. Chen can be used again or whether significant redesign efforts are required.

Acknowledgments The authors thank the OTLG GmbH & Co. KG for the permission to use the company materials in this textbook.

Bibliography

Aldrighetti R, Battini D, Ivanov D, Zennaro I (2021) Costs of resilience and disruptions in supply chain network design models: a review and future research directions. Int J Prod Econ 235:108103

Aldrighetti R, Battini D, Ivanov D (2023) Efficient resilience portfolio design in the supply chain with consideration of preparedness and recovery investments. Omega 117:103841

Amiri-Aref M, Klibi W, Babai MZ (2018) The multi-sourcing location inventory problem with stochastic demand. Eur J Oper Res 266(1):72–87

Amiri-Aref M, Farahani RZ, Hewitt M, Klibi W (2019) Equitable location of facilities in a region with probabilistic barriers to travel. Transp Res Part E: Logist Transp Rev 127:66–85

Arampantzi C, Minis I (2017) A new model for designing sustainable supply chain networks and its application to a global manufacturer. J Clean Prod 156:276–292

Askin RG, Baffo I, Xia M (2014) Multi-commodity warehouse location and distribution planning with inventory consideration. Int J Prod Res 52(7):1897–1910

Baghalian A, Rezapour S, Farahani RZ (2013) Robust supply chain network design with service level against disruptions and demand uncertainties: a real-life case. Eur J Oper Res 227:199–215

Baker KR (2015) Optimization modeling with spreadsheets, 3rd edn. Wiley, New York

Baron O, Milner J, Naseraldin H (2011) Facility location: a robust optimization approach. Prod Oper Manag 20(5):772–785

Benyoucef L, Xie X, Tanonkou GA (2013) Supply chain network design with unreliable suppliers: a Lagrangian relaxation-based approach. Int J Prod Res 51(21):6435–6454

Bierwirth C (2000) Adaptive search and management of logistics systems: base models for learning agents. Springer Science & Business Media

Brandenburg M, Rebs T (2015) Sustainable supply chain management: a modeling perspective. Ann Oper Res 229:213–252

Brusset X, Ivanov D, Jebali A, La Torre D, Repetto M (2023) A dynamic approach to supply chain reconfiguration and ripple effect analysis in an epidemic. Int J Prod Econ 263:108935

Chopra S, Meindl P (2012) Supply chain management. Strategy, planning and operation, 5th edn. Pearson, Harlow

Cousins PD (2002) A conceptual model for managing long-term inter-organisational relationships. Eur J Purchasing Supply Manage 8(2):71–82

Cousins PD, Lamming R, Lawson B, Squire B (2012) Strategic supply management – principles, theories and practice, 2nd edn. Pearson, Harlow

Daskin MS (1995) Network and discrete location: models, algorithms, and applications. Wiley, Hoboken

Dolgui A, Gusikhin O, Ivanov D, Li X, Stecke K (2024) A network-of-networks adaptation for cross-industry manufacturing repurposing. IISE Trans 56(6):666–682

Domschke W, Drexl A (1985) Logistik: standorte, 2nd edn. Oldenbourg Wissenschaftsverlag, Munich

Drezner Z (ed) (1995) Facility location: a survey of applications and methods. Springer, New York

Fahimnia B, Sarkis J, Eshragh A (2014) A tradeoff model for green supply chain planning: a leanness-versus-greenness analysis. Omega 54:173–190

Fahimnia B, Jabarzadeh A, Sarkis J (2018) Greening versus resilience: a supply chain design perspective. Transp Res E 119:129–148

Fattahi M, Govindan K, Keyvanshokooh E (2017) Responsive and resilient supply chain network design under operational and disruption risks with delivery lead-time sensitive customers. Transp Res Part E: Logist Transp Rev 101:176–200

Geoffrion AM, Graves GW (1974) Multicommodity distribution system design by benders decomposition. Manag Sci 20(5):822–844

Govindan K, Fattahi M, Keyvanshokooh E (2017) Supply chain network design under uncertainty: a comprehensive review and future research directions. Eur J Oper Res 263:108–141

Günther H-O, Tempelmeier H (2009) Produktion und Logistik, 8th edn. Springer, Heidelberg
Homayoonia Z, Pishvaee MS, Jahani H, Ivanov D (2023) A robust-heuristic optimization approach to a green supply chain design with consideration of assorted vehicle types and carbon policies under uncertainty. Ann Oper Res 324:395–435
http://www.truckinginfo.com/channel/fleet-management/news/story/2014/04/amazon-testing-delivery-fleet.aspx. Accessed 2 Jan 2025
Ivanov D (2021) Supply chain viability and the COVID-19 pandemic: a conceptual and formal generalisation of four major adaptation strategies. Int J Prod Res 59(12):3535–3552
Ivanov D (2022) Viable supply chain model: integrating agility, resilience and sustainability perspectives – lessons from and thinking beyond the COVID-19 pandemic. Ann Oper Res 319:1411–1431
Ivanov D (2024) Introduction to supply chain analytics. Springer Nature, Cham
Ivanov D (2025a) When is the supply chain resilient? Customer and operational perspectives. Int J Prod Res. https://doi.org/10.1080/00207543.2025.2454331
Ivanov D (2025b) Introduction to Supply Chain Resilience and Viability, 2nd edn. Springer Nature, Cham
Ivanov D, Dolgui A (2020) Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability. A position paper motivated by COVID-19 outbreak. Int J Prod Res 58(10):2904–2915
Ivanov D, Sokolov B, Pavlov A (2014) Optimal distribution (re)planning in a centralized multistage network under conditions of ripple effect and structure dynamics. Eur J Oper Res 237(2):758–770
Ivanov D, Pavlov A, Pavlov D, Sokolov B (2017) Minimization of disruption-related return flows in the supply chain. Int J Prod Econ 183:503–513
Jabbarzadeh A, Fahimnia B, Sabouhi F (2018) Resilient and sustainable supply chain design: sustainability analysis under disruption risks. Int J Prod Res 56(17):5945–5968
Klibi W, Martel A, Guitouni A (2010) The design of robust value-creating supply chain networks: a critical review. Eur J Oper Res 203(2):283–293
Li Q, Zeng B, Savachkin A (2013) Reliable facility location design under disruptions. Comput Oper Res 40(4):901–909
Lim MK, Bassamboo A, Chopra S, Daskin MS (2013) Facility location decisions with random disruptions and imperfect estimation. Manuf Serv Oper Manag 15(2):239–249
Manzini R, Bindi F (2009) Strategic design and operational management optimization of a multistage physical distribution system. Transp Res Part E-Logist Transp Rev 45(6):915–936
Melo MT, Nickel S, Saldanha-da-Gama F (2009) Facility location and supply chain management – a review. Eur J Oper Res 196(2):401–412
Miehle W (1958) Link-length minimization in networks. Oper Res 6(2):232–243
Padovano A, Ivanov D (2025) Towards resilient and viable supply chains: a multidimensional model and empirical analysis. Int J Prod Res 1–39. https://doi.org/10.1080/00207543.2025.2470350
Pavlov A, Ivanov D, Pavlov D, Slinko A (2019) Optimization of network redundancy and contingency planning in sustainable and resilient supply chain resource management under conditions of structural dynamics. Ann Oper Res 1:30
Ruel S, El Baz J, Ivanov D, Das A (2024) Supply chain viability: conceptualization, measurement, and nomological validation. Ann Oper Res 335:1107–1136
Sawik T (2020) Supply chain disruption management, 2nd edn. Springer, New York
Schönberger J (2011) Model-based control of logistics processes in volatile environments: decision support for operations planning in supply consortia. Springer, New York
Simchi-Levi D, Schmidt W, Wei Y, Zhang PY, Combs K, Ge Y, Gusikhin O, Sander M, Zhang D (2015) Identifying risks and mitigating disruptions in the automotive supply chain. Interfaces 45(5):375–390
Snyder LV, Daskin MS (2005) Reliability models for facility location: the expected failure cost case. Transp Sci 39(3):400–416

Stadtler H, Kilger C (2008) Supply chain management and advanced planning, 4th edn. Springer, Berlin
Vidal CJ, Goetschalckx M (1997) Strategic production-distribution models: a critical review with emphasis on global supply chain models. Eur J Oper Res 98(1):1–18
Wernerfelt B (1984) A resource-based view of the firm. Strateg Manag J 5(2):171–180

Distribution and Transportation Network Design

8

In Chap. 7, we discussed the identification of appropriate facility locations. The interactions of several facilities and their common activities have a crucial impact on the success of a supply network. Therefore, the linkage of these facilities also plays an important role in the SC configuration phase. In this phase, key decisions related to the availability of facilities have already been made, but their physical connections through the installation of appropriate transportation links have not been addressed yet.

In this chapter, we will discuss basic strategies to configure distribution and transportation networks so that given supply locations are physically connected with demand locations. In Sect. 8.1, we introduce a case study to discuss the challenges related to the installation of transportation links. Section 8.2 introduces generic types of transportation network setups as a starting point for more sophisticated network layouts. Section 8.3 introduces the concept of shipment consolidation as the key approach to the realization of economies of scale in transportation. Section 8.4 is dedicated to the presentation of different transport network configuration objectives and their balancing so that efficient and profitable network layouts can be set up. Section 8.5 surveys the optimization of incorporating transport service providers (freight carriers). Section 8.6 is devoted to different case studies of distribution network design.

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_8) contains supplementary material, which is available to authorized users.

Find additional case studies and videos on information technology for SCOM in the *E-Supplement* to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_8.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_8

Learning Objectives for This Chapter
This chapter will familiarize you with the most important aspects of configuring transportation networks. In particular, you will:

- Learn typical aspects and decision problems relevant to configuring today's distribution and transportation networks.
- Become familiar with generic types of distribution and transportation network structures.
- Understand the basic trade-offs in distribution and transportation network design with regard to costs and responsiveness.
- Understand what consolidation is and how consolidation can be used to achieve economies of scale in the context of distribution and transportation.
- Optimize the engagement of a logistics service provider by assigning different shipping quantities to appropriate transportation links.

8.1 Introductory Case Study: Designing a Distribution Network

Consider a high-quality furniture manufacturer A. The product portfolio is mainly comprised of all types of furniture made from natural wood. A typical customer of A is willing to pay quite a high price to get a fairly high-quality product. Figure 8.1 depicts the supply network for A.

Since wood is the primary material of A products, the procurement of different kinds of wood is critical to the performance and success of A operations. A large amount of the wood is bought from Asian suppliers that meet the necessary sustainability criteria and regulations. After seeing the trees, deputies from A's procurement department selected those that would be used as raw material "suppliers." Local farmworkers chop the trees, clean them roughly, and load them in containers. These containers are forwarded to the nearest container port, where they are shipped directly or with transshipments from different carriers to the container ports of Hamburg, Bremerhaven, or Wilhelmshaven in northern Germany. Due to weather conditions in the Asian supply regions and the incorporation of different container vessel lines, it is impossible to even out and balance the number of containers bringing supplies for A.

After a container for A arrives in one of the three container terminals, a wood expert inspects raw materials. Once a container load has passed quality control, the wood is shipped on rented trucks, which bring it to A's manufacturing site near Munich, approximately 800 km away from the coast (see Fig. 8.1).

A was founded in 1875 and has grown continuously with stable economic success. However, during the economic crisis between 2008 and 2010, A was in danger of becoming bankrupt. It was able to survive only with the help of additional investments. Recently, the new stakeholders have been analyzing and evaluating all of A processes in order to protect their investments and achieve long-term profitability. These evaluations are executed by external consultants. In their final evaluation

Fig. 8.1 The European procurement network (screenshot from anyLogistix software)

report, they mentioned that transportation costs for the raw materials are quite high. They recognize that imported wood is not available in Europe, so the international procurement strategy is necessary. However, experts recommend reconsidering the process of bringing imported wood from the ports in northern Germany to the manufacturing site in Bavaria.

Having received the evaluation reports from the stakeholders, the management board of A discusses possible implications. Regarding the "transportation problem," they are concerned about insufficient logistical knowledge to assess and potentially restructure the transportation processes. They discuss the situation with stakeholder representatives and get the OK to extend the management board to another member who will be in charge of the transportation and logistics processes. After a careful selection procedure, A hires Mike as the transportation and logistics manager. Mike has worked for several years as a transport consulting engineer and on several developments and restructuring projects for national and international transport networks.

His first duty with A is to check the current setup of the transportation process system between the ports and the manufacturing site of A.

(a) Mike discovers that A does not possess own trucks. Whenever a certain transport quantity has to be moved, A hires a truck from a freight forwarding company at a fixed daily rate. Is this a good idea? Please discuss the pros and cons of this setup.

Having managed the critical years, A wants to start growing again. Currently, they are preparing the integration of two other furniture factories into their company. One of these production facilities is located close to Prague, the capital of the Czech Republic. The second additional production site is in Austria, close to Salzburg, directly beside the Austrian-German border. After the integration of these two additional production sites has been completed, imported quantities of wood will increase by approximately 150%. To ensure the availability of sufficient quantities of wood, imports will also be processed via the Dutch port of Rotterdam.

(b) Should the previous method of transport be retained? What are the possible modifications required for hinterland transportation? Should A buy own trucks and set up own transport department? Which production site should be supplied from which port in order to keep transportation costs to a minimum? Should all necessary transport services be given away to a transport service provider?

8.2 Generic Transport Network Structures

Mike prepares a short summary about the generic *transport network* structures from the perspective of the so-called origin-to-destination (OD) demand. The OD-demand expresses the need to move a given quantity of goods from a location of supply (the origin) to a location of demand (the destination).

In the simplest setup, the supply quantities are available at a single location (the origin), and the available supply quantity is requested completely or partially from a single location (the destination) as outlined in Fig. 8.2. There is no opportunity to optimize (reduce) the transportation efforts if the destination wants to receive the requested quantity immediately. If a split delivery is allowed, then it is possible to divide the total quantity into smaller shipments, which can be picked up and delivered one after another. Time of delivery of individual shipments can be arranged.

Decisions about the distribution network setup outlined in Fig. 8.3 are the same as in the link setup described previously. Only the time of delivery can be decided.

In Fig. 8.4, the destination requests a certain supply quantity, but the demand quantity is distributed over several origin locations. In general, the total available quantity exceeds the quantity demanded. Besides responding to the question of when to deliver, another important decision must be made. It is necessary to decide which origin should provide which supply quantity. It is important to take

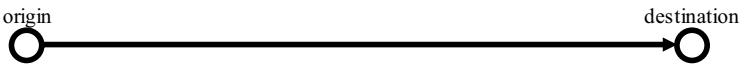


Fig. 8.2 Transportation link

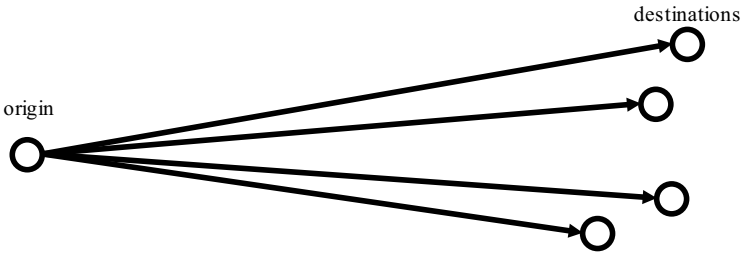


Fig. 8.3 Distribution network

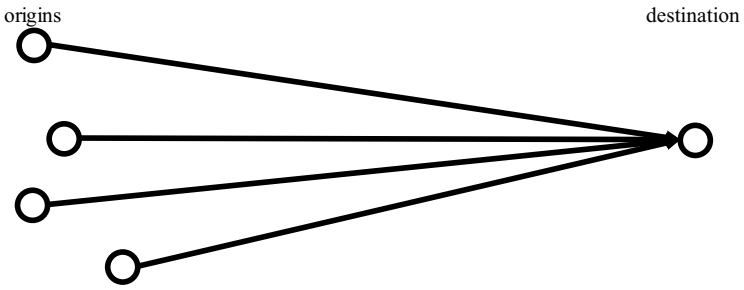


Fig. 8.4 Supply network

care that the cheapest transportation links are used. However, if the relevant origin does not offer sufficient quantities, it is necessary to use a second (and maybe further) transportation links to satisfy demand from the destination. From the perspective of the location planning framework introduced in Chap. 7, answering the question as to which supply location contributes which quantity to fulfill the total demand would be a fourth-phase decision: this is called the capacity allocation decision.

The most complex situation is represented by the *many-to-many network* structure shown in Fig. 8.5. Here, several origins provide supply quantities, but there are also several demanding destination locations. Besides decisions about delivery times, it is necessary to make two additional decisions. First, it is necessary to decide from which location(s) an individual destination location is supplied. Second, for each origin location, a decision is required regarding the quantities and destinations that are supplied. Finally, the decision becomes even more complex if the costs for moving goods along the available transportation links are different.

Mike concludes his introduction into generic transportation network structures by applying the new vocabulary to the A case. Currently, there is a supply network structure available with the three ports, Wilhelmshaven, Bremerhaven, and Hamburg, forming the set of origins and a single destination in Munich. However, as soon as the second manufacturing facility (either in the Czech Republic or in Austria) is integrated into the wood supply network, the network type changes to a many-to-many network structure.

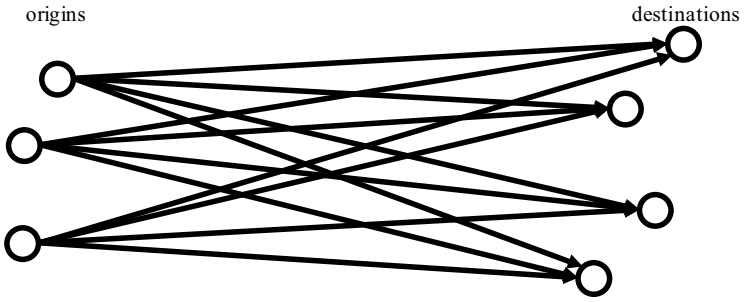


Fig. 8.5 Many-to-many network structure

► **Practical Insights** New Big Data-driven marketing strategies influence distribution network design. One example is the omnichannel strategy. Omnichannel is the cross-channel business model to increase the customer's experience by organizing promotions on-site, using social networks, apps, and other channels. This diversity increases demand unpredictability and distribution network coordination complexity.

8.3 Realizing Economies of Scale in Transportation

Walter, who is A's finance officer, is very interested in Mike's statements. He is the first one to ask Mike a question. He wants to know how much money can be saved in the transportation business and how the efficiency of the process can be improved. From his daily business, he knows that economies of scale are typically realized if large quantities of a product are bought or if unnecessary setups of machines can be avoided. Walter wants to know from Mike if similar principles are exploited in transportation.

Mike is happy about this question. It offers an entry point for his explanations about how money can be saved through sophisticatedly organized transportation systems and improved transportation processes. Walter asks: "What is the general logic behind reducing costs in transportation? What are the general issues to be considered here for a "good transportation" system configuration?"

8.3.1 Consolidation of Shipments

Mike wants to hire trucks from a trucking company to carry out the necessary transportation services between Hamburg and Munich, since A does not run its own trucks. After a telephone call to an old friend in the forwarding company TransRoad, he receives an offer for forwarding services. According to the offer, the provision of a truck costs 850 € per day, including all mileage and the costs for the driver, insurances, etc. The truck provides a payload of at most 25 tons, and the aforementioned

amount is due independently from the actual loaded weight. If Mike wants to ship more than 25 tons a day, he should hire and pay for a second truck. The rental conditions for the second and a conditional third truck are identical to those already outlined.

An evaluation of this offer reveals the following interesting cost structures and properties. If a truck is hired for the first ton payload, the remaining 24 tons can be transported without additional costs. This means that the proposed tariff contains a "bounded flat rate" up to 25 tons (see Fig. 8.6). The continuous line represents the costs if one truck is hired, the dotted line represents the situation with two hired trucks, and the dashed line applies to the costs if the payload is so high that a third truck must be hired. There are discontinuities of the total costs whenever another truck is needed.

Some conclusions can be formulated after some basic variable name conventions have been introduced. Let F be the price for hiring a truck and Q be the payload of the hired truck. Let d_R be the corresponding OD-demand for $\left\lceil \frac{d_R}{Q} \right\rceil$ the considered OD relation R. The number of required trucks $N(R; Q)$ equals then and the total transportation costs for this relation R sum up to $C(R; Q; F) := F \cdot N(R; Q)$.

If the costs for the truck are equally split over the complete payload assigned to this truck, then the average costs per ton decrease with increasing payload as long as the total payload does not exceed 25 tons (the payload capacity provided by the truck). The carriage of 1 ton leads to a cost factor of 850 € per ton, while the carriage of 10 tons reduces this to 85 € per ton (Fig. 8.7). If the payload exceeds 25 tons, it becomes necessary to hire an additional truck which causes a discontinuity in the total costs C, leading to an implied increase in the costs $c(d_R;F;Q)$ per ton payload.

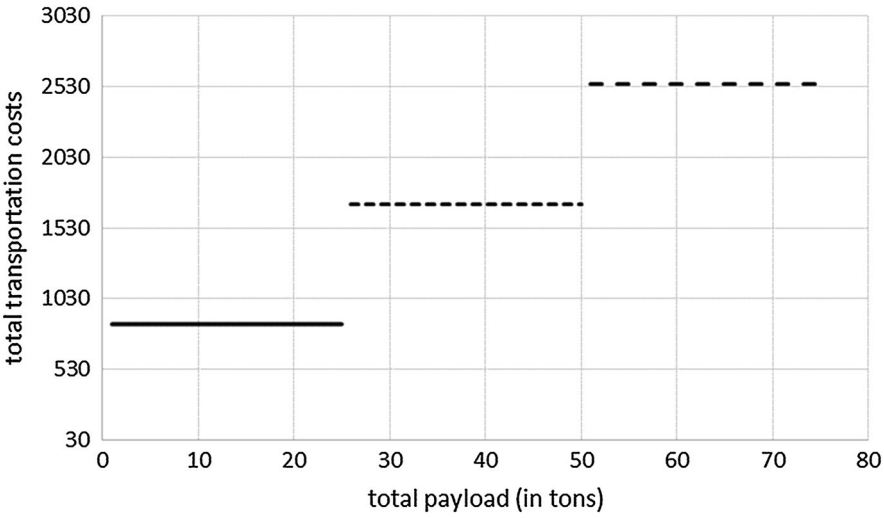


Fig. 8.6 Piece-wise constant function of the total transportation costs for given payload quantity

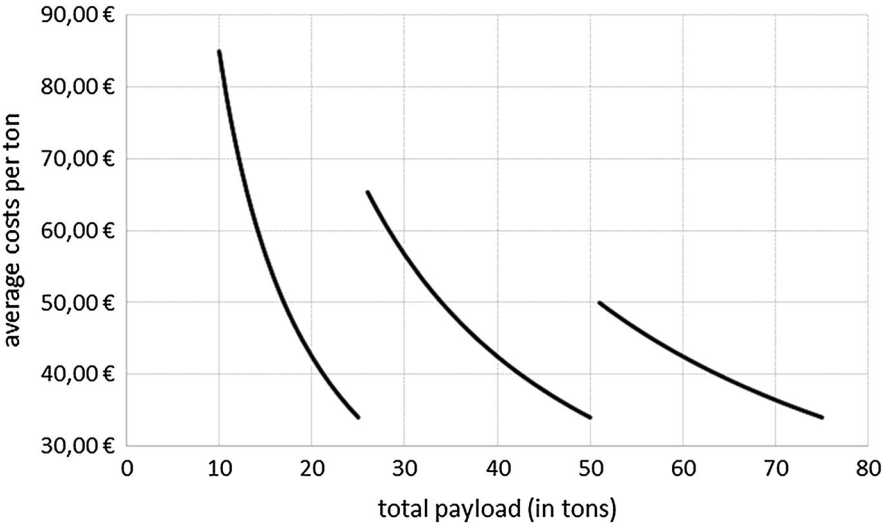


Fig. 8.7 Costs per ton of payload (the cost-per-ton-values for fewer than 10 tons of payload are not shown to preserve the readability of the graphs)

▶ **Practical Insights** Transportation costs typically comprise 6–8% of the company's costs. In most cases, transportation costs can be reduced by 10–15%. For example, a company with a total cost of 1.0 billion euros can cut transport costs by around 8.0 million euros.

Mike concludes from the aforementioned facts that he should attempt to fill a hired truck to minimize the costs per moved ton of payload. Furthermore, a potentially necessary second truck should also be filled as much as possible to reduce transportation costs.

Consolidation is the combination of two or more quantities to be shipped between a given origin and a given destination. Mike thinks about ways to consolidate several quantities to fill up a partially filled truck. In the selected week shown in Table 8.1, Mike detects that the stated demand for Monday (the production output from the concrete factory) is 32 tons, so that two trucks are needed to carry the complete production output to Munich. This results in total truck hiring costs of 1700 €, which leads to 1700 €/32 tons = 53.13 €/ton costs for each carried ton, but the average fill rate is only 64%. The immediate transport of the total quantity without the postponement of any quantities requires hiring 15 trucks, so that the weekly transportation costs account for 15 × 850 € = 12,750 € at an average fill rate of 71.5%.

8.3.2 Postponement

Analyzing the daily transport demand for a week as shown in Table 8.1, Mike comes up with the idea of postponing the transfer of a fraction of the daily transportation

Table 8.1 Daily transportation demand within a week without consolidation

	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Sums
OD-demand d_R (in tons)	32	53	18	34	43	60	30	270
Fulfilled demand	32	53	18	34	43	60	30	270
Hired trucks	2	3	1	2	2	3	2	15
Costs C (€)	1700	2550	850	1700	1700	2550	1700	12,750
Costs c (per ton) (€)	53.13	48.11	47.22	50.00	39.53	42.50	56.67	

Table 8.2 Daily transportation demand within a week with postponement of fractional truck loadings

	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Sums
Incoming OD-demand d_R	32	53	18	34	43	60	30	270
Postponed demand	0	7	10	3	12	5	15	
Available demand	32	60	28	37	55	65	45	
Now fulfilled demand	25	50	25	25	50	50	45	270
Hired trucks	1	2	1	1	2	2	2	11
Costs C (€)	850	1700	850	850	1700	1700	1700	9350
Costs c (per ton) (€)	34.00	34.00	34.00	34.00	34.00	34.00	37.78	

demand to avoid the inclusion of a truck that is only partially filled. Mike knows that there is no urgency related to the shipment of demand as long as at least one truck travels from Hamburg to Munich each day. His idea is now to send out only completely filled trucks (except on Sunday when the residual quantity of the weekly demand must be moved to Munich).

Table 8.2 shows the resulting transport activities. The incoming demand is merged with postponed quantities to the recently available demand. *Full truckloads* (FTL) are derived from the available demand quantities, but residual quantities are postponed until a later day of the week. This strategy results in completely filled trucks Monday to Saturday, but on Sunday, the average fill rate falls to 90%. However, only 11 trucks are needed, costing 9350 €, representing a reduction in transportation costs of 26.67% and an average capacity utilization rate greater than 98%.

Mike is very surprised at how much money he can save using the postponement strategy to send out only fully loaded trucks to realize the largest possible economies of scale. However, Mike is aware that this postponement strategy is fragile. If the demanding construction site requests immediate transportation of the produced quantities, this strategy can no longer be applied. Therefore, Mike will look for other opportunities to increase the fill rate of the hired trucks.

8.3.3 Milk-Runs

Mike analyzes the daily supply quantities from the ports of Hamburg, Bremerhaven, and Wilhelmshaven, all located close together in northern Germany. Currently, A

Table 8.3 Daily transportation demand within the three German ports

	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Sums
Demand waiting in Hamburg								
OD-demand d_R (in tons)	32	53	18	34	43	60	30	270
Fulfilled demand	32	53	18	34	43	60	30	270
Hired trucks	2	3	1	2	2	3	2	15
Costs C (€)	1700	2550	850	1700	1700	2550	1700	12,750
Costs c (per ton) (€)	53.13	48.11	47.22	50.00	39.53	42.50	56.67	48.17
Utilization (%)	64	71	72	68	86	80	60	72
Demand waiting in Bremerhaven								
OD-demand d_R (in tons)	23	42	31	28	38	50	20	232
Fulfilled demand	23	42	31	28	38	50	20	232
Hired trucks	1	2	2	2	2	2	1	12
Costs C (€)	850	1700	1700	1700	1700	1700	850	10,200
Costs c (per ton) (€)	36.96	40.48	54.84	60.71	44.74	34.00	42.50	44.89
Utilization (%)	92	84	62	56	76	100	80	79
Demand waiting in Wilhelmshaven								
OD-demand d_R (in tons)	13	12	25	13	26	38	11	138
Fulfilled demand	13	12	25	13	26	38	11	138
Hired trucks	1	1	1	1	2	2	1	9
Costs C (€)	850	850	850	850	1700	1700	850	7650
Costs c (per ton) (€)	65.38	70.83	34.00	65.38	65.38	44.74	77.27	60.43
Utilization (%)	52	48	100	52	52	76	44	61

hires trucks for all three links, Hamburg-Munich, Bremerhaven-Munich, and Wilhelmshaven-Munich. Each of these trucks picks up the assigned payload quantity at the selected port and brings it directly to the Munich area. For the example week given in Table 8.3, this approach leads to a total of 36 hired trucks. The costs to be paid to the forwarding companies add up to 30,600 €. Average capacity utilization between 61% and 79% are observed, and the average costs per moved ton vary between 44.89 € and 60.43 €.

A first analysis of the data from Table 8.3 reveals that only 17 of the 36 hired trucks are fully loaded. Over half of the hired trucks are operated with only a partial load. In addition, the quantities carried by partially loaded trucks are often quite small. Mike analyzes the situation on Thursday. From Hamburg, the second hired truck carries only 9 tons. The second truck hired for the Bremerhaven to Munich service picks up only 3 tons, while the only truck going from Wilhelmshaven loads only 13 tons. Mike recognizes that these three "excess" quantities add up to 25 tons so that one truck could move all three shipments. For this reason, Mike decides to evaluate a so-called milk-run service (Chopra 2019) from the ports to Munich. In a milk-run, one truck visits two or more pickup locations before bringing the whole load to a common destination.

In the A setup, there are four different milk-runs possible: (1) Hamburg, Bremerhaven and Wilhelmshaven are served as pickup locations by the same truck; (2) Hamburg and Bremerhaven are served in one milk-run; (3) demand quantities

waiting in Hamburg as well as in Wilhelmshaven are consolidated; (4) one truck loads demand quantities in Bremerhaven as well as in Wilhelmshaven. The visiting sequence depends on the initial position of the truck as well as on the pickup locations and the common drop-off location (see Chap. 14). Milk-runs represent the location-based consolidation of shipments.

In order to achieve the largest cost reduction possible with milk-runs, Mike employs the following approach. He starts determining the minimal number of fully loaded trucks for each day and port. As an example, Mike considers Monday. Hamburg expresses a demand of 32 tons so that one full truckload of 25 tons can be achieved. Mike puts the remaining 7 tons in the "milk-run quantity." From Bremerhaven, 23 tons must be carried away, but no full truckload can be generated, so Mike puts all 23 tons into the milk-run quantity. From Wilhelmshaven, Mike puts 13 tons into the milk-run quantity. Overall, 43 tons were selected for a milk-run service.

One option (Table 8.4) is to hire two trucks for milk-runs. The first truck among them starts in Hamburg, picking up 7 tons. It proceeds to Bremerhaven, where it loads an additional 13 tons; next, it continues to Wilhelmshaven, where 5 tons are picked up. The second hired milk-run truck starts in Bremerhaven and loads ten trucks. Additionally, it picks up 8 tons in Wilhelmshaven. The incorporation of the milk-run services needs two trucks instead of three if no milk-run is used, so that one truck is saved on Monday. Similarly, Mike compiles milk-runs for the remaining days (see Table 8.4). The daily saving of trucks for hire finally leads to a reduction of trucks required to 28 (compared to 36 trucks without milk-runs). This implies a reduction of hiring costs by 22.2% down to 23,800 €. The average capacity utilization rate climbs up to 92%, and the average costs for moving a 1-ton payload fall to 37.38 € when milk-run operations are performed.

8.3.4 Transshipment

Mike has explained the benefits of the milk-run services. However, he also mentions the shortcomings and deficiencies. The major challenge is to ensure that the milk-run services fit daily demand. From day to day, the milk-runs have to be adapted to recent demand. This makes milk-run planning quite complicated. If the total quantities from the milk-run locations are slightly above the full truck load quantity, then the incorporation of an almost empty truck cannot be avoided. Furthermore, in the case of the prospective inclusion of the port of Rotterdam as well as the two manufacturing sites in Austria and the Czech Republic, milk-runs seem to become less attractive due to the long distances involved. Because of these challenges, Mike proposes a third consolidation approach to the management board of A.

As can be observed from the left-hand map in Fig. 8.8, a major part of the three truck routes from the seaports to the production site is the same. All three pickup locations Wilhelmshaven, Bremerhaven, and Hamburg "share" a large part of their connection path to their common destination, Munich. To increase the fill rate of trucks traveling long distances, Mike proposes installing a transshipment facility in

Table 8.4 Daily transportation with milk-runs

	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Sums
Demand waiting in Hamburg								
OD-demand d_R (in tons)	32	53	18	34	43	60	30	270
Fulfilled demand	25	50	0	25	25	50	25	200
Hired trucks	1	2	0	1	1	2	1	8
Costs C (€)	850	1700	–	850	850	1700	850	6.800
Utilization (%)	100	100	0	100	100	100	100	86
Demand waiting in Bremerhaven								
OD-demand d_R (in tons)	23	42	31	28	38	50	20	232
Fulfilled demand	0	25	25	25	25	50	0	150
Hired trucks	0	1	1	1	1	2	0	6
Costs C (€)	–	850	850	850	850	1700	–	5100
Utilization (%)	0	100	100	100	100	100	0	71
Demand waiting in Wilhelmshaven								
OD-demand d_R (in tons)	13	12	25	13	26	38	11	138
Fulfilled demand	0	7	25	0	25	25	11	93
Hired trucks	0	1	1	0	1	1	1	5
Costs C (€)	–	850	850	–	850	850	850	4250
Utilization (%)	0	28	100	0	100	100	44	53
Remaining demand fulfilled in milk-runs								
Milk-run demand in Hamburg (HH)	7	3	18	9	18	10	5	70
Milk-run demand in Bremerhaven (BHV)	23	17	6	3	13	0	20	82
Milk-run demand in Wilhelmshaven (WHV)	13	5	0	13	1	13	0	45
Total demand	43	25	24	25	32	23	25	
Milk-run HH-BHV-WHV	25	25		25				
Milk-run HH-BHV			24		25		25	
Milk-run HH-WHV						23		
Milk-run BHV-WHV	18				7			
Fulfilled demand	43	25	24	25	32	23	25	
Hired trucks	2	1	1	1	2	1	1	9
Costs C (€)	1700	850	850	850	1700	850	850	7650
Utilization (%)	86	100	96	100	64	92	100	91
Overall number of hired trucks	3	5	3	3	5	6	3	28
Costs C (€)	2550	4250	2550	2550	4250	5100	2550	23,800
Moved quantity	68	107	74	75	107	148	61	640
Costs per ton (€)	38	40	34	34	40	34	42	37.38
Utilization (%)	91	86	99	100	86	99	81	92

Hanover, where trucks coming from the three seaports enter the same major north-to-south motorway, A7. In this transshipment facility, the inbound trucks are unloaded and sent back to the ports in order to bring further quantities to the facility. Other trucks are then loaded with the quantities that recently arrived, and these trucks then travel to Munich (Fig. 8.9).

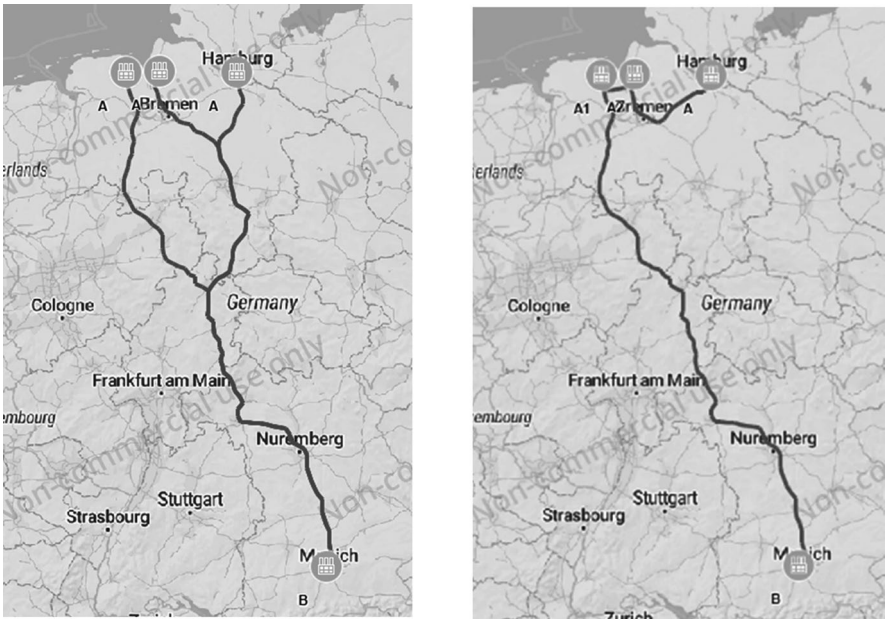


Fig. 8.8 Direct delivery network for A (left) and milk-run service-based network (right) (screenshots from anyLogistix software)

Mike outlines two major benefits of this type of shipment consolidation. First, the trucks traveling from the ports to the transshipment facility travel relatively short distances, between 150 and 200 km. Therefore, they can shuttle two or even three times between their pickup port and Hanover, so a single truck is enough to forward all quantities from the port to the transshipment facility in the port hinterland. Second, due to the consolidation of the inbound flow of quantities originating from the three seaports, there are huge opportunities to compile full truck loads for the main-haul service between Hanover and Munich. Both aspects contribute to the reduction of transportation costs.

Mike explains the idea of transshipment-based consolidation according to the following setup. A hired truck is available and paid for 24 h. It can execute three feeder cycles within this period between a port and the transshipment facility and a subsequent main-haul trip from Hanover to Munich (type-A service pattern). Alternatively, it can complete six feeder cycles and no main-haul trip (type-B service pattern); or, as the third option, it can execute one main-haul trip only (type-C service pattern). A truck that is operated only in the feeder service in northern Germany can be hired for a reduced tariff of 750 € since the truck operator incurs fewer costs for overnight stays for the vehicle and crew. The reduced tariff of 700 € applies to a truck executing one main haul only due to reduced travel distances, as in the type-C-pattern. In the first two consolidation approaches (postponement and milk-runs), Mike was unable to exploit the reduced tariffs since each hired truck was assigned to a long-haul operation. The separation of feeder operations from

Fig. 8.9 Transportation network with transshipment point in Hanover (screenshot from anyLogistix software)

main-haul operations results in the possibility of agreeing on different tariffs for trucks assigned to different service patterns.

Table 8.5 and model (Eqs. 8.1–8.4) contain calculations for the number of trucks needed as well as the associated costs. Determining the least-cost number of trucks operating type-A, type-B, or type-C patterns per day is an optimization problem with constraints. The least-cost number of hired trucks for the three patterns should be found, but it must also be ensured that the minimal number of required feeder cycles and main-haul services are provided. However, this problem can be modeled as an integer linear program as follows. Let y_A be the integer decision variable representing the number of trucks hired for executing a type-A service pattern. For the number of hired type-B pattern trucks and type-C pattern trucks, the integer decision variables y_B as well as y_C are used. Let k^{feed} denote the minimal feeder service number required and let k^{mh} denote the minimal main-haul services required.

$$Z = 850y_A + 750y_B + 700y_C \tag{8.1}$$

Table 8.5 Daily transportation with transshipments

	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Sums
Demand waiting in Hamburg								
OD-demand d_R (in tons)	32	53	18	34	43	60	30	270
Fulfilled demand	32	53	18	34	43	60	30	270
Required feeder services	2	3	1	2	2	3	2	15
Demand waiting in Bremerhaven								
OD-demand d_R (in tons)	23	42	31	28	38	50	20	232
Fulfilled demand	23	42	31	28	38	50	20	232
Required feeder services	1	2	2	2	2	2	1	12
Demand waiting in Wilhelmshaven								
OD-demand d_R (in tons)	13	12	25	13	26	38	11	138
Fulfilled demand	13	12	25	13	26	38	11	138
Required feeder services	1	1	1	1	2	2	1	9
Total demand Hanover to Munich	68	107	74	75	107	148	61	640
Required feeder cycles	4	6	4	5	6	7	4	36
Required feeder cycles	3	5	3	3	5	6	3	28
Hired trucks for type-A-pattern	2	5	2	2	2	3	2	18
Hired trucks for type-B-pattern	0	0	0	0	0	0	0	0
Hired trucks for type-C-pattern	1	0	1	1	3	3	1	10
Available feeder cycles	6	15	6	6	6	9	6	54
Available main-haul services	3	5	3	3	5	6	3	28
Hiring costs (€)	2400	4250	2400	2400	3800	4650	2400	22,300
Hired trucks	3	5	3	3	5	6	3	28

$$3y_A + 6y_B \geq k^{\text{feed}} \tag{8.2}$$

$$1y_A + 1y_C \geq k^{mk} \tag{8.3}$$

$$y_A, y_B, y_C \in \{0;1;2;\ldots\} \tag{8.4}$$

The goal is to minimize the total hiring charges (Eq. 8.1), but the minimal number of required feeder services (Eq. 8.2) and main-haul services (Eq. 8.3) must be respected. Only complete trucks can be hired (Eq. 8.4). This model can easily be solved using Excel Solver or any other solver software.

Mike has determined the best number of trucks as shown in Table 8.5. The incorporation of trucks assigned to main-hauls only (type-C pattern) realizes costs reductions of more than 6% compared to the milk-run situation, although the number of incorporated trucks remains the same.

Although the transshipment strategy requires the least weekly transportation costs if no postponement is possible, Mike is aware of some shortcomings. The transshipment facility must be constructed, hired, or rented. The transshipment activities and the necessary storage processes incur costs. As long as the additional costs are less than the transportation costs savings, Mike suggests using a

transshipment facility. However, Mike also warns managers that the complexity of the transport network configuration would increase.

So far, Mike has introduced the concept of shipment consolidation as a strategy to avoid partially filled truck trips. Consolidation intends to utilize economies of scale for transportation. Three consolidation strategies have been proposed. Postponement compiles shipments from different time slots together. Milk-runs aim to load shipments from different pickup operations on one truck to lift the fill rate of this truck. Transshipment exploits different hiring tariffs for different kinds of truck operations (depending on regional services).

Mike concludes that the ideas presented here generally also apply to other means of transport, such as trains, plans, vessels, or barges. In the case of transshipment consolidation, feeder services, and main-haul operations can be assigned to resources from different means of transport (see Sect. 8.6).

8.4 Trade-Off-Based Transportation Network Design

After Mike finished his presentation, a discussion is initiated about the right distribution network strategy for A. The board members like the idea of redesigning the supply network based on transshipment. However, they do not feel comfortable with the necessary investment nor with the annual costs for running a transshipment point. With respect to the prospective extension by the inclusion of Rotterdam, Salzburg, and Prague, they cannot estimate the associated costs and benefits. Therefore, they ask Mike to deliver further explanations about this issue.

Mike expected such a question. He has prepared an executive summary of the section "Factors Affecting Distribution Network Design in the Supply Chain" from Chopra's (2019) SCM book, where the aforementioned topic is discussed in detail. He distributes these copies and provides an oral survey.

There are two general directions for designing and evaluating transportation networks. First, *customer demand* must be identified. Demand estimates will be the starting point for the setup of the transportation network. Second, the *costs* of the installation and deployment of the transportation network have to be determined. These costs are subject to uncertainty regarding the future under which the transport network will be used. On the one hand, customer needs are unknown, hidden, or will change. On the other hand, the costs for running a certain network are subject to external impacts, possibly from oscillating customer demand, varying fuel prices, labor costs, and so on. Therefore, only a rough estimation of benefits with regard to customer satisfaction and costs is possible.

The major driver of customer satisfaction is the ability to react immediately and with high reliability to customer demand. If a customer recognizes the need for transportation, this customer wants to receive the services as soon as possible. Here, the urgency is related to either the quickest possible pickup time or the earliest delivery time. With respect to A's situation, the following might occur. A manufacturing site (Munich, Salzburg, or Prague) recognizes that additional material is needed as soon as possible. If the material is stocked (even for a short time, but

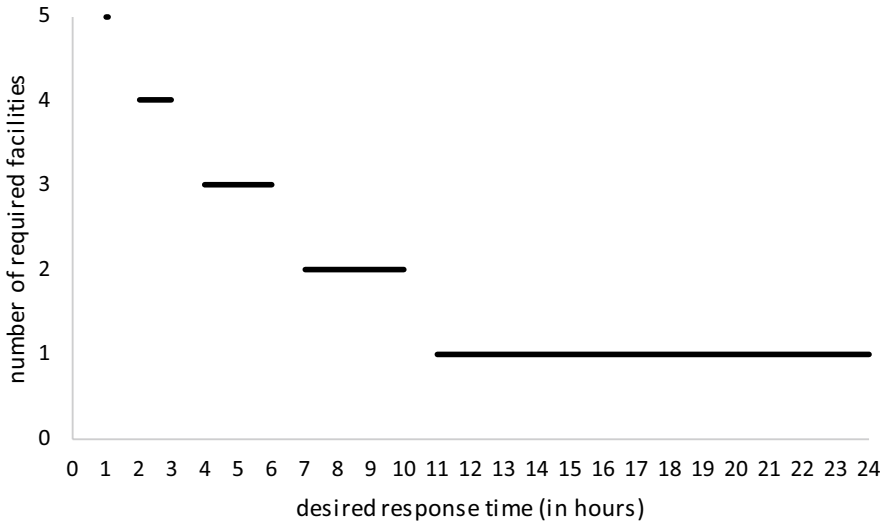


Fig. 8.10 Relation between desired customer response time (*x*-axis) and the number of facilities required to ensure this response time

regularly) in one of the import or transshipment facilities contained in the A supply network, the average re-supply time tends to decrease with the increasing number of network nodes. Depending on the spatial distribution of the nodes where materials are stocked, the re-supply time (which equals the difference between the time of arrival of the additional material and the time when the need is expressed) decreases proportionally, degressively, or progressively (see Fig. 8.10).

Figure 8.10 clearly shows that there is a strong relationship between the *number of facilities* for storing supply materials and the customer *service level*. In the A case, each transshipment point will hold some quantities of different types of wood that are in transit from the import ports in Germany and (prospectively) in the Netherlands to the manufacturing sites. Mike emphasizes that storing even small quantities in a transshipment yard can contribute to ensuring a quick re-supply. The major reason is the reduced distance from a transshipment yard to the factories compared to the average distance from the seaport to the manufacturing sites. However, Mike also explains that the number of facilities plays a higher role in the distribution of goods with late customer orders.

To achieve a higher level of material availability and a quicker re-supply time, Mike proposes discussing the installation of two transshipment points. Besides the one in Hanover, he proposes establishing a transshipment facility in the region of Wuerzburg, which lies just between Hanover and Munich, close to the major motorway A7. In this transshipment facility, the inbound flow of wood from the German seaport can be merged with the inbound flow from Rotterdam. Furthermore, from the Wuerzburg facility, it is possible to send materials to all three manufacturing sites. A small warehouse here can be used to maintain a centralized safety stock of

all material for all three production sites at each transshipment yard. Of course, costs for keeping all facilities running increase if the number of operated facilities is increased.

After Mike has clarified the interrelations between the number of considered network nodes and the service to be offered to the customer, he continues to explain the main cost drivers in a transportation network. Mike points out that there are three major cost drivers to be considered during the determination of the right network design (Fig. 8.11). First, there are *inventory costs*. Inventory costs tend to grow if the number of networked facilities is increased because each facility holds a certain quantity of inventory. Besides the fact that there is planned inventory already stocked, there might also be inventory just waiting to be forwarded. Second, the costs directly associated with the *usage of the facilities* have to be considered: costs increase if the number of facilities is increased. Third, there are the *transportation costs* that contribute to the overall expenditures.

In Fig. 8.11, the typical development of the transportation costs is represented by the dashed-dotted graph. Up to a certain number of maintained transshipment facilities, additional economies of scale can be realized by consolidation, as discussed in Sect. 8.3. If more transshipment facilities are incorporated into the network, then the number of transport connections exceeds a threshold value. Beyond this threshold value, it is no longer possible to completely fill a truck. The average quantity to be moved by truck decreases since total transport demand is distributed over a higher number of transport options so the costs per moved ton increase further. Totaling two increasing cost functions for facility and inventory costs and of the convex transportation cost function leads to a convex function for the total costs. Such a function has an optimal number of facilities with minimal total costs.

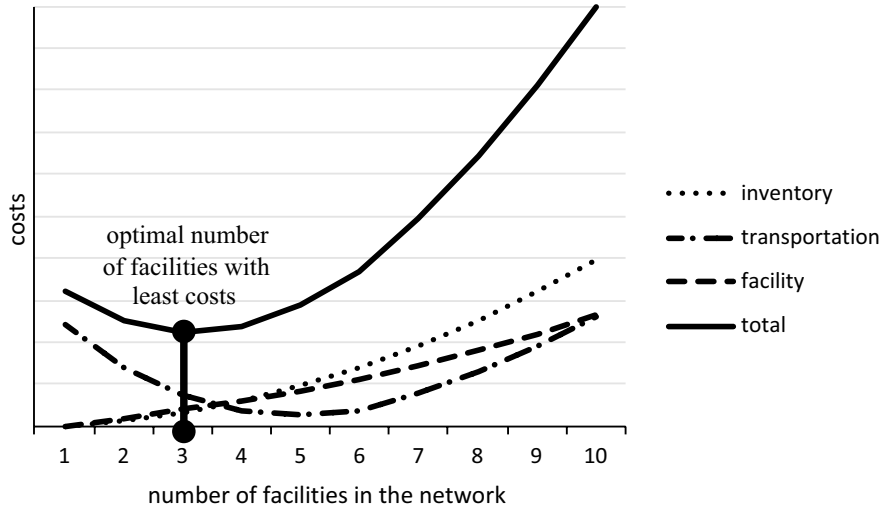


Fig. 8.11 Relation between desired customer response time (*x*-axis) and the number of facilities required to ensure this response time

Incorporating this optimal number of facilities minimizes the total costs of the transportation system and thus contributes to balancing costs and customer satisfaction.

8.5 Capacity Allocation in a Many-to-Many Network

After Mike has completed the scheme outline to find the appropriate transportation network, another discussion among the board members of A is initiated. Robert, the human resource manager, wants to know if it is possible to establish and operate a highly sophisticated transportation network without well-trained and educated staff members. Mike is sure that this is impossible since transportation management requires in-depth knowledge about transport technologies, processes, and regulations and laws.

This clear statement is considered by the board members in the subsequent discussions. Since it seems impossible to operate a complex transportation network efficiently without a well-trained group of transportation managers, there are two options left: either a well-trained, highly specialized group of transportation managers must be established (hired or trained), or transportation activities must be outsourced.

Robert wants to know Mike's opinion. Mike replies as follows. So far, A has followed a strategy somewhere in between the two options. Since A has no trucks of its own, these vehicles are hired. However, the risk of failing to achieve the required high fill rate to gain economies of scale is completely with A. If no economies of scale can be realized for any reason, the transportation costs will be quite high, as seen in the examples in Sect. 8.3. For this reason, Mike agrees that they should think about outsourcing all transportation. In this context, it can be useful to look for transport service providers who are paid only for an executed transport service, as in the letter and parcel service industry. From calculations similar to those conducted in Sect. 8.3, our company A knows the costs per ton if they decided to hire again.

Mike suggests contacting several forwarding companies to collect information about their tariffs for all possible import connections between A's four seaports and three manufacturing facilities. One week later, Mike and the board members have another meeting. Mike wants to use this meeting to explain how transportation costs can be minimized if the transport service providers are paid by carried tons only.

During the last week, Mike contacted several freight carriage companies and asked for their tariffs to move freight between the seaports and the manufacturing sites. More specifically, he had asked for the price per ton moved between a given seaport as the loading location and a given factory as the unloading location. Mike collected all the information. For each of the 12 possible transportation links, he selected the cheapest freight carrier. The associated tariffs (per ton) are given in Fig. 8.12. Additionally, Mike has indicated the supply quantities available at each seaport and the quantities ordered by each factory per week.

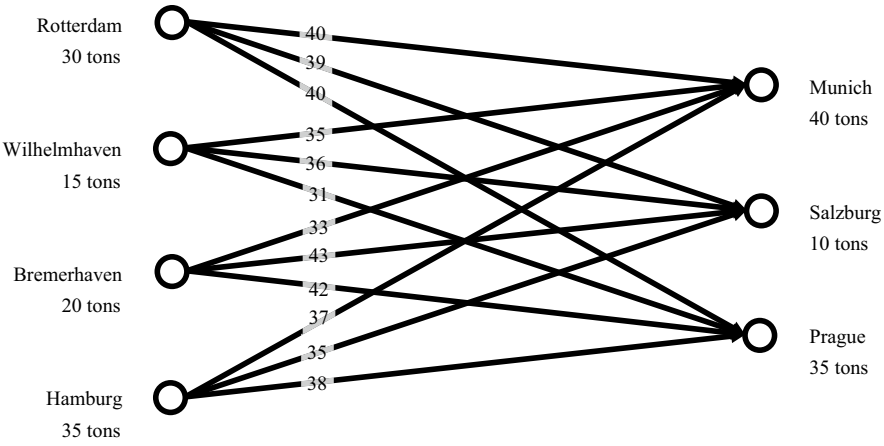


Fig. 8.12 Data of the transportation analysis for A

8.5.1 The Transportation Problem

The *transportation problem* [or the Hitchcock distribution problem (Hitchcock 1941)] addresses the situation outlined in Fig. 8.12 and aims to minimize the total sum of transport expenses while the individual demand from each receiving node is fulfilled and no supply node supplies more quantities than it has in stock. As can be seen from the example in Fig. 8.12, solving the transportation problem is not a trivial task. For each supply node it must be decided which quantity is assigned to which receiving node and vice-versa for each receiving node, as it has to be decided which quantity is ordered from which supply node.

The core to solving the transportation problem is to assign the shipped quantities to each available transport link connecting a seaport (supply node) with a manufacturing site (receiving node). If zero quantity is assigned to a link, then this link is not used. It is necessary to select those shipment quantities for each link where the sum of costs overall transportation links is minimal, but where the demand for each receiving node is fulfilled (restriction A) and no supply node stock is exceeded (restriction B). Obviously, all assigned quantities must be zero or larger than zero (restriction C).

8.5.2 Decision Model

Similar to the warehouse location problem (WLP) in Chap. 7, it is possible to represent (to "model") the decision situation associated with the Hitchcock distribution problem in terms of a *linear optimization model*. The parameters of this model (Eqs. 8.5–8.8) are as follows. The set S contains all of the supply nodes, and for each element $s \in S$, the available (stored) supply quantity q_s is known. All receiving nodes are collected in the set R, and node $r \in R$ requests the quantity d_r. All quantities are expressed in tons per week. The costs for moving one ton from supply node $s \in S$ to

the receiving node $r \in R$ is written over the corresponding arc in Fig. 8.12, and this cost factor is c_{sr}. There are no discounts available if the transported quantity is increased.

The decision variables are x_{sr}, representing the transport quantity assigned to the transportation link originating from the supply node s and terminating in the receiving node r. All non-negative real values are allowed for x_{sr}.

$$\min \; Z = \sum_{s \in S} \sum_{r \in R} c_{sr} \cdot x_{sr} \tag{8.5}$$

$$\sum_{s \in S} x_{sr} \geq d_r \forall r \in R \tag{8.6}$$

$$\sum_{r \in R} x_{sr} \leq q_s \forall s \in S \tag{8.7}$$

$$x_{sr} \geq 0 \forall s \in S, r \in R \tag{8.8}$$

It is necessary to minimize the total transportation costs (8.5). The individual demand of all receiving nodes must be covered (8.6), but the available stock is to be considered (8.7). All quantities assigned to a transportation link must be non-negative (8.8). This linear program formulation of the Hitchcock distribution problem was originally introduced by Dantzig (1951), who has also proposed an elegant algorithm to determine optimal transport quantities for each link. This method is known today as the modified distribution *(MODI) method*. However, the MODI method presumes that a feasible initial solution for the model (8.5)–(8.8) is already available. Therefore, such a construction procedure is discussed first in Sect. 8.5.3, before the MODI methodology is introduced. The MODI algorithm is then used to iterate and improve the initial solution, taking care that, for each iteration, feasibility is maintained and the costs are reduced. This two-phase concept, comprised of a construction phase and a subsequently processed improvement phase, is an important and well-investigated approach to the identification of optimal or sub-optimal solutions to an optimization model. Mike is going to explain the computation of the cost of the minimal transport quantity assignment using the MODI approach. He uses the example data given in Fig. 8.12.

▶ **Practical Insights** In practice, the transportation problem can also be solved for cases with capacity bottlenecks. In this situation, the capacity can be extended by the so-called over-costs. In addition, the transportation problem can be extended to the multi-stage SC setting, such as cross-docking or hub-and-spoke systems. This form is called a transshipment problem.

8.5.3 Finding the First Feasible Model Solution

Mike starts the construction of the first feasible assignment by analyzing the total quantities demanded as well as the total available quantities. To apply the MODI approach, it is necessary that these two sums are equal ("balanced transportation

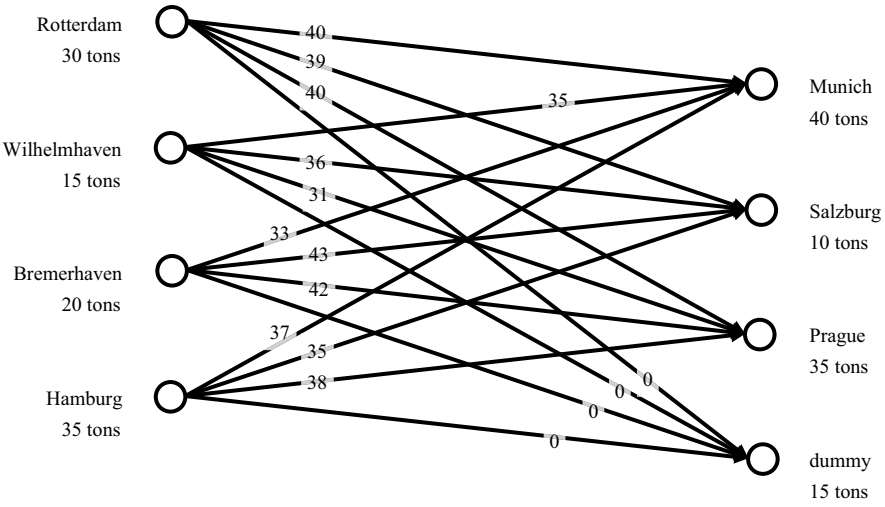


Fig. 8.13 The transportation problem data for A including the dummy customer ("balanced transportation problem")

problem"). Since equality is not observed in our example (The sum of supply quantities is 100 tons, but total demand is only 85 tons), Mike introduces an artificial customer ("dummy demand node") that requests the residual supply quantity. The artificial transportation costs towards this artificial demand node are set to 0. With this modification, the transportation problem is now "*balanced*" with respect to demand and supply in the SC (Fig. 8.13).

In order to keep the description of the construction algorithm as compact as possible, Mike presents all data in the so-called *transportation matrix* shown in Fig. 8.14. The heads of the rows contain the seaport names and the available supply quantities, but the column heads carry the names of the demanding nodes (factories) and the quantities demanded. The spanned matrix consists of 4 × 4 cells. In the upper left corner of the cell in row i and column j Mike writes the costs c_{ij} for shipping 1 ton along the link originating from i and terminating in j.

Currently, all cells are empty but Mike is going to fill them consecutively with the quantities that are forwarded along the corresponding transportation links. Mike starts with the upper left-hand cell that represents the transportation link from Rotterdam to Munich (1 → 1). This cell (connection) is marked with a bold frame. Mike assigns as much quantity to this link as possible. The maximal quantity is 30 tons since Rotterdam does not offer a larger quantity. Mike assigns 30 tons to the labeled link and writes this quantity in the marked cell. After this assignment, Mike concludes with two issues. First, the demand from Munich is partially unsatisfied (10 tons have to be carried there from another seaport), but Rotterdam cannot be used to supply any other location. Therefore, Mike can close all remaining connections starting in Rotterdam. He indicates this closure by writing a "-" into the affected cells (Fig. 8.15).

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30				
Wilhelmshaven 15	35	36	31	0
Bremerhaven 20	33	43	42	0
Hamburg 35	37	35	38	0

Fig. 8.14 Empty transportation matrix (start of construction)

Mike is going to fill the next cells. He tries to satisfy the demand of the only partially served site in Munich. This means that he assigns 10 tons to the link Wilhelmshaven to Munich with a cost factor of 35 €/ton. Now the demand from Munich is completely satisfied, and Mike closes all connections terminating here. They cannot be used further (Fig. 8.16).

Wilhelmshaven still has some quantity available, and the remaining 5 tons are now assigned to the next available demand location after Munich. This is Salzburg (see Fig. 8.17).

The stock size at Wilhelmshaven is exhausted, but demand from Salzburg is not yet completely fulfilled. Mike selects Bremerhaven as the next available supply node after Wilhelmshaven (Fig. 8.18). Bremerhaven provides the remaining 5 tons for Salzburg.

The demand from Salzburg is completely fulfilled, but Bremerhaven still has 15 tons available. This quantity is given to the next demand location, which is Prague (Fig. 8.19).

However, Bremerhaven is unable to completely fulfill the demand from Prague. The remaining quantity of 20 tons is provided by the next supply location after Bremerhaven, which is Hamburg (Fig. 8.20).

Figure 8.21 shows the completed first transportation matrix which represents the first constructed feasible solution of the transportation model. The residual supply quantity of 15 tons is assigned to the link originating from Hamburg and heading to the dummy demand node. All used transportation links are shaded in gray.

This construction scheme is called the north-west corner rule (Gass 2003). Mike determines the costs associated with the instantiated transport quantities by totaling products of assigned quantities and the cost coefficients. Here, Mike gets 40 €/t_0 × 30 t_0 + 35 €/t_0 × 10 t_0 + 36 €/t_0 × 5 t_0 + 43 €/t_0 × 5 t_0 + 42 €/t_0 × 15 t_0 + 38 €/t_0 × 20 t_0 + 0 €/t_0 × 15 t_0 = 3335 € as total forwarding costs.

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30	40 30	39 -	40 -	0 -
Wilhelmshaven 15	35	36	31	0
Bremerhaven 20	33	43	42	0
Hamburg 35	37	35	38	0

Fig. 8.15 Partially filled transportation matrix after assignment of first quantity to the transportation link Rotterdam to Munich

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30	40 30	39 -	40 -	0 -
Wilhelmshaven 15	35 10	36	31	0
Bremerhaven 20	33 -	43	42	0
Hamburg 35	37 -	35	38	0

Fig. 8.16 Partially filled transportation matrix after assignment of quantity to the transportation link Wilhelmshaven to Munich

8.5.4 Optimality Check

After the presentation of the construction procedure, Mike is asked by Walter if the proposed solution is optimal and how Mike can check if it is optimal. In general, the application of the matrix minimum method does not lead to the optimal solution of the model, Mike replies. However, there is an intuitive way to check if and how the assignment can be modified so that costs are reduced.

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30	40 30	39 -	40 -	0 -
Wilhelmshaven 15	35 10	36 5	31 -	0 -
Bremerhaven 20	33 -	43	42	0
Hamburg 35	37 -	35	38	0

Fig. 8.17 Partially filled transportation matrix after assignment of quantity 5 to the transportation link Wilhelmshaven to Salzburg (*third step*)

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30	40 30	39 -	40 -	0 -
Wilhelmshaven 15	35 10	36 5	31 -	0 -
Bremerhaven 20	33 -	43 5	42	0
Hamburg 35	37 -	35 -	38	0

Fig. 8.18 Partially filled transportation matrix after assignment of quantity 205 to the transportation link Bremerhaven to Salzburg (*fourth step*)

The basic idea is to find out whether shifting some of the quantities between the transportation links will lead to a cost reduction. If such a shift is impossible, then the available solution is optimal, as no further cost reductions are possible. On the other hand, if a cost-reducing shift is identified and applied immediately, this leads to a new and updated assignment with lower total costs.

Total costs of 3335 € are associated with the transportation links used so far. The used transportation link is called a base link, and a currently unused link is called a

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30	40 30	39 -	40 -	0 -
Wilhelmshaven 15	35 10	36 5	31 -	0 -
Bremerhaven 20	33 -	43 5	42 15	0 -
Hamburg 35	37 -	35 -	38	0

Fig. 8.19 Partially filled transportation matrix after assignment of quantity 15 to the transportation link Bremerhaven to Prague (*fifth step*)

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30	40 30	39 -	40 -	0 -
Wilhelmshaven 15	35 10	36 5	31 -	0 -
Bremerhaven 20	33 -	43 5	42 15	0 -
Hamburg 35	37 -	35 -	38 20	0

Fig. 8.20 Partially filled transportation matrix after assignment of quantity 20 to the transportation link Hamburg to Prague (sixth step)

non-base link. Mike is going to check whether the usage so far of a non-base line will lead to a cost reduction. Therefore, Mike first distributes the current cost amount overall network nodes.

$$u_1 + v_1 = 40 \tag{8.9}$$

$$u_2 + v_1 = 35 \tag{8.10}$$

$$u_2 + v_2 = 36 \tag{8.11}$$

$$u_3 + v_2 = 43 \tag{8.12}$$

$$u_3 + v_3 = 42 \tag{8.13}$$

$$u_4 + v_3 = 38 \tag{8.14}$$

$$u_4 + v_4 = 0 \tag{8.15}$$

Therefore, he introduces the so-called potential values u_i for supply node $i \in S$ and v_j for receiving node $j \in R$ and splits the costs c_{ij} for all currently used transportation links $i \rightarrow j$ so that $c_{ij} = u_i + v_j$. This leads to the system of linear equalities shown in Eqs. (8.9)–(8.15). There is a unique solution if we set $v_1 = 0$ (Without this setting, there would be more than one solution). This solution can be derived by the application of the Gaussian elimination method. In the example considered here, we get the potential values $u_1 = 40$, $u_2 = 35$, $u_3 = 42$, $u_4 = 38$, $v_1 = 0$, $v_2 = 1$, $v_3 = 0$, and $v_4 = 38$. These values are written in brackets (see Fig. 8.22).

Mike is now able to calculate imputed cost factors c'_{ij} per ton for the currently unused links by adding the potential u_i for the starting location i and the potential v_j for the terminating location j ($c'_{ij} := u_i + v_j$). If he compares the imputed current costs with the real costs computing $c'_{ij} - c_{ij,}$ which apply if a currently unused link is used, he can see if the incorporation of this leads to savings. These savings/values are written in the upper right corners of the cells representing the recently unused cells (small black rectangles). All connections with the exceptions of Rotterdam-Prague and Wilhelmshaven-dummy will lead to a cost reduction. Therefore, the initial solution is not an optimal solution.

If Mike finds out for a given solution that there is no non-base connection associated with a positive (>0) saving $c'_{ij} - c_{ij}$ per ton, then the solution currently available is optimal. Therefore, Mike can use the implied savings $c'_{ij} - c_{ij}$ associated with

	Munich 40	Salzburg 10	Prague 35	dummy 15
Rotterdam 30	40 30	39 -	40 -	0 -
Wilhelmshaven 15	35 10	36 5	31 -	0 -
Bremerhaven 20	33 -	43 5	42 15	0 -
Hamburg 35	37 -	35 -	38 20	0 15

Fig. 8.21 Completely filled transportation matrix after assignment of the residual quantity of 15 tons to the transportation link Hamburg to the dummy demand node (*seventh step*)

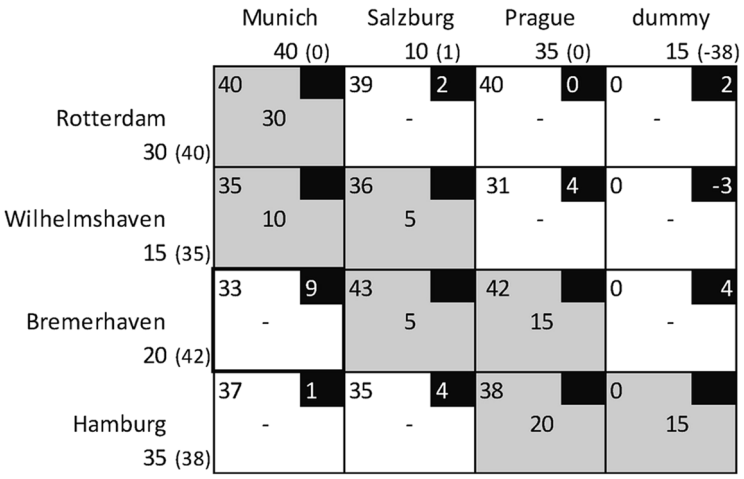


Fig. 8.22 Initial solution proposal with potential values and savings per ton

non-base connections as a general optimality criterion to check whether a given feasible solution is optimal or not.

8.5.5 Solution Improvement

Since Mike has found out that the constructed solution is not optimal, he has to modify it. Therefore, he first identifies the non-base connection with the highest saving per ton. This is the link Bremerhaven to Munich with a saving of 9 € per assigned ton (3 → 1). Mike indicates this link with a black bold frame (see Fig. 8.22).

Since each ton assigned to this link reduces the total transportation costs by 9 €, Mike wants to assign the highest possible quantity Y to this link. Therefore, he has to move some quantities between the links already used. When Mike assigns Y tons to the delivery from Bremerhaven to Munich, then Munich receives Y tons above the quantity demanded. Therefore, the quantity associated with another link used terminating in Munich must be reduced by Y tons. In the example shown in Fig. 8.23, the quantity to be shipped from Wilhelmshaven to Munich is reduced by Y tons. However, the reduction of the quantity to be shipped from Wilhelmshaven reduces the total outbound quantity by Y tons. This implies the need to increase the quantity to be shipped from Wilhelmshaven to another location by Y tons in order to prevent the reduction of the total outbound quantity associated with Wilhelmshaven. Here, it is only possible to increase the quantity assigned to the link Wilhelmshaven to Salzburg by Y units. Now, Salzburg receives too much quantity, so the inbound quantity provided by another supply node must be reduced by Y tons. Here, only the quantity assigned to the link Bremerhaven to Salzburg is reduced by 5 tons. After this reduction, all inconsistencies related to the assigned quantities are solved. The sequence of the arcs shown in Fig. 8.23 represents the sequence of modified

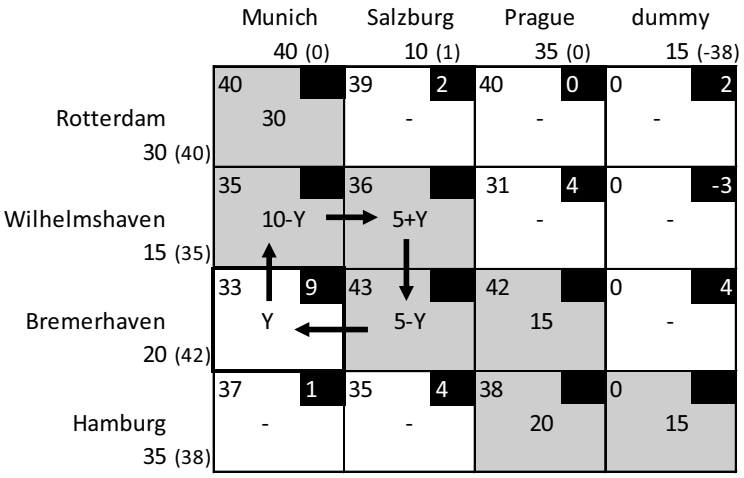


Fig. 8.23 First solution with zig-zag-path (costs are 3335 €)

shipment quantities. This is known as the "zig-zag path." For its construction, only base connections are allowed to be modified.

Following the zig-zag path, Mike can determine the maximal feasible value that can be assigned to *Y*. Therefore, he records all original quantities from which he has subtracted *Y*. Since the updated quantities must not become negative, Mike assigns the minimal recorded value to *Y*. In the example, it is $Y := \min\{10;5\} = 5$. The resulting updated solution is shown in Fig. 8.24. The costs associated with this new solution are 3335 € – 5 t_0 × 9 €/t_0 = 3290 €.

Mike summarizes the steps to update the solution for the transportation problem.

1. Determine the potential values u_i and v_j for all involved locations.
2. Determine the imputed costs $c'_{ij} = u_i + v_j$ for all current non-base links (connections).
3. Compare the imputed costs with the real costs. If $u_i + v_j - c_{ij} > 0$ for at least one non-base link, the given solution is not optimal (Otherwise, it is an optimal solution).
4. Select the non-base link i^*–j^* with the maximal savings value $u_i + v_j - c_{ij}$ per ton.
5. Determine the quantity *Y* to be assigned to the link originating from i^* and termination in j^* by the determination of a zig-zag-path through the current base links.

These five steps are repeated until in step (3) it is recognized that there is no non-base link whose incorporation will reduce the total cost. Figures 8.24, 8.25, 8.26, 8.27, 8.28, 8.29, and 8.30 contain the intermediate transportation matrices that are consecutively generated by the repeated application of steps (1)–(5).

The third transportation matrix is non-optimal, so that a zig-zag-path is determined (Fig. 8.25). Here, both base links Wilhelmshaven to Salzburg and Bremerhaven

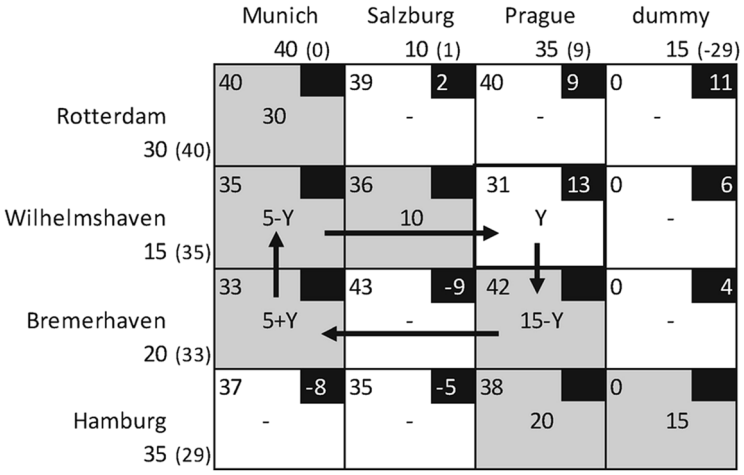


Fig. 8.24 Second transportation matrix with costs = 3290 € and $Y = \min\{10;5\} = 5$

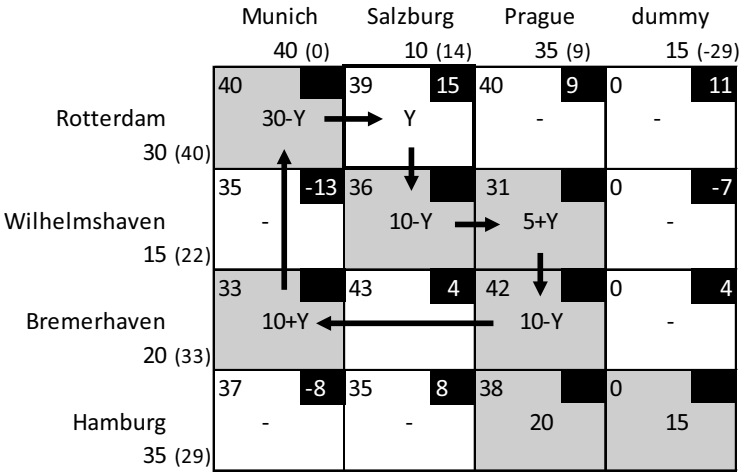


Fig. 8.25 Third transportation matrix with costs = 3225 € and $Y = \min\{10;10;30\} = 10$

to Prague carry the same minimal quantity of 10 tons. The first of both base links Wilhelmshaven to Salzburg, becomes a non-base link ("–") while the other link remains a base link, and the quantity is set to 0 (Fig. 8.26).

The MODI method can be used to optimize the assignment of quantities to different transportation links in a many-to-many-network setup. Besides this north-west-corner algorithm, there are other procedures to generate an initial solution: the matrix minimum algorithm, as described in Vahrenkamp and Mattfeld (2007), and Vogel's approximation algorithm, as described in Domschke (2007). Also, alternative improvement algorithms, like stepping-stone algorithms (Charnes and Cooper 1954), can be used to identify optimal solutions for a transportation model.

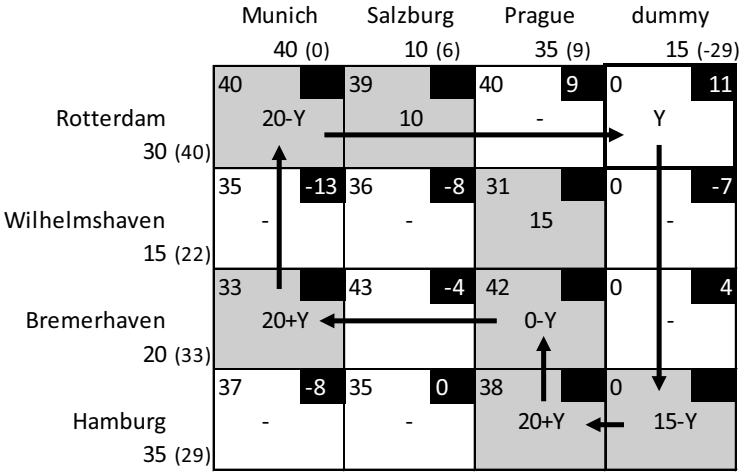


Fig. 8.26 Fourth transportation matrix with costs = 3075 € and $Y = \min\{15;0;20\} = 0$

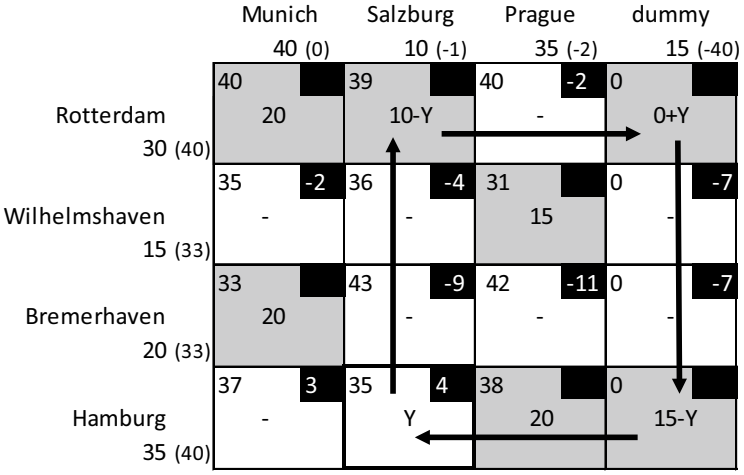


Fig. 8.27 Fifth transportation matrix with costs = 3075 € and $Y = \min\{10;15\} = 10$

8.6 Distribution Network Design

8.6.1 Types of Distribution Networks

According to the transportation network design, different *distribution network structures* can be distinguished (Fig. 8.31).

In the case of direct shipments, distribution is organized as a direct network, with storage on the manufacturing side. Alternatively, if the production strategy is

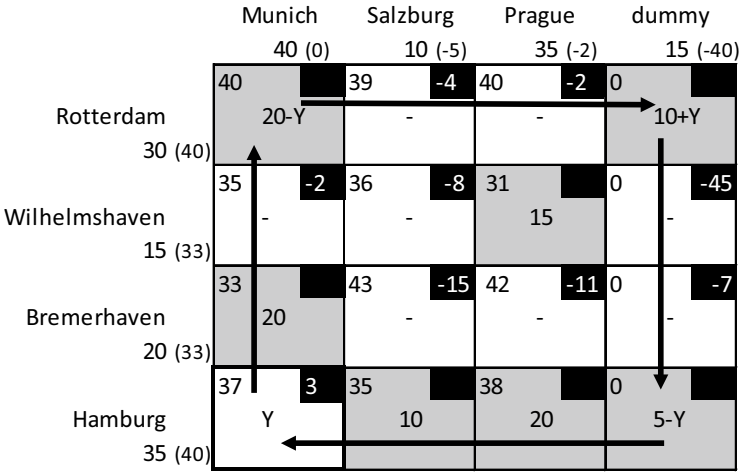


Fig. 8.28 Sixth transportation matrix with costs = 3035 € and $Y = \min\{20;5\} = 5$

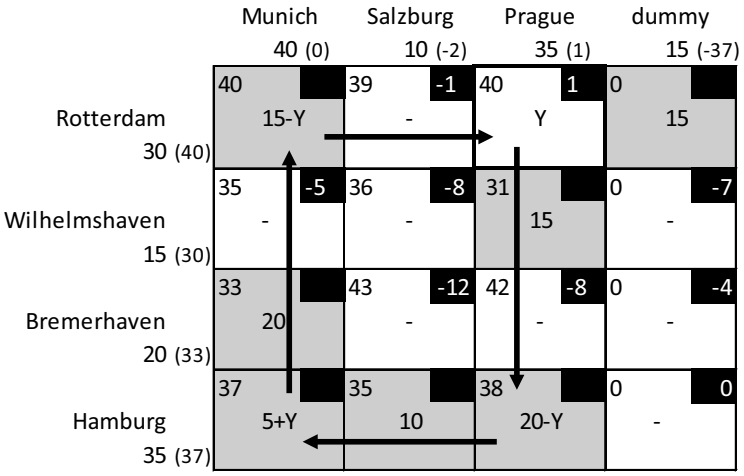


Fig. 8.29 Seventh transportation matrix with costs = 3020 € and $Y = \min\{15;20\} = 15$

assemble-to-order (ATO) or make-to-order (MTO), the products can be delivered directly to customers (for example, this has been DELL's strategy).

In retail, most distribution networks are organized as multi-stage systems with intermediate warehouses, distribution centers, cross-docking centers, and hub terminals. Such complex systems are typically managed by logistics service providers. For examples, consider cross-docking and hub-and-spoke networks.

Cross-docking implements the idea of the consolidation of goods from inbound trucks to outbound trucks via an intermediate transit point (i.e., the cross-docking center). The special feature of cross-docking is that goods do not remain in the warehouse longer than 24 h (see Fig. 8.32).

	Munich 40 (0)	Salzburg 10 (-2)	Prague 35 (1)	dummy 15 (-39)
Rotterdam 30 (39)	40 -1 -	39 -2 -	40 15	0 15
Wilhelmshaven 15 (30)	35 -5 -	36 -8 -	31 15	0 -9 -
Bremerhaven 20 (33)	33 20	43 -12 -	42 -8 -	0 -6 -
Hamburg 35 (37)	37 20	35 10	38 5	0 -2 -

Fig. 8.30 Eighth transportation matrix with costs = 3005 € (optimal)

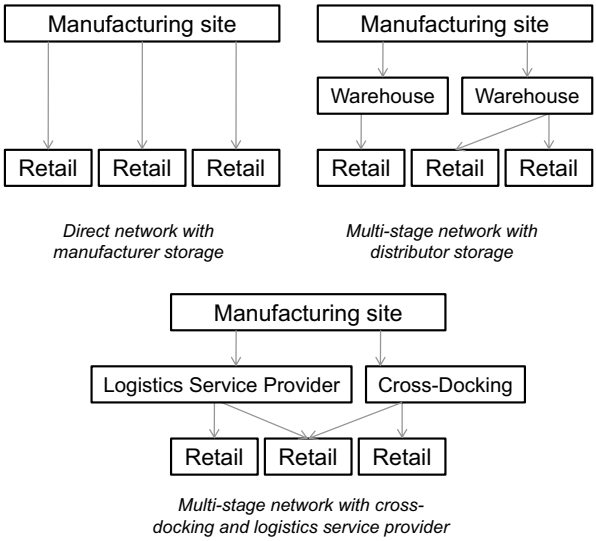


Fig. 8.31 Types of distribution networks

Some *advantages* of cross-docking are the realization of economies of scale in transportation, frequently bundled deliveries, inventory reduction, faster product flow, reduction of errors, savings in material handling and labor costs, increase in fleet capacity utilization, and mitigation of the bullwhip effect. At the same time, a high investment is required to establish a cross-docking terminal. The coordination complexity and risks of stock-out/disruptions in the SC also increase. Finally, data security issues become crucial. Examples of successful cross-docking realization can be found in many retailing companies, e.g., Walmart, E. LeClerc, etc.

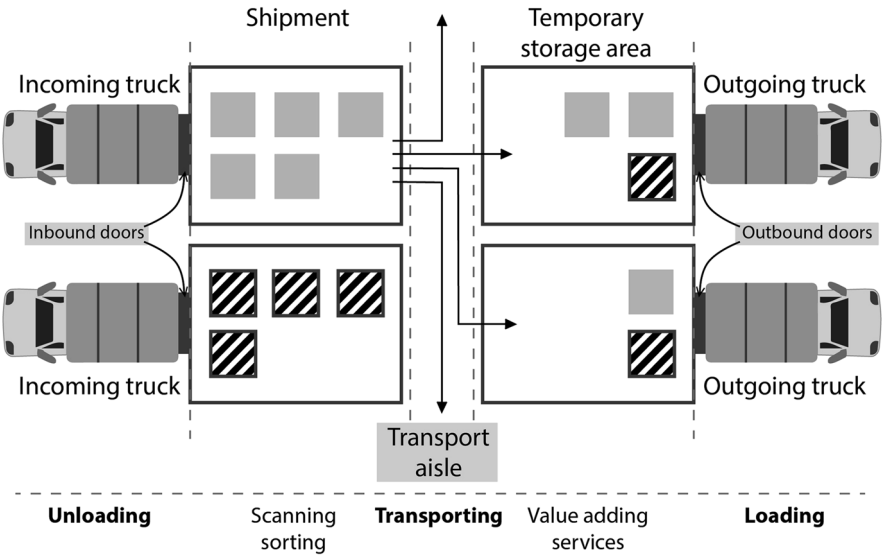


Fig. 8.32 Cross-docking terminal [based on Stephan and Boysen (2011)]

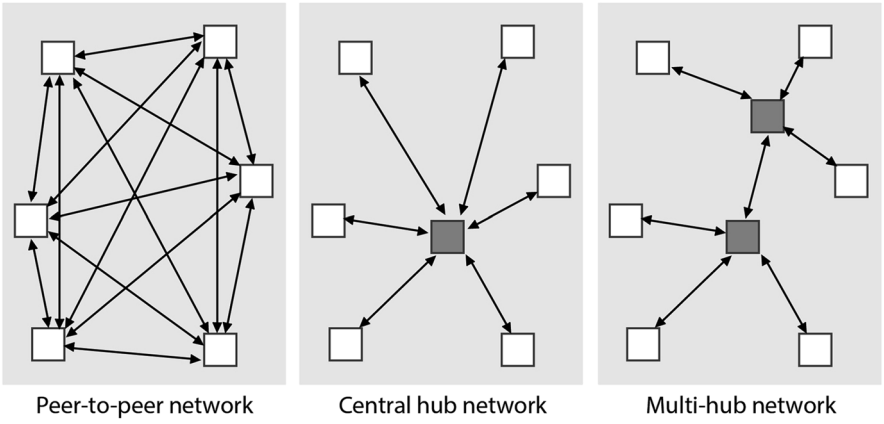


Fig. 8.33 Types of hub-and-spoke networks

Hub-and-spoke network also implements the idea of consolidation (Fig. 8.33).

A hub represents the consolidation element in the distribution network. Spokes are the regional warehouses and customers. This concept is used for shipment consolidation with similar advantages and disadvantages as cross-docking. For example, the airport Leipzig-Halle is DHL's largest hub in Germany. In the further course of the chapter, you will find case studies on different distribution networks.

▶ **Practical Insights: Role of Back-Up Distribution Centers and Multiple Routes in SC Resilience** If there is a disruption at a central warehouse location or a distribution center, alternative distribution centers (i.e., a back-up facility) can help keep the operations running. The number of backup facilities to maintain should be based on a detailed risk assessment of a particular facility location. Furthermore, to avoid a complete halt of supply chain operations during a crisis, it is important to have alternate routes identified well in advance as part of the risk mitigation strategy. Even if these routes are longer or more expensive, they can help firms survive during a more prolonged disruption. The lead time buffer in such conditions thus has to be appropriately defined based on the possible alternative routes.

8.6.2 Case Study: Seven–Eleven Japan

The Seven–Eleven Japan Co. Ltd was founded in 1974 and grew to be one of the largest retail networks, with more than 50,000 stores worldwide with an average area of 100–150 m^2 per store. At first, the company's SC was straightforward, oriented from producers to the stores (Fig. 8.34).

This type of structure resulted in significant shortcomings. Due to the high number of facilities, high total facility/handling, inventory, and logistics costs (more than 70 deliveries a day to the stores) were encountered. The information line was too long, which negatively impacted the service level because of low response times and inventory turnover (about 25 turns a year in 1970s).

In redesigning the distribution network, Seven–Eleven Japan (SEJ) undertook some crucial changes. The key objectives of the redesign were to further enhance customer satisfaction and profitability. The objectives were reducing costs, preventing stock-outs, and ensuring high-quality products. The company launched several actions, such as high investment in information and communication systems, identification of an adequate partner for each region, and assurance of long-term relationships. In addition, a combined delivery system (CDS) was developed, which was based on the absence of direct store delivery (Fig. 8.35).

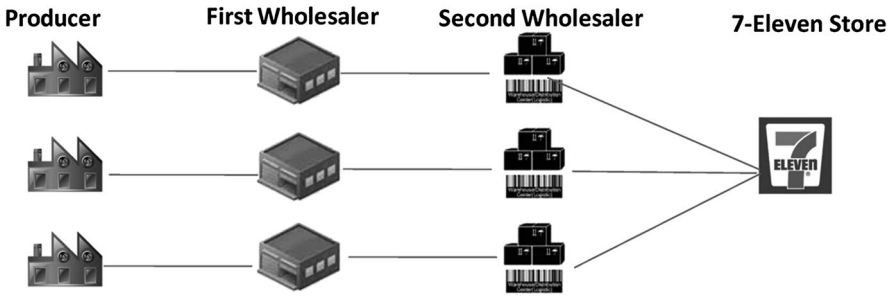


Fig. 8.34 Straightforward-oriented supply chain

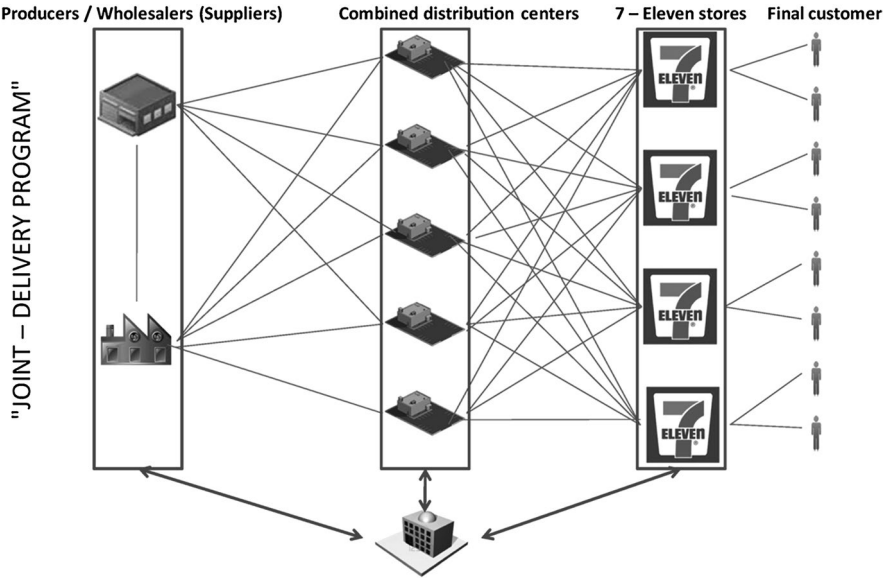


Fig. 8.35 Redesigned supply chain

Different manufacturers and distributors from the same geographical regions consolidate their shipments to minimize the number of trucks going to the combined distribution centers (CDC). The five CDCs (i.e., frozen products, chilled products, fresh foods, drinks and noodles, and magazines) are sorted by product category and required storage temperature. From here, different suppliers' products are delivered to the shops. For some products (e.g., fresh boxed products), deliveries happen several times a day. For frozen products, three deliveries a week are enough. Each truck's tour covers more than one store, preferably during off-peak hours. By omitting the need for the inspection of incoming goods, the costs of the whole SC operations can be further decreased.

A changed distribution network design affects all stages in the SC and the SC costs (inventory, transportation, facilities, and information). Stores have to reconfigure store shelves at least three times daily, and storefronts cater to different consumer segments and demands at different times. SEJ headquarters should design and manage location, transportation, inventory, and information systems to support their objectives. They are also responsible for planning new stores locations to ensure a strong presence. On-time delivery is now the supplier's responsibility. In the event of non-compliance with agreed delivery dates, the supplier has to pay a penalty equivalent to the gross margin of the product being delivered. In return for this, suppliers are rewarded with long-term business relationships and adoption in new projects, such as the e-commerce project "7dream.com." In conclusion, awareness of customer needs and management's ability to react and respond to market preferences by simultaneously considering costs ensure SEJ's long-term economic success.

Discussion

- Identify impacts on different actors in the SC after a redesign of the distribution system.
- Why has Seven–Eleven chosen off-site preparation of fresh foods and subsequent delivery to stores?
- How did the following SC costs change: inventories/transportation/facilities and handling/information?
- What is the name of the transportation strategy when the tour of each truck starts from a distribution center and covers more than one store?
- Which transportation network design is used in the SEJ case? What are the advantages and limitations of this network type?

8.6.3 Transportation Modes

8.6.3.1 Case Study: Transportation of Shoes from China to Austria via Hamburg

Diary of a container full of shoes from the company Deichmann (see Fig. 8.36).

In the following, the journey of a container from Wenzhou to Vienna is described, based on the article "Schuhe unterwegs" in HHLA-Messe-Magazine, 2011.

Ningbo, China, 7 November, 11:30: One of the largest container ships is loaded with different containers. One of them is container CMAU 146090–0, which is loaded with the Deichmann shoe collection. The shoes are produced in Wenzhou and transported to Ningbo, where the container is loaded onto a container ship.

Suez Canal, 28 November, 0:57: The container ship sails the 160 km-long Suez Canal at 7.5 knots. The Suez Canal is one of the most important shipping routes in the world.

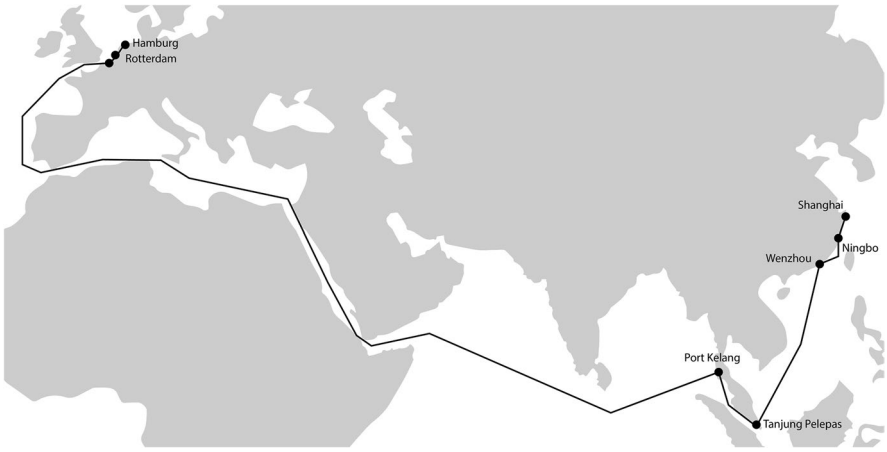


Fig. 8.36 Deichmann's transportation chain

Hamburg, Burchardkai, 29 November, 8:00: The container ship has capacity for 14,000 standard boxes. This means that good planning of the loading process is essential in the different ports. At the HHLA Container Terminal Burchardkai, loading preparations for the port of Le Havre are underway, overseen by a responsible ship planner.

Le Havre, December, 7, 6:00: The container ship leaves Le Havre after a short-stop. The ship planner is organizing the dispatching and loading of the ship in Hamburg. The ship planner uses planning software to decide where each container will be stacked.

Hamburg, Burchardkai, December 10, 8:45: Tugboats help the container ship dock. Giant, detailed dispatching machinery starts. 9:00: The responsible ship planner decides together with the container ship's officer when the planned loading and reloading will be done.

12:14: From 9 o'clock, dispatching starts. Several thousands of containers have to be reloaded.

12:15: For such cases, the HHLA has invested in berths and efficient bridges.

12:16: The bridge driver has an inside monitor so he can see which containers he has to take. At the same time, another ship planner controls the number of unloaded containers.

12:17: The driver of the van-carrier gets an automatic message when the containers have been put down on the quayside.

12:20: The driver of the van-carrier is putting down the container full of shoes.

The cranes are computer-assisted, and they move the container into an optimal position for interim storage.

Hamburg, Burchardkai, December, 11, 10:50: The container is loaded onto the train by the company Metrans going towards Prague. The train is fully loaded with 60 containers.

11:25: Loading from interim storage and the train is done by ten van-carriers. The van-carrier drivers obtain their information about the loading process from the computerized networks. The goal is to achieve a smooth loading and unloading.

14:31: Metrans employees are organizing processes for the 1781 Metrans trains, which are transporting the goods from Prague to Hamburg. The container with the shoes is loaded onto the first section of the train.

15:48: The 600 m-long train is leaving the port of Hamburg. The computerized network informs the Metrans dispatcher in Prague and the logistics experts from Deichmann that the container is on the way to Prague.

Decin, goods rail depot, December, 12, 6:10: A change of locomotive drivers. Metrans has a special logistic plan of their capacity utilization so that they are able to transport more containers than their competitors. Of the international traffic overseas on the rail networks, 75% is done by Metrans.

Prague, December, 12, 8:57: The train is arriving at one of the largest transshipment centers in Europe. A total of 15,000 containers can be stored, and ten trains can be loaded and reloaded at the same time.

Dunajska Streda, December, 13, 9:11: The company Deichmann opened their new distribution center next to the terminal area. The terminal area offers space for

Mode of transport	Typical usage	Advantages	Disadvantages
Road	Door-to-door Ideal for mixed cargo. Typically used for first and final leg	Most flexible for door-to-door. cheap	Limited to continental transport Urban congestion Damage to roads
Rail	For domestic. continental and inter-continental transport	Ideal for heavy goods and long distances Environmental-friendly	Connection to rail system required. Complete trains require large volumes (thus low frequency), otherwise handling in yards (low transport speed)
Air	To ensure fast transport	Fast and safe	Expensive Limits for size and weight Typically as part of multi-modal transport
Sea	bulk shipments, where long lead time is ok	Ideal for bulky and heavy goods Highly standardized sea containers worldwide Less costly than air for inter-continental transports	inflexible routes Long lead time Inflexible timetables (ship will not wait for missing container)

Fig. 8.37 Comparison of different transportation modes (based on SupplyOn)

25,000 containers. The shoes arrive at the distribution center, where they are arranged into various transport units for the different cities.

Vienna, the permanent establishment of Schenker, December, 14, 13:45: The container destined for Vienna is leaving the distribution center. The container arrives at the warehouse in the Gottfried-Schenker street at 13:45.

Vienna, Mariahilfer Street 41, December, 15, 8:12: The trip from China to Austria ends at Deichmann's subsidiary company, in the old Vienna shopping street. The journey from Ningbo to Vienna has taken 37 days, 20 h, and 42 min.

As we can see from this example, the modern transportation chains are comprised of different transportation modes (i.e., the *intermodal transportation*). The main transportation modes, their advantages, and disadvantages are summarized in Fig. 8.37.

Figure 8.37 shows that no single transportation mode provides the ideal solution. The selection of the mode is based on prioritizing criteria such as speed, costs, networking, security, and schedules.

8.7 Key Points

The fundamental questions in distribution and transportation network design are as follows:

- When and where do goods in required quantity and quality need to be picked up and consolidated in order to have them in time at the destination?

- Which service levels have to be respected (on-time delivery, tolerances)?
- Which modes and means of transport are possible and the right choice for the whole or a defined part of the transport?
- Who organizes and executes (which part of) the transport?
- Who pays for which part of the transport?
- How can the promised cost for efficient and reliable transport be minimized?

We have discussed the typical aspects and decision problems of configuring distribution and transportation networks as they appear today. Thereby, we have discussed general challenges associated with the setup of the right and appropriate transportation network; in addition, we have provided an in-depth presentation of particular decision tasks.

At the end of this chapter, we have become familiar with several generic types of distribution and transportation network structures. Starting from simple direct delivery layouts, we are able to extend these structures by postponement strategies, milk runs, and transshipments in order to achieve economies of scale in the context of transportation planning through consolidation. However, we have seen that the success of consolidation depends on the availability of adequate shipments that can be grouped together to make full truckloads. The risk associated with shipment consolidation is quite high if the transportation demand varies over time. In such a situation, the incorporation of a logistics service provider can be beneficial if it is paid only per shipped quantity. The assignment of quantities to be shipped to different links in a many-to-many transport network is a complicated decision problem known as the Hitchcock distribution problem. We have discussed and applied procedures to solve this important decision task in configuring a transportation network.

Bibliography

Aldrighetti R, Calzavara M, Martignago M, Zennaro I, Battini D, Ivanov D (2024) A methodological framework for the design of efficient resilience in supply networks. Int J Prod Res 62(1–2):271–290

Akkermann R, Farahani P, Grunow M (2010) Quality, safety and sustainability in food distribution: a review of quantitative operations management approaches and challenges. OR Spectr 32:863–904

Amiri A (2006) Designing a distribution network in a supply chain system: formulation and efficient solution procedure. Eur J Oper Res 171(2):567–576

Barnhart C, Laporte G (eds) (2007) Transportation. In: Handbooks in operations research and management science (vol 14). Elsevier, Amsterdam

Button K (2010) Transport economics, 3rd edn. Edward Elgar, Cheltenham

Charnes A, Cooper WW (1954) The stepping-stone method for explaining linear programming: calculation in transportation problems. Manag Sci 1(1):49–69

Chen Z-L (2010) Integrated production and outbound distribution scheduling: review and extensions. Oper Res 58(1):130–148

Chopra S (2019) Supply chain management: strategy, planning and operation, 7th edn. Pearson, Harlow

Dantzig GB (1951) Application of the simplex method to a transportation problem. In: Koopmans TC (ed) Activity analysis of production and allocation. Wiley, New York, pp 359–373

Domschke W (2007) Logistik: transport, 5th edn. Oldenbourg Wissenschaftsverlag, München
Echefaj K, Charkaoui A, Cherrafi A, Ivanov D (2024) Design of resilient and viable sourcing strategies in intertwined circular supply networks. Ann Oper Res. https://doi.org/10.1007/s10479-024-05873-1
Gass SI (2003) Linear programming: methods and applications. Dover, New York
Hitchcock FL (1941) The distribution of a product from several sources to numerous localities. J Math Phys 20:224–230
Ivanov D, Sokolov B, Pavlov A (2014) Optimal distribution (re)planning in a centralized multistage network under conditions of ripple effect and structure dynamics. Eur J Oper Res 237(2):758–770
Ivanov D (2024) Introduction to supply chain analytics. Springer Nature, Cham
Kunnumkal S, Topaloglu H (2011) Linear programming based decomposition methods for inventory distribution systems. Eur J Oper Res 211(2):282–297
Li J, Chu F, Chen H (2011) A solution approach to the inventory routing problem in a three-level distribution system. Eur J Oper Res 210:736–744
Manzini R, Bindi D (2009) Strategic design and operational management optimization of a multistage physical distribution system. Transport Res E Log 45(6):915–936
de Palma A, Lindsey R, Quinet E, Vickerman R (eds) (2011) A handbook of transport economics. Edward Elgar, Cheltenham
Stephan K, Boysen N (2011) Cross-docking. J Manag. Control 22:129–137
Vahrenkamp R, Mattfeld DC (2007) Logistiknetzwerke: modelle für Standortwahl und Tourenplanung. Gabler, Wiesbaden

References for Sect. 8.6.1

BBC (2011) Shopping by phone at South Korea's virtual grocery. http://www.bbc.co.uk/news/business-15341910. Accessed 2 Jan 2025
Coe NM, Lee Y-S (2006) The strategic localization of transnational retailers: the case of Samsung-Tesco in South Korea. Econ Geogr 82(1):61–88
Kumar N (2006) Strategies to fight low cost rivals. Harv Bus Rev 84:104–112
Park N, Kim Y-C, Shon HY, Shim H (2013) Factors influencing smartphone use and dependency in South Korea. Comput Hum Behav 29:1763–1770
Slack N, Chambers S, Johnston R (2010) Operations management, 6th edn. Pearson Education, London
TESCO PLC (2012) Tesco Homeplus expands number of virtual stores. https://www.tescoplc.com/tesco-homeplus-expands-number-of-virtual-stores/. Accessed 2 Jan 2025
The Telegraph (2011) Tesco builds virtual shops for Korean commuters. https://www.telegraph.co.uk/technology/mobile-phones/8601147/Tesco-builds-virtual-shops-for-Korean-commuters.html. Accessed 2 Jan 2025

References for Sect. 8.6.3

Bensaou B (1996) 7-Eleven Japan: creating the virtual corporation. In: Proceedings of the 4th European conference on information systems. Lisbon, pp 1275–1289
Zentes J, Morschett D, Schlramm-Klein H (2012) Strategic retail management: text and international cases. Gabler Verlag, Springer Fachmedien, Wiesbaden

Factory Planning and Process Design

9

Learning Objectives for This Chapter

- Role of factory planning in SCOM
- Factory planning processes
- Role and methods of capacity planning
- Options for process flow design
- Lean production systems
- SCOM excellence

9.1 Introductory Case Study: Factory Planning at Tesla

The location of a new factory can provide a competitive advantage and is strategically relevant for every company. For once, location influences cost structures. For example, producing in a low-wage country influences the production price of the product. Politics, exchange rates, and currency swings create risks. Many companies operate in various countries, taking advantage of the advantages in each location.

On the other hand, innovation plays another important role in business success. Proximity to universities, research areas, or other enterprises operating in the

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_9) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_9.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_9

business can enhance innovation incentives. Here, qualified employees are also easier to reach, for example, through recruiting from universities. Governments will often support particular areas with tax advantages to trigger the development of innovation clusters. In these areas, the infrastructure is well-developed. Therefore, supplies are easily delivered to the area, and finished products are shipped to the retailer or end customer. The location of the customer's market is also relevant.

E-mobility is the new trend in the car industry. With decreasing amounts of fossil fuel, electricity is an attractive option to pursue while helping to protect the environment. However, there is criticism that some factories are not zero-emission producers but simply reallocate the emissions elsewhere. Batteries should be charged by power plants and are difficult to produce. Nevertheless, researchers found that electric cars save energy and are more environmentally friendly than gasoline or diesel-operated vehicles. According to experts, e-mobility could reduce CO_2 emissions by 47% by 2030. Companies have discovered this trend and have been trying to find the most efficient versions of e-mobility. Batteries still present a critical obstacle. Many are expensive to produce or do not last very long. Moreover, safety issues have to be resolved. The development of lithium-ion batteries has brought major advancement. Many governments support the implementation of non-fuel systems. The so-called original equipment manufacturers (OEM) are constantly developing new electric vehicles.

Tesla is one of the new-generation enterprises on the e-mobility scene. Having started with the Tesla Roadster, the company currently promotes the models 3, Cybertruck, S, X, and Y on its website (https://www.tesla.com/de_de/). The cars have a relatively low noise level and zero emissions, which constitutes serious competition for long-established OEMs.

Let's take an example of Tesla's production plant, which is located in Fremont, California. The city is located right in the heart of the prestigious Silicon Valley research area. Here, many companies have settled, producing everything technology has to offer. Silicon Valley became popular in the 1980s and 1990s and is called a post-fordistic complex with special dynamics. There are various advantages to this location. Its proximity to technology-savvy Stanford University is just one of them. So-called knowledge dynamics and well-educated, highly qualified employees, along with a fiscal-friendly policy, attract enterprises. Additionally, military and aircraft industries are close by, which supports location factors that are already beneficial. A supplier network has been developed and established. Moreover, Silicon Valley is located near San Francisco, which offers a good working atmosphere for its employees.

Recently, Tesla decided to create a new factory close to Berlin in Grünheide, Germany. Try to find out more about the factory, location, layout, and infrastructure.

9.2 Factory Planning

9.2.1 Role of Factory Planning in SCOM

Factory planning and *process design* problems occur when, for example, an existing factory requires adaptation to ensure that new products can be produced using the existing infrastructure. Planning and design could also be required to implement new technology into the production process to reduce throughput times. These are just two typical problems when the factory has to be modified to generate a competitive advantage.

However, problems also arise when creating entirely new factories. A completely new manufacturing facility is also considered the "greenfield approach" compared to the "brown-field approach" when modifying existing plants.

Sometimes, it might also be necessary to centralize the so-called *production footprint*, which means that factories might be adjusted in size (expanded or reduced) in cases of closure or partial reactivation. Current challenges in the business require dynamic, adaptable, scalable, and modular factories (Cañas et al. 2022; Wiendahl et al. 2005; Schenk et al. 2014). This is related—among many other influencing factors—to shorter product life cycles, shorter time-to-market requirements, and improvement in production technology. Additionally, a fast reaction to changing market needs and technological conditions requires the design of new and modular factories. Grundig (2021) provides a very structured and systematic approach to the processes and steps in factory planning.

A *modular factory* was created by BMW in Leipzig, Germany. This facility covered the three main production areas of the car—body shop, paint shop, and final assembly. The important aspect is the scalability and the expansion potential the factory has to enable adjustments to its layout in accordance with market development. "The BMW Group plant in Leipzig is one of the most modern car factories in the world. Series production began in March 2005. Around 1300 vehicles roll off the production line here daily, including the BMW 1 Series, the BMW 2 Series Gran Coupé, the BMW 2 Series Active Tourer, and the MINI Countryman. The BMW Group has already invested more than four billion euros in the Leipzig location. The core workforce currently comprises around 6800 employees. Green energy for the production of these cars is provided by four purpose-built wind turbines on site. (www.bmw.de, January 2025)"

It becomes evident that the spectrum of "factory planning and process design" covers a wide variety of practical problems that are approached in a structured way. It is the purpose of this chapter to elaborate on the approach and steps taken to solve the challenges summarized above.

9.2.2 Processes of Factory Planning

Systematic factory planning is related to a planning cascade, also referred to as a "*planning pyramid,*" from the initial idea, through the process of the ideal factory,

to the detailed elaboration, to when the factory finally operates. In general, one will need to consider the steps of target planning, concept development, elaboration and evaluation of alternatives, and execution of the selected alternative leading to the final factory building.

Three basic *processes in factory planning* can be classified as in Table 9.1.

Consider these processes in more detail.

Planning pre-requisites and factory planning preparation consist of the collection of the relevant short-, medium-, and long-term inputs for the factory planning project. This *target planning* is the input for structuring the process and for creating the planning documentation. The results from this first block create the precise formulation of the tasks and targets/objectives of the factory planning process and have contributed to a first or pre-feasibility study.

9.2.2.1 Factory Structure Planning

This complex deals with the creation of the first high-level factory layout designs, which support the evaluation of alternative concepts for the factory. It is important to mention that in this case, an ideal view is taken, which means that potential constraints regarding the available infrastructure or pieces of land are not considered. The reason for this disconnected view is to develop the most appropriate structure (*principle planning*), the identification of the main production areas, the necessary assembly processes, and also the required logistical flows and setup.

Table 9.1 Factory planning processes

Phase	Main contents or results to be achieved
Planning	Collection of requirements (e.g., production program); Analysis of tasks that need to be performed; Health, Safety, and Environment (HSE)/feasibility studies that need to be considered; Description of functional areas and expected output, layout alternatives; Layout models/site layout plans to be presented; Required documents to obtain authority approval.
Realization	Statements of work (e.g., also for subcontracting purposes); Screening of quotations/offers to be evaluated; Detailed project plans and related information on realization and capacity planning (who/which company is going to work when on what to be finalized by when - this is a very challenging task for the project management team); Purchasing plans for heavy equipment (e.g., robots, transportation devices, machines, logistical equipment, cranes …); Plans for the IT installations such as terminals for shop floor management, etc.; Plans for departmental commuting; Plans for the training or hiring workforce.
Operation	Plans on the final approval and commissioning of the factory and its functional areas to make sure it can be operated; Documents to be obtained from external approval bodies, quality certificates, HSE, and construction approval; Planning of the ramp-up phase including training and education of the workforce, Preparation for start of volume production.

In other words, this planning complex has the objective to evaluate and select the most appropriate structure. It also aims to dimension the functional areas. Furthermore, the connection and logistical linkages, as well as the process sequencing, are advanced by the factory planning team. As a result, an *ideal layout* of the factory can be presented. Following this step, adaptation to the real existing or available conditions is made. This transfer from the ideal scenario to what can be achieved is leading to the generation of the so-called *realistic layout.* The different variants presented (scenarios) can be mentioned as well as the related *feasibility study.*

Still, it might be necessary to de-couple functional areas due to vibrations (e.g., a press shop), and to separate buildings in order to comply with local construction rules and regulations (e.g., fire protection, noise emission, pollution) or to consider a special foundation. These aspects differ from case to case and from country to country. A detailed analysis is performed once a decision for a new factory is made in practice.

9.2.2.2 Detail Planning and Project Preparation for Realization

After management has decided on the preferred factory layout variant, detailed planning takes place. Starting with the block layouts (rough layouts), the exact allocation and space for machines, positioning of supporting equipment, planning of additional space and office infrastructure (locker rooms, medical support, offices, sanitary space), but also planning of media (gas, water, electricity, information technology …) have to take place. One could say that the "empty facility" of the rough planning is now filled in with its exact functional content (fine planning) so that the factory can finally be created. The deliverable of this process is the factory planning project, including all detailed plans.

9.2.2.3 Project Realization

When details are defined, the suppliers and partners must be involved in line with the specified project plan. In parallel to all these planning activities, a team must make the necessary cost calculations (business case development and validation) followed by running stringent cost controlling. Additionally, the planning should be properly set up by a project management team to track progress, visualize it, and alert in case of deviations. These could be cost, time, or quality-related deviations. In principle, these are typical tasks that will be run under the guidance of the factory planning project management team.

In order to make the factory planning process with typical use case scenarios more transparent for the reader, the short cases are presented as follows:

- An existing factory is regularly adjusted to current production needs.
- An existing factory is completely redesigned to meet strategic objectives.
- A new factory is completely built from scratch in order to generate a competitive advantage in a new market.
- Temporary factory with a partner.

9.2.2.4 An Existing Factory Is Regularly Adjusted to Current Production Needs

The factories are planned from a layout and process point of view in a way that ensures flawless execution of all necessary production and assembly steps. This is also done in consideration of the corresponding production system and its underlying lean principles. The company needs to adjust its production and thus the factory layout whenever a new product is launched. That means the team dealing with factory planning reviews the existing factory, the material flow from the incoming area over the storage area, considers the involvement and allocation of suppliers into the production process, and deals with questions as to how to test the product and how to ship it to the customer. As we can see, factory planning is linked to layout planning, production sequencing and also recognizing the importance of a streamlined material flow. If there are new production or assembly steps, if there are more suppliers that need to be involved, if the existing buildings need to be adjusted, or if there is a need to create a new building, this is all done by the factory planning department. As this first case shows, even without creating a completely new factory, the aspect of factory planning plays a crucial role during the normal production of the product portfolio and thus represents a regular task to ensure continuous improvement.

9.2.2.5 An Existing Factory Is Completely Redesigned in Order to Meet Strategic Objectives

In contrast to the above summarized continuous adjustment of the factory to meet current production needs, a complete factory makeover is a task where the location of the factory on the plot of land remains the same, but where fundamental changes are implemented. Such a complete factory makeover is furthermore based on a strategic decision as it involves a significant investment.

Based on inputs from the sales team regarding the expected production program for the next few years, the resulting capacity needs are derived. That means an understanding is required of what type of product and quantity are expected to be placed on the market. This information is provided to the factory planning team on an aggregated level. The factory planners then review the existing setup of the facility and determine which production steps are necessary, what kind of technology is required, and what it means in terms of manufacturing hours, stations, and number of buildings.

For example, a strategic decision was made to almost double the production volume to satisfy global markets. This would require a complete redesign of the layout, allowing, for example, mixed-model lines with shortened setup times; it was also strongly linked to the setup of a completely new logistics concept to feed the assembly lines continuously with material from the warehouse. The warehouse would be completely newly built and so-called logistics trains to be installed to provide the parts and components to the assembly stations. That means that the shape of the building remains the same. However, the internal allocations of the stations, the material flow, and the allocation of the warehouse have to be completely made over to satisfy the demands of growing markets.

9.2.2.6 A New Factory Is Completely Built from Scratch in Order to Generate Competitive Advantage in a New Market

Factory planning also plays a significant strategic role with respect to the generation of competitive advantage. For example, the Indian market might require new products (e.g., investment goods). To win the corresponding commercial bid, it would be necessary to ensure that the products are made to a very high degree locally in India itself. This is also referred to as "local content." That means a location needs to be identified to set up a completely new factory, in which local staff would create the final product with a high proportion of locally sourced materials. For this case of factory planning, the aspect of location planning (as described in Chap. 7) would also be important because the company needs to identify a region in which to create the new factory on the greenfield site. Under consideration of site selection criteria (e.g., existing infrastructure, availability of potential workforce, appropriate suppliers, or capable logistics service providers), a factory location has to be identified. The plot of land also has to provide enlargement potential for possible later expansion of the factory. Finally, a completely new facility with internal suppliers on site would need to be created in India.

9.2.2.7 Temporary Factory with a Partner

This is a special case of factory planning, where local content requirements have to be met but where no new factory has to be created. Imagine that investment goods have to be made for a customer in Romania. Assume that this would be related to a one-time contract (products are produced, but then no further products have to be made for that customer). In such a situation, there would be no business case justifying the creation of a new factory. In order to produce in Romania with local staff for the local customer, the existing infrastructure would be analyzed to find already existing facilities that need to meet the necessary requirements (such as infrastructure, size and height of buildings, handling equipment, and floor density). In a traditional industrial zone, appropriate facilities could be identified. The available alternatives will be reviewed regarding their strategic and operational fit. Finally, the investment goods might be produced in a factory that originally generated, for example, heavy power plant transformers. Of course, that factory will need to be replanned with consideration for the assembly and logistical processes required to ensure on-time and on-quality delivery of products to the final customer. This example shows that factory planning has to be considered a strategic decision to be responsive and flexible regarding customer focus.

9.2.2.8 Health, Safety, and Environment (HSE)

It is important to emphasize that the aspect of HSE is the highest priority in all examples and activities related to factory planning. HSE deals with the proper working conditions that have to be ensured by the company so that employees can contribute to value-adding under safe and healthy conditions. Thus, the workforce has to be provided with personal protection accessories, such as safety shoes, gloves, hard hats, and ear plugs. Furthermore, the ergonomic aspects should be considered, and corresponding tools (e.g., lifting or transportation devices) have to be made

available. People need to be safe when they perform their work. Safety training and education is a mandatory objective. Also, safety and work instructions should be available for all employees, including instructions on how to proceed in case of an injury.

Factory planning and HSE are strongly connected, as HSE is also related to ensuring proper daylight or illumination, as well as to keeping working areas free from dust (e.g., from welding), smells (e.g., from glues), or noise of machines or processing. Also, emissions must be avoided when, for example, fork lift trucks are charging the batteries: proper emission or smoke evacuation technology must be installed. A safe workplace also means that emergency exits need to be clearly marked, fire extinguishers are available within easy reach, and corresponding visual aids show the shortest path to extinguishers or to the emergency exit. Therefore HSE related topics always have to be on top of the agendas in running operations but also when new operations are planned.

9.3 Capacity Planning

Under *capacity*, we understand the throughput, or the quantity of units a facility can hold, receive, store, or produce in a period of time (Cheong and Laik 2024; Heizer and Render 2014). That means there is a constraint or a limit on production to satisfy demand. This limit can be theoretical and effective.

The theoretical limit is called *design capacity,* which is the maximum theoretical output of a system. *Effective capacity* is the capacity a firm expects to achieve given current operating constraints (Heizer and Render 2014). In practice, effective capacity is usually lower than design capacity.

Let us look at capacity using a very simple example. Assume that we have a small bakery shop in a big railway station. If it takes a single person 2 min to prepare a sandwich, then that person can produce 30 sandwiches (60 min per hour/2 min preparation time per sandwich = 30 sandwiches) per hour. So, looking at the aforementioned definition, the total amount that can be produced per hour is 30 sandwiches, in other words, the capacity is equal to 30 sandwiches per hour. In this theoretical example, we might have analyzed that there are peak demands in the morning hours (8.00 o'clock), during lunchtime (12.00 o'clock), and in the evening (at 17.00 o'clock), when people are traveling back to their homes. During these three peak times, the demand might be 90 sandwiches per hour. That shows two typical operational problems. First, we have a capacity issue at the peak times. Second, we need to *balance capacity according to demand*. What we could do, for example, is to increase the workforce during times of high demand, or we might use technology to shorten the time to create one product. We might pre-produce the components of our sandwiches (this could be pre-slicing of the bread or slicing the cheese). This simple example indicates that a company needs to know its existing capacity and what it is able to perform per period. The company also needs to adjust its capacity and clearly know where the *constraints or bottlenecks* are.

Now, let us translate this way of thinking into the operational world. In order to determine the capacity, it is necessary to know the products that have to be placed on the market. This understanding of demand could be order-driven, and/or it could be forecast driven. In our bakery example, it could be that there is a bank building nearby that places a fixed order of 50 sandwiches every day. That means that there is an existing contract that specifies exactly the quantity and the composition of the goods. On top of that, we know the regular daily demand patterns, including the three peaks per day explained at the beginning of this section. The forecasting aspect (see Chap. 11) plays a role when we consider vacation periods or other seasonal aspects. It can also happen that we recognize that daily demand increases about a certain percentage when there is a football game nearby.

All this is very important information for us to understand the demand of products per time unit and how to balance that against our existing capacity. This also tells us that we need to be able to adjust our capacity to cope with swings on the demand side. As a consequence, we need to have space, the workforce, and the machinery and tools to meet the demand. Of course, this information on future demand is not only required to determine capacities, but it is also relevant to managing the SC, as the corresponding need for materials needs to be communicated to suppliers. In the simplified bakery example for better illustration purposes, this might be bread, butter, and cheese. In real large factory operations, raw materials, purchased parts, and components need to be in line with foreseen demand as well as with the required technical and human capacities.

9.3.1 Queuing Theory

Queuing theory is an important tool for capacity planning in services and industries. We find queues, for example, in supermarkets or in waiting rooms; or there are queues of trucks in front of distribution centers or of several orders that have to be processed in production.

Queuing theory helps to analyze the *performance of a process* such as that of the temporal arrival of goods in a distribution center. Performance generally can be measured in different dimensions such as capacity, time, cost, quality, or flexibility and these factors can influence each other. The queuing model is a good tool for the analysis of time dimension in a process. In Fig. 9.1, a simple queuing model (one-channel model) is presented.

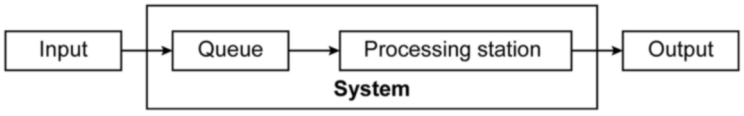


Fig. 9.1 Queuing model

A queuing model comprises the following elements:

- *Input* (e.g., patients).
- *Queue* (e.g., patients in the waiting room).
- *Processing station* (e.g., a doctor).
- *Output* (patients after the treatment).

In queuing theory, different options exist to describe these four elements. To describe the queues, we use the notation

$$a / b / c,$$

where
a is the probability distribution of arrival rate, e.g.:

- D—degenerate distribution (deterministic)
- M—Markovian distribution (Poisson process)
- G—general distribution;

b is the probability distribution of service time

- D—degenerate distribution
- M—exponential distribution
- G—general distribution;

c is the number of servers (processing stations)

In regard to the number of servers, there are two basic types of models. A *one-channel model* considers only one queue and one server that processes the input. A *multiple-channel model* is involved with one queue and several servers or several queues and several servers.

Consider a system where there is no constraint in the queue, and we follow the rule "first come first serve" for defining the processing sequence. To analyze this case, we introduce some notations that describe the system components (Fig. 9.2).

We consider the so-called M/M/1 queue. We determine the mean number of arrivals per time period as λ and the mean number of people or items served per time period (i.e., average service rate) as μ. It means that the arrival rate is Markovian distributed, service time exponentially distributed, and there is one server. We can find this queue, for example, in post offices or small supermarkets with only one cashier or in waiting rooms for only one doctor's surgery.

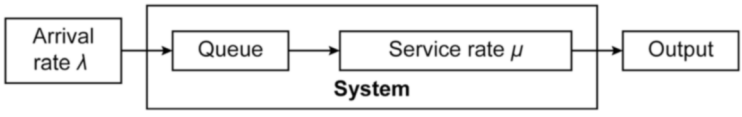


Fig. 9.2 Notations for queuing theory

Consider the following formulas to understand how to analyze the performance of the queuing process. To determine the time between the arrivals of two customers, e.g., patients, we use Eq. (9.1):

$$\text{Estimated inter - arrival time} = \frac{1}{\lambda} \tag{9.1}$$

To analyze the service time, we use Eq. (9.2):

$$\text{Estimated service time} = \frac{1}{\mu} \tag{9.2}$$

To analyze the *utilization* factor for the system, we use Eq. (9.3):

$$\rho = \frac{\lambda}{\mu}; \rho \leq 1 \tag{9.3}$$

To calculate the *probability* of more than *n*-units in the system, we can use Eq. (9.4):

$$P_n = \rho^n \cdot (1 - \rho) \tag{9.4}$$

To find out the *length of the queue* (e.g., the estimated number of patients in the waiting room), we use Eq. (9.5):

$$L_Q = \frac{\rho^2}{1 - \rho} \tag{9.5}$$

The *length of the system* (e.g., the estimated number of patients in the waiting room *and* in the doctor's surgery) is determined with the help of Eq. (9.6):

$$L_s = L_Q + \rho = \frac{\rho}{1 - \rho} \tag{9.6}$$

Waiting time in the queue (e.g., time spent in the waiting room) can be found using Eq. (9.7):

$$T_Q = \frac{\rho^2}{\lambda \cdot (1 - \rho)} \tag{9.7}$$

Waiting time in the whole system (e.g., time spent in the waiting room and doctor's surgery) is calculated with the help of Eq. (9.8):

$$T_s = T_Q + \frac{1}{\mu} = \frac{1}{\mu \cdot (1 - \rho)} \tag{9.8}$$

Task 9.1 Queuing Theory M/M/1 System

There is only one cashier in a bookshop in the city center because the other two are off with influenza. Normally, the cashier is able to cash up an average of 70 customers per hour, but on average, there are only 55 customers per hour in the bookshop. In addition to cashing up customers, the cashier also has to unpack deliveries.

Analyze the performance of the queue and check if there is enough time to unpack deliveries.

$$\text{Inter-arrival time} = \frac{1}{55} = 0.018\,\text{h} = 1.08\,\text{min} = 64.8\text{s}$$

Every 65 s, there is a new customer who wants to buy books.

$$\text{Service time} = \frac{1}{70} = 0.0143\,\text{h} = 0.86\,\text{min} = 51.4\text{s}$$

$$\frac{1}{70} = 0.86\,\text{min} \cdot 55\,\text{customers} = 47.3\,\text{min}$$

The cashier needs at average 51.5 s to cash up one customer and 47.3 min for all 55 customers per hour, which means the cashier has 12.7 min per hour to unpack deliveries.

$$\rho = \frac{55}{70} = 78.6\%$$

And so the cashier is occupied 78.6% of the time.

Now we want to analyze the queue as seen from the customer's perspective.

$$L_{\text{Q}} = \frac{0.786^2}{1-0.786} = 2.89; L_{\text{s}} = 2.89 + 0.786 = 3.68$$

There are, on average, around three people waiting in the queue and approximately four people in the system.

$$T_{\text{Q}} = \frac{0.786^2}{55 \cdot (1-0.786)} = 0.0525\,\text{h} = 3.15\,\text{min}$$

$$T_{\text{s}} = 0.0525 + \frac{1}{70} = 0.0668\,\text{h} = 4.01\,\text{min}$$

Customers are waiting 3.15 min on average in the queue and 4.01 min in the bookshop in total before they can leave the shop with their purchase.

Task 9.2 Queuing Theory M/M/c System

In the case of an M/M/c queue, formulas are different. We can find M/M/c queues, for example, in a distribution center. There are several trucks in waiting zones to unload their goods onto loading docks or to get loaded. In the case of M/M/c queues, we have a Markovian distribution of arrival rate, an exponential distribution of service time, and c servers (>1).

Consider the following example for M/M/c queues. On average, there are 40 trucks arriving per hour, and one loading dock is able to unload and load 12 trucks/h. Furthermore, there are four loading docks. Assume that every loading dock has the same capacity and needs the same time. There is only one waiting queue.

To calculate the occupancy rate of the system, we use Eq. (9.9):

$$\rho = \frac{\lambda}{c \cdot \mu} = \frac{40}{4 \cdot 12} = 83.3\% \tag{9.9}$$

The distribution center loading docks are occupied for 83.3% of the time.

To calculate the estimated number of trucks waiting in the queue, we need to find the probability that the system is completely empty (Eq. 9.10).

$$P_0 = \frac{1}{\sum_{i=0}^{c-1} \frac{(c \cdot \rho)^i}{i!} + \frac{(c \cdot \rho)^c}{c!} \cdot \frac{1}{(1-\rho)}} \tag{9.10}$$

$$P_0 \frac{1}{\frac{(4 \cdot 0.833)^0}{0!} + \frac{(4 \cdot 0.833)^1}{1!} + \frac{(4 \cdot 0.833)^2}{2!} + \frac{(4 \cdot 0.833)^3}{3!} + \frac{(4 \cdot 0.833)^4}{4!} \cdot \frac{1}{(1-0.833)}} = 2.14\%$$

The probability that the system is empty is 2.14%. Now it is possible to calculate the number of trucks in the queue [Eq. (9.11)].

$$L_Q = \frac{c^c \cdot \rho^{c+1}}{c! \cdot (1-\rho)^2} \cdot P_0 = \frac{4^4 \cdot 0.833^{4+1}}{4! \cdot (1-0.833)^2} \cdot 0.0214 = 2.183 \tag{9.11}$$

In the case of four loading docks and 40 estimated trucks per hour, 2.2 trucks are waiting in the queue.

The estimated number of trucks in the whole system is calculated using Eq. (9.12):

$$L_s = L_Q + c \cdot \rho = 2.183 + 4 \cdot 0.833 = 5.515 \tag{9.12}$$

On average, there are 5.5 trucks in the system.

Waiting time in the queue is defined using Eq. (9.13):

$$T_Q = \frac{L_Q}{\lambda} = \frac{2.183}{40} = 0.0546\,\text{h} = 3.276\,\text{min} \tag{9.13}$$

Waiting time in the whole system is defined using Eq. (9.14):

$$T_s = T_Q + \frac{1}{\mu} = 0.0546 + \frac{1}{12} = 0.1379\,\text{h} = 8.274\,\text{min} \tag{9.14}$$

As you can see, the waiting time in the system is 8.3 min.

9.3.2 Little's Law

In 1961, John D.C. Little provided the mathematical evidence for the rule that puts the inventory L (= average number of objects in a system) in relation to the throughput rate λ (= average arrival rate of objects) and the waiting time W (= average time

an object spends in the system). Little's Law says "that, under steady state conditions, the average number of items in a queuing system equals the average rate at which items arrive multiplied by the average time that an item spends in the system." (Little and Graves 2008). Letting

L—average number of items in the queuing system (e.g., work-in-progress inventory in the system)

W—average waiting time in the system for an item

λ—average number of items arriving per unit time the law can be represented as Eq. (9.15):

$$L = \lambda \cdot W \tag{9.15}$$

Little's Law demonstrates that when we reach a steady state, i.e., the system states are stationary and the arrival rate is smaller than the service rate (cf. Fig. 9.2), the average number of objects arriving in the system during a time period (e.g., 1 h or 1 day) are equal to the ones that are leaving the system during that time period. Elegant in its simplicity, Little's Law represents one of the basics in capacity analysis. The following adapted example is related to the application of Little's Law in a manufacturing plant described by Bozarth and Handfield (2019).

Consider an application of the Little's Law to inventory and lead time analysis in a manufacturing system (Fig. 9.3).

The upper part of Fig. 9.3 presents the dynamics of incoming and outgoing orders in a manufacturing system. At the point of time 1, two orders arrive in the system, and the number of jobs in the system, i.e., the inventory $L(t)$ is 2 at $t = 1$. At $t = 2$, one more order arrives, and first two orders are still processed. Therefore, $L(t) = 2$ at $t = 2$. At $t = 3$, two new orders arrive, and the first order exit the system. Therefore, $L(t) = 4$ at $t = 3$. This inventory dynamics is shown in the bottom part of Fig. 9.3. Moreover, the upper part of Fig. 9.3. shows lead time $(s_j - c_j)$ for each job, where s_j is the completion time of the job and c_j is the start time of job processing. For example, the lead time for job no. 1 is two-time units.

Finally, we merge the previous discussions on the waiting line theory and the Little's Law (Fig. 9.4).

Four major events, i.e., the arrival of a customer order in the manufacturing system queue, production start, production end, and the customer order delivery build the framework of the analysis. The time interval between the first and the last event is called *lead time*. *Waiting time* is the time interval between the first and the second event. *Flow time* is the time interval between the second and the third event. During the waiting time, backlog costs and lost order costs occur. During the flow time, processing costs and WIP inventory costs are considered. The delivered orders generate revenues and profits.

Task 9.3 Little's Law

Let us assume that in a car garage, two standard repairs are offered in a special express service zone at extremely competitive prices. Batteries can be exchanged in workstation 1, and oil is changed in workstation 2. Prior to this, customers need to register and receive a general car inspection. Then they are assigned to either station

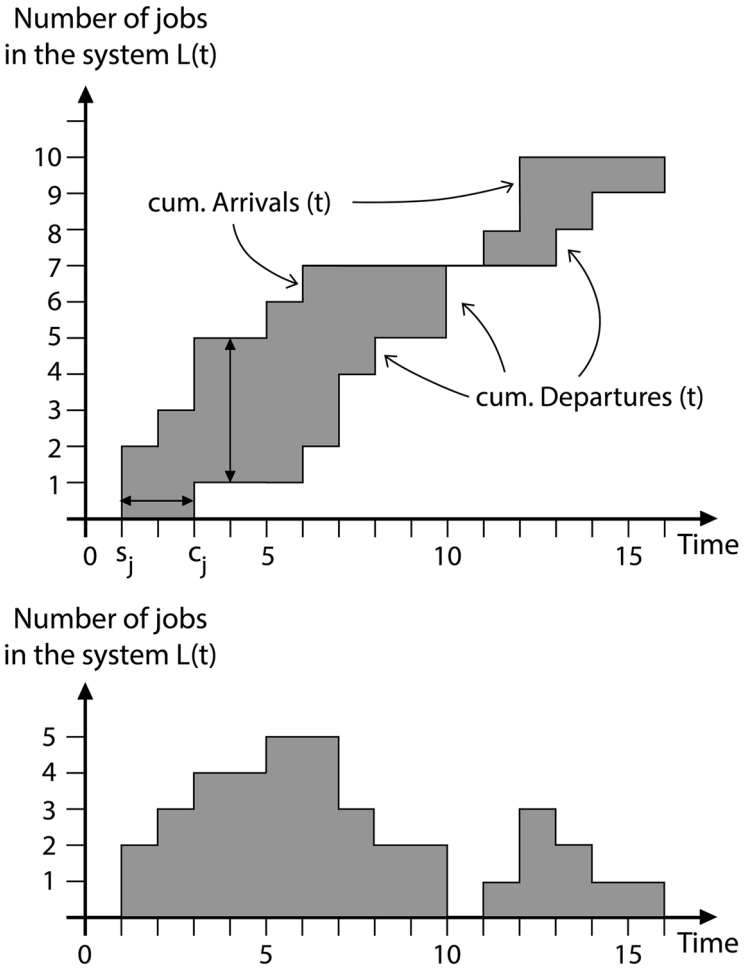


Fig. 9.3 Inventory and lead time dynamics analysis

1 or 2 depending on the repair that is required (either battery or oil service). As shown in Fig. 9.5, 200 customers per day arrive at that garage.

- As stated, all customers are passing through the general inspection where, on average, 20 cars are in the system for inspection.
- Sixty percent of clients request an oil service, and the average inventory is seven cars.
- The other 40% want their batteries to be exchanged, and the average inventory is three cars.

Looking at this scenario, we can calculate the following metrics:
The waiting time W for the general inspection is

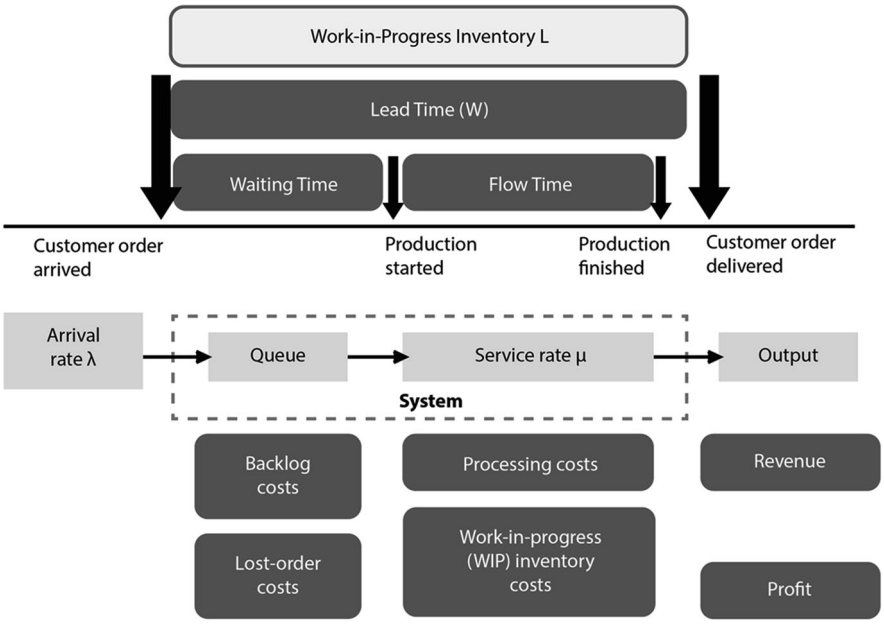


Fig. 9.4 Major components of the waiting line and Little's Law model

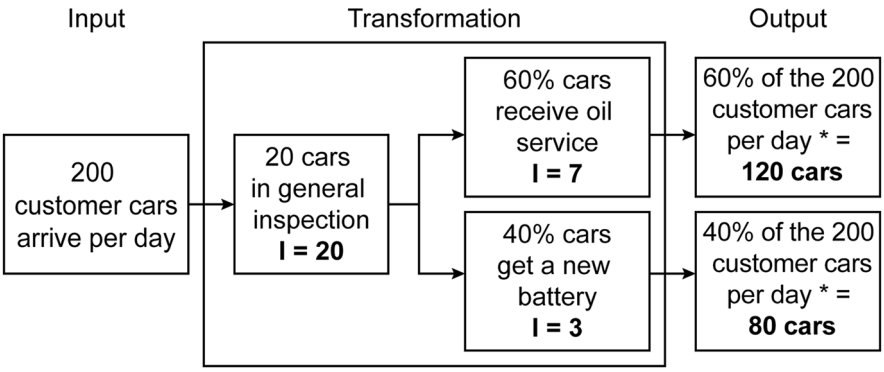


Fig. 9.5 Little's Law—example of usage in a car garage

$W = L/\lambda$ = (20 cars inventory in inspection)/(200 customer cars per day) = 0.1 day in general inspection = 48 min waiting time, if we assume that the company operates 8 h (= 480 min) per day.

We can observe that 120 cars receive an oil service per day, and on average, there are seven cars in the oil service area. Applying the same calculation principle as before, this leads us to the waiting time for the oil service of:

$W = L/\lambda$ = (7 cars inventory in oil service)/(120 cars per day) = 0.0583 per day = 28 min waiting time for a day with 480 min working time and per car.

However, we also have a total of 80 cars per day that will receive a new battery. On average, there are just three cars in the inventory. Taking the formula according to Little's Law, this leads us to the waiting time for the battery exchange of:

$W = L/\lambda$ = (3 cars inventory for a new battery)/(80 cars per day) = 0.0375 per day = 18 min of waiting time for a day with 480 min working time.

What is the waiting time for the cars receiving either an oil service or a new battery? In total, the cars require 48 min for the general inspection +28 min for the oil service = 76 min of waiting time. In the case of the general inspection plus installation of a new battery, it is 48 min + 18 min = 66 min of waiting time.

As we have learned from that example, the simple but powerful "Little's Law" formula can be used for just one single process, but can also be applied for a multi-process-step transformation system.

9.3.2.1 Learning Take Away on Little's Law

One can also use that simple formula for the quick assessment of a manufacturing facility, for example, in connection with a *rapid plant assessment* (RPA). During a plant tour, simply consider Eq. (9.15). If you ask during a visit to the factory what the average throughput rate per day is and what the throughput time is, you can simply calculate the average inventory that is carried. On the other hand, if you are an operations manager in a factory and you know Little's Law, you should try to identify the key drivers for the throughput time, i.e., the key processes with their durations that cause delays or prolong the throughput time. Try to understand these and try to identify actions to reduce these delaying factors. The overall idea is to keep the same output throughput rate, but to reduce the throughput time in order to bring down the inventory carried. If inventory is not a concern for the company, but achieving a higher output per day is the priority, a throughput time reduction will also help here. If inventory remains stable and throughput time can be reduced as a consequence, the throughput rate will increase. That means we can produce more, and that is the reason why Little's Law is also connected to the management of capacity.

Discussion Points

- Where else could you think about applying the simple formula of "Little's Law"? Think about usage in a manufacturing environment but also in a service environment.
- Consider yourself the manager of an insurance company. You would like to increase customer satisfaction by promising all customers that they will be served within 15 min of arrival. How could "Little's Law" help you?
- If you would like to increase the performance of the value-added system, thus reducing the throughput time, where would you prioritize your observations?

9.3.3 Bottleneck Analysis/Theory of Constraints

The *bottleneck* determines the capacity of the transformation system. To make it more transparent, just assume that there is a product (e.g., a chair) that requires, for example, four simple production steps (see Fig. 9.6).

In this simplified production process, painting and drying require the longest duration as it requires 10 min per chair (10 min per chair/60 min per hour = 0.167 h per chair). Or putting it more simply, six chairs per hour can be painted, and thus the system capacity is six chairs as the bottleneck determines this. After the identification of the bottleneck (step 1), the operations manager's task would be to seek opportunities to reduce this bottleneck time.

Another frequently used term for the bottleneck is *constraint*, as it is the process that prevents the enterprise from achieving its objectives (Goldratt 1990). If we want to produce ten chairs per hour, something needs to be done in order to limit the impact of that constraint or bottleneck, the performance manager needs to find a way (most probably with the help of a manufacturing engineer) to reduce the bottleneck time from 10 to just 6 min per chair. A possible solution might be to improve the spray process, or to use a different type of paint, or to increase the temperature for the paint to dry faster. That means a plan has to be created to mitigate the impact of the constraint and to reduce the bottleneck time (step 2). The operations manager and the manufacturing engineer are the dedicated resources that will need to work on the implementation of the identified measures (step 3).

Furthermore, one should better balance the work and try to do some necessary preparatory work required for the painting, such as cleaning after the back of the body of the chair was installed. If this is also done, we are trying to spread the work from the bottleneck as well as to try to increase capability in related production processes (step 4). If this has been successfully implemented, the performance manager will start again to identify new constraints and bottlenecks as part of this continuous search for improvements (step 5).

9.3.4 Drum, Buffer, Rope

The drum is the instrument responsible for the *beat* (beat rate or *takt time*) of the production system. This can also be considered as the pace or production schedule and thus provides information regarding the items that need to be produced per time unit.

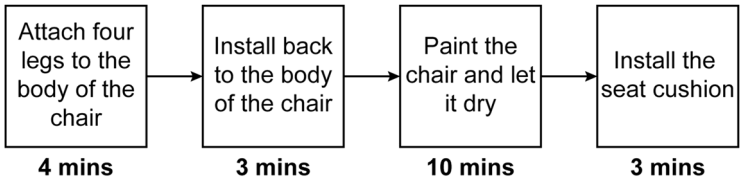


Fig. 9.6 Process chain

The buffer can also be considered as the *inventory* (or another type of resource) of the system, which allows the bottleneck to perform at the beat of the drum. The last symbol is the rope. This is the means to exchange information, or it enables necessary communication between the different parties. It refers to the need to synchronize the different elements or workstations in the system to operate in alignment and thus has the function of sending signals.

Assume there is a motorway that has five lanes, and at a particular location (e.g., a bridge), it is narrowed to three lanes. Then, there is the capacity limitation or constraint, as cars can only use these three lanes. According to the theory of constraints (ToC), you only *achieve an improvement or advantage when it is possible to increase the capacity of the bottleneck*. That means, for our example, that you can only improve the system by creating a broader bridge with four or even five lanes.

The second principle of the ToC is that *it does not improve the performance of the system if you try to work on a non-bottleneck area*. For our example, if you create a better motorway with six lanes but you keep the old bridge with three lanes, this will not improve the system, as the bottleneck remains the same.

The third principle says that the *time that you spend at the bottleneck*, the bridge in our case, *is lost*. In real operations, you could try to mitigate the impact of the bottleneck by using well-trained staff or by putting in a strict management system to lower the impact of the bottleneck. In a real factory, the paint shop or a crane might be a bottleneck. In the case of the crane, you might schedule its usage tightly and should use a skilled crane operator who can maneuver goods with the highest precision, so that the time consumed for picking the goods, moving them and to position them at their destination is strictly limited.

We can summarize that bottleneck management and dealing with constraints is a daily task for operations managers. Furthermore, a constraint will always exist, and due to expected or unexpected occurrences, new bottlenecks will arise (key people leave the company, workers are sick or on strike, a machine breaks down, or a supplier cannot deliver on time). Other drivers for bottlenecks could also be changing product mixes, using new technology, and training people to create new products using new technology. Knowing and dealing with such constraints is part of the day-to-day business in SCOM.

9.3.5 Break-Even Analysis

In the previous sections dealing with Little's Law or the ToC, we looked only at capacity without consideration of profit and loss aspects. The break-even analysis is an approach that also looks at costs, respectively revenues. Slack et al. (2010, p. 658) summarize that the break-even analysis is "the technique of comparing revenues and costs at increasing levels of output in order to establish the point at which revenue exceeds cost, that is, the point at which it 'breaks even.'"

The *break-even point* is the point at which the total costs of the number of units that need to be produced are equal to the total revenues that are generated (Krajewski et al. 2021). That means that from a capacity point of view, a firm needs to operate

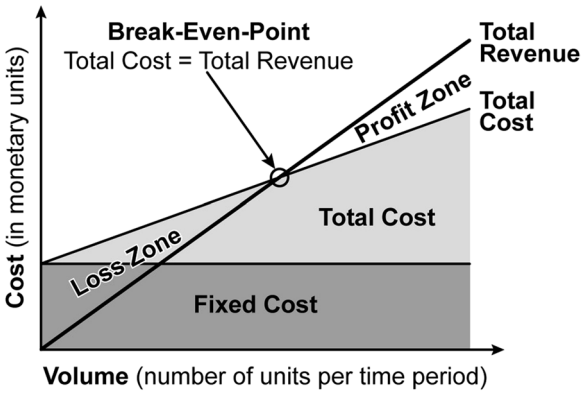


Fig. 9.7
Break-Even-Point

above the break-even point, or graphically speaking on the right-hand side of that point, as Fig. 9.7 depicts.

With the number of units that lead to a surplus on the revenue side, i.e., where the revenues are higher than the cost, the company will enter the profit zone. Below that production volume, graphically speaking, the firm is running left from the break-even point and thus is only generating losses, as the costs are higher than the revenues.

Task 9.4 Break-Even-Point

Put yourself in the position of a production manager who needs to evaluate if it is reasonable to purchase a new machine for the production of certain goods. The underlying assumptions are that costs and revenues are linear functions and that we know these exact costs. Furthermore, interest rates or cost of inventory is excluded from our break-even analysis model. It will be your task to calculate the break-even point in the number of units but also in terms of money, i.e., in $US.

Consider the following notations:

- BP_x represents the *break-even point in units x*
- $BP_\$$ represents the *break-even point in $US*
- p represents the *price per unit*

- x represents the *number of units produced*
- vc represents the *variable costs*
- fc represents the *fixed costs.*

Equation (9.16) is used to determine the total revenues (TR):

$$\mathrm{TR} = p \cdot x \tag{9.16}$$

Equation (9.17) is applied to determine the total cost (TC):

$$\mathrm{TC} = \mathrm{fc} + \mathrm{vc} \cdot x \tag{9.17}$$

Table 9.2 Initial data for break-even point analysis

Fixed cost (FC)	$20,000
Material cost per unit	1 $ per unit
Direct labor per unit	2 $ per unit
Selling price per unit	5 $ per unit

According to the definition of the break-even analysis, the *break-even point* is reached when TRs are equal to TC. That means for you as the production manager, you will need to set:

TR = TC or considering the details from above, we can use Eq. (9.18)

$$p \cdot x = \mathrm{fc} + \mathrm{vc} \cdot x \tag{9.18}$$

to identify the *break-even point in units.*

The *break-even point in units* thus equals (see Eq. 9.19):

$$\mathrm{BP}_x = \mathrm{fc} / (p - \mathrm{vc}) \tag{9.19}$$

To convert this into monetary values (e.g., $US), the result BP_x needs to be multiplied by the price per unit, i.e.,

$$\mathrm{BP}_\$ = \mathrm{BP}_x \cdot p = \left[fc / (p - vc) \right] \cdot p = fc \cdot p / (p - vc) \tag{9.20}$$

Based upon Eq. (9.20), the profit can also be calculated, because profit (simplified) equals TR–TC, which translates in Eq. (9.21):

$$\text{Profit } P = \mathrm{TR} - \mathrm{TC} = p \cdot x - (\mathrm{fc} + \mathrm{vc} \cdot x) \tag{9.21}$$

Consider a numerical example. You received the following annual values from your controlling department (see Table 9.2):

In order to calculate the break-even point in $US $BP_\$$, you will need to use formula (9.20):

$$\mathrm{BP}_\$ = 20{,}000 \cdot 5 / \left[5 - (1+2) \right] = 100{,}000 / 2 = 50{,}000\$ / \text{year}.$$

That means the break-even point in $US $BP_\$$ is reached, once the value of $50,000 is exceeded. As a production manager, you would like to know the quantity in units that needs to be produced to reach the break-even point in units BP_x. Therefore, you now need to use formula (9.20):

$$\mathrm{BP}_x = 20{,}000 / \left[5 - (1+2) \right] = 20.000 / 2 = 10{,}000 \text{ units} / \text{year}.$$

Of course, the break-even point in $US is directly linked by the sales price per unit with the break-even point in units, as you can easily see in the example above. You can simply test the result by taking the BP_x (which is equal to 10,000 units in our task) and multiplying it by the selling price (which is here equal to $5/unit). The result is 10,000 units · $5/unit = $50,000, as calculated above.

▶ **Practical Insights** In practice, the break-even point in $US is more relevant for the sales or the controlling department, whereas the break-even point in units "talks" more appropriately to the managers responsible for production, purchasing, or logistics. This is because

production or logistics managers need to consider not only the size or weight of the products in order to determine space requirements on the production floor or in the warehouse but also to identify the correct lifting or transportation devices. That means that the break-even analysis is a simple calculation method that provides a wide variety of people inside the organization with information essential for their purposes.

9.3.6 Decision Trees

In practice, *market uncertainty* and factors such as production cost, demand, prices, inflation, or exchange rates influence managerial decisions. Should we expand our production or not? Is it a profitable choice to redevelop the old distribution center if demand for the following years is uncertain? Should we sign a long-term contract with our supplier if we do not know if customers like our new product?

Decision tree analysis helps in making decisions in uncertain environments. A decision tree can be used to visualize possible strategies and calculate the revenue, cost, and expected monetary value of several alternatives. Decision tree analysis offers a combination of strategic decisions, such as expanding a production site or not, and financial planning. It evaluates strategic decisions by their profitability to support the decision-making process. For that reason, it can help in making decisions and aid in the design of processes and networks.

Task 9.5 Decision Tree Analysis
A company thinks about expanding their production site assuming a favorable market development. To calculate *expected monetary value (EMV),* we consider probability P for both a favorable (0.35) and an unfavorable (0.65) market development. Option A means a large expansion, option B—a medium expansion, option C—a small expansion, and option D—no expansion. At the end of the tree, you can see the cost or gain of each opportunity (Fig. 9.8).

To calculate the EMV, we have to use Eq. (9.22):

$$\text{EMV} = P(\text{favorable}) \cdot \text{gain} + P(\text{infavorable}) \cdot \text{losses} \tag{9.22}$$

Using Eq. (9.22), it becomes possible to decide which expansion would be the most profitable:

$$\begin{aligned}
&\text{EMV}(\text{A}) = 0.35 \cdot 100{,}000€ + 0.65 \cdot (-80{,}000€) = -10{,}000€ \\
&\text{EMV}(\text{B}) = 0.35 \cdot 75{,}000€ + 0.65 \cdot (-25{,}000€C) = 10{,}000€ \\
&\text{EMV}(\text{C}) = 0.35 \cdot 30{,}000€ + 0.65 \cdot (-3000€) = 8550€ \\
&\text{EMV}(\text{D}) = 0€
\end{aligned}$$

We can observe that medium expansion would be the best financial decision in a case where market development will be similarly predicted. Option D, no expansion, would make no difference because the company would not be changing anything.

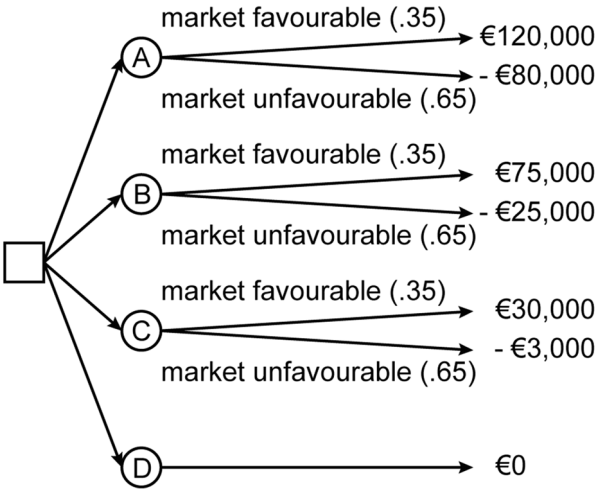


Fig. 9.8 Decision tree expansion production site

▶ **Practical Insights** It is important in decision tree analysis to identify all factors that influence the cost and to decide which factors are uncertain in the calculation. The critical issue is how to assess the probability of relevant uncertain factors and consider their influence. Moreover, a company should not focus only on financial analysis. Other factors may also be important, and the financial part can only be a fraction of the decision-making process. In practice, decision tree analysis is frequently supported by the development of *business cases* for possible future expansion. Discussion Questions

- Why do you think capacity management is important for globally active companies with large production networks and multiple production sites?
- Why is capacity management very important from a strategic point of view?
- What are the consequences when it is identified that the existing capacity of a firm is too high as opposed to when it is too low? What are the corrective measures that you would suggest to adjust the existing available capacity so that it can meet future demand?
- Is it the right strategy to aim for almost 100% capacity utilization, or do you think one should also consider some idle capacity for the unpredictable?

9.3.7 Simulation: Case-Study AnyLogic

9.3.7.1 Problem Description

Centrotherm Photovoltaics AG is a global technology and equipment supplier for the photovoltaics, semiconductor, and microelectronics industries. The company

needed to identify the best-automated production line and factory configuration to minimize costs and maximize throughput and reliability. The special focus was to:

- Determine the type and amount of equipment required to fulfill production plans
- Evaluate layout alternatives to improve the throughput and utilization rate
- Inspect possible bottlenecks in the material flow
- Evaluate the impact of operator behavior on factory output
- Test the consequences of maintenance within various time frames
- Identify system behavior in case of breakdowns
- Identify scrap probability
- Evaluate change in performance during planned downtimes

9.3.7.2 Model Development

The consultants used AnyLogic simulation and modeling technology to provide the answers to these problems. Using AnyLogic Professional's ability to create and save custom object libraries, the consultants encapsulated their vast simulation experience in the photovoltaic and semiconductor manufacturing industries and created their own libraries, which they reused in many projects, including this one.

9.3.7.3 Solution Development

These libraries featured elements built on top of the AnyLogic model development environment. They allowed the consultants to easily model different kinds of equipment, material handling systems inside factories, personnel, and production control systems, all specific to the photovoltaic and semiconductor industries. In the Centrotherm project, this solution helped quickly reproduce various aspects of the client's production system behavior.

9.3.7.4 Testing the Solution

Once the model was built, the experimental phase of the project began. The consultants tested many parameters to find the best solutions. The input data, which included layout configurations and various parameters, was taken directly from Excel and Access files. Each simulation run reproduced 1 year of factory operation.

First, the consultants experimented with the overall production line design, using parameter variation and optimization. They tested many parameters, including those concerning capacity and the number of various pieces of equipment, cassettes, and bins to see which configurations would work best in terms of throughput, reliability, and scrap rate. Then the consultants worked to optimize transportation policies, buffer zone allocation, and watermark control processes.

9.3.7.5 Analyzing the Results

Some of the proposed layout variants were rejected during this stage due to their poor performance. Finally, the few resulting solutions were tested manually to investigate each one's benefits and drawbacks and to find out how they could be further improved.

9.3.7.6 Implementation

The solutions proposed provided Centrotherm Photovoltaics AG with the opportunity to significantly improve the production line design and choose the best solution in terms of throughput, reliability, and scrap rate at a low cost. Additionally, at the end of the project, the model was released to the client so that it could be used for the analysis of future changes in the factory. The model allows the customer to carry out their own experiments, playing with parameters and layout, and changing input data. This manufacturing facility simulation model will serve as a decision support tool at the factory for a long time to come.

9.4 Process Flow Structures

It is intuitively clear that not all products are produced in the same way. In order to enlarge on this for you, we need to think through another hypothetical case, in which you would be the producer and the product would be a chair. It is a very simple example, but it makes it easy to grasp the underlying ideas and concepts.

9.4.1 Job Shop

Just put yourself into the situation where you plan to produce your own little chair for your room or apartment. You would go to the closest do-it-yourself center, would buy some pieces of wood, some nails, and your favorite color of paint (e.g., red). These are the materials you would need for the chair. Furthermore, you need some tools, maybe a brush, a saw, and a hammer. These standard tools and your materials plus your creativity and manual work are everything that you require in order to build your own, fully individualized piece of furniture (maybe you would produce and assemble it in your kitchen or in the garage).

The result would be an absolutely unique product, and nobody else would have the same one! Maybe one of your friends will ask you to produce a second one, but with some changes regarding the size and also regarding the color (your friend wants to have a bigger chair, painted in green). With your equipment, you would also be able to create such a product for your friend.

What we have just described is a typical setup of the so-called "*job shop*" process flow structure. The secret behind it is that with such a process flow structure, absolutely unique products are made under the usage of standard tools in very low quantities. Also, the manual effort is quite high, in respect of the automation level, which is in general very low.

As you produced it manually and bought only a small amount of material, the unit price is quite high, but so is the flexibility to adjust your production setup and to meet changing requirements. If you followed the hypothetical wooden chair example carefully from above, you would now understand that a job shop is a process flow structure that is used to create products at very low volume (just this one

chair for you and the adjusted one for your friend). At the same time, we can understand that the job shop is suitable for creating products with great variation.

It is obvious that, even with this very unusual and hypothetical example, we still have developed the concept of the "job shop." In reality, this is used for the creation of, e.g., replacement teeth, medical eye-glasses, or maybe exclusively styled motorcycles or exclusively tuned cars. Another example is the traditional tailor who produces a dress or suit exclusively for you. These are all individual products with great variety and very low volume.

9.4.2 Batch Shop

Now let us continue with the hypothetical example of our chair, which is made out of wood. Let us assume that you would go to a market to sell your products. While you are selling and people come to you stating that they would like to have the small chair and the big one with arm rests in various colors, the result would be increasing the volume of similar types of products that need to be created.

Most probably, the best idea would be to offer two standard chairs (a small one and a big one) that could be painted in five different colors, for example. Because of that selected product portfolio, the natural reaction would be that you would produce a certain lot size of the small and then another lot size of the bigger one. As products are processed in *lots or batches*, this flow structure is called batch processing or batch production (i.e., products are produced in a batch shop). In the last production step, you would then do the customization, i.e., the individual painting on the market using a spray can.

Again, right now, let us switch back to reality. Of course, not only exclusive suits are produced as you know, but also other exclusive products in small series like agricultural or construction machines, helicopters, or ambulances. In these cases, batches of similar products would be created. For example, the producer of special construction equipment would produce a small series of 20 excavators, which would all have the same communication equipment on board and would all be painted in blue for a specific customer and bearing the company's logo. The idea is to reduce the impact of setup times and setup costs and also to introduce stability into the value-creation process.

9.4.3 Assembly Line

Let us continue the hypothetical chair journey. Your chairs are superb, everyone in your neighborhood loves them, and now even shops ask you to produce them in higher quantities. That means the required volume is significantly increasing. For example, manually cutting the wood for the chairs and running a manual assembly would no longer be efficient.

The consequence then might be to look at the possible usage of more machines for cutting the wood, drilling holes, putting the parts together, painting, and even

packaging them. So, we would consider higher machine utilization and even think about more automation. Furthermore, we would think about the definition of work packages or focus on work specialization (e.g., cutting and drilling), leading to the identification of workstations in the *value creation process*. In order to ensure a constant *product flow*, the different stations would also need to be balanced to avoid *idle times*. The product being created moves at a defined *takt time* from station to station. This is also referred to as the product following a defined *pacing* through the workstations. In an ideal case, one would employ a *conveyor belt* or similar for the movement of the product throughout the production facility.

And this is what is applied in the real world. *Assembly lines* are used for the mass production of articles with relatively high volumes and a well-defined level of variants. The most known examples of products created on an assembly line are cars (e.g., Henry Ford's assembly line for mass-produced automobiles of the famous T-Model). Other examples produced on assembly lines are household devices such as washing machines, refrigerators, or microwave ovens, as well as consumer goods such as TVs or computers. All these have in common that the item produced is made in *large volume* and is moving at a constant speed through the factory from work station to *work station.* This makes the production quite efficient, but it is also vulnerable if a station may not be working or the wrong (or no) parts have been supplied to the station. The result could be a stopping line.

Task 9.6 Assembly Line Balancing

The planner in charge of assembly line balancing requires some inputs. One fundamental input is the manufacturing bill of materials (BoM) that provides the necessary information regarding the composition of the single parts that will be assembled in order to create the final product. To ensure this, the logical sequence of the different steps (so-called *precedence relation*) also needs to be known (just imagine the sequence in which a pizza is created: first comes the dough, then the tomato sauce, followed by mushrooms, then the cheese, then it is placed in the oven, etc.). Additionally, the *task times* for different activities have to be known along with demand and available capacity per period of time.

Let us look at the example of how to assemble a simple manual scooter. You are in charge of balancing the assembly line according to the following inputs from the manufacturing engineers (Table 9.3):

Table 9.3 Initial data for assembly line balancing

Activity	Task	Predecessor	Task time (min)
A	Install brake in board	None	1.5
B	Install rear wheel in board	A	1.0
C	Glue anti-slip covering on top of the board	B	1.8
D	Pre-assemble Front Fork-Module	None	2.5
E	Install Front wheel in Fork-Module	D	0.6
F	Insert Front Fork-Module into Board	C, E	1.3
G	Install Steering Bar-Module	F	1.5
H	Test the correct functionality	G	1.4
	Sum of total Task Times		11.6

9.4.3.1 Step 1: Precedence Diagram

The first step is to create the so-called "precedence diagram" based upon the inputs received. Table 9.3 indicates the precedence relationships, and the corresponding diagram can be created. A simplified illustration is shown in Fig. 9.9:

9.4.3.2 Step 2: Identify the Number of Followers

According to the precedence diagram, we can identify the number of the following activities. Just count the respective following tasks. This is needed to apply heuristics for the later assignment of the tasks to workstations (WS).

- Activity A has five followers.
- Activities B and D have four followers.
- Activities C and E have three followers.
- Activity F has two followers.
- Activity G has one follower.
- Activity H has 0 followers.

9.4.3.3 Step 3 Defining the Assignment Heuristic

We will apply the heuristic to start loading the assembly line, beginning with the activity with the highest number of the following activities. The precedence diagram needs to be respected. If there are activities that have the same number of followers, the one with the higher number of task times will be used.

▶ **Practical Insights** Heuristics can be applied to obtain a reasonable result, but sometimes, it might be required to review and re-do the exercise in case the heuristic might not lead to a satisfactory result. In addition to the heuristic presented here (Jacobs and Aquilano 2023), Heizer and Render (2014) suggest further heuristics, but these are not considered for our example. In real business, more sophisticated mathematical methods are used, which are supported by the software. These methods allow considering different cases with parallel machines, uncertain processing times, etc. (Dolgui and Proth 2010).

9.4.3.4 Step 4: Identify the Bottleneck Activity and Determine Maximal Production Rate

In our example, activity "D" has the longest processing time of 2.5 min. This activity is also considered as the bottleneck and determines the maximal production rate.

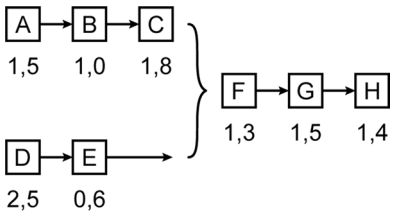


Fig. 9.9 Precedence diagram for the scooter assembly

Let us assume that we work 8 h per day in our scooter manufacturing facility. Considering 60 min per hour, we get

$$\text{available production time per day of } 8\,\text{h/day} \times 60\,\text{min/h} = 480\,\text{min/day}$$

In order to identify the maximal production rate per day, we need to divide the available production time per day by the bottleneck time. In our example, the calculation as follows

$$\text{max.production rate} = \left(480\,\text{min/day}\right) / \left(2.5\,\text{min/product}\right) = 192\,\text{products/day}$$

9.4.3.5 Step 5: Calculate the Takt Time Based on the Required Output

Let us assume that we have to balance our assembly line for a daily production rate of 150 scooters. In this case, we divide the available production time per day by the required output. We get:

$$\text{Takt time} = \left(480\,\text{min/day}\right) / \left(150\,\text{scooters/day}\right) = 3.2\,\text{min/scooter}$$

9.4.3.6 Step 6: Calculate Theoretical Minimal Number of Workstations

As we can observe from Table 9.3, the sum of the task times equals 11.6 min. Takt time is 3.2 min/scooter. That means we need to distribute the total task time over several workstations. This is exactly the task right now. In order to identify the minimum number of WS, we divide the total task time by the identified takt time:

$$\text{min number of WS} = \left(11.6\,\text{min/scooter}\right) / \left(3.2\,\text{min/scooter/WS}\right) = 3.63\,\text{WS}$$

We cannot have fragments of WS, so we should round up. For our example, at least four WSs will need to be considered.

9.4.3.7 Step 7: Loading the Workstations According to Step 3

Workstation 1

- 3.2 min – A (1.5 min) = 1.7 min
- 1.7 min – B (1.0 min) = 0.7 min (= remaining idle time)

Workstation 2

- 3.2 min – D (2.5 min) = 0.7 min
- 0.7 min – E (0.6 min) = 0.1 min (= remaining idle time)

Workstation 3

- 3.2 min – C (1.8 min) = 1.4 min
- 1.4 min – F (1.3 min) = 0.1 min (= remaining idle time)

Workstation 4

- 3.2 min – G (1.5 min) = 1.7 min
- 1.7 min H (1.4 min) = 0.3 min (= remaining idle time)

The clustering of the activities per workstation (WS) is shown in Fig. 9.10.

Figure 9.11 illustrates the clustering of the tasks per workstation well, but it does not provide a good overview of the takt time and efficiency of the assembly line. Therefore, *stacked bar charts* are used in practice to demonstrate the loading per station, including the corresponding idle time. In Fig. 9.11, the corresponding stacked bar chart is presented. In practice, this is also called *Yamazumi chart* ("*Yama*" means *mountain,* and "*zumi*" means to *pile up* in Japanese). Thus, it is a graphical interpretation of the piled-up work per station.

9.4.3.8 Step 8: Calculate the Efficiency of the Assembly Line

In the last step, the efficiency of the newly balanced assembly line will need to be determined. As the stacked bar chart (Yamazumi) shows (Fig. 9.11), there are idle

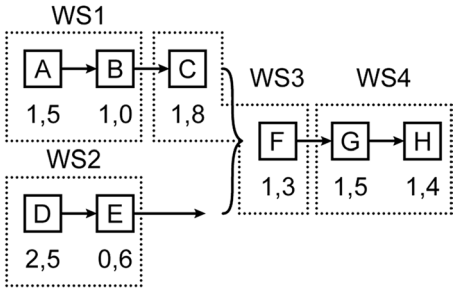


Fig. 9.10 Precedence diagram with indication of workstations

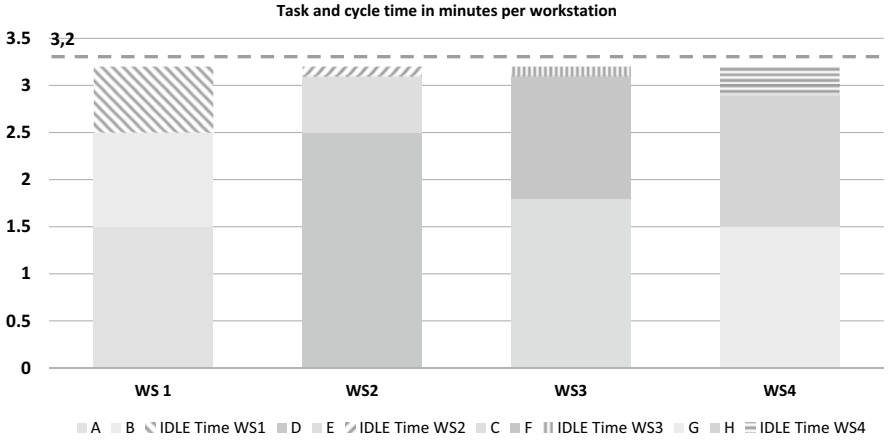


Fig. 9.11 Yamazumi Chart for the balanced assembly line

times on each WSs, resulting in inefficiency. The planners of the assembly line will try to find ways to reduce these inefficiencies in numerous iterations. If we consider the loading of our assembly line for the scooters, we can calculate the efficiency using Eq. (9.23):

$$\text{Efficiency}\left[\%\right] = \text{total task times} / \left(\text{takt time} \bullet \text{number of workstations}\right) \quad (9.23)$$

For our example, efficiency **[%]** = 11.6 min/(3.2 min × 4) = 11.6 min/12.8 min = 0.906, which means 91%.

The planners would now try to break down the individual tasks into smaller portions in order to reduce the idle times and increase efficiency.

9.4.4 Continuous Flow

Now, should demand for chairs be very high, i.e., many thousand items per day are required, one would even rethink the chair's design and materials. The question is: how can we simplify the design in order to reduce the number of parts and the *throughput time* of assembly? One could argue that a product should no longer be made out of wood but maybe from plastics. Instead of cutting pieces of wood, putting them together on the assembly line, and then painting them, one could suggest that the chair is done in one continuous outflow of plastics in various colors. Production would take place following the plastic molding process.

Instead of having multiple workstations, the process would take place in one continuous flow, a value-added process starting with filling the plastic granulate, heating it up, pressing it into the form required, cooling it, and then removing it from the mold. Human intervention would be involved for the supervision of the machines and to ensure quality, but the creation of the product would be achieved through the utilization of machines, using the highest automation. The cost per product would remain relatively low. However, on the other hand, the cost of the facility and machines would be very high because of the enormously high level of automation. Furthermore, complete customization is impossible because the molds are specially designed to create one product and thus are very expensive. This will result in very low flexibility when adjusting the product to possible changes from the customer side.

9.4.5 Product-Process Matrix

In essence, the considered product flow structures have different advantages and disadvantages. The relationship between products and processes can be summarized in a *product-process matrix* (see Fig. 9.12).

Let us get back to the initial hypothetical example of our chair. If you want to produce your chair exclusively and in small quantities, you will select the job shop; if, on the other hand, the task is to produce the chair in the highest quantities, the continuous flow would be a recommended structure. Overall, the question of

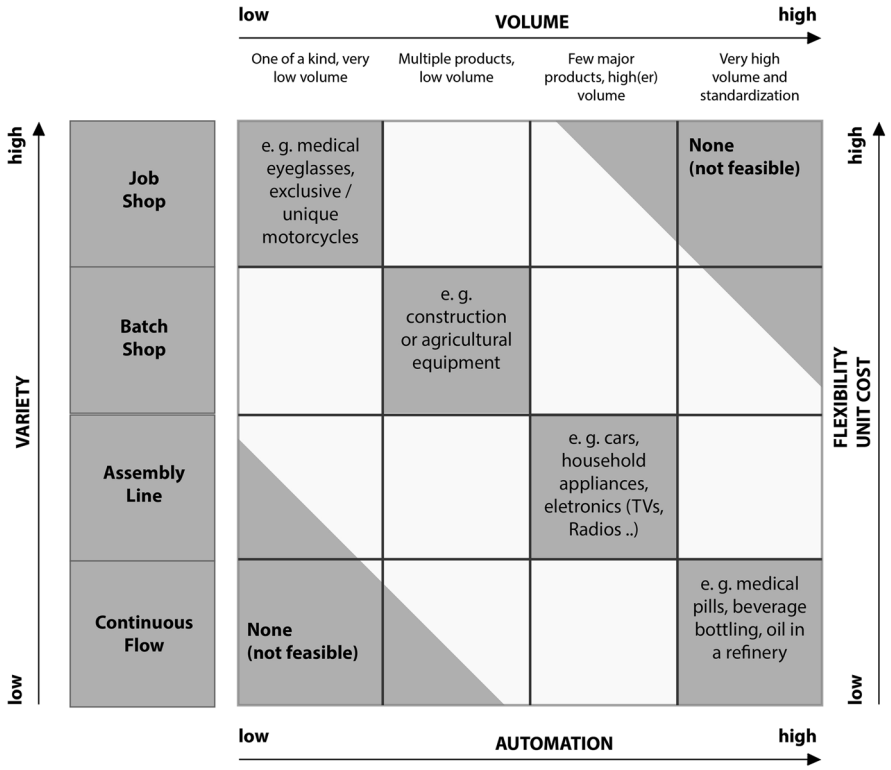


Fig. 9.12 Product-process matrix [based on Chase et al. (2006)]

volume and variety are the factors that will drive your decision toward the most appropriate process flow structure.

9.5 Lean Production Systems

9.5.1 Lean Thinking

"*Lean*" means the continuous process of increasing efficiency and improvements in the organization, which requires teamwork, respect, and taking over responsibility. Team members are also responsible for detecting defects, striving for their root cause, and identifying a solution for future avoidance. Lean also requires a good relationship with the unions accompanied by trust.

The lean organization is characterized by a set of the following five key principles (Womack and Jones 1996; Erne 2022; Bicheno and Holweg 2024):

- *Value* from the customer point of view
- *Value stream* and waste ("*muda*") elimination

- Smooth *flow* throughput in production
- *Pull*-control policies, e.g., *Kanban*
- Perfection and *continuous improvement (Kaizen)*

The underlying philosophy of all the above-mentioned principles is *respect* for people and *learning* from mistakes.

9.5.1.1 Value

Identifying the "*value*" of a product or service means understanding and specifying what the customer is expecting to receive. Understanding what the customer wants and does not want are key aspects in lean thinking. Womack and Jones (2005) described value as a capability provided for a customer at the right time at an appropriate price, as defined in each case by the customer.

9.5.1.2 Value Stream and Value Stream Mapping

The *value stream* presents the specific activities required to design, order, and provide a specific product, from concept to launch, order to delivery, raw materials into the hands of the customer. The most effective process is achieved by performing the minimal required number of value-added steps and no unnecessary steps. This is the ideal goal that is almost impossible to achieve in practice. A method to maximize the volume of the value-added steps in lean practice is called value stream mapping and analysis.

VSA (*value stream analysis*) is a tool to connect process flow and costs analysis. It is a suitable tool for redesigning value creation systems. The availability of a well-trained team is essential as a key success factor. In VSA, different process steps in the value creation process are identified, listed, sequenced, and visualized. Then the value-added times are measured. For comparison, the non-value added times are also measured.

Value stream mapping (VSM) provides an understanding of the current process by:

- Visualizing multiple process levels
- Highlighting waste and its sources
- Rendering "hidden" decision points apparent

VSM also provides a communication tool for stimulating ideas by capturing critical organizational knowledge and identifying locations for data gathering and process measurement.

9.5.1.3 Flow

Womack and Jones (2005) described *flow* as "the progressive achievement of tasks along the value stream so that a product proceeds from design to launch, order to delivery, and raw materials into the hands of the customer with no stoppages, scrap or backflows." The concept of flow is focused on the movement and value-adding processes of the product as such but also involves the material, information, and

financial flow. The flow principle can also be extended to the non-production area, which is related to information flow. The flow of information among the parties involved in a value-creation network is as important as the physical flow of the materials. According to Modig and Ählström (2016, p. 5), "flow efficiency focuses on the amount of time it takes from identifying a need to satisfying that need."

9.5.1.4 Pull

The "*pull*" principle considers a system from downstream to upstream activities in which the upstream supplier produces nothing until the downstream customer signals a need. Pull means that one should start the creation of a product or service when the customer asks for it. In other words: in a pull system, "a workstation pulls output from the preceding station as it is needed" and the "output of the final operation is pulled by customer demand or the master schedule (Stevenson 2018, p. 624)."

9.5.1.5 Seek Perfection

Seeking perfection is the element of *continuous improvement* in the lean philosophy. Perfection does not only mean quality. Perfection should be considered as being able to create exactly what the customer wants when it is needed and at an acceptable, fair price with a minimum of waste. It is important to understand that a process will not be perfect and that there is a need for continuous re-examination of the process to remain competitive and lean. It is required to repeat this optimization cycle to get closer to the optimum with every loop taken. Womack and Jones (2005) summarized perfection as "the complete elimination of *muda* so that all activities along the value stream create value."

9.5.2 Lean Production Principles

For an all-embracing view on the origin of lean, please consider "The Machine that Changed the World" by Womack et al. (2007). In 1950, Eiji Toyoda, a member of the Toyoda family who founded the Toyota Motor Company, visited Ford's Rouge factory in Detroit. From 1937 to 1950, i.e., over a period of 13 years, Toyota produced a total of 2685 cars in contrast to the 7000 automobiles that Ford produced at the Rouge site in a single day (Womack et al. 1990). Eiji Toyoda assessed in detail the mass production principles applied in Detroit and concluded that the performance of the Toyota facilities could also be improved, and he exchanged information on that with Taiichi Ohno, Toyota's leading production engineer in Nagoya. The Toyota staff initially produced looms, then military trucks and some handcrafted automobiles, but today, the company is recognized as one of the world's most efficient and highest-quality producers of motor vehicles.

9.5.2.1 Lean Production/Toyota Production System

As a result of the research at the MIT International Motor Vehicle Program (IMVP), the expression "lean production" was coined, and Holweg (2007, p. 426) points out that: "The term 'lean production' was first used by Krafcik in 1988, and

subsequently, Womack et al. of course used the term 'lean production' to contrast Toyota with the Western 'mass production' system in the 'Machine' book."

Ohno (1988) identified that the US mass production principles would not be applicable to the craft production-oriented Japanese industry. Thus, he created a new approach. In Western industries, multiple machines were used, and each one had to produce enormous amounts of the same pressed parts, as the changeover required specialists and took a lot of time.

Taiichi Ohno developed presses in which the dies were slid in (every couple of hours instead of months) via rollers and fixed by simple, quick mechanisms to use a few machines to press with multiple dies. This saved *time, cost, and space*. Also, instead of using specialist changeover teams, Taiichi Ohno suggested that the workers should be capable of and responsible for running the exchange of dies; after continuous efforts on improvement, changeover times were reduced down to just 3 min by the end of the 1950s. He also identified that it was more cost-efficient to press small lot sizes of parts instead of running press batches of large quantities, resulting in (a) a reduction in carried inventories and (b) immediate identification of defective pressed metal sheets.

The result was an increased awareness regarding quality and early identification of flaws. It also avoided large quantities of wrongly produced parts that would have needed to be repaired, reworked, or even scrapped—if they had followed the system of producing *large batches* according to the mass production philosophy. In order to further perfect this approach, no stupid exchangeable de-humanized workers would be required; instead, a highly skilled, motivated, and proactive workforce would be needed!

Toyota also suffered from the financial crisis when it was subject to post-war American occupation. Toyota intended to lay off 25% of its workforce, which resulted in turmoil leading to the strengthened rights of employees and strong unions. The compromise was that the workforce was reduced by a quarter. However, President Kiichiro Toyoda resigned, and the remaining workers received a lifetime working guarantee at Toyota accompanied by seniority-based wages and receipt of bonus payments. In other words, the *employees became part of the Toyota community* with the right to use the available facilities.

Ohno understood that workers could remain at Toyota for life because of this agreement. Toyota and the workers became "one community." So, he intended to strive for *continuous improvement* of their skills and competencies as they became part of the "human assets." This is in contrast to Ford's perception of using the workers as "human equipment" to perform one or two tasks repeatedly, supervised by a foreman, managed by industrial engineers, and so on along the hierarchy. This also contributes to the establishment of a lean culture (Mann 2014).

One striking finding for Ohno was that in Ford's approach, cars would be moved along the line, even when there were defects because these were corrected in an especially dedicated rework area before being shipped to dealers. Back in Japan, Taiichi Ohno formed teams of skilled workers led by a capable team leader. The team leader was a manager and worker at the same time and was also utilized as a "jumper" to fill in the gap of an absent team colleague. These teams were

furthermore autonomously responsible for numerous assembly processes and for a certain area in the plant, so to say, their working domain. They were also responsible for quality control, housekeeping, practical problem solving, and cleaning, as well as conducting small repairs. A key element of the teams was *continuous process optimization*—the teams were at the frontline, so they felt the pain and knew how to improve things. This also emphasizes the attitude concerning "respect for people" (Liker and Trachilis 2014, p. 21) as well as the philosophy that there should be "no culture of blaming for mistakes, but to learn from mistakes."

9.5.2.2 Kaizen

An approach of *continuous improvement* is called "*Kaizen*"in Japanese. MacDuffie et al. (1996) summarized that "under the philosophy of continuous improvement that characterized lean production, problems identified through the minimization of buffers are seen as opportunities for organizational learning." In contrast to mass production in the Western world, where mistakes or parts installed incorrectly would move throughout the entire line, and by the time mistakes had possibly been identified, many faulty cars would have been created; Taiichi Ohno did something completely different. Ohno made another simple but revolutionary invention. He attached a cord to the assembly line, and workers were encouraged to pull the cord, thus stopping the entire assembly line if a mistake was identified by a worker (Ohno 1988). Then, the workers in their community looked for a solution to the problem, which gave them a lot of responsibility to make sure they did it right the first time.

9.5.2.3 Waste Reduction

In order to convert to lean principles in other functions of the organization, such a process can also be implemented in non-production processes: the cord or *andon*[1] sign is there to highlight mistakes when they appear (Dennis 2010). Ohno also tried to assess the reason for mistakes occurring by asking the "Five Whys," which is a structured approach to get to the so-called "root cause" of problems (Ohno 1988). Initially, the assembly lines stopped very often. However, the more mature this error analysis and prevention attitude became, the more failure rates dropped significantly to almost no defects: this is a 100% yield at Toyota compared to 90% in the mass production plants. This is the reason why in today's Toyota plants, there is almost no space planned for rework, because there is almost no rework compared to today's mass production plants, which devote 20% of their space to quality corrections. Close to the waste reduction idea is also the Six Sigma concept that set up special requirements for quality management and control (Chiarini 2012).

"Waste," which are the non-value-added activities (Cachon and Terwiesch 2019, p. 152), or *muda*, can be classified in two ways.

[1] Andon is Japanese and means "lantern" and thus is a visual, such as a traffic light that highlights in the production process anything going according to plan (green = OK) or if there are problems/disturbances (red = NOK) that would require immediate intervention and mitigation.

- Type I Muda is found in activities that add no value for the customer, but are necessary, to deliver the product.
- Type II Muda is found in activities that do not create value and can be eliminated immediately, such as waiting and unnecessary transport.

Type I muda needs to be reduced to the minimum, and type II muda needs to be eliminated.

This list of *7 wastes* is one of the key elements in Lean Thinking. Womack and Jones (2005) stated that Taiichi Ohno's original enumeration of the wastes was commonly found in production. The term "*seven plus x*" wastes comes from the fact that initially, seven wastes were identified, but then it became obvious that "not meeting the requirements of the customer," the "waste of resources" or "not utilizing talent, skills or knowledge" inside the organization (Dennis 2010) was also recognized as waste, hence "7 + x wastes:"

- *Overproduction* ahead of demand
- *Waiting* for the next processing step
- *Unnecessary transport* of materials (for example, between villages, suppliers, facilities, etc.)
- *Over-processing* of parts due to, e.g., poor tool and/or product design
- *Inventory* that exceeds what is required
- *Unnecessary motion and movement* by the employees during the performance of their work (for example, looking for parts, tools, and documentation)
- Production of *defective parts plus*
- *Not meeting customer requirements*
- *Wasting resources*
- *Not utilizing skills, talent, or knowledge*

Toyota's case clearly demonstrates how aspects of the process flow structures, layout and factory planning, and the evolution from craft to lean production are connected. Furthermore, it has shown that the principles of continuous improvement and thinking in terms of quality are all closely connected. It has also shown its initial limitations, for example, the need for a total system view regarding the lean enterprise and the intensive need for cooperation with the SC partners involved (no matter if they are internal or external).

To summarize the principles of lean production, it can be stated that *lean* strives for combining the advantages of both craft and mass production, trying to avoid the high unit cost of craftsmanship while avoiding the rigid, boring, and inflexible processes of mass production. This has led to the ability to reduce costs per unit and dramatically improved quality while providing a wider range of products to the customer and more challenging work for the staff. Regarding the composition of workers' teams, lean production is striving for the creation of flexible teams consisting of multi-skilled staff that operates highly flexible machines to produce volumes of products in high variety. These principles are applied to the tiers of suppliers, but they also look at the integration of dealers and buyers into the lean concept.

To summarize, the *Toyota Production System* covers, in principle the following key areas (see also Heizer and Render 2014, p. 661):

1. *Pull-System*: Parts are only produced, when they are needed for further production. This is contrary to a Push-System, in which material is fed into the production area or warehouse. The purpose of the pull principle is, e.g., to increase efficiency and to avoid overproduction, unnecessary transportation, and too high levels of inventory.
2. *Just in Time (JIT)*: Following the Pull-System, parts and modules are delivered to the station in the right quantity and at the right time (thus just in time), when they are needed. It is to avoid that the facility is flooded with unnecessary material.
3. *Kanban*: Kanban means a "sign" or a "card." A Kanban signal is used to indicate that a small batch of components needs to be produced. It is a practical approach based on real material consumption that initiates the replenishment process (unlike as material requirements planning (MRP) that presumes the planned replenishments).
4. *Standard Work Practices*: These are agreed upon documented procedures for the production process. This is to secure that everyone follows the same steps and sequence in the value creation. Thus, the timing per operation will be met, and the quality to complete the work will be the same for all similar products.
5. *Minimal Machines*: Proprietary machines are designed for specific Toyota applications. However, it might also be a good idea to use general multi-purpose machines with certain adapters to simplify usage and ease maintenance.
6. *Respect for People*: The employees are treated and respected as capable workers with knowledge and empowered to use their capabilities.
7. *Empowered Employees*: The employees have responsibility. For example, they can stop production in urgent cases, might evaluate improvement ideas, and are also actively involved in quality control circles.
8. *Kaizen Area*: This is an area, where the employees are involved in order to make suggestions. Furthermore, the ideas or suggestions are tested, discussed, and evaluated.
9. *Jidoka*: Machines are equipped with built-in devices for monitoring performance and making "judgements." If a problem occurs, the machine will interrupt the process. That is why Jidoka is also considered a machine or automation with a "human touch." This avoids processing defective components and allows for analyzing the root causes of the defects.
10. *Andon System*: Initially, in case of a problem, a cord was pulled or a light flashed at a station. Nowadays, problems are also openly displayed on a board or a screen to highlight deviations from the standard. It is part of the idea to visualize problems.
11. *Level Schedules*: Models on the assembly line are mixed in order to meet customer demands and to avoid huge variation in the loading of the assembly line. The work needs to be balanced or leveled, focusing on the processing of smaller batches.

12. *Assembly Components*: Parts or modules may be placed inside the car for easier access instead of placing the parts on shelves next to the assembly line. Bigger modules are pre-assembled close to the main assembly line and are then fed into the main assembly line ("fishbone line feeding").

9.5.3 Lean Supply Chain

9.5.3.1 Lean Thinking in the Supply Chain

Anker (2013) pointed out that a car can comprise 25,000 single parts and the related material cost represents approximately 70% of the total production cost. In relation to the price the customer pays, the material cost accounts are in a range of 50%. The level of supplies provided by external parties is at approximately 75%, which means the in-house production is roughly 25% (Anker 2013). In the case of the Smart production facility in Hambach (France), the external contribution is close to 90%, which shows that in that case, the production depth is mainly with the suppliers, and the original equipment manufacturer (OEM) accounts for approximately 10%.

If we compare this with the examples given earlier, Ford initially strived for 100% at the Rouge Complex but reduced it back to 50% after World War II. For Taiichi Ohno, the question was how suppliers and producers could efficiently work together to jointly reduce cost and increase quality.

The principle process from design to production in Ford's or GM's approach was that their own engineers designed the parts (previously purchased externally) completely on their own. Then, the blueprints were sent to potential part suppliers who were supposed to submit their tenders. Following that, the supplier was selected in principle on price, but also considering the quality and delivery reliability—with the side effect that suppliers were often switched, when others made better offers. Also, one can see the change in mindset regarding the criteria for selecting suppliers.

Lean SCM recognizes the need for strong cooperation with suppliers and their integration into the value chain. The potential of continuous improvement and their contribution regarding best practices remained unutilized in the mass production approach. Also, suppliers followed the principle of producing large batch sizes that were shipped to the car producers. If a defect was discovered, many faulty parts had already been installed in cars, and at the same time, a high number of defective parts (inventory) were produced that required rework or scrapping.

Lean SCM identifies the potentials of the suppliers that were—comparable to the knowledge capabilities of the shop floor workers—not utilized. Therefore lean SCM encourages suppliers to contribute their own expertise, meaning that Toyota specified the technical requirements that the part needed to fulfill, but the detailed design was developed by the suppliers (who were able to optimize the components they offered according to their own manufacturing).

Then, Ohno classified suppliers into multiple tiers. The *tier 1 suppliers* were the ones who autonomously developed parts based on key requirements (e.g., deceleration parameters, diameters, and measures for brake systems). The suppliers created prototypes that became serial components when they successfully passed

functionality tests. The tier 1 suppliers were, in principle, not in competition with each other. Thus, they openly shared their information at the supplier level. Taiichi Ohno also requested that the tier 1 suppliers should, on their own, establish a second line of suppliers for themselves according to a similar principle. These suppliers were, in principle, manufacturing specialists but not engineering experts.

These *second-tier* suppliers were clustered and organized in non-competitor supplier associations, which also contributed to openly sharing information among suppliers. As a side effect, "Toyota spun its in-house supply operations off into quasi-independent first-tier supplier companies in which Toyota retained a fraction of the equity and developed similar relationships with other suppliers who had been completely independent" (Womack et al. 2007, p. 60). Toyota also supported its suppliers with financial loans and workforce and encouraged the supplier base to offer their products to other companies and industries.

Lean SCM attempts to eliminate waste, e.g., by continuously striving for better inventory control and smooth processes. Furthermore, the processing steps in a lean SC should be assessed regarding non-value-added steps. This might cover the planning of production processes, as well as an analysis of warehousing locations, stock levels, replenishment orders and material buffering, lot sizes, shipment processes, modes and frequency of transportation, and the management of returns. It is obvious that the complexity is increasing and that the overview might get fragmented (see also Chap. 16) because multiple enterprises are involved in the supply network. This is a reason why SCs are supported by the usage of supplier collaboration portals or so-called SC towers to obtain a holistic view and end-to-end visibility in order to increase transparency via information technology. The increasing usage of technology converts traditional SCs into digital SCs (Tsipoulanidis 2017a, b; Ivanov et al. 2019, 2021; Dolgui et al. 2020; Fragapane et al. 2022; Ivanov and Dolgui 2021; Sokolov et al. 2020). While we refer to the lean principles in SCM, the attributes described in Chap. 4 regarding resilient or viable SCs need to be strongly considered (Ivanov 2022; Ivanov and Dolgui 2020; Ruel et al. 2024).

9.5.3.2 Kanban

Lean thinking in the SC has been frequently considered in light of SC risk management. It has been recognized that typical lean techniques such as single sourcing or low inventory make the SC fragile in the case of disruptions at factories and suppliers due to human-made and natural catastrophes (see Chap. 4, and Ivanov et al. 2014; Dolgui et al. 2018; Li et al. 2021; Dolgui and Ivanov 2021).

The next idea of lean SCM is simplifying logistical processes and material call-offs. In principle, a connected chain, like one in a huge factory, has to be created, in which the previous production process feeds material in limited quantities to the following process step after the material has been called off.

The idea was to employ *circulating containers or boxes*. Each container or box was equipped with a *card* that signals the need for another container or material. This card was called "*kanban.*" The containers and boxes were filled with parts and then sent to the next production step and—when completely empty—returned to their origin and thus acted as a call-off sign. The advantage was that inventory was

reduced throughout this SC. In an unfavorable situation where one container carried defective parts, the entire system would stop as soon as the defective part was detected, since the circulation of the boxes would also be interrupted.

In essence, the following five elements are crucial for a lean SC.

- *Perfect first-time quality* is achieved through the objective of creating zero defects, revealing and solving problems at their ultimate source (e.g., the shop floor), achieving higher quality and productivity simultaneously through teamwork or worker empowerment.
- *Waste minimization* is achieved by removing all non-value-added activities and by, making the most efficient use of scarce resources (e.g., capital, people, and space), and applying JIT deliveries as well as eliminating any safety buffers for inventory reduction.
- *Continuous improvement* (reducing costs, improving quality, and increasing productivity) is achieved through a dynamic process of change, simultaneous and integrated product/process development, rapid takt time and time-to-market, openness, and information sharing.
- *Flexibility* is achieved through producing different mixes or greater diversity of products quickly, without sacrificing efficiency at lower volumes of production, through rapid setup and manufacturing in small lot sizes.
- *Long-term relationships* between suppliers and primary producers (assemblers, system integrators) are established through collaborative risk sharing, cost sharing, and information sharing arrangements.

9.6 SCOM Excellence and Digitalization

9.6.1 Operational Excellence

The term "excellence" is widely used in business. This section focuses on SCOM excellence. Companies are striving to achieve excellence in SCOM, but they increasingly face severe pressure from their business environments. Outside the enterprises, numerous changes occur in the markets and with competition. In addition, there are various types of risks, such as strikes, natural disasters, political instability, or terrorist attacks that significantly influence SC resilience. An additional factor is fast-changing customer needs and expectations: it is a challenge to manage and adapt continuously changing SCs, and, as a result, also necessary to redesign those SCs and operations.

Operational Excellence is often mentioned in close connection with Lean Thinking or Six Sigma. Martin (2008, p. 605) defined Operational Excellence as "an umbrella initiative including Lean, Six Sigma, and Total Productive Maintenance (TPM)," which focuses on the systematic usage of these philosophies and instruments to provide customer value. The significant difference of operational excellence in addition to the classical Lean and Six Sigma philosophies are the mindset, culture, and organizational behavior for improving performance with the objective

of securing financial growth. In this sense, Operational Excellence is an element that secures a company's growth strategy and, thus, an incremental part of Strategic Management. Operational Excellence is a dynamic and continuous ability that opens the door to competitive leadership and high profits in continually changing business environments.

9.6.2 From Operational Excellence to SCOM Excellence

Operational Excellence means searching for opportunities to accelerate value-adding processes, to eliminate wasted efforts, or to identify opportunities for quality improvements. SCOM Excellence is:

- A change enabler for dramatic performance improvements and financial growth in line with the strategic direction.
- A philosophy of providing superior value for the customer by designing attractive products and services, as well as highly efficient value-adding processes throughout global SCs.
- A cultural mindset that exists across an agile organization.
- A systematic approach including Lean, Six Sigma, and Total Productive Maintenance (TPM) focusing on the continuous pursuit of doing things better.

9.6.3 Digitalization as New Driver in SCOM Excellence

Case Study: Smart Logistics—How Artificial Intelligence helps Amazon to deliver faster and more reliable

Amazon has been a leading player in the e-commerce industry for a long time. When Amazon started "Prime" shipping, it was a significant challenge for competitors to offer that same level of responsiveness. Later, Amazon started offering 1–2 h delivery in some areas for daily essentials and groceries. Using artificial intelligence (AI), Amazon anticipates the products that customers are likely to buy and ships these items to the nearest warehouses well in advance of getting orders. In fulfillment centers, autonomous driving units are employed to bring inventory, whether to the storage area after sorting the received products or to the workstations for picking and packaging to satisfy customer orders. Scanning barcodes of products while receiving or sending them helps to identify the inventory levels and provides a smooth flow of products in the supply chain. Optimization of the delivery activities also relies on AI to calculate the required number of vehicles, their types, capacity, and routes. Amazon also applies machine learning algorithms to provide a personalized shopping environment on its website by using recommendation engines and to forecast demand for specific items by looking at the transaction databases (Amazon 2020).

Anticipatory Shipping and Product Recommendations—The "First Mile"

Amazon provides recommendations for its customers to offer a personalized purchase experience. With these recommendations, customers are made aware of the products that are suitable for their needs. For two decades now, Amazon.com has been building a store tailored to every customer's unique interests. To deliver in short time slots, Amazon needs accurate forecasting to predict product demand in a given market and to locate these products as close to customers as possible. This comes with high warehousing, handling, and transportation costs. Paired with predictive analytics tools and a massive amount of Amazon customer data, the anticipatory shipping process helps to ensure product deliveries in a much shorter timeframe than more traditional methods.

Warehousing Activities—The "Middle Mile"

Inside Amazon's warehouses, AI powers the Kiva Robots, which take the goods to the picker. The fulfillment centers are heavily involved with high-tech cyber-physical systems, which are manufactured by a subsidiary of Amazon. Thanks to the robots carrying shelves to the pickers, employees do not need to walk long distances while carrying products. AI systems support the warehouse management software to ensure that the orders are processed, and the required products are moved to the workstations where pickers can select and package them for delivery.

Delivery of the Products—The "Last Mile"

Amazon has many logistics contracts for deliveries, e.g., with DHL, FedEx, UPS, Hermes, Uber, and other small companies. The process counts on AI to continually calculate how many drivers are needed at any given time. The app considers the weight and number of packages headed to the same neighborhoods and whether packages can fit in one of the available cars. The app, called "Flex," even recommends the order that boxes should be packed into a vehicle to guarantee the most efficient delivery.

When SCM was being introduced to the management practices, it was popular to say that "the company is as good as the SC behind it." Christopher formulated the proposition that the competition is not between firms, but rather between SCs. Today and in the near future, specialists say that "the SC is as good as the digital technology behind it." Moreover, a new proposition can be formulated: competition is not between SCs, but rather between the information services and analytics algorithms behind them.

► **Practical Insights** With the help of smart sensors and plug-and-produce cyber-physical systems, the stations in the assembly system are capable of changing operation processing and setup sequences according to the actual order of incoming flows and capacity utilization (Theorin et al. 2017). In the front-opening, unified pods technology in the semiconductor industry, robots are used in real-time operation sequencing. Collaborative robots read the information about the products from sensors and tags and decide flexibly where to forward a

wafer batch next (Mönch et al. 2012). Amazon recently applied for a patent for a self-learning robot, including an automatic packaging system. This robot will be able to pick up items ordered and package them appropriately, while secure data provides the opportunity to save the ideal packaging strategy of a product. Blockchain applications to SCs are becoming more and more important. The central idea is to increase visibility and efficiency based on record-keeping in the SC. New cloud-based analytics platforms such as SupplyOn Industry 4.0 Sensor Clouds make it possible to control the SC in real-time and plan and adjust processes using up-to-date information. By simply clicking on a container type, the graphs indicate whether there has been a violation of the defined temperature or humidity limits along the time axis. The data analysis in this chart allows quick identification of all orders where lead time was exceeded, allowing for quick identification of questionable transports. Resilience360 at DHL allows comprehensive disruption risk management by mapping the SC end to end, building risk profiles, and identifying critical hotspots to initiate mitigation activities and alert in near-real-time mode on incidents that could disrupt the SC.

We are living in times characterized by market dynamics as well as technological and organizational changes. Competition is growing rapidly on a global scale: customers expect individualized products in ever shorter frequencies. Innovations are disrupting the current players and business models in all industries and services, but what does it mean when we use the expression "digital revolution" or "digital transformation"? Digitalization is a game changer. It will lead to significant changes in business and society in terms of work conditions and consumption of goods as well as cooperation and communication in value chains.

To create a starting point for the development of a digital SCOM strategy, a guiding framework ("CORRIDOR Model") has been suggested by Kiehne and Tsipoulanidis (2016). The framework was successfully used to assess many industries, such as food, retail, automotive, fashion, aircraft producers, general merchandise, and electronics. It was further developed under the consideration of lean thinking (Tsipoulanidis 2017b). The CORRIDOR Model therefore can serve as a structured approach for practitioners to develop an aligned SC strategy that focuses on SCOM Excellence using digital technologies. The main components of the CORRIDOR checklist are as follows:

- Complexity is growing: the increasing number of products, variants, markets, information, and communication channels require a holistic view. The broader / larger an SC, the higher the complexity. → How do you manage or orchestrate SC complexity?
- Overview (overall transparency) is fragmented: the SC and all relevant processes of SC partners cannot be determined. Furthermore, not all influencing factors are known. → Where do we lack transparency, and how can we gain SC visibility?

- Risk and Vulnerability: Some external factors can easily be *predicted*, and some others cannot. Some have a significant impact, and others have a rather low *impact*. → What are SC risks, and where are we vulnerable? How can we proactively reduce SC risk and develop mitigation strategies for SC resilience?
- Responsibility concerning the environmental and ecological impact and adherence to the code of ethics is key for all involved players. → What should be done to focus on sustainability and CSR?
- Information Flow/Information (Data) Availability: Supply flow and IT are closely connected, which is relevant for data analysis on the part of suppliers, producers, and logistics service providers to meet customers' needs. → What are the sources of SC data? Is it structured and usable, or what should we do to convert the vast amount of data into usable information?
- Dynamics: Digital Transformation and Technology leads to the involvement of the customer/customer focus - fast-changing customer needs. Changing product requirements, changing components, new suppliers, new locations, new technology, new communication and distribution channels require SC adaptability →How do we manage the digital transformation and implement digital technologies, and where it provides the most benefits?
- Organisational structure: roles and responsibilities per function and involved party need to be clear, and talents should be developed according to the changes in the business environment. → How do we ensure that everyone, internally and externally, clearly knows their roles and responsibilities? How to focus on human-centricity?
- Re-Design: of the SC is frequently required due to the reasons above → How do we configure our SC to be well prepared for the future?

With the help of the CORRIDOR, a checklist for an SC assessment and recurring turbulence drivers can be developed. SCOR analyses the primary value-adding processes for improvement areas. Lean reveals efficiency potentials in value creation. A combination of the CORRIDOR, SCOR, and lean allows for a holistic approach, i.e., analyzing the SC bottlenecks and taking specific improvement measures or checking where digital solutions could help generate competitive advantage. The CORRIDOR Model is a good starting point for checking the SC regarding transformation drivers, identifying wastes, and thus drafting a digital SC strategy with a focus on excellence.

9.7 Key Points and Discussion Questions

The decision to build a new or modify an existing factory is strategically important and impacts competitive advantage. Factory planning also considers a successful partnership with the involved parties in the SC and aims at efficient operations management. It is recommended to consider the following questions in factory planning activities to elaborate a successful approach to placing a well-operating facility with the appropriate process flow structures and capacity at the right location.

How to Plan a Factory in a Structured Way?

Factory planning is, in general, related to significant investments (e.g., the four billion euros investment for the BMW Leipzig location). That is the reason why the planning needs to take place in a structured and systematic way. A wrong approach can result in an enormous loss of capital. Thus, a logical connection between the initial planning phase of the facility, the realization of the new or adjusted factory, and the later operation (running the factory) needs to be established. Between these different steps, careful milestone checks need to be made in order to obtain approval by the decision-makers to move into the next phase.

Which Different Options Could Be Considered?

In factory planning, it is not always necessary to create a completely new facility. It is part of the general, regular tasks of the planners to redesign existing facilities in order to ensure that new models can be produced at existing sites. Furthermore, the existing plants can be expanded because of increasing demands. They can also be reduced (or even closed) when the markets show decreasing demands. Another option is the complete new creation of a factory on the "green field" to be close to the new market, or local temporary production partnerships might also be an option to meet possible local content requirements.

How to Support the Factory Capacity Planning Process?

In order to ensure the appropriate output of the facility, the demand needs to be known or estimated in a substantiated way so that the (design and effective) capacity can be determined. Useful tools that help to identify the capacity aspects are, for example, related to the application of Little's Law, to run a bottleneck analysis (theory of constraints), or to apply the principles of a systematic break-even-analysis, decision trees, and waiting line theory.

What Are the Generic Process Flow Structures That Need to Be Considered?

When we talk about process flow design, the immediate relation to the volume of demand and the variety of the goods that will be produced needs to be made. When products with very low volume and a significant variety have to be produced, the principle of the "job shop"will be the right decision. If products need to be produced in small lot sizes, for example, 10 exactly the same fire trucks for the regional fire brigade, these ten products will most probably be produced following the "batch shop"approach, as these items are periodically produced in small batches or lots. Assuming that, for example, 700 cars or 2000 microwave ovens will need to be created daily, the "assembly line"is a suitable process flow structure as it allows mass production of standard goods following a regular pace while moving the item from work station to work station. Finally, for the production of very large volumes of standard products such as medical pills, bottles of mineral water, or sugar, the production following the concept of a "continuous flow"will be the fitting strategy.

How Do We Identify Non-Value-Adding Activities and What to Focus On?

One of the fundamental principles in running the operations is the application of "lean thinking." According to the idea of "lean," a clear statement of what the customer expects and what is considered as "value" is needed. Therefore, the value stream has to be mapped, and value-adding and non-value-adding processes have to be analyzed so that unnecessary efforts or activities, such as waiting,

overproduction, and defects can be identified. This results in the identification and elimination of the so-called 7 wastes. Recently, wasted talent or knowledge has been added to the 7 + *x* wastes. Another aspect of "lean" is the smooth flow of the value-adding processes without obstacles or buffers according to the so-called "pull" principle. Lean thinking also emphasizes to seek perfection by continuous improvement. A fundamental lean consideration is respect for people.

What Will Be Necessary for the Future?

Besides the focus on operational factors such as the design or the application of lean principles, the rapid development of intelligence and internet technology should be mentioned. For example, Industry 4.0 is related to the fourth industrial revolution following firstly mechanization, secondly electrification, and thirdly, the use of information technology in the production domain (see Chap. 16). Using the *Internet of Things*, machines and materials can exchange information in so-called cyber-physical-systems. It is expected to gain additional efficiency potentials of approximately 30% due to the intelligent combination of machines, materials, and human beings (Lemmer 2014).

Discussion Points

- Discuss in groups which external factors might influence companies to search for new manufacturing or service sites.
- Elaborate on the advantages and disadvantages and which types of risks and opportunities you can see when firms regularly adjust their industrial footprint.
- Analyze the impact of lean thinking on SC risk and vulnerability.
- What are the new requirements for human beings in the production environment considering the development towards Industry 4.0?
- Will new additive manufacturing technologies, such as 3D printing, in combination with new lightweight materials, such as carbon composites, impact process design, and flow structures in the future?

Bibliography

Amazon (2020). https://blog.aboutamazon.com/amazon-ai. Accessed 10 July 2020

Anker S (2013) Wie viel Luft ist noch im Autopreis?. VDI Nachrichten, 01 Jun 2013, p 4

Belekoukias I, Garza-Reyes JA, Kumar V (2014) The impact of lean methods and tools on the operational performance of manufacturing organisations. Int J Prod Res 52(18):5346–5366

Bicheno J, Holweg M (2024) The lean toolbox: a sourcebook for process improvement, 6th edn. Picsie Books, Buckingham

BMBF – Federal Ministry of Education and Research (2014) Project of the future: industry 4.0. http://www.bmbf.de/en/19955.php?hilite=industry+4.0. Accessed 16 Jun 2014

Bozarth CC, Handfield RB (2019) Introduction to operations and supply chain management, 5th edn. Pearson, Harlow

Cachon G, Terwiesch C (2019) Matching supply with demand – an introduction to operations management, 4th edn. McGraw-Hill, New York

Cañas H, Mula J, Campuzano-Bolarín F, Poler R (2022) A conceptual framework for smart production planning and control in Industry 4.0. Comput Ind Eng 173:108659

Chase RB, Jacobs FR, Aquilano NJ (2006) Operations management for competitive advantage, 11th edn. McGraw-Hill, Boston
Cheong MLF, Laik MN (2024) Capacity planning. In: Data and decision analytics for business operations. Springer, Cham. https://doi.org/10.1007/978-3-031-72255-4_5
Chiarini A (2012) From total quality control to lean six sigma: evolution of the most important management systems for the excellence. Springer, Milan
Dennis P (2010) The remedy: bringing lean thinking out of the factory to transform the entire organization. Wiley, Hoboken, NJ
Dolgui A, Proth J-M (2010) Supply chain engineering: useful methods and techniques. Springer, Berlin
Dolgui A, Ivanov D, Potryasaev S, Sokolov B, Ivanova M, Werner F (2020) Blockchain-oriented dynamic modelling of smart contract design and execution control in the supply chain. Int J Prod Res 58(7):2184–2199
Dolgui A, Ivanov D (2021) Ripple effect and supply chain disruption management: new trends and research directions. Int J Prod Res 59(1):102–109
Dolgui A, Ivanov D, Sokolov B (2018) Ripple effect in the supply chain: an analysis and recent literature. Int J Prod Res 56(1-2):414–430
Erne R (2022) How to apply lean thinking in project management. Springer, Wiesbaden
Fragapane G, Ivanov D, Peron M, Sgarbossa F, Strandhagen JO (2022) Increasing flexibility and productivity in Industry 4.0 production networks with autonomous mobile robots and smart intralogistics. Ann Oper Res 308:125–143
Goldratt EM (1990) What is this thing called theory of constraints and how should it be implemented? North River Press, Croton-on-Hudson
Grundig CG (2021) Fabrikplanung: Planungssystematik – Methoden – Anwendungen, 7th edn. Carl Hanser Verlag, München
Heizer J, Render B (2014) Principles of operations management: sustainability and supply chain management, 9th edn. Pearson, Boston
Holweg M (2007) The genealogy of lean production. J Oper Manag 25(2):420–437
Ivanov D, Dolgui A (2021) A digital supply chain twin for managing the disruptions risks and resilience in the era of Industry 4.0. Prod Plan Cont 32(9):775–788
Ivanov D, Tang CS, Dolgui A, Battini D, Das A (2021) Researchers' perspectives on Industry 4.0: multi-disciplinary analysis and opportunities for operations management. Int J Prod Res 59(7):2055–2078
Ivanov D, Dolgui A, Sokolov B (2019) The impact of digital technology and Industry 4.0 on the ripple effect and supply chain risk analytics. Int J Prod Res 57(3):829–846
Ivanov D (2022) Viable supply chain model: integrating agility, resilience and sustainability perspectives – lessons from and thinking beyond the COVID-19 pandemic. Ann Oper Res 319:1411–1431
Ivanov D, Dolgui A (2020) Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability. A position paper motivated by COVID-19 out-break. Int J Prod Res 58(10):2904–2915
Ivanov D, Sokolov B, Dolgui A (2014) The ripple effect in supply chains: trade-off 'efficiency-flexibility-resilience' in disruption management. Int J Prod Res 52(7):2154–2172
Jacobs FR, Aquilano RB (2023) Operations and supply chain management, 17th edn. McGraw-Hill, Berkshire
Jasti NVK, Kodali R (2015) Lean production: literature review and trends. Int J Prod Res 53(3):867–885
Kiehne B, Tsipoulanidis A (2016) Berufskompetenzen entwickeln durch Perspektivübernahme im Supply Chain und Operations Management. DUZ medienhaus
Knotte D, Senger E, Tsipoulanidis A (2020) Operational Excellence in Ihrem Purchase-to-Pay Prozess – Ein Guide zur Realisierung von Potenzialen im operativen Einkauf. Capgemini Invent
Krajewski LJ, Ritzman LP, Malhotra MK (2021) Operations management – processes and supply chains, 13th edn. Pearson, Harlow
Lemmer R (2014) Inselhopping in der Werkshalle. WirtschaftsWoche, 13 Oct 2014, 94–96

Li Y, Chen K, Collignon S, Ivanov D (2021) Ripple effect in the supply chain network: forward and backward disruption propagation, network health and firm vulnerability. Eur J Oper Res 291(3):1117–1131

Liker JK, Trachilis G (2014) Lean Leader auf allen Ebenen entwickeln. Lean Leadership Institute Publications

Little JDC, Graves SC (2008) Little's law. In: Chhajed D, Lowe TJ (eds) Building intuition: insights from basic operations management models and principles. Springer, New York, pp 81–100

MacDuffie JP, Sethuraman K, Fisher ML (1996) Product variety and manufacturing performance: evidence from the international automotive assembly plant survey. Manag Sci 42(3):350–369

Mann D (2014) Creating a lean culture: tools to sustain lean conversions, 3rd edn. Productivity Press, New York

Marodin GA, Saurin TA (2013) Implementing lean production systems: research areas and opportunities for future studies. Int J Prod Res 51(22):6663–6680

Martin JW (2008) Operational excellence – using Lean Six Sigma to translate customer value through global supply chains. Auerbach Publications, New York

Modig N, Ählström P (2016) This is Lean – Resolving the efficiency paradox. Rheologica Publishing, Stockholm

Mönch L, Fowler JW, Mason S (2012) Production planning and control for semiconductor wafer fabrication facilities: modeling, analysis, and systems. Springer, New York

Ohno T (1988) Toyota production system: beyond large-scale production. Productivity Press, New York

Ruel S, El Baz J, Ivanov D, Das A (2024) Supply Chain Viability: Conceptualization, Measurement, and Nomological Validation. Ann Oper Res 335:1107–1136

Schenk M, Wirth S, Müller E (2014) Fabrikplanung und Fabrikbetrieb* Methoden für die wandlungsfähige, vernetzte und ressourceneffiziente Fabrik, 2nd edn. Springer, Berlin

Slack N, Chambers S, Johnston R (2010) Operations management, 6th edn. Pearson, Harlow

Sokolov B, Ivanov D, Dolgui A (eds) (2020) Scheduling in Industry 4.0 and cloud manufacturing. Springer, New York. 978-3-030-43176-1

Stevenson WJ (2018) Operations management, 13th edn. McGraw-Hill, New York

Theorin A, Bengtsson K, Provost J, Lieder M, Johnsson C, Lundholm T, Lennartson B (2017) An event-driven manufacturing information system architecture for Industry 4.0. Int J Prod Res 55(5):1297–1311

Tsipoulanidis A (2015) Von lean thinking zu Supply Chain & Operational Excellence durch die digitale Transformation. Der Tagesspiegel 2(11):2015

Tsipoulanidis A (2017a) Leandustry 4.0 – the efficient way from company to customer at times of the digital transformation – how lean thinking is leading to smart supply chain management and operational excellence. In: ICCMI conference 2017, Thessaloniki

Tsipoulanidis A (2017b) With lean thinking and industry 4.0 to operational excellence. Supply Chain Manag III/2017:13–19

Tsipoulanidis A (2020) Digital & lean, così la supply chain diventa più efficiente. Logistica Management 308:86–92

Wiendahl H-P, Nofen D, Klußmann JH, Breitenbach F (eds) (2005) Planung modularer Fabriken – Vorgehen und Beispiele aus der Praxis. Carl Hanser Verlag, München

Wilson JM (2014) Henry Ford vs. assembly line balancing. Int J Prod Res 52(3):757–765

Womack JP, Jones DT (1996) Lean thinking – banish waste and create wealth in your corporation. Touchstone Books, London

Womack JP, Jones DT (2005) Lean solutions: how companies and customer can create wealth together. Simon & Schuster, New York

Womack JP, Jones DT, Roos D (1990) The machine that changed the world – the story of lean production. HarperCollins, New York

Womack JP, Jones DT, Roos D (2007) The machine that changed the world: the story of lean production – Toyota's secret weapon in the global car wars that is revolutionizing world industry. Simon & Schuster, New York

References for Sect. 9.1

Bresnahan T, Gambardella A (2004) Building high-tech clusters. Cambridge University Press, Cambridge

Dickerman L, Harrison J (2010) A new car, a new grid. IEEE Power Energy Mag 10:55–61

Eberhard M, Tarpenning M (2006) The 21st century electric car tesla motors. http://idc-online.com/technical_references/pdfs/electrical_engineering/Tesla_Motors.pdf. Accessed 4 Aug 2014

Husmann N (2014) Mitten im Strom. http://www.focus.de/magazin/archiv/fotoreportage-mitten-im-strom_id_3969468.html. Accessed 4 Aug 2014

Markoff J (2012) Skilled work, without the worker. The New York Times:18

Layout Planning

10

Learning Objectives for This Chapter

- Role of layout planning in SCOM
- Layout planning in manufacturing
- Layout planning in warehouses
- Methods of layout planning.

10.1 Introductory Case Study: OTLG Ludwigsfelde

Volkswagen Original Teile Logistik GmbH & Co. KG (OTLG) is a service partner for the spare parts of Volkswagen, Volkswagen Nutzfahrzeuge, Audi, Seat, and Škoda. OTLG is part of the SC's wholesale level and is responsible for sales and marketing in Germany and Denmark. At present, OTLG operates in seven distribution centers in Germany. After locating a new distribution center, the layout planning has been done as follows (Fig. 10.1).

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_10) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_10.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_10

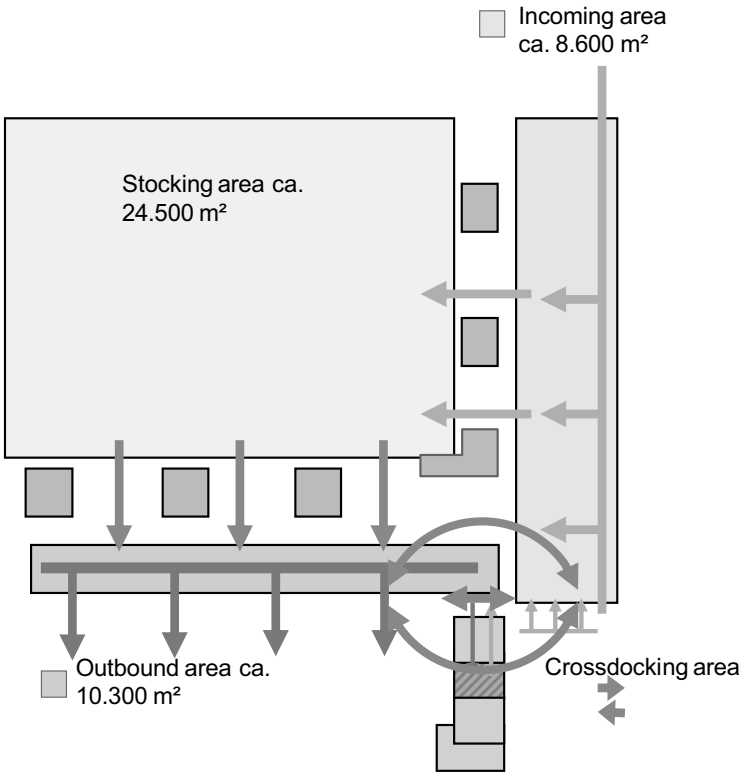


Fig. 10.1 Layout planning at OTLG in Ludwigsfelde

The warehouse layout includes an incoming area for receiving and inspecting goods. If an order for the supplied item already exists, the item is moved via the cross-docking area directly to the outbound area. Otherwise, the items are first stocked, put away, and picked up after an order is received. Such a layout ensures the shortest transportation routes and the efficient placement, put-away, and pick-up of items with the help of sophisticated software and automated material handling technologies.

On the basis of this case study, describe the fundamental processes and areas in the warehouse logistics layout.

10.2 Layout Planning in Manufacturing

In Chap. 9, the process flow structures were presented, and it became evident that the underlying parameters determining which process flow structures should be used are volume and variety. The process flow structures do not, as such, indicate what type of manufacturing layout is best, but layout types are closely linked to the processes applied to create a product.

A layout is the physical arrangement of objects in space. Fundamental objectives related to this are as follows:

- To limit travel distances between machines or departments
- To allocate departments or functional areas/main production areas in a reasonable pattern
- To avoid crossing flows (material, transportation devices, human beings, etc.)
- To prepare for later adaptation or expansion

In general, there are four basic types of layouts in manufacturing, and these are:

- Fixed position layout
- Process flow layout
- Product flow layout
- Cell-based layout

10.2.1 Fixed Position Layout

We start with the *fixed position layout*. As the name indicates, in this type of layout, the product or object remains in a fixed position while the value is being added. Imagine construction work in a city, e.g., in a subway station: all workers, tools, machines, and parts are brought and placed at the location where the work is performed. The same is also valid for a movie set, where the technicians, actors, camera teams, and directors come to the location where the movie is shot.

Now that we understand the meaning of a fixed position layout, we discuss the manufacturing environment. A similar approach to the movie example can also be applied in operations where the volume of goods produced is rather low and where the product is extremely heavy or bulky, i.e., difficult to move around. Examples are locomotives/trains, airplanes, or large-scale electrical power wind turbines. In all of these cases, the product remains at the same location for a very long period of time, and the materials should be brought closer to the point of value creation.

On the *production floor*, certain designated areas need to be defined where the parts or components will be placed by logistics/material movement staff. Footprints (= marked areas on the factory floor) should be defined where the necessary material will be buffered. The workers will need to walk back and forth from the material buffer to the product to move the material to the product. The same transportation effort also applies to tools, machines, or media (such as electricity, gas, or water).

To give a practical example, imagine the assembly of a locomotive in a factory facility. All necessary parts are buffered outside, so that the workers need to pick them up and carry them inside the locomotive to conduct the final assembly, which is inside the locomotive. As this example implies, the flexibility to, e.g., change the sequence of assembly work packages is quite high in a fixed position layout setup.

In addition, the impact of defective tools or machines is not that intense, as the layout provides an opportunity for replacing the machine and then continuing to

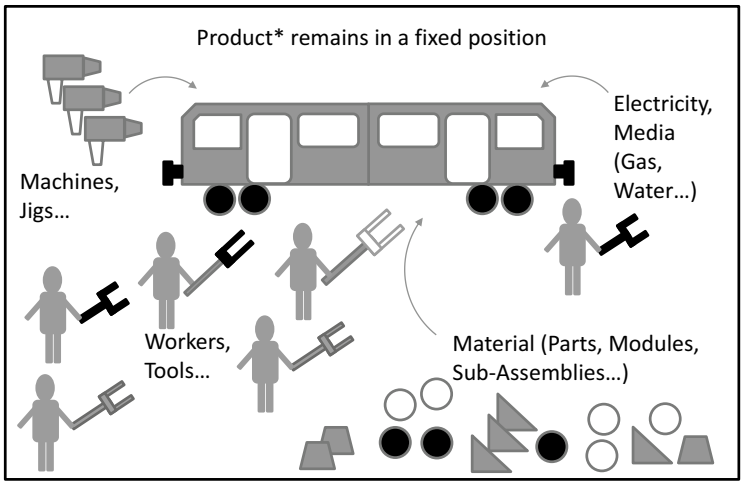


Fig. 10.2 Principles of the fixed position layout

work. At the same time, aspects of motion and movement (walking, waiting, searching) or the relatively high possible levels of buffered inventories are disadvantages of this layout system. The symbolic illustration below shows the motion and movement of the media, workers, tools, etc., in a simplified way (see Fig. 10.2).

▶ **Practical Insights** *In essence*, for the production of a few heavy, bulky products, the fixed position layout is a good choice since it provides the advantages of flexibility and adaptability. On the other hand, the disadvantages are the high amount of unnecessary motion and movement.

10.2.2 Process Flow Layout

The process flow layout is another basic type of layout for further processing of metal parts or products. The key principle is that the operational environment is split into specialized functional areas, such as a department where metal is cut, another department that specializes in drilling, another one specialized in milling, etc.

Such a layout of specialized functional areas in manufacturing is quite similar to the layout of a hospital, where you have first aid, the ophthalmologist, internist, or the cardiologist ward.

In metal manufacturing, product 1 might first require a drilling operation, then some milling, and finally some welding. Product 2 might first require bending, drilling, milling, and then welding. What becomes obvious is that the sequence of processes that need to be performed determines the product flow between departments. This is also the reason why this layout type is called a "process flow layout." The

advantages are that the layout is quite flexible since it allows the production of a very high number of very different products (of course, they need to be created under the usage of similar production technologies and machines, as explained above) in relatively high numbers.

The disadvantages are production in particular lot sizes, the high transportation effort, and the danger of having quite a high number of crossing/contradicting material flows. As the departments are in principle specialized, multiple machines are also likely to be available in, e.g., the drilling department (see Fig. 10.3).

The process flow layout is not very vulnerable to the breakdown of machines. If drilling machine 4 brakes down, the product can be produced alternatively using drilling machine 5, for example, as Fig. 10.3 depicts. In order to understand Fig. 10.3, the following short explanation regarding the hypothetical case above is given. Product A is first processed on drilling machine D2, then it is moved to milling machine M4, and then it is welded on W2. After this, the product is painted in paint shop P1. Finally, the packaging and shipping preparation takes place. Next or possible simultaneously, product B is moved through the facility and processed on machines B6, D1, M2, W1, and P2. As indicated above, products A and B flow through the manufacturing department according to the relevant process steps.

As illustrated in Fig. 10.3, the flow might become almost chaotic. This is referred to as a "spaghetti flow illustrated as a spaghetti diagram" in lean thinking terminology.

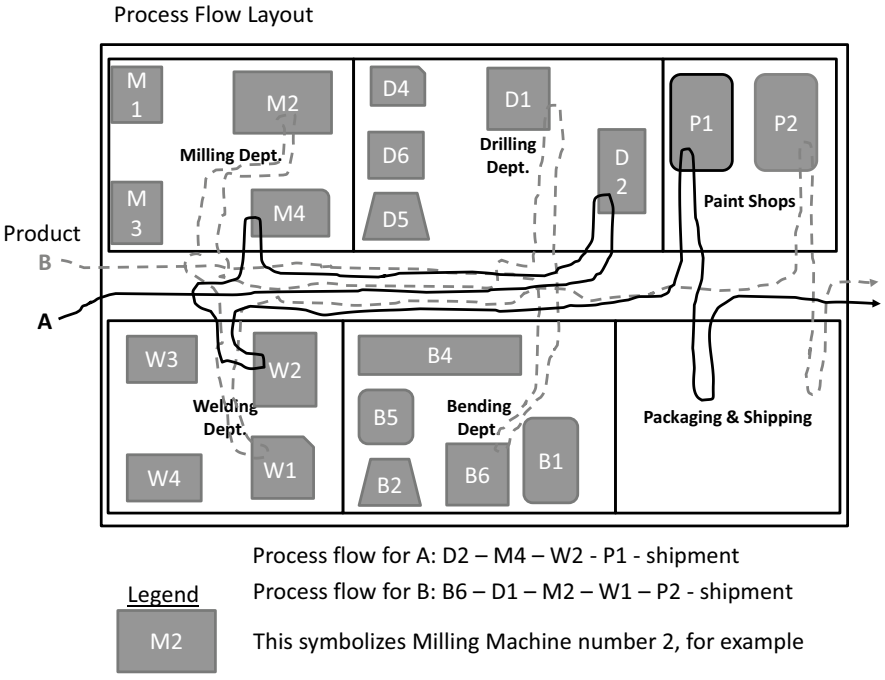


Fig. 10.3 Principles of the process flow layout

▶ **Practical Insights** *In essence*, for producing hundreds or thousands of similar (metal) parts, the process flow layout is a good choice, since it provides flexibility, general usability, and a relatively low level of vulnerability. The disadvantages are the high amount of unnecessary motion and movement and crossing material flows.

Process Flow Layout: A Practical View

Figure 10.4 shows the bay of the factory where machine centers for the creation of high-tech brake disks are installed. The minimization of the transportation distances between the machines was the fundamental principle of layout planning. The machine allocation ensures a flow for the performance of a series of specialized processes. In this specific case, the flow is structured in the following way: turning—milling—drilling. Non-destructive testing is also one of the processes that can be performed along with the in-process removal of burrs or the balancing of the brake disks.

"High flexibility" is a key criterion to this layout design. The production processes that are performed on multiple machines require the object to be moved from one machine center to the other centers. This can be implemented in the layout developed. In principle, all the product flows according to the production processes are possible.

Figure 10.4 depicts the major lean principles. For example, visualization boards showing material availability, Pareto charts on quality KPIs, check sheets, or charts

Fig. 10.4 Machine-centers for turning, milling, and drilling (© Knorr-Bremse Berlin SfS GmbH)

showing adherence to production progress is placed on the shop floor. The objective is to obtain transparency regarding the planned and actual production progress, to respect the *5S principles* (sort, systemize, sweep, standardize, sustain—in Japanese *seiri*, *seiton*, *seiso*, *seiketsu,* and *shitsuke,* see George et al. 2016, p. 202) and to secure TPM (*Total Productive Maintenance,* see Brunner 2011, p. 81). Furthermore, the standardized color-coding clearly indicates the footprints for raw materials and finished goods.

The brake disks are produced in specified lot sizes (batches) (see also Chap. 13) and are subject to the storage capacity of the shelves on the shop floor that define the respective batch sizes. For example, if the storage capacity of a shelf equals 20 units, a batch sizebatch size of 30 units would be impossible.

The setup of the manufacturing area in Fig. 10.4 is very flexible. It makes it possible to implement alternative flow patterns according to the required processes for the creation of the respective product. This is a very good example of a "process-flow layout."

10.2.3 Product Flow Layout

The key idea of this type of layout is that the product being created moves through a series of specially designed and equipped workstations. It could be that the product will be moved with a conveyor system (such as an assembly line for the production of cars or household devices), but it could also be that the product is moved through the defined workstations manually (e.g., production of converter boxes for the railway industry).

Usually, this type of layout is considered when a very high number of quite similar products are produced, for example, 1000 cars per day. It is important for a properly working product flow layout that the product moves through a facility at the same speed (see Fig. 10.5).

For example, the product moves at a beat (or pace) of, e.g., 90 s from workstation to workstation. In principle, the different stations are equipped with all necessary machinery and media, and the material required to be installed into the product at that station is also buffered at the station. This means each and every station is designed for a special purpose. If you think about car assembly, you have an early station where the cable harnesses are placed into the car, after which the carpets are installed, followed by the seats, and so on. The sequence is important, but so is the duration of the work that needs to be performed, as well as supporting technology and the necessary space for the material that is required at the respective station. With respect to the logistics and the placement of materials, the exact number of parts needs to be identified prior to planning the layout since they must be buffered right next to the assembly station at the line, so that everything (the product, the parts to be installed, the tools, and media) is within easy reach. These parameters, parts, processes, tools, media, etc., have a significant influence regarding the dimensioning of the layout.

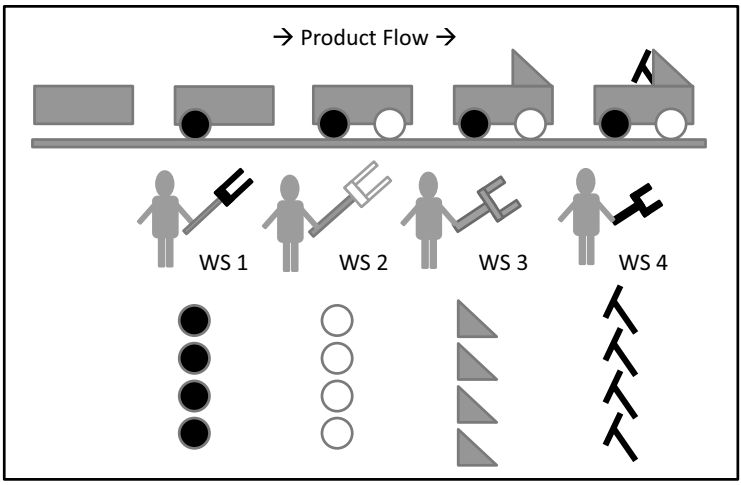


Fig. 10.5 Principles of the product flow layout

The product flow layout can also be applied to creating household appliances such as washing machines, refrigerators, microwave ovens, or sports shoes, etc., and the production volume needs to justify the special arrangements of the workstations and the necessary tooling. The advantage is that the products can be assembled in a very efficient way: the product flow is longitudinal, and the timing is very well balanced, according to the defined pace (beat rate) of the product that moves through the production facility in a one-piece single flow mode. Besides these obvious advantages, there are also disadvantages: if parts are missing or defective, or when a machine or tool breaks down, the product flow will be interrupted. While the layout results in high efficiency, it is also highly vulnerable.

▶ **Practical Insights** *In essence*, for the production of thousands of similar products (such as cars or household appliances) per day, the product flow layout might be a good choice, since it provides the advantages of efficiency and smooth flow at a given pace along specially defined, designed, and technically equipped workstations. The product flows over these workstations at a constant cycle time. The disadvantage is the high probability of interruptions or stoppages in the event of missing parts or defective machines.

Product Flow Layout: A Practical View

In the labor-intensive after-sales and services business areas, with their rather low production volumes and high variety, the fundamental manufacturing layout with a steady *product flow* combined with lean principles can also be implemented (see Fig. 10.6).

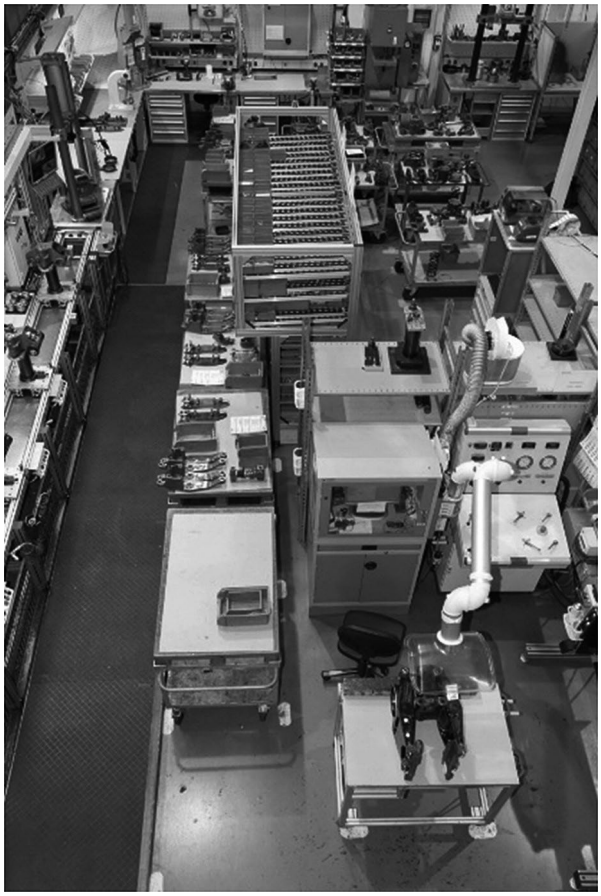


Fig. 10.6 Product flow layout/assembly line for refurbishment processes [front view] (© Knorr-Bremse Berlin SfS GmbH)

In this specific case, the used brake systems are refurbished in a special area of the factory. Although no conveyor belt exists, the general assembly line principles (see Chap. 9) were applied when the layout was designed. Figure 10.6 depicts the final assembly of the refurbished brake systems as the product moves through a set of specially designed workstations following a steady timing. This is also called *takt time* (see also Chap. 9), which "is the rate (time per unit) at which the system must produce a product to meet external customer demand" (Martin 2014).

Prior to the final assembly, the used brake systems received are checked in the initial goods reception processes. Following this, the disassembly and cleaning of the received systems take place (Fig. 10.7).

Pre-kitted sets of new replacement parts and other required assembly materials are provided to the assembly line along with the cleaned and disassembled components on trolleys. Visual media and footprints on-the-floor mark the individual

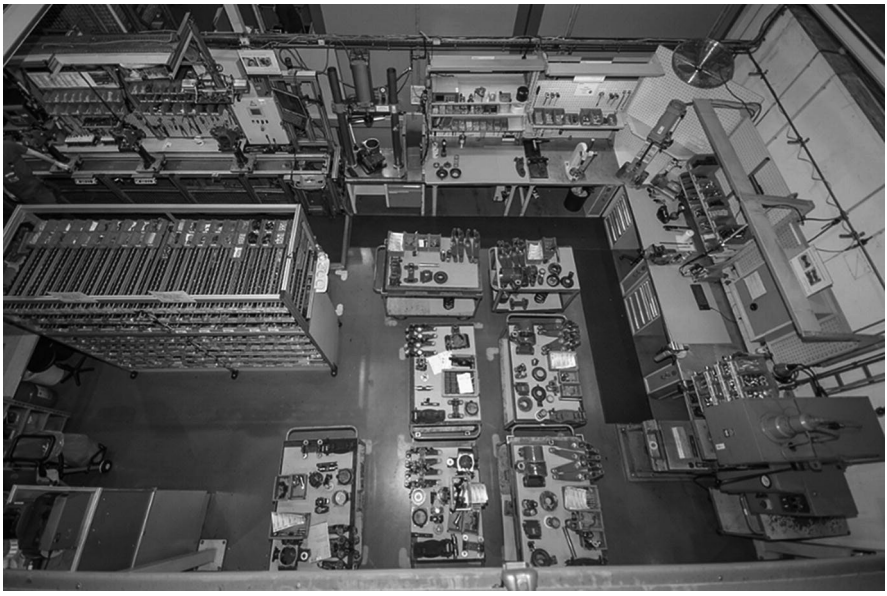


Fig. 10.7 Product—flow layout/assembly line for refurbishment processes [top view] (© Knorr-Bremse Berlin SfS GmbH)

positions for the trolleys, as shown in Fig. 10.7. Furthermore, tools and jigs are placed within easy reach, respecting the *best point philosophy* for an *ergonomic workplace*. Figure 10.7 clearly shows that for smaller series with high manual efforts, the fundamental principles of the "product flow layout" can be effectively applied.

10.2.4 Cell-Based Layout

In the case of the *cell-based layout*, the advantages of the product flow layout (smooth flow) and process flow layout (flexibility) are combined. This type of layout is also called a hybrid layout or is referred to as the grouping of technology. The main idea is to group the machines, tools, and equipment required in the exact sequence of the production steps so that there will be no crossing of the materials and transportation distances are minimized.

In such a case, the product is created by moving through the cell. Most likely, the pallets with the in-production products (semi-finished goods) are moved from one machine to another manually, i.e., in most cases, there will be batches of products produced per machine, and when they are finished on that particular machine, the lot is moved to the next machine or station (see Fig. 10.8).

In most cases, this is not a single piece flow, but most often a flow of batches. Despite this inefficiency, there will still be a smooth flow. In contrast to a real

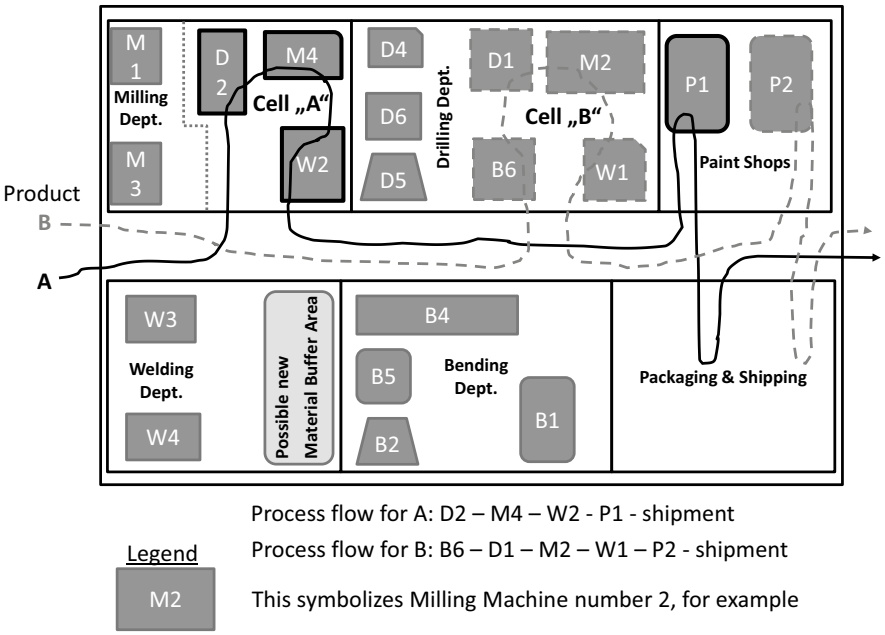


Fig. 10.8 Principles of the cell-based layout

assembly line, the alignment of the machines in the cell is cheaper, and with reasonable effort, the cells can be rearranged to make other products.

Figure 10.8 shows products A and B and their production sequence as it was introduced for the process flow layout. The following illustration considers the generation of two cells. One for the creation of product A and the second for product B. As one can see, the flow of products becomes almost one directional, and the crossing of the material flows is significantly reduced because of the rearrangement of the required machines.

▶ Practical Insights Please note that for some machining, such as welding, an exhaust evacuation system might be required. Space constraints or certain media connections will need to be assessed in this regard.

In essence, for the production of similar products (such as computer chips, sub-assemblies, lamps, etc.), the cell-based layout is a good choice, since it provides the advantages of a smooth product flow layout and the flexibility of a process flow layout.

Cell-Based Layout: A Practical View

In Fig. 10.9, the assembly of valves in a *cell-based layout* is depicted. The developed layout considers a balanced one-piece-single-flow that runs in a

Fig. 10.9 Assembly of valves in a cell (© Knorr-Bremse Berlin SfS GmbH)

counter-clockwise manner. The operational processes start on the upper left-hand side. It can be observed that the cell has a typical "*U-shape.*" After the product has moved through the cell and reached the upper right-hand point, all necessary production steps are finalized, and the handover to the inspection department takes place.

The applied layout design strictly separates the operators and their value-adding working environment inside the very well illuminated U-shape from the material replenishment processes which take place outside the cell. The workers continue their work in the center of the cell while new material is fed into the cell and empty bins are collected from the outside. This ensures disturbance-free operation on the process side and the availability of secured material.

All tools, machinery, and material can be easily reached. *Lean ergonomics* criteria, adjustable best point locations, and positions, as well as the *5S principles,* have also been considered in the creation of the cell-based layout. Furthermore, the screens show the standard operating instructions. The torques of the machines and the usage of lubricants are automatically supervised in order to secure *zero defects* throughout all the production steps.

Such a concept eliminates non-value-adding times, securing the highest quality outputs while the operating processes are balanced and conducted in the most ergonomic way all at the same time. The manufacturing area shown in Fig. 10.9 is a very good example of a "*cell-based layout.*"

10.3 Layout Planning in Warehouses

The processes in warehouses can be done manually or be automated. Automation is not always the best option. Automation requires investments in software. Manual process organization can be the more flexible option. However, automation can help in increasing efficiency and reducing lead time.

The basic warehouse processes are represented in Fig. 10.10. Consider these processes in detail.

10.3.1 Incoming Area

After the receipt and control of goods in the incoming area, items can be either moved to storage or directly to the outbound area if, e.g., customer orders already exist or because of a cross-docking principle, which means the in-transit merging of different incoming bins to some outgoing bins without intermediate storage.

In many cases, returns are also processed in the incoming area. For example, in the warehouses of E-Commerce retailers, the return rate can be up to 70% of sales.

10.3.2 Storage Area

Next, in the storage area, several storage strategies can be applied:

- Dedicated storage
- Cluster storage and class-based storage
- Random storage

In *dedicated storage*, each item is assigned exactly to a predetermined storage place. *Cluster storage* presumes building some zones for items with common features, e.g., items that are frequently ordered together; or items from the same supplier. *Class-based storage* means that items are stored on the basis of ABC

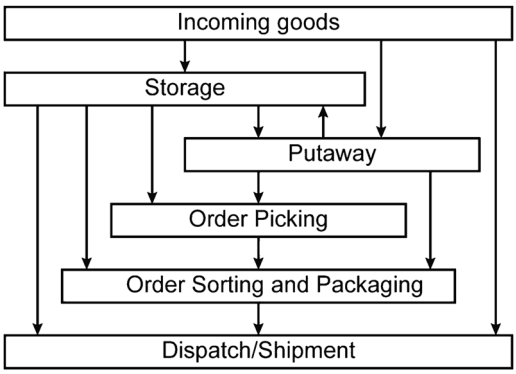


Fig. 10.10 Warehouse processes

Table 10.1 Advantages and shortcomings of storage strategies

Strategy	Advantages	Shortcomings	Applications
Dedicated storage	Item positions are exactly known	Low capacity utilization	Spare part warehouse Workforce fluctuations
Cluster storage/ class-based storage	Productivity increase Transportation reduction	Decrease in capacity utilization	Retail warehouses
Random storage	High capacity utilization	Control complexity; investments in IT	Automated high-level warehouse

classification (see Chap. 13). In random storage, a chaotic warehouse policy is applied.

The advantages and shortcomings of each strategy are summarized in Table 10.1.

10.3.3 Outbound and Order Pick-Up

FIFO (first-in-first-out) and LIFO (last-in-first-out) are two basic strategies for put-away organization. Two order picking strategies are "person-to-goods" and "goods-to-person." In the "person-to-goods" strategy, there are several options for picking: Printed pick-up list, Electronic messaging, Pick by Light, Pick by Voice.

It should be noted that even if new technologies like pick-by-voice present advantages, many logistics managers consider them to have shortcomings from an ergonomic point of view.

10.3.4 Layout Concepts

Two layout concepts, straightforward and L-shaped, can be distinguished from one another. In the introductory case study, we considered the OTLG distribution center. This company applied the L-shaped layout. In Fig. 10.11, the L-shaped layout of the OTLG warehouse is presented.

Consider another example, the company REWE, which is active in the food retailing industry. Specific for the food retailing industry is the fast processing times for fresh goods. This is why a straightforward layout was selected (Fig. 10.12).

According to this layout, the warehouse is divided into four temperature zones: dry assortment, fruits and vegetable area, fresh meat area (0 °C), and a deep-freeze area (−24 °C). The work in the deep-freeze area is very hard since in the summer, the difference between the outside and inside temperatures can be more than 60 °C. The trucks from the suppliers arrive at the gates and are unloaded. The pallets (mostly plastic pallets for hygiene and weight reasons) are moved to the storage area. The random storage principle is applied to the placement of pallets on the shelves. With the help of lift trucks, the pallets are picked up and delivered to the outbound area. The plastic pallets are used to simplify handling at the stores since they only weigh 6 kg. In addition, heavier items are typically located on the bottom

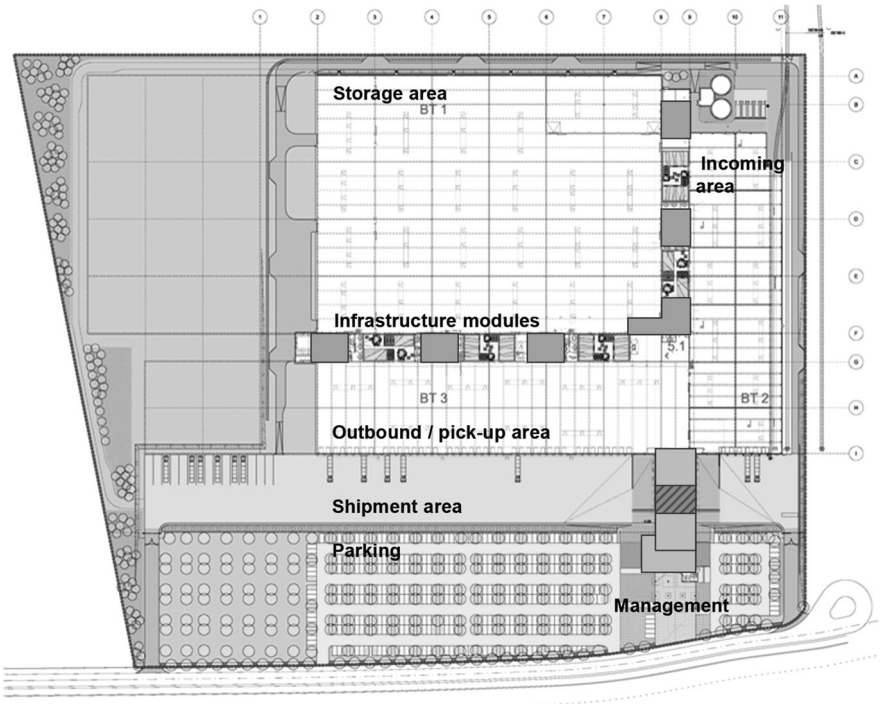
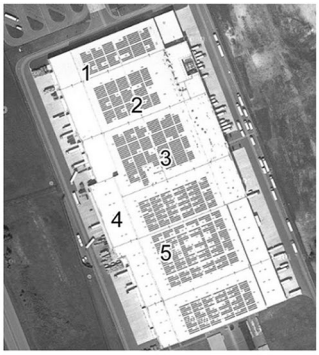


Fig. 10.11 L-shaped layout of a warehouse

1: Outbound area 3: Incoming area
2: Storage area



1: Deep-freezing area
2: Fresh food area
3: Fruits and vegetables
4: Disposal area
5: Dry assorment area

Fig. 10.12 Straightforward layout form of a warehouse

of the pallet, while lighter items come on the top. The warehouse operates from Sunday 4:00 am to Saturday 6:00 pm. Since demand constantly increases, automation can reduce the lead time and adjust process capacity.

About 380 employees in Oranienburg (Germany) ensure the replenishment of REWE supermarkets with about 11,000 SKU (Stock Keeping Units). One hundred trucks deliver products to 330 supermarkets from Sassnitz to Jüterbog daily. This means the optimization of 240 routes daily. With the help of modern software using a scientific methodology for vehicle routing and advanced navigation information technology, 95% capacity utilization for trucks is achieved. In the first half of 2012, 35 million incoming items of freight, which arrived on 958,398 euro-pallets, were processed in the incoming area. Thirty-six million items of freight were delivered to markets. The average inventory in the warehouse is valued at about 30.6 million euros. The inventory turnover for dry assortment is about 12; for fresh meat—2100 (on average, a package of fresh meat is processed at the warehouse within 4 h). This example provided insights from 2012 to obtain a feeling for the handled volumes. Of course, such numbers are subject to change, and they may differ from location to location.

10.4 Methods of Layout Planning

Methods of layout planning can be divided into qualitative and quantitative. In this chapter, we consider the qualitative method "REL-charts" and a quantitative method "quadratic assignment problem."

10.4.1 REL-Charts

10.4.1.1 Problem Statement

REL stands for relationship. With the help of this chart, the activities are illustrated according to their relationship (activity relationship charts, ARC). Now let us consider a specific problem. A layout for a retail warehouse has to be developed with consideration of desirable/undesirable proximity of some areas. The process is presented in Fig. 10.13.

The following notations are used:

- WE—incoming and inspection area
- LA, LB, LC—storage areas for A-items, B-items, and C-items, respectively
- WA—outbound shipment area
- KO—pick-up area (or kitting area)
- RE—receiving area for returns
- RS—storage area for returns
- RA—outbound shipment area for returns.

10.4.1.2 Solution Method

The REL-chart method is based on the identification of desirable/undesirable proximity relations for each pair of warehouse areas. This identification is subjective and based on expert estimations. In our case, the following REL-chart can be developed (Fig. 10.14).

The following notations are used:

- A—proximity is absolutely mandatory
- E—proximity is very desirable
- I—proximity is important

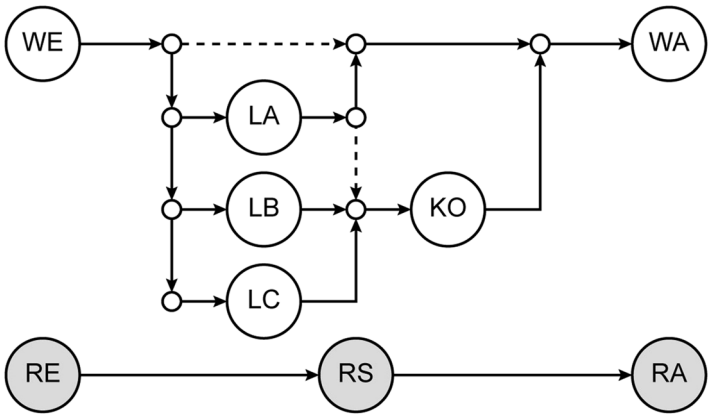


Fig. 10.13 Warehouse process

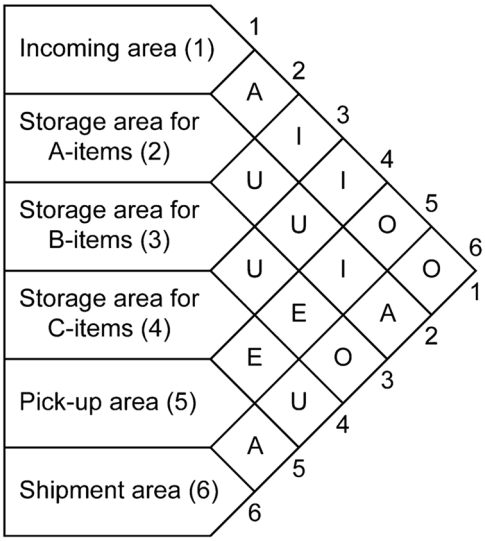


Fig. 10.14 REL-chart matrix

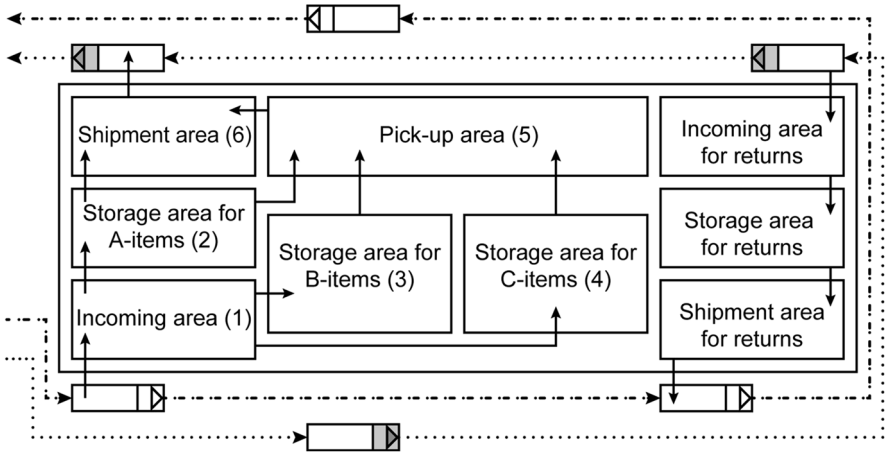


Fig. 10.15 Warehouse layout

- O—proximity is possible
- U—proximity is undesirable

For example, the relation between the pick-up and shipment areas is set up as "A" since these areas are to be located close to each other according to the process flow in Fig. 10.10.

Taking into account the process flow from the problem statement and the developed REL-matrix, the following layout can be suggested (Fig. 10.15).

Note that since the REL-chart is a qualitative method-based expert estimations, the suggested solution is only one possible option.

10.4.2 Quadratic Assignment Problem

10.4.2.1 Problem Statement

Some machines have to be assigned to some available locations. For each pair of locations, a *distance* is known. Similarly, for each pair of facilities, a *material flow* is specified (Fig. 10.16).

Assign machines to locations so that total transportation cost is minimized.

10.4.2.2 Model

Consider the following notations according to Fig. 10.16.

$M = \{M_j, j, k \in M, j, k = (1, \ldots M)\}$ is a set of locations.

$N = \{N_h, h, i \in N, h, i = (1, \ldots N)\}$ is a set of machines.

$d_{j,k}$ is a distance between two locations.

$t_{h,i}$ is material flow intensity between two machines.

Assume that transportation costs are linear and therefore proportional to $d_{j,k} \times t_{h,i}$ for the assignment $h \to j$ and $i \to k$.

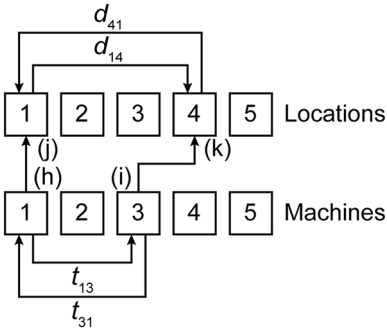


Fig. 10.16 Problem statement for the assignment problem

We introduce an auxiliary variable x_{hj}, which is equal to 1 if the h-machine is assigned to the j-location, and $x_{hj} = 0$ if it is not. Then the model can be defined in the following way:

Objective function

$$F(x) = \sum_{h=1}^{p} \sum_{\substack{i=1, \\ i \neq h}}^{p} \sum_{j=1}^{p} \sum_{k \neq j}^{p} t_{hi} \dot{c} d_{jk} \dot{c} x_{hj} \dot{c} x_{ik} \Rightarrow \min! \tag{10.1}$$

Constraints

$$\sum_{j=1}^{p} x_{hj} = 1 \sum_{h=1}^{p} x_{hj} = 1 \tag{10.2}$$

$$x_{hi} \in \{0,1\} \tag{10.3}$$

Constraints (10.2) dictate that each machine should be assigned to only one location. Constraint (10.3) dictates that the auxiliary variable x_{hj} is a Boolean variable. In reality, additional technological and technical constraints may exist, which are difficult to formalize.

10.4.2.3 Model Analysis and Solution Methods

The model (Eqs. 10.1, 10.2 and 10.3) is a linear integer programming model. Such a class of models is NP-hard. This means the optimal solution for reasonable problem size and in reasonable time is difficult to obtain with the help of classical methods for solving such problems (e.g., branch & bound). This is why different heuristics are used in practice for such solutions.

10.4.3 Simulation: Modeling Operations at Pharmaceutical Distribution Warehouses with AnyLogic

Problem Description

Cardinal Health, a billion-dollar pharmaceutical distribution and logistics firm, manages multiple products, from brand-name pharmaceuticals and generic drugs to over-the-counter drugs, health and beauty items, and their own private label. They

face a multitude of typical distribution warehouse challenges that are further complicated by the nature of pharmaceutical products, which are smaller in size, consumable, expensive, and possibly life-critical. Cardinal Health is an essential link in the healthcare SC, offering next-day delivery to over 30,000 locations, including hospitals, retail pharmacies, physicians' offices, and direct-to-consumer.

Other value-added services include efficiency and demand management, working capital management, and contract credit management. These services, along with poor manufacturing reliability and supply disruptions in the market due to Food and Drug Administration (FDA) and Disability Discrimination Act (DDA) regulations, add complexity to decision-making. Cardinal Health must keep up with the variability in pharmaceutical distribution management. Cardinal Health must consider facility layout, flow of product, order picking, labor planning and scheduling, customer order requirements, and day-to-day operations management.

Model Development
Traditional analysis tools, such as empirical trial and error, are risky, expensive, and difficult to use for making changes. Industrial engineering and operations researchers suggested mathematical models which were inexpensive but did not capture unexpected dynamics. If anything is open or has emergent behaviors, such as congestion, a standard mathematical model is unable to solve the problem. Process-driven or discrete event modeling was not advantageous because of its inability to represent a facility naturally. This led Brian Heath (Director of Advanced Analytics at Cardinal Health between 2013 and 2017) and Cardinal Health to explore alternative analysis options. Agent Based Modeling (ABM) with AnyLogic Simulation and Modeling software gave Cardinal Health the tools required to tackle many distribution warehouse issues without the restrictions of traditional tools.

Solution Development
ABM represents the abstraction of distributed autonomous entities that can interact with each other and their environment through space and time, allowing Cardinal Health to capture work time allocation, congestion waiting time, cycle times, distance traveled, worker variability, and other important metrics.

Testing the Solution
The model built was ultimately concerned with the activities of employees and their interaction with each other during the day, making it necessary to import data such as picking time and performance standards into the model.

Analyzing the Results
Cardinal Health could gather congestion waiting time data and see how much of a problem it was causing in the warehouse since "agents" were modeled as individuals with special relationships to each other. Additional parameters included in the model were several worker speeds, worker behavior, learning curves, cycle times, product turn-around, and distance covered walking or driving. The ability to import Excel files was also imperative as Cardinal Health has numerous warehouses, and it

is mandatory to test multiple layouts. Using AnyLogic, a change is as simple as updating the Excel file, importing it into the model, and running the model again.

Implementation
The ABM built with AnyLogic software allowed Cardinal Health to compare layouts, picking technology, and product slotting strategies. In addition, they could evaluate different methods of picking to update staffing models and for on-the-floor support if a workload changed, as orders vary on a day-to-day basis. Statistics were also gathered, such as takt time, how many batches are completed in an hour, truck unloading time, and sequencing of events. Despite the clarity provided by the above metrics, the model revealed a problem because of the random distribution of work. Employees' workloads were uneven, with one being faster and one slower. By balancing the workload, employees began working at a similar pace, and congestion decreased dramatically. By minimizing congestion using AnyLogic software, Cardinal Health was able to decrease the average shift length from 10.5 to 7.25 h and increased employee capacity. Cardinal Health saves over $three million annually using ABM with AnyLogic Simulation technology. "AnyLogic's agent libraries, flexible architecture, and integrated animation enables the continuing success of this project," declared Brian Heath, Director of Advanced Analytics at Cardinal Health.

10.5 Key Points

A layout is the physical arrangement of objects in a space. Fundamental objectives of layout planning are cost minimization for internal transportation, lead time reduction, or reasonable arrangement of departments in facilities. In addition, layout planning can be used to reduce the risk of good damages because of accidents in the warehouse, e.g., a fire.

In general, there are four basic types of layouts in manufacturing:

- Fixed position layout
- Process flow layout
- Product flow layout
- Cell-based layout

These forms of layouts depending on the process flow design. In warehouses, the layout is determined by the process flow, which includes the following stages:

- Incoming and inspection area
- Storage area
- Put-away area
- Pick-up area
- Packing area
- Shipment area

Methods of layout planning can be divided into qualitative and quantitative. We considered a qualitative method "REL-charts" and a quantitative method "quadratic assignment problem." REL-charts can be applied to cases where considerations of desirable/undesirable proximity of some areas are important. The quadratic assignment problem is an optimization method for optimal layout planning with the objective of minimizing total transportation costs.

10.6 Discussion

In our examples, production facilities were described (e.g., the creation of brake systems) or the layout of retail warehouses for distribution centers for spare parts and food retailing. Please discuss how the layout planning consideration might be transferred to layout problems in the following cases:

- Offices in an insurance company
- Departments at your local authorities
- Structure of supermarkets or shopping malls
- Organization of functional areas at hospitals or clinics

What do you think are the relevant criteria which influence decisions on these layouts?

Acknowledgments **Introductory Case-Study "OTLG Ludwigsfelde"**

The authors thanks the OTLG GmbH & Co. KG for the permission to use the company materials in this textbook.

Layout Planning in Warehouses

The authors thank the logistics direction of REWE East for the permission to use the company data in this case study.

Layout Examples in Production

The authors thank the *Knorr-Bremse Berlin Systeme für Schienenfahrzeuge GmbH* for the permission to use the company materials in this textbook.

REL-Charts

We thank Prof. Dr. Joachim Daduna for preparing materials for Sect. 10.4.1.

Bibliography

Bicheno J (2004) The new lean toolbox. PICSIE Books, Buckingham

Brunner FJ (2011) Japanische Erfolgsrezepte. Hanser, München

Dolgui A, Proth J-M (2010) Supply chain engineering: useful methods and techniques. Springer, Berlin

George ML, Rowlands D, Price M, Maxey J (2016) Das Lean Six Sigma Toolbook. Verlag Franz Vahlen, München

Gogate AS, Pande SS (2008) Intelligent layout planning for rapid prototyping. Int J Prod Res 46(20):5607–5631

Günther H-O, Tempelmeier H (2012) Produktion und logistik, 9th edn. Springer, Heidelberg

Heizer J, Render B, Munson C (2023) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow

Ioannou G (2006) Time-phased creation of hybrid manufacturing systems. Int J Prod Econ 102(2):183–198

Martin JW (2014) Lean six sigma for supply chain management – the 10-step solution process, 2nd edn. McGraw Hill, New York

Nee AYC, Ong SK, Chryssolouris G, Mourtzis D (2012) Augmented reality applications in design and manufacturing. CIRP Ann Manuf Technol 61(2):657–679

Okulicz K (2004) Virtual reality-based approach to manufacturing process planning. Int J Prod Res 42(17):3493–3504

Pérez-Gosende P, Mula J, Díaz-Madroñero M (2021) Facility layout planning: an extended literature review. Int J Prod Res 59(12):3777–3816

Saad SM, Lassila AM (2004) Layout design in fractal organizations. Int J Prod Res 42(17):3529–3550

Schenk M, Wirth S, Müller E (2014) Fabrikplanung und Fabrikbetrieb. Methoden für die wandlungsfähige, vernetzte und ressourceneffiziente Fabrik, 2nd edn. Springer, Berlin

Schonberger RJ (2008) Best practices in lean six sigma process improvement – a deeper look. Wiley, Hoboken, NJ

Slack N, Chambers S, Johnston R (2010) Operations management, 6th edn. Pearson, Harlow

Part III

Matching Demand and Supply: Tactical and Operative Planning

Demand Forecasting

11

Learning Objectives for This Chapter

- Understand the role of demand forecasting in SCOM
- Understand the forecasting process and methods in SCOM
- Understand the role of expert methods in forecasting
- Apply statistical methods to forecasting
- Calculate forecasts based on statistical methods
- Understand and apply measures for forecast quality assessments
- Role of data analytics in demand forecasting in SCOM

11.1 Introductory Case Study

Good forecasting is vital for all business planning. If estimates are too high, a company overproduces goods that cannot be sold. This is unnecessary production and increases costs for resources such as raw materials, workforce, or storage space. On the other hand, underproduction can result in shortages, lost business opportunities, and a potential loss of customers who cannot be served.

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_11) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_11.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_11

Good forecasting can be a competitive advantage. For example, Walt Disney Entertainment Parks put extremely high emphasis on predicting the next hours', days', and weeks' demand. Since they have been managing to almost always accurately predict future demand, revenue has been increasing immensely. Moreover, forecasting can be useful for analyzing areas for improvement.

Predictions about demand need to be made. However, forecasts are almost always wrong, as operations to calculate them are rather complex. Demand sometimes changes without prior warning and usually the effects set in later, making it difficult for operations managers to react.

However, some demand swings can be taken into consideration with relative reliability, such as seasonal changes. These include weather or particular holidays. For example, a city hotel, a supermarket, a metal producer, and a knitwear shop differently depend on these external factors. A city hotel has the most customers during the holiday season, whereas a knitwear shop sells primarily in autumn and winter. The supermarket may face peak times, for example, during lunch and after work hours. The factory producing metals is the most likely to be independent of seasonal swings and to have a steady demand throughout the year. However, many businesses have realized the correlation between weather and sales, which is why demand for meteorological forecasting services has increased rapidly.

Forecasts are generally expressed in terms that are also useful for capacity management, production, and SC planning. The aim is not only to be as accurate as possible but also to know the difference between the plan and actual data. This range can help to improve forecasts in the future and to estimate which forecasting techniques perform better. Moreover, forecasts are made for the close and long-term future. Generally, short-term forecasts tend to be more accurate.

There are various approaches to generate a demand forecast:

- Expert estimates
- Time series analysis
- Regression analysis

An expert estimate is a qualitative analysis tool that is drawn from historic demand and expert opinions on future development. In one case, Sport Obermeyer, a sports clothing retailer, applied this method. Initially, the whole team was involved in a decision about the next forecast. However, management noticed that some people would always dominate the decision process, so they decided that decisions would be made independently and the results would be compared and clustered. If most team members had forecast a similar result the product was categorized as easy to forecast. Otherwise, it was labeled "hard to forecast." For these "hard to forecast" products, the sports retailer delayed forecasting in order to reduce errors. In the case of Sport Obermeyer, using expert opinions worked extremely well with the new method.

Time series analysis, for instance exponential smoothing, takes sales data from the past to estimate new demand in the future. The restaurant chain "Red Lobster" uses this analysis tool with demand forecasting software. Each evening the software

creates a new forecast for the next day listing exactly which meals need to be prepared. Managers then adjust these forecasts based solely on local specialties.

Regression analysis can be applied to estimate the dependence between two factors. For example, in a telecommunication company, monthly repair efforts can be analyzed to estimate the number of service employees needed. There are various other options for predicting future demand. These include simple probability, neutral networks, and fuzzy logic, to name a few.

Demand forecasting in medical care has an extraordinary role. Unlike for production companies, an insufficient supply of resources does not mean an unhappy customer but affects the condition of a patient's health. Even though hospitals serve the public, they also need to make a profit. Hence, the sector faces the same dilemma in forecasting demand to match supply and resources: lost sales opportunities equal lost treatment opportunities. On the other hand, the aim is to minimize excess capacities and to minimize costs.

Forecasts in hospitals can be useful when applied to human resources such as nurses and doctors, customers (patients needing treatment), and operational goods such as beds and medicines. For staffing beds, queuing models can be applied. These methods help to analyze the capacity utilization of beds. This can be relevant for long-term planning to determine how many rooms and beds are needed. These are then fixed costs, which are difficult to reduce in the future. Average methods are not an appropriate measurement here. If the mean of all flu patients is taken over the year, there would be empty beds in summer and not nearly enough beds in winter.

In medical care, emergency departments face the most extreme situation. Patients arrive randomly at any time of the day. This means the workload is highly variable. The severity of sicknesses also varies. To manage patient arrivals, hospitals apply systems that categorize patients as seriously ill, non-urgent, and routine. The latter are likely to be turned away from the emergency department and asked to arrange a doctor's appointment. For the first two categories, a schedule is developed. Crowding in emergency departments endangers healthcare provision. Hence, forecasting plays an important role, and analytical models are used to ensure patients' needs are met more efficiently. Researchers have spent much effort in finding the best forecasting methods for this field.

In a sample emergency department, researchers came to the conclusion that time series analysis can provide a good method for indicating future needs. Daily patient attendances can be computed. For estimates, the autoregressive integrated moving average (ARIMA) method can be used, including such variables as holidays, air pollution, temperature, and humidity. The variables are then correlated to arrivals in the emergency department. The application of the ARIMA method can improve resource availability, especially staff planning.

Questions

In which fields is (demand) forecasting used?

Why is demand forecasting in medical care particularly challenging?

Why are means or averages not sensible measures for planning bed capacities?

Which analysis methods can be used to evaluate crowding problems in emergency departments?

11.2 Forecasting Process and Methods

Forecasting future demand is one of the major decisions in SCOM. Forecast sales data represent the input for further production, transportation, sourcing, and inventory planning. Moreover, many strategic decisions, such as facility location planning, revenue management, and process design, interact with demand forecasts (see Fig. 11.1).

Forecasts can be made for operative or strategic issues. Similarly, different methods can be applied to forecast demand. In this section, we consider the process of forecasting, different time horizons, and basic forecasting methods.

11.2.1 The Forecasting Process and Time Horizons

The forecasting process begins with the definition of the objective. The second step is to determine the time horizon over which the forecasts should be performed. For example, for an airline, the objective might be to define how many airplanes should be purchased over the next 10 years. The next step is to select the method of forecasting. This depends on both the defined horizon and the available data. After data acquisition, forecasting is performed, typically using software.

The results should be controlled continuously since any SCOM system and its environment is highly dynamic and subject to many external and internal changes.

The forecasting process is presented in Fig. 11.2 using an example for forecasting the number of printed materials for a university lecture based on the prediction of student attendance.

A professor must define every week how many handouts should be printed for the next lecture. In this case, the time horizon is a week. For such short-term issues, time series analysis is the preferred method. Based on the attendance list from the last lectures, exponential smoothing can be performed to predict demand.

Other time horizons may include anything from 3 months to 3 years. Examples of decisions for this medium-range horizon are sales and operations planning or

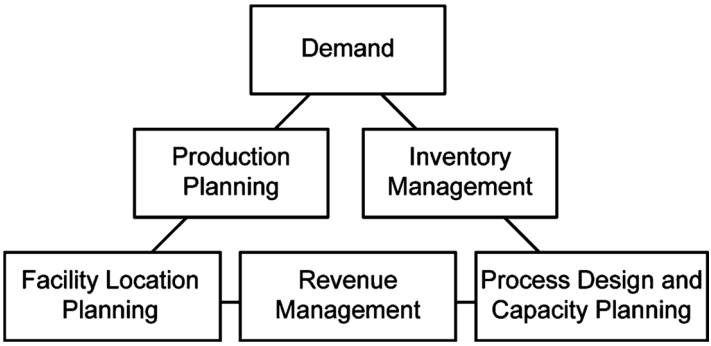


Fig. 11.1 Role of demand forecasting in SCOM

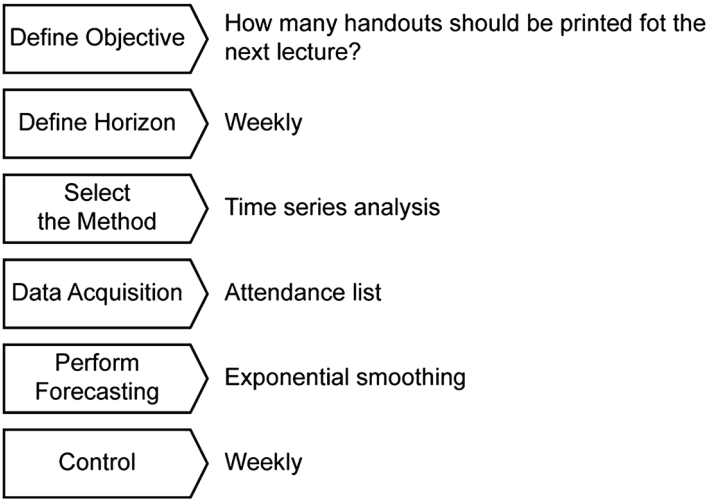


Fig. 11.2 Forecasting process

budgeting. *Long-term forecasting* typically refers to a period of over 3 years, e.g., new product planning, facility location planning, and research and development (R&D). *Medium-range* forecasts are used to deal with more comprehensive issues and support management decisions regarding planning and products, plants, and processes. *Short-term forecasts* tend to be more accurate than longer-term forecasts.

Case study dm: Accurate Demand Forecasts improve staff planning and Product Availability

dm is a drugstore company founded in 1973 in Karlsruhe, Germany. The company operates a very large network of warehouses and stores. Demand forecasting, inventory management, and staff allocation are, therefore, very challenging tasks, especially when cost and profit are taken into consideration. Inventory shortages result in lost sales while overstocking translates to high storage costs and other risks, such as product obsolescence. Another question is related to how many employees to assign to a store. Too many employees can reduce efficiency, while too few can create long queues and generally reduce customer satisfaction.

Ordering directly from manufacturers just as needed (i.e., just-in-time) is usually done to reduce storage costs. This is not applicable for dm's case since stores require a short delivery time. Instead, stores must rely on distribution centers for faster delivery. The question of how much and when to order is then passed to the warehouse. To optimize this decision, demand forecasting is carried out by Machine Learning algorithms with specific features/parameters. Multiple data are needed to train the algorithm. For a demand forecast, data for the last 2.5 years from distribution centers are collected. At the individual store level, this data includes daily sales up to the last 10 years. Seasonal products and holidays are also considered. When determining the number of employees, additional adjustable parameters such as

operating hours and store size are incorporated. The resulting demand forecasts are precise for a period of 6 months on the individual product level. As for staff planning, the weekly prediction has a low deviation of 2%. Furthermore, the solution is scalable as new stores can be included in the calculations without many adjustments. These precise forecasts, thus, benefit all involved parties. Manufacturers can plan their capacity better and optimize logistics costs, while dm can improve their product availability while also reducing and preventing excess inventory. Employees can also plan their working hours more easily, as staff allocation is conducted beforehand, and changes are rare. All this contributes to satisfied customers, better working conditions, and lower costs (Blue Yonder 2019).

11.2.2 Forecasting Methods

Forecasting methods can be divided into qualitative and quantitative methods (see Fig. 11.3).

Qualitative methods are used when the situation is vague and little data exists. Typically, such situations have a long-term nature or include decisions that are to be taken about new technology or products. Forecasting for such cases involves intuition and experience. In many cases, the forecasting process relies on historical data such as demand numbers or turnovers. This works well as long as a company has reliable data which is available for the required forecasting. However, how does this work if a company wants to launch a new technology? In this case, they must rely on expert views and customers' opinions.

This expert type of forecasting is normally based on professional knowledge and experience. A company uses input from different experts to develop forecasting for the next period. The problem with the expert methods is that they are based on opinions, experiences, and knowledge of human beings and are therefore subjective.

There are different methods for qualitative forecasting. One of the methods is *sales estimation*. For this method, a company asks their sales expert about the latest turnover figures and aggregates the data for overall turnover, using the data to develop a forecast. The next qualitative method is the *customer survey*. This is a good tool to use when a company launches a new product, and there are no historical data or expert experience. A company can ask its customers directly what they think of the new product and try to develop forecasts for the product launch.

The third method is the basic idea of qualitative forecasting—the *expert view*. A company asks proven experts about new technologies or their experiences with

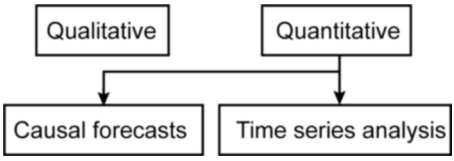


Fig. 11.3 Forecasting methods

similar products in the past. These experts typically work in different departments of the same company.

The best-known qualitative forecasting method is the *Delphi Method*. It is an extension of the expert view and the feature is a structured multiple expert survey using specialists from different companies and institutions. The Delphi Method runs in two or more iterations. A moderator develops a questionnaire and all the experts reply anonymously. After the first round, the moderator creates a new questionnaire based on the information he got. Answers to the first round are published to all other experts so everybody can rethink their views. Next, they fill out the new questionnaire and so on until it leads to a consensus.

Quantitative methods are used when historical data exists, e.g., existing products, existing technology, or existing markets. These methods are based on statistics. Statistical methods can be classified into two different forecasting methods:

- Causal forecast
- Time series analysis

In *causal forecasting*, the demand (y) for a product depends on a known factor (x) and because of this, it can be forecast (see Fig. 11.4). For example: Demand for axis (y) depends on the number of citizens in a district (x). Linear trends can be found using the least squares technique that results in Eq. (11.1):

$$\hat{y} = a + bx \tag{11.1}$$

where $\hat{y}$ is the computed value of the variable to be predicted (dependent variable); a is y-axis intercept; b is the slope of the regression line; and x is the independent variable.

Nonlinear regression is more complicated, but also more realistic. For a causal forecasting method, it is important to have reliable historical data for the known factor (the independent variable).

On the other hand, there is *time series analysis*. It is based on historical sales data. Sets of evenly spaced numerical data can be obtained by observing response variables at regular time periods. This method actually forecasts on the basis of the past values and assumes that factors influencing past and present will continue to influence in the future.

There are different methods to forecast demand depending on the temporal progress of demand (see Fig. 11.5).

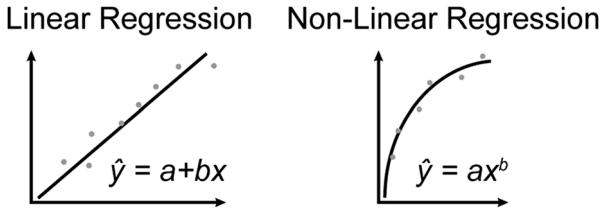


Fig. 11.4 Regression analysis

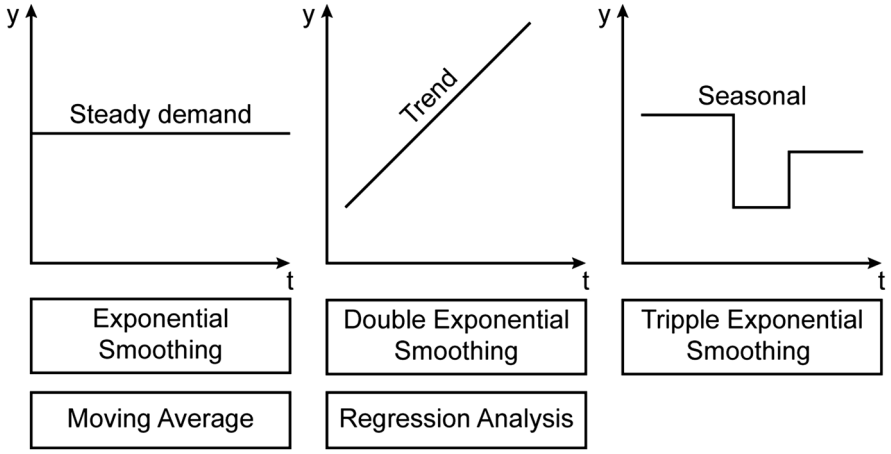


Fig. 11.5 Time series analysis

If a product's demand is constant, we can use the *Moving Average* or *Simple Exponential Smoothing*. For products with a demand that is based on trends, *Regression Analysis* or *Double Exponential Smoothing* is a more suitable choice. Products with seasonal demand can be forecast using *Triple Exponential Smoothing*.

11.2.3 Forecasting Quality

Different forecasting methods can provide different *forecasting qualit*ies. In order to estimate the quality of a forecast, some measures are used in practice, including:

- Mean absolute deviation (MAD)
- Mean squared error (MSE)
- Mean absolute percentage error (MAPE)

MAD measures the absolute deviation (ε) of forecast and actual data over T-periods according to formula (11.2):

$$\text{MAD} = \frac{1}{T}\sum_{t=1}^{T}\left|\varepsilon_t\right| \tag{11.2}$$

MSE measures the quadratic deviation of forecast and actual data according to formula (11.3):

$$\text{MSE} = \frac{1}{T}\sum_{t=1}^{T}\varepsilon_t^2 \tag{11.3}$$

MAPE measures the percentile deviation of forecast and actual data according to formula (11.4):

$$\text{MAPE} = \frac{1}{T}\sum_{t=1}^{T}\left|\frac{\varepsilon_t}{y_t}\right| \tag{11.4}$$

The practical problem with the MAD measure is that it expresses the absolute values and is, therefore, quantity-dependent. This makes it difficult to apply MAD to compare forecasts for different products. MSE is frequently used because of its good theoretical properties, but may be negatively affected by so-called outliers or freak values. MAPE is intuitively good and understandable since it shows the forecasting error as a percentage, and can be used for comparison of different forecasts where MAD/MSE values differ.

▶ **Practical Insight** In practice, statistically computed measures for forecast quality analysis should be checked carefully. For example, consider actual sales data for five periods: 10–10–11–10–11 and two forecasts: (a) 8–8–9–8–9 and (b) 8–12–9–12–9. From a statistical point of view, both forecasts have the same MAD = 2. However, from a management point of view, the first forecast (a) is biased since we always produce less than the demand. For managers, it is more important to identify bias rather than strive for minimal forecast error.

11.3 Statistical Methods

In this section, we learn in more detail how to compute demand forecasts with the help of different statistical methods.

11.3.1 Linear Regression

Linear regression is normally used as a causal forecasting method. The procedure will be explained by taking sales in a newly established, small tea shop that depends on the number of customers as an example, as shown in Table 11.1.

The first step is to calculate the regression function $\hat{y}(x) = \hat{a} + \hat{b}x$ with the parameter $\hat{a}$ and $\hat{b}$. It is important to find the $\hat{a}$ and $\hat{b}$ where the MSE has its minimum. To find the parameters, we fill out Table 11.2.

In order to calculate the coefficient $\hat{b}$, formula (11.5) can be used:

$$\hat{b} = \frac{(6 \cdot 185{,}956) - (330 \cdot 3020)}{(6 \cdot 20{,}884) - (330)^2} = \frac{119{,}136}{16{,}404} \approx 7.263 \tag{11.5}$$

The computed value 7.263 means that when there is one more customer per week, sales income rises by approximately 7.26 €.

There are two options to calculate $\hat{a}$. The first option is the usage of Eq. (11.6):

Table 11.1 Initial data for linear regression

Week n	Customer x_n	Sales (in €) y_n
1	26	269
2	35	389
3	43	432
4	67	536
5	78	685
6	81	709

Table 11.2 Results for calculating parameters

Week n	Customer x_n	Sales y_n (in €)	x_n^2	$x_n \cdot y_n$
1	26	269	676	6994
2	35	389	1225	13,615
3	43	432	1849	18,576
4	67	536	4489	35,912
5	78	685	6084	53,430
6	81	709	6561	57,429
$\sum$	330	3020	20,884	185,956

$$\hat{a} = \frac{\sum_{n=1}^{N} x_n^2 \sum_{n=1}^{N} y_n - \sum_{n=1}^{N} x_n \sum_{n=1}^{N} \left(x_n \cdot y_n \right)}{N \sum_{n=1}^{N} x_n^2 - \left(\sum_{n=1}^{N} x_n \right)^2} \tag{11.6}$$

The easier way is as follows (Eq. 11.7):

$$\hat{a} = \overline{y} - \hat{b}\overline{x} \tag{11.7}$$

When using formula (11.7) we have to calculate the average values:

$\overline{y} = \frac{\sum_{n=1}^{N} y_n}{N}$ and $\overline{x} = \frac{\sum_{n=1}^{N} x_n}{N}$. For our example:

$$\hat{a} = \frac{3020}{6} - \left(7263 \cdot \frac{330}{6} \right) \approx 103.868$$

The regression function now reads as follows: $\hat{y}\left(x\right) = 103.868 + 7.263x$.

The next step is to calculate the MSE using the regression function (see Table 11.3).

To calculate the forecast error ε_n we use formula (11.8):

$$\varepsilon_n = y\left(x_n\right) - y_n \tag{11.8}$$

To calculate the MSE we use formula (11.3):

$$\text{MSE} = \frac{5234.773}{6} = 872.462$$

Table 11.3 Results for calculating MSE

Week n	Customer x_n	Sales y_n (in €)	$y(x_n)$	ε_n	ε_n^2
1	26	269	292.706	23.706	561.974
2	35	389	358.073	−30.927	956.479
3	43	432	416.177	−15.823	250.367
4	67	536	590.489	54.489	2969.051
5	78	685	670.382	−14.618	213.686
6	81	709	692.171	−16.829	283.215
∑	330	3020			5234.773

11.3.2 Moving Average

The method of *moving average* is based on computing the average demand from the previous periods and using these values for forecasting. It is called "moving" because with every period the average value changes or moves onward.

Consider the following example and function to understand forecasting with the help of the moving average. It is also important to find the result with the lowest MSE here. The demand for each subsequent period is computed subject to formula (11.9):

$$\hat{y}_{t+1} = \frac{1}{T} \sum_{\tau=t-T+1}^{t} y_t \tag{11.9}$$

where $\hat{y}_{t+1}$ is the forecast for the next period, y_τ is the demand in period t, and T is the number of periods. It is important to choose a T that is not too large and not too small. Normally, you calculate with different T values to find the one with the lowest MSE.

Consider this example. Company X wants to find its weekly average demand for Cupcakes. Table 11.4 illustrates the last weeks' demand.

Three different numbers of periods to be averaged are given:

$T_1 = 2$; $T_2 = 3$; and $T_3 = 4$. The calculation results are presented in Table 11.5. For example, to compute the forecast for week #3: 31 = (26 + 35)/2 → we add demand from the previous two periods and divide it using T_1.

To check which T is the one with the smallest MSE we have to calculate the forecast error. It is important to start with a period where for all T values the sales have been forecast in order to make the MSE values comparable. In this case, it is week #5 (see Table 11.6).

Using the error calculation formula $\varepsilon_t = \hat{y}(T) - y_t$ and Eq. (11.3), we get:

$$\mathrm{MSE}(T_1) = \frac{99}{4} = 24.75; \mathrm{MSE}(T_2) = \frac{122}{4} = 30.5; \mathrm{MSE}(T_3) = \frac{171}{4} = 42.75$$

As we can see, the smallest T is the best one because the MSE is at its minimum there. For this reason, the recommendation for forecasting the next periods will be using $T = 2$.

Table 11.4 Initial data for moving average

Week t	1	2	3	4	5	6	7	8
Demand y_t	26	35	28	42	38	45	46	39

Table 11.5 Results calculating moving average with three different Ts

T	y_t	$\hat{Y}(T_1 = 2)$	$\hat{y}(T_2 = 3)$	$\hat{y}(T_3 = 4)$
1	26	–	–	–
2	35	–	–	–
3	28	31[a]	–	–
4	42	32	30	–
5	38	35	35	33
6	45	40	36	36
7	46	42	42	39
8	39	46	43	43

[a]Forecasted demand from this period onwards

Table 11.6 Results calculating MSE

t	y_t	$\hat{y}(T_1 = 2)$	ε_t	ε_t^2	$\hat{y}(T_2 = 3)$	ε_t	ε_t^2	$\hat{y}(T_3 = 4)$	ε_t	ε_t^2
1	26	–	–	–	–	–	–	–	–	–
2	35	–	–	–	–	–	–	–	–	–
3	28	31	–	–	–	–	–	–	–	–
4	42	32	–	–	30	–	–	–	–	–
5	38	35	3	9	35	−3	9	33	−5	25
6	45	40	5	25	36	−9	81	36	−9	81
7	46	42	4	16	42	−4	16	39	−7	49
8	39	46	7	49	43	4	16	43	4	16

In some cases, a weighted moving average can be used. In this case, older data usually becomes less important, and the weights are assigned to different periods based on experience and intuition.

11.3.3 Simple Exponential Smoothing

This method is similar to the weighted moving average. The weightings decline exponentially, and the most recent data are weighted higher. For sales forecasts, average demand/sales values of previous periods are exponentially weighted to forecast the next periods. The simple exponential average means working with a smoothing parameter α. The idea is to weigh current demand higher and assign lower weights to previous demand.

In this method we use formula (11.10) to forecast the demand for the following periods:

$$\hat{y}_{t,t+1} = \alpha \cdot y_t + (1-\alpha) \cdot \hat{y}_{t-1,t} \tag{11.10}$$

where $\hat{y}_{t,t+1}$ is the forecast that we are making in the current period for the following period, y_t is the current demand, and $\hat{y}_{t-1,t}$ is the forecast we made in the last period for the current period.

Consider the following example as an aid to understanding simple exponential smoothing. Demand for a product during the last six periods is provided (Table 11.7).

The smoothing parameters are $\alpha_1 = 0.3$ and $\alpha_2 = 0.5$. The idea is to choose the smoothing parameter, which leads to a smaller MSE (see Table 11.8).

For example, to compute the forecast sales for period #3: 54.5 = 0.3·44 + (1–0.3) · 59 → we multiply the demand in week #2 by α and the forecast demand for week #2 by $(1-\alpha)$ to get the forecast for week #3. In this case, $\text{MSE}(\alpha_1) = \frac{338.75}{6} \approx 56.46$.

Consider the calculations for Table 11.9:

Table 11.7 Initial data for single exponential smoothing

Week t	1	2	3	4	5	6
Demand y_t	59	44	61	58	49	52

Table 11.8 Results for single exponential smoothing with $\alpha_1 = 0.3$

T	y_t	$\hat{y}_{t,t+1}$	ε_t	ε_t^2
0		59	–	–
1	59	59	0	0
2	44	59	15	225
3	61	54.5[a]	−6.5	42.25
4	58	56.45	−1.55	2.40
5	49	56.92	7.92	62.65
6	52	54.54	2.54	6.45
$\sum$				338.75

[a]Forecasted demand from this period onwards

Table 11.9 Results for single exponential smoothing with $\alpha_2 = 0.5$

t	y_t	$\hat{y}_{t,t+1}$	ε_t	ε_t^2
0		59	–	–
1	59	59	0	0
2	44	59	15	225
3	61	51.5[a]	−9.5	90.25
4	58	56.3	−1.75	3.06
5	49	57.1	8.13	66.02
6	52	53.1	1.06	1.13
$\sum$				385.46

[a]Forecasted demand from this period onwards

MSE(α_2) = 385.46/6 ≈ 64.24. As we can see, the second smoothing parameter leads to a higher MSE. For this reason, it is advisable, in this case, to forecast demand for the following periods with the smoothing parameter of 0.3 instead of 0.5.

This method helps to forecast the demand for products with a steady demand. But we need other methods to forecast the sales for products with a trend-shaped or seasonal demand.

11.3.4 Double Exponential Smoothing

Double exponential smoothing is used for situations where a trend already exists. This method works with two smoothing factors, α and β. The basic idea is that demand is usually overrated when a trend is positive and underrated when the trend is negative. Consider the formula (11.11):

$$\hat{y}_{t,t+\tau} = a_t + b_t \cdot \tau \tag{11.11}$$

where $\hat{y}_{t,t+\tau}$ is the forecast for the following period performed in the current period, a_t is the smoothed forecast, b_t is the smoothed trend parameter (slope), and τ is the parameter that describes the number of periods for which we are making the forecast. We calculate a_t and b_t as follows from Eqs. (11.12) and (11.13):

$$a_t = \alpha \cdot y_t + (1-\alpha) \cdot (a_{t-1} + b_{t-1}) \tag{11.12}$$

$$b_t = \beta \cdot (a_t - a_{t-1}) + (1-\beta) \cdot b_{t-1} \tag{11.13}$$

Assume that we have the demand data of gluten-free bread in a small grocery shop for the last seven periods (Table 11.10).

To forecast the demand for 7 months in the following year we use the following parameters: $\alpha = 0.2$; $\beta = 0.35$; $\tau = 1$; $b_1 = 4$. The computational results are presented in Table 11.11:

Table 11.10 Initial data for double exponential smoothing

Months t	1	2	3	4	5	6	7
Demand y_t	249	257	269	285	298	302	312

Table 11.11 Results for double exponential smoothing

t	y_t	a_t	b_t	$\hat{y}_{t,t+1}$
1	249	249	4	
2	257	253.80	4.28	253
3	269	260.26	5.04	258.08
4	285	269.25	6.42	265.31
5	298	280.14	7.99	275.67
6	302	290.90	8.96	288.12
7	312	302.28	9.81	299.85

By changing the smoothing factors, we can obtain a more reliable forecast for the next periods in which a trend exists.

To handle forecasts for seasonal products we can use *triple exponential smoothing*. In this case, the third parameter, for seasonal fluctuations, is added to double exponential smoothing.

11.4 Case study: Demand forecast using machine learning

Background:

You are a new demand planner in a consumer goods company. Your main responsibility in this new position is to forecast future monthly demand for one product using historical data. Accurate demand forecasting is critical for effective inventory management, as it helps avoid both stock-outs and overstocking. During your onboarding and job training, you discover that your new team consistently uses Simple Exponential Smoothing (SES) to forecast demand. On the other hand, your leader is interested in exploring alternative methods. You are eager to utilize a *machine learning* technique to demonstrate to your team that it can outperform the current forecasting approach

Dataset:

Below is the historical monthly demand (in units) for the product over the past 72 months, starting from January 2016

120, 134, 148, 142, 138, 154, 170, 168, 158, 152, 136, 144,
128, 150, 140, 148, 160, 172, 190, 178, 165, 154, 142, 150,
140, 160, 155, 152, 145, 185, 210, 195, 170, 165, 150, 160,
125, 148, 162, 158, 140, 170, 185, 180, 172, 160, 146, 155.
135, 145, 160, 155, 150, 180, 195, 190, 175, 165, 148, 160,
140, 155, 172, 165, 155, 200, 220, 210, 180, 175, 158, 165.

Objective:

1. Use SARIMA (Seasonal Autoregressive Integrated Moving Average) modeling to forecast the next 12 months of demand. Evaluate the model performance using a train-test split. Use the first 60 months of data for training and the last 12 months for testing.
2. Compare the results with the old practice using SES (alpha = 0.2).

Solution:

In the following steps, we guide you in solving the task using Python.

Step 0. Initial setup. We import the required packages and import data as below:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.holtwinters import SimpleExpSmoothing
```

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
from sklearn.metrics import mean_absolute_error,
mean_absolute_percentage_error
from pmdarima import auto_arima

# Step 0: import dataset
data = {
   "Month": pd.date_range(start="2016-01-01", periods=72,
freq="MS"),
   "Demand":[120,134,148,142,138,154,170,168,158,152,136,144, #Y1
             128,150,140,148,160,172,190,178,165,154,142,150, #Y2
             140,160,155,152,145,185,210,195,170,165,150,160, #Y3
             125,148,162,158,140,170,185,180,172,160,146,155, #Y4
             135,145,160,155,150,180,195,190,175,165,148,160, #Y5
             140,155,172,165,155,200,220,210,180,175,158,165, #Y6
            ]
        }
df = pd.DataFrame(data)
df.set_index("Month", inplace=True)

#df.head()
```

Step 1. Visualize the data. We plot the time series to understand its trends and seasonality.

```
# Step 1: Visualize the data
plt.figure(figsize=(10, 5))
plt.plot(df, label="Monthly Demand")
plt.title("Monthly Product Demand")
plt.xlabel("Month")
plt.ylabel("Demand")
plt.legend()
plt.grid()
plt.show()
```

The results from step 1 are presented in Fig. 11.6. We notice a seasonal pattern, with demand higher in the summer than in other seasons. Overall, demand has trended upward over the past 6 years, although there was a decline in 2019, after which it has gradually increased.

To confirm our observations, we could use some commands in Python to perform the analysis.

Step 2: Decompose the time series. We use the following code to decompose the series into trend, seasonality, and residual components to understand its behavior better.

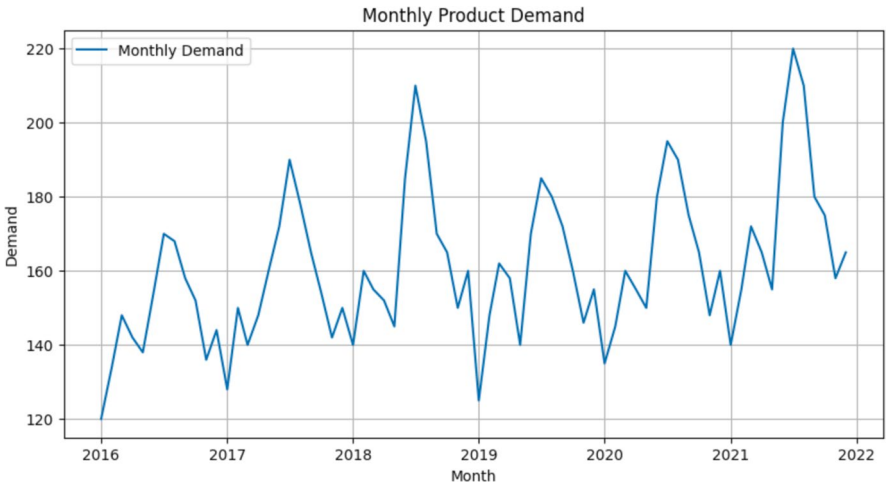


Fig. 11.6 Monthly product demand over 6 years

```
# Step 2: Decompose the time series
decompose = seasonal_decompose(df["Demand"], model="additive",
period=12)
decompose.plot()
plt.show()
```

Figure 11.7 presents the results of the code and confirms our observations. We can see a clear upward trend in the data, along with identifiable seasonal patterns in demand. At this point, it is evident that SES techniques are not a suitable choice, as they assume the time series is non-seasonal. We keep working to forecast and find additional evidence to position our choice better.

Step 3. Check for stationarity. We use the Augmented Dickey-Fuller (ADF) test to check the stationarity of the demand series. Additionally, the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test can also be considered. In this case, we are using the ADF test. The p-value obtained is 0.7856, which indicates that the original demand series data is not stationary.

```
# Step 3: Check for stationarity
result = adfuller(df["Demand"])
print(f"ADF Statistic: {result[0]}")
print(f"p-value: {result[1]}")
if result[1] > 0.05:
    print("The series is not stationary.")

else:
    print("The series is stationary.")
```

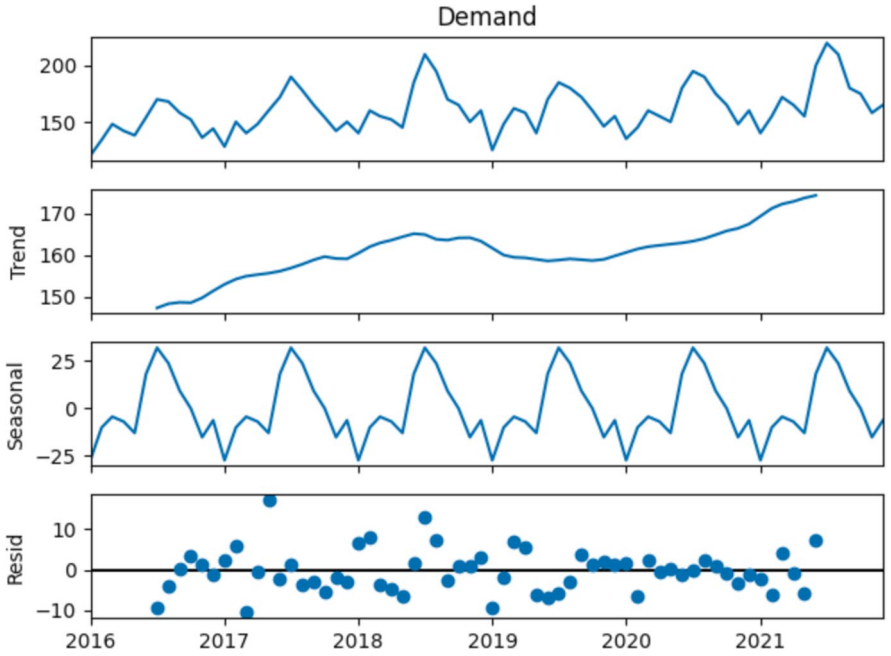


Fig. 11.7 Results of decomposing demand data

Let us analyze the first difference using the code below. The p-value is 0.0036, indicating that the first-order differenced series is stationary. This also determines the value of the parameter "d" in step 4:

```
# Step 3-continue: Check for stationarity - 1st diff.
df_diff1 = df["Demand"].diff().dropna()
result_diff1 = adfuller(df_diff1)
print("First-order Differencing:")
print(f"ADF Statistic: {result_diff1[0]}")
print(f"p-value: {result_diff1[1]}")
if result_diff1[1] > 0.05:
    print("The first-order differenced series is not stationary.")
else:
    print("The first-order differenced series is stationary.")
```

Step 4. Model selection. To identify the best SARIMA model, we use the following code. We recommend that readers take the time to familiarize themselves with the parameters outlined in the documentation for the `auto_arima` function. Although we can set the parameter (d = None), doing so may take longer to identify the optimal SARIMA model. Additionally, we divided the original data into two parts: the training set, which includes data from the first 5 years, and the testing set,

which contains the demand data from the last year. The resulting model is "SARIMAX(0, 1, 1)x(2, 1, [], 12)".

```
# Step 4: Model selection using auto_arima
train = df.iloc[:60]
test = df.iloc[60:]

model = auto_arima(train["Demand"],
                   start_p=1,
                   start_q=1,
                   max_p=3, max_q=3, #maximum p and q
                   m=12, # frequency of series
                   test='adf', #use ADF test to find optimal 'd'
                   d=1,
                   D=None, #order of the seasonal differencing
                   seasonal=True,
                   trace=False,
                   suppress_warnings=True,
                   stepwise=True)
print(model.summary())
```

Step 5. Train the model. We fit the SARIMA model using the training dataset using the following code.

```
sarima_model = SARIMAX(train["Demand"],
                       order=model.order, #(0, 1, 1)
                       seasonal_order=model.seasonal_order,
                       #(2,1,0,12)
                       enforce_stationarity=False,
                       enforce_invertibility=False)
sarima_result = sarima_model.fit(disp=False)
print(sarima_result.summary())
```

Step 6. Make predictions. We forecast the demand for year 6 and compare the predictions with the actual values in the test set. Additionally, we forecast the next 12 months beyond the available data for future demand planning.

```
# Step 6: Forecasting
forecast = sarima_result.forecast(steps=12)       # predict
demand Y6
future_forecast = sarima_result.forecast(steps=24)# predict
demand Y7
```

Step 7. Evaluate the model. We calculate metrics such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) to assess the accuracy of our

model. These metrics are essential in persuading our team to adopt the updated forecasting methods.

```
SARIMA_mae = mean_absolute_error(test["Demand"], forecast)
SARIMA_mape = mean_absolute_percentage_error(test["Demand"],
forecast)
print(f"SARIMA MAE: {SARIMA_mae}")
print(f"SARIMA MAPE: {SARIMA_mape:.2%}")
```

Table 11.12 compares the SARIMA model with the SES model (where alpha = 0.2). To determine the optimal alpha value, we conducted a for loop ranging from 0.1 to 1.1 in increments of 0.1. Based on our demand data, an alpha value of 0.2 yields the best MAE for the SES method. The result in Table 11.12 is not surprising, as the SES model tends to perform poorly in the presence of seasonality.

Step 8. Visualize the results. We plot the actual values against the predicted values for the test set and the 12-month future forecast using the code below. The results are displayed in Figs. 11.8 and 11.9.

Table 11.12 Comparison model accuracy of the proposed method and the current method

	SARIMA	SES
MAE	6.29	18.33
MAPE	3.43%	9.81%

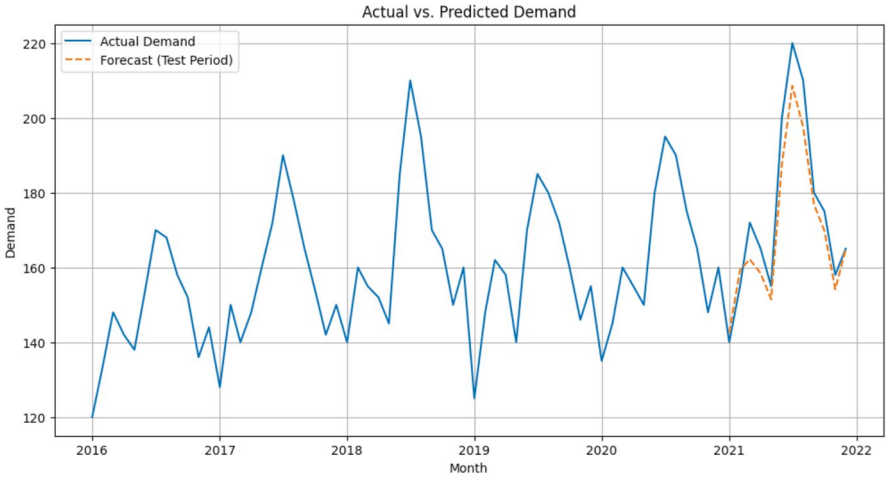


Fig. 11.8 Actual demand vs forecasted demand in year 6

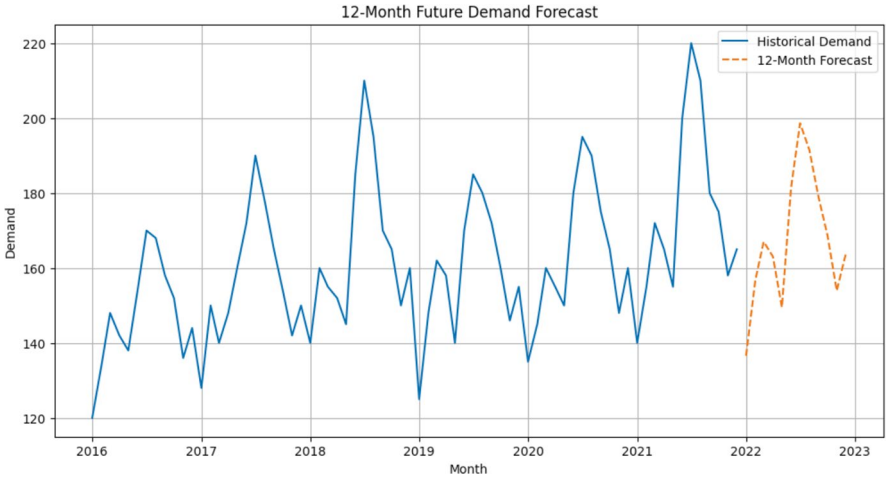


Fig. 11.9 Forecasted demand in year 7

```
# Step 8: Visualize results - Actual vs. Predicted Demand
plt.figure(figsize=(12, 6))
plt.plot(df.index, df["Demand"], label="Actual Demand")
plt.plot(test.index, forecast, label="Forecast (Test Period)",
linestyle="--")
plt.title("Actual vs. Predicted Demand")
plt.xlabel("Month")
plt.ylabel("Demand")
plt.legend()
plt.grid()
plt.show() # Figure 3

# Step 8 - continue: Visualize results - Forecast year 7
visualization
future_index = pd.date_range(start="2022-01-01", periods=12,
freq="MS")
plt.figure(figsize=(12, 6))
plt.plot(df.index, df["Demand"], label="Historical Demand")
plt.plot(future_index, future_forecast[12:], label="12-Month
Forecast", linestyle="--")
plt.title("12-Month Future Demand Forecast")
plt.xlabel("Month")
plt.ylabel("Demand")
plt.legend()
plt.grid()
plt.show()# Figure 4
```

11.5 Key Points and Outlook

In this chapter, we learned methods and practical tools for demand forecasting. Let us summarize the key points of this chapter as follows. For SCOM, it is important to understand the role, process, and methods of demand forecasting, since forecasts are used for both tactical and strategic decisions. Production planning, inventory management, facility location, and capacity planning rely on decisions that depend on demand forecasting.

The forecasting process consists of six steps: (1) define the objective; (2) determine the time horizon; (3) select the method of forecasting; (4) gather data; (5) perform the forecasting; and (6) validate and control the results. Forecasting methods can be divided into qualitative and quantitative methods. Qualitative methods are used either for new products and technology or for long-term decisions where data from the past cannot be considered a reliable source. Qualitative methods use expert knowledge and experience and include such techniques as *sales estimation, customer surveys*, *expert estimation*, and the *Delphi Method*.

We also learned several quantitative methods which are used when historical data exists and statistical tools can be applied. Regression analysis and time series analysis, including the moving average and exponential smoothing (simple, double, triple), belong to quantitative methods. We learned how to apply statistical methods for forecasting and to calculate the forecasts based on statistical methods.

For quantitative methods, it is important to understand the measures for forecasts' quality assessment, which are based on error estimation. The most popular measures are MSE, MAD, and MAPE.

Finally, it should be noted that a common problem with statistical methods is the reliability of historical data and future developments. If forecasting is only based on numeric data, it is not safe. Numeric data often does not consider external features such as ecological, economic, or political factors. Assume that demand for ice cream has increased over the last few years by 10% because of hot summers. The conclusion would be that the forecast for the following year would also be high because the company assumes the weather will be hot again. If the company does not consider weather forecasts in its calculations or consider expert views, the forecast based only on historical data could be wrong. Another critical area is forecasting product returns in the online retail industry. For this reason, it is common practice to combine expert and statistical methods to get safer forecasts. Such methods are known as ARIMA (autoregressive integrated moving average model), SARIMA, and ANOVA (analysis of variance).

Finally, we note the increasingly important role of data analytics in demand forecasting. Machine learning techniques, social network data analysis, and advanced predictive and prescriptive analytics make it possible to improve forecast accuracy in both short—and long-term perspectives.

Bibliography

Chase CW (2013) Demand-driven forecasting: a structured approach to forecasting, 2nd edn. Wiley, Hoboken
Chase CW (2016) Next generation demand management: people, process, analytics, and technology. Wiley, Hoboken, NJ
Choi TM, Wallace SW, Wang Y (2018) Big data analytics in operations management. Prod Oper Manag 27(10):1868–1883
Chopra S (2019) Supply chain management. Strategy, planning and operation, 7th edn. Pearson, Harlow
Günther H-O (2005) Supply chain management and advanced planning systems: a tutorial. In: Günther H-O, Mattfeld DC, Suhl L (eds) Supply chain management und logistik. Springer, Heidelberg, pp 3–40
Heizer J, Render B, Munson C (2022) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow
Jackson I, Ivanov D (2023) A beautiful shock? Exploring the impact of pandemic shocks on the accuracy of AI forecasting in the beauty care industry. Trans Res E Logist Transp Rev 180:103360
Johnson K, Lee ABH, Simchi-Levi D (2016) Analytics for an online retailer: demand forecasting and price optimization. Manuf Serv Oper Manag 18(1):69–85
Render B, Stair RM, Hanna ME, Hale TS (2021) Quantitative analysis for management, 13th edn. Pearson, Harlow
Slack N, Brandon-Jones A, Burgess N (2022) Operations management, 10th edn. Pearson, Harlow
Stadtler H, Kilger C, Meyr H (2015) Supply chain management and advanced planning, 5th edn. Springer, Berlin

References for Sect. 11.1

Hoot NR et al (2008) Forecasting emergency department crowding: a direct event simulation. Ann Emerg Med 52(2):117–125
Sun Y, Heng BH, Seow YT, Seow E (2009) Forecasting daily attendances at an emergency department to aid resource planning. BMC Emerg Med 9(1)
Thonemann U (2015) Operations management, 3rd edn. Pearson Studium, München

References for Sect. 11.2.1

Blue Yonder (2019). https://blueyonder.com/knowledge-center/collateral/dm-drugstore-case-study. Accessed 2 Jan 2025

Production and Material Requirements Planning

12

Learning Objectives for This Chapter

- Understand different planning horizons
- Understand the role of aggregate planning
- Explain different options of matching demand and supply at an aggregate level
- Understand the rolling planning concept
- Explain the master production schedule concept
- Apply linear programming to production planning
- Understand the principles of exploiting the bill-of-materials
- Compute dependent material requirements while taking lead time into account

12.1 Introductory Case Study: Integrated Operations and Supply Chain Planning

Let us look at the following practical case study to better understand the necessity for production and material requirements planning. Suresh Chandra has just received the demand forecast data for next year. He is new to the company and is responsible for operations planning. His previous experiences in different companies provided

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_12) contains supplementary material, which is available to authorized users.
Find additional case studies, Excel spreadsheet templates, and video streams in the *E-Supplement* to this book on www.global-supply-chain-management.de!
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_12.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_12

him with valuable knowledge and tools for aggregate planning, master production scheduling, and material requirements planning. In the new position, his responsibility was to integrate all these activities into the operations planning.

The previous operations planning manager applied the following concept. First, sales and production plans were set up very early (typically in September of the previous year), and then agreements with customers and suppliers were signed. The problem with this system was that production and material procurement were not flexible enough to react to demand fluctuations. In addition, since the structure of the products was set up far in advance, the possibility of product individualization according to customer orders was missing.

Janna, who is a trainee in the SCOM department, should support Suresh in the redesign of the operations planning system. Janna and Suresh understand that matching demand and supply should take place as close as possible to the date when a customer order or demand is known. Janna makes a suggestion: "We could make production planning on a weekly basis each Friday for the following week when customer orders and demands are known." Suresh saw two problems here. First, it would be difficult to plan material requirements since many purchased items had quite a long lead time. Second, frequent production setups are expensive and not always possible for technological reasons.

Then, Janna proposed analyzing which data should be set up for a long period of time (so-called design variables) and which data can be adjusted during the planning process (so-called control variables). This was a really good idea. Suresh and Janna invited marketing, sales, manufacturing, and procurement managers to a meeting and jointly analyzed the as-is planning process which can be described as follows.

A mid-term forecast is available for 18 months, making it possible to start operations planning along the SC. Operations planning is based on the rolling planning procedure. Before planning production and material requirements, capacity, budget, and further constraints should be taken into account. After planning the master production schedule (MPS) and transportation, products are delivered to customers.

Since demand forecasts, production, and supply plans are updated frequently, a rolling planning procedure is implemented. The rolling planning concept is based on creating a detailed MPS for the whole SC on a daily basis for 30–45 days. This MPS is updated every 15 days to take into account the dynamics of real execution.

They found that rough demand data and production quantities for each month must really be known in advance. It is necessary to plan capacities and personnel and to make basic agreements with customers and suppliers. At the same time, such data are not required in detail for each product. It would be enough to aggregate products into product families according to marketing and sales interests.

For manufacturing, it is important to know detailed data on production quantities at least 3–5 days before the beginning of the next month. This would be the basis for

calculating master production schedules among all the production factories in the SC, which are interconnected in the technological process. For procurement, such a timeline would also be okay for calculating material requirements and replenishing items from suppliers, while taking lead times into account.

Then, daily production plans can be set up. Suresh asked if it could be demonstrated that it would be possible to update production plans twice a month using new demand information. Janna also noticed that exact material specifications were not available for all products. This could create difficulties in procurement planning and result in missing items in production, and costly interruptions in manufacturing. Indeed, each product was composed of 2000 parts on average. Even if one part was missing, delivery delays would occur. Suresh asked if this problem could be rectified, and all the BOM (Bill-of-Materials) were entered into the ERP system.

In this chapter, we learn basic methods of operations planning, which will help Janna and Suresh increase profits and reduce costs.

12.2 Planning Horizons/MRP-II

Planning decisions can be divided into:

- Strategic
- Tactical
- Operative

Strategic decisions are taken for a long period of time, typically for many years. *Tactical* decisions use information from strategic decisions and determining planning policies for some months. *Operative* decisions are taken in a weekly or daily manner for executing processes. Such a decomposition is necessary since some data, which can be adjusted during the planning process (so-called control variables), should be set up for a long period of time (so-called design variables). Some examples of different decisions at the tactical and operative levels are presented in Fig. 12.1.

Manufacturing resource planning—MRP-II—means that, from a management point of view, tools are being used for planning the activities of all functions of a manufacturing company. In the planning process, MRP II means the translation of business plans and demand forecasts, including customer orders, into production plans and purchasing plans using planned orders in a real-time multilevel scheduling environment (see Fig. 12.2).

MRP-II is a *hierarchical planning* approach. This means that decisions from the upper levels are used as input data for the lower levels (as opposed to *simultaneous planning* which is more complex).

Based on requirements and constraints from finance, marketing, and manufacturing on the one side and demand forecasts along with capacity decisions on the other,

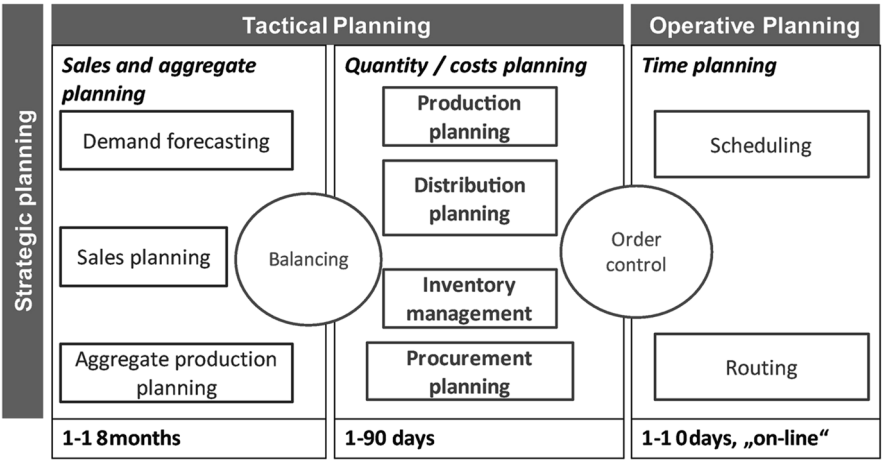


Fig. 12.1 Interrelations of strategic, tactical, and operation planning decisions

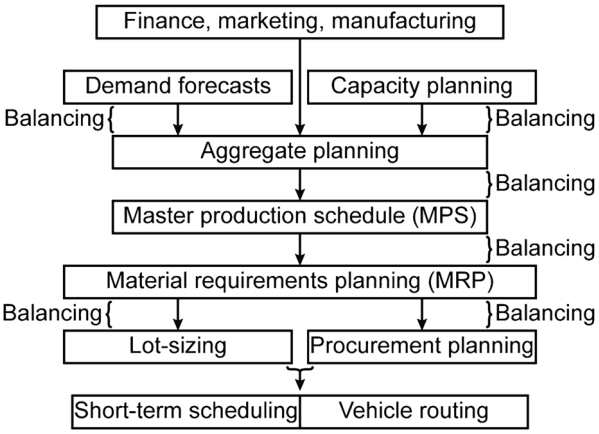


Fig. 12.2 MRP II process

aggregate planning is achieved. The results of aggregate planning are further used for master production scheduling (MPS), which is the base for material requirements planning (MRP), lot-sizing/procurement order quantity calculation, and finally for short-term scheduling and routing.

At each stage, *balancing* is mandatory to ensure that the results of the upper level may be implemented at the next level. Otherwise, incorrect calculations will be made throughout the planning levels. This could result in a situation where short-term daily schedules would have nothing in common with real demand; the wrong material could be purchased. In the following, we consider some important elements of the MRP-II concept.

12.3 Sales and Operations Planning

12.3.1 Role of Sales and Operations Planning

Production planning is usually done at an aggregate level, for both products and resources. *The objective of aggregate planning or so-called sales and operations planning (S&OP)* is to meet forecasted demand while minimizing cost over the planning period. The concept of S&OP is aimed at achieving a balance between demand and supply. Distinct but similar products are combined into aggregate product families that can be planned for together so as to reduce planning complexity.

Consider some examples. Zara produces more than 12,000 different product designs each year. Barilla produces more than 800 different kinds of pasta weekly. At BMW, more than 1000 different car variants are assembled daily. For such product variety, the first planning decisions should be made at the aggregate level in order to manage complexity.

Similarly, production resources, such as distinct machines or labor pools, are aggregated into an aggregate machine or labor resource. Care is required when specifying these aggregates to assure that the resulting aggregate plan can be reasonably disaggregated into feasible production schedules.

The quality of an aggregate plan has a significant impact on a firm's profitability. A poor aggregate plan can result in lost sales and profits if the available inventory and capacity are unable to meet demand. It may also result in a large amount of excess inventory and capacity, thereby raising costs.

The aggregate planner's main objective is to identify the following operational parameters over the specified time horizon (Chopra 2020; Heizer et al. 2023):

- Production rate: the number of units to be completed per unit time (such as per week or per month)
- Workforce: the number of workers/units of capacity needed for production
- Overtime: the amount of overtime production planned
- Machine capacity level: the number of units of machine capacity needed for production
- Subcontracting: the subcontracted capacity required over the planning horizon
- Backlog: the demand not satisfied in the period in which it arises, but carried over to future periods.
- Inventory on hand: the planned inventory carried over the various periods in the planning horizon

An aggregate planner requires the following information:

- A logical overall unit for measuring sales and output
- A forecast of demand for an intermediate planning period in these aggregate terms
- A method for determining costs

- A model that combines forecasts and costs so that scheduling decisions can be made for the planning period
- Capacity planning results

Using this information, a company makes the following determinations through aggregate planning:

- Production quantity from regular time, overtime, and subcontracted time: used to determine the number of workers and supplier purchase levels
- Inventory held: used to determine the warehouse space and working capital required
- Backlog/stock-out quantity: used to determine customer service levels
- Workforce Hired/Laid Off: used to determine any labor issues to be encountered
- Machine capacity increase/decrease: used to determine if new production equipment should be purchased or idled

In order to match demand and supply at the aggregate level, companies can:

- Use inventories to absorb changes in demand
- Accommodate changes by varying workforce size
- Use part-timers, overtime, or idle time to absorb changes
- Use subcontractors and maintain a stable workforce
- Change prices or other factors to influence demand

The application of these methods results in different options for aggregate planning.

12.3.2 Options for Aggregate Planning

In general, a company attempts to use its capacity and inventory to meet demand best. Therefore, the fundamental trade-offs relevant for a planner include:

- Capacity (regular time, overtime, subcontracted)
- Inventory
- Backlog/lost sales because of delay

There are three distinct aggregate planning options for achieving a balance between these costs. These options involve trade-offs between capital investment, workforce size, work hours, inventory, and backlogs/lost sales. Most options that a planner actually uses are a combination of these three and are referred to as tailored strategies. The three *options* are as follows:

- Capacity options
- Demand options
- Mixed options

Capacity options are based on the assumption that production capacity can be changed according to demand fluctuations. *First*, inventory levels can be changed to increase inventory in low-demand periods to meet high demand in the future. The shortcoming of such a policy is that it increases costs associated with storage, insurance, handling, obsolescence, and capital investment. In addition, shortages may mean lost sales due to long lead times and poor customer service.

Second, it is possible to change capacity by varying workforce size by hiring or layoffs. This allows the production rate to be matched to demand but creates training and separation costs from hiring and laying off workers. New workers may have lower productivity.

Third, varying the production rate through overtime is possible. This will allow for a constant workforce, but overtime can be costly and may drive down productivity. In addition, overtime space is very limited; it may be difficult to meet large increases in demand. *Fourth*, subcontracting can be used as a temporary measure during periods of peak demand. Usually, it is a costly option with coordination problems in terms of quality and delivery time.

Demand options are based on the assumption that it is possible to shift demand to slow periods or to use advertising or promotion to increase demand in low periods. Demand options include (1) longer lead times during high-demand periods and (2) mixing counter-seasonal products and services. In other words, demand options are used to find a balance between supply and demand by influencing demand. An anecdotal example can be given here as an illustration. In a Russian restaurant on Brighton Beach, the cook cries to the waiter "Pelmeni are ready!" The waiter responds "Nobody has ordered Pelmeni?!" The cook replies "Then go and ask the guests: maybe somebody would like to order them!"

Mixed options combine elements from capacity and demand options. The common mixed options are as follows:

- Chase strategy
- Level strategy

Chase strategy is based on using capacity as a lever. With the help of this strategy, we match output rates to the demand forecast for each period. Typically, we can vary workforce levels or production rates. Such a strategy is common in service organizations.

Level strategy is based on using inventory as a lever. With the help of this strategy, a stable workforce and machine capacity are maintained with a constant output rate. While daily production is steady, we use inventory as a buffer. Stable production tends to lead to better quality and productivity.

12.3.3 Methods for Aggregate Planning

Methods for aggregate planning can be based on the empirical knowledge of the planners or on mathematical models. Since the number of possible combinations of different methods is very high, a mathematical model would tend to lead to better results. However, in many cases, the construction of such models is difficult since many "soft" factors have to be taken into account. One basic quantitative method for S&OP is linear programming, which allows the optimal allocation of resources to demand subject to capacity constraints. We will consider linear programming in the course of this chapter. Here, we limit ourselves to consideration of empirical methods.

Task 12.1 Sales and Operation Planning
You have to prepare an aggregate plan for the production of skateboards which will cover nine periods (Table 12.1).

You have 20 workers with a capacity (output) of 10 skateboards each month. The production cost is \$6, the inventory holding cost is \$5, and the backlog cost is \$10 a unit. Overtime and subcontracting are not used. However, in this case, existing capacities cannot cover demand. You have the possibility to employ two additional workers. In order to avoid surplus production of 40 skateboards, you decide to hire the second worker for the first 5 months only. The cost of these two workers is \$500.

We make an aggregate plan and the costs, as shown in Table 12.2, are calculated under the following assumptions:

- Inventory costs are calculated for average inventory on hand; average inventory on hand is calculated as 50% from the sum of inventory at the beginning and end of each month.
- Our customer can wait: We can sell products in subsequent periods to cover the demand of the previous periods, but we calculate backlog costs for each period.

The total cost of this plan is \$14,790. In order to make any judgments on its quality, you have to consider at least two other options.

Option (1): You can introduce flexible capacities to cover monthly discrepancies between output and demand. This will cost you \$4000 additional. (Note: here, you do not need the table calculation.)

Answer: In this case, demand and production volumes are equal, and neither backlogs nor inventory exist. However, the total cost in this case is \$15,640. Therefore, this flexibility option costs more than the first option.

Table 12.1 Demand forecast

Month	1	2	3	4	5	6	7	8	9	Total
Demand forecast	190	230	260	280	210	170	160	260	180	1940

Table 12.2 Sales and operations plan

Month	1	2	3	4	5	6	7	8	9	Total
Demand forecast	190	230	260	280	210	170	160	260	180	1940
Production										
Regular Overtime Subcontract	220	220	220	220	220	210	210	210	210	1940
Shortage/ overage	+30	−10	−40	−60	+10	+40	+50	−50	+30	0
Inventory										
Beginning	0	30	20	0	0	0	0	20	0	
Ending	30	20	0	0	0	0	20	0	0	
Average	15	25	10	0	0	0	10	10	0	70
Backlog	0	0	20	80	70	30	0	30	0	230
Costs										
Production	1320	1320	1320	1320	1320	1260	1260	1260	1260	11,640
Inventory	75	125	50	0	0	0	50	50	0	350
Backlog	0	0	200	800	700	300	0	300	0	2300
Total costs	1395	1445	1570	2120	2020	1560	1310	1610	1260	14,290

Option (2) You decide on subcontracting instead of hiring two workers. The supplier is able to deliver 50 units in months 3, 5, and 7 at $7 a unit.

Answer: see Table 12.3.

This option is less efficient compared to the first one. In order to make subcontracting profitable, it should be used for the periods with the greatest backlogs (4)–(6).

12.4 Sales and Production Planning with Linear Programing

12.4.1 Problem Description

In S&OP, overall production quantities and capacity have been defined. The next step is to decide on the product mix within these aggregate volumes. The *objective function* is to maximize profit. The capacity is limited and demand is known: these are *constraints*.

Consider an example. The demand in January for plastic windows of types A and B is forecasted at 100 and 150 units, respectively. The available capacity is 1000 time units. The price is $60 for windows of type A and $50 for windows of type B. For one unit of A and B, five and four capacity units are needed. For production of A and B, components #1 and #2 are required in the proportion 40/60 and 70/30, respectively. These components are available in stock with 130 and 90 units, respectively. The production of A and B implies material and manufacturing costs of $48 and $35, respectively, for each unit.

Table 12.3 Results of S&OP calculation

Month	1	2	3	4	5	6	7	8	9	Total
Demand forecast	190	230	260	280	210	170	160	260	180	1940
Regular production	200	200	200	200	200	200	200	200	200	1800
Subcontracting production			50		50		50			150
Shortage/overage	+10	−30	−10	−80	+40	+30	+90	−60	+20	+10
Beginning inventory	0	10	0	0	0	0	0	50	0	
Ending inventory	10	0	0	0	0	0	50	0	10	
Average inventory	5	5	0	0	0	0	25	25	5	65
Backlog	0	20	30	110	70	40	0	10	0	280
Regular production costs	1200	1200	1200	1200	1200	1200	1200	1200	1200	10,800
Subcontracting production costs	0	0	350	0	350	0	350	0	0	1050
Inventory costs	25	25	0	0	0	0	125	125	25	325
Backlog costs	0	200	300	1100	700	400	0	100	0	2800
Total costs	1225	1425	1850	2300	2250	1600	1675	1425	1225	14,975

Consider another example. A wholesaler deals with only two sorts of juice. Sort A is handled in barrels of 20 L. Sort B is handled in full boxes of 10 L per box. Since no chemical additives are used, refrigerated storage is needed. The capacity of this refrigerator is 10,000 units. The maximal storage time is 1 month. The producer of juice A is able to deliver 450 barrels a month max. The producer of juice B is able to deliver 400 boxes a month, at a maximum. The profit from one barrel sold is $1 and for one box sold is $2.

Both companies have to decide on quantities A and B in order to maximize their profit subject to demand and capacity constraints. Such problems can be represented as linear optimization problems with some decision variables and constraints. Our objective in the paragraph is to learn how to:

- Represent such kinds of management problems in the form of an optimization model
- Solve this model analytically and graphically

Of course, in real life, the number of decision variables (i.e., the number of products) is very high. To simplify the explanations, we will consider cases with two variables (i.e., products A and B).

12.4.2 Method: Linear Programming

Linear programming (LP) is a widely used quantitative analysis tool that supports SCOM decisions about allocating available resources subject to demand in order to maximize profit or minimize costs (e.g., in the transportation problem). The following is characteristic for LP models and describes its requirements and assumptions:

- An objective function that describes the management goal (e.g., profit maximization)
- Constraints that limit the degree to which we can pursue our objective
- Alternative ways for resource allocation
- Both objective and constraints must be expressed as linear functions.

Let us formulate the LP models for the two cases described above. For the case with plastic windows, the objective is to maximize the profit. We can state the following general objective function (Eq. 12.1):

$$Z = \sum_{i=1}^{n} p_i \cdot x_i \rightarrow \text{ or } Z = p_A \cdot x_A + p_B \cdot x_B \rightarrow \max! \tag{12.1}$$

where p_A and p_B and x_A and x_B are the profits and quantities of products A and B, respectively.

Since the price for A is \$60 and the variable costs are \$48, p_A = \$12. Similarly, p_B = \$15. We can rewrite the objective function as follows (Eq. 12.2):

$$Z = 12x_A + 15x_B \rightarrow \max! \tag{12.2}$$

What is the maximal possible value of the objective function? Without having any constraints, the profit can be unlimited if we decide to produce an unlimited quantity of product B. However, we have limited demand and capacity and have, therefore, to consider a constraint system. The solution to this constraint system will allow us to find quantities such as x_A and x_B so that the Z-value will be the maximum possible profit under the considered constraints, prices, and costs.

The constraint system ~~is~~ in general form can be written as follows (Eqs. 12.3–12.5):

$$\text{Production capacity}: \sum_{i=1}^{n} c_i \cdot x_i \leq C_{\text{total}} \text{ or } C_A \cdot x_A + C_B \cdot x_B \leq C_{\text{total}} \tag{12.3}$$

$$\text{Demand}: x_A \text{ " } D_A; x_B \text{ " } D_B \tag{12.4}$$

$$\text{Non-negativity constraint}: x_A, x_B \geq 0 \tag{12.5}$$

where:

C_A and C_B are the number of capacity units required for the production of one unit of A and B, respectively.

C_{total} is the overall available capacity.

D_A and D_B are the demand for A and B, respectively.

The constraint system for the case with plastic windows can be, therefore written in the following way (Eqs. 12.6–12.9):

$$\text{Production capacity}: 5x_A + 4x_B \leq 1000 \tag{12.6}$$

$$\text{Demand}: x_A \text{ ″ } 100; x_B \text{ ″ } 150 \tag{12.7}$$

$$\text{Non-negativity constraint}: x_A, x_B \geq 0 \tag{12.8}$$

Additional constraints for components:

$$\text{Component } 1: x_A + 0.7x_B \leq 130; \text{Component } 2: 0.6x_A + 0.3x_B \leq 90 \tag{12.9}$$

Similarly, we can formulate the LP model for the case with the juice wholesaler (Eqs. 12.10–12.14):

$$\text{Objective function}: Z = 1x_A + 2x_B \rightarrow \max! \tag{12.10}$$

$$\text{Storage capacity}: 20x_A + 10x_B \leq 10{,}000 \tag{12.11}$$

Supply:

$$x_A \text{ ″ } 450 \tag{12.12}$$

$$x_B \text{ ″ } 400 \tag{12.13}$$

$$\text{Non-negativity constraint}: x_A, x_B \geq 0 \tag{12.14}$$

▶ **Practical Insights** For such LP models, different solution techniques exist. In real-life cases, special software, such as CPLEX, Lindo, AMPL, Matlab, GAMS, Gurobi, XPRESS, etc., are used. All these software apply solution algorithms based on the *simplex method* and modern advanced computational techniques. LP is really a working tool used in practice.

12.4.3 Graphical Solution

In this paragraph, we will consider a graphical solution method for LP models. First, we must identify a set of feasible solutions. To do this, we plot the constraints on a graph. The variables x_A and x_B are plotted as the horizontal and vertical axes, respectively. Consider the solution to the case with plastic windows. To plot the line for Eq. (12.6), we need to define the points where the line $5x_A + 4x_B \leq 1000$ intersects the x_A and x_B axes. When $x_A = 0$, $x_B = 250$. When $x_B = 0$, $x_A = 200$. Similarly, we do this for the lines subject to Eq. (12.9), see Fig. 12.3.

It can be observed that four lines form a *feasible region*: demand A, demand B, component #1, and component #2. According to the *corner-point solution* method, the optimal solution belongs to one of the corner points in the feasible regions. For example, take point (100;0). For this point, the profit is $12 \cdot 100 + 15 \cdot 0 = \$1200$. We have to calculate the profit at different corner points. Having done this, we will

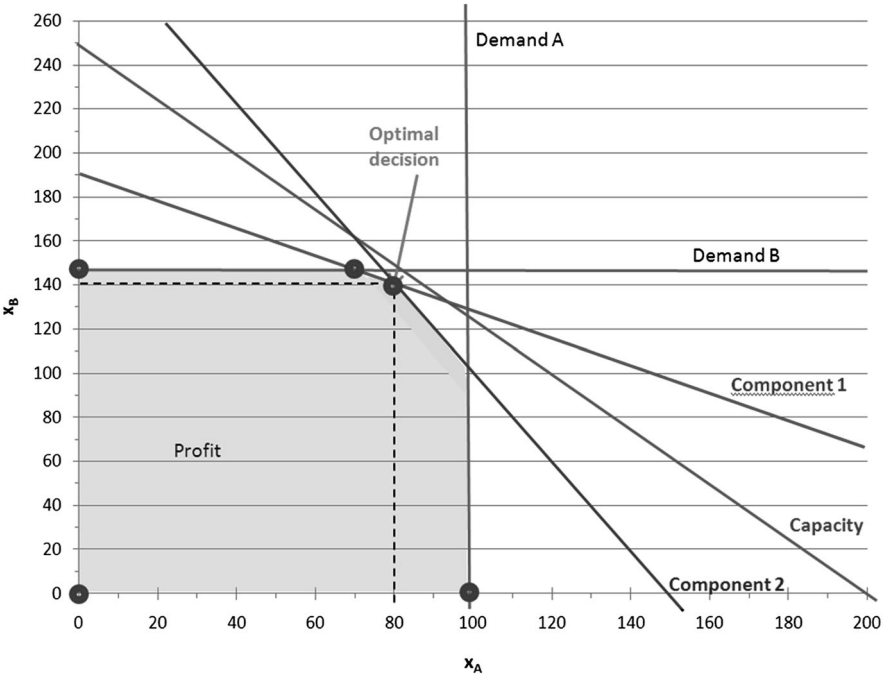


Fig. 12.3 Graphical solution to the LP problem with the help of the corner-point method

get maximal profit at $x_A = 80$ and $x_B = 140$ units. The objective function is, in this case, $12 \cdot 80 + 15 \cdot 140 = \$3060$. We can also observe that only 960 of 1000 available capacity units will be used. This is because of the limited component quantities.

Consider the second case about the juice wholesaler. We plot the constraints in the same way as for the first case but solve this task with the help of another method, namely the *iso-profit line* method (see Figs. 12.4 and 12.5). We define the decision variables for products A and B as x_1 and x_2, respectively. The constraints (12.11)–(12.13) are marked as NB1–NB3, respectively.

Having defined the feasible region, the *iso-profit line* is now used to find the optimal solution. We start by letting the profit be equal to some arbitrary amount, that is, selecting a relatively small value. In our example, letting the profit be \$500, we get the iso-profit line G_{500} (see Fig. 12.5).

The farther we move the iso-profit line from the origin, the higher our profit will be. In our case, the highest possible profit of \$1100 will be achieved at $x_1 = 300$ and $x_2 = 400$.

▶ **Practical Insights** In SCOM, managers are typically interested not only in the optimal solution, but also in an assessment of its robustness, since many risks and uncertainties can influence the input parameters, such as

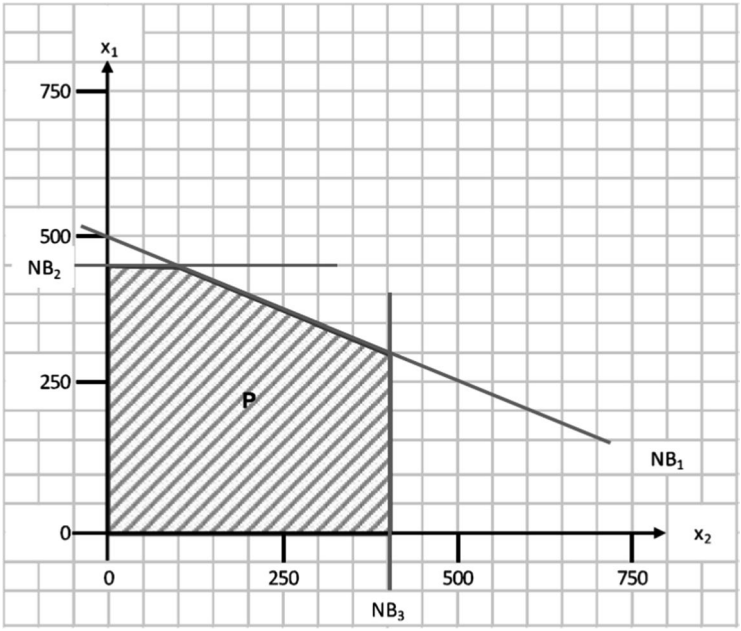


Fig. 12.4 Feasible region in the LP problem

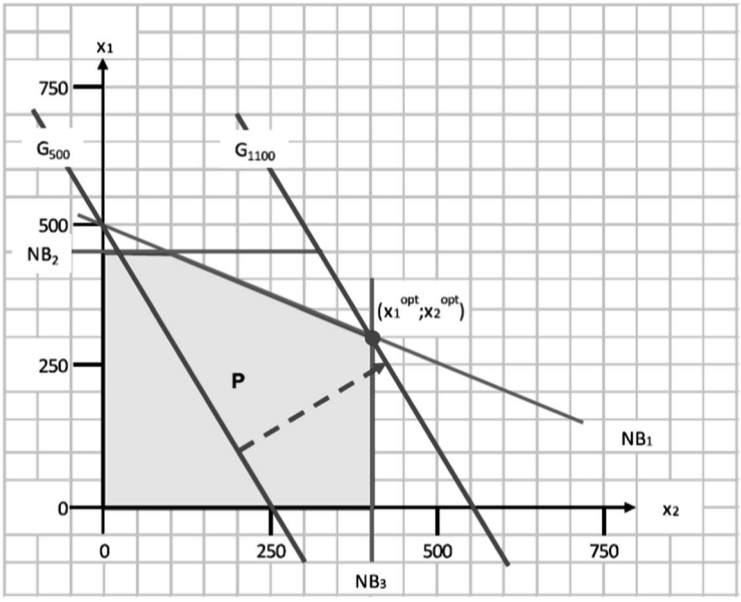


Fig. 12.5 Solution to the LP problem with the help of iso-profit line

capacity, material supply, and demand. An analysis of how sensitive the optimal solution is regarding changes in demand and supply is called a *sensitivity analysis*. Sensitivity analysis is a standard tool in LP solvers.

12.5 Master Production Schedule and Rolling Planning

12.5.1 Master Production Schedule

Master production schedule (MPS) specifies what is to be made and when in accordance with the aggregate production plan. The MPS is the main result of the production planning process. If aggregate planning works with product families, MPS disaggregates them. MPS is established in terms of specific products. For example, aggregate planning determined that 2000 tables should be produced in August. MPS determines that in the first week 600 tables, in the second week 400 tables, in the third week 700 tables, and in the fourth week 300 tables will be produced.

▶ **Practical Insights** In practice, the MPS is quite often fixed or "frozen" in the near-term part of the plan. For example, in the automotive industry, a so-called *pearl chain* concept is used. The sequence of car production is fixed for 7 days and remains unchanged throughout the whole production cycle. This allows the efficiency of manufacturing to be increased, capacity balancing to be improved, and the planning process made more convenient for managers.

12.5.2 Rolling Planning

Planning at the MPS level is typically done in a rolling or (adaptive) horizon fashion. *Rolling planning* is a continuous, event-driven, real-time (re)planning, and control process. It uses not only simple open time slots (in contrast to incremental planning) but also employs conflict-driven plan changes during system execution (see Fig. 12.6). Rolling (or adaptive) planning is a method of planning in which a plan is modified periodically by a change in the system parameters or the characteristics of control influences the basis of information feedback about a current system state, the past, and the updated forecasts for the future.

In *adaptive planning*, the precision of planning decisions decreases moving away from the decision point. Planning decisions become fuzzier with increasing distance from the decision-making point. This may be interpreted as a sequence of inter-inserted funnels. At certain intervals, the plans are updated, the "fuzzy" part of the plans becomes precise, and a new funnel for the further process course appears. Such an approach provides the required flexibility with regard to dynamics and uncertainty. Hence, adaptive planning implies problem resolution and redefinition through the learning process, rather than problem-solving. This allows us to interpret planning and scheduling not as discrete operations, but as a continuous adaptive

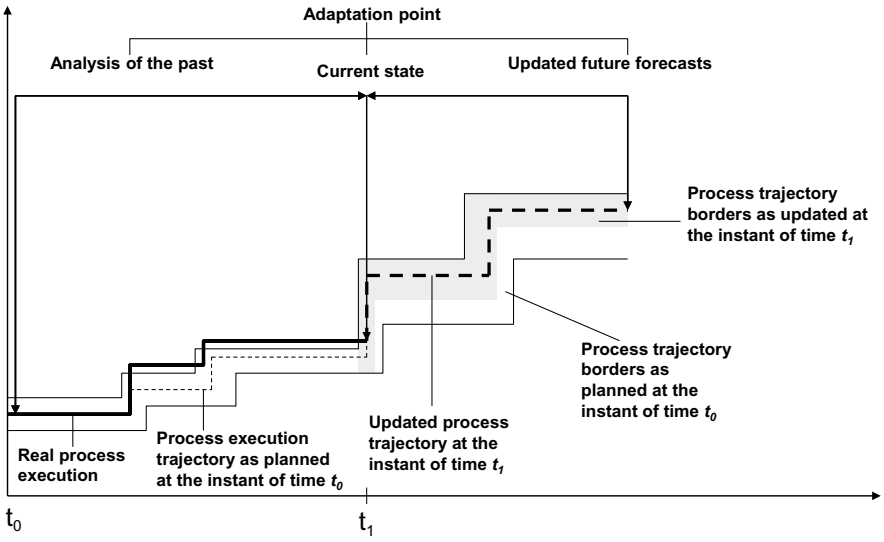


Fig. 12.6 Adaptive planning logic

process. The adaptive planning procedure can be interconnected with the adaptive procedure. So we can improve reactiveness in disruption management in SCs.

If sufficiently full information about the system and operation dynamics is not available, certain key operations and events must be determined. The uncertainty of states does not allow operation plans to be formed. As a result, a number of alternative operation dynamics scenarios can be elaborated. Further, with time, the uncertain (at the time of planning) operations' states become certain and possess certain characteristics (parameters).

12.6 Material Requirements Planning

Each of us sometimes buys furniture which should be assembled at home. For assistance, we get directions that explain to us:

- Which items are contained in the package and in what quantity
- How these items can be assembled
- Which items are necessary for subassembly
- In which sequence we should perform subassemblies and final assembly
- How much time is needed for subassemblies and final assembly

You can observe from the assembly instructions that:

- We have a list of materials needed and their interrelations.
- We can determine the quantity of items required for furniture assembly.

In terms of operations management, we would describe this case as follows:

- We have a *bill-of-materials* (BOM).
- We have performed *material requirements planning* (MRP).
- We have *standard operating instructions*.

If the demand for one item is related to the demand for another item, we call it *dependent demand*. Given a quantity for the end item, the demand for all parts and components can be calculated. For any product for which a schedule can be established, dependent demand techniques should be used. The most common techniques in this area are BOM and MRP.

12.6.1 Bill-of-Materials

First, the BOM is a list of the components, ingredients, and materials needed to make a product. Second, the BOM provides a product structure (see Fig. 12.7). This structure can be rewritten more simply if we depict the number of items required for one unit at the upper level (see Fig. 12.7). From the product structure, we can determine how many items are required at different levels of the BOM (Fig. 12.8) (see Table 12.4).

This is important for synchronizing manufacturing and procurement since we can produce items at the upper level only after the items from the lower level have been purchased or manufactured.

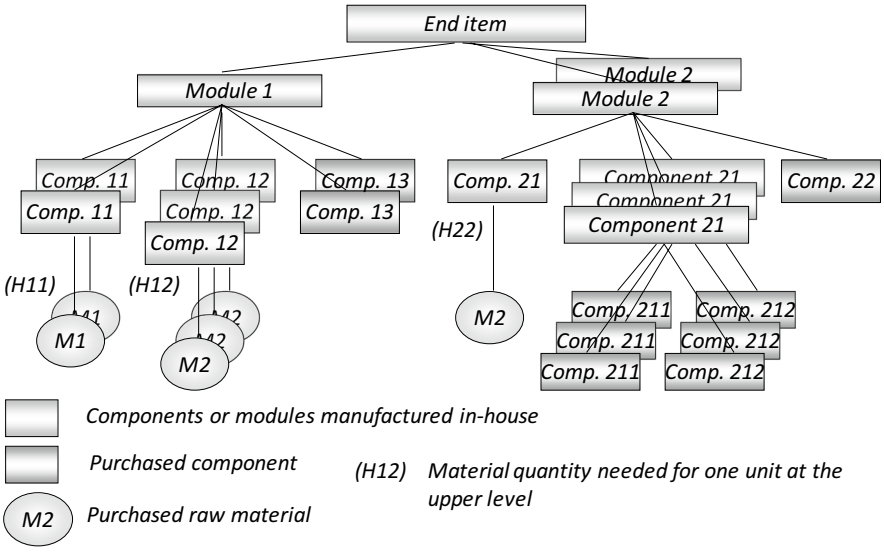


Fig. 12.7 Structure of an end product

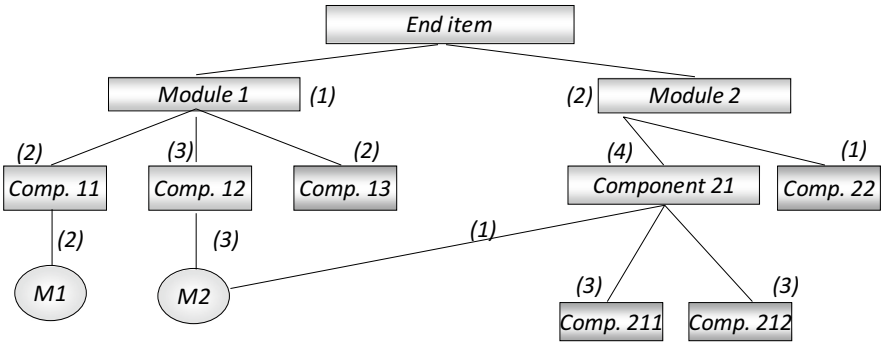


Fig. 12.8 Structure of an end product with item quantities

From the product structure, we can also determine the total quantity of all items. If we perform this procedure, we *explode the BOM*. For the given structure, we can explode the BOM as shown in Table 12.5.

Consider as an example the total requirements for material #2 (M2). Three pieces are needed for one component #12. In turn, three pieces of component #12 are required for one module #1, and one module #1 is used for one end item. So the dependent demand on M2 is now $3 \cdot 3 \cdot 1 = 9$ pieces. In addition, one piece of M2 is needed for one component #21, and four components #21 are used for one module #2. Two modules #2 are required for one end item. So the dependent demand on M2 is now $9 + 1 \cdot 4 \cdot 2 = 17$ pieces.

Exploding the BOM is important for manufacturing and procurement and determining the aggregate quantities for production and purchasing. This data is needed for production and procurement planning.

Table 12.4 Level BOM

Level				Item	Quantity
0				End item	1
	1			Module 1	1
		2		Component 11	2
			3	Material 1	2
		2		Component 12	3
			3	Material 2	3
		2		Component 13	2
	1			Module 2	2
		2		Component 21	4
			3	Component 211	3
			3	Component 212	3
			3	Material 2	1
		2		Component 22	1

Table 12.5 Explode the BOM

Item	Quantity
End item	1
Module 1	1
Module 2	2
Component 11	2
Component 12	3
Component 13	3
Component 21	8
Component 22	2
Component 211	24
Component 212	24
Material 1	4
Material 2	17

▶ **Practical Insights** In practice, BOM calculation is a complex task. A car could be composed of nearly 9100 items. Six hundred kilograms of steel and 25 kg of color are needed. An aircraft is composed on average of 3–six million items. Even if only one item is missing, delivery and production delays can occur. BOM and MRP calculations are automated in the ERP system. For really complex products, these procedures are integrated within CAD (computer-aided design) and PDM (product data management) systems. In addition, so-called *module BOMs* are typical for complex products. This means that the BOM is created separately for each module.

12.6.2 MRP Calculation

With the help of the BOM, the product structure is defined. However, the manufacturing and procurement of items do not happen immediately. This requires a certain *lead time*. For production, lead time is the sum of the order, wait, move, setup, store, and run times. For purchased items, lead time is the time between the recognition of a need and the availability of the item for production. The lead time is taken into account to calculate MRP.

For MRP calculations, the following data are required:

- MPS
- BOM
- Inventory availability
- Purchase orders outstanding (scheduled receipts)
- Lead/manufacturing times for each item

On the basis of this data, the MRP system calculates orders for manufacturing and purchasing (see Fig. 12.9).

The *MRP calculation* is based on the following algorithm:

1. From exploding the BOM, so-called *gross requirements* on items are defined beginning with the upper level and going down to the lower levels.
2. Taking into account inventory availability and purchase orders outstanding (scheduled receipts), *net requirements* can be determined.
3. Net requirements = Gross requirement (inventory availability + scheduled receipts).
4. Finally, net requirements are used to determine the actual order release time, which is subject to lead time. This step is often called "lead time offset" or "time phasing."

Consider an example.

Task 12.2 MRP Calculation

An MPS for an end product is composed in Table 12.6.

We are given the following BOM (Fig. 12.10):

We assume that no inventory is available and no scheduled receipts are planned. Lead time for all items is 1 month. In this case, *gross requirements equal net requirements* and the following represents the results of the MRP calculations (see Table 12.7).

We start with the net requirements for an end item. This data is taken from the MPS. The next step is to determine the net requirements for the items at the first level. Since for all items exactly one piece is needed for one end item, their net requirements are identical to those of the end item. Net requirements for the items

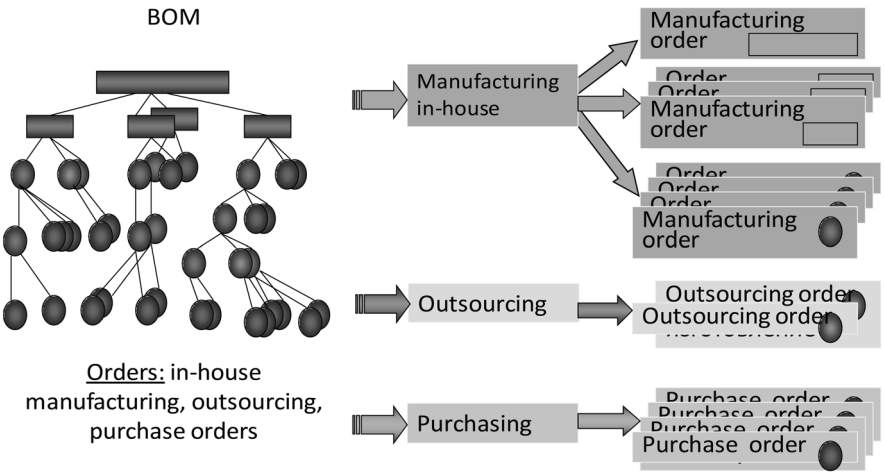


Fig. 12.9 Structure of MRP system

Table 12.6 MPS for an end product

April	May	June	July
10	10	20	50

Fig. 12.10 BOM

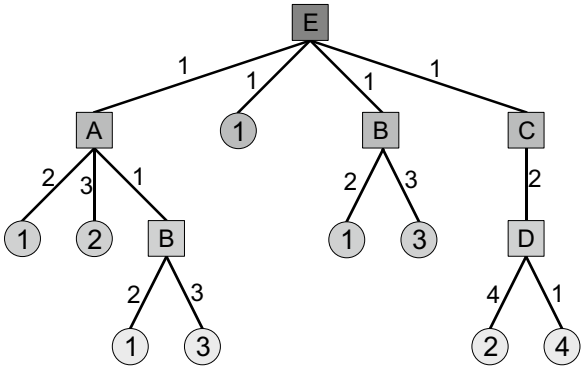


Table 12.7 MRP calculation

Level	Item	Months						
		1	2	3	4	5	6	7
0	E				10	10	20	**50**
1	A			10	10	20	**50**	
	B			10	10	20	**50**	
	C			10	10	20	**50**	
	1			10	10	20	**50**	
2	B		10	10	20	**50**		
	D		20	20	40	**100**		
	1		40	40	80	**200**		
	2		30	30	60	**200**		
	3		30	30	60	**150**		
3	1	20	20	40	**100**			
	2	80	80	160	**400**			
	3	30	30	60	**150**			
	4	20	20	40	**100**			

at the first level should be shifted according to 1 month of lead time. Similarly, net requirements for the items at the second level are determined subject to the net requirements of the items at the first level and lead-time shifting.

For example, 20 units of item D in period 2 means that we have to manufacture 20 units of item D in February in order to be able to assemble 10 units of end item E in April. By processing the BOM by level, items with multiple parents are typically only processed once, saving time and resources and reducing confusion. We transform Table 12.7 as in Table 12.8.

Table 12.8 MRP calculation

Item		Months						
	1	**2**	**3**	**4**	**5**	**6**	**7**	
E				10	10	20	50	
A			10	10	20	50		
B		10	20	30	70	50		
C			10	10	20	50		
D		20	20	40	100			
1	20	60	90	190	220	50		
2	80	110	190	460	150			
3	30	60	90	210	150			
4	20	20	40	100				

Table 12.9 Initial data

Period	8	9	10
Demand	125	90	110

The above transformation is based on aggregating the net requirements of the same items over different levels in each period. For example, in period #2, we aggregated net requirements for item #2 from the second level (30) and third level (80). The net requirement for item #2 in the second period is 110 pieces. This information is important for purchasing planning for February.

Now we are going to include inventory and schedule receipts into consideration. Assume the following problem statement.

Task 12.3 MRP Calculation

You are given the MPS for table production as in Table 12.9.

Lead time for the table production is one period. To produce one table we need four legs and one plate. For tables, you have received an additional demand of 25 units in period 8, 10 units in periods 9 and 10, respectively. We have 35 tables, 25 plates, and 32 legs as beginning inventory. The retailer announced that it will send back seven tables in period 8 and five tables in period 10. For plate production, we need two periods, and one period is needed for leg production.

First, we perform the calculations for the tables (lead time = 1) (see Table 12.10).

Managers frequently use so-called closed-loop MRP systems. These systems allow production planners to move work between time periods to smooth the load or at least bring it within capacity.

In practice, managers need to know how to group "planned order releases." Indeed, we can order 400 legs for June as one large order, or split this order into four orders of 100 units in each order. Here the trade-off is between *ordering fixed costs* and *inventory holding costs*. We will consider order quantity planning and *lot-sizing* decisions in Chap. 13.

Table 12.10 MRP calculation

Period	4	5	6	7	8	9	10
Demand					125	90	110
Additional demand					25	10	10
Gross requirement					150	100	120
On-hand inventory					35	0	0
Scheduled receipts					7	0	5
Net requirement					108	100	115
Planned order release				108	100	115	

Consider the calculations for plates (lead time = 2)

Period	4	5	6	7	8	9	10
Gross requirement				108	100	115	
On-hand inventory				25	0	0	
Scheduled receipts				0	0	0	
Net requirement				83	100	115	
Planned order release		83	100	115			

Consider the calculations for legs (lead time = 1):

Period	4	5	6	7	8	9	10
Gross requirement				432	400	460	
On-hand inventory				32	0	0	
Scheduled receipts				0	0	0	
Net requirement				400	400	460	
Planned order release			400	400	460		

12.7 Key Points

Planning decisions can be divided into strategic, tactical, and operative. MRP II means the translation of business plans and demand forecasts, including customer orders, into production and purchasing plans. The objective of aggregate planning or so-called sales and operations planning (S&OP) is to meet forecasted demand

while minimizing costs over the planning period. The concept of S&OP is aimed at achieving a balance between demand and supply.

In general, a company attempts to use its capacity and inventory to best meet demand. Therefore, the fundamental trade-offs relevant to a planner include capacity, inventory, and backlog/lost sales because of delay. Three *options of S&OP* are capacity options, demand options, and mixed options. Mixed options combine elements from capacity and demand options. Common mixed options are a chase strategy and level strategy.

MPS specifies what is to be made and when in accordance with the aggregate production plan. Planning at the MPS level is typically done in a rolling or (adaptive) horizon fashion. If demand for one item is related to the demand for another item, we call it dependent demand. The most common techniques in this area are BOM and MRP. BOM represents a list of components, ingredients, and materials needed to make the product. Second, in the BOM, the product structure is defined. However, the manufacturing and procurement of items do not happen immediately. This requires a certain *lead time*. Lead time is taken into account while calculating MRP. The MRP system calculates orders for manufacturing and purchasing.

Bibliography

Chopra S (2020) Supply chain management. Strategy, planning and operation, 7th edn. Pearson, Harlow

Heizer J, Render B, Munson C (2023) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow

Hendricks KB, Singhal VR (2014) The effect of demand-supply mismatches on firm risk. Prod Oper Manag 23(12):2137–2151

Mangan J, Lalwani C, Butcher T, Javadpour R (2012) Global logistics and supply chain management, 2nd edn. Wiley, New York

Silver EA, Pyke DF, Thomas DJ (2017) Inventory and production management in supply chains, 4th edn. Taylor & Francis Inc, New York

Thomé AMT, Sousa RS, do Carmo LFRRS (2014) The impact of sales and operations planning practices on manufacturing operational performance. Int J Prod Res 52(7):2108–2121

Wallace TF, Stahl RA (2008) Sales and operations planning, 3rd edn. TF Wallace, Cincinnati

Inventory Management

13

Learning Objectives for This Chapter

- Understand the trade-off "service level vs. costs" in inventory management
- Understand the role of inventory in the supply chain
- Conduct ABC and XYZ analyses
- Explain and use the EOQ/EPQ models for independent inventory demand
- Compute a reorder point
- Calculate service levels and probabilistic inventory models
- Explain and use the dynamic lot-sizing models
- Understand and compute the effects of inventory aggregation
- Explain ATP/CTP concept
- Explain inventory turnover and on-time delivery (OTD) performance indicators

13.1 Introductory Case Study: Amazon, Volkswagen, and DELL

The trade-off "service level vs. costs" is one of the most important issues in inventory management. Its resolution strongly depends on manufacturing, sourcing, distribution, and SC strategies. Consider three examples.

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_13) contains supplementary material, which is available to authorized users.
Find additional case studies, Excel spreadsheet templates, and video streams in the E-Supplement to this book on www.global-supply-chain-management.de!
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_13.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_13

Amazon
Founded in 1994, Amazon.com, Inc. is the world's largest online retailer. It started as an online bookstore but has diversified over the last 20 years selling DVDs, CDs, MP3, software, games, furniture, food, toys, and more. The basic idea is that everybody with access to the internet can buy a massive range of products at any time and get them delivered wherever they want from (nearly) anywhere in the world. Amazon apps make it possible to buy their products easily with smartphones or tablets. Their inventory management for books is based on the number of sales for each book. Best-selling books are stocked in many regional warehouses to enable fast responsiveness for customers in any region. Books with a smaller number of sales are stocked in only a few regional warehouses to reduce inventory costs. The slow-moving books are not even held in stock but are obtained directly from the publisher or distributor. Through this strategy, it is possible for Amazon to reduce inventory costs for slow-moving products and to have the best possible service level and reliability for their customers.

Volkswagen
Founded in Germany in 1937, Volkswagen implemented the JIT-based SC strategy in Germany. This means that components and items are delivered to the assembly line when they are needed. Only a small amount of safety stock is held for critical parts. This strategy helps to reduce inventory, transport, labor, and administration costs. Often, suppliers locate their plants close to the manufacturer to deliver the required components and items at the right moment and reduce their transport costs. This choice of location reduces the risk of delivery delays and supply shortages. Examples of these JIT components are dashboards with all the electronic devices or seats. JIT is normally used for A-, B-, X-, and Y-items. For C-items, Volkswagen uses a Kanban replenishment system so that they are in stock at any time. These items normally have low inventory holding costs. Based on the JIT strategy and Kanban replenishment system, Volkswagen is able to minimize inventory costs.

DELL
Founded in the USA in 1983, DELL implemented an MTO production strategy. The assembly starts and materials are replenished only after a customer orders. In such a business model, DELL is able to drastically reduce its inventory. This allows Dell to respond to a wide range of demand quantities, meet short lead times, handle a large variety of products, build innovative products, and handle supply uncertainty.

13.2 Role, Functions, and Types of Inventory

The *role of inventory management* is to strike a balance between inventory investment and customer service. Inventory is one of the most expensive assets of many companies, representing as much as 50% of total invested capital. In SCOM we must therefore balance inventory investment and customer service level.

Inventory has different *functions*, e.g.:

- To decouple the company and the SC from fluctuations in demand and hold a stock of goods that will provide a selection for customers
- To increase SC flexibility by placing inventory in the right places
- To hedge against facility disruptions in the event of natural catastrophes, strikes, or supplier bankruptcies.
- To decouple or separate various parts of the production process
- To take advantage of quantity discounts and hedge against inflation

Typically, inventory is classified according to the following *types*:

- *Raw material* (items which are purchased but not processed)
- *Work-in-process (WIP)* (items which underwent some changes, but are not completed)
- *Maintenance/repair/operating (MRO)* (items which are necessary to keep machinery and processes productive)
- *Finished goods* (completed product awaiting shipment)

In making decisions in the scope of inventory management, the following two *basic questions* are put to the forefront for consideration:

- How much should I replenish?
- When should I replenish?

In calculating inventory amounts, the following *costs* are typically considered:

- Holding costs (variable)—the costs of holding inventory over time
- Ordering costs (fixed)—the costs of placing an order and receiving goods
- Setup costs (fixed)—the costs of preparing a machine or process for manufacturing an order
- Stock-out costs (variable)—the costs of lost customer orders resulting from product shortage, loss-of-goodwill costs

Holding costs include operating costs, labor costs, and material handling costs, and vary considerably depending on the business, location, and interest rates. Generally, these costs are greater than 15% of the item price, but some items have holding costs higher than 40%. Holding costs depend on the order quantity.

Setup costs (in manufacturing) and ordering costs (in procurement) are fixed and do not depend on the order quantity. These costs tend less for e-business developments. At the same time, setup/ordering costs can be increased in the event there is a high variety of assortment and market dynamics.

According to inventory functions and types, inventory can be used to *manage:*

- Economy of scale—this is *cycle inventory*.
- Uncertainty—this is *safety inventory*.

Cycle inventory exists as a result of producing or purchasing in large *lots* or *batches*. A lot or *batch size* is the quantity that a stage in the SC either produces or purchases at a time. The SC can exploit the economy of scale and order in large lots to reduce fixed costs. With the increase in lot size, however, comes an increase in carrying costs. As an example of a cycle stock decisions, consider an online book retailer. This retailer's sales average around ten truckloads of books per month. The cycle inventory decisions the retailer must make include how much to order for replenishment and how often to place these orders. We will consider cycle inventory optimization in the "Deterministic models" section (for one period) and "Dynamic lot-sizing models" section for many periods.

Safety inventory is carried to satisfy demand subject to unpredictable demand fluctuations and to reduce product shortages. Safety inventory can help the SC manager improve product availability in the presence of uncertainty. In the presence of safety inventory, shortage costs or overage costs can occur. The calculation of safety inventory is based on a predetermined *service level*. Choosing safety inventory involves making a *trade-off* between the costs of having too much inventory and the costs of losing sales due to inventory shortage. We will consider methods to support decisions on safety inventory in the "Stochastic models" section.

For many industries, some products are sold in high quantities in the summer, and lower quantities in the winter (e.g., mineral water). *Seasonal inventory* is built up to counter predictable variability in demand. Companies use seasonal inventory to increase product availability in periods of low demand and store it for periods of high demand when they will not have the capacity to produce all that is demanded. Managers face key decisions in determining whether to build up seasonal inventory and, if they do build it up, in deciding how much. If a company can rapidly change the rate of its production system at a very low cost, then it may not need seasonal inventory, because the production system can adjust to a period of high demand without incurring large costs. The basic trade-off SCOM managers face in determining how much seasonal inventory to build up is the cost of carrying the additional seasonal inventory vs. the cost of having a flexible production rate.

▶ **Practical Insights** It is intuitively clear that car tires and pins have different inventory management policies. However, in a distribution center, which runs over 2 million items, it can be quite a complicated task to find the right policy for each item. Many companies that ask for inventory optimization want to start immediately with software and mathematical models. But before starting any calculations, items should be properly analyzed and classified. False organization of inventory management is in many cases the key point of problems in optimization.

13.3 Material Analysis

13.3.1 ABC Analysis

The first step in item classification is the *ABC analysis*. ABC analysis divides inventory into three classes based on annual dollar volume:

- Class A—high annual dollar volume
- Class B—medium annual dollar volume
- Class C—low annual dollar volume

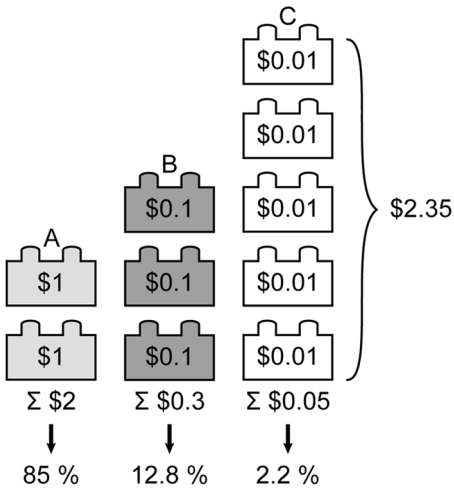
ABC analysis is used to establish policies that focus on the few critical parts and not the many trivial ones. This method implements the Pareto Principle which states that "there are few critical and many trivial." Consider a simple example. Imagine that you have ten Lego blocks: two of them are red, three green, and five blue (see Fig. 13.1). To perform the ABC analysis, we measure the annual demand of each inventory item times the cost per unit. Assume that one red block costs \$1, one green block costs \$0.1, and one blue block costs \$0.01.

You can easily calculate that total inventory costs are \$2.35. Two red blocks take only 20% of the total inventory amount, but they create 85% of inventory costs. These are critical items in the A group. A-items have to be managed especially carefully. First, the relationships with suppliers of A-items must be managed, i.e., supplier development. Second, tighter physical inventory control for A-items is necessary. Third, we need more care in forecasting demand for A-items.

Consider a numerical example for ABC analysis.

Task 13.1 ABC Analysis
Quarted Ltd. is a company which sells table lamps.

Fig. 13.1 ABC analysis



In Tables 13.1 and 13.2, a numerical example is provided.

We are given annual demand and the costs. To determine which table lamps are in the A, B, or C category, we need to calculate the annual expenditure:

$$\text{Annual expenditure} = \text{annual demand} \cdot \text{costs per lamp} \tag{13.1}$$

The next step is to arrange the different types of table lamps according to their annual expenditure. After calculating cumulative expenditure, we have to compute the percentage of cumulative expenditure. Then we can classify the A, B, and C items. Assuming that the classification is 80:15:5 we can label the different table lamps as follows (see Table 13.2).

We can observe that 150 table lamps generate around 80% of inventory costs (A), whereas 600 table lamps generate only 5% of inventory costs (C). This relationship can be presented in graphic form as shown in Fig. 13.2.

From Fig. 13.2 it can be observed that 80% of expenditure is created by only 20% of demand.

13.3.2 XYZ Analysis

The monetary value of capital commitment is not the only criterion for classifying materials. Other criteria include the physical volume of items, demand patterns, or delivery times. For example, in terms of the physical volume of items, it is possible to classify unwieldy items in group X and small and handy items in groups Y and Z. The principle is the same as for the ABC analysis. Another option for XYZ analysis is to divide inventory into three classes based on different demand patterns:

- Class X: constant, non-changing demand
- Class Y: neither constant nor sporadic demand (fluctuating demand)
- Class Z: sporadic or strongly fluctuating demand

Changes in demand make it possible to determine the prediction accuracy of each inventory class. Information on XYZ analysis provides an opportunity to

Table 13.1 Initial data for ABC analysis

Table lamp	Annual demand	Cost per unit	Annual expenditure
X1	100	0.5	50
X2	200	0.05	10
X3	50	1.65	82.5
Y1	40	10.75	430
Y2	200	0.11	22
Y3	200	0.19	38
Y4	50	2.4	120
T1	90	0.6	54
T2	10	13.6	136
T3	60	1.35	81

Table 13.2 Results of ABC analysis

Table lamp	Annual demand	Cost per unit	Annual expenditure		Cumulative expenditure	Percentage expenditure	Category
Y1	40	10.75	430		430	42.0	A
T2	10	13.6	136		566	55.3	A
Y4	50	2.4	120		686	67.0	A
X3	50	1.65	82.5		768.5	75.1	A
T3	60	1.35	81		849.5	83.0	B
T1	90	0.6	54		903.5	88.3	B
X1	100	0.5	50		953.5	93.2	B
Y3	200	0.19	38		991.5	96.9	C
Y2	200	0.11	22		1013.5	99.0	C
X2	200	0.05	10		1023.5	100.0	C

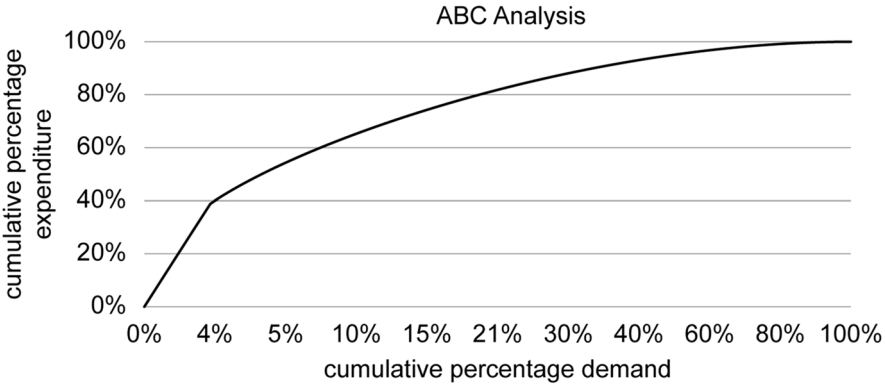


Fig. 13.2 ABC analysis in graphical form

develop strategies concerning alternative stocking arrangements, reorder calculations, and intervals of inventory control. Consider Fig. 13.3 to understand the connection between the ABC and XYZ analyses.

XYZ analysis can be used to enhance ABC analysis. A combined ABC/XYZ analysis helps to define the purchasing method for different inventory classes and to determine which parts must be in stock when they should be ordered JIT, and when it is only viable for them to be ordered on a forecast basis.

A Hollywood star is constantly complaining of the lack of space in her wardrobe, represented as Tables 13.3 and 13.4.

Perform a combined ABC/XYZ analysis to help the Hollywood star to gain at least 50% of space and to reduce the expenditure by 70%! *Solution*. First, we sort the items according to their volume (see Table 13.5).

Second, we perform the ABC analysis (Table 13.6).

Finally, we sort the items according to the XYZ classification in order to see how much potential there is for space reduction. Of course, we will have to identify X-items and the percentage of never used X-items, which are primary candidates for leaving the wardrobe. In parallel, we will have to identify A-items, since our second goal in this task is also to reduce expenditure to 70% (Table 13.7).

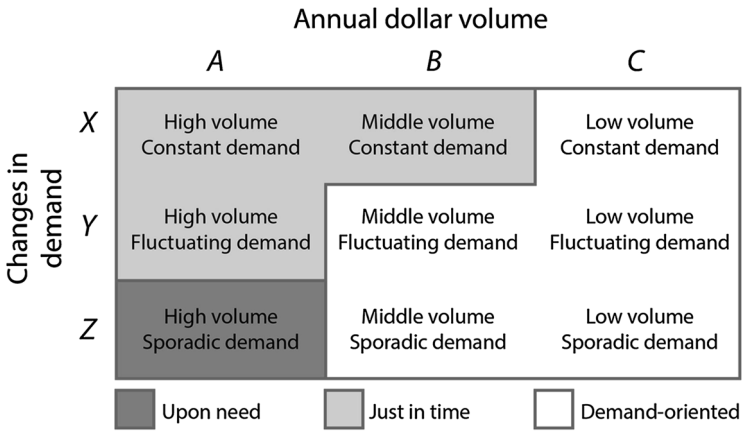


Fig. 13.3 ABC/XYZ analysis

Table 13.3 Initial data for ABC/XYZ analysis

	Quantity	Cost per unit ($)	Annual expenditure	Volume per unit (dm^3)	Total volume (dm^3)
Blouses	120	200	24,000	1.00	120
Pantsuits	420	200	84,000	1.00	420
Jeans	50	200	10,000	2.00	100
Dresses and skirts	450	500	225,000	2.00	900
Costumes	280	1000	280,000	2.00	560
Fur coats	120	10,000	1,200,000	10.00	1200
Sport pants	10	80	800	2.00	20
T-shirts	1200	80	96,000	0.50	600
Scarfs	100	50	5000	0.40	40
Underwear	500	75	37,500	0.02	10
Belts	600	95	57,000	0.05	30
Total	3850		2,019,300		4000

Table 13.4 Initial data for ABC/XYZ analysis

	Quantity	Frequent use	Seldom use	No use	% of no use from quantity
Blouses	120	30	30	60	50.00
Pantsuits	420	100	120	200	47.62
Jeans	50	20		30	60.00
Dresses and skirts	450	10	20	420	93.33
Costumes	280	20	30	230	82.14
Fur coats	120	5	15	100	83.33
Sport pants	10	4	2	4	40.00
T-shirts	1200	300	400	500	41.67
Scarfs	100	30	30	40	40.00
Underwear	500	200	100	200	40.00
Belts	600	200	200	200	33.33

Table 13.5 XYZ analysis

	Quantity	Volume per unit (dm^3)	Total volume (dm^3)	Percentage	Cumulative percentage
Fur coats	120	10	1200	30	30
Dresses and skirts	450	2	900	22.5	52.5
T-shirts	1200	0.5	600	15	67.5
Costumes	280	2	560	14	81.5
Pantsuits	420	1	420	10.5	92
Blouses	120	1	120	3	95
Jeans	50	2	100	2.5	97.5
Scarfs	100	0.4	40	1	98.5
Belts	600	0.05	30	0.75	99.25
Sport pants	10	2	20	0.5	99.75
Underwear	500	0.02	10	0.25	1
Total	3850	20.97	4000	100.00	100

Table 13.6 ABC analysis

	Quantity	Cost per unit ($)	Annual expenditure	Cumulative expenditure	Percentage expenditure	Category
Fur coats	120	10,000	1,200,000	1,200,000	59.4	A
Costumes	280	1000	280,000	1,480,000	73.3	A
Dresses and skirts	450	500	225,000	1,705,000	84.4	B
T-shirts	1200	80	96,000	1,801,000	89.2	B
Pantsuits	420	200	84,000	1,885,000	93.4	B
Belts	600	95	57,000	1,942,000	96.2	C
Underwear	500	75	37,500	1,979,500	98.0	C
Blouses	120	200	24,000	2,003,500	99.2	C
Jeans	50	200	10,000	2,013,500	99.6	C
Scarfs	100	50	5000	2,018,500	99.7	C
Sport pants	10	80	800	2,019,300	100	C
Total	3850		2,019,300	2,019,300	100	

Table 13.7 Integrated ABC–XYZ analysis

	Quantity	% of never used items	Cumulative percentage (volume)	Space saving (%)	ABC	Cost saving
Fur coats	120	50.00	30	25.00	A	1,000,000
Dresses and skirts	450	47.62	52.5	21.00	B	210,000
T-shirts	1200	60.00	67.5	6.25	B	40,000
Costumes	280	93.33	81.5	11.50	A	230,000
Pantsuits	420	82.14	92	5.00	B	40,000
Blouses	120	83.33	95	1.50	C	12,000
Jeans	50	40.00	97.5	1.50	C	6000
Scarfs	100	41.67	98.5	0.40	C	2000
Belts	600	40.00	99.25	0.25	C	19,000
Sport pants	10	40.00	99.75	0.20	C	3200
Underwear	500	33.33	100	0.10	C	15,000
Total	3850		100	100		1,577,200

We thank Mr. Martin Pruy for preparing this task

It can be observed that due to the high value, high volume, and high percentage of never used items, such as fur coats, dresses, skirts, and costumes, are key in achieving both the objectives, i.e., value reduction of 70% and space reduction of 50%. Clearing the wardrobe of never used fur coats, dresses, skirts, and costumes will enable the Hollywood star both to reduce the value of the wardrobe items to 70% and space to 50%.

13.4 Deterministic Models

After classifying the items, the next step is to determine order quantities. In this section, we consider items with independent deterministic demand in the setting of the economy of scale, i.e., the *cycle inventory*. Why do we need to determine optimal order quantities? Theoretically, we could order exactly the quantity that corresponds to the daily demand each day. However, then we would also have ordering costs each day. We would pay each day for transportation.

In most cases, it is reasonable to exploit the *economy of scale* and order in large lots to reduce fixed ordering costs. We are going to learn how to determine when and how much to order. We will consider the following methods:

- Basic economic order quantity (EOQ)
- Economic production order quantity (EPQ)
- Quantity discount model
- Reorder point (ROP)

13.4.1 EOQ Model

Let us start with the *EOQ model*. It is simple and helps us to understand the relationship between ordering and holding costs. Consider the system that exhibits the following characteristics (see Fig. 13.4):

- Demand and lead time are known and constant.
- Receipt of inventory is instantaneous and complete.
- Quantity discounts are not possible.
- The only variable costs are setup and holding.
- Stock-out can be avoided.

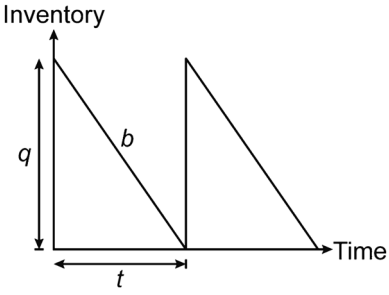


Fig. 13.4 Inventory consumption and replenishment pattern in EOQ model

We introduce the following notations:

q is the number of units per order.
q^* is the optimal number of units per order (EOQ).
b is the annual demand in units for the inventory item.
f is the setup or ordering cost for each order.
c is the holding or carrying cost per unit per year.

Under the assumption of linear inventory consumption, cycle inventory, and lot-sizes are related as follows in Eq. (13.2):

$$\text{Cycle inventory} = \frac{q}{2}. \tag{13.2}$$

This means that the average inventory on hand is equal to 50% of the order quantity.

$$\text{Then annual inventory holding costs is} c \cdot \frac{q}{2}. \tag{13.3}$$

In order to calculate ordering costs, we have to know the number of orders per year. This number can be easily calculated as shown in Eq. (13.4):

$$\text{Number of orders per year} = \frac{b}{q} \tag{13.4}$$

$$\text{Then, annual fixed ordering costs is} = f \cdot \frac{b}{q} \tag{13.5}$$

Optimal order quantity is found when annual ordering costs equal annual holding costs (see Eq. 13.6):

$$c \cdot \frac{q}{2} = f \cdot \frac{b}{q} \tag{13.6}$$

In solving Eq. (13.6) for q^*, we get the EOQ formula as follows (Eq. 13.7):

$$q^* = \sqrt{\frac{2b \cdot f}{c}} \tag{13.7}$$

Consider graphical representation (see Fig. 13.5).

It can be observed from the graph that the lowest total cost (the top curve) is the sum of the two curves below it. Minimal total costs are achieved at the intersection point of the fixed and variable costs curves. This corresponds to the EOQ point q^*. It can also be observed that the total cost function is relatively flat in the minimal region. It means that moderate EOQ changes will not influence a significant increase in total costs by tendency. This observation exemplifies the useful *robustness property* of this method.

We can also determine the expected number of orders per year (Eq. 13.8) and the expected time between orders (Eq. 13.9):

$$N = \frac{b}{q^*} \tag{13.8}$$

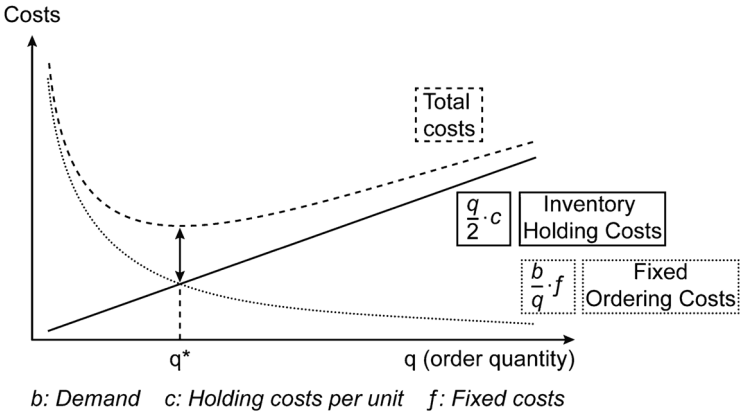


Fig. 13.5 Graphical representation of EOQ model

$$T = \frac{365}{N} \tag{13.9}$$

The total annual cost is calculated as shown in Eq. (13.10):

$$\text{TC} = c \cdot \frac{q^*}{2} + f \cdot \frac{b}{q^*} \tag{13.10}$$

Consider an example.

Task 13.2 EOQ

Demand for the Sakri2 LED TV at Amillos is 3200 units per quarter. Amillos charges fixed costs of $2500 per order. Annual holding costs per LED TV are $80. Calculate the number of LED TVs that the store manager should order per refill.

Solution

Annual demand: b = 3200 × 4 = 12,800 units; Ordering cost/order: f = $2500 Holding cost per unit per year: c = $80.

Using the EOQ formula, the optimal order quantity is

$$q^* = \sqrt{\frac{2 \cdot 12{,}800 \cdot 2500}{80}} = 895$$

To minimize total costs at Amillos, the store manager orders 895 LED TVs per refill. The cycle inventory is the average resulting inventory and is calculated as follows: cycle inventory = 895/2 ≈ 448 units.

For an order size of q^* = 895, the store manager evaluates:

Number of orders per year: N = 12,800/895 = 14.3
Expected time between orders: T = 365/14.3 = 25.5 days
Total cost: TC = 80 · 895/2 + 2500 · 12,800/895 = $71,554

Total cost for 1 year is $71,554 for 14.3 orders with 895 LED TVs in each order.

▶ **Practical Insights** The EOQ model was developed around the beginning of the twentieth century in the era of mass production and economy of scale. By that time, procurement processes were being performed manually and were quite costly and time-consuming. Today, an economy of flexibility and small lot-sizes exists in many industries and services. Transportation costs and SC coordination has become more and more important. Procurement processes have been automated, many of them performed via the Internet. A significant part of fixed ordering costs (telephone, fax, etc.) has been cut.

In applying EOQ computation results and inventory control policies, the practical reality must also be considered. For example, in some cases, suppliers fix the dates for new orders (e.g., each Friday) or only allow orders at fixed quantities (e.g., 50, 150, or 300 units) only.

Deliberate the following questions:

- Can we include transportation costs in the EOQ model?
- Is the EOQ of each firm really an optimal solution for the whole SC?
- For which department is EOQ really "optimal"? (Hint: what happens to transportation costs if EOQ decreases?)

Consider an example. EOQ is 40 units. This corresponds to 25 deliveries a year subject to the annual demand of 1000 modules. Consider the following procedure for determining transportation costs: 400 € per delivery +4 € per module. It is possible to transport up to 100 units at a time. Inventory holding unit cost is 29 € and fixed ordering cost is 23.2 €.

Costs Analysis for 100 Units:

- Transportation costs = 10 deliveries · (400 + 4 · 100) = 8 € per unit
- Cycle inventory costs = (100/2) · 29 + (1000/100) · 23.2 = 1682 €
- Total costs per module: 9.68 €

Costs Analysis for 40 Units:

- Transportation costs = 25 deliveries · (400 + 4 · 40) = 14 € per unit
- Cycle inventory costs = (40/2) · 29 + (1000/40) · 23.2 = 1160 €
- Total costs per module: 15.16 €

Conclusion: EOQ is not optimal for the integrated inventory-transportation setting. At lot-sizes of 100 units, total costs are reduced from 15.16 € to 9.68 € per unit. For an annual demand of 1000 modules, the cost savings is 5480 €.

13.4.2 EOQ Model with Discounts

Inventory costs can also be calculated on the basis of the unit prices p (i.e., the actual costs of the material purchased) as shown in Eq. (13.11):

$$\text{Costs}(q) = p_1(1-r_1)b + \frac{q}{2}p_1(1-r_1)I + \frac{b}{q}f \tag{13.11}$$

In this case for calculating EOQ, Eq. (13.12) is used:

$$q^* = \sqrt{\frac{2 \cdot b \cdot f}{p \cdot I}}, \tag{13.12}$$

where I is the *interest rate* (*capital commitment*) and p is the unit price.

As such, different prices can be included in the analysis. This allows us to apply the EOQ model for situations with quantity discounts. Reduced prices are often available when larger quantities are purchased. In this case, the *trade-off* is between reduced item costs and increased holding costs.

The *algorithm of calculating EOQ with discounts* involves the following steps:

- For each discount, calculate q^*.
- If q^* does not qualify for a discount, choose the smallest possible order size to get the discount.
- Compute the total cost for each q^* or adjusted value from Step 2.
- Select the q^* that gives the lowest total cost.

Task 13.3 EOQ with Discounts

Carlo Inc. operates a chocolate shop in New York. The chocolate is ordered from a supplier in Switzerland. Normally, the cost for one unit of chocolate is $5.00, but a quantity discount is provided by the manufacturer (see Table 13.8).

Carlo Inc.'s annual demand for chocolate is 10,000 units and the setup cost per order is $50. The interest rate is 20%.

Solution:

$$\text{For discount } 0\%, q^* = \sqrt{\frac{2 \cdot 10{,}000 \cdot 50}{5 \cdot 0.2}} = 1000.$$

Table 13.8 Initial data for EOQ calculation with discounts

Discount quantity in units	Discount (%)	Discount price p ($)
0–999	0	5.00
1000–1999	4	4.80
2000–10,000	10	4.50

$$\text{Costs}(1000) = 5 \cdot (1-0) \cdot 10{,}000 + \frac{1000}{2} \cdot 5(1-0) \cdot 0.2 + \frac{10{,}000}{1000} \cdot 50 = \$51{,}000$$

$$\text{For discount } 4\%, q* = \sqrt{\frac{2 \cdot 10{,}000 \cdot 50}{4.8 \cdot 0.2}} = 1021$$

$q^* = 1021$ units qualify for the interval [1000–1999] units and therefore can be used for costs calculation:

$$\text{Costs}(1021) = 5 \cdot (1-0.04) \cdot 10{,}000 + \frac{1021}{2} \cdot 5(1-0.04) \cdot 0.2 + \frac{10{,}000}{1021} \cdot 50 = \$48{,}980$$

$$\text{For discount } 10\%, q^* = \sqrt{\frac{2 \cdot 10{,}000 \cdot 50}{4.5 \cdot 0.2}} = 1054$$

$q^* = 1054$ units do not qualify for the interval [2000–10,000] units and therefore, the smallest possible order size to get the discount of 10% should be used for cost calculations:

$$\text{Costs}(2000) = 5 \cdot (1-0.1) \cdot 10{,}000 + \frac{2000}{2} \cdot 5(1-0.1) \cdot 0.2 + \frac{10{,}000}{2000} \cdot 50 = \$46{,}150$$

It can be observed that we have the lowest total costs with 2000 units of chocolate per order. We can also determine the expected number of orders per year $N = 10{,}000/2000 = 5$ and the expected time between orders: $T = 365/5 = 73$.

13.4.3 EPQ Model

The EPQ model is fairly similar to the EOQ model, but it is applied to manufacturing. This model is used when:

- Inventory builds up over a period of time after an order is placed.
- Units are produced and sold simultaneously.

In the EOQ model, we assume that the receipt of inventory is instantaneous and complete. Now, we allow the receipt of inventory *over a period of time* and introduce three new parameters:

- r is the daily production rate.
- d is the daily demand.
- t is the length of the production run in days.

Setup costs remain unchanged; holding costs are now calculated subject to the relation of production and demand as follows (Eq. 13.13):

$$c \cdot \frac{q}{2}\left[1-\left(d/r\right)\right] \tag{13.13}$$

For example, if we produce the quantity that corresponds exactly to the daily demand each day, we will not have any holding costs (the expression in brackets will always equal zero), but we will have very high set-up costs.

Then, the following EPQ formula (13.14) can be stated as follows:

$$q^* = \sqrt{\frac{2b \cdot f}{\left(1-d/r\right) \cdot c}} \tag{13.14}$$

Maximal inventory level in the system can be calculated as per Eq. (13.15):

$$I_{\max} = q\left(1-\frac{b}{r}\right) \tag{13.15}$$

Note: In EPQ, we consider daily demand. If we are given annual demand, it should be divided into the corresponding number of working days.

Consider an example.

Task 13.4 EPQ

Natural Inc. produces high-quality food processors. It sells 18,000 processors per year and is able to produce 125 machines per day. Natural Inc. works 250 days per year. Annual holding costs per food processor is $18 and setup costs are $800. Calculate the economic production quantity for Natural Inc. and the maximal inventory level.

Solution

d = 18,000/250 = 72 units per day

$$q^* = \sqrt{\frac{2 \cdot 18{,}000 \cdot 800}{\left(1-72/125\right) \cdot 18}} = 1943 \quad I_{\max} = 1943\left(1-\frac{72}{125}\right) = 824$$

It can be observed that the optimal lot-size is 1943 units. At this quantity, minimal setup and holding costs can be achieved. Taking into account the actual consumption rate of 72 units a day and the production rate of 125 units a day, the maximal inventory level in the system is 824 units.

13.4.4 Reorder Point

The EOQ model answers the "how much" question. The *reorder point* (ROP) tells "when" to order. ROP is introduced to take into account the so-called *lead time*, i.e., the time between placement and receipt of an order (see Fig. 13.6).

With the assumption of constant demand and a set lead time, ROP is calculated as in Eq. (13.16):

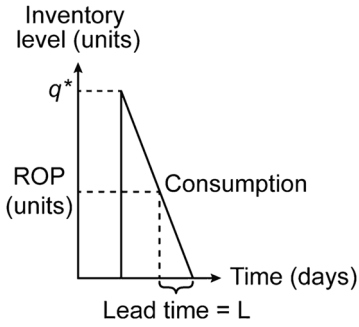


Fig. 13.6 Reorder point

$$\text{ROP} = d \cdot L, \tag{13.16}$$

where d is daily demand and L is lead time.

Note: In ROP, we consider daily demand. If we are given annual demand it should be divided into the corresponding number of working days in a year. Consider an example.

Task 13.5 Reorder Point
A company experiences an annual demand of 8500 cheese knives per year (250 working days). The lead time for an order is five working days. Calculate the ROP and explain its meaning.

Solution

1. Daily demand = 8500/250 = 34 units
2. ROP = 34 · 5 = 170 units

If our inventory reaches 170 cheese knives, then we have to place a new order.

13.5 Stochastic Models

We already know how to determine order quantities and ROPs for situations where demand and lead time are deterministic. However, in many practical cases, both demand and lead time fluctuate. We do not know their values but can only estimate them based on probability. For such cases, *stochastic* (*probabilistic*) models are needed.

13.5.1 Service Level and Safety Stock

Imagine that one Saturday evening you are sitting at home and notice that you have only 16 nappies left for your baby. You are in a province city in Germany where all stores are closed on Sunday. You recall that normally you need eight nappies a day. Normally, you would need three nappies for the rest of Saturday, eight for Sunday,

and three for Monday morning until the stores open again. So normally, 16 nappies would be enough, and you would not need to go out now to buy a new pack of nappies (the stores close in 1 hour, and it is rainy). But what will happen if the demand for nappies increases and deviates from the mean value? So here, the trade-off in your decision is to balance the risk of being out of stock and the additional costs of driving the car to the store and spending your time buying nappies.

Uncertainty in demand makes it necessary in SCOM to maintain a certain customer service level or *level of product availability* to avoid stock-outs. The level of product availability is the fraction of demand that is served on time from a product held in inventory. A high level of product availability provides a high level of *responsiveness,* but increases costs because much inventory is held, but rarely used. In contrast, a low level of product availability lowers inventory holding cost but results in a higher fraction of customers who are not served on time. The basic trade-off when determining the level of product availability is between the cost of inventory to increase product availability and the loss from not serving customers on time.

For example, a 0.05 probability of stock-out corresponds to a 95% service level (see Fig. 13.7).

▶ **Practical Insights** In practice, different service level estimations are used. For example, anyLogistix software uses three service level indicators (Ivanov 2024):

- The Alpha service level measures the probability that all customer orders that arrive within a given time interval will be completely delivered from stock on hand. In other words, a lack of stock will not delay the deliveries.
- The Beta service level is a quantity-oriented service level with consideration of backordering.
- The Lead Time service level is the ratio of orders delivered to the customers within the expected lead time to total orders.

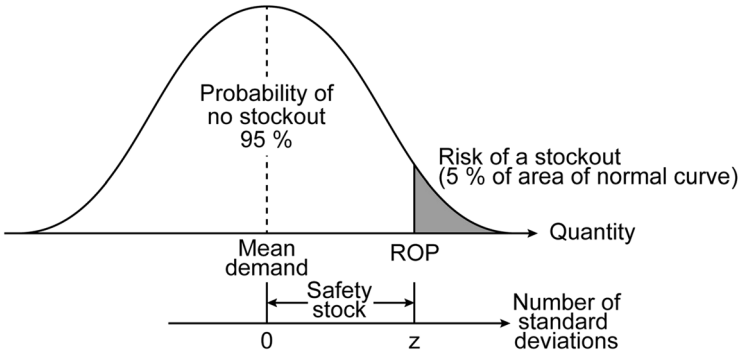


Fig. 13.7 Interrelation of demand distribution, ROP, service level, and safety stock. Adapted from Heizer et al. (2023)

In the further course of this chapter, we will focus on the Alpha service level. In a situation of demand uncertainty, *safety inventory* is introduced with the objective to ensure product availability even in the case of demand fluctuations. Consider an example in Figs. 13.8, 13.9, and 13.10 based on practice of inventory control in software anyLogistix.

Assume that we use Eqs. (13.12) and (13.16) to compute EOQ and ROP, respectively. The dashed line in Fig. 13.8 reflects the inventory dynamics in the case of using optimal EOQ and ROP and can be named as an ideal inventory behavior. The ideal inventory behavior means in this case that all assumptions of EOQ and ROP models subject to Eqs. (13.12) and (13.16) are met, i.e., demand and lead time are

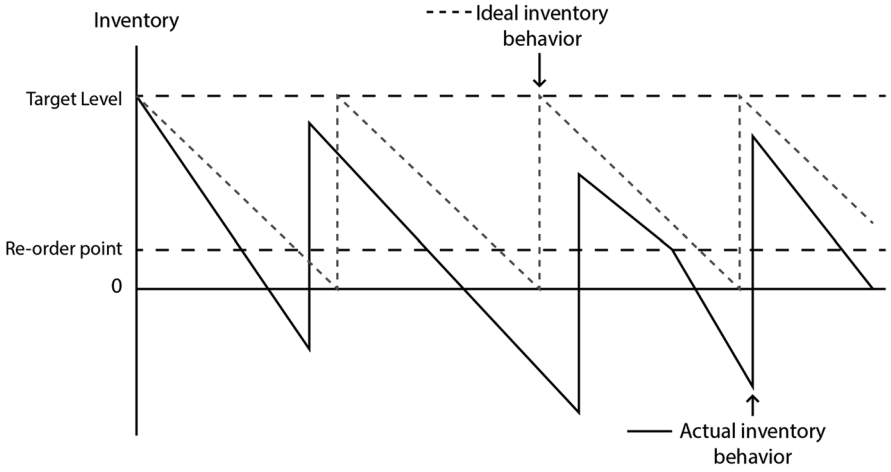


Fig. 13.8 Actual and ideal inventory behavior

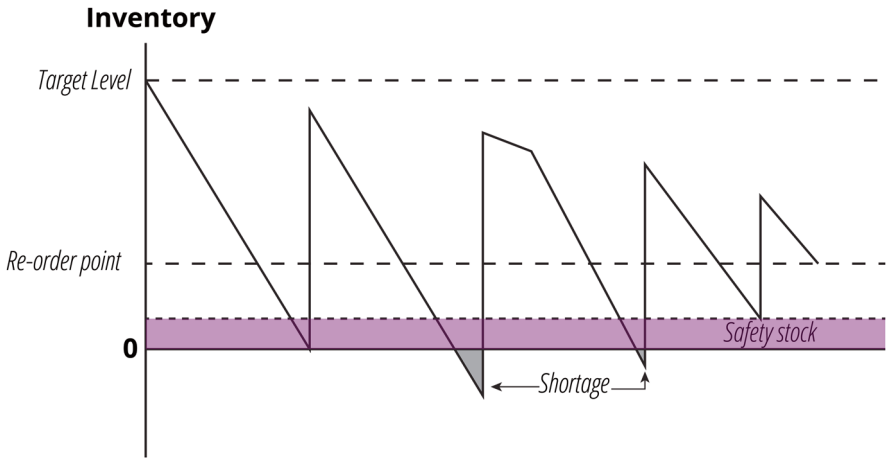


Fig. 13.9 ROP with safety stock and shortage

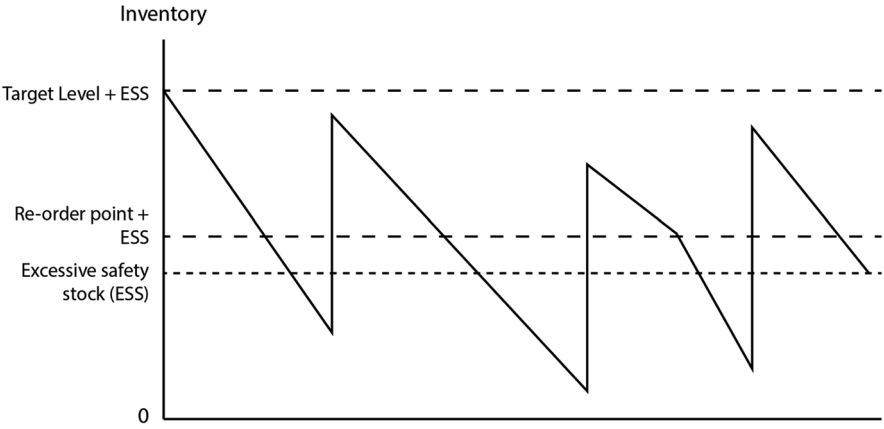


Fig. 13.10 ROP with excessive safety stock and without backlogs

constant. In reality, this is not the case. Both demand and lead time fluctuate, resulting in actual inventory behavior, which is different from the ideal one.

In order to cope with this situation, the ROP should be increased by the safety stock. Consider Figs. 13.9 and 13.10.

In Fig. 13.9, the ROP from Fig. 13.8 is increased by safety stock. It considers an example where safety stock allows to cope with demand fluctuations in some cases. However, in other cases, there exists a shortage (backlog). Figure 13.10 shows an example where ROP from Fig. 13.9 is increased by an excessive safety stock (ESS). The ESS is so high that demand fluctuations would never result in a backlog, which means a 100% product availability on stock, resulting in a 100% service level. However, the inventory level in Fig. 13.10 is much higher as compared to Figs. 13.8 and 13.9 resulting in higher inventory costs.

The question is, *how much safety stock should we plan to find the right balance between inventory investment and customer satisfaction?* Technically, the safety stock computation is based on the desired service level and demand volatility (see Fig. 13.7 and Eq. 13.17).

It can be observed from Fig. 13.9 that ROP is enlarged by the safety stock subject to a 95% service level.

In order to compute safety stock subject to a desired service level, Eq. (13.17) is used:

$$\text{ss} = z \cdot \sigma_{\text{dLT}}, \tag{13.17}$$

where ss is safety stock, σ_{dLT} is the standard deviation of demand during lead-time, and z is the number of standard deviations.

Demand deviation can be gleaned, e.g., from analysis of demand forecasts and actual sales in the past. For example, $\sigma = 1.25$ MAD (for the MAD, see Chap. 11) is a typical value. The Z-value can easily be determined (see Table 13.9).

For example, $z = 1.65$ for a service level of 95%, $z = 2.33$ for a service level of 99%, and $z = 1.28$ for a service level of 90%. If the standard deviation of demand

Table 13.9 Table of normal distribution

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

during lead time is 10, then safety stocks equals $1.65 \cdot 10 = 16.5$; $2.33 \cdot 10 = 23.3$; $1.28 \cdot 10 = 12.8$. We can observe that the increase of service level from 90% to 99% results in a doubling of inventory.

Another important issue is that stock-outs can be objective and subjective. For example, the absence of a certain sort of milk in a supermarket does not automatically mean all milk is missing since other kinds are available. In retail, 3–8% of stock-out is typical and results not only from false inventory planning, but could for many other reasons, e.g., false shelf placement.

► **Practical Insights** The service level is determined subject to SC strategy (efficient vs responsive). The higher the service level, the higher SC responsiveness, but also the higher inventory costs. If managers were to set the service level to 100%, this would mean that each customer order would be satisfied with the inventory without delay. However, a 100% service level would result in a huge inventory. This cost can scatter the positive effects of high customer satisfaction. That is why the service level is typically set at less than 100%.

The inclusion of safety stock changes the calculation of ROP (see Eq. 13.18):

$$\text{ROP} = \bar{d} \cdot L + \text{ss}, \tag{13.18}$$

where $\bar{d}$ is average daily demand.

In order to calculate ROP, four situations are possible:

- Demand is assumed to be normally distributed during the lead time.
- Daily distribution of demand is given (i.e., demand is variable) and lead time is constant.
- Daily demand is constant and lead time is variable.
- Both demand and lead time are variable.

In order to calculate ROP if demand is assumed to be normally distributed during the lead time, formula (13.19) can be used:

$$\text{ROP} = \bar{d} \cdot L + z \cdot \sigma_{\text{dLT}} \tag{13.19}$$

If daily distribution of demand is given (i.e., demand is variable) and lead time is constant, formula (13.20) can be used:

$$\text{ROP} = \bar{d} \cdot L + z \cdot \sigma_{\text{d}} \cdot \sqrt{L} \tag{13.20}$$

If daily demand is constant and lead time is variable, formula (13.21) can be used:

$$\text{ROP} = \bar{d} \cdot L + z \cdot \bar{d} \cdot \sigma_L \tag{13.21}$$

▶ **Practical Insights** Equation (13.21) nicely provides evidence of the importance of reducing lead time variability. We can observe that lead time variability reduction directly influences safety stock levels. This observation depicts the integration of supplier selection, contracting, and inventory management decisions.

Finally, if both demand and lead time are variable, formula (13.22) can be used:

$$\text{ROP} = \bar{d} \cdot L + z\sqrt{L \cdot \sigma_d^2 + \bar{d}^2 \cdot \sigma_L^2} \tag{13.22}$$

Consider an example.

Task 13.6 ROP with Safety Stock

The average demand for toothbrushes is 35 units per day. The standard deviation of normally distributed demand during lead time is ten toothbrushes per day. Lead time is 3 days. The service level is 95%. Calculate ROP and safety stock.

Solution

$$\text{ROP} = 35 \cdot 3 + 1.65 \cdot 10 = 122 \,\text{units};$$

$$\text{Safety stock is } 1.65 \cdot 10 = 16.5 \,\text{units}$$

Now we assume that we are given daily distribution of demand instead of standard deviation of normally distributed demand during lead time. Consider ten units as daily standard deviation of demand and calculate ROP:

Solution

$$\text{ROP} = 35 \cdot 3 + 1.65 \cdot 10 \cdot \sqrt{3} = 134 \,\text{units}$$

Next, consider a situation where demand is constant, but lead time may fluctuate with a standard deviation of 1 day. Calculate ROP.

Solution

$$\text{ROP} = 35 \cdot 3 + 1.65 \cdot 35 \cdot 1 = 163$$

Finally, we assume that both demand and lead time are variable. Calculate ROP. *Solution*

$$\text{ROP} = 35 \cdot 3 + 1.65\sqrt{3 \cdot 10^2 + 35^2 \cdot 1^2} = 170$$

▶ **Practical Insights** Practical implementation of the statistical methods for inventory management is not easy. In many companies, decisions have been taken manually for many years on the basis of expert knowledge. In this case, it can be reasonable to allow manual rewriting of the calculated results in software to start working with new technology. As practice shows, in a short period of time, inventory managers will see the advantages of the new system and accept 95% of automatically generated orders.

13.5.2 Single Period Systems ("Newsvendor Problem")

The newsvendor problem is a mathematical model for calculating the optimal inventory level for one single period. It is called the newsboy or newsvendor problem. A newspaper vendor who must decide every day how many daily newspapers he wants to stock for the next day is faced with uncertain demand and the knowledge that unsold copies will be almost worthless the next day.

The newsvendor problem is characterized by the following conditions:

- Fixed price for each unit
- Perishable product
- Uncertain demand
- No additional delivery in period t
- Short purchase time

Consider the following notation for single period model:

c is the purchase price
r is the retail price
v is the salvage price
c_o is overage cost

c_u is underage cost
z is the number of standard deviations
σ is the standard deviation of demand
μ is the expectation of demand
S is the order quantity
S^* is the optimal order quantity
$Z(S^*)$ is the expected cost for optimal order quantity
$\prod(S^*)$ is the expected profit for optimal order quantity
$f_{01}(z)$ is the density function.

To calculate the *overage* and *underage costs* we can use Eqs. (13.23) and (13.24):

$$c_o = c - v \tag{13.23}$$
$$c_u = r - c \tag{13.24}$$

Then, we use the *critical ratio* (CR) to find the z-value from the table of normal distribution (Eq. 13.25):

$$\text{CR} = \frac{c_u}{c_u + c_o}; F(\text{CR}) = z \tag{13.25}$$

For example, $\text{CR} = \dfrac{0.75}{0.75 + 0.25}; F(0.75) = 0.68 = z.$

The next step is the calculation of S^* according to Eq. (13.26):

$$S^* = \mu + z \cdot \sigma \tag{13.26}$$

In order to calculate the expected cost and profit for optimal order quantity, Eqs. (13.27) and (13.28) can be used:

$$Z(S^*) = (c_o + c_u) \cdot f_{01}(z) \cdot \sigma \tag{13.27}$$
$$\Pi(S^*) = c_u \cdot (\mu - Z(S^*)) \tag{13.28}$$

Consider an example.

Task 13.7 Newsvendor Problem
Coff&Co., a coffee shop, sells vegan chocolate croissants. They purchase the croissants from a small bakery at the end of the street. The bakery sells chocolate croissants to Coff&Co. for $0.70 each. Coff&Co. sells them for $2.40 to their customers. Unsold croissants can be returned to the bakery for $0.15 each. On the basis of the last few months, Coff&Co. expects a normal distributed demand for chocolate croissants. The expectation of demand is 14 croissants per day with a standard deviation of 4 per day. Calculate the optimal order quantity, and expected costs and profit for the chocolate croissants.

Solution

$$\text{Overage cost}: c_o = c - v = 0.70 - 0.15 = \$0.55$$
$$\text{Underage cost}: c_u = 2.40 - 0.70 = \$1.70$$
$$\text{CR} = \frac{1.70}{1.70 + 0.55} = 0.75; F\left(0.7556\right) = 0.7 = z$$

$S^* = 14 + 0.7 \cdot 4 = 17$ units; therefore, Coff&Co. should order 17 vegan chocolate croissants per day.

$Z(S^*) = (0.55 + 1.70) \cdot f_{01}(0.7) \cdot 4 = \$2.81$/day; so expected cost is \$2.81 per day.

$\Pi(S^*) = 1.70 \cdot (14\ 2.81) = \$20.99$/day; so expected profit is \$20.99 per day.

Note: $f_{01}(z)$ value can be taken from the full version of a normal distribution table. $f_{01}(z)$ is the density function.

13.5.3 Safety Stock and Transportation Strategy

This section focuses on the impact of a global transportation concept on inventory management. It illustrates how the sea-leg part of SCs relates to inventory management at destination countries. It shows that an increase in shipment frequency can offer substantial benefits in terms of safety stock reduction. Consider an example (Fig. 13.11).

In the example of Fig. 13.11, the sea journey from Yantian to Felixstowe takes 30 days on average; now, one departure of a vessel is offered each day instead of once a week. We assume a constant demand of 10 units a day, 98% service level, and a normally distributed lead time from Yantian to Felixstowe. The safety stock (ss) at destination (Felixstowe) can be then defined as:

$$\text{ss} = z \cdot \bar{d} \cdot \sigma_{\text{LT}}, \tag{13.29}$$

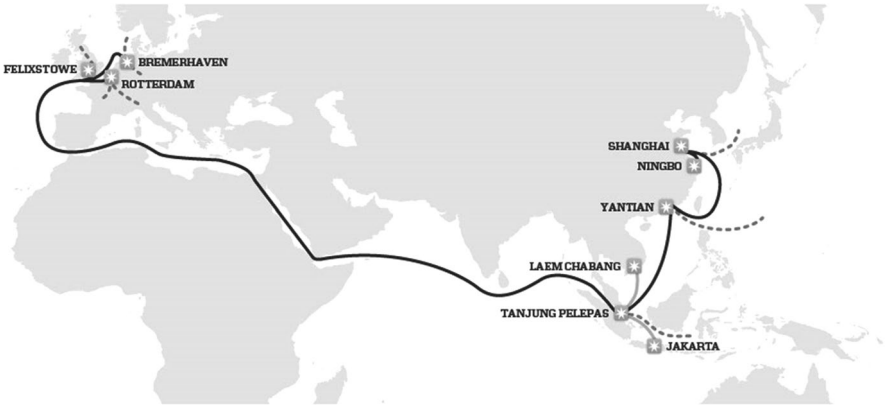


Fig. 13.11 A transportation network

where z is the number of standard deviations, d is daily demand, and σ_{LT} is the standard deviation of lead time.

First, an analysis of lead time was performed for weekly departures. It showed that the standard deviation of lead time for weekly departures is 2 days. Subsequently, an analysis of lead time was performed for daily departures: standard deviation for daily departures of 0.3 days.

Assuming a service level of 98% and daily demand of 10 units, safety stock can be calculated as follows:

$$\text{ss}(\text{ weekly departure }) = 2.055 \cdot 10 \cdot 2.0 = 41.1 \,\text{units}$$
$$\text{ss}(\text{ daily departure }) = 2.055 \cdot 10 \cdot 0.3 = 6.2 \,\text{units}$$

The calculation above shows that the safety stock level falls significantly as a result of the daily shipment concept to only 6.2 units required in stock vs. 41.1 units in a weekly service. This translates into lower inventory costs for direct customers or intermediaries, such as forwarders, at destination points and offers a measurable benefit to customers. Note: this example is based on the premium "Daily Maersk" service that was launched in 2011 and has been abandoned in 2015.

Discussion Questions

What impact does daily vessel departure have on inventory?

Daily vessel departures have two impacts: (1) lower average lead time (due to shorter average waiting time at the port of origin); and (2) a lower standard deviation of lead time (as the increased sailing frequency leads to a high probability of catching a vessel within 24 h, even if a vessel is missed for 1 day).

Which other factors could be included in this analysis?

We restricted our analysis only to sailing lead time and included no other elements of lead time (thus ignoring, e.g., hinterland transportation).

13.6 Inventory Control Policies

Inventory control policy is a managerial procedure that helps to define how much and when to order. The review may happen periodically (e.g., at the end of a month) or continuously (i.e., tracking each item and updating inventory levels each time an item is removed from inventory). Four parameters are important in the setting up of inventory control policies:

- t is the replenishment interval (also known as replenishment cycle time)
- q is the order quantity
- s is the reorder point
- S is the target inventory level

Since order quantity and replenishment intervals may be both fixed and variable, four basic inventory control policies can be classified (see Fig. 13.12).

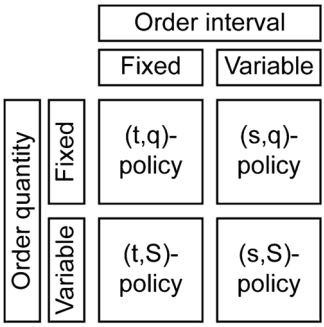


Fig. 13.12 Inventory control policies

If the period between two orders is always the same, we talk about *periodic review systems*. If the point of time of the next replenishment depends on the ROP, we talk about the ROP method of stock control or a *continuous review system*. The above-mentioned four parameters can be fixed or changed (adjusted) in dynamics according to changes in demand and supply. Therefore, *static* and *dynamic* views on inventory control policies can be considered.

13.6.1 Fixed Parameters

When the replenishment interval, order quantity, ROP, and target inventory levels are fixed, the following policies can be classified.

Policy 1: *t,q*

- *t*: fixed time between two orders
- *q*: fixed order quantity

In (*t,q*) policy, a fixed amount (*q*) is ordered for a fixed period of time (*t*). The dashed line represents available inventory, which is measured as inventory-on-hand plus planned future receipts minus open orders (shortage) (see Fig. 13.13).

(*t,q*) is a simple policy for handling the ordering process. This policy opens possibilities to further automatic control, which improves quality and saves resources, such as labor, energy, or materials. However, the timing and quantities following the (*t,q*)-policy will be contractually specified. This implies that the replenishment is simple but also inflexible. Should uncertainty or fluctuation in demand exist, this policy cannot be adjusted. In addition, shortage or overstocking make the (*t,q*)-policy an unattractive tool for many companies. Thus, it is recommended to implement this policy under constant demand.

Policy 2: *t,S*

- *T*: fixed time between two orders
- *Q*: variable order quantity to stock up to the target level *S*

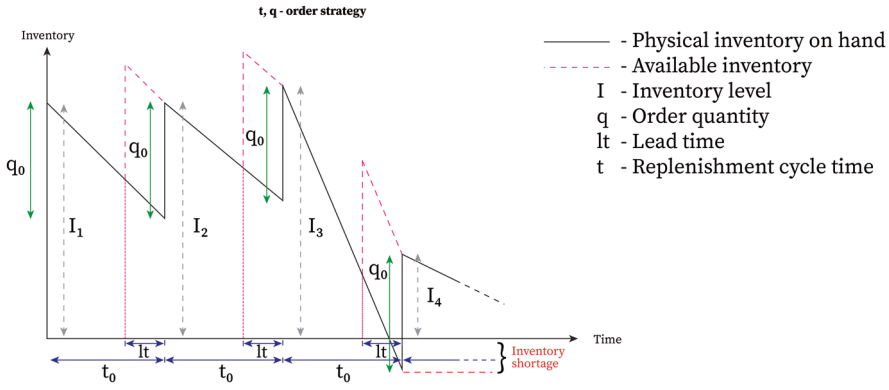


Fig. 13.13 (*t*,*q*)-inventory control policy

In the (*t*,*S*)-policy, the order quantity (*q*) is variable, and *q* is placed at a fixed time (*t*). We need to order a certain amount of inventory to reach the desired quantity *S* subject to lead time (lt). Order quantity is calculated as the target level *S*—stock on hand (see Fig. 13.14).

This policy avoids excessive inventory, which cannot be used for any other purpose and thus involves opportunity costs. The model is easy to use for control of orders. However, the physical control of the inventory could be so expensive that the exact count is only performed once a month, for example. In certain cases, the (*t*, *S*)-policy can lead to relatively high capital commitment because of the high average inventory. This policy also implies high ordering costs because we might not place a large order on a fixed day. At the same time, we might need to wait too long to fulfill our target inventory, and thus, a shortage can occur. The (*t*,*S*)-policy is recommended for use in companies with cycled replenishment.

Policy 3: *s*,*q*

- *t*: variable time between two orders
- *q*: fixed order quantity

This model operates when order quantity (*q*) is fixed and the interval (*t*) between orders can vary. In this case, the order point (*s*) is defined as ROP (see Fig. 13.15). Every order arrives to replenish inventory after a lead time. The lead time is assumed to be known and constant. The only uncertainty is associated with demand. In the following analysis, one should be most concerned with the possibility of shortage during an order cycle, that is, when the inventory level falls below zero. This is also called a stock-out event. Every time we extract inventory, we compare what is left with *s*.

Note: for calculating the ROP, Eq. (13.20) should be slightly modified to take into account the replenishment interval (see Eq. 13.30):

$$\mathrm{ROP} = d \cdot \left(T + L\right) + z \cdot \sigma \cdot \sqrt{\left(T + L\right)} \tag{13.30}$$

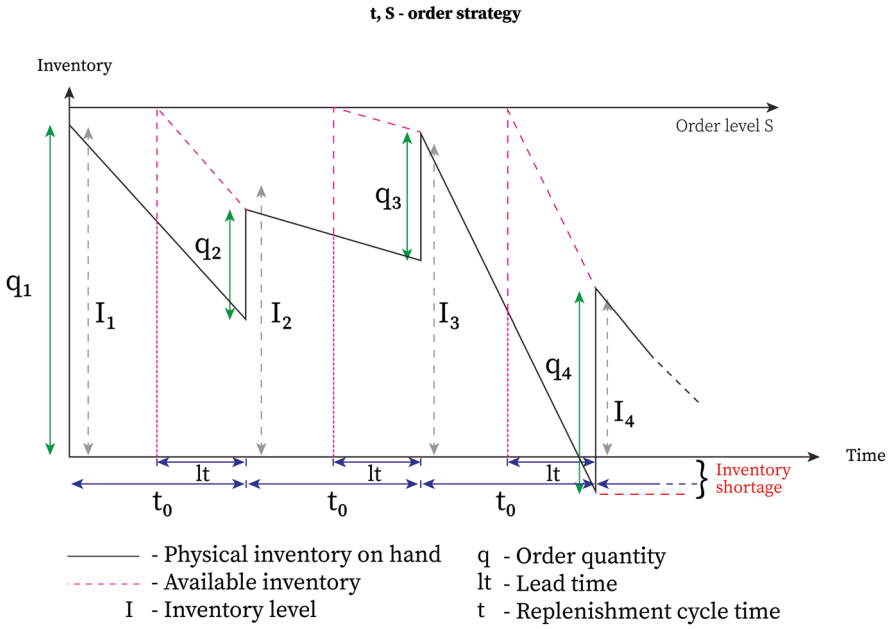


Fig. 13.14 (t,S)-inventory control policy

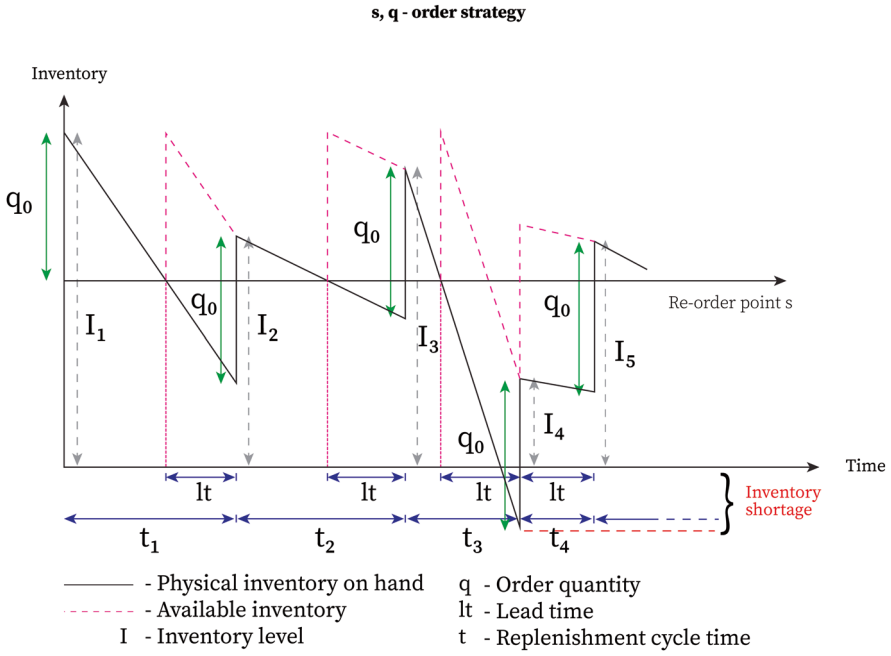


Fig. 13.15 (s,q)-inventory control policy

If the stock level is less than s, then we place an order at the rate of q. Similar to the (t,q)-policy, in the (s,q)-policy, q also refers to the optimal order quantity. The policy (s,q) results in a shorter time between orders if there is an inventory shortage. Because of its simple operation and full control over results, this policy is widely used in organizations. An advantage of the model is that it considers demand fluctuations. Disadvantages lie in regular inventory control.

Policy 4: s,S

- t: variable time between two orders
- q: variable order quantity between the order level S and ROP s

This strategy is used to define the drop of order quantity s after every inventory usage. Should this be the case, a manager should refill inventory to raise the inventory position to level S, which is desirable property (see Fig. 13.16).

Therefore, both order quantity and the time interval between orders is variable. This system can handle any level of demand and at any time, and includes demand fluctuations in planning. Target level is calculated as Eq. (13.31):

$$S = \left(\text{ROP} + q\right) \tag{13.31}$$

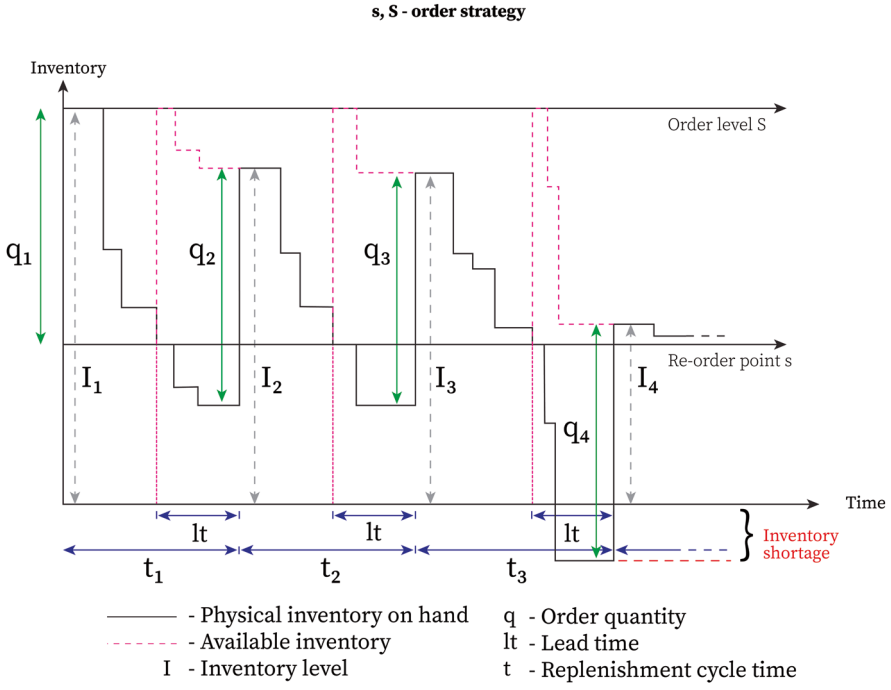


Fig. 13.16 (s,S)-inventory control policy

Order policy (*s,S*) avoids an excessive level of inventory and ensures that the business has the right goods on hand to avoid stock-outs. However, this policy requires much effort and high control. It is used in industrial and commercial areas of business, given the fact that flexible order quantity is possible and a target quantity can be predetermined.

Task 13.8 Inventory Control Policy
Consider the given data and determine parameters and annual holding costs for (*s*,*S*)-policy for 95% and 98% service levels respectively:

- Demand per day (*d*): 100 units
- Standard daily deviation of demand (σ): 20 units
- Annual holding costs (*h*): $10 per unit
- Fixed ordering costs (*f*): $100 per order
- Order interval (*T*): 4 days
- Lead time (*L*): 2 days

Solution

1 Find *z*-values for 95% and 98% service level; we get 1.65 and 2.05, respectively.

$$\text{ss} = z \cdot \sigma \cdot \sqrt{(T+L)} = 1.65 \cdot 20 \cdot \sqrt{4+2} = 81\,\text{units}$$

$$\text{ROP} = 100 \cdot (4+2) + 81 = 681\,\text{units}$$

2 $S = \text{ROP} + q$; $q^* = \sqrt{\frac{2.36,500 \cdot 100}{10}} = 855$ units; $S = 681 + 855 = 1536$ units

3 The policy is (681;1536); Average inventory position is (681 + 1536)/2 = 1108.

4 Costs = 1108 · 10 = $11,082.

5 ss = 2.05 · 20 · $\sqrt{4+2} = 100$; ROP = 100 · (4 + 2) + 100 = 700 units

6 $S = \text{ROP} + q$; $q^* = \sqrt{\frac{2.36,500 \cdot 100}{10}} = 855$ units; $S = 700 + 855 = 1555$ units.

7 The policy is (700; 1555); Average inventory position is (700 + 1555)/2 = 1128.

8 Costs = 1128 · 10 = $11,282.

13.6.2 Dynamic View

When the replenishment interval, order quantity, ROP, and target inventory levels are not fixed, but changes in dynamics subject to changes in demand, the following changes to the above-mentioned policies must be considered. We have to take into account demand, current and projected inventory, and in-transit quantities as well as planned deliveries (see Fig. 13.17).

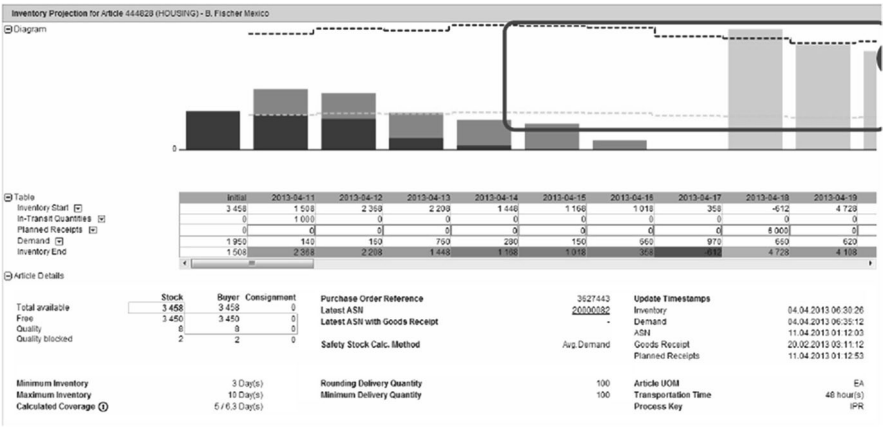


Fig. 13.17 Dynamic view of inventory control (based on SupplyOn VMI solution, used with permission)

It can be observed from Fig. 13.17 that both the target inventory level and the reorder point change in dynamics subject to changes in demand. The calculation basis is the planned days of supply.

13.7 Dynamic Lot-Sizing Models

We already know how to manage inventory with both deterministic and stochastic demand. However, our previous discussion addressed one-period problems. Now we will consider *multi-period* problems. Assume that you need tickets for the tram, say 20 tickets for a month. You can buy all of them right at the beginning of the month. In this case, you spend time for buying tickets only once, but you invest quite a lot of money at one time. Alternatively, you can buy tickets every day. Your capital commitment will be lower, but you will spend more time buying tickets.

In multi-period problems, the parameters (e.g., demand, lead time, costs) can vary in different periods. In addition, inventory from the previous periods can be used to cover demand in the future. This is why the basic *trade-off in multi-period models* is how to balance inventory holding costs and ordering costs over time.

Dynamic lot-sizing models help us decide in every period if we should order/produce only for one period or build lot sizes over several periods together to minimize order and holding costs. These models can be divided into two groups: optimization and heuristics. An example of optimization is the Wagner–Within model, which is a generalization of the EOQ model, that considers changes in demand over time. The second group contains the heuristics, e.g., the Silver-Meal heuristic, the least unit cost heuristic, and the part-period heuristic.

To explain the different dynamic lot-sizing models and show the different procedures and results, we use the following example.

Table 13.10 Demand data

t	1	2	3	4	5	6
b_t	500	200	600	300	200	100

Task 13.9 Dynamic Lot-Sizing Problem
Sheeran Ltd. has the following periodic demand for a material they need for their production (Table 13.10):

Orders are placed at the beginning of each period, and the product is immediately available. Every order generates a fixed cost f of \$100, and there are holding costs c per unit per period of \$0.5.

13.7.1 Least Unit Cost Heuristic

The least unit cost heuristic is based on an average unit cost per period K_1^{unit}. For the calculation of the average costs for the first period we can use Eq. (13.32):

$$K_1^{\text{unit}} = \frac{f_1}{b_1} \tag{13.32}$$

In the second step, the task is to find out if it is more efficient to order the material for period 1 and period 2 jointly in the first period. To find out the average unit cost jointly for the first and second period we use Eq. (13.33):

$$K_{1,2}^{\text{unit}} = \frac{f_1 + b_2 \cdot c}{b_1 + b_2} \tag{13.33}$$

We continue with this extension until we reach the minimum of the average costs. This means that as soon as the cost increases we stop the calculation and start from the beginning with the next period onwards.

For example:

$$K_{1,1}^{\text{unit}} = \frac{100}{500} = 0.2 \quad K_{1,2}^{\text{unit}} = \frac{100 + 200 \cdot 0.5}{500 + 200} = 0.286$$

We can see the average cost per unit increases, so we decide not to order the demand for period 1 and period 2 together. Instead of ordering them together, we start the calculation with period 2 again and order in period 1 only the material for this single period:

$$K_{2,2}^{\text{unit}} = \frac{100}{200} = 0.5 \; K_{2,3}^{\text{unit}} = \frac{100 + 600 \cdot 0.5}{200 + 600} = 0.5$$

Because the average unit cost remains at the same level we continue the calculation:

$$K_{2,4}^{\text{unit}} = \frac{100 + 600 \cdot 0.5 + 300 \cdot 2 \cdot 0.5}{200 + 600 + 300} = 0.636$$

Note: since the inventory for period 3 is held over two periods, we multiply (300 · 0.5) by 2.

In this case, we might want to order material in period 2 for periods 3, 4, and 5 jointly and start the calculation for period 4 again.

$$K_{4,4}^{\text{unit}} = \frac{100}{300} = 0.33 \; K_{4,5}^{\text{unit}} = \frac{100 + 200 \cdot 0.5}{300 + 200} = 0.4$$

The average cost increases again, so we decide to order only the material for period 4 and not for periods 4 and 5 together.

$$K_{5,5}^{\text{unit}} = \frac{100}{200} = 0.5 \; K_{5,6}^{\text{unit}} = \frac{100 + 100 \cdot 0.5}{200 + 100} = 0.5$$

In the result, order quantities for each period are determined as follows:

$$q_1 = 500; q_2 = 800; q_3 = 0; q_4 = 300; q_5 = 300; q_6 = 0$$

In order to calculate the total cost, we add up the fixed ordering costs for every order and the holding costs for every period in which we generate them:

$$K^{\text{total}} = 4 \cdot 100 + \left(600 \cdot 0.5\right) + \left(100 \cdot 0.5\right) = \$750.$$

We order in four periods and in periods 2 and 5 we order material for the next period as well. The total cost is $750.

13.7.2 Silver-Meal Heuristic

The Silver-Meal method uses the average cost per period instead of the average unit cost. As long as the cost decreases, the lot size is extended similarly to the least unit cost heuristics.

For period 1 Eq. (13.34) can be used:

$$K_1^{\text{period}} = \frac{f_1}{1} \tag{13.34}$$

If we want to find out if it is less expensive to order the material for period 2 in period 1, we use Eq. (13.35):

$$K_{1,2}^{\text{period}} = \frac{f_1 + b_2 \cdot c}{2} \tag{13.35}$$

Consider an example:

$$K_{1,1}^{\text{period}} = \frac{100}{1} = 100 \quad K_{1,2}^{\text{period}} = \frac{100 + 200 \cdot 0.5}{2} = 100$$

Now, we expand the first calculation with the material holding costs for period 2. Because they do not increase, we continue with period 3.

$$K_{1,3}^{\text{period}} = \frac{100 + 200 \cdot 0.5 + 600 \cdot 2 \cdot 0.5}{3} = 267$$

As you can see, the average period cost increases remarkably. For this reason, we order only material for periods 1 and 2 together and start our calculation again with period 3.

$$K_{3,3}^{\text{period}} = \frac{100}{1} = 100 \; K_{3,4}^{\text{period}} = \frac{100 + 300 \cdot 0.5}{2} = 125$$

The cost increases again and so we order the material for period 3 only.

$$K_{4,4}^{\text{period}} = \frac{100}{1} = 100 \; K_{4,5}^{\text{period}} = \frac{100 + 200 \cdot 0.5}{2} = 100$$

$$K_{4,6}^{\text{period}} = \frac{100 + 200 \cdot 0.5 + 100 \cdot 2 \cdot 0.5}{3} = 100$$

As we can see we can order the material for periods 4, 5, and 6 jointly, resulting in the following solution.

$$q_1 = 700; q_2 = 0; q_3 = 600; q_4 = 600; q_5 = 0; q_6 = 0$$

$$K^{\text{total}} = 3 \cdot 100 + \left(200 \cdot 0.5\right) + \left(200 \cdot 0.5 + 100 \cdot 2 \cdot 0.5\right) = \$600$$

The total cost is lower with Silver-Meal heuristic than with the least unit cost heuristic. In this case, the Silver-Meal order strategy is the preferred method because we can save \$150.

▶ **Practical Insights** Silver-Meal heuristic is most preferable in cases with sporadically fluctuating demand and it is quite simple to calculate. On the other hand, Silver-Meal does not consider potential capacity limitations, i.e., it assumes that we have unlimited physical storage capacity for the preordered material in the warehouse for multiple periods, for example as for $K_{4,6}$.

13.7.3 Wagner–Whitin Model

The *Wagner–Whitin model* considers deterministic demand that changes over several periods for one product. This model helps us to find the optimal solution that will tend to be (but not necessarily!) better than Silver-Meal and offer the least unit cost heuristic solutions. The Wagner–Whitin method provides optimal results for the periods under consideration, but not beyond them. The basic idea of the model is to minimize holding and order costs. Unlike the heuristics, the Wagner–Whitin method does not stop when costs are increasing in one period, because the optimization method compares different periods with each other.

For the first period, we only have the fixed ordering cost as $K_1^{\text{WW}} = f_1$. In the second step, we want to find out the costs for ordering periods 1 and 2 jointly in the first period instead of ordering them separately.

Consider the basic Wagner–Whitin formula for costs calculation (Eq. 13.36):

$$K_{t,t+1} = \min\left[\min\left(1 \le t < j\right)\left[f + c \cdot \sum_{t=t+1}^{j} b_t \cdot n + K_{t-1}\right]; f + K_{j-1}\right] \quad (13.36)$$

According to formula (13.36), for each period t, all possible options to order for n-periods in this period until the n-period or to not order at all are considered. The option with minimal costs is selected.

The calculation for the first period is performed as $K_{1,2}^{\mathrm{WW}} = f_1 + b_2 \cdot c$. Normally, this calculation is continued for all possible options. This is remarkable for the Wagner–Whitin method and differentiates it from the heuristics. Calculating one or two more costs is safer than building the wrong lot sizes. In case we want to calculate costs for the next period, we do not start only with the fixed ordering costs in this period. We have to add up the ordering costs from this period and the total costs from the previous ordering period.

Consider the following example to understand better the procedure of the Wagner–Whitin method (Table 13.11).

$$K_{1,1}^{\mathrm{WW}} = 100;$$
$$K_{1,2}^{\mathrm{WW}} = 100 + 200 \cdot 0.5 = 200; K_{1,3}^{\mathrm{WW}} = 100 + 200 \cdot 0.5 + 600 \cdot 2 \cdot 0.5 = 800$$
$$K_{2,2}^{\mathrm{WW}} = K_{1,1}^{\mathrm{WW}} + 100 = 200; K_{2,3}^{\mathrm{WW}} = K_{1,1}^{\mathrm{WW}} + 100 + 600 \cdot 0.5 = 500$$

In this case, the costs for ordering the materials for periods 1 and 2 jointly (strategy 1) are similar to ordering them separately (strategy 2). We can follow both ways and decide which is the preferred strategy at the end.

$$K_{2,2}^{\mathrm{WW}} = K_{1,1}^{\mathrm{WW}} + 100 + 600 \cdot 0.5 + 300 \cdot 2 \cdot 0.5 = 800$$
$$K_{3,3}^{\mathrm{WW}} = K_{1,2}^{\mathrm{WW}} + 100 = 300$$

If we decide to order in period 1 for the first and second periods jointly, the calculation we use for period 3 is the total cost of $K_{1,2}^{\mathrm{WW}}$. If we decide to order for the first and second periods separately, we use the total cost of $K_{2,2}^{\mathrm{WW}}$ for the calculation of $K_{3,3}^{\mathrm{WW}}$. We will continue with strategy 1.

Table 13.11 Wagner–Whitin calculation

	Demand b_t					
	500	200	600	300	200	100
Period t	1	2	3	4	5	6
1	100	200	800			
2		200	500	800		
3			300	450	600	
4				400	500	600
5					500	**550**
6						600
K_t^{WW}	100	200	300	400	500	550

The bold value indicates minimal total costs

$$K_{3,4}^{WW} = K_{1,2}^{WW} + 100 + 300 \cdot 0.5 = 450$$
$$K_{3,5}^{WW} = K_{1,2}^{WW} + 100 + 300 \cdot 0.5 + 200 \cdot 2 \cdot 0.5 = 4650$$
$$K_{4,4}^{WW} = K_{3,3}^{WW} + 100 = 400; K_{4,5}^{WW} = K_{3,3}^{WW} + 100 + 200 \cdot 0.5 = 500$$
$$K_{4,5}^{WW} = K_{3,3}^{WW} + 100 + 200 \cdot 0.5 + 100 \cdot 2 \cdot 0.5 = 600$$
$$K_{5,5}^{WW} = K_{4,4}^{WW} + 100 = 500; K_{5,6}^{WW} = K_{4,4}^{WW} + 100 + 100 \cdot 0.5 = 550$$
$$K_{6,6}^{WW} = K_{5,5}^{WW} + 100 = 600$$

The costs K_{min}^{WW} at the end of the table in the last period are the total costs. When trying different strategies, we realize that the total cost can never be below \$550.

▶ **Practical Insights** The dynamic lot-sizing models considered are basically applied to make-to-stock (MTS) production strategies (see Chap. 6). In small series manufacturing and assemble-to-order (ATO) production strategies, other methods are typically used. These methods are based on bottleneck-based manufacturing control such as *EPEI* (Every Part Every Interval) lot-sizing, *heijunka* or DBR (*drum-buffer-rope*). Lot size can also be constrained by limited storage capacity within the production processes. In addition, the integration of manufacturing lot-sizing and vehicle scheduling belong to practical trends, e.g., the so-called BIB (*batch-in-batch*) lot size.

13.8 Aggregating Inventory

In several cases, *many markets* are replenished from the same warehouse. In this context, interesting *trade-offs* between the number of facilities, inventory, and transportation costs may arise when comparing different options, e.g.:

- Building warehouse in each market or
- Centralizing inventory.

Some important questions can be discussed regarding these options.

How does the replenishment interval affect the safety stock requirement in warehouses?

A larger replenishment interval results in higher safety stock requirements because safety stock is often used at the end of replenishment intervals; if replenishment intervals are larger, it is important that safety stock is available to be used.

How does the replenishment interval affect the level of inventory (and thus the size of the warehouse)?

A smaller replenishment interval means a lower level of inventory within the warehouse, and thus, a smaller warehouse is useful. On the other hand, a higher replenishment interval leads to a higher level of inventory, and thus, a bigger warehouse must be provided.

How does the replenishment interval affect other warehouse costs (such as labor cost)?

A shorter replenishment interval probably results in higher labor costs because of greater material handling.

Furthermore, it is necessary to evaluate other factors that affect the decision, such as the market in which the firm produces. Perhaps in a smaller market, the firm would focus on responsiveness rather than efficiency, and in a larger market the opposite. Other SC drivers affect the decision. Since sustainability is becoming more important, firms also have to find a trade-off between transportation costs and responsiveness while considering *sustainability*.

Another specific case is a situation where two warehouses have to be merged in order to save warehousing fixed and operating costs. How can we determine the right level of safety stock in the new larger warehouse? Here are two possible situations:

- Two warehouses served the same market previously.
- Two warehouses served different markets previously.

When two warehouses served the same market previously, safety stock should remain unchanged, according to the formula $ss = z \times \sigma_{dLT}$. Indeed, if the deviation of demand for tomatoes in Berlin is 100,000 tons, it will not change only because we merge our warehouses. However, if two warehouses served different markets previously, the following holds true: $\sigma_{new} = \sqrt{\sigma^2_{new}}$ where $\sigma^2_{new} = \sigma^2_1 + \sigma^2_2$.

Note: If average demand and demand deviation differ significantly from each other in two warehouses that previously served the same market, the formula for different markets has to be applied.

Task 13.10 Merging Warehouses

Mr. Tsching has two wholesale flower markets and his most important products during the Christmas period are Christmas stars. Mr. Tsching has to order Christmas stars for the next season from his supplier. His markets are located in Berlin and Cologne and provide Christmas stars for different regions in Germany, Switzerland, and Austria. The warehouses serve different markets. Cologne serves west and south Germany and Switzerland, and Berlin serves north and east Germany and Austria. Based on sales data from last year, Mr. Tsching assumes demand is to be evenly distributed. Demand for his market in Cologne is 12,000 Christmas stars with a standard deviation of 4600 stars. The market in Berlin has a demand for 14,300 Christmas stars with a standard deviation of 6200 stars. The purchase price for one Christmas star is 1.12 € and the retail price is 3.65 €. Christmas stars that are not sold can be sold on to the textile industry for 0.31 € per star. The textile industry can use the red pigments for dying. Mr. Tsching thinks about merging his markets. Merging will involve extra transport costs of 22,500 €, but a reduction in fixed costs of 20,000 €.

Calculate optimal order volume, and expected costs and profit for every market, and decide if a merging of Mr. Tsching's markets is a profitable idea.

Solution.

We refer to the newsvendor model considered in Sect. 13.5.2. First, the expected costs and profit for the non-merged case are as follows

$$c_{\mathrm{o}} = 1.12 - 0.31 = 0.81€; c_{\mathrm{u}} = 3.65 - 1.12 = 2.53€$$
$$\mathrm{CR} = \frac{2.53}{2.53 + 0.81} \approx 0.76; F\left(0.76\right) = z = 0.71$$
$$f\left(0.71\right) = 0.31 S^*_{\mathrm{Berlin}} = 14,300 + 0.71 \cdot 6200 = 18,702$$
$$S^*_{\mathrm{Cologne}} = 12,000 + 0.71 \cdot 4600 = 15,266$$

Expected cost for the Christmas season:

$$Z\left(S^*_{\mathrm{Berlin}}\right) = \left(0.81 + 2.53\right) \cdot 0.31 \cdot 6200 = 6420€$$
$$Z\left(S^*_{\mathrm{Cologne}}\right) = \left(0.81 + 2.53\right) \cdot 0.31 \cdot 4600 = 4763€$$

Expected profit for the Christmas season:

$$\Pi\left(S^*_{\mathrm{Berlin}}\right) = 2.53 \cdot 14,300 - 6420 = 29,759€$$
$$\Pi\left(S^*_{\mathrm{Cologne}}\right) = 2.53 \cdot 12,000 - 4763 = 25,597€$$
$$\Pi\left(\text{ total }\right) = 25,597 + 29,759 = 55,356€$$

Merging the markets:

$$\mu_{\mathrm{new}} = \mu_{\mathrm{Berlin}} + \mu_{\mathrm{Cologne}} = 14,300 + 12,000 = 26,300$$
$$\sigma_{\mathrm{new}} = \sqrt{\sigma^2_{\mathrm{Berlin}} + \sigma^2_{\mathrm{Cologne}}} = \sqrt{6200^2 + 4600^2} = 7720$$
$$S^*_{\mathrm{new}} = 26,300 + 0.71 \cdot 7720 = 31,781$$

Expected cost for the merged the market:

$$Z\left(S^*_{\mathrm{new}}\right) = \left(0.81 + 2.53\right) \cdot 0.31 \cdot 7720 = 7993€$$

Compare expected costs with total costs for two markets:

$$Z\left(S^*_{\mathrm{Berlin}}\right) + Z\left(S^*_{\mathrm{Cologne}}\right) - Z\left(S^*_{\mathrm{new}}\right) = 6420 + 4763 - 7993 = 3190€$$

Do not forget to add the costs of merging (22,500 € = 20,000 €) when calculating the expected profit:

$$\Pi\left(S^*_{\mathrm{new}}\right) = 2.53 \cdot 26,300 - \left(7993 + 2500\right) = 56,046€$$

We opt for merging the warehouses because the expected profit of 56,046 € is higher than in the initial situation (55,356 €). In practice, the second scenario might be considered as well since the saving potential in this example is quite low.

13.9 ATP/CTP: Available-To-Promise / Capable-To-Promise

John was browsing the Internet to find new furniture for his flat. After entering the desired parameters, he had an overview of beds. The overview contained offers from four suppliers. The prices were quite similar, but lead times differed significantly. Two suppliers indicated 3 weeks of lead time, one supplier indicated 2–3 days, and the last indicated 8–10 days since it does not have the item in stock but will be able to produce it and deliver within a week. John selected the offer of the third supplier, since the prices were almost equal and lead time became the competitive advantage. His friend Alex selected the offer of the fourth supplier since the price was a bit lower than that of the third supplier, but lead time was shorter than for suppliers 1 and 2.

Indication of a lead time of "3 weeks" is a signal that the supplier works with standard lead times. This means that it does not take into account actual inventory or production capacity loads. If there is inventory or slack in production capacity, delivery time can be significantly shorter than 3 weeks. Otherwise, it may be longer.

Suppliers 3 and 4 apply so-called ATP/CTP (available-to-promise/capable-to-promise) systems. This means that they have introduced IT [e.g., SAP APO (advanced planning and optimization)], which has the ability to check actual inventory and production capacity along the SC and to define an exact lead time and delivery date (see Fig. 13.18).

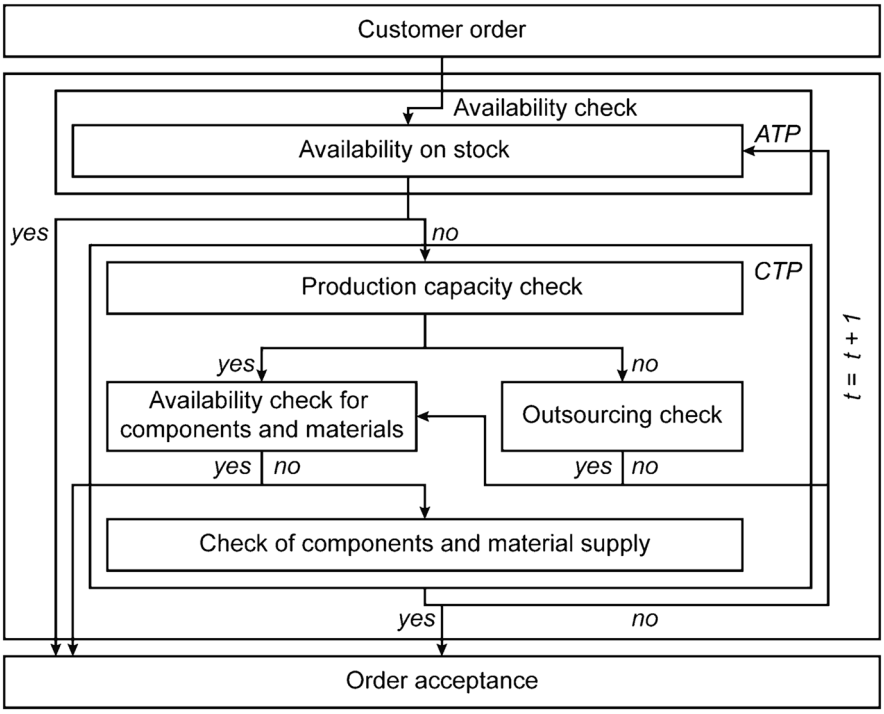


Fig. 13.18 ATP/CTP process. Based on Teich (2003)

At the order fulfillment stage, customer orders are matched with quantities available in stock and from scheduled receipts. In addition, customer requests for delivery of a product with regard to quantity, time, and location have to be answered.

The investigation into whether a delivery can actually be made is called ATP. The standard method of ATP is to search for available stocks, which can be promised for delivery.

CTP can be applied when conventional ATP checks fail, i.e., the requested product is not in stock at the respective locations and not available from scheduled receipts. CTP refers to the planning and scheduling run with regard to work-in-progress (WIP) during an ATP check. CTP checks whether it is possible to modify existing scheduled production orders with regard to time and order size or whether to create a new production order, taking production capacities and component availability into account.

Such concepts are used by many online retail companies such as Amazon. Manufacturing companies apply such systems to manage complexity, which frequently involves thousands of orders per day, which may be placed for hundreds of thousands of items. Investment in IT for ATP/CTP can be high but will be paid for through the achievement of competitive advantages.

13.10 Measuring Inventory Performance

To measure inventory performance, we consider three key performance indicators (KPI)—*inventory turnover*, *days of supply,* and *on-time delivery (OTD)*.

Inventory turnover is used to analyze the efficiency of the firm's operations in the area of inventory management. This KPI is computed, as shown in Eq. 13.37:

$$\text{Inventory turnover} = \frac{\text{Costs of goods sold}}{\text{Inventory investment}} \tag{13.37}$$

Inventory turnover KPI shows how many times the firm's inventory is renewed, or in other words, turned over. For example, Apple's inventory turnover in 2019 was 40.1, wherein their Costs of Goods Sold was $161.8 B, and inventory investment accounted for $4.1 B. What is the meaning of these numbers? On the one hand, it shows that Apple renewed its inventory 40 times per year. Breaking this down into days, we can conclude that one item (e.g., an iPhone) spends 365/40.1 = 9.1 days on average in a warehouse. We can also see that, on average, 40 different iPhones come through a single storage bin in an Apple warehouse per year.

We now turn to Days of Supply (Eq. 13.38).

$$\text{Days of supply} = \frac{\text{Inventory investment}}{\text{Average weekly costs of goods sold}} \tag{13.38}$$

Days of Supply KPI shows how many days the firm's inventory-on-hand would allow them to maintain manufacturing or delivery operations. For example, Samsung's annual costs of goods sold accounted for $151 B in 2019, and their

annual inventory investment has been $29.6 B. Considering 365 days per year, this implies the following computation of Days of Supply = 29.6/(151/365) = 72.2 days. This number shows that inventory at Samsung would reach for a period of 72 days.

On-Time Delivery (OTD) KPI (also known as Lead-Time Service Level) has been developed to analyze the inventory-related degree of customer satisfaction. OTD is the ratio of finished goods or shipments delivered on time to customers as a percentage of total units delivered or shipped (Eq. 13.39).

$$\text{OTD} = \frac{\text{Quantity of on} - \text{time delivered products}^{*}}{\text{Total quantity of delivered products}} 100\% \tag{13.39}$$

For example, a firm with 95 on-time delivered orders out of 100 orders in total would have an OTD = 95%. The higher OTD, the higher the customer satisfaction and service level will be.

What conclusions can be made from an analysis of inventory management KPIs? We organize our debate around efficiency, responsiveness, and resilience. The higher the inventory turnover, the more efficient the firm's inventory is. Higher inventory turnover also indicates a lower capital commitment (or so-called frozen capital). Higher values of Days of Supply translate to a longer product availability period. Looking at our previous computational example, we can observe that if there were a production or supply stop, Samsung would be able to continue delivering its product for 72 days. Longer product availability and higher inventory-on-hand quantities would also imply higher responsiveness in terms of reaction to unexpected demand and supply changes. At the same time, higher inventory turnover (and subsequent shorter inventory availability times) could positively affect flexibility and product variety. For example, it would be easier for Apple to accelerate the time-to-market for new products or updated products with some new components due to lower inventory-on-hand.

▶ **Practical Insights** How can we use the inventory performance KPIs in practice? First, we can use them to compare the performance of our own firm with an industry average. For example, inventory turnover in 2019 was 5.1 at Samsung, 40.1 at Apple, and the industry average accounted for 6.59. Second, we can use these KPIs to compare the efficiency and responsiveness of different companies in the area of inventory management. Looking at the numbers of Apple and Samsung, we can see that Apple's inventory is more efficient, while Samsung's inventory management is characterized by longer product availability. Note: be careful when comparing companies in different sectors and with different business models. For example, it might be difficult to compare McDonald's and Apple since they are operating under different business models and are involved with different business sectors (industrial production vs. food services).

Task 13.11 Inventory Performance Calculation
You have the task of assessing the inventory performance of "Smartphone Ltd." The company's new annual report states that the costs of goods sold for the year are 220 million euros. Their total average inventory consists of 14 million euros of production materials, 12.3 million euros of work-in-process, and 16 million euros of finished goods. Other companies in this industry normally have an inventory turnover of 9.5 turns. The company Smartphone Ltd. had in the past years, an inventory turnover of 5.8 turns.

(a) Calculate the Inventory Turnover.

$$\frac{220}{(14+12.3+16)} = 5.2$$

(b) Calculate the Weeks of Supply.

$$\frac{(14+12.3+16)}{220/52} = 10$$

(c) What do these values mean compared with the company's past year's Inventory Turns (are we improving in terms of inventory efficiency)?

No, we are not improving our inventory efficiency. Inventory turnover has been decreased from 5.8 to 5.2, which means slower inventory and higher inventory costs.

(d) What does it mean compared with the average Inventory Turnover in the industry?

We are still below the industry average, which is 9.5 turns per year.

(e) How would the Inventory Turnover and the Weeks of Supply develop if we could cut the total average inventory by half through optimizing our SCM? Calculate the optimized Inventory Turnover & Weeks of Supply again. Compare it with the industry average and the company's past. Would we improve? New inventory = 21.15 million euros

New Inventory Turnover:

$$\frac{220}{21.15} = 10.4$$

New Weeks of Supply:

$$\frac{21.15}{220/52} = 5$$

The reduction of our inventory by 50% would improve our efficiency, seen in an increase in inventory turnover from 5.2 to 10.4. Our supply availability (or

inventory on stock) would be reduced from 10 to 5 weeks. We would thus be above the industry average in terms of inventory turnover.

13.11 Key Points and Outlook

In this chapter, we learned about methods and practical tools for inventory management. Let us summarize the key points of this chapter as follows.

The role of inventory management is to strike a balance between inventory investment and customer service. Inventory is one of the most expensive assets of many companies representing as much as 50% of total invested capital. Operations and SC managers must balance inventory investment against levels of customer service. Inventory has different functions and types. Trade-off of "service level vs costs" is one of the most important issues in inventory management. Inventory creates costs, but at the same time, it can be used to increase SC flexibility. The most important costs in inventory optimization are holding, ordering, and stock-out costs.

In making decisions in the scope of inventory management, the following two basic questions are put to the forefront for consideration:

- How much should I replenish?
- When should I replenish?

According to the inventory functions and types, inventory can be used to manage:

- Economy of scale—this is cycle inventory.
- Uncertainty—this is safety inventory.

Basic methods for item classification are the ABC and XYZ analysis. ABC analysis divides inventory into three classes A-B-C based on annual dollar volume. ABC analysis is used to establish policies that focus on the few critical parts and not the many trivial ones. XYZ analysis is used to classify items according to their demand dynamics.

For cycle inventory, the basic methods are as follows:

- Basic EOQ and EPQ
- Quantity discount model
- Reorder point (ROP).

The EOQ method helps us to understand the relationship between ordering and holding costs. EOQ finds the optimal order quantity for deterministic demand in one-period model. The EOQ model answers "how much."

In the quantity discount model, inventory costs can also be calculated on the basis of different unit prices. The EPQ model is fairly similar to the EOQ model but is applied to production. This model is used when:

- Inventory builds up over a period of time after an order is placed.
- Units are produced and sold simultaneously.

The reorder point (ROP) tells a manager "when" (i.e., at what remaining quantity) to order. ROP is introduced to take into account the so-called lead time, i.e., the time between order placement and receipt.

However, in many practical cases, both demand and lead time can be fluctuating. We do not know their actual values, but can only estimate them on the basis of probability. For such cases, stochastic (probabilistic) models are needed. Uncertainty of demand makes it necessary to maintain certain customer service levels to avoid stock-outs. In the presence of demand uncertainty, safety inventory is introduced.

The essential difference between multi-period problems and single-period problems is that all the relevant parameters (e.g., demand, lead time, costs) can vary in different periods. In addition, inventory from previous periods can be used to cover demand in the following periods. This is why the basic trade-off in multi-period models is how to balance inventory holding costs and ordering costs over time. Basic methods for multi-period problems are the Wagner–Whitin method (optimization) as well as Silver-Meal and unit cost methods (both methods are heuristics).

In many practical cases, EOQ and ROP cannot be applied because of business policies. For example, many companies in the electronics retail industry have to place their orders on Fridays on the basis of demand forecasts since this is required by OEMs. In other cases, deliveries may only be possible on certain dates (e.g., on Monday). In these cases, inventory control policies are applied.

In many cases, interesting trade-offs between the number of facilities, and inventory and transportation costs may arise when comparing different options. This is why integrated inventory-transportation models are considered. In order to take into account actual inventory in stock production capacities, the ATP/CTP concept is used, which allows the right delivery dates to be determined.

In order to measure inventory performance, inventory turnover, days of supply, and on-time delivery (OTD) indicators can be used. These KPIs allow us to analyze efficiency, responsiveness, and resilience aspects of SC management (Ivanov 2017, 2018, 2020; Dolgui et al. 2020).

Many additional problems and methods of inventory management exist in practice, e.g., multi-echelon inventory management in the SC. Multi-echelon inventory management is a way to cut costs and increase customer service levels. A multi-echelon SC strategy has the objective to distribute and manage inventory across the SC. Inventory is kept at different SC partners or storage areas, each with their own inventory levels / volumes, replenishment policies and control mechanisms.

Single-echelon inventory management has some major problems. The SC carries excess inventory in the form of redundant safety stock. End customer service failures occur even when adequate inventory exists in the network. Customer-facing locations experience undesirable stock-outs, even while service between echelons is more than acceptable. External suppliers deliver unreliable performance when they have received unsatisfactory demand forecasts. Shortsighted internal allocation decisions are made for products with limited availability.

The objective of multi-echelon inventory management is to deliver the desired end customer service levels with minimum network inventory, with the inventory divided among the various echelons. However, multi-echelon techniques are more complex. The complexity of managing inventory increases significantly. All locations should be under the internal control of a single enterprise. Instead of simply replenishing the warehouse that sits between supplier and end customer, as in the single-echelon situation, we also need to contend with the problems of replenishing the multi-stage inventory system.

Additional problems and methods of inventory management that could also be considered for practical issues are as follows:

- Different kinds of service level (alpha, beta, and gamma)
- Dynamic safety stock calculations
- Inventory–lot-size problems
- Inventory–routing problems, especially in the context of vendor-managed inventory (VMI)
- Production–inventory–transportation problems
- Stochastic economic lot-sizing and scheduling problems with constraints on budget, capacity, etc.

These and a wide range of other problems of inventory management in the SC are truly fascinating and deserve much attention.

Bibliography

Axsater S (2015) Inventory control, 3rd edn. Springer, New York

Babai MZ, Ivanov D, Kwon OK (2023) Optimal ordering quantity under stochastic time-dependent price and demand with a supply disruption: a solution based on the change of measure technique. Omega 116:102817

Bowersox DJ, Closs DJ, Copper MB (2019) Supply chain logistics management, 5th edn. McGraw-Hill, Boston

Chopra S (2019) Supply chain management: strategy, planning and operation, 7th edn. Pearson, Harlow

Christopher M (2023) Logistics and supply chain management, 6th edn. FT Press, Upper Saddle River, NJ

Dolgui A, Ivanov D, Rozhkov M (2020) Does the ripple effect influence the bullwhip effect? An integrated analysis of structural and operational dynamics in the supply chain. Int J Prod Res 58(5):1285–1301

Drexl A, Kimms A (1997) Lot sizing and scheduling—survey and extensions. Eur J Oper Res 99(2):221–235

Günther H-O (2005) Supply chain management and advanced planning systems: a tutorial. In: Günther H-O, Mattfeld DC, Suhl L (eds) Supply chain management und logistik. Physica-Verlag, Heidelberg, pp 3–40

Heizer J, Render B, Munson C (2023) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow

Hwang H-C (2010) Economic lot-sizing for integrated production and transportation. Oper Res 58(2):428–444

Ivanov D (2017) Simulation-based ripple effect modelling in the supply chain. Int J Prod Res 55(7):2083–2101
Ivanov D (2018) Revealing interfaces of supply chain resilience and sustainability: a simulation study. Int J Prod Res 56(10):3507–3523
Ivanov D (2019) Disruption tails and revival policies: a simulation analysis of supply chain design and production-ordering systems in the recovery and post-disruption periods. Comput Indus Eng 127:558–570
Ivanov D (2020) Predicting the impact of epidemic outbreaks on global supply chains: a simulation-based analysis on the coronavirus outbreak (COVID-19/SARS-CoV-2) case. Transp Res E 136:101922
Ivanov D (2024) Supply chain optimization and simulation with anyLogistix. https://www.anylogistix.com/%2Fresources%2Fbooks%2Falx-textbook%2Findex.php
Ivanov D (2024b) Introduction to supply chain analytics. Springer Nature, Cham
Ivanov D, Rozhkov M (2020) Coordination of production and ordering policies under capacity disruption and product write-off risk: an analytical study with real-data based simulations of a fast moving consumer goods company. Ann Oper Res 291(1–2):387–407
MacCarthy B, Ivanov D (eds) (2022) The digital supply chain. Elsevier, Amsterdam
Render B, Stair RM, Hanna ME (2012) Quantitative analysis for management, 11th edn. Pearson, Harlow
Sainanthuni B, Parikh PJ, Zhang X, Kong N (2014) The warehouse-inventory-transportation problem for supply chains. Eur J Oper Res 237(2):690–700
Stadtler H (1996) Mixed integer programming model formulations for dynamic multi-item multi-level capacitated lot-sizing. Eur J Oper Res 94(3):561–581
Stadtler H, Kilger C, Meyr H (2015) Supply chain management and advanced planning, 5th edn. Springer, Berlin
Teich T (2003) Extended value chain management (in German). GUC Verlag, Chemnitz
Tempelmeier H (2018) Bestandsmanagement in supply chains, 6th edn. Springer, Berlin
Thonemann U (2015) Operations management: Konzepte, Methoden und Anwendungen, 3rd edn. Pearson, New Jersey
Wagner HM, Whitin TM (1958) Dynamic version of the economic lot-size model. Manag Sci 5(1):89–96
Waters D (2008) Supply chain management: an introduction to logistics, 2nd edn. Palgrave Macmillan, Basingstoke
Waters D (2013) Inventory control and management, 2nd edn. Wiley, Chichester
Wolsey LA (2006) Lot-sizing with production and delivery time windows. Math Program 107(3):471–489
Zipkin PH (1991) Computing optimal lot sizes in the economic lot scheduling problem. Oper Res 39(1):56–63
Zipkin PH (2000) Foundations of inventory management. McGraw-Hill, Boston

Routing and Scheduling

14

Routing and scheduling are part of the operative decisions required for running *daily operations*. The prerequisites are as follows. Typically, we are given known capacities that cannot be extended at short notice so that resource scarceness cannot always be prevented. Furthermore, detailed information about the demand to be fulfilled is available.

The central challenge of operative decision-making is now to match demand with the available capacities on a daily basis. Here, two basic decision tasks have to be solved (separately or in parallel): (1) demand has to be assigned to resources and (2) schedules have to be set up. A *schedule* describes the sequence in which the assigned tasks are executed and at which starting points individual operations are initiated. The major difficulty is to ensure that the available capacities are not exceeded and that resources are deployed at the highest efficiency.

In Sect. 14.1 we introduce a typical case for operative decision-making. Section 14.2 introduces *mathematical graphs* as a tool for representing decision scenarios in a network structure. Additionally, the first insights into the algorithmic processing of graph data as the basic ingredient for decision-making in network structures are provided. The consideration of complex restrictions during the deployment of a resource is discussed in Sect. 14.3 by means of the so-called *traveling salesman problem* (TSP) in which the sequencing of operations to build a schedule for a

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_14) contains supplementary material, which is available to authorized users.

Find additional case studies, Excel spreadsheet templates, and video streams in the E-Supplement to this book on www.global-supply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_14.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_14

resource is the focus of decision-making. The integrated consideration of *assignment and scheduling/sequencing* decision problems under limited resource availability is addressed in Sect. 14.4 in the context of the *capacitated vehicle routing problem* (CVRP). In Sect. 14.5 a short introduction to the scheduling of the production machines is given. Finally, Sect. 14.6 summarizes the key aspects of this chapter.

The Learning Objectives for This Chapter

- Learn about typical operative process planning tasks from transportation as well as production.
- Understand the representation of networks in mathematical graphs.
- Define decision models in mathematical graphs.
- Select adequate algorithms for solving optimization models defined in graphs.
- Understand basic ingredients of heuristic algorithms for solving complex routing and scheduling models with several constraints.
- Understand basic objectives and trade-offs in routing and scheduling.
- Compute optimal and sub-optimal (heuristic) production schedules.

14.1 Introductory Case Study: RED SEA BUS TRAVEL

The leading actor in this chapter is Christina. She has just completed her study program in Global Supply Chain and Operations Management. Equipped with a recently acquired degree, she wants to gain some job experience abroad. So, she applies for several jobs in Egypt's well-known Hurghada Red Sea area. After several interviews, she finally gets a job at RED SEA BUS TRAVEL (RSBT). This company offers transport services in the Hurghada region. International travel agencies book RSBT service operations in order to ensure that their tourists are moved to their preferred holiday region.

On her first working day, Christina is assigned to the planning and dispatching office of RSBT in downtown Hurghada. After several complicated years of business due to a lack of tourists, RSBT has detected increasing bookings from travel agencies and individual customers. These bookings can be classified into the following four categories:

– Limousine Services (LS)
– Sightseeing Tours (SST)
– Airport Arrival Transfer (AAT)
– Airport Departure Transfer (ADT)

RSBT receives all bookings at short notice, e.g., on the day before the transport service is requested. Service (transport) processes have to be set up to cover customer demand. RSBT operates a fleet of (mini)buses of different sizes and has

qualified drivers. Assignment of bookings to buses has to be decided, and for each bus, the daily schedule has to be determined.

Christina's task is now to analyze the four products, LS, SST, AAT, and ADT, and to make suggestions about how the day-to-day deployment of the available buses (which are the available resources) must be carried out to use these resources most efficiently while fulfilling all booking requests. Having accompanied several transport services during her first week at RSBT, Christina has identified the core planning challenges in the four business sectors:

LS: What is the *shortest (quickest) path* through the local street network to travel from the limousine service headquarters to the airport? How can this path be identified?
SST: In which *sequence* should all the tourist attractions be visited so that visitors have enough time to enjoy their leisure time in the hotel?
AAT: What is the minimal travel *distance* to bring all inbound passengers associated with an inbound flight to their hotels? What is the minimum number of required buses?
ADT: Which bus should *pick up* which guests from which hotel in order to take them to the airport on time if customers do not accept a pickup more than 5 h earlier than their scheduled flight departure?

Christina is now looking for suitable planning models for the four types of transportation service situations. She intends to make the determination of the processes traceable, but, at the same time, she wants to ensure that the efficiency of the deployment resources is as high as possible.

14.2 Shortest Paths in a Network

This section addresses the optimization of travel paths in a network structure. In particular, we are looking for the shortest (or quickest) connections between a given start location and a given destination location. These two locations are both parts of a network, but there is no direct connection available between them. Instead, it is necessary to determine a sequence of concatenated direct connections between intermediately passed connections/points that connect the start and terminus locations.

14.2.1 Outline of the Shortest Path Problem in a Network

Christina first turns towards the LS business. In the first step, she asks the booking department for data about the usual and most frequently booked places for limousine transport. Her survey showed that six locations contribute to the daily LS operations (see Table 14.1).

Table 14.1 Locations for LS

Notation	Locations
A	Reefside—The Fisherman's Village
B	The Palace Hotel
C	Downtown 5star
D	Mermaid's Inn
E	Red Sea Diver's Base
F	Hurghada International Airport

The fare for an LS booking is calculated based on the travel distance between the booked pickup location and the booked delivery location. From discussions with the RSBT customer care manager, Christina has learned that there are several complaints from customers who think that they have to pay too much, since drivers do not follow the shortest travel route.

Christina wants to find out if these complaints are justified, so she decides to talk to some of the drivers. After her interviews, she was convinced that the complaints were not justified. All of the complaints were a result of a misunderstanding: the infrastructure in the area is quite different from the typical European street network. Since all hotels have been built around the populated area downtown, there are only streets connecting the most prominent hotels. This means going from a pickup point to a delivery point often requires visiting several hotels.

To provide more transparency for customers and instruct the limousine drivers, Christina decides to create a list for each pair of pickup and delivery point combinations. She wants to add the shortest route to each pair (to inform the drivers) and the corresponding travel distance (to inform the customers).

In the first attempt to compile the desired list, Christina collects data from the street map. She marks the six locations A to F (Fig. 14.1). Then, she looks for streets connecting these locations. Whenever she finds out that there is a street connecting a hotel with the airport or the airport with a hotel or a hotel with another hotel, she connects the origin location with the terminal location by an arc; that is, an arc represents an existing street connecting these two points where the arc direction indicates the direction of travel (Fig. 14.1). Finally, she appends the length (given in kilometers) of such a street (section) between two points. From the compiled simplified map, she learns, for example, that there is a street that goes from A (Reefside—The Fisherman's Village) to D (Mermaid's Inn), but this street cannot be used to go from D to A. The length of the street section from A to D is 3 km. But what is the distance of a trip from E to F? Christina is looking for a comfortable method to determine the least travel distance (i.e., to solve the shortest path problem [SPP]) between a pair of pickup and delivery locations not directly connected by an arc.

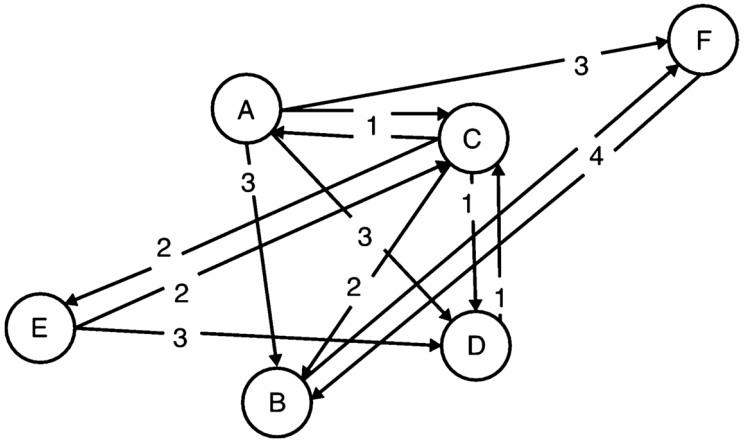


Fig. 14.1 Street network connecting the Airport (F) with the most important hotels (A–E) in the Hurghada area

14.2.2 Mathematical Graphs

A mathematical *graph* is a tool to describe network structures. A *network* consists of several points and connections between these points. In a mathematical graph, these points are represented by *nodes*, and all nodes are collected in a *node set*, which is often labeled N. If there is a connection between two points, then there is also a connection (i.e., an *arc*) between the two corresponding nodes, say i and j, taken from the node set N ($i, j \in$ N). If it is only necessary to know that these two nodes are connected, then this connection is expressed by putting the two connected nodes into a set $\{i; j\}$. This set is called an *edge*. In most situations, it is necessary to save information about the type of connection between the nodes i and j. For example, if i as well as j are cities served by an airline, it is necessary to know whether an offered flight goes from i to j and/or from j to i. It is, therefore, necessary to distinguish a connection from i to j and a connection from j to i. In such a situation, the connection is established by the ordered pair $(i; j)$, indicating a connection originating from i and going to j. If there is also a connection from j to i then a second ordered pair $(j; i)$ is used to represent this additional connection. An ordered pair $(i; j)$ connecting two nodes is called an *arc*. All required arcs and/or edges are saved in another set, called the arc (or edge) set typically labeled E. Information associated with a node or an arc (edge) can be stored by mappings f and g, so that $f(i)$, $g(i; j)$, or $g(\{i; j\})$ respectively are kept as stored information. The quadruple (N; E; f; g) is called a *weighted mathematical graph.*

14.2.3 The SPP as Graph-Based Optimization Model

Christina wants to represent the LS problem setting in a mathematical graph. She starts by setting up a graph-based optimization model for the network structure shown in Fig. 14.1. Therefore, it is necessary to represent the given information as a mathematical graph G = (Ω; Θ; d) with evaluated arcs. The node set Ω consists of the six nodes Ω = {A; B; C; D; E; F}. The arc set Θ comprises exactly those arcs connecting the nodes as shown in Fig. 14.1, so that Θ:={(A; B), (A; C), (A; D), (A; F), (B; F), (C; A), (C; B), (C; D), (C; E), (D; C), (E; C), (E; D), (F; B)}. We incorporate the real-valued mapping d to assign the travel distance $d(i; j)$ to each arc $(i; j) \in \Theta$.

In order to prepare the analysis of the derived graph, we introduce several definitions. These definitions help us to discuss the specific properties of a graph-based decision model.

Let $(i; j)$ as well as $(k; l)$ be two arcs in a given graph G. We call arc $(k; l)$ a "successor of arc $(i; j)$" if and only if the two nodes j and k coincide, i.e., if and only if $j = k$. In such a situation, we also say that arc $(k; l)$ follows arc $(i; j)$ if and only if $(k; l)$ is a successor of $(i; j)$ in the given graph G. In our example arc (F; B) is a successor of arc (A; F), which means that arc (F; B) follows arc (A; F).

We are now prepared to introduce the term "*path*" into a network G. Let s and t be nodes from the node set Ω of G. In our example, s might be node A and t could be node E. Let $(i_0; i_1)$, $(i_1; i_2)$, …, $(i_{k-1}; i_k)$ be a finite sequence of following arcs with the properties i_0 = s and i_k = t. We call the arc sequence $(i_0; i_1)$, $(i_1; i_2)$, …, $(i_{k-1}; i_k)$ a path in G from s to t. For short, we call this arc sequence an s-t-path in G.

$$L\left(\left(i_0;i_1\right),\left(i_1;i_2\right),\ldots,\left(i_{k-1};i_k\right)\right) := d\left(\mathrm{s},i_1\right)+d\left(i_1;i_2\right)+\ldots+d\left(i_{k-1};\mathrm{t}\right) \qquad (14.1)$$

The length of a given s-t-path $(i_0; i_1)$, $(i_1; i_2)$, …, $(i_{k-1}; i_k)$ in G is defined by Eq. (14.1). In a case where all nodes contained in the path $(i_0; i_1)$, $(i_1; i_2)$, …, $(i_{k1}; i_k)$ are pairwise different, we call the s-t-path a "simple path."

The property "pairwise different" refers to a situation in which each two nodes of a given subset of the node set Ω are not the same. A simple path in the graph shown in Fig. 14.1 is, for example, the path (A; D), (D; C), (C; E). In contrast, the path (A; D), (D; A), (A; F) is not a simple path since node A is visited more than once.

Using the aforementioned definitions, we are now prepared to describe Christina's task to determine the shortest path between a pair of nodes in the given network structure formally. Let s be the pickup point of an LS booking, and let t be the planned drop-off point of this LS booking. Christina is looking then for a simple s-t-path (s; i_1), $(i_1; i_2)$, …, $(i_{k-1}$; t) in G with minimal length L((s; i_1), $(i_1; i_2)$, …, $(i_{k-1}$; t)).

14.2.4 Dijkstra's Algorithm for the Identification of a Shortest S-T-Path

There is a very efficient and quick algorithm available to identify the shortest s-t-path in a given graph G. The only requirement for the applicability of this algorithm is that all arcs $(i; j)$ of G have a non-negative length $d(i; j)$. Applied to determine a shortest s-t-path in G, this algorithm does a little more than needed: it calculates the shortest paths from s to every other node in G. The basic idea of this algorithm was a contribution of E.W. Dijkstra (1959). For this reason, the algorithm is often called the "Dijkstra Algorithm."

Dijkstra's algorithm works as follows. Let $|\Omega|$ be the number of nodes in the given graph G, in which we are going to identify the shortest s-t-path.

Algorithm Start
In the beginning, each node is assigned a so-called temporary label value $l(v)$. This value equals the shortest path length known so far (not the path itself!) from start node s to node v, i.e., it carries the length of the shortest s-v-path found so far. Since, in the beginning, no s-v-path is known, we set $l(v)$ equal to infinity ($l(v):=\infty$), for all $v \varepsilon \Omega$, $v \neq s$, but we set $l(s) = 0$.

Algorithm Iteration
Dijkstra's algorithm iterates the following steps $|\Omega|$ times.

1. Each iteration starts with the selection of a node k with a temporary label value $l(k)$. If such a node exists, we select node k with a minimal temporary label value among all temporarily labeled nodes. If no temporarily labeled node is left, then the algorithm terminates.
2. The label value $l(k)$ of the selected node k is declared to be "permanent" and will not be changed anymore in later iterations.
3. For all remaining nodes $w \in \Omega \backslash \{k\}$ with a label value $l(w)$ which is not yet permanently labeled, we check the following: if the arc (k; w) is contained in G and if $l(k) + d(k; w) < l(w)$, then we update $l(w)$ by $l(w):=l(k) + d(k; w)$. We have found a shorter s-w-path in G. If there is no such arc (k; w) or if $l(k) + d(k; w) \geq l(w)$, then we do not change $l(w)$ and check the next node $w \neq k$.

The following definitions help to ease the execution of the Dijkstra algorithm for even larger networks. Let v be a node from the node set Ω of graph G. In step (2) of the iteration of Dijkstra's algorithm, we need to analyze all those nodes in Ω that are directly connected with v by an arc that originates from v. The termination node of such an arc is called a "successor of v." Let SUCC(v) be the set that contains all temporarily labeled successors of v, i.e., $SUCC(v):=\{k \in \Omega | (v; k) \in \Theta$ and k is temporarily labeled$\}$.

Let $(i; j)$ be an arc taken from the arc set Θ of graph G. We define i as a "predecessor" of j in G. For short, we write pred(j): = {i}if $(i; j) \in \Theta$. In the Dijkstra algorithm, we use the predecessor convention to recursively store the shortest s-t-path, e.g., the sequence of visited nodes in the shortest s-t-path. Starting from terminal node t, we store pred(t), and for the node pred(t), we store pred(pred(t)) and so on (right-to-left path determination).

Let Ω^T be the subset of the node set Ω that contains all nodes that are currently labeled temporarily, and let Ω^P be the subset of Ω formed by the nodes that are already labeled permanently.

Dijkstra algorithm	
(a)	Set Ω^T:=V and Ω^P:= $\varnothing$
(b)	Set l(v):= ∞ and pred(v):= $\varnothing$ for all v $\in \Omega \backslash$ {s}, l(s):=0 and pred(s):= $\varnothing$, l(s) = 0
(c)	Proceed with steps (d)–(g) if $\Omega^T \neq \varnothing$, otherwise go to (h)
(d)	Select a node k $\in \Omega^T$ so that l(k) is minimal over Ω^T.
(e)	Declare the label l(k) as permanent and update Ω^T:= $\Omega^T \backslash$ {k} and Ω^P:= $\Omega^P \bigcup$ {k}
(f)	For all w $\in$ SUCC(k) that fulfill l(k) + d(k; w) < l(w) update l(w):=l(k) + d(k; w) and pred (w):={k}
(g)	Goto (c)
(h)	Stop the algorithm

Such an algorithm description is quite close to a computer programming code. It is written in pseudocode. Pseudocode descriptions are often used to make the "real" implementation (coding) with a programming language like BASIC, Java, or C++ easier.

The Dijkstra algorithm starts with the initialization of the two node sets Ω^T, as well as Ω^P (the symbol $\varnothing$ represents an empty set) in the steps (a) and (b). The steps (c)–(g) are repeated as long as there is still a node with a temporary label available. In each repetition, exactly one node is moved from the set of temporarily labeled nodes into the set of permanent nodes (d), (e), and the temporary labels are updated if a shorter s-w-path to a node w is found (f).

Christina has read about the Dijkstra algorithm and the shortest path problem. She applies it now to the problem situation shown in Fig. 14.1. She starts with the identification of the shortest path between point E and point F.

In order to simplify the repetitive applications of steps in the Dijkstra algorithm, it seems to be appropriate and helpful to represent the working data and intermediate results in a table as shown in Table 14.2. The recent label values are contained in the columns headed by the nodes. The recent pred(X)-value of node X is given in brackets to the right of the recent label value. Label values declared as permanent are marked by *. Each row represents one iteration of the Dijkstra algorithm.

Iteration 0 (Start): $\Omega^P = \varnothing$ and Ω^T = {A; B; C; D; E; F}. The label value l(E) of the starting node E is set to 0. All remaining nodes are temporarily labeled by ∞.

Iteration 1: $\Omega^P = \varnothing$ and Ω^T = {A; B; C; D; E; F}. Node E is labeled with the minimal temporary label value 0. For this reason, E is moved from Ω^T to Ω^P ("perma-

Table 14.2 Structured table-based presentation of the iterations of the Dijkstra algorithm

	Node recently selected and labeled as permanent	Recent label values					
Iteration		A	B	C	D	E	F
0 (start)	–	∞	∞	∞	∞	0 (−)	∞
1	E	∞	∞	2 (E)	3 (E)	0* (−)	∞
2	C	3 (C)	4 (C)	2* (E)	3 (E)	0* (−)	∞
3	A	3* (C)	4 (C)	2* (E)	3 (E)	0* (−)	6 (A)
4	D	3* (C)	4 (C)	2* (E)	3* (E)	0* (−)	6 (A)
5	B	3* (C)	4* (C)	2* (E)	3* (E)	0* (−)	6 (A)
6	F	3* (C)	4* (C)	2* (E)	3* (E)	0* (−)	6* (A)

nently labeled"). SUCC(E) = {C; D}. Since l(E) + d(E; C) = 0 + 2 < ∞, we update l(C):=2 and set pred(C):={E}. Since l(E) + d(E;D) = 0 + 3 < ∞, we update l(D):= 3 and set pred(D):={E}.

Iteration 2: Ω^P = {E} and Ω^T = {A; B; C; D; F}. Node C is labeled with the smallest temporary label value. For this reason, C is moved from Ω^T to Ω^P ("permanently labeled"). SUCC(C) = {A; B; D}. Since l(C) + d(C; A) = 2 + 1 < ∞, we update l(A):=3 and update pred(A):={C}. Since l(C) + d(C; B) = 2 + 2 < ∞, we update l(B):=4 and update pred(B):={C}. Since l(C) + d(C; D) = 2 + 1 ≥ 3, we do not update l(D) and pred(D).

Iteration 3: Ω^P = {C; E} und Ω^T = {A; B; D; F}. Both nodes A and D are labeled with the same minimal temporary label value 3. Node A is selected since it comes first in a lexicographic order. For this reason, we move A from Ω^T to Ω^P and update SUCC(A):={B; F}. Since l(A) + d(A; B) = 3 + 3 ≥ 4, we preserve l(B) and pred(B). Since l(A) + d(A; F) = 3 + 3 < ∞, we update l(F):=6 as well as pred(F):={A}.

Iteration 4: Ω^P = {A; C; E} and Ω^T = {B; D; F}. Node D has the minimal temporary label value 3. Therefore, we move D from Ω^T to Ω^P and update SUCC(D):= ∅. No temporary label value must be checked to be updated.

Iteration 5: Ω^P = {A; C; D; E} and Ω^T = {B; F}. Node B is labeled with the smallest temporary label value, which is 4. Thus, we move B from Ω^T to Ω^P and we set SUCC(B) = {F}. Since l(B) + d(B; F) = 4 + 4 ≥ 4, we preserve l(F) and we do not update prec{B}.

Iteration 6: Ω^P = {A; B; C; D; E} and Ω^T = {F}. Node F is labeled with the minimal temporary label value 6. We, therefore, move F from Ω^T to Ω^P and set SUCC (B):= ∅. No further checks are necessary.

Iteration 7: Ω^P = {A; B; C; D; E; F} and Ω^T = ∅. There is no further node available that has been labeled temporarily. The Dijkstra algorithm terminates here.

The permanent label value l(F) of the designated goal node gives the shortest E-F-path length in graph G. The path is determined by a recursive backtrack-analysis of the final pred-values. We have found that pred(F):=A, pred(A):=C, pred(C):=E. The shortest E-F-path in G is ((E; C), (C; A), (A; F)).

There is a chance that more than one E-F-path with the same minimal distance exists. The Dijkstra algorithm can find only one shortest E-F-path. Christina can use the Dijkstra algorithm to determine the shortest distances between each pair of nodes in the Hurghada network.

14.3 Round Trip Planning/Traveling Salesman Problem

Now that Christina has successfully solved the planning challenge for the LS operations, she turns towards the next open challenge. In the SST department of RSBT, it is necessary to determine round trips of buses that contain exactly one visit for each location contained in a given list of sightseeing objects.

Christina examines the SST situation, and she finds out that there are some similarities with the shortest path identification problem, which she solved as explained in Sect. 14.2. First, the SST challenge also exists in a network structure. Second, for a given list of points of interest, a connecting path in the given network structure is required. Third, in the SST situation, a path with minimal length is also searched. However, Christina discovers an important difference between the LS and SST situations. In the LS challenge, only the starting, as well as the terminating node of the requested s-t-path, is given. There are no requests to visit other nodes. In contrast, the necessity to visit all locations contained in the aforementioned list of points of interests is postulated. For this reason, Christina concludes that the Dijkstra algorithm is not a sufficient tool to solve the SST challenge, since this algorithm does not provide the opportunity to manage the constraint that each node contained in the network is visited exactly once.

In this section, we accompany Christina while she learns how optimization problems on network structures can be solved if restrictions on the solution must be considered. In Sect. 14.3.1, the underlying basic decision problem of the SST situation is introduced and explained in detail. In Sect. 14.3.2, a mathematical optimization model for a restricted network optimization problem is introduced. Techniques to determine feasible solutions for these models are presented in Sect. 14.3.3.

14.3.1 Traveling Salesman Problem

Christina wants to know if her SST challenge has already been solved by anybody else. She remembers a lecture she visited during her studies. It was called "Operations Research Methods for Management." She contacts one of her friends from the course and asks him to send her a copy of the lecture materials. Several days later, she receives these printouts and starts scanning the documents.

After a few minutes, she discovers the picture of a network (Fig. 14.2) that attracts her attention. The slide that contains this graphic is headed by the words "Travelling Salesman Problem."

Christina starts reading the remarks given by the instructor. She reads that the *Traveling Salesman Problem (TSP)* represents the basic decision task of many vehicle deployment problems. It is quite similar to the shortest path problem since, in the TSP, it is also necessary to fulfill a transportation task in a network structure with minimal resource utilization.

However, the TSP is distinct from the shortest path problem since all nodes in the network have to be visited, but no node is allowed to be visited more than once (which means that each available node must be visited *exactly once*). There is also a distinct node, called the starting node or the depot, from which the traveling salesman starts his journey and returns after all other nodes have been visited exactly once. The salesman wants to keep his total sum of travel distances between the visited nodes as small as possible.

Christina summarizes these findings and transfers them to the SST challenges. She is convinced that the TSP represents exactly the SST problem: there is a set of nodes given, and these nodes (in the SST case: the given points of interests) must all be visited by a vehicle, but the length of the travel path must be minimal.

Christina continues to read the instructor's notes. Without going into detail, she understands that the TSP can be modeled as a *linear program* (cf. Chap. 12), but that

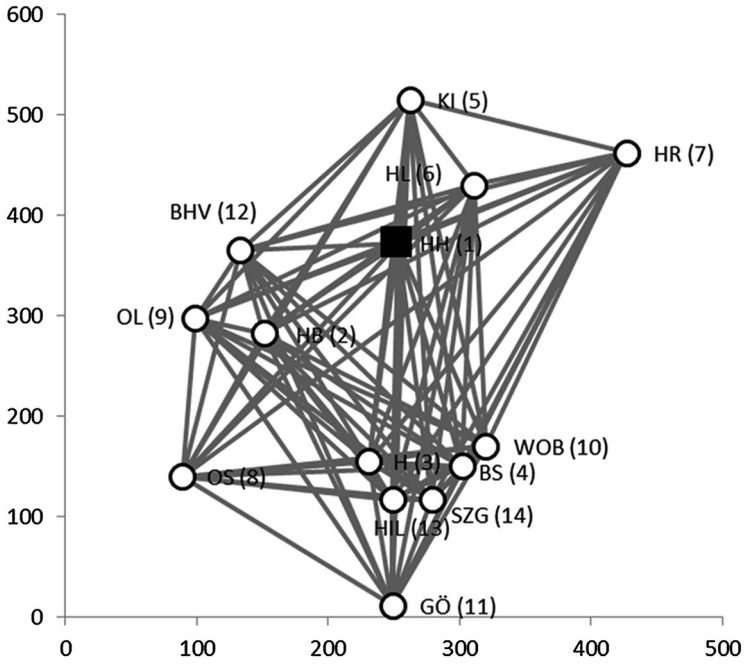


Fig. 14.2 Problem setting of the traveling salesman problem (TSP)

it is impossible to apply the standard approach for linear programs, which is the Simplex algorithm, to solve this model. For this reason, her instructor proposes applying other decision-supporting methods called heuristics. Now, Christina's attention is awakened and she decides to study the information given in the course material about the TSP, its modeling, and the heuristic model solving techniques in detail.

First, Christina wants to understand the case that motivates Fig. 14.2. The outline of the case study is as follows. A sales representative lives in Hamburg (city numbered 1 according to Table 14.3). Each week, she has to visit the remaining 13 cities in Northern Germany, which are given in Table 14.3 (cities 2–14). She wants to visit all these cities on one trip (If necessary, she will have an overnight stay in one of the cities) in order to visit all the retail outlet stores of her company. According to her contract with this company, she has to pay for the fuel costs on her own. In order to keep the fuel expenses as low as possible, the sales representative searches for a least-distance round trip that starts in Hamburg, terminates in Hamburg, and visits all remaining 13 cities exactly once.

Christina continues to learn more about the case. The sales representative (who is in the role of the traveling saleswoman) can travel directly from each given city and each given other city as indicated by the graph in Fig. 14.2. Such a graph in which each node has a connection to each other node is called a complete graph. At first look, such a network structure seems to make the determination of a round trip easier than the determination of a round trip in a general graph like the one given in Fig. 14.1: Detours on the way from one node to another node are not necessary.

Practical Insights

A high number of nodes and arcs in the network let the determination of the shortest path becomes a management problem of high *complexity*. The following calculation

Table 14.3 Coordinates of major cities in Northern Germany

Name of location (abbreviation)	i	Coordinates	
		x_i	y_i
Hamburg (HH)	1	253	372
Bremen (HB)	2	151	282
Hannover (H)	3	231	155
Braunschweig (BS)	4	302	150
Kiel (KI)	5	262	515
Lübeck (HL)	6	311	425
Rostock (HR)	7	427	462
Osnabrück (OS)	8	89	140
Oldenburg (OL)	9	99	297
Wolfsburg (WOB)	10	320	170
Göttingen (GÖ)	11	249	12
Bremerhaven (BHV)	12	133	365
Hildesheim (HIL)	13	249	117
Salzgitter (SZG)	14	279	117

emphasizes this statement. How many round trips are available that visit each node exactly once? For the first customer node, the sales representative can select from 13 so far unvisited nodes. For the second customer node to be visited, she can still select from 12 remaining customer locations and so on. In the given example, this leads to $13 \cdot 12 \cdot \ldots \cdot 2 \cdot 1 = 13! = 6{,}227{,}020{,}800$ possible trips that visit each node exactly once. Such a path that contains a node exactly once and that is closed (start and end nodes coincide) is called a Hamiltonian path. Now assume that the identification and the calculation of the length of each individual Hamiltonian path lasts 1 s. The compilation of a list of all evaluated Hamiltonian paths would last around 1200 days. For this reason, this so-called complete enumeration is not an appropriate approach to identify the shortest Hamiltonian path due to practicality.

Christina is happy. She knows now that the SST challenges match the TSP challenges. Therefore, she is sure she can solve her SST task after learning how the TSP is solved. She continues to study the lecture notes about the TSP and finds out that, again, a model-based approach is proposed to identify a solution for the given decision task. First, a model for the TSP is needed, and second, an algorithm for identifying a solution of the model is required.

14.3.2 A Mixed-Integer Linear Program for TSP-Modeling

All $N = 14$ locations to be visited are numbered by $1, 2, \ldots, N$ in order to ease further description of the problem (cf. Table 14.3). These locations form the node set. An arc $(i; j)$ connects two nodes and indicates that it is possible to travel from i to j. The node set in the TSP is equal to $\{1, \ldots, N\} \times \{1, \ldots, N\} \backslash \{(1; 1), \ldots, (N; N)\}$. Therefore, the graph in the TSP is complete.

The basic idea for modeling the TSP is that each arc $(i; j)$ contained in the aforementioned node set is either contained in the Hamiltonian path (the round trip through all N nodes) or not. In the first case mentioned, node j is visited immediately after node I, but in the latter case, node j is not visited immediately after i has been left. The TSP decision problem can be reduced to the question of which arcs form the requested Hamiltonian path and which arcs are ignored. In order to represent these binary decisions, the family of binary decision variables x_{ij} ($i \in \{1, \ldots, N\}$, $i \in \{1, \ldots, N\}$) is introduced. Each decision variable x_{ij} is either 0 or 1, and x_{ij} represents the decision whether arc $(i; j)$ is contained in the Hamiltonian path or not. We declare x_{ij} to be 1 if and only if arc $(i; j)$ is included in the Hamiltonian path. It equals 0 if this is not true.

Let $d(i; j)$ be the distance between node i and node j if the arc $(i; j)$ is used.

$$Z\left(x_{11}, x_{12}, \ldots, x_{NN}\right) = \sum_{i=1}^{N} \sum_{j=1}^{N} d\left(i; j\right) \cdot x_{ij} \rightarrow \min \tag{14.2}$$

Equation (14.2) determines the sum of travel distances of all arcs that are selected to be included in the Hamiltonian path. This value $Z(x_{11}, \ldots, x_{NN})$ must be minimized.

$$\sum_{j=1}^{N} x_{ij} = 1 \quad \forall i \in \{1,\ldots,N\} \tag{14.3}$$

$$\sum_{j=1}^{N} x_{ji} = 1 \quad \forall i \in \{1,\ldots,N\} \tag{14.4}$$

In order to ensure that the selected arcs form a Hamiltonian path through the node set $\{1, \ldots, N\}$ it is necessary to restrict the selection of arcs. First, each node must be left once (Eq. 14.3), and the salesman has to go to each node exactly once (Eq. 14.4).

It is not sufficient (but it is necessary) that the two restrictions (Eq. 14.3) as well as (Eq. 14.4) are fulfilled in order to ensure that a Hamiltonian path through $N = \{1, \ldots, N\}$ is determined. Figure 14.3 shows a selection of arcs that fulfill Eqs. (14.3) and (14.4), but a Hamiltonian Path is not achieved. Instead, two so-called short cycles are generated. One of these two short cycles does not contain the start node 1. Therefore, it is necessary to add other constraints to ensure that a Hamiltonian path is generated.

$$u_i - u_j + N x_{ij} \leq N - 1 \quad \forall i, j \in \{2,\ldots,N\} \tag{14.5}$$

Desrochers and Laporte (1991) proposed the constraint family (Eq. 14.5) as short-cycle elimination constraints. They demonstrate that these constraints and the

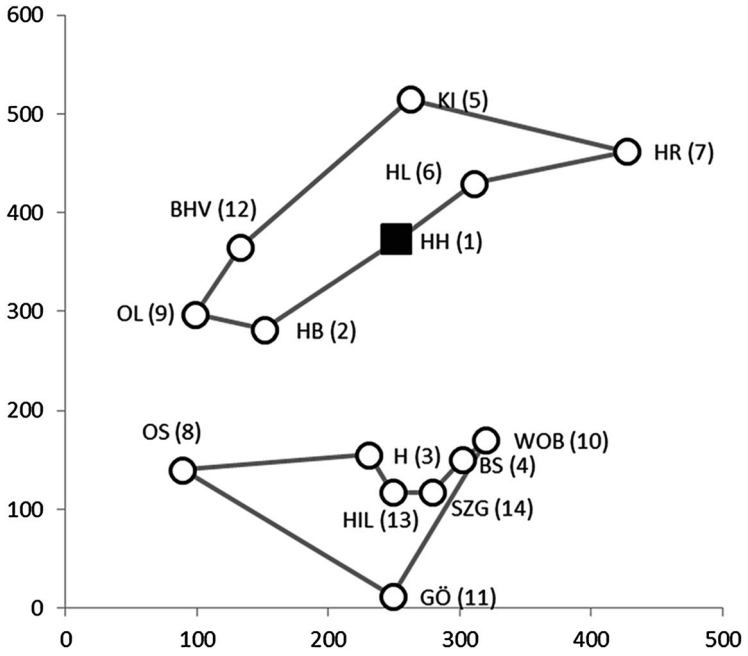


Fig. 14.3 Infeasible TSP solution with too short cycles

incorporation of the real-valued u_i-decision variables can prevent short cycles that do not contain the start node 1.

$$x_{ij} \in \{0;1\}\ \forall i,j \in \{1,\ldots,N\} \text{ and } u_i \text{ real - valued for all } i \in \{1,\ldots,N\} \tag{14.6}$$

Jointly, Eqs. (14.5) and (14.6) ensure that only round trips that contain the designated starting point 1 are allowed.

The TSP is now represented by the linear program (Eqs. 14.2–14.6). Since some of the decision variables included in this model are limited to integer (binary) values while other decision variables are of real value, this model falls into the class of a mixed-integer linear program (MILP). A linear program containing only integer-valued decision values falls into the class of integer linear programs (ILP), according to Grünert and Irnich (2005).

Having studied the derivation of the mathematical model and learned that the model consists of a linear objective function and a collection of linear constraints, Christina re-implements the reported problem instance in an Excel spreadsheet model. She tries to solve this model with the Excel Solver add-in. Quite rapidly, the solver returns a feasible solution. Christina studies the returned decision variable values. She is worried by the proposed values: she finds the proposal to set an x-decision variable to the value 0.25. She asks herself how she can interpret this value. After a while, she finds out the reason for this unexpected proposal: she forgot to declare the x-decision variables to be binary. She corrects this shortcoming and restarts the solver. Unexpectedly, the solver is unable to return a feasible solution. The calculation lasts more than 20 min, so she decides to interrupt it, being disappointed that there is no straightforward way to solve the TSP.

From her previous experience, Christina has learned that the Simplex algorithm fails to return a feasible solution since this algorithm is unable to ensure that integer values are returned. An explicit limitation of the decision variable domains to binary or integer values significantly prolongs the model solving times. Christina supposes that in the latter situation, another algorithm is used. Her interest is aroused. She wants to know if there are other algorithmic approaches to solve mathematical optimization models. She returns to the textbook and continues to read the instructor's ideas about solving a TSP-like problem.

14.3.3 Heuristic Search for High-Quality Round Trips

An algorithmic approach for solving a decision model by exploiting a rational strategy of planning without mathematical proof that the proposed (return) feasible model solution cannot be dominated by another feasible solution is called a *heuristic* or a non-optimizing algorithm. In contrast, an algorithm for which the returned solution proposal is proven to be minimal (maximal) is called an *exact* or optimization algorithm or approach. The Simplex algorithm is an example of an exact algorithm. Heuristics are incorporated into model solvers in case no exact algorithm is available or if the processing times provided by an exact algorithm are expected to be prohibitively long. Since the *processing time* for solving relatively small instances

of a TSP is, in general, quite long, it seems reasonable to think about a heuristic approach for the TSP.

This paragraph introduces heuristics for the management of TSP-like planning tasks. Two types of algorithms are distinguished. First, we discuss the construction of the first feasible solution for the TSP, i.e., we explain the construction of the first Hamiltonian path in the given network. Second, we discuss approaches for the improvement of the Hamiltonian path, i.e., we want to reduce the travel length associated with the Hamiltonian path. Therefore, we need to analyze the existing Hamiltonian path in order to find out which modifications are promising for a reduction in the length of the Hamiltonian path. The most promising modifications are tentatively implemented and, if they lead to a reduction in travel distance, finally implemented.

Algorithms for the Construction of an Initial Feasible Model Solution

Joereßen and Sebastian (1998) describe an intuitive approach for the construction of a Hamiltonian path. The central idea is to consecutively insert the nodes after the last visited node until all nodes are contained in the path. A Hamiltonian path is created. This approach is also known as a "*nearest neighborhood heuristic*."

It starts with an empty path. In the first step, the starting node (in our example node 1) is inserted. Next, the node with the shortest distance from the starting node is inserted immediately after the starting node. This path extension is iterated until no further node remains. This path construction procedure follows the rational strategy of keeping the additional travel distance associated with serving the next customer after the previously inserted customer as low as possible. Therefore, the applied rational strategy can be characterized as "*greedy*." Consequently, this construction heuristic is called the "greedy construction procedure." A major shortcoming of the greedy construction procedure is its myopic insertion strategy. After all nodes have been inserted into the node sequence, it is necessary to return to the start node. The travel distance for the return to the depot is not considered in the calculation of the additional travel distance to serve the next customer. Consequently, the greedy strategy cannot be proven to return the shortest possible Hamiltonian path. However, it returns a first Hamiltonian path.

Task 14.1 TSP Solution with Greedy Heuristic

According to the TSP, consider a transportation company that has to deliver products from Hamburg to 13 cities in Germany. The data refer to Table 14.3, and the distances are reflected in Fig. 14.4.

Of course, the lengths $d(i, j)$ of the arcs are very important for the decision about the inclusion of the next node. Typically, these values are composed in the distance matrix, as shown in the example from Table 14.3 in Fig. 14.4. The entry in the ith row in column j corresponds to the length of the arc $(i; j)$. Headers are not considered in this numbering.

This pseudocode represents the greedy construction procedure that consists of an initialization phase (a), iteration loops (b)–(i), and a termination phase (j)–(l). During the initialization phase, the travel distance is set to 0, and the first node in the

i \ j	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1		136	218	227	143	79	196	284	171	213	360	120	255	256
2	136		150	201	258	215	330	155	54	203	287	85	192	209
3	218	150		71	361	282	364	143	194	90	144	232	42	61
4	227	201	71		367	275	336	213	251	27	148	273	62	40
5	143	258	361	367		102	173	413	272	350	503	198	398	398
6	79	215	282	275	102		122	361	248	255	418	188	314	310
7	196	330	364	336	173	122		467	367	311	484	310	388	375
8	284	155	143	213	413	361	467		157	233	205	229	162	191
9	171	54	194	251	272	248	367	157		255	322	76	234	255
10	213	203	90	27	350	255	311	233	255		173	270	89	67
11	360	287	144	148	503	418	484	205	322	173		372	105	109
12	120	85	232	273	198	188	310	229	76	270	372		274	288
13	255	192	42	62	398	314	388	162	234	89	105	274		30
14	256	209	61	40	398	310	375	191	255	67	109	288	30	

Fig. 14.4 Distance matrix for the TSP instance defined by the nodes in Table 14.3 (values for $d(i; j)$ in kilometers)

Greedy construction procedure	
(a)	**Initialize: L = 0, start = 1, k = 1;**
(b)	**Repeat steps (c)–(i) until all columns are blocked**
(c)	**DEST:=min{d(START, q)| column q is not blocked}**
(d)	**L:=L + *d*(START, DEST);**
(e)	**Block row of START as well as column associated with START;**
(f)	**Stop[k]:=START;**
(g)	**START:=DEST;**
(h)	**k = k + 1;**
(i)	**Goto (b);**
(j)	**Stop[k]: = 1;**
(k)	**L:=L + *d*(START, 1);**
(l)	**End;**

path is set to the starting node start: =1 (home of the salesman). The iteration counter k is initialized.

Exactly one node is appended to the already existing partial path in each iteration. Each iteration starts with checking whether there are still unblocked columns. Such an unblocked column represents a node that has not yet been appended. Among all unblocked columns (available nodes), a node DEST with the least distance d (START, DEST) from the last appended node START is chosen (c).

Afterwards, the path length L traveled so far is increased by L(START; DEST) (d). Next, the row, as well as the column associated with DEST, is blocked (e). The recently selected node DEST is appended to the already existing partial path (f), and DEST will be set to the last inserted node (g). The iteration counter is increased by 1 (h), and the procedure jumps back to the beginning of the loop (i). If no further unblocked column is available, the path is closed, i.e., the arc from the last appended node to the starting node is selected (j), and the travel distance is updated (k). Further details can be found in the examples in the E-Supplement.

In the example, we have framed the cells representing the arcs (1; 6), (2; 3), (3; 13), (4; 10), (5; 7), (6; 5), (7; 12), (8; 1), (9; 2), (10; 11), (11; 8), (12; 9), (13; 14) as well as (14; 4). These arcs lead to the Hamiltonian path 1, 6, 5, 7, 12, 9, 2, 3, 13, 14, 4, 10, 11, 8, 1. The length is 1746 km. This path can be found in Fig. 14.5.

Figure 14.5 visualizes the proposal generated by the greedy construction procedure. The generated Hamiltonian path is not optimal, i.e., it is possible to reduce the travel length. Within the route segment $1 \rightarrow 6 \rightarrow 5 \rightarrow 7 \rightarrow 12$, as well as in the route segment $9 \rightarrow 2 \rightarrow 3 \rightarrow 13 \rightarrow 14 \rightarrow 4 \rightarrow 10 \rightarrow 11 \rightarrow 8 \rightarrow 1$, a modification of the visiting order leads to a shorter Hamiltonian path. This is not a surprising observation since the primary goal of the greedy construction procedure is the generation of a feasible (but not necessarily a least-distance) Hamiltonian path. The idea of inserting one customer after another, even with the least marginal costs, does not result in a least-distance Hamiltonian path.

The phenomenon observed here refers to a general shortcoming of the heuristic search for solutions if several constraints must be considered while trying to optimize the objective function. It either tries to fulfill the optimization property, or it tries to achieve feasibility of the generated solution proposals. At first glance, it seems that the greedy construction procedure fails for the TSP, but it is worth getting

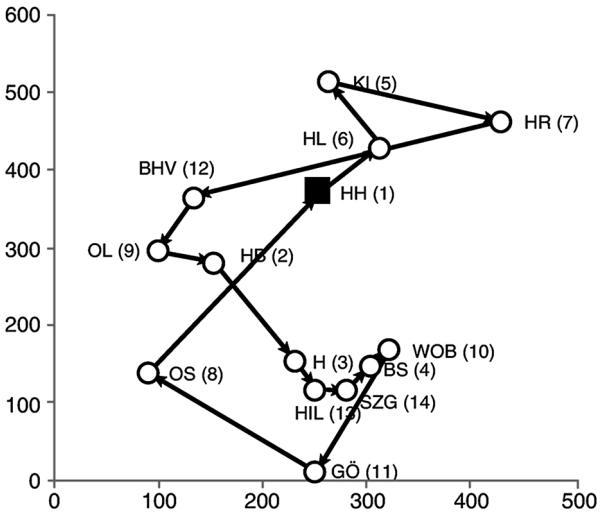


Fig. 14.5 Visualization of the greedy construction procedure proposal

a first feasible Hamiltonian path, since it is possible to improve the objective function using so-called improvement heuristics that modify the initial feasible solution. These improvement heuristics focus on the minimization of the objective function value, but retain the feasibility of the modified solution.

Improvement Heuristics for Travel Distance Reduction

We design and apply improvement heuristics in order to update an existing, feasible decision problem solution. During the update, it is necessary to preserve feasibility, but the objective function value should be brought closer to the (unknown) optimization goal. For a given Hamiltonian path in the TSP, an improvement heuristic tries to find another Hamiltonian path with reduced travel distance by modifying the visiting sequence as proposed by the construction heuristic. Detours should be eliminated. The 2-opt-improvement heuristic presented here is based on ideas discussed by Croes (1958). In its most simple realization, the 2-opt heuristic tries to reduce the length of a given Hamiltonian path by swapping two adjacent nodes in the path. Therefore, the swap is applied tentatively. Then, it is checked to see whether length reduction is achieved. In this case, the tentative swap is confirmed. Otherwise, the swap is canceled (Fig. 14.6).

We now apply the 2-opt-improvement logic to reduce the travel distance by swapping adjacent nodes in the available Hamiltonian path proposed by the greedy construction heuristic. We start with the partial path 1; 6; 5; 7 and try to swap nodes 6; 5, i.e., we tentatively replace 1; 6; 5; 7 with 1; 5; 6; 7 (cf. Fig. 14.6).

Iteration 1: The sub-path 1; 6; 5; 7 has a length of 354 km. Swapping 6 and 5 will lead to the sub-path 1; 5; 6; 7 with a length of 367 km. This swap is therefore not beneficial, and it is not confirmed.

Iteration 2: The sub-path 6; 5; 7; 12 has a length of 605 km. Swapping 5 and 7 will lead to the sub-path 6; 7; 5; 12 of length 493 km. Since the executed swap leads to a length reduction, it is confirmed, and the sub-path 6; 5; 7; 12 is replaced by the sub-path 6; 7; 5; 12.

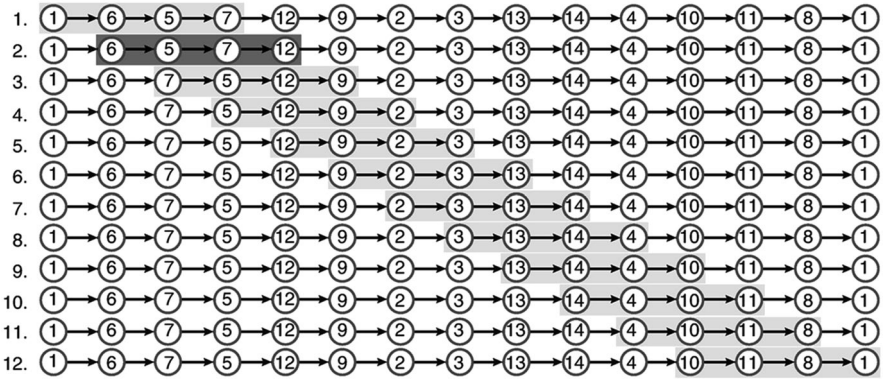


Fig. 14.6 Processing of 2-opt-improvement-algorithm

Iteration 3: The sub-path 7; 5; 12; 9 has a length of 447 km. Swapping nodes 12 and 5 lead to the sub-path 7; 12; 5; 9 of length 780 km. This swap is therefore not beneficial, and it is not confirmed.

Iteration 4: The sub-path 5; 12; 9; 2 has a length of 328 km. Swapping nodes 12 and 9 lead to sub-path 5; 9; 12; 2 with a length of 433. This swap is not confirmed.

Iteration 5: The sub-path 12; 9; 2; 3 has a length of 280 km. Swapping nodes 2 and 9 lead to sub-path 12; 2; 9; 3 with a length of 333 km. This swap is not confirmed.

Iteration 6: The sub-path 9; 2; 3; 13 has a length of 264 km. Swapping nodes 2 and 3 lead to sub-path 9; 3; 2; 13 with a length of 536 km. This swap is not confirmed.

Iteration 7: The sub-path 2; 3; 13; 14 has a length of 222 km. Swapping nodes 3 and 13 lead to sub-path 2; 13; 3; 14 with a length of 295 km. This swap is not confirmed.

Iteration 8: The sub-path 3; 13; 14; 4 has a length of 112 km. Swapping nodes 13 and 14 lead to sub-path 3; 14; 13; 4 with a length of 153 km. This swap is not confirmed.

Iteration 9: The sub-path 13; 14; 4; 10 has a length of 97 km. Swapping nodes 14 and 4 lead to sub-path 13; 4; 14; 10 with a length of 169 km. This swap is not confirmed.

Iteration 10: The sub-path 14; 4; 10; 11 has a length of 240 km. Swapping nodes 4 and 10 lead to sub-path 14; 10; 4; 11 with a length of 242 km. This swap is not confirmed.

Iteration 11: The sub-path 4; 10; 11; 8 has a length of 405 km. Swapping nodes 10 and 11 lead to sub-path 4; 11; 10; 8 with a length of 554 km. This swap is not confirmed.

Iteration 12: The sub-path 10; 11; 8; 1 has a length of 662 km. Swapping nodes 11 and 8 lead to sub-path 10; 8; 11; 1 with a length of 798 km. This swap is not confirmed (Fig. 14.7).

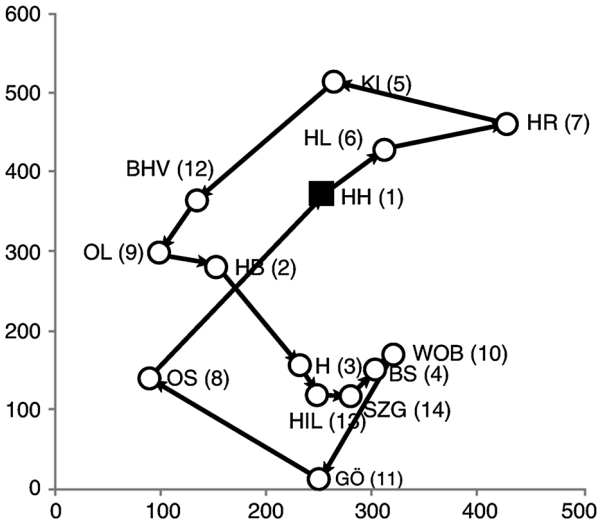


Fig. 14.7 Hamiltonian path after the first application of the 2-opt-improvement heuristic

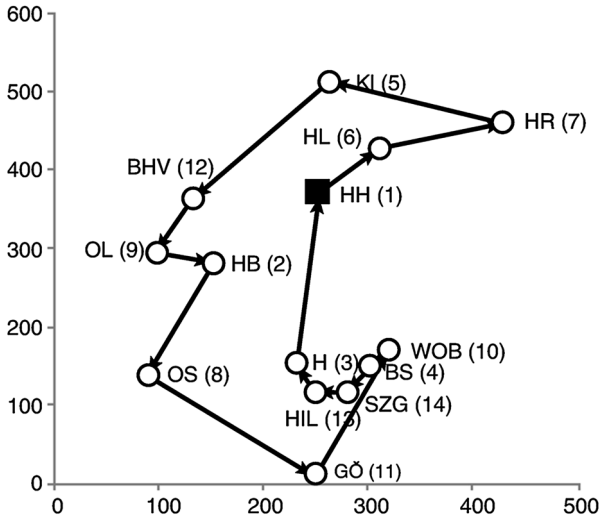


Fig. 14.8 Solution of inversion of the sub-path from 3 to 8

The existing Hamiltonian path is now completely processed. A length reduction of 112 km has been achieved. The improved Hamiltonian path is shown in Fig. 14.7. Despite the significant improvement, we see immediately that further improvements are needed. However, the required improvement cannot be achieved by interchanging adjacent nodes, so a 2-opt improvement procedure call is useless. In the situation reported here, it is beneficial to invert the sub-path 3; 13; 14; 4; 10; 11; 8, i.e., to replace this sub-path by 8; 11; 10; 4; 14; 13; 3 (see Fig. 14.8). However, the identification of a sub-path whose inversion is beneficial is a challenging task in itself.

In general, reapplication of the 2-opt procedure to the already improved solution can be useful for identifying further length-saving swapping. In our example, swapping of nodes 6 and 7 is now possible (but was not possible in the first application of the 2-opt procedure). Sometimes nodes are moved from a position at the beginning of a Hamiltonian path to a later position or vice versa.

14.4 Vehicle Routing

Having solved the first two operative vehicle deployment tasks (LS and SST) successfully, Christina turns to the next large business area, which is the local transfer of inbound tourists. RSBT provides this service to international travel agencies. These agencies sell all-inclusive travel packages to customers. Such a package includes

- A return flight ticket from the selected origin airport to Hurghada international airport (HRG).

- Accommodation in the selected hotel in the Hurghada region including breakfast or half board or full board or all-inclusive food and beverage.
- Transfer on the day of arrival with a pickup at the airport and a drop-off at the selected hotel.
- Transfer from the hotel to the airport on the day of departure.

Christina finds out that RSBT generates significant income from the transfer services, but the profits are pretty small. She discusses this observation with the responsible division manager. The manager mentions that the transfer of incoming passengers (AAT) is easier to plan than the transfer of outbound passengers (ADT) since in the ADT setting up pickup times at the hotels has to be considered. In the AAT setup, such time windows do not play any role.

14.4.1 Case Study ORION: Vehicle Routing at UPS

ORION (**O**n-**R**oad **I**ntegrated **O**ptimization and **N**avigation) is a routing optimization system that suggests an optimized route to get from point A to point Z and all potential stops in between while also considering specified delivery windows. It aims to determine routes by a heuristic that uses predominantly right-hand-turns in order to increase efficiency by maximizing both time savings and safety (therefore saving on repairs, etc.). ORION uses a database of 250 million addresses, customized map data and takes into account different variables such as distances, delivery times, different types of customers, and package types. ORION constantly re-evaluates alternative routing options, up to the last moment before a driver leaves the depot and thereby provides optimal routing instructions to UPS drivers—in the second phase planned for the ORION project, UPS plans to enable live recalculations even during delivery.

What Has Been the Business Value of ORION?
The optimization of routes at UPC has had effects on three dimensions: efficiency, sustainability, and service level. By optimizing routes, UPS managed to reduce the total distance traveled by trucks by 20 million miles and to deliver 350,000 additional packages. Estimations indicate that a reduction of 1 mile per day per driver for 1 year results in savings amounting to $50 M.

These efficiency gains in turn positively affect the use of resources, as a reduction in miles traveled causes a diminution of CO_2 emissions (14,000 metric tons) and leads to gasoline savings (5.7 million liters of fuel). At the same time, UPS also increased its service called “My Choice,” which allows consumers to actively choose delivery preferences, reroute shipments, and adjust delivery locations and dates as needed.

What Are the Challenges of ORION?
The costs connected with the development of soft- and hardware and the setup of vehicle routing solutions such as ORION can be extremely high. UPS dedicates 700

employees to the operation and maintenance of ORION, even though there are no official comments regarding the costs directly attributable to ORION.

UPS's overall annual budget of $1.0B on technology suggests that the investment in innovation and technology improvement is material. Also, the lead time needed for the setup can often be considerable. In UPS's case, the ORION algorithm was initially developed in a laboratory and tested at various UPS sites from 2003 to 2009. In order to gather the huge amount of data necessary to develop the ORION system, UPS installed GPS tracking equipment and vehicle sensors on UPS delivery trucks in 2008. The company prototyped ORION at eight sites between 2010 and 2011 before deploying the system to six beta sites in 2012.

Furthermore, the *complexity* of vehicle routing problems in the real world is often very high, requiring special experts for the implementation of VRP solutions. The complexity can easily be understood when considering the possibilities for determining only one route: an average UPS driver has about 125 customers to reach a day. The number of options for scheduling a route for 125 different locations is 125! which is a number with 210 digits, which by far exceeds the number of nanoseconds the earth has existed. Another hint for the complexity of the problem is the fact that after 6 years of research, UPS was able to determine an algorithm that takes 90 pages to describe and 1000 pages of code to implement.

In 2020, UPS stated: "Dynamic Optimization, which will roll out over 12 months, is part of the third generation of ORION. In addition to more accurate delivery time estimates, it will enable new products and services that require higher levels of accuracy, reliability and responsiveness. The new dynamic functionality will help drivers manage their daily commitments more effectively and efficiently and better meet customers' needs, which often evolve by the hour." (https://pressroom.ups.com/pressroom/ContentDetailsViewer.page?ConceptType=PressReleases&id=1580323236977-290).

14.4.2 Decision Situation Outline

Christina and the manager decide together that because of time, they have to separate the AAT from the ADT operations planning. Furthermore, they decide to analyze the AAT setup first. Since Christina is not familiar with the AAT business, she asks for an example to gain insight into the preparation and execution of some AAT services. The manager selects a certain day from the last month. On that day, there was one evening flight that carried overall 29 passengers to the HRG airport. Different travel agencies have booked the corresponding AAT service at RSBT.

Overall, 14 hotels have to be served. The number of passengers q_i to be brought to a hotel H_i varies between 1 and 4, as shown in Table 14.4. The hotels are distributed around the airport (Fig. 14.9).

Next, Christina wants to know how the passengers are transported to the hotels. She learns that four minibusses are available to fulfill the AAT services. Each minibus offers ten passenger seats. All minibusses start their operations from a large parking lot at the airport. From there, a minibus moves to the exit of HRG's arrival

Table 14.4 Hotels to be served in the AAT example

Hotel (code)	Hotel number i	Coordinates		Number of passengers
		x	y	q_i
Marina (H1)	1	103	30	1
Blue Lagoon (H2)	2	143	75	2
Three Palms (H3)	3	123	113	2
Golf Resort (H4)	4	126	88	4
Delphin (H5)	5	117	81	2
Pyramid (H6)	6	38	102	2
Inn at the Beach (H7)	7	81	53	2
Sharky Corner (H8)	8	120	41	1
Diver's Paradise (H9)	9	97	101	1
Red Sea Exclusive (H10)	10	60.75	60.75	3
Water Palace (H11)	11	8.75	78.75	2
Grand Hotel Landside (H12)	12	103	58.5	4
Desert Castle (H13)	13	101.25	87.75	2
Wonder World (H14)	14	87.5	22	1

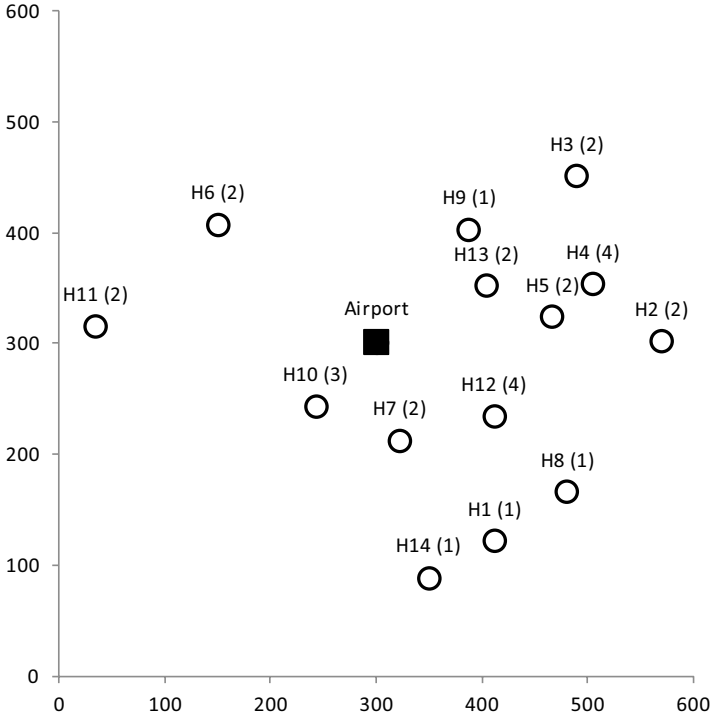


Fig. 14.9 Map of hotels in the example

hall, where the passengers to be served enter the minibus. Next, a minibus goes to the requested hotel(s), where passengers are dropped off. Within one service trip, a minibus regularly visits one or more hotels before it returns empty to the parking lot at HRG.

The AAT manager tells Christina that he has the feeling that the performance of the AAT operations is quite low. In particular, he thinks that the distances traveled (and therefore trip durations) are quite high. As a consequence, the number of deployed vehicles seems to be quite high. Overall, AAT operations seem to produce unnecessary travel costs (fuel costs resulting from detours) as well as personal costs (for the drivers). To demonstrate this, the AAT manager produces Table 14.5 showing the trips executed by the four vehicles. Four vehicles are deployed, and the total distance traveled is 232.42 km + 285.77 km + 204.08 km + 150.81 km = 873.07 km.

14.4.3 Current Approach for the Route Compilation

Having seen only the vehicle trips already executed, Christina feels uncomfortable in giving a statement about the suitability of the proposed trips. She looks for hints that support the evaluation of the proposed service processes (vehicle trips). In a first attempt to estimate the quality of the proposed trips, she visualizes the proposed solution (the trips) in the map provided in Fig. 14.9. Therefore, she connects the hotels by lines in a sequence proposed by the visiting sequences shown in Table 14.5. As a result, she gets the route map shown in Fig. 14.10.

Vehicle 1 follows a continuous line, but vehicle 2 travels along the dotted line. In addition, vehicle 3 operates along the dashed line, while the fourth vehicle follows the gray line. Christina thinks about generating trips in the AAT business and tries to uncover similarities and differences compared to the two previous investigations. Here is what she finds out:

1. Since several vehicles are available, it is necessary to distribute all nodes representing locations to be visited among the available vehicles. No node is allowed to be left uncovered.
2. In order to save traveled distances, each node must be served by one visit, i.e., by exactly one vehicle.
3. Since vehicle capacity is limited, a constraint must be considered: The maximal payload capacity may not be exceeded.
4. For each vehicle, a TSP has to be solved (a Hamiltonian path through the common starting point and the locations assigned to this vehicle).

According to the first two properties of AAT's current situation, it is necessary to make a decision about the assignment of customer locations to the vehicles, i.e., the set of customer nodes is clustered into pairwise disjoint sets, which are called tours. Each tour is extended by the central position (the depot) from which all vehicles are located before the trip execution starts and to which all vehicles return after they have served all assigned customers (all deliveries have been executed).

Table 14.5 Vehicle trips in the AAT example

t	X	y	Occupied seats at departure	Route of
Airport	300	300	9	Vehicle 1
H1	412	121	8	Trip length: 232.42
H2	570	301	6	
H3	490	450	4	
H4	505	353	0	
Airport	300	300		
Airport	300	300	8	Vehicle 2
H5	467	323	6	Trip length: 285.77
H6	150	407	4	
H7	322	212	2	
H8	481	165	1	
H9	388	402	0	
Airport	300	300		
Airport	300	300	9	Vehicle 3
H10	243	243	6	Trip length: 204.08
H11	35	315	4	
H12	412	234	0	
Airport	300	300		
Airport	300	300	3	Vehicle 4
H13	405	351	1	Trip length: 150.81
H14	350	88	0	
Airport	300	300		

The third property implies that clustering is a decision that is constrained, e.g., some clustering decisions are not allowed (they are infeasible). The fourth observation demonstrates that there is also a sequencing problem (a TSP) that must be solved.

Christina leans back and summarizes her findings. The current AAT situation is a constrained combined clustering and sequencing problem. To find out if such a decision situation has been discussed before, she starts an Internet search and learns that the current AAT decision was a problem that had been quite well and intensively studied: It is called the *capacitated vehicle routing problem* (CVRP).

14.4.4 Capacitated Vehicle Routing Problem

Solving a CVRP is the process of finding optimal transportation routes for a given fleet of vehicles under capacity constraints. There are different variations of the CVRP regarding the specific constraints that an organization faces. In Fig. 14.11, the most important CVRP determinants are summarized.

Consider the most important CVRP determinants. First, *objectives* may relate to time, distance, costs, or sustainability. Second, the consideration of *customers* may be subject to different numbers of customers, demand patterns, and time windows.

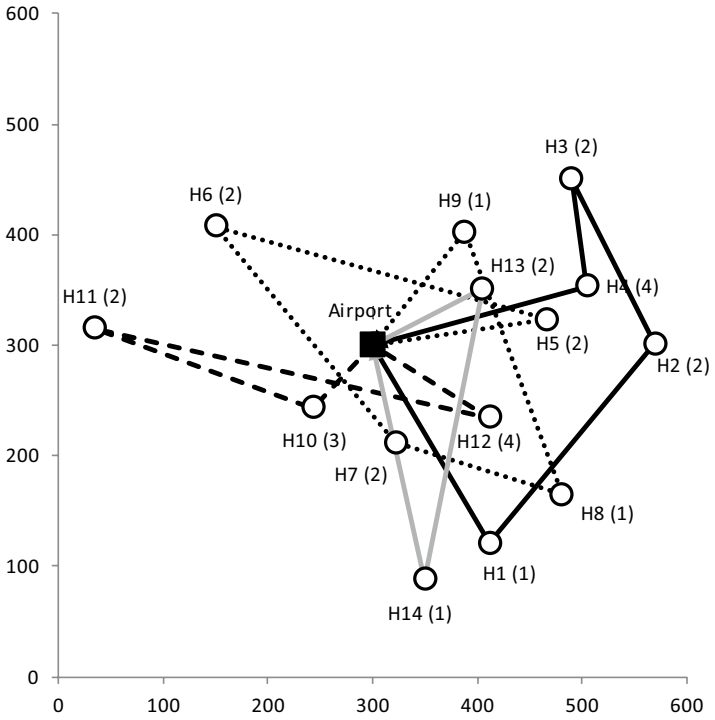


Fig. 14.10 Vehicle trips in the AAT example

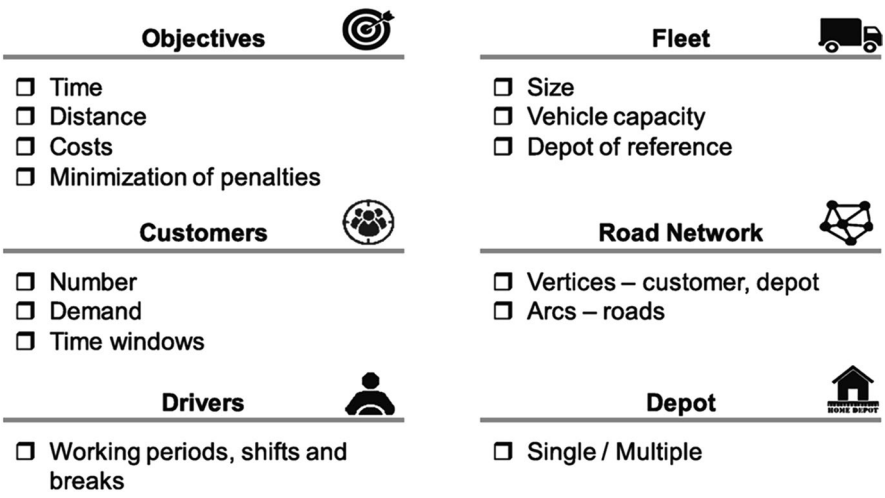


Fig. 14.11 CVRP characteristics

Third, the *drivers* can be classified regarding different working periods and regulations for working conditions and time.

Fourth, the *fleet* can be considered in different ways depending on the size of vehicles (homogenous or heterogeneous), capacity utilization rules (e.g., LTL: less than truckload or FTL: full truckload), and depot of references (e.g., the requirement on returning to the start depot). Fifth, the representation of the *road network* may be different for different CVRP cases. Finally, the *depots* can be considered as single or multiple. The solution methods used for the CVRP consider *heuristic* and *exact* approaches.

Practical Insights

CVRP solutions are typically based on heuristic methods since CVRP is even more complex than TSP. In addition, up-to-date information technology allows the use of real-time data about traffic jams, and weather conditions. This makes it possible to implement *dynamic* or *adaptive* vehicle routing models in combination with sophisticated software to increase flexibility and customer service level.

Having learned that the AAT challenge can be represented as CVRP, Christina wants to know how the vehicle trips (routes) are derived from the requests submitted by the travel agencies. She discusses this issue with the AAT manager. First, she learns that all requests are submitted from the travel agencies to AAT 3 days prior to the due day, which is the execution day. Second, she is told that the consecutively arriving requests are processed in the order of their arrival ("first come/first serve"). Hotel H1 belongs to the first request received, H2 to the second, and so on. The request received first is assigned to the first vehicle, i.e., vehicle 1 visits H1. The second request received is appended to this route after the first request, i.e., H2 is visited by vehicle 1 immediately after H1 has been visited. The next requests are processed similarly as long as the residual capacity (number of free seats) of the vehicle currently under consideration is large enough to serve the next request, i.e., as long as adding the next request would not result in exceeding the capacity of the vehicle. The route of this vehicle is then closed, i.e., the return of the vehicle to the central depot (parking lot) is added, and the route of this vehicle is not changed any further. A new vehicle route is opened, and the drop-off location for the request that has caused excess capacity is inserted as the first customer location in the route of the second vehicle. Altogether, a greedy strategy similar to the strategy introduced for the TSP is applied to the CVRP. Two differences must be mentioned.

1. The next request is not determined by the smallest distance increase but by the arrival time of the request.
2. A capacity limit assigned to each vehicle is observed. When the addition of the next request would lead to a capacity excess, a new vehicle route is started.

The pseudocode description of the applied construction procedure for the CVRP is as follows.

Construction Procedure for the CVRP	
(a)	Initialize: START = 0; DEST = 1; k = 1; v = 1; cap_used(v) = 0; L(v) = 0; PATH[v,k] = START;
(b)	Repeat steps (c)-(p) until DEST > CUSTOMERS
(c)	If CAP_used(v) + q(DEST) ≤ CAP(v) then goto (j)
(d)	PATH[v;k + 1]: = 0;
(e)	L:=L + d(START,0);
(f)	v:=v + 1;
(g)	L(v):=0;
(h)	cap_used(v):=0;
(i)	k = 0; START:=0;
(j)	L(v):=L(v) + d(START,DEST);
(k)	cap_used(v): = cap_used(v) + q(DEST);
(l)	k:=k + 1;
(m)	PATH[v;k]:=DEST;
(n)	START:=DEST;
(o)	DEST:=DEST +1
(p)	Goto (b);
(q)	Return PATH;
(r)	End;

All vehicles start at the depot node 0 and have the remaining payload capacity CAP[v]. The vehicle currently under consideration is represented by v. The assigned payload of vehicle v is labeled by cap_used(v), and L(v) carries the length of the route of vehicle v. PATH[v;k] contains the node that is the kth customer location in the route of vehicle v. START gives the last visited node and DEST contains the node associated with the request at position DEST in the sequence of requests.

The CVRP construction procedure starts with the initialization of the used variables (a). In each iteration, the steps (b)–(p) are processed. Iteration starts by checking whether there are still unassigned customer locations (b). After that, the assignment of the next request to vehicle v is tested to see if it will lead to an excess in capacity in this vehicle (c). If this is not the case, then the trip length is updated (j), the used capacity is increased (k), the vehicle customer location is appended to the existing partial route (m), and the next customer to be considered is selected (n)–(o). If the capacity test fails, i.e., if the next customer cannot be assigned to the current vehicle, then the depot is appended as the last visited location to the route (d), and the return distance is added (e). Afterwards, the route of the next vehicle is initialized (f)–(i).

Christina asks for the ratio behind the planning logic represented by the CVRP construction procedure. She learns that this planning procedure is very simple and can be applied very quickly. In studying this information, Christina is sure that she can improve the route plan, because the proposed CVRP construction procedure does not consider any information about the geographic location of the hotels to be visited. This results in unnecessary detours. For example, she refers to the dashed

and to the dotted vehicle routes in Fig. 14.10. Here, the two vehicles are "zig-zagging" through the area; traveling back and forth between different regions causes detours.

14.4.5 The Sweep Algorithm

Task 14.2 Solving the CVRP by the Sweep Algorithm
Consider the locations with transformed coordinates (Table 14.6).

To overcome the shortcomings associated with the CVRP construction procedure, Christina thinks about opportunities to identify and exploit similar locations of different hotels. One characteristic of a customer location (hotel) is its direction in relation to the central reference point, which is the airport parking lot at point (75; 75). Another characteristic of a hotel location is the distance between the airport parking lot and the hotel. If and only if both values, direction as well as distance from the airport of different hotels, are similar, then both hotels are situated close together. However, if the direction components are similar, then it is perhaps beneficial to assign both customers to the same vehicle since detours are unlikely and the vehicle is already heading in the same direction for both customers (hotels) (Fig. 14.12).

The analysis of the x-y-coordinates that describe the hotel locations does not help Christina to find similarities between the hotel sites. For this reason, she thinks about transforming the x-y-coordinates into polar coordinates. Therefore, she defines the gray vertical reference line as shown in Fig. 14.12 and determines the direction of a hotel, e.g., H6, by drawing a line from the airport's coordinates to H6. Each hotel, i.e., H6, is now defined by the distance r_i between the airport and the

Table 14.6 Locations with transformed coordinates

	No.	(x; y)-coordinates			Polar coordinates		Angle
Location	i	x	y	q_i	r_i	$\deg_i$	α_i
Airport	0	75	75	0	0	0	0
H6	6	37.50	101.75	2	46.06	0.95	54.51
H11	11	8.75	78.75	2	66.36	1.51	86.78
H10	10	60.75	60.75	3	20.15	2.36	135.01
H14	14	87.50	22.00	1	54.45	3.37	193.27
H7	7	80.50	53.00	2	22.68	3.39	194.04
H1	1	103.00	30.25	1	52.79	3.70	212.04
H8	8	120.25	41.25	1	56.45	4.07	233.29
H12	12	103.00	58.50	4	32.50	4.18	239.50
H2	2	142.50	75.25	2	67.50	4.72	270.21
H5	5	116.25	80.75	2	42.14	4.85	277.84
H4	4	126.25	88.25	4	52.94	4.96	284.50
H13	13	101.25	87.75	2	29.18	5.16	295.91
H3	3	122.50	112.50	2	60.52	5.38	308.30
H9	9	97.00	100.50	1	33.68	5.57	319.22

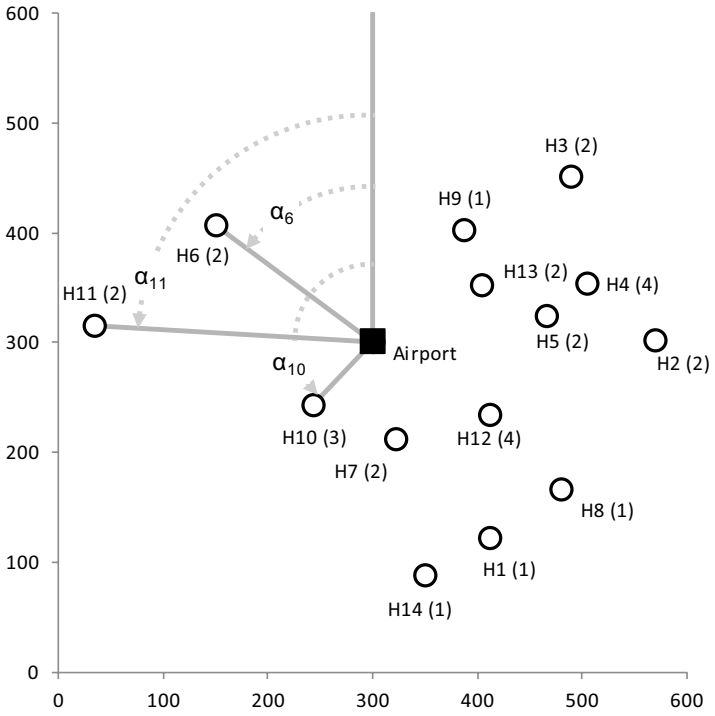


Fig. 14.12 Determination of polar coordinates of locations in relation to the *vertical gray line*

hotel ("radius") plus the direction as defined by the angle α_i between the vertical reference line and the direction line to the hotel side. In column 7 of Table 14.6, this angle is given by $\deg_i$ of the gray arc (the length of the circumference of a complete circle = 6.282). If $\deg_i$ is multiplied by 360°/6.282, then we get the angle α_i.

Christina now sorts the requests according to increasing α_i-values. She gets the request sequence shown in columns 1 and 2 in Table 14.6. She saves this sequence in the list SEQ, so that SEQ[1] = H6, SEQ[2] = H11, SEQ[3] = H10, SEQ[4] = H14, SEQ[5] = H7, SEQ[6] = H1, SEQ[7] = H8, SEQ[8] = H12, SEQ[9] = H2, SEQ [10] = H5, SEQ[11] = H4, SEQ[12] = H13, SEQ[13] = H3, SEQ[14] = H9.

Sweep procedure for the CVRP	
(a)	Initialize: START = 0; REQ = 1; DEST = SEQ[REQ]; k = 1; v = 1; cap_used(v) = 0; *L* (v) = 0; PATH[v,k] = START;
(b)	Repeat steps (c)–(q) until DEST > CUSTOMERS
(c)	If CAP_used(v) + *q*(DEST) ≤ CAP(v) then goto (j)
(d)	PATH[v;k + 1]: = 0;
(e)	L:=L + *d*(START,0);
(f)	v:=v + 1;
(g)	*L*(v):=0;
(h)	cap_used(v):=0;

Sweep procedure for the CVRP	
(i)	k = 0; START:=0;
(j)	L(v):=L(v) + d(START,DEST);
(k)	cap_used(v): = cap_used(v) + q(DEST);
(l)	k:=k + 1;
(m)	PATH[v;k]:=SEQ[REQ];
(n)	START:= SEQ[REQ];
(o)	REQ:=REQ + 1;
(p)	DEST:=SEQ[REQ];
(q)	Goto (b);
(r)	Return PATH;
(s)	End;

The aforementioned pseudocode depicts the modified construction procedure that now incorporates the determined sequence of requests ordered by increasing angle α_i. This procedure appends requests in the sequence SEQ to a vehicle as long as this vehicle has enough capacity. When vehicle capacity is not sufficient to serve the next request according to the SEQ-order, it continues with the next vehicle. If we fix the lower corner of the vertical gray line at the depot at (75; 75) and if we move the other corner counterclockwise around the depot, then the gray line "sweeps" consecutively over all locations in the sequence SEQ. For this reason, the procedure represented in the pseudocode is called the "sweep algorithm for the capacitated vehicle routing problem" (Gillett and Miller 1974).

Christina applies the sweep algorithm to the AAT example and gets the route as shown in Table 14.7. She adds up the travel distances of the three generated routes and finds out that they add up thus: 239.45 km + 209.02 km + 172.42 km = 620.90. Compared to the travel distance achieved from the application of the construction procedure shown in Fig. 14.12, which was 873.07 km, Christina calculates a travel distance saving of ≈29%. This results from the variation of the sequences in which the request locations are appended to the existing routes. Shorter distances also mean higher sustainability in terms of emission reductions. She learns that a careful analysis of the locations to be inserted next at the end of a tentative route significantly contributes to saving travel distances. The resulting vehicle routes are plotted in Fig. 14.13.

The number of vehicles required is also reduced if the sweep procedure is applied. However, Christina concludes that this is an achievement that has not been addressed by the sweep procedure. Christina realizes that a reduction in the number of vehicles used has the potential to contribute to the reduction of the total sum of traveled distances. Her statement is mainly inspired by the following observation. If there are two vehicles following two routes, then each vehicle must travel to the first customer in its route, and each vehicle has to travel back from the last visited customer location to the depot. We have two outbound route segments from the depot with length $d1$ (to node p) and $d2$ (to node q) and the inbound travel distances from the nodes s and t to the depot of length $d3$ and $d4$. If both routes are integrated, then we can save one outbound segment (say of the second vehicle) and one inbound

Table 14.7 Vehicle trips in the AAT example proposed by the sweep algorithm

Location	x	Y	Occupied seats at departure	Route of
Airport	75.00	75.00	10	Vehicle 1
H6	37.50	101.75	8	Trip length: 239.45
H11	8.75	78.75	6	
H10	60.75	60.75	3	
H14	87.50	22.00	2	
H7	80.50	53.00	0	
Airport	75.00	75.00		
Airport	75.00	75.00	10	Vehicle 2
H1	103.00	30.25	9	Trip length: 209.02
H8	120.25	41.25	8	
H12	103.00	58.50	4	
H2	142.50	75.25	2	
H5	116.75	80.75	0	
Airport	75.00	75.00		
Airport	75.00	75.00	9	Vehicle 3
H4	126.25	88.25	5	Trip length: 172.42
H13	101.25	87.75	3	
H3	122.50	112.50	1	
H9	97.00	100.50	0	
Airport	75.00	75.00		

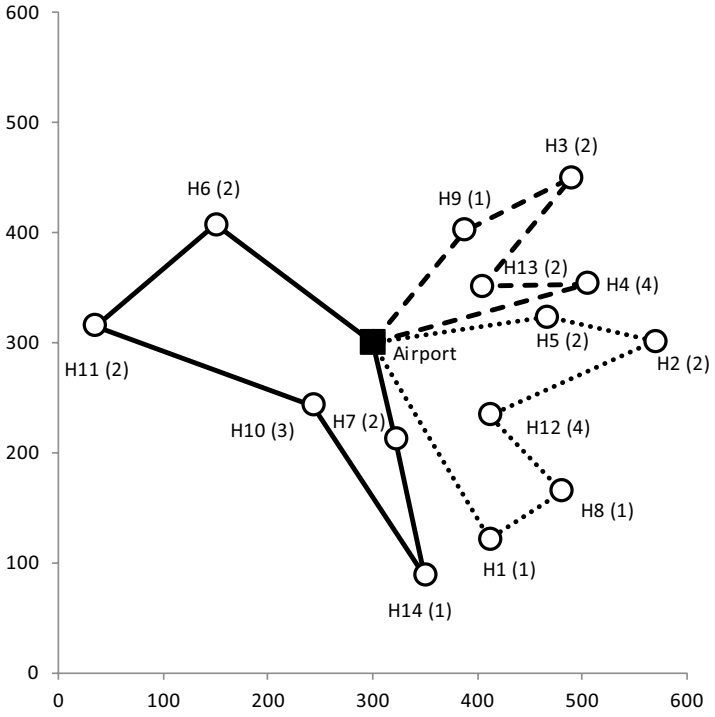


Fig. 14.13 Routes proposed by the sweep algorithm

segment, which summarize to $d2 + d4$. If this sum is larger than the distance of the additional route segment from s to p of length $d(s, p)$, i.e., if $d2 + d4\ d(s, p) > 0$, then the combination of the two trips realizes travel distance reductions. Clarke and Wright (1964) have proposed a so-called saving algorithm that starts with pendulum routes between the depot and a single customer location and continues to combine two routes to save travel distances. Further details on the ADT scenario can be found in the example in the E-Supplement.

14.5 Machine Scheduling

Christina wants to thank her study colleague (his name is Mark) for providing all the information about the TSP that enabled her to understand the algorithmic ideas and improve the operations at RSBT. She calls him one evening, and they talk for a long time about their current professional involvement. Mark mentions that he is working on scheduling problems, but in the manufacturing business, he is responsible for setting up machine schedules. Christina is interested and invites him to talk about his experiences.

14.5.1 The Problem of Scheduling a Machine

Mark explains the general setup of a simple one-machine scheduling challenge in a manufacturing environment (Fig. 14.14).

There is a machine that is able to process one manufacturing task at a time. A manufacturing task is part of a customer *order*. All received orders are collected in the order or job backlog. At a specific time, all jobs are prepared for fulfillment. Fulfilling a job means using the machine to execute the manufacturing tasks requested by the customer that is associated with a job.

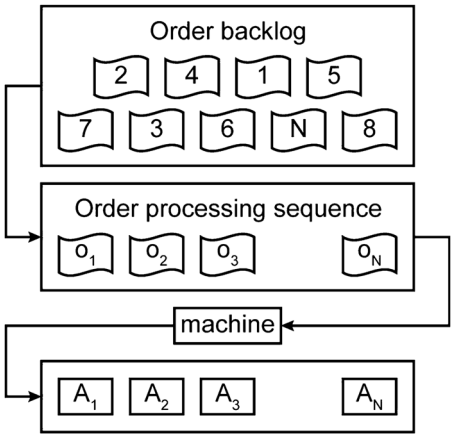


Fig. 14.14 Machine scheduling problem

All jobs from the job backlog are ordered and jobs from the backlog are consecutively processed by the machine in the order determined by the job sequence o_1, o_2, …, o_N. The executed tasks create the assets A_1, A_2, …, A_N requested by the customers. The basic decision problem associated with the outlined machine scheduling scenario is to determine the job *processing sequence* that determines the *schedule of the machine*.

Christina wants to know how a manufacturing request can be evaluated in order to find an appropriate job processing sequence. Mark explains that each individual job is associated with different attributes. According to these attributes, jobs can be compared and prioritized. Thonemann (2015) proposed the following three attributes to compare jobs in a job backlog. First, the *due date* d_i of job i is used.

The due date determines the latest permitted completion time associated with a job. Second, each job has a specific *processing or flow time* p_i, which is the time needed to process job i on the machine. Third, the *release time* t_i of job i is used to characterize a job. The release time is the time at which the job is completely specified so that its processing can start.

Christina interrupts Mark and tells him that she would have been faced with a similar situation if she had handled the TSP. She had experienced that finding good sequences was a very challenging task due to the strong increase in possible permutations of operations. Furthermore, she wants to know about the planning objectives in machine scheduling. Mark continues and states that there are several commonly used *performance indicators*:

The *completion time* of job i determines the time when the job is completely fulfilled, i.e., it is the time when the job "leaves" the machine. The completion time is typically denoted by c_i.

The *flow (lead time) time* f_i of job i is the difference between the completion time c_i and the release time r_i of a job i. It can be interpreted as the time span necessary to respond to a customer job release.

Lateness or *tardiness* TD_i of job i measures the amount of time in excess of the due date associated with a job i.

Work-in-process inventory and *capacity utilization.*

Lead time is composed of processing time, setup time, waiting time, lying time, and transportation time (Fig. 14.15).

Setup time is needed to prepare the machine for processing the next job. *Laying time* is needed because of technological restrictions (e.g., drying). *Transportation*

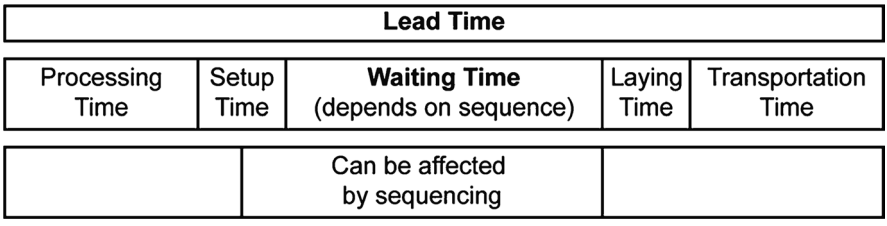


Fig. 14.15 Lead time

time reflects the physical movement of jobs through the machines. *Waiting time* exists because a job cannot be processed at the next machine since this machine is occupied with another job. Even the waiting time and sometimes the setup time can be affected by sequencing.

Completion times, flow times, and tardiness are performance indicators for quantifying scheduling decisions from the perspective of an individual request. In order to evaluate the quality of a complete job processing sequence, additional performance indicators are used.

The *makespan* MS(S) associated with a job processing sequence S is defined as the length of the period that starts with the release of the first job and ends with the completion of the last job in sequence S.

The *average completion time* ACT(S) associated with sequence S is the average value of the completion times of the processed jobs if the jobs are processed in sequence S by the machine.

The *maximal completion time* MCT(S) is the maximal value observed among all completion times after the application of the job processing sequence S.

Mark concludes his short introduction to machine scheduling with the remark that all the performance indicators can be used as primary planning goals. This creates *trade-offs* or goal conflicts since some of the goals contradict each other.

A typical trade-off in production scheduling is the lead time and capacity utilization. Just imagine that you are in a supermarket on a Tuesday morning. Probably, you are the only person there at this time. So you will not have to wait anywhere and will exit the supermarket in a short time. The time of your shopping will be short. But what about the supermarket? If you are the only customer, their capacity utilization is very low. The same effect can be encountered in manufacturing and service systems in regard to customer order lead time and resource capacity utilization. This typical *trade-off in scheduling* is called *the scheduling dilemma* (Fig. 14.16).

If the number of orders or waiting customers increased, this would lead to a higher capacity utilization. At the same time, lead time for each individual order or customer would increase due to longer waiting times. According to the scheduling dilemma, we cannot achieve a reduction in lead time and an increase in capacity

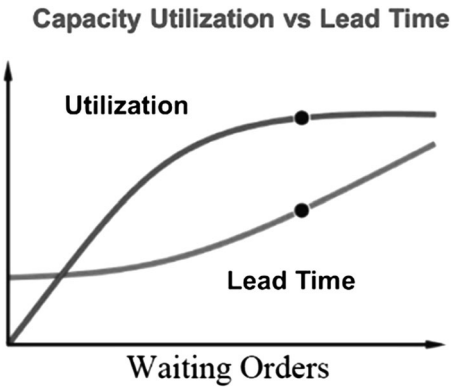


Fig. 14.16 Scheduling dilemma: capacity utilization vs. lead time

utilization at the same time. Mark gives an example. Students at the Berlin School of Economics and Law arrive at the subway station Berliner Strasse, which is about a 10 min walk from the university. They can also use bus 104 to get to the school from the subway station. Assume that the buses run every 10 min, and every minute, one student comes to the bus stop. In this case, the waiting times and the lead times will be quite long, but the bus will take ten students and have good capacity utilization. Should the buses run every minute, the lead times would be very short (no waiting time), but the capacity utilization of each bus would be low.

14.5.2 Priority Rule-Based Scheduling

After she has learned that the schedule generation is also based on the sequencing of operations, as is the visit sequencing in the TSP, Christina wants to know how appropriate job processing sequences are determined. She asks Mark whether he also uses "greedy" strategies to add operation by operation to a partial sequence. Mark confirms her speculation, but tells Christina that there are some sort of *priority* or *dispatching rules* whose applications make the sequencing straightforward.

Mark explains that a priority rule is a "rule of thumb" to sort a set of jobs according to increasing or decreasing values of the release times r_i, the flow times f_i, or the due dates d_i. The application of such a "simple" rule seems to be promising since the evaluation value of an individual job is not affected by the earlier or the subsequently processed job. This is a major difference from sequencing in the TSP when the travel distances between consecutively visited nodes impact the overall objective, which is the total sum of traveled distances. Priority rules can be applied both for *job shop* and *flow shop* (cf. Chap. 9).

Task 14.3 Scheduling by Priority Rules
To demonstrate the effects of determining a job processing sequence using a priority rule, Mark introduces an example with eight jobs. The attributes of the jobs from the job backlog are summarized in Table 14.8.

The most intuitive approach for the determination of the job processing sequence is to process the jobs in the sequence of their arrival or release time. Those jobs arriving first will be processed first. This sequencing rule is called *First Come/First Serve (FCFS).* It does not apply to any "intelligent" sorting, but it is a fair approach.

Table 14.9 shows the machine schedule resulting from the FCFS-processing job. The makespan is 44–1 = 43 time units. The average completion time equals 27 time units, but the maximal completion time is 44 time units. Five jobs are completed

Table 14.8 Job backlog

Job i	1	2	3	4	5	6	7	8
Release time r_i	1	2	3	5	6	7	8	1
Duration p_i	4	2	3	7	2	8	3	4
Due date d_i	14	29	16	23	24	18	31	14

with delays. The average tardiness is 6.75 time units, and the maximal tardiness is 23 time units. On average, the flow time is 22.5 time units.

The application of FCFS does not consider processing times or due dates while determining the sequence of processing. In order to enable a quick delivery of jobs, it is useful to try to minimize the completion times of jobs. Here, it is beneficial to process those jobs first that can be processed rapidly in order to make the machine available for the next job as quickly as possible. According to the *Shortest-Processing-Time rule (SPT-rule)* the jobs are sorted by increasing processing time. The job with the shortest processing time in the job backlog is processed first.

The SPT-generated job processing sequence is listed in column 1 of Table 14.10. Again, the makespan is 43 time units, but the average completion time is reduced from 27 time units to 23.25 time units. The maximal completion time remains 44 time units. Four jobs are still delayed, with a slightly reduced tardiness of 6 time units. However, the maximal tardiness is prolonged to 26 time units, but the average flow time is reduced to 18.75 time units.

Neither the application of the SPT-rule nor the application of FCFS is able to control the maximal tardiness since neither of these two rules considers the due date information associated with the jobs. In order to minimize maximal tardiness, the *Earliest Due Date rule (EDD-rule*) is proposed. According to the EDD-rule, the most urgent job is processed first, followed by the job with the second-highest urgency.

Table 14.9 Job processing sequence and machine schedule generated by FCFS

Job i	Release time r_i	Duration p_i	Due date d_i	Start time	Finish time	Tardiness	Flow time	Delayed?
1	1	4	14	8	12	0	11	0
2	2	2	29	12	14	0	12	0
3	3	3	16	14	17	1	14	1
4	4	7	28	17	24	0	20	0
5	5	7	23	24	31	8	26	1
6	6	2	24	31	33	9	27	1
7	7	8	18	33	41	23	34	1
8	8	3	31	41	44	13	36	1

Table 14.10 Job processing sequence and machine schedule generated by SPT

Job i	Release time r_i	Duration p_i	Due date d_i	Start time	Finish time	Tardiness	Flow time	Delayed?
2	2	2	29	8	10	0	8	0
6	6	2	24	10	12	0	6	0
3	3	3	16	12	15	0	12	0
8	8	3	31	15	18	0	10	0
1	1	4	14	18	22	8	21	1
4	4	7	28	22	29	1	25	1
5	5	7	23	29	36	13	31	1
7	7	8	18	36	44	26	37	1

Table 14.11 Job processing sequence and machine schedule generated by EDD

Job i	Release time r_i	Duration p_i	Due date d_i	Start time	Finish time	Tardiness	Flow time	Delayed?
1	1	4	14	8	12	0	11	0
3	3	3	16	12	15	0	12	0
7	7	8	18	15	23	5	16	1
5	5	7	23	23	30	7	25	1
6	6	2	24	30	32	8	26	1
4	4	7	28	32	39	11	35	1
2	2	2	29	39	41	12	39	1
8	8	3	31	41	44	13	36	1

The makespan remains 43 time units (since there is no slack time in the schedules) as shown in Table 14.11. The average completion time is 29.5 time units, and the maximal completion time equals 44 time units. An average flow time of 25 time units is observed. The application of the EDD-rule leads to a significant reduction in maximal tardiness from 23 (FCFS) and 26 (EDD), respectively, down to 13 time units. Unfortunately, the number of delayed requests is increased (six jobs). The same development is observed for the average tardiness, which is increased by 7 time units compared to 6 time units (SPT) and 6.75 time units (FCFS).

▶ **Practical Insights** The application of a priority rule enables the minimization of a single performance indicator value. There is no priority rule known that minimizes all three performance indicators (1) number of delayed requests (2) average completion time, and (3) maximal delay. However, priority rules are easy to apply. This is why these simple heuristics are widely used in practice. In sophisticated scheduling software, so-called *meta-heuristics* such as *genetic algorithms* and *ant colony optimization* (ACO) are used. They provide solutions of very high quality.

14.5.3 Scheduling Algorithm of Moore

Moore (1968) published a paper about a one-machine scheduling algorithm that minimizes the number of delayed jobs. The basic idea is to extend the EDD-rule by identifying those jobs that block the machine. Those jobs are then moved to the end of the sequence with the hope that now one or even more jobs can be started earlier so that the number of delayed jobs can be reduced. Mark is going to explain to Christina how the Moore-algorithm works by means of the example previously introduced.

First, the job backlog is scheduled according to the EDD-rule. If there are no delayed jobs, then the scheduled construction is finished. Otherwise, the first delayed job according to the EDD-induced schedule is marked; here job 7 is the first delayed job (Table 14.12).

The basic idea of the Moore-algorithm is to find out the cause of this delay. Moore supposed that the marked or a previously processed job is responsible for the delay. To this end, Moore suggested moving the labeled job 7 or one of the previously scheduled jobs (job 1 or job 3) to the end of the sequence so that all subsequent jobs can be started earlier. In order to identify the most appropriate job to be moved to the end, Moore proposed comparing the processing times p_i of the marked job and of the earlier scheduled jobs and selecting the job among the aforementioned jobs with the longest processing time in order to achieve a maximal left shift of the job starting times. In the example (Table 14.12), job 7 exhibits the longest processing time $p_7 = 8$ so that it is moved to the end of the sequence. This job is marked by * in Table 14.13.

In the next iteration, the first delayed job (job 4) is marked. This job and job 5 have the same maximal processing time of 7 time units among all requests started after the marked job. Job 5 is selected (Since it appears first in the intermediate sequence), and job 5 is moved to the end of the sequence (Table 14.14).

Table 14.14 contains the final scheduling sequence (1; 3; 6; 4; 2; 8; 7; 5). None of the so far unmoved jobs is delayed so that the algorithm terminates. The minimum number of delayed jobs is two.

Table 14.12 Moore-algorithm (start): job processing sequence and machine schedule generated by EDD and marked first delayed job 7

Job i	Release time r_i	Duration p_i	Due date d_i	Start time	Finish time	Tardiness	Flow time	Delayed?
1	1	4	14	8	12	0	11	0
3	3	3	16	12	15	0	12	0
7	**7**	**8**	**18**	**15**	**23**	**5**	**16**	**1**
5	5	7	23	23	30	7	25	1
6	6	2	24	30	32	8	26	1
4	4	7	28	32	39	11	35	1
2	2	2	29	39	41	12	39	1
8	8	3	31	41	44	13	36	1

Table 14.13 Moore-Algorithm (second iteration): Job processing sequence and machine schedule generated by EDD and marked first delayed job 4

Job i	Release time r_i	Duration p_i	Due date d_i	Start time	Finish time	Tardiness	Flow time	Delayed?
1	1	4	14	8	12	0	11	0
3	3	3	16	12	15	0	12	0
5	5	7	23	15	22	0	17	0
6	6	2	24	22	24	0	18	0
4	**4**	**7**	**28**	**24**	**31**	**3**	**27**	**1**
2	2	2	29	31	33	4	31	1
8	8	3	31	33	36	5	28	1
7*	7	8	18	36	44	26	37	1

14.5.4 Scheduling Two Machines in a Flow Shop

Christina detects similarities between the machine scheduling problem outlined by Mark and the TSP. There are the jobs that correspond to the requests in the TSP and there are resources: in the TSP, there is a vehicle, but in Mark's setup, there is a manufacturing machine. Christina wonders if there is a machine scheduling setup with two or even more machines that correspond to the CVRP (Fig. 14.17).

She asks Mark about this question. Mark affirms but tells Christina that there are several multi-machine setups. Christina wants to know what schedule generation is like in such a multi-machine scenario. Mark starts with a description of the system setup (Fig. 14.17). Each job requires two production steps. In the first step, the workpiece is painted, and in the second step, the painted piece is protected by a coating. Of course, it is necessary that painting precedes coating for all jobs. Each job i is characterized by the two processing times p_i^{paint} for the painting procedure and p_i^{coat} for the coating procedure (Table 14.15).

Table 14.14 Moore-Algorithm (third iteration): Final job processing sequence and machine schedule with minimum number of delayed requests

Job i	Release time r_i	Duration p_i	Due date d_i	Start time	Finish time	Tardiness	Flow time	Delayed?
1	1	4	14	8	12	0	11	0
3	3	3	16	12	15	0	12	0
6	6	2	24	15	17	0	11	0
4	4	7	28	17	24	0	20	0
2	2	2	29	24	26	0	24	0
8	8	3	31	26	29	0	21	0
7*	7	8	18	29	37	19	30	1
5*	5	7	23	37	44	21	39	1

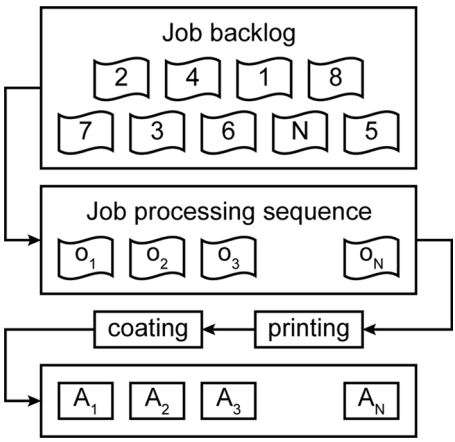


Fig. 14.17 Two machine scheduling situation

Table 14.15 Job backlog for the two machine setup

Job i	1	2	3	4	5	6	7	8
Release time	3	4	5	6	7	8	9	10
Paint duration p_i^{paint}	4	2	3	7	2	8	3	4
Coating duration p_i^{coat}	7	6	1	5	4	3	4	2

Mark explains that there is a famous approach to this setting for minimizing the completion time of the complete job backlog. This algorithm was originally proposed by Johnson (1954) and is hence called *Johnson*'s *algorithm*. This algorithm follows the principle of keeping the *idle times* of the second machine as short as possible. This is achieved by starting those jobs first on the first machine with the shortest p_i^{paint}-values. These jobs are quickly processed by the first machine and then forwarded to the second machine in order to avoid idle times on the last-mentioned machine. The starting sequences of the jobs on the two machines should be kept unchanged. Therefore, it is sufficient to define the job processing sequence only for the first machine. The determined sequence then also applies to the second machine.

Johnson's algorithm starts by splitting up the job backlog into two subsets J_1 and J_2. All jobs i whose processing times p_i^{paint} are shorter than their processing times p_i^{coat} are put into J_1. The remaining jobs are collected in set J_2. In the example setting, the first set is J_1:= {1; 2; 5; 7} and the second set is J_2:={3; 4; 6; 8}.

In the second step, both sets are ordered. The set J_1 is sorted by increasing processing times $p_i^{\text{paint,}}$ and the set J_2 is sorted by decreasing processing times p_i^{coat}. In the example, we have the sequence S_1:=(2; 5; 7; 1) as well as the sequence S_2:= (4; 6; 8; 3).

In the third step, sequence S_1 is combined with sequence S_2 by appending S_2 to S_1. In the example, we get the complete job backlog sequence S:=(S_1,S_2) = (2; 5; 7; 1; 4; 6; 8; 3).

Mark outlines the generated schedule in a *Gantt-chart* (i.e., a special form of schedule representation) and compares it with the schedule that results from starting the jobs in the order in which they have been released (Fig. 14.18). It can be observed that makespan has been reduced with the help of Johnson's algorithm.

14.5.5 Further Challenges in Machine Scheduling

In reality, there is a variety of other more complicated machine scheduling situations (Kovalyov et al. 2007; Dolgui et al. 2010; Berrichi and Yalaoui 2013; Ren et al. 2022). Li and Chen (2010) investigated a one-machine setup in which the exact processing times of the jobs in the backlog are unknown, but estimated by a distribution function. Albers (1997) analyzed a setup in which the job backlog is not available. Instead, no arriving job is scheduled before it arrives or without considering a job arriving subsequently.

If two or more machines must be scheduled in order to fulfill a set of jobs, two general setups are distinguished. The case outlined by Mark falls into the category

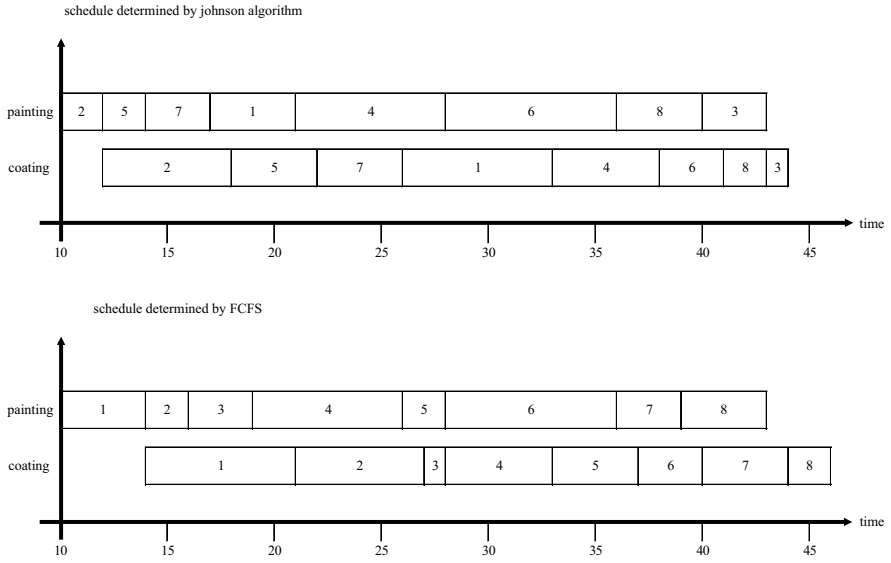


Fig. 14.18 Two machine schedules generated by FCFS and Johnson's algorithm

of *flow production* (Dolgui and Proth 2010; Pinedo 2022): all jobs are processed in the same sequence on all available machines. In the *job shop* setup (Mattfeld 1996; Werner 2013), each job maintains an individual machine processing sequence. In many industries, flexible flow and job shops with alternative parallel machines are used (Kyparisis and Koulamas 2006; Ivanov and Sokolov 2012). In the process industry, continuous flows represent an additional challenge (Shah 2004; Bożek and Wysocki 2015; Ivanov et al. 2016a).

Next, SC coordination belongs to future trends in scheduling research and practice. In this area, *integration of scheduling and routing* problems is studied (Agnetis et al. 2006; Chen 2010). In addition, integration of *inventory-routing* decisions and supplier selection and scheduling is a promising research area (Sawik 2013).

Uncertainty challenges scheduling and routing decisions. In this setting, the issues of schedule *stability and robustness* along with the development of rescheduling policies have become more and more important (Kolisch and Hess 2000; Vieira et al. 2003; Artigues et al. 2005; Aytug et al. 2005; van de Vonder et al. 2007; Hazir et al. 2010; Gomes et al. 2013; Sotskov et al. 2013; Harjunkoski et al. 2014; Liu and Ro 2014; Zhang et al. 2022; Zheng et al. 2022).

Finally, *scheduling in Industry 4.0 and Cloud Manufacturing* are new and exciting areas for research and practice (Liu et al. 2019; Sokolov et al. 2020; Ivanov et al. 2021; Mahmoud et al. 2022; Destouet et al. 2023).

14.6 Key Points

In this chapter, you have been introduced to typical operative process planning tasks from transportation as well as production. In particular, you are now familiar with the shortest path problem, the TSP problem, the CVRP, with the single machine scheduling problem, and a simple two machine flow shop problem.

You have learned the basics about the formulation of decision models on mathematical graphs, and you are now familiar with the concept of modeling operative decision tasks as graph-based optimization models. Decision variables are used to represent atomic decision tasks, and constraints must be considered in order to compile a feasible solution for a complex decision problem from the atomic decisions.

You are aware now that a careful selection of adequate algorithms for solving optimization models defined on graphs is important. For some decision problems, such models as the one presented for the shortest path problem, there are exact algorithms. As an example, you have been introduced to the Dijkstra algorithm.

Christina leans back and recapitulates what she has learned. First, a network-based optimization model for TSP was introduced. This model falls into the category of a mixed-integer linear problem since some decision variables must be integers or even binary. Within such a model, it is possible to add constraints to control the decision variable value determination.

Compared to the shortest path problem in a network, which does not have any constraints about nodes to be visited, the complexity of the TSP is significantly increased. Therefore, it is reasonable to refrain from using exact algorithms for the identification of the best possible Hamiltonian path (with the least travel distance). Instead, heuristics are proposed to approximate an optimal Hamiltonian path. There are different types of heuristics. Construction heuristics are used to set up an initial feasible solution for a constraint optimization model. Improvement heuristics (e.g., a 2-opt improvement procedure) are used to improve the objective function value of the best-found feasible solution, i.e., to replace an existing feasible solution with another feasible solution having an improved objective function value). Christina is happy since she can use the proposed techniques to model and solve RSBT's SST challenge.

Section 14.4 introduced the decision problem class of combined assignment and sequencing problems from operation fleet management. The basic decision task is described in the CVRP. We started the discussion of the CVRP with an outline of a typical planning situation. By means of this example, we discussed the sophisticated decision challenges of a simultaneously conducted partition of the set of requested locations and the sequencing of the locations to determine vehicle routes. We have seen that failures in the assignment of requests to vehicles typically result in detours and an increase in the number of required vehicles. Both issues contribute to additional costs for fulfilling customer requests.

In order to contribute to keeping the fulfillment costs as low as possible, we proposed analysis of customer locations, i.e., the locations that require a visit. For this reason, we proposed sorting all locations by means of their angle relative to a reference line. We then proposed the sweep algorithm, which exploits information about

the vicinity of different locations. The sweep algorithm tries to compile closely situated locations into one route in order to keep the sum of traveled distances low. We are now prepared to manage all decision situations that are in the form of the AAT challenge.

Finally, we have learned how to design simple heuristics for complex routing and scheduling models. But you are also aware that the computation of (sub)-optimal model solutions is often very complicated. Nevertheless, you understand the importance of formulating an appropriate decision model as the interface between applications and computers. The formulation is the most important ingredient for the setup of computerized decision support systems.

Toth and Vigo (2002) discussed model formulations for the CVRP. These models are used to apply special solver tools like CPLEX or LINGO in order to derive proven optimal solutions to the CVRP. However, depending on the actual data of a CVRP scenario and depending on the number of available vehicles and requests to be served, the processing times are often prohibitive so that for practical, real-world problem settings, heuristics are applied preferentially (Gendreau et al. 2002).

For other decision models, it is necessary to incorporate heuristics like the greedy heuristic or the sweep heuristic. There are also priority rules that can be applied to sequence tasks in machine scheduling. You can distinguish between construction and improvement procedures as basic ingredients of heuristic algorithms for solving complex models with several constraints. Fleet routing and machine scheduling belong to the most exciting tasks in SCOM!

Bibliography

Agnetis A, Hall NG, Pacciarelli D (2006) Supply chain scheduling: sequence coordination. Discret Appl Math 154(15):2044–2063

Albers S (1997) Better bounds for online scheduling. SIAM J Comput 29(2):459–473

Andersson A, Hoff A, Christiansen M, Hasle G, Løkketangen A (2010) Industrial aspects and literature survey: combined inventory management and routing. Comput Oper Res 37:1515–1536

Artigues C, Billaut J-C, Esswein C (2005) Maximization of solution flexibility for robust shop scheduling. Eur J Oper Res 165(2):314–328

Aytug H, Lawley MA, McKay K, Mohan S, Uzsoy R (2005) Executing production schedules in the face of uncertainties: a review and some future directions. Eur J Oper Res 161(1):86–100

Berrichi A, Yalaoui F (2013) Efficient bi-objective ant colony approach to minimize total tardiness and system unavailability for a parallel machine scheduling problem. Int J Adv Manuf Tech 68(9–12):2295–2310

Blazewicz J, Ecker K, Pesch E, Schmidt G, Weglarz J (2001) Scheduling computer and manufacturing processes, 2nd edn. Springer, Berlin

Blazewicz J, Ecker KH, Pesch E, Schmidt G, Sterna M, Weglarz J (2019) Handbook on scheduling. From theory to practice, 2nd edn. Springer, Berlin

Bożek A, Wysocki M (2015) Flexible job shop with continuous material flow. Int J Prod Res 53(4):1273–1290

Chen Z-L (2010) Integrated production and outbound distribution scheduling: review and extensions. Oper Res 58(1):130–148

Clarke G, Wright JW (1964) Scheduling of vehicles from a central depot to a number of delivery points. Oper Res 12(4):568–581

Croes G (1958) A method for solving traveling-salesman problems. Oper Res 6(6):791–812
Desrochers M, Laporte G (1991) Improvements and extensions to the Miller-Tucker-Zemlin subtour elimination constraints. Oper Res Lett 10:27–36
Destouet C, Tlahig H, Bettayeb B, Mazari B (2023) Flexible job shop scheduling problem under industry 5.0: a survey on human reintegration, environmental consideration and resilience improvement. J Manuf Syst 67:155–173
Dijkstra EW (1959) A note on two problems in connexion with graphs. Numer Math 1:269–271
Doerner KF, Gronalt M, Hartl RF, Kiechle G, Reimann M (2008) Exact and heuristic algorithms for the vehicle routing problem with multiple, interdependent time windows. Comput Oper Res 35:3034–3048
Dolgui A, Proth J-M (2010) Supply chain engineering: useful methods and techniques. Springer, Berlin
Dolgui A, Eremeev AV, Kovalyov MY, Kuznetsov PM (2010) Multi-product lot-sizing and scheduling on unrelated parallel machines. IIE Trans 42(7):514–524
Dolgui A, Ivanov D, Sethi SP, Sokolov B (2019) Scheduling in production, supply chain and industry 4.0 systems by optimal control. Int J Prod Res 57(2):411–432
Dolgui A, Ivanov D, Potryasaev S, Sokolov B, Ivanova M, Werner F (2020) Blockchain-oriented dynamic modelling of smart contract design and execution control in the supply chain. Int J Prod Res 58(7):2184–2199
Gendreau M, Laporte G, Potvin J-Y (2002) Metaheuristics for the capacitated VRP. In: Toth P, Vigo D (eds) The vehicle routing problem. Society for Industrial and Applied Mathematics, Philadelphia, pp 129–154
Gillett BE, Miller LR (1974) A heuristic algorithm for the vehicle-dispatch problem. Oper Res 22(2):340–349
Gomes MC, Barbosa-Póvoa AP, Novais AQ (2013) Reactive scheduling in a make-to-order flexible job shop with re-entrant process and assembly a mathematical programming approach. Int J Prod Res 51(17):5120–5141
Grünert T, Irnich S (2005) Optimierung im transport—band I: grundlagen. Shaker, Aachen
Harjunkoski I, Maravelias CT, Bongers P, Castro PM, Engell S, Grossmann IE, Hooker J, Méndez C, Sand G, Wassick J (2014) Scope for industrial applications of production scheduling models and solution methods. Comput Chem Eng 62:161–193
Hazir O, Haouari M, Erel E (2010) Robust scheduling and robustness measures for the discrete time/cost trade-off problem. Eur J Oper Res 207(2):633–643
Ivanov D, Sokolov B (2012) Dynamic supply chain scheduling. J Sched 15(2):201–216
Ivanov D, Sokolov B, Dolgui A, Werner F, Ivanova M (2016a) A dynamic model and an algorithm for short-term supply chain scheduling in the smart factory industry 4.0. Int J Prod Res 54(2):386–402
Ivanov D, Dolgui A, Sokolov B (2016b) Robust dynamic schedule coordination control in the supply chain. Comput Ind Eng 94(1):18–31
Ivanov D, Dolgui A, Sokolov B (2018) Scheduling of recovery actions in the supply chain with resilience analysis considerations. Int J Prod Res 56(19):6473–6490
Ivanov D, Sokolov B, Chen W, Dolgui A, Werner F, Potryasaev S (2021) A control approach to scheduling flexibly configurable jobs with dynamic structural-logical constraints. IISE Trans 53(1):21–38
Joereßen A, Sebastian H-J (1998) Problemlösung mit Modellen und Algorithmen. Teubner, Stuttgart
Johnson SM (1954) Optimal two- and three-stage production schedules with setup times included. Nav Res Logist Q 1(1):61–68
Kolisch R, Hess K (2000) Efficient methods for scheduling make-to-order assemblies under resource, assembly area and part availability constraints. Int J Prod Res 38(1):207–228
Kovalyov MY, Ng CT, Cheng TCE (2007) Fixed interval scheduling: models, applications, computational complexity and algorithms. Eur J Oper Res 178:331–342
Kyparisis GJ, Koulamas CP (2006) Flexible flow shop scheduling with uniform parallel machines. Eur J Oper Res 168:985–997

Li Y, Chen R (2010) Stochastic single machine scheduling to minimize the weighted number of tardy jobs. In: Cao B-Y, Wang E, Guo S-Z, Chen S-L (eds) Fuzzy information and engineering 2010. Springer, Berlin, pp 363–368
Liu Z, Ro YK (2014) Rescheduling for machine disruption to minimize makespan and maximum lateness. J Sched 17(4):339–352
Liu Y, Wang L, Wang XV, Xu X, Zhang L (2019) Scheduling in cloud manufacturing: state-of-the-art and research challenges. Int J Prod Res 57(15–16):4854–4879
Maccarthy BL, Liu J (1993) Addressing the gap in scheduling research: a review of optimization and heuristic methods in production scheduling. Int J Prod Res 31(1):59–79
Mahmoud H, Thabet M, Khafagy MH, Omara FA (2022) Multiobjective task scheduling in cloud environment using decision tree algorithm. IEEE Access 10:36140–36151
Mattfeld DC (1996) Evolutionary search and the job shop. Physica-Verlag, Heidelberg
Moore JM (1968) An n job, one machine sequencing algorithm for minimizing the number of late jobs. Manag Sci 15(1):102–109
Pinedo ML (2022) Scheduling: theory, algorithms, and systems, 6th edn. Springer, New York
Ren W, Yan Y, Hu Y, Guan Y (2022) Joint optimisation for dynamic flexible job-shop scheduling problem with transportation time and resource constraints. Int J Prod Res 60(18):5675–5696
Ritzinger U, Puchinger J, Hartl RF (2016) A survey on dynamic and stochastic vehicle routing problems. Int J Prod Res 54(1):215–231
Sawik T (2013) Integrated selection of suppliers and scheduling of customer orders in the presence of supply chain disruption risks. Int J Prod Res 51(23–24):7006–7022
Shah N (2004) Process industry supply chains: advances and challenges. Comput Aided Chem Eng 18:123–138
Sokolov B, Ivanov D, Dolgui A (eds) (2020) Scheduling in industry 4.0 and cloud manufacturing. Springer, New York
Solomon MM (1987) Algorithms for the vehicle routing and scheduling problems with time window constraints. Oper Res 5(2):254–265
Sotskov YN, Lai T-C, Werner F (2013) Measures of problem uncertainty for scheduling with interval processing times. OR Spectr 35(3):659–689
Thonemann U (2015) Operations management: konzepte, methoden und anwendungen, 3rd edn. Pearson, München
Toth P, Vigo D (2002) Models, relaxations and exact approaches for the capacitated vehicle routing problem. Discret Appl Math 123(1–3):487–512
Van de Vonder S, Demeulemeester E, Herroelen W (2007) A classification of predictive-reactive project scheduling procedures. J Sched 10(3):195–207
Vieira GE, Herrmann JW, Lin E (2003) Rescheduling manufacturing systems: a framework of strategies, policies and methods. J Sched 6(1):35–58
Werner F (2013) A survey of genetic algorithms for shop scheduling problems. In: Siarry P (ed) Heuristics: theory and applications. Nova Science, New York, pp 161–222
Zhang Z, Zhao M, Wang H, Cui Z, Zhang W (2022) An efficient interval many-objective evolutionary algorithm for cloud task scheduling problem under uncertainty. Inf Sci 583:56–72
Zheng P, Zhang P, Wang J, Zhang J, Yang C, Jin Y (2022) A data-driven robust optimization method for the assembly job-shop scheduling problem under uncertainty. Int J Comput Integr Manuf 35(10–11):1043–1058

References for Sect. 14.4.1

Konrad A (2013) Meet ORION, software that will save ups millions by improving drivers' routes. http://www.forbes.com/sites/alexkonrad/2013/11/01/meet-orion-software-that-will-save-ups-millions-by-improving-drivers-routes/. Accessed 19 Jan 2015

Noyes K (2014) The shortest distance between two points? At UPS, it's complicated. Fortune Online, July 25. http://fortune.com/2014/07/25/the-shortest-distance-between-two-points-at-ups-its-complicated/. Accessed 2 Dec 2014

Shontell A (2011) Why UPS is so efficient: "our trucks never turn left". Business Insider, March 24. http://www.businessinsider.com/ups-efficiency-secret-our-trucks-never-turn-left-2011-3. Accessed 28 Nov 2014

Wohlsen M (2013) The astronomical math behind UPS' new tool to deliver packages faster. Wired Magazine Online, December 13. http://www.wired.com/2013/06/ups-astronomical-math/. Accessed 9 Dec 2014

Part IV

Advanced Topics in Supply Chain and Operations Management

Supply Chain Risk Management and Resilience

15

Learning Objectives for This Chapter

- What are uncertainty and risks?
- What types of risks are encountered in the supply chain?
- Which decisions are in the scope of supply chain risk management?
- Why do the risks appear in the supply chain?
- What is the difference between the operational and disruption risks?
- What is the ripple effect in the supply chain?
- How to mitigate the disruption risks and the ripple effect?
- What is supply chain resilience?
- Which objectives are used to measure the performance of supply chain and operations management in terms of risks?
- How to measure supply chain resilience and exposure to the ripple effect?
- What is supply chain viability?

15.1 Introductory Case-Study: Toyota's Resilience to the Global Semiconductor Shortage

Toyota Motor Corporation, a globally recognized leader in the automotive industry, produces vehicles in 28 countries and serves over 170 markets worldwide (Toyota 2021). In 2023, the company achieved a worldwide production volume of 9.1 million vehicles despite challenges in the global supply chain (Reuters 2023). Toyota is

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_15.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_15

also famous for its Toyota Production System (TPS) and lean management (see Chap. 9, Sect. 9.5), which include several key pillars such as automation (Jidoka) and just-in-time. The primary goal of TPS is to eliminate waste and redundancies, such as excessive inventory, which are considered wastes (Toyota 2021b). Contrary to conventional beliefs about a trade-off between lean and resilience, Toyota has argued that TPS provided them with resilience during the global chip shortage in 2021.

The global chip shortage caused chaos in automotive supply chains in 2021. Automakers, including Toyota and Volkswagen, were forced to cut production due to the acute shortage of semiconductors (The Guardian 2021). Toyota identified two significant reasons for this crisis: the growing usage of chips in modern cars and inaccurate demand forecasting caused by the COVID-19 pandemic. Indeed, chip shortage is an excellent example of the ripple effect. The disruption at semiconductor manufacturers in Southeast Asia cascaded downstream, creating a ripple effect that severely impacted automotive production. The automotive industry alone has potential sales losses of up to $210 billion (AlixPartners 2023).

According to Toyota, TPS played a critical role in mitigating the impacts of the chip shortage. Shih (2022) explains Toyota's approach through the following four key principles.

1 Lean does not mean zero inventory. Rooted in Japan, Toyota has a rich experience in disruption management. Understanding its products and its suppliers, the company does not treat all materials in the same way. Toyota differentiates between materials based on their vulnerability to disruption. For instance, semiconductors, identified as fragile due to the complexity of their manufacturing process, are stockpiled (Shih 2022). Toyota dynamically adjusts inventory levels based on contextual factors like weather patterns. Inventory, in this case, should be understood as a risk mitigation inventory rather than a safety stock.
2 Multiple suppliers as a resilience strategy. Toyota leverages multiple sources not because it brings financial benefits at first. Multiple suppliers foster healthy competition in the supply base, leading to technological advancements. From a resilience perspective, having multiple suppliers provides structural redundancy, which Toyota strategically leverages (Shih 2022).
3 Supplier relationships are built on trust. The success of TPS relies heavily on strong, trust-based relationships with suppliers. According to Bonini, president of the Toyota Production System Support Center, "If you are going to run a tight just-in-time system, you have to have the expertise as well as the trusting relationship to go into a supplier and help out when there is a problem"(Shih 2022). Besides actively supporting its Tier 1 suppliers, Toyota requests Tier 1 suppliers to extend similar collaborative practices with Tier 2 suppliers (Toyota 2023).
4 TPS is a learning system and an organizational culture. It embodies a culture of continuous learning and problem-solving. Daily problem-solving practices at various organizational levels foster rapid response during high-pressure situations. Toyota's support for its suppliers goes beyond transactions and deep engagement in capacity building and problem resolution. This collaborative

approach extends throughout the supply chain and creates a collaborative environment, which amplifies Toyota's ability to manage disruptions more effectively than competitors (Shih 2022).

The 2021 chip shortage underscores aggravated disruption overlays between operational and supply chain disruption risks. While many automakers struggled, Toyota leveraged TPS principles smartly to remedy the ripple effect. The company stockpiles vulnerable materials, maintains structure redundancy, and fosters trust-based relationships with suppliers. Despite these successes, challenges remain. Implementing such strategies demands substantial organizational culture, information sharing, and trust.

15.2 Uncertainty and Risks

Uncertainty is a system property characterizing the incompleteness of our knowledge about the system and the conditions of its development. The concept of *risk* is subject to various definitions. Knight (1921) classified under "risk" the "measurable" uncertainty. From the financial perspective of Markowitz (1952), the risk is the variance of return. From a project management perspective, the risk is a measure of the probability and consequence of not achieving a defined project goal. According to March and Shapira (1987), the risk is a product of the probability of occurrence of a negative event and the resulting amount of damage.

Generally, in decision theory, the risk is a measure of the set of possible (negative) outcomes from a single rational decision and their probabilistic values. In SCOM literature, the term "risk" is also replaced with "vulnerability," which means "at risk."

In analyzing uncertainty and risks, four aspects are usually encountered. The first is uncertainty itself, the second is risks, the third is perturbation influence (disturbances), and the last is the perturbation impact influences (deviations). In the further course of this and the following chapters, we will frequently encounter this constellation (see Fig. 15.1).

Uncertainty is the general property of a system environment that exists independent of us for any system of a sensible complexity degree. As shown in Fig. 15.1, we can broaden and narrow the uncertainty space.

Risk arises from uncertainty. Risks can be identified, analyzed, controlled, and regulated. We consciously talked about uncertainty factors and the appearance of risks such as the risk of demand fluctuation as a result of the environmental uncertainty.

A *disturbance* (perturbation impact) is the consequence of risks. It may be purposeful (i.e., thefts) and non-purposeful (i.e., demand fluctuations or the occurrence of some events that may necessitate adapting the SC). It may cause a deviation (disruption) in the SC or not (E.g., an SC can be robust and adaptive enough to overcome the disturbance).

Operational *deviations* (or severe *disruptions*) are the result of perturbation influences. They may affect operations, processes, plans, goals, or strategies. To adjust the SC in the case of deviations, adaptation measures need to be taken.

For the SCOM domain, uncertainty factors and their handling measures can be distinguished as follows (see Table 15.1).

In Table 15.2, some examples of disturbances and disruptions in SCs in recent years are presented.

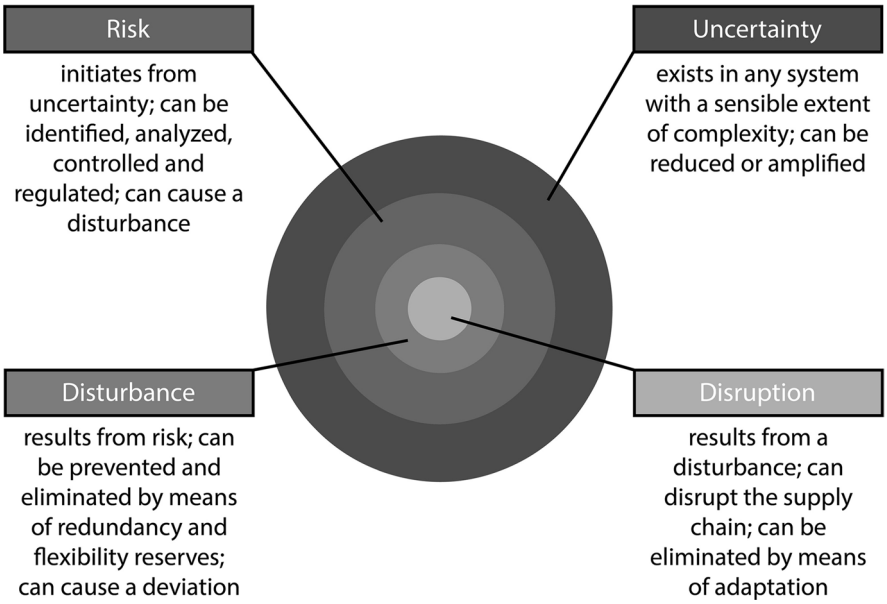


Fig. 15.1 Interrelations of uncertainty, risk, disturbance and disruption (based on Ivanov and Sokolov 2010; Ivanov 2018)

Table 15.1 Uncertainty factors and measures for their handling in SCs

Decision-making level	Uncertainty factors	Handling measures
Strategic	Multiple management goals Terrorism, piracy Financial and political crises	Multi-criteria analysis techniques SC risk management Liquid assets reserves Strategic material inventories market diversification and outsourcing
Tactical and operational	Natural disasters Weak coordination Stockless processes Weak control of cargo security Technological breaks Human errors	Product lines' flexibility and modularity Safety stocks and time buffers Reserves of SC capacities SC coordination, monitoring, and event management

Table 15.2 Examples of disturbances and disruptions in SCs

Factor	Example	Impacts
Terrorism piracy	September 11 Somali, 2008	Five Ford plants have been closed for a long time Breaks in many SCs
Natural disasters	Earthquake in Thailand, 1999; Flood in Saxony, 2002; Earthquake in Japan, 2007	Apple computers' production in Asia has been paralyzed Significant production decrease at VW, Dresden Production breakdown in Toyota's supply chains amounted to 55,000 cars
	Hurricane Katrina, 2006	This storm halted 10–15% of total US gasoline production, raising both domestic and overseas oil prices
	Earthquake and tsunami in Japan, 2011	Massive collapses in global automotive and electronics SCs; Toyota lost its market leadership position
	Floods in Chennai, India, in 2015	Production of academic literature has been stopped at many international publishing houses
Human-made disasters	Explosion at BASF plant in Ludwigshafen in 2016	15% of raw materials were missing for the entire SC Production of some products at BASF has been stopped for many weeks
	Fire at distribution center of e-commerce retail company ASOS in 2005	Delivery stop for a month
	A fire in the Phillips Semiconductor plant in Albuquerque, New Mexico in 2000	Phillips's major customer, Ericsson, lost $400 million in potential revenue
Political crises	"Gas" crisis 2009	Breaks in gas supply from Russia to Europe, billions of losses to GAZPROM and customers
Financial crises	Autumn 2008	Production decrease or closing; breaks in SCs throughout
Strikes	Strikes at Hyundai plants in 2016	Production of 130,000 cars has been affected
Legal contract disputes	Volkswagen and Prevent Group contract dispute in summer 2016	Six German factories face production halt on parts shortage; 27,700 workers are affected, with some sent home, and others moved to short-time working
Epidemics and pandemics	COVID-19 global pandemic	Worldwide disruptions in supply and demand, devastating effects in many global and local SCs; ripple effects; SC collapses, and long-term performance degradation
Global shortage of critical components	Semiconductor crisis 2021–2024	Sony's Play Station supply chain became prone to semiconductor shortages. The reduced resilience has translated in sales decreases and product shortages in 2021 and 2022. It took Sony three years until the end of 2023 to recover and completely fix its Play Station supply chain worldwide

Extended from Ivanov and Sokolov (2010), Ivanov (2018)

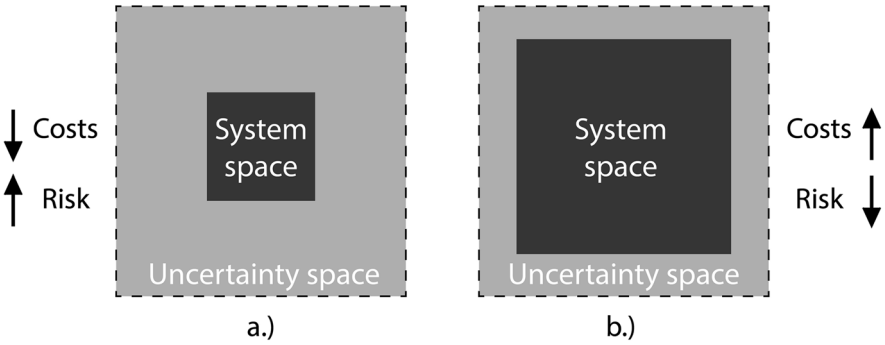


Fig. 15.2 System space and uncertainty space (Ivanov 2018)

Complexity management and system modeling can be considered as a theoretical basis for handling uncertainty in SCs. From the perspective of complexity management, the problem of a system under control and uncertainty is related to an area under control and an area under uncertainty. This idea is based on Ashby's principle (Ashby 1956) of requisite variety (Fig. 15.2).

By broadening the control area (Fig. 15.2b) and narrowing the uncertainty area or reverse (Fig. 15.2a), the system control can be adapted. Hence, the mutual relations between the system and environment spaces fall into the categories of amplification of a *control variety* and attenuation of an *environmental variety* (see Fig. 15.2). Thus, amplifying the variety of our control area and reducing the area of uncertainty, (1) a balance of control and perturbed impacts as well as (2) the maintenance of the planned execution processes and a quick cost-efficient process recovery once disturbed can be reached (Ivanov 2022a).

15.3 Risk Management in the Supply Chain

15.3.1 Risk Classification

Risk management in the SC became one of the most important topics in research and practice over the last decade. A number of books (Handfield and McCormack 2008; Kouvelis et al. 2012; Waters 2011; Gurnani et al. 2012; Heckmann 2016; Ivanov 2018) and literature review papers (Blackhurst et al. 2005; Wagner and Bode 2008; Klibi et al. 2010; Simangunsong et al. 2012; Ho et al. 2015; Fahimnia et al. 2015; Gupta et al. 2016; Dolgui et al. 2018) provide insightful overviews and introductions to different aspects of this exciting field. In Fig. 15.3, we summarize major SC risks.

▶ **Practical Insights** SC risks are highly intertwined and can induce each other. Consider an example based on (Manager Magazine 2021). In January 2021, production stops at automotive assembly plants have

been observed due to the supply shortages of semiconductors. The shortages were caused by surges in demand at automotive firms that recovered after the pandemic shock in 2020. However, this fast demand recovery was not well anticipated by semiconductor suppliers, which, in turn, have reduced or re-allocated their capacities by cooperating with the electronics industry and healthcare SCs in order to substitute the missing demand from the automotive industry, which suffered from severe shocks in the wake of the COVID-19 pandemic. On the other hand, silicon as a raw material for the semiconductor industry is associated with high sourcing risks. Silicon production has been reduced in China in 2020 due to the shortages of electricity generated by hydropower stations which, in turn, were caused by long periods of drought. This example illustrates interconnections between different SC risks shown in Fig. 15.3.

SC risks stem from different areas related to material flows such as supply, demand, production, and logistics process. Along with material flow-related risks, risks arise in financial and information flows, e.g., due to cyber-attacks or financial crisis (Ghadge et al. 2019). Moreover, SCs are subject to macro-risks such as climate changes and shortages of critical resources.

Recent literature introduced different classifications of SC risks (Chopra and Sodhi 2004; Tang and Musa 2011; Ho et al. 2015; Quang and Hara 2017). For example, Chopra and Sodhi (2004) categorized potential SC risks into nine categories:

- Disruptions (e.g., natural disasters, terrorism, war, etc.)
- Delays (e.g., inflexibility of supply source)
- Systems (e.g., information infrastructure breakdown).
- Forecast (e.g., inaccurate forecast, bullwhip effect, etc.)

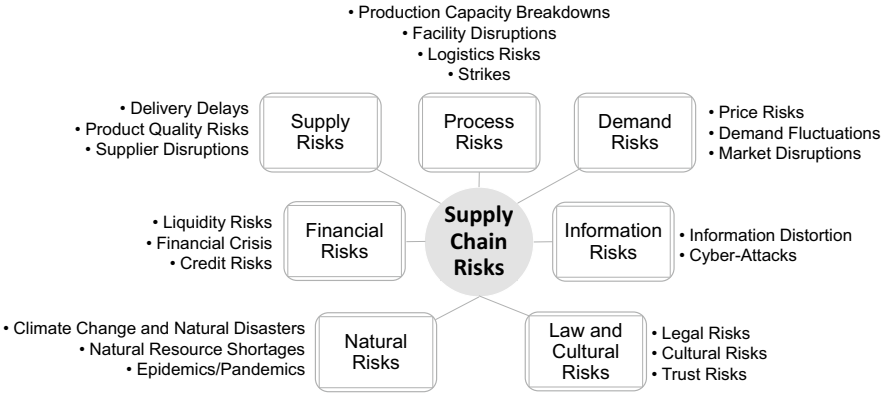


Fig. 15.3 Supply chain risks

- Intellectual property (e.g., vertical integration)
- Procurement (e.g., exchange rate risk)
- Receivables (e.g., number of customers)
- Inventory (e.g., inventory holding cost, demand and supply uncertainty, etc.)
- Capacity (e.g., cost of capacity)

Quang and Hara (2017) classify the following seven groups of risks:

- *External risks* which "deal with threats from an external perspective of SC that can be caused by economical, sociopolitical or geographical reasons. Examples are fire accidents, natural catastrophes, economic downturn, external legal issues, corruption, and cultural differentiation."
- *Time risks* referring to delays in SC processes.
- *Information risks*, e.g., communication breakdown within the project team, information infrastructure complications, distorted information, and information leaks.
- *Financial risks*, e.g., inflation, interest rate level, currency fluctuations, and stakeholder requests.
- *Supply risks*, i.e., risks related to suppliers, e.g., supplier bankruptcy, price fluctuations, unstable quality, and quantity of inputs.
- *Operational risks,* caused by problems within the organizational boundaries of a firm, e.g., changes in design and technology, accidents, and labor disputes.
- *Demand risks* that refers to demand variability, high market competition, customer bankruptcy, and customer fragmentation.

Another classification of SC risks has been proposed by Ivanov and Dolgui (2019) as follows:

- Network risks, i.e., risks imposed by specific types of SC networks (e.g., scale-free vs. decentralized networks).
- Process risks, i.e., specific parameters of SC processes such as inventory and capacity, may influence SC proneness to disruptions, and.
- Supplier risks, i.e., risks induced by specific suppliers, e.g., those located in geographical zones with high probability of natural disasters.

15.3.2 General Framework of Risk Management in the Supply Chain

Risk management is a methodological approach to managing uncertainty outcome. A particular feature of risk management in SCs (unlike in technical systems) is that people do not strive for a *100% guarantee* of the result: they consciously tend to take risks. There is a contradiction between *objective risks, those* determined by experts applying quantitative scientific means and *perceived risk, those which include managers*' *perceptions*.

Actually, the objective risk treatment is rooted in technical science, where 100% reliability is mandatory. In socio-economic systems, like SCs, a value of 95% as an orientation for SCs is empirically suggested (e.g., Sheffi 2005). Different managers perceive the risk to different extents, and these perceptions can change in the same manager due to changes in his environment. That is why the models for SCs should not strive for a unique optimal solution but allow the formation of a number of alternative solutions with different degrees of potential economic performance and risk. Summarizing, we will note that risk can be considered from three basic positions:

1. Risk is a likelihood estimation of a negative outcome of the event leading to losses/losses (the technological approach).
2. Risk is an individual estimation by the person of the danger of a negative outcome of the event leading to losses/losses; the risk is ultimately a property of any entrepreneurship (the psychological approach).
3. Risk is an integral property of any process or system, the management of which is a key problem in economic performance and stability maintenance (the organizational approach).

In SC risk management, four key steps to reduce SC vulnerability to disruptions are usually considered as follows: (1) identifying potential risks, (2) measuring and assessing risks, (3) mitigating risks, and (4) responding to risks (see Fig. 15.4).

The first stage is to identify the risks related to network design structure, integrated supply chain processes, and individual supplier risks. Once the risks are identified, they need to be quantified. Similarly, resilience and ripple effect are assessed. The next stage is to protect the SC by inventory, capacity agility, and backup suppliers to mitigate the negative impact of possible disruptions. If a disruption happens, the objective is to respond by the deployment of contingency plans, recovery strategies, and situational adaptations.

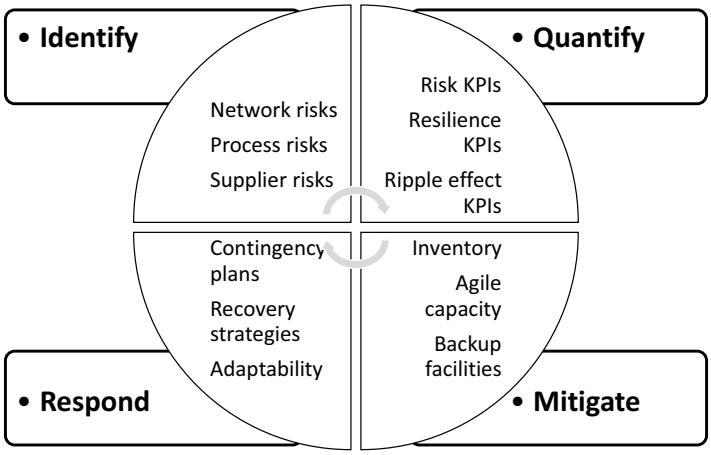


Fig. 15.4 Supply chain risk management framework

In summarizing the practical advancements in uncertainty handling within an SC environment, the following can be concluded:

1 Uncertainty space may be reduced by means of, e.g.,

   (a) Introducing excessiveness in SC structures (e.g., time buffers, safety stocks, additional resources, capacity reserves, etc.).
   (b) Improving coordination and information flows to make better quality, timeliness, and accessibility.
   (c) Introducing SC monitoring and event management systems to react quickly to disturbances and disruptions; and.
   (d) Forming a set of not final decisions, i.e., postponement and rolling/adaptive planning.

2 It is impossible to avoid uncertainty.

15.4 Operational and Disruption Risks

Klibi et al. (2010) classify uncertainties and risks in the SC as follows:

- Random uncertainty (demand fluctuation risks)
- Hazard uncertainty (risk of unusual events with high impact)
- Deep uncertainty (severe disruption risks)

Random uncertainty belongs to the category of *known-known*, i.e., we know which events can happen, when, and how likely they are. The hazard uncertainty is close to "*known-unknown,*" i.e., we know which events can happen, but we do not know when they would happen and what their impact is. The deep uncertainty is related to the "unknown-unknown," i.e., we do not know what can happen, when, and what the consequences could be. Deep uncertainty represents the most complex case in decision-making. One example of the deep uncertainty is the COVID-19 pandemic.

The different types of risks in the SC can be classified into demand, supply, process, and structure areas (see Fig. 15.5).

Risks of demand and supply uncertainty are related to random uncertainty and business-as-usual situation. Such risks are also known as *recurrent* or *operational risks.* SC managers achieved significant improvements at managing global SCs and mitigating recurrent SC risks through improved planning and execution (Chopra and Sodhi 2014).

Disruption risks are severe, exceptional events of low probability but the high impact (Kinra et al. 2020). We define *SC disruptions* as *unexpected events with severe negative impacts such as tsunamis, fires, or strikes.*

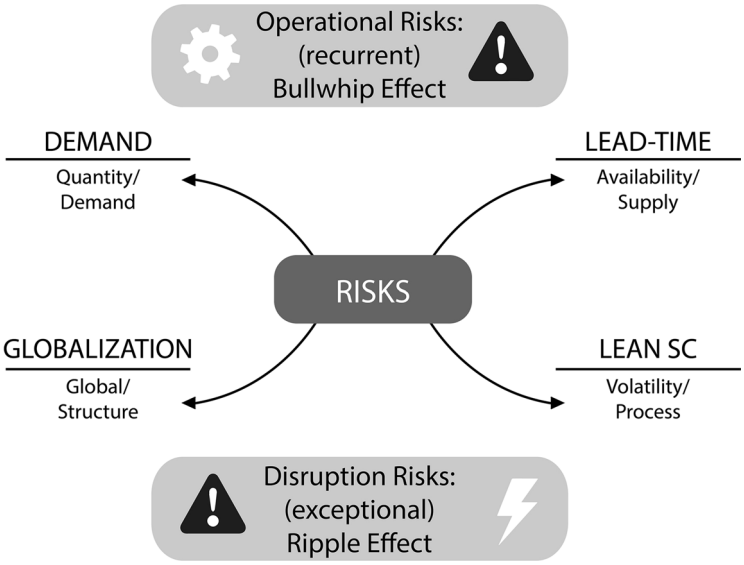


Fig. 15.5 Supply chain operational and disruption risks (Ivanov 2018)

SC disruptions occurred in greater frequency and intensity, and thus with greater consequences (Chopra and Sodhi 2014; Simchi-Levi et al. 2014). Hendricks and Singhal (2005) quantified the negative effects of SC disruption through empirical analysis and found 33–40% lower stock returns relative to their benchmarks over a 3-year time period that started 1 year before and ended 2 years after a disruption. *Disruption risks* represent a fundamental challenge for SC managers who face the *ripple effect* (Ivanov et al. 2014a, b, 2017; Ivanov 2017; Dolgui et al. 2018; Ivanov and Dolgui 2021) subject to *structural disruptions* in the SC, unlike the *parametrical deviations* in the bullwhip effect (Fig. 15.6).

In the last two decades, considerable advancements have been achieved in research regarding the mitigation of inventory and production shortages and response to demand fluctuations. In particular, the *bullwhip effect* in the SC has been extensively considered in this domain, subject to *random uncertainty* with the help of stochastic and simulation models. The differences between the bullwhip effect and ripple effect are presented in Table 15.3.

The Bullwhip effect considers weekly/daily demand and lead-time fluctuations as primary drivers of the changes in the SC, which occur at the parametric level and can be eliminated in a short-term perspective. In recent years, the research community has started to investigate severe SC disruptions with long-term impacts that can be caused, for example, by natural disasters, political conflicts, terrorism, maritime piracy, economic crises, sabotage, cyber-attacks or transport infrastructure failures. We refer to these severe natural and human-made disasters as the ripple effect in the

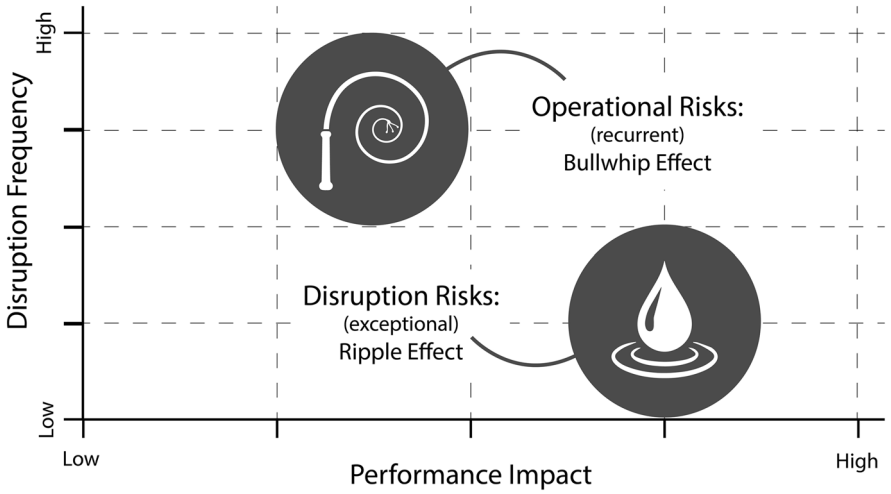


Fig. 15.6 Operational and disruption risks in supply chains

Table 15.3 Ripple effect and bullwhip effect (Dolgui et al. 2018)

Feature	Ripple effect	Bullwhip effect
What uncertainty?	Hazard, deep uncertainty	Random uncertainty
What risks?	Disruption, exceptional risks (e.g., a plant explosion)	Operational, recurrent risks (e.g., demand fluctuation)
What can be disturbed?	Structures and critical performance (such as supplier unavailability or revenue)	Operational parameters such as lead time and inventory
How are deviations prevented?	Proactive redundancy and flexibility	Information coordination, see CPFR
What happens after the disturbance?	Short-term stabilization and middle- and long-term recovery; high coordination efforts and investments	Short-term coordination to balance demand and supply
What is performance impact?	Output performance can decrease, such as in annual revenues or profits	Current performance can decrease, such as in daily or weekly stock-out/overage costs

SC where changes in the SC occur at the structural level, and recovery may take mid- and long-term periods of time with significant impact on output performance such as annual revenues. In this setting, SC disruption management can be considered a critical capability that helps to create cost-efficient SC protection and implement appropriate actions to recover SC disruptions and performance.

Most studies on SC disruption consider how changes to some variables are rippling through the rest of the SC and impacting performance. Studies by Ivanov et al. (2014a, b) and Dolgui et al. (2018) suggest considering this situation as *the ripple*

effect in the SC, as an analogy to computer science, where the ripple effect determines the disruption-based scope of changes in the system.

15.5 Ripple Effect in the Supply Chain

The *ripple effect* in the SC occurs if a disruption cannot be localized and cascades downstream impacting SC performance such as sales, stock return, service level, and costs (Ivanov et al. 2014a; Dolgui et al. 2018; Ivanov 2018; Pavlov et al. 2018; Ojha et al. 2018; Ivanov and Dolgui 2021). The methodical elaborations on the evaluation and understanding of low-frequency-high-impact disruptions are therefore vital for understanding and further development of network-based supply concepts (Tomlin 2006; Liberatore et al. 2012; Ghadge et al. 2013; Mizgier et al. 2013; Garvey et al. 2015; Sawik 2016; Ivanov et al. 2019; Garvey and Carnovale 2020; Dolgui and Ivanov 2021; Li and Zobel 2020).

Details of empirical or quantitative methodologies differ across the works on SC disruption management (Sawik 2020), but most share a basic set of attributes:

- A disruption (or a set of disruptions)
- Impact of the disruption on operational and strategic economic performance
- Stabilization and recovery policies

Within this set of attributes, most studies on SC disruption consider how changes to some variables are rippling through the rest of the SC and impacting performance. We suggest considering this situation *the ripple effect in the SC*. The ripple effect is a phenomenon of disruption propagations in the SC and their impact on output SC performance (e.g., sales, on-time delivery, and total profit). It may have more serious consequences than just short-term performance decrease. It can result in market share losses (e.g., Toyota lost its market leader position after tsunami in 2011 and needed to redesign the SC coordination mechanism). The ripple effect is also known as "domino effect" or "snowball effect." The reasons for the ripple effect are not difficult to find. With increasing SC complexity and consequent pressure on speed and efficiency, an ever-increasing number of industries come to be distributed worldwide and concentrated in industrial districts. In addition, globalized SCs depend heavily on permanent transportation infrastructure availability.

According to Dolgui et al. (2020), the ripple effect "*refers to structural dynamics and describes a downstream propagation of the downscaling in demand fulfilment in the supply chain as a result of a severe disruption*." Ivanov et al. (2014b) state that the "*Ripple effect describes the impact of a disruption on supply chain performance and disruption-based scope of changes in the supply chain structures and parameters*." These definitions imply that the ripple effect refers to multi-stage networks and triggering failures in the network elements as a domino effect.

Following a disruption, its effect ripples through the SC. The missing capacities or inventory at the disrupted facility may cause missing materials and production decrease at the next stages in the SC. Should the SC remain in the disruption mode

longer than some critical period of time [i.e., *time-to-survive* (Simchi-Levi et al. 2015)], critical performance indicators such as sales or stock returns may be affected.

▶ **Practical Insights** Earthquakes and tsunami in Japan in 2011 have disrupted multiple suppliers in the automotive industry and lead to production breaks and material shortages worldwide (Ivanov 2018).

In May 2017, production at BMW was disrupted as a consequence of a supply shortage of steering gears. BMW's first-tier supplier, Bosch, was unable to deliver the steering gears since an Italian second-tier supplier experienced production delays for certain steering parts due to internal machine breakdowns (Moetz et al. 2019). The COVID-19 pandemic has caused numerous ripple effects. Haren and Simchi-Levi (2020) observed two examples of a ripple effect triggered by COVID-19 immediately after the epidemic outbreak. Fiat Chrysler Automobiles NV halted production at a car factory in Serbia in response to being unable to receive parts from China. As Hyundai stated, it had "decided to suspend its production lines from operating at its plants in Korea … due to disruptions in the supply of parts resulting from the coronavirus outbreak in China." While these observations were made in the second half of February 2020, the scaling of the ripple effects between March and May 2020 has been exponential, driven by the closures of manufacturing facilities, stores, and logistics activities, and adversely affecting almost all industries and services worldwide (Ivanov and Dolgui 2020).

The scope of the rippling and its impact on economic performance depends both on robustness reserves (e.g., redundancies like inventory or capacity buffers) and speed and scale of recovery measures. Therefore, the risks and SC resilience should be estimated at the design and planning stages in the proactive mode.

At the control stage in the reactive mode, contingency plans (e.g., alternative suppliers or shipping routes) must happen quickly to expedite stabilization and recovery in order to ensure continuity of supply and avoid long-term impacts. In implementing such recovery policies, companies need a tool supported by collaboration and SC visibility solutions for assessing the disruption impact on the SC and the effects and costs from redirecting material flows. In SCOM settings, the ripple effect should also include recovery strategies that may compensate for disruptions and avoid their rippling (Ivanov and Dolgui 2022a, b; Ivanov 2024; Dolgui and Ivanov 2021; Ghadge et al. 2021; Hosseini and Ivanov 2021; Burgos and Ivanov 2021).

It has been extensively documented in the literature that severe disruptions may ripple quickly through global SCs and cause losses in supply chain performance that can be measured by such key performance indicators as revenues, sales, service level, and total profits (Schmitt and Singh 2012; Simchi-Levi et al. 2015; Snyder et al. 2016). Such risks are new challenges for research and industries that face the *ripple effect* arising from vulnerability, instability, and disruptions in SCs (Liberatore et al. 2012; Ivanov et al. 2014a, b). As opposed to the well-known bullwhip effect that considers high-frequency-low-impact *operational risks*, the ripple effect

studies low-frequency-high-impact *disruptive risks* (Fahimnia et al. 2015; Simchi-Levi et al. 2015; Sokolov et al. 2016; Snyder et al. 2016). Along with the identification of the reasons for the disruption risks, risk mitigation strategies have been developed with regard to proactive and reactive control stages (Bode et al. 2011; Ivanov et al. 2017). Let us consider them in the next section.

15.6 Supply Chain Resilience

15.6.1 Resilience Framework

SC *resilience* is the "*ability to maintain, execute and recover (adapt) planned execution along with achievement of the planned (or adapted, but yet still acceptable) performance is, therefore, the next objective property of the supply chain*" (Ivanov 2018). According to Hosseini et al. (2019b), SC resilience is "*the firm's capability to withstand, adapt, and recover from disruptions to meet customer demand, ensure target performance, and maintain operations in vulnerable environments.*"

SC resilience management was developed in response to more and more frequent natural and human-made disasters during the first decade of the 2000s by Christopher and Peck (2004), Blackhurst et al. (2005), Ponomarov and Holcomb (2009), Pettit et al. (2010), Bode et al. (2011), Jüttner and Maklan (2011). Wieland and Wallenburg (2013), Brandon-Jones et al. (2014), Melnyk et al. (2014), Ambulkar et al. (2015), Chowdhury and Quaddus (2017), Altay et al. (2018), Dubey et al. (2019) extended SC resilience understanding by linking it to other theoretical frameworks such as contingency theory, relational view and dynamic capability view and exploring its interplay with other notions such as disruption, collaboration or even performance. Ivanov et al. (2014a) coined the definition of the ripple effect in SC resilience context, i.e., disruption propagation and cascading through the networks.

Put simply, SC resilience has been considered an ability to *bounce-back and recover* (Fiksel 2006). SC *resilience* is a multi-facet property that comprises a number of components in both internal SC processes and in interaction with the environment (Fig. 15.7).

SC resilience looks at maintaining some desired performance despite disruptions (Ivanov 2025d). Using some proactive capabilities (i.e., inventory), an SC can absorb negative disruption impacts (e.g., supply unavailability) without performance degradation. However, if proactive capabilities do not help, performance (e.g., on-time delivery or revenue) can decline. In this case, reactive capabilities should be employed to restore the performance and operations. This takes time and creates costs. Balancing vulnerabilities and capabilities is, therefore, a major concern in SC resilience management.

Recall that one of the main objectives of SCOM is to increase total SC output performance, which is basically referred to as SC effectiveness (i.e., sales and service level) and efficiency (SC costs). At the same time, the achievement of planned performance can involve the impact of perturbations in a real-time execution environment. SC execution is subject to uncertainty at the planning stage and disruption

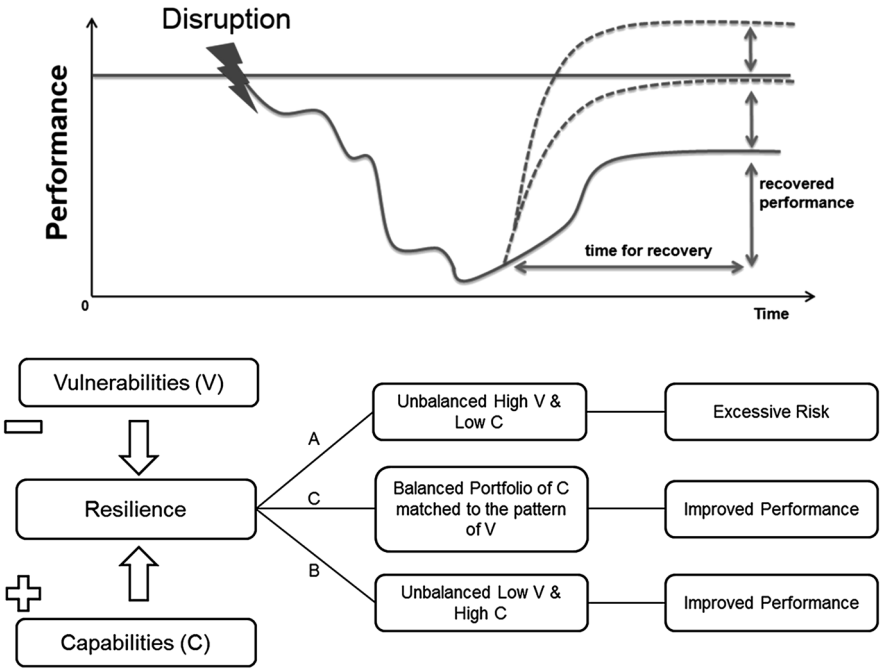


Fig. 15.7 Supply chain resilience concept (based on Sheffi 2005; Pettit et al. 2010)

at the execution stage. Cost efficiency comes with a substantial hidden expense should a major disruption (i.e., a more severe impact than a routine disturbance) occur. This requires SC protection against and efficient reaction to disturbances and disruptions. Therefore, SCs need to be planned to be *stable, robust, and resilient* enough to (1) maintain their basic properties and ensure execution; and (2) be able to adapt their behavior in the case of disturbances in order to achieve planned performance using recovery actions.

In Fig. 15.8, we summarize relations of redundancy, robustness, resilience, and flexibility (see also Ivanov and Sokolov 2013; Ivanov 2018).

First, different *reserves* (material inventory, capacities) can be referred to. Tomlin (2006) and Chopra and Sodhi (2004) analyzed the issues of *risk mitigation inventory* and reserve capacity on SC resilience. It should be noted that risk mitigation inventory differs from classical safety stock and is considered to decouple from disruptive risks. For this issue, valuable approaches and models for SC design and planning under uncertainty were elaborated.

Second, new *strategies* such as leagile, agile, and responsive SCs can be applied to make SCs more flexible in the broader sense of the word. The third method is related to better *coordination* in SCs and refers to concepts like collaborative planning, forecasting, and replenishment. Fourth, a set of *postponed decisions* (product postponement, rolling/adaptive planning) can be used. All these approaches can be referred to as SC *redundancy*.

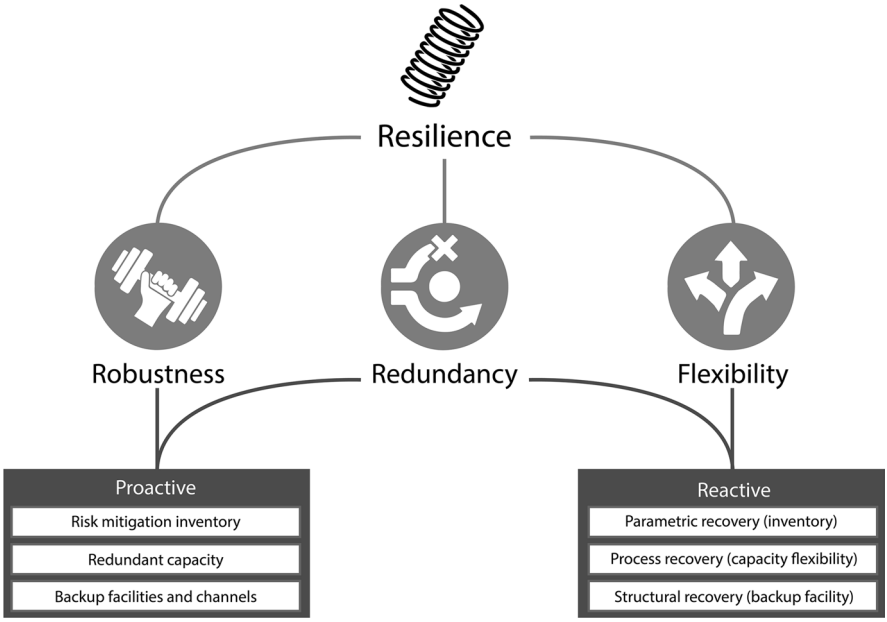


Fig. 15.8 Resilience control elements (Ivanov 2018)

The above-mentioned redundancies generally serve for two problem areas. First, they are intended to protect the SC against perturbation impacts based on certain reserves. This issue is related to SC *robustness*. Second, redundancies are created to amplify the fork variety of SC paths to react quickly and flexibly to changes in a real execution environment. This issue is related to SC *flexibility*.

The *robustness* of SCs is a complex characteristic of a non-failure operation, durability, recoverability, and the maintaining of SC processes and an SC as a whole. This is connected with the creation of a reserves system (the introduction of resource excessiveness) for the prevention of failures and deviations in SC processes.

Recent literature has identified different methods to strengthen SCs to mitigate uncertainty impacts and ensure SC robustness. Different robustness reserves can include material inventory and capacity buffers. For this issue, valuable approaches and models for SC design and planning under uncertainty were elaborated.

The flexibility of SCs is a property concerning its ability to change itself quickly, structurally and functionally depending on the current execution state and reaching SCOM goals by a change in SC structures and behavior. This is connected with the creation of an *adaptation system* (with regard to operations and resources) for the prevention, improvement, or acquisition of new characteristics for the achievement of goals under the current environmental conditions varying in time.

▶ **Practical Insights** The outbreak of COVID-19 and the associated global pandemic have clearly shown the key role of SC resilience SCOM. The

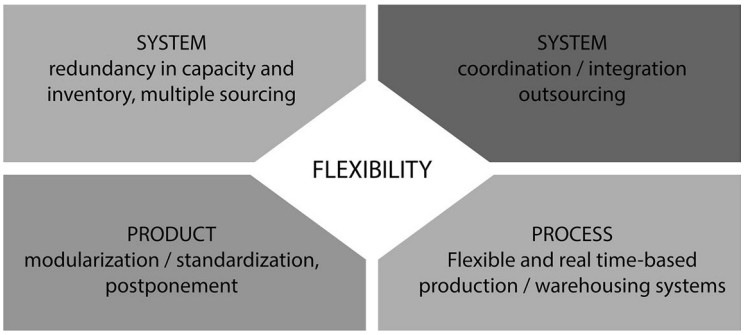


Fig. 15.9 Supply chain flexibility (adopted with changes from Ivanov et al. 2014b)

pandemic became a test for SCs regarding their robustness (i.e., the ability to withstand), flexibility (i.e., the ability to adapt), and recovery (i.e., the ability to restore operations and performance after a disruption), pointing to the central role of resilience in managing the SCs in this volatile world.

Basic areas of flexibility include system, process, and product flexibility (Fig. 15.9).

System flexibility is composed of structural and strategy components. Companies implement product and process flexibility extensively. Coordination and sourcing strategies in SCs are also typical in practice. Many companies invest in structural redundancy (e.g., Toyota extends its SC subject to multiple sourcing and building new facilities on the supply side). All these four elements of flexibility can be seen as strategies for mitigating the ripple effect at the mitigation stage and reacting at the post-disruption stage.

15.6.2 Costs of Resilience

Increase in inventory, additional production capacities, and alternative transportation methods or backup facilities would increase costs (Aldrighetti et al. 2021). At the same time, these so-called redundant elements would potentially lead to an increase in sales and service levels. The robustness elements would also reduce the risk of perturbations, which may influence schedule execution. Therefore, target objectives (e.g., on-time delivery) can be better achieved. This will positively influence sales and service levels. Redundancy elements may also increase SC flexibility and have positive effects on both service level and costs. The resilient state of an SC requires balanced robustness and flexibility, which allows for achieving maximum service level with disruption risk considerations at acceptable redundancy costs. These ideas are simplified in Figs. 15.10 and 15.11.

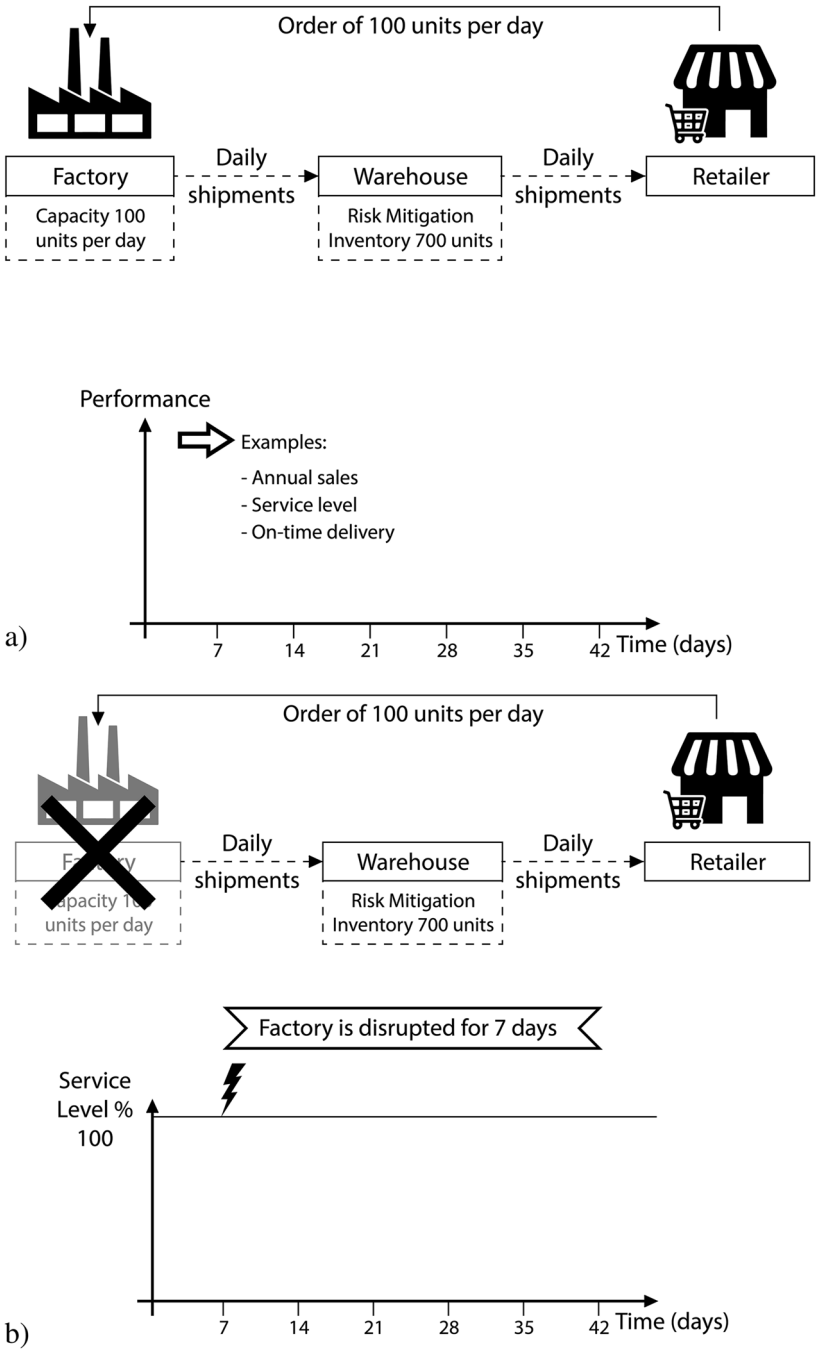


Fig. 15.10 Supply chain with a proactive policy (risk mitigation inventory) (Ivanov 2018). (**a**) Supply chain without disruptions. (**b**) Supply chain with a disruption

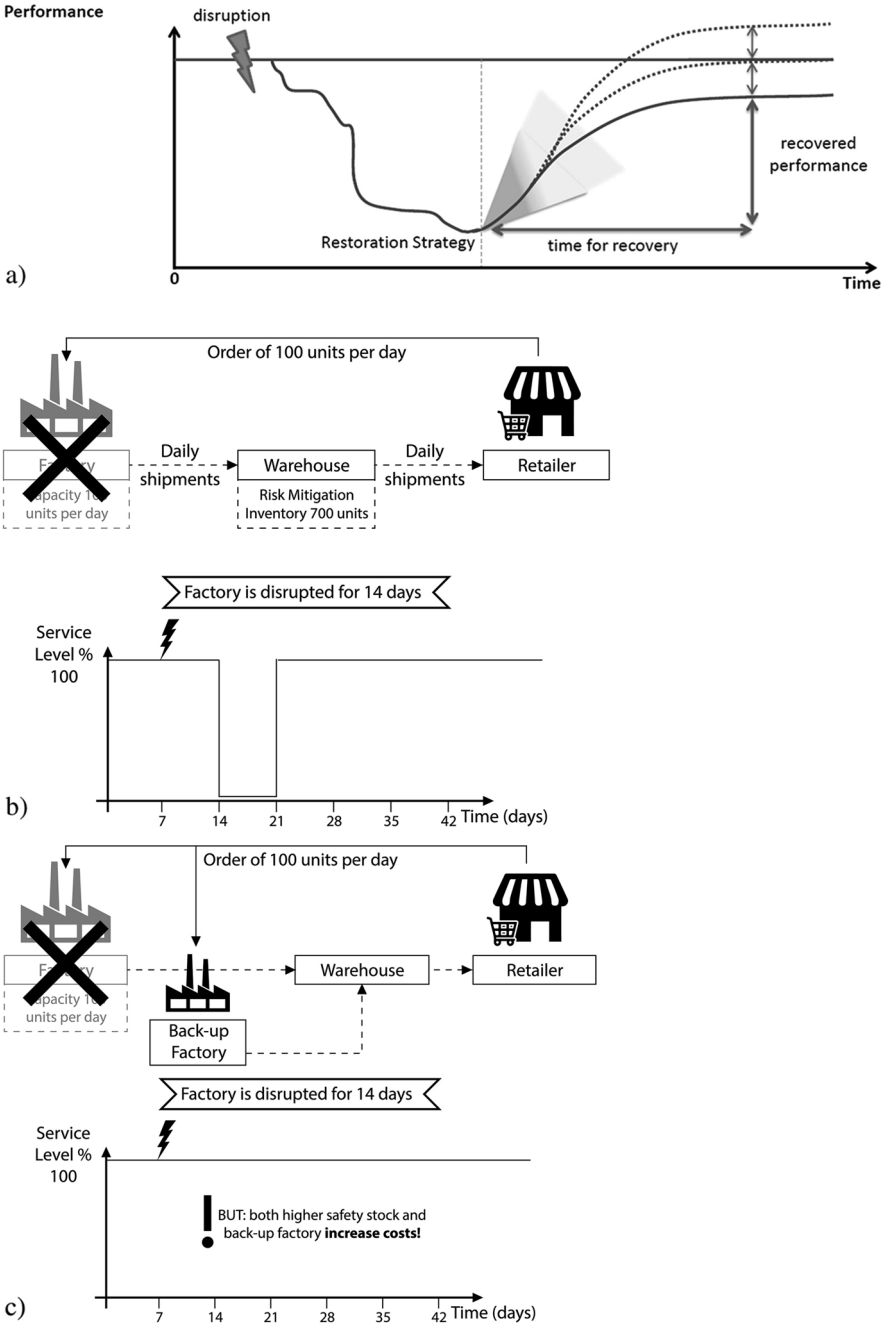


Fig. 15.11 Supply chain with a proactive policy (backup facility) (Ivanov 2018). (**a**) General performance and recovery analysis framework. (**b**) Disrupted supply chain performance. (**c**) Backup facility as a reactive flexibility strategy

In Fig. 15.10a, an SC of a retailer, a warehouse, and a factory is presented. The factory orders 100 units of a product every day from the factory that is aligned with the daily production capacity of 100 units. No batching is considered in this simple example. Daily shipments are assumed. The warehouse holds a risk mitigation inventory of 700 units as a proactive resilience policy. In Fig. 15.10b, factory capacity is disrupted for 7 days. Considering the risk mitigation inventory of 700 units and daily demand of 100 units, this disruption does not affect SC performance in terms of service level, i.e., the ratio of on-time delivered orders to all placed orders.

Now let us consider Fig. 15.11.

The disruption profile constituted by Sheffi (2005) includes eight phases: preparation actions, the disruptive event, the first response, the initial impact, the full impact, the recovery preparations, and the recovery and long-term impact. In Fig. 15.11, a general framework of SC performance and recovery analysis is depicted. The major idea here is to observe the impact of different recovery policies on SC performance. Figure 15.11b depicts that if a factory stops producing for 14 days, service levels will be disrupted since risk mitigation inventory would help for 1 week only. In Fig. 15.11c, it can be observed that the usage of a backup factory mitigates the ripple effect and performance decrease. However, both risk mitigation inventory and a backup capacity increase SC costs.

It is evident that SC flexibility and robustness are interrelated through adaptation. From the dynamics point of view, the robustness elements can also be considered as flexibility elements, and the flexibility elements can also be considered as robustness drivers. This is quite natural since both robustness and flexibility serve as "uncertainty cushions" of an SC. Balancing the elements of flexibility and robustness at proactive and reactive control loops, different constellations of service level, costs, and stability can be analyzed methodically and well-founded and with regard to a risk covering strategy and SCM strategy (Fig. 15.12).

Figure 15.12 depicts an example of a typical multi-objective analysis with regard to different proactive and reactive policies. We are interested in investigating the impacts of different combinations of disruptions (i.e., two scenarios in Fig. 15.12), proactive mitigation strategies, and reactive recovery policies (A, B, C, and D) on the SC performance in the form of service level and costs. The combinations of proactive mitigation strategies and reactive recovery policies (e.g., A—higher risk

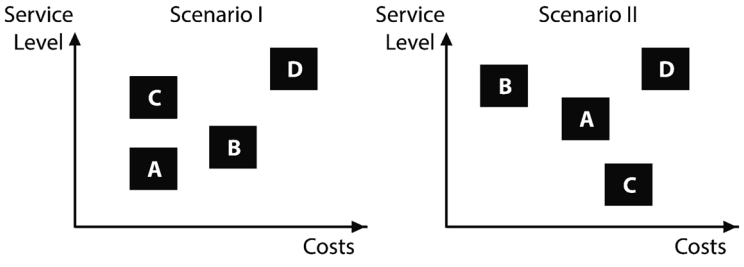


Fig. 15.12 Efficiency vs. effectiveness trade-off in supply chain resilience management (Ivanov 2018)

mitigation inventory, and B—a backup facility) may differ in costs and service level impacts for different disruption scenarios. The task of quantitative analysis methods is to decide what proactive and reactive policies must be selected.

The study by Ivanov and Dolgui (2019) suggested a new approach to SC disruption risk management where SC behavior is less dependent on the certainty of our knowledge about the environment and its changes. The unpredictability of the occurrence of disruption and its magnitude suggests that designing SCs with a low need for "certainty" may be as important, if not more so, than predetermined disruption control strategies. In this setting, this study calls for the development of a new perspective in SC disruption management, i.e., low-certainty-need (LCN) SCs. Structural variety, process flexibility, and parametrical redundancy are key LCN SC characteristics that ensure efficient disruption resistance and recovery resource allocation. Two efficiency capabilities of the LCN SC are low need for uncertainty consideration in planning decisions and low need for recovery coordination efforts based on a combination of lean and resilient elements.

15.7 Measuring Supply Chain Risks, Resilience, and the Ripple Effect

Quantification of SC risk is based on KPIs (key performance indicators) that can be classified into operational and disruption risk KPIs regarding bullwhip and ripple effects, respectively (cf. Fig. 15.4).

15.7.1 Operational Risks

Disney and Towill (2003) suggested the use of *inventory variance ratio* (InvVarR) to quantify the magnitude of bullwhip effect as the relation of inventory variance and demand variance ratio at a generic node (Eq. 15.1).

$$\mathrm{InvVarR} = \frac{\sigma^2_{\mathrm{inventory}} / \mu_{\mathrm{inventory}}}{\sigma^2_{\mathrm{demand}} / \mu_{\mathrm{demand}}} \tag{15.1}$$

where variances of demand and inventory are denoted by σ, and μ are the average values of demand and inventory.

The *order rate variance ratio* (OrdRVarR) operates in a similar way, see Eq. (15.2).

$$\mathrm{OrdVarR} = \frac{\sigma^2_{\mathrm{order}} / \mu_{\mathrm{order}}}{\sigma^2_{\mathrm{demand}} / \mu_{\mathrm{demand}}} \tag{15.2}$$

where variances (σ) are computed according to Eq. (15.3):

$$\sigma^2_{\mathrm{demand}} = \frac{\sum\left(x_{\mathrm{demand}} - \mu_{\mathrm{demand}}\right)^2}{\left(n-1\right)}; \sigma^2_{\mathrm{orders}} = \frac{\sum\left(x_{\mathrm{orders}} - \mu_{\mathrm{orders}}\right)^2}{\left(n-1\right)} \tag{15.3}$$

where x is the demand or orders in a period, μ is the average demand or orders, and n is the number of periods. If the variance ratio measure is >1, then variance amplification is present, and the risk of the bullwhip effect occurring arises. If the variance ratio measure is =1, then no bullwhip effect amplification is present. If the variance ratio measure is <1, then smoothing or dampening is occurring (cf. Heizer et al. 2022, p. 508). Some extensions of the metrics (15.1) and (15.2) for multi-echelon SCs have been proposed, e.g., in Vicente et al. (2018).

15.7.2 Disruption Risks

Ivanov (2018) suggested the use of ripple effect *performance impact* (PI) (Eq. 15.4).

$$\text{PI} = \frac{\text{Sales}_{\text{plan}}}{\text{Sales}_{\text{disruption}}} \tag{15.4}$$

The PI index represents the relationship between the KPI planned (e.g., revenue or sales) in a disruption-free mode and the real KPI in the case of disruption. Ideally, PI = 1. If the PI measure is >1, then disruption reduces SC performance. If PI measure is <1, then the SC planning was performed in a non-optimal manner.

Consider an example of PI computation for the six SC designs (SCD) in Table 15.4.

It can be observed in Table 15.4 that SC designs no. 4 and 6 can be considered the most robust since their PIs are the lowest among the SC designs considered and = 1.13.

Supplier reliability is a further important disruption risk KPI and can be computed as shown in Eq. (15.5)

$$\text{SR} = \frac{\text{Quantity}_{\text{received}}}{\text{Quantity}_{\text{ordered}}} \tag{15.5}$$

Equation (15.5) can be modified by considering on-time deliveries only. Other relevant metrics are order fill rate and lead-time variance.

Time-To-Recovery (TTR) and *Time-To-Survive* (TTS) also belong to disruption risk KPIs proposed by Simchi-Levi et al. (2015). TTR is the time elapsed from an instant disruption (e.g., an earthquake) that immediately shuts down some capacity until the recovery of an original capacity or its substituting by a backup source. TTR is supplemented by TTS, which shows how long an SC can keep operating without

Table 15.4 PI computation results

	Initial	SCD 1	SCD 2	SCD 3	SCD 4	SCD 5	SCD 6
Maximum revenue in disruption-free scenario	82,550	82,550	82,550	82,550	82,550	82,550	82,550
Revenue in the disruption scenario	54,600	67,600	67,600	70,200	72,800	70,200	72,800
PI	1.5119	1.2211	1.1759	1.1759	1.1340	1.1759	1.1340

the disrupted capacity (e.g., by using some pre-positioned inventory). TTR/TTS is used to "identify bottleneck suppliers for which it is critical to obtain accurate TTR information and distinguish them from other suppliers where even plus or minus 30% error in TTR information will have very little impact on the supply chain." (Simchi-Levi 2015). If TTS > TTR, then an SC can be considered resilient. In other words, the existing resilience capabilities such as risk mitigation inventory, capacity agility, and backup sourcing allow maintaining SC operations and surviving during the recovery time (Ivanov 2025a, 2025b).

For particular analysis purposes, additional KPIs for disruption risks can be used, e.g., VAR (Value-At-Risk), CVAR (Conditional Value-At-Risk), resilience, robustness, flexibility, adaptability, and reliability of SCs (Goh and Meng 2009; Schmitt and Singh 2012; Kim et al. 2015; Chen et al. 2017; Mizgier 2017; Ivanov 2018; Pavlov et al. 2018; Sokolov et al. 2016; Behzadi et al. 2020; Hosseini and Ivanov 2022; Dixit et al. 2020). Software such as anyLogistix, DHL Resilience 360, and RiskMethods facilitate the computation of these and further KPIs at the risk assessment, monitoring, identification, recovery, and control stages. We refer to Ivanov (2025d) for further details on resilience measurement.

15.7.3 Resilience Quantification

By definition, SC resilience is 'the firm's capability to withstand, adapt, and recover from disruptions to meet customer demand, ensure target performance, and maintain operations in vulnerable environments' (Hosseini et al. 2019a, b). This definition shows three major dimensions forming the resilience notion, namely, performance impact, an ability to withstand/absorb disruptions, and an ability to adapt/recover after disruptions. Fig. 15.13 summarizes resilience indicators accordingly.

The *first* group of indicators used in stress tests aims at measuring the impact of disruptions SC performance. This impact has been frequently measured by three major service level indicators, i.e., on-time delivery, fill rate, and demand fulfilment. While the on-time delivery and demand fulfilment indicators helps to detect a disruption in the supply chain from the customer perspective, fill rate allows identifying a disruption from the operational perspective. For example, fill rate immediately informs about a disruption based on missing inventory in a warehouse while on-time delivery is measured at the customer and so indicated a disruption with some time delay because of some inventory pipeline and system inertia (Ivanov 2025a). Another important perspective of performance impact-based indicators is the financial one, for which such indicators as annual revenues, costs and sales have been used in extant literature. The concept of possible maximum losses has been used by Kinra et al. (2020) for quantifying the ripple effect in supply chains. This method is based on a monetary quantification of possible losses in sales and revenue due to disruptions upstream and associated material/component shortages which translate in the sales losses. Interestingly, indicators of supplier performance such as delivery

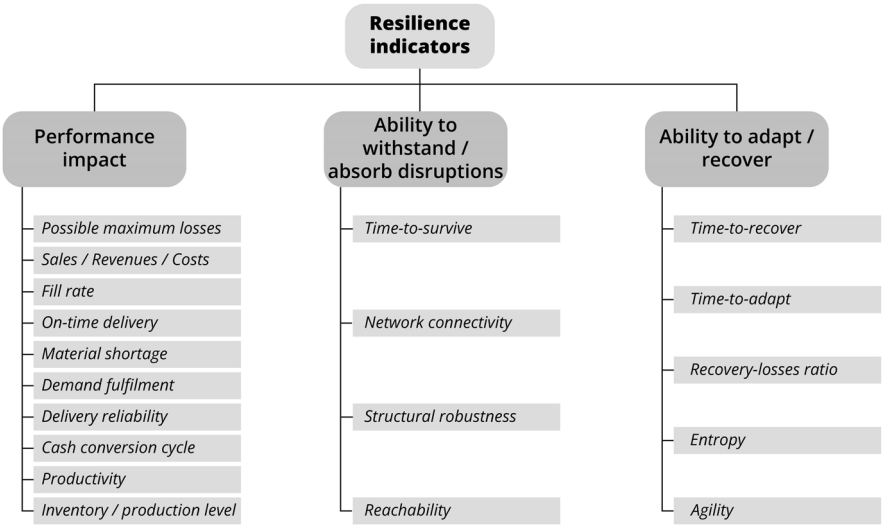


Fig. 15.13 A taxonomy of supply chain resilience indicators (Ivanov 2025c)

reliability and delivery delays have been frequently applied in disruption predictions using AI methods.

The *second* group of indicators characterizes the ability to withstand/absorb disruptions. Network theory has been frequently applied using such metrics like spatial complexity, network density, degree and betweenness centrality, connectivity, reachability, and structural robustness to analyse structural integrity (Li and Zobel 2020). Network indicators focus on quantifying the ability of a supply chain to maintain its structural connectivity and adaptability. When using network metrics, the focus is on a structure connectivity rather than a disruption impact. For example, a supplier disruption can result in some temporary discontinuity of a network. However, if a recovery is deployed effectively, the disruption can be mitigated and not seen at the performance level. This separation of network and process levels is one of the shortcomings to use graph-theoretical metrics for practical stress tests.

The *third* group of indicators focuses on the ability to recover. Here, two major approaches exist. The first one is based on measuring time. Simchi-Levi et al. (2015) introduced TTR (time-to-recover) indicator which measures the period between a disruption-driven capacity shutdown and the recovery of original capacity or its substitution by a back-up source. TTR has been used in stress tests against delivery time variability at. The second approach is based on a ratio of disruptions and recovery efforts. Focusing on TTR and the time a supply chain is able to cope with shortages, Simchi-Levi et al. (2015) measured resilience as a relation of TTR and time-to-survive (TTS). TTS shows how long a supply chain can operate without its disrupted capacity (e.g., using some pre-positioned inventory). In short, if TTS is longer than TTR, a supply chain is considered resilient. However, there is no explicit quantification of resilience in this method.

Resilience measuring can also be based on the intensity of performance loss and recovery where resilience or vulnerability indicators are bounded between 0 and 1. Resilience is calculated by 1 minus fraction of loss, so both metrics are bounded between 0 and 1. Dolgui et al. (2024) proposed a time-to-adapt (TTA) metric which measures time between disruption recognition and begin of adaptation deployment. Table 15.5 summarizes major resilience indicators described above.

Table 15.5 Resilience indicators (Ivanov 2025a)

Indicator	Definition	Practical example
On-time delivery	Ratio of on-time delivered orders to the total number of orders placed	Lockdowns and high infection rates during COVID-19 pandemic interrupted logistics operations resulting in delivery delays
Fill rate/ material shortage	Ratio of orders successfully fulfilled from available inventory-on-hand to the total number of orders placed	A fire at a warehouse of UK online retailer Asos damaged 20% of stock leading to a reduced inventory availability in the supply chain
Sales/possible maximum loss	Monetary quantification of possible losses in sales and revenue due to upstream disruptions and associated material/component shortages, which translate to sales losses.	Semiconductor shortage resulted in sales decreases in Sony's Play Station supply chain in 2021 and 2022; A fire at a warehouse of UK online retailer Asos caused a sales loss during the quarter of between £25m and £30m
Time-to-survive	It shows how long a supply chain can operate without its disrupted capacity (e.g., using some pre-positioned inventory) before its performance begins declining	During the COVID-19 outbreak, automotive companies worldwide could keep producing for about 2–5 weeks after production stops at suppliers in China because their carry 15 to 30 days' worth of inventory
Network connectivity/ density	Ratio of number of existing (healthy) links to total number of links in a network	After earthquake and tsunami in Japan in 2011, many Toyota's factories became disconnected from their suppliers
Structural Robustness	Minimal number of nodes and arcs, the unavailability of which makes the graph disconnected.	Explosion at BASF plant in Ludwigshafen in 2016 led to production stops of some products at BASF for many weeks
Reachability/ centrality	A sum of maximum in-degree and out-degree centralities of nodes displaying the relationships between neighbour nodes in a network	Apple computers' production in Asia has been paralysed because of disruptions at their Tier-1 suppliers in Thailand after an earthquake in 1999
Time-to-recover	Time elapsed between a disruption-driven capacity shutdown and the recovery of original capacity or its substitution by a backup source	BMW needed 1 week to resume production which was stopped as a consequence of a supply shortage of steering gears. BMW's Tier-1 supplier, Bosch, was unable to deliver the steering gears and it took a week to recover.
Time-to-adapt	Time between disruption recognition and onset of adaptation/recovery deployment	Ford has adapted its supply chain from car manufacturing to ventilator production within a week in spring 2020

(continued)

Table 15.5 (continued)

Indicator	Definition	Practical example
Resilience/ vulnerability indexes	Recovery-loss ratios, e.g., resilience quantification as a ratio of recovery and loss levels associated with a service level	UK online retailer Asos was able to restart operations 48 hours after a fire at a warehouse. The disruption caused a sales loss during the quarter of between £25 m and £30 m
Entropy	Measures adaptability as the structural property which characterizes the decoupling or branching degree of supply chain processes	AGGO was capable of adapting its logistics in the wake of the COVID-19 pandemic, anticipating lockdowns and re-routing goods using rail across Russia rather than traditional sea freight, despite the 85% higher costs. This approach ensured European sites had a steady supply despite China's shutdown.

15.7.4 Ripple Effect Quantification

Kinra et al. (2020) proposed a new model to measure the ripple effect of a supplier disruption based on possible maximum loss. The constructed *ripple effect exposure* (REE) model quantifies the ripple effect, comprehensively combining features such as financial, customer, and operational performance impacts, consideration of multi-echelon inventory, disruption duration, and supplier importance.

Figure 15.14 shows the computational logic of the REE model.

We introduce the following notations in line with Kinra et al. (2020):

REE is the ripple effect exposure of the supply chain.
SE_i is supplier exposure.
i is a supplier index, $i \in [1, \ldots, n]$.
j is a product index, $j \in [1, \ldots, m]$.

l_j is a part needed for product j, $l \in [1, \ldots, r_j]$.
q_{li} is the number of units of part l sourced from supplier i at stage k or the number of missing parts resulting from supplier disruption when considering stage $k + 1$ according to downstream disruption propagation in the supply chain.
Q_l is the total number of units of part l.
m is the total number of units of part l_j sourced across all suppliers.
V_j is the demand for j.
π_j is the profit margin for product j in % of revenue.
p_j is the sales price of product j.
c is the business interruption time.
I_k is inventory-on-hand (measured, e.g., as weeks or days of supply) held at the SC echelon k.
k is the supply chain echelon, $k \in [1, \ldots, s_j]$.

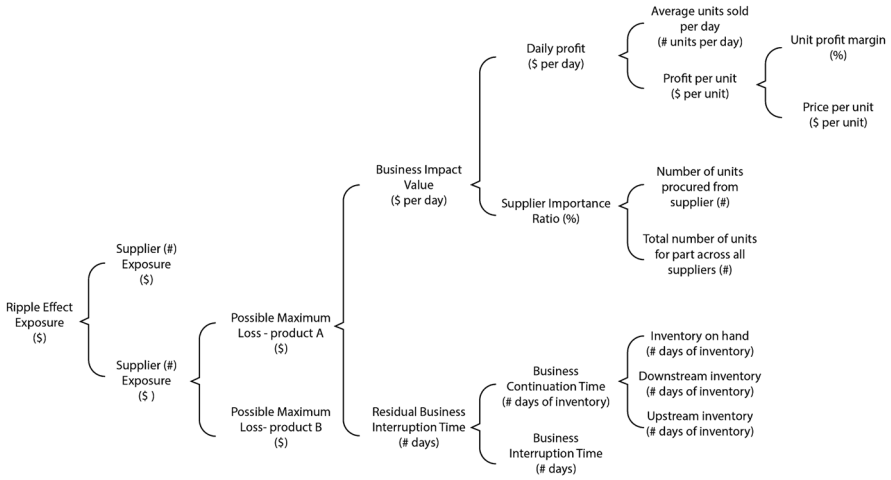


Fig. 15.14 REE model for Ripple effect measurement (Kinra et al. 2020)

REE is considered a function of SE to risk. Assessment of SE is based on the analysis of "Possible Maximum Loss" (PML), resulting from upstream disruptions in the supply chain (Eq. 15.6):

$$\mathrm{PML}_{lj} = \mathrm{BIV}_{lj} \cdot \mathrm{RBIT}_{lj} \tag{15.6}$$

PML is a compounding function of two variables: residual business interruption time (RBIT) and "Business Impact Value" (BIV), a financial impact variable. The BIV computation is shown in Eq. (15.7).

$$\mathrm{BIV}_{lj} = \frac{q_{li}}{Q_l} \cdot V_j \cdot \left(\pi_j \cdot p_j\right) \tag{15.7}$$

The term $\frac{q_{li}}{Q_l}$ in BIV shows the supplier importance ratio (SIR). The higher supplier spending, the higher the SIR, and the higher the BIV. The second part of the BIV computation is related to the lost sales due to supplier disruptions.

The second factor in determining the PML in the model is RBIT. RBIT consists of two separate parts, i.e., the BIT (Business Interruption Time) and BCT (Business Interruption Time). Their meaning is close to TTR/TTS time introduced by Simchi-Levi et al. (2015). The BIT is the amount of time for which the supplier is expected to be unavailable due to a disruption and during which no supply substitute can be established. The BCT is the amount of time the focal company can continue to meet demand in spite of a disruption in supply, e.g., using some inventory on-hand. Thus, RBIT considers the maximum time the system is affected, given some recovery strategies. The model sets negative RBIT values to 0 to ensure that instances with BCT > BIT, i.e., making the RBIT negative, are not included as negative PML values, i.e., profit gains. The computation of the RBIT is outlined in Eq. (15.8).

$$\mathrm{RBIT}_{lj} = \max\left(\left(\mathrm{BIT}_{lj} - \mathrm{BCT}_{lj}\right), 0\right) \tag{15.8}$$

Now we compound the Eqs. (15.6–15.8) to the SE and REE indexes (Eq. 15.9).

$$\begin{aligned} \mathrm{REE} &= \sum_{k=1}^{s} \mathrm{SE}_k; \\ \mathrm{SE}_j &= \sum_{j=1}^{m} l_{j\in(1,\ldots,r_j)} \left(\left(\frac{q_{li}}{Q_l} \cdot V_j \cdot \pi_j \cdot p_j \right) \cdot \max\left(0; \left(c - \sum_{k=1}^{s_j} I_k \right) \right) \right), \end{aligned} \tag{15.9}$$

SE_i are computed at each supplier. SE_k are computed at each echelon in the SC. REE extends the analysis toward a multi-echelon setting. REE is considered as a compounding function for the disruption propagation (i.e., the ripple effect) downstream in the supply chain, taking into account individual SE assessments. Depending on the decision-maker's risk-aversion, REE can reflect either the total impact, as a worst-case scenario where all suppliers would experience a disruption, or the average impact among all possibly disrupted suppliers.

Practical Example

A three-stage retail supply chain with five suppliers, one distribution center (DC), and two customers with equal demand is considered. Each supplier delivers exactly one product (i.e., Water, Drinks, Juice, Yogurt, and Milk) to the DC, and the DC delivers these products to customers. Analyze the REE and PML with consideration of the total impact and the importance of each homogenous node in the supply network. Data for the analysis and computations of REE/SE are shown in Fig. 15.14.

Figure 15.15 displays five supplier-product combinations with different average disruption durations (BITs), supplier importance ratios, average inventory in terms of days of supply, and profits based on analysis of historical company data over 3 years. The average daily demand of the DC is 1310 boxes for water, 1900 boxes for drinks, 980 boxes for juices, 1200 boxes for yogurt, and 850 boxes for ESL (extended shelf life) milk. Average inventory in supplier data refers to the inventory of the supplier's products at the DC. Analogously, average inventory in the DC data refers to the inventory of the DC's products at the customers.

We first compute SEs on the supplier level. PML equals 0 for the two cases (water and yogurt) because average inventory enables compensation after the disruption during the BIT (i.e., average inventory in days >BIT, and RBIT = 0). For drinks, juices, and ESL-milk, PML is positive, and its total is $769,200. As such, the total SE (i.e., the case if all suppliers would be disrupted simultaneously) is $769,200.

Considering the ripple effect of the supplier disruptions, we now move to the DC level. The total daily demand at the customers (i.e., the supermarket stores) is 2620; 3800; 1960; 2400; and 1700 boxes for water, drinks, juices, yogurt, and ESL-milk, respectively. Since, in reality, two DCs deliver to the stores, the average daily demand at the DC considered is lower (1310; 1910; 980; 1200; and 850 boxes, respectively). The disruption-driven shortage is identical to those in the upper part of the computational sheet. The REE of the supply chain is $1,027,650, meaning the

Level Supplier	Water	Drinks	Juices	Yogurts	ESL-Milk		SE for product j
Average demand per day	1310	1900	980	1200	850	Level Supplier	769200
Average Profit per unit	60	50	30	40	30	Level DC	258450
Number of units procured from supplier per day	262	480	442	386	440		
Total number of units for part across all suppliers (#)	1310	1900	980	1200	850		
Average Inventory in days	30	30	30	20	20		
Maximum BIT	10	40	50	15	40		
Daily profit	78600	95000	29400	48000	25500		
SIR	0.2	0.252632	0.45102	0.321667	0.517647		
BIV	15720	24000	13260	15440	13200		
RBIT	0	10	20	0	20		
PLM	0	240000	265200	0	264000		
RE exposure	1027650						
Level DC	Water	Drinks	Juices	Yogurts	ESL-Milk		
Average demand per day	1310	1900	980	1200	850		
Average Profit per unit	60	50	30	40	30		
Number of units missing due to supplier disruption, per day	262	480	442	386	440		
Total demand across all DCs (#)	2620	3800	1960	2400	1700		
Average Inventory in days	5	5	5	5	5		
Maximum BIT	10	40	50	15	40		
Daily profit	78600	95000	29400	48000	25500		
SIR	0.1	0.126316	0.22551	0.160833	0.258824		
BIV	7860	12000	6630	7720	6600		
RBIT	0	5	15	0	15		
PLM	0	60000	99450	0	99000		

Fig. 15.15 Input data and REE/SE computation example (Kinra et al. 2020)

maximum possible financial loss that our supply chain can suffer from supplier disruptions.

A number of management recommendations can be derived using the REE and SE indicators. In terms of sourcing strategy, identifying critical suppliers and/or replacing/re-planning inventory in the supply chain can be recommended in cases with high REE and SE. At a strategic level, the REE exposure analysis can be used to determine an improved supplier strategy (e.g., dual/backup supply) by identifying which suppliers will have the greatest impact on performance when disrupted. The model can also be applied at the operative level in two ways. First, it can provide critical information for adjusting inventory placements and levels in the supply chain, subject to RBIT. Second, using the REE index can enable quick estimations of possible impacts of disruptions on supply chain performance. Such estimation can also be useful when deciding on the timing and scale of deployment of a contingency policy, e.g., the timing of installation of a backup source and its scope.

15.8 Super Disruptions and Supply Chain Viability

15.8.1 Super Disruptions

To recapitulate, SC resilience is considered the SC's ability to bounce back from a severe disruption and recover to an "old normal" (Fiksel 2006; Pettit et al. 2010; Hosseini et al. 2019a, b). Notably, resilience theory has been developed for managing disruptions, which are usually defined as events. As a new super disruption, the

Table 15.6 Instantaneous SC disruptions and super disruptions (SC crises)

Criteria	Instantaneous SC disruption, e.g., an earthquake or fire	Super disruption (SC crises), e.g., a pandemic
Impact	Instant impact	Long-lasting impact with hardly predictable scaling
Scope	Single supply chain echelon (with possible propagations)	Simultaneous disruptions in supply, demand, and logistics
Recovery	Begins when disruption is over	It is performed in the presence of a disruption and its unpredictable scaling
Timing	A single disruptive event	Simultaneous and/or sequential openings and closures of suppliers, facilities, and markets

COVID-19 pandemic raised novel questions and decision-making contexts, which frequently exceeded the resilience scope.

The pandemic represents as specific type of disruption risks, i.e., a *super disruption* which is characterized by four major aspects which differentiate the pandemic disruption from all other "instantaneous" (i.e., an event of an immediate impact) disruptions, as shown in Table 15.6 (Ivanov 2020, 2022b; Ivanov and Dolgui 2020; Ivanov and Das 2020).

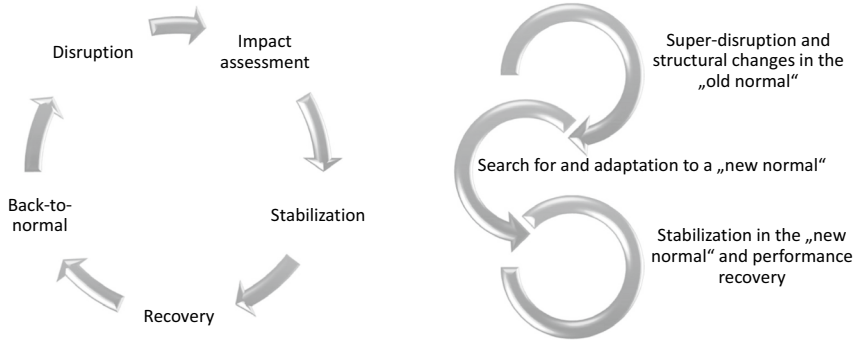
Indeed, SC resilience theory has been developed to manage disruptions that are considered events. With the COVID-19 pandemic, some novel context has been unveiled, which goes beyond an instantaneous event-driven understanding and can be described as a *supply chain crisis*. *A supply chain crisis* is a long-term disrupted state that is characterized by the unstable current situation and uncertainty of future developments in the markets, supply base, and capacities, entailing a danger of SC collapses and interruption of market provision with goods and services (Queiroz et al. 2022).

15.8.2 Supply Chain Viability

The concept of *viability* in SCOM was coined by Ivanov (2022b) and Ivanov and Dolgui (2020). They defined viability as "*the ability of an supply chain to maintain itself and survive in a changing environment through a redesign of structures and replanning of performance with long-term impacts*" (Ivanov and Dolgui 2020; Ivanov 2022b). Viability can be considered an extended resilience perspective, which extends the SC resilience notion from a closed system, "bounce-back" view to an open system, "bounce-forward-and-adapt" notion (Fig. 15.16).

Despite several common features such as disruption-orientation, recovery, and system view, viability and resilience have some key differences, as shown in Table 15.7.

SC resilience is usually considered the ability to bounce back once disrupted and recover to a known normal state. Viability looks at the ability to bounce forward and adapt once disrupted when a normal state cannot be reached at a reasonable time scale, and a search for an unknown "new normal" is required. Speaking in terms of



Supply Chain Resilience: Closed-System View
bouncing back to an „old normal“ after a disruption or the ripple effect to recover a planned performance

Supply Chain Viability: Open System View
adaptation to a „new normal“ to survive in radically changed internal and external conditions

Fig. 15.16 Viability and resilience concepts (Ruel et al. 2024)

Table 15.7 Differences between SC resilience and SC viability

	Resilience	Viability
Time horizon of disruptions	Short-term (single or correlated disruption events)	Long-term (radically changed external and internal environments)
Notion	Ability to bounce back once disrupted and recover to a known normal state	Ability to bounce forward and adapt once disrupted when a normal state cannot be reached in a reasonable time scale, and a search for an unknown “new normal” is required. In terms of chaos theory, we have a case of searching for new attractors to stabilize the system.
Profile	Closed system view: Disruption → performance degradation → restoration actions → performance recovery	Open system view: Disruption → performance and structure degradation → adaptation-to-survive → search for “new normal” → stabilization in the “new normal” → restoration actions → performance recovery
Assessment criterion	Degree of performance degradation and recovery	Survivability of an SC
Management objective	Maintaining the performance of an SC or a firm	Maintaining provisions for a society with critical services such as mobility (survival of car manufacturing companies), communication (survival of electronic companies), and food (survival of food SCs)
Preparedness	One can prepare a contingency and recovery plan to overcome the disruption period and survive in the disrupted mode	One cannot plan for super disruptions; adaptation is the only way to survive
Object	Individual SCs	Individual SCs and intertwined supply networks (ISN)

chaos theory, with the SC viability, we have a case of searching for new attractors to stabilize the system (Ivanov 2021). Next, resilience is related to the level of individual SCs or supply networks, whereas viability is concerned with intertwined supply networks "*that encapsulate entireties of interconnected SCs, which, in their integrity, secure the provision of society and markets with goods and services.*" Further, we retain the focus of SC resilience towards performance in a fixed time window and in a closed system setting, while SC viability considers the structural dynamics of continuous system change in an open system context wherein the time window is open to reach survivability. In SC resilience, one can prepare a contingency and recovery plan to overcome the disruption period and survive in the disrupted mode. In SC *viability*, one cannot plan for super disruptions, and adaptation is the only way to survive. We refer to Chap. 4 of this textbook for more details on SC viability and the Viable SC model.

15.9 Key Points and Discussion Questions

In this chapter, SC risk management has been introduced. Fundamentally, there are four important categories to be analyzed in this domain, i.e., uncertainty, risk, disturbance (the so-called perturbation impact), and disruption (operational deviations).

- Can you explain how these four elements are mutually connected?
- Can you summarize uncertainty factors and the corresponding handling measures?
- Can you explain different SC risks?

We have seen that demand, globalization, lead-time, and lean SCs can be considered as reasons for risks in SCs. SCs are exposed to recurrent operational risks and exceptional disruption risks. Summarize in your own words factors relevant to operational risks under the Bullwhip Effect. Do the same for disruption risks and, explain the Ripple Effect providing some examples of the Bullwhip and the Ripple Effect.

- Where does information coordination help, and when would you suggest building up inventory buffers?
- Is the explosion of a factory an operational or a disruptive risk? Is an explosion an exceptional or a recurrent risk?
- What are low-frequency-high-impact risks? Are these operational or disruptive risks?
- What are high-frequency-low-impact risks? Are these operational or disruptive risks?
- What are the consequences of the Ripple Effect that can be diagnosed in reduced SC performance?
- What are two other good expressions that you know for the Ripple Effect?
- Can you summarize major reasons for the Ripple Effect, and can you give examples for recovery strategies or policies?
- What KPIs can we use to quantify operational and disruption risks?

SC managers do not only deal with SC risks. They also need to put strategies in place to manage SC resilience. SCs need to be planned such that they are stable, robust, and resilient, i.e., so they can cope with uncertainty in the planning stage and are able to deal with disturbances and disruptions during SC execution.

- Can you give examples of proactive and reactive measures that will lead to SC resilience?
- What are approaches to support SC redundancy?
- What are the four elements that enable companies to implement SC flexibility?

SC flexibility and robustness are interrelated. As we have seen, risk mitigation inventory or considering a backup location, and purchasing items from an alternative supplier, respectively, are useful strategies. These robustness and flexibility strategies have certain costs. Companies should therefore balance their robustness and flexibility investments and possible losses from disruptions to reduce the Ripple Effect and mitigate decreased SC performance, which would lead to lost sales, penalty payments, and is significantly negative from a reputational point of view.

Viability is the SC's ability to adapt and survive during severe crisis. This is an extended property related to resilience and severe disruptions such as pandemics which radically change operational conditions for a long period.

- Elaborate on the interrelations of SC viability and resilience.
- What differences and commonalities can you see between instant disruptions (i.e., singular event disruptions with an immediate impact such as fires or earthquakes) and a pandemic? In particular, think over the timing of disruptions, their scaling and impacts; preparedness and recovery strategies for SCs; and disruption propagation (i.e., the ripple effect).

Bibliography

Aldrighetti R, Battini D, Ivanov D, Zennaro I (2021) Costs of resilience and disruptions in supply chain network design models: a review and future research directions. Int J Prod Econ 235:108103

Aldrighetti R, Battini D, Ivanov D (2023) Efficient resilience portfolio design in the supply chain with consideration of preparedness and recovery investments. Omega 117:103841

Aldrighetti R, Calzavara M, Martignago M, Zennaro I, Battini D, Ivanov D (2024) A methodological framework for the design of efficient resilience in supply networks. Int J Prod Res 62(1–2):271–290

AlixPartners (2023). Shortages related to semiconductors to cost the auto industry $210 billion in revenues this year, says new AlixPartners forecast. AlixPartners https://www.alixpartners.com/media-center/press-releases/press-releaseshortages-related-to-semiconductors-to-cost-the-auto-industry-210-billion-in-revenuesthis-year-says-new-alixpartners-forecast/. Accessed 6 Jan 2025

Altay N, Gunasekaran A, Dubey R, Childe SJ (2018) Agility and resilience as antecedents of supply chain performance under moderating effects of organizational culture within the humanitarian setting: a dynamic capability view. Prod Plan Control 29(14):1158–1174
Ambulkar S, Blackhurst J, Grawe S (2015) Firm's resilience to supply chain disruptions: scale development and empirical examination. J Oper Manag 33(34):111–122
Ashby WR (1956) An introduction to cybernetics. Chapman & Hall, London
Behzadi G, O'Sullivan MJ, Olsen TL (2020) On metrics for supply chain resilience. Eur J Oper Res 287(1):145–158
Blackhurst J, Craighead CW, Elkins D, Handfield R (2005) An empirically derived agenda of critical research issues for managing supply-chain disruptions. Int J Prod Res 43(19):4067–4081
Blackhurst J, Dunn KS, Craighead CW (2011) An empirically derived framework of global supply resiliency. J Bus Logist 32(4):374–391
Bode C, Wagner SM, Petersen KJ, Ellram LM (2011) Understanding responses to supply chain disruptions: insights from information processing and resource dependence perspectives. Acad Manag J 54(4):833–856
Brandon-Jones E, Squire B, Autry CW, Petersen KJ (2014) A contingent resource-based perspective of supply chain resilience and robustness. J Supply Chain Manag 50(3):55–73
Burgos D, Ivanov D (2021) Food retail supply chain resilience and the COVID-19 pandemic: a digital twin-based impact analysis and improvement directions. Transp Res E Logist Transp Rev 152:102412
Casti JL (1979) Connectivity, complexity and catastrophe in large-scale systems. Wiley, New York
Chen X, Xi Z, Jing P (2017) A unified framework for evaluating supply chain reliability and resilience. IEEE Trans Reliab 66:1144
Chervenkova T, Ivanov D (2023) Adaptation strategies for building supply chain viability: a case study analysis of the global automotive industry re-purposing during the COVID-19 pandemic. Transp Res E 177:103249
Chopra S, Sodhi MS (2004) Managing risk to avoid supply-chain breakdown. MIT Sloan Manag Rev 46:52–61
Chopra S, Sodhi MS (2014) Reducing the risk of supply chain disruptions. MIT Sloan Manag Rev 55(3):73–80
Chopra S, Reinhardt G, Mohan U (2007) The importance of decoupling recurrent and disruption risks in a supply chain. Nav Res Logist 54(5):44–555
Chowdhury MMH, Quaddus M (2017) Supply chain resilience: conceptualization and scale development using dynamic capability theory. Int J Prod Econ 188:185–204
Christopher M, Peck H (2004) Building the resilient supply chain. Int J Logist Manag 15(2):1–14
Craighead CW, Ketchen DJ Jr, Darby JL (2020) Pandemics and supply chain management research: toward a theoretical toolbox. Decis Sci 51(4):838–866
Disney SM, Towill DR (2003) The effect of vendor managed inventory (VMI) dynamics on the bullwhip effect in supply chains. Prod Econ 85:199–215
Dixit V, Verma P, Tiwari M (2020) Assessment of pre and post-disaster supply chain resilience based on network structural parameters with CVaR as a risk measure. Int J Prod Econ 227:107655
Dolgui A, Ivanov D (2021) Ripple effect and supply chain disruption management: new trends and research directions. Int J Prod Res 59(1):102–109
Dolgui A, Ivanov D, Sokolov B (2018) Ripple effect in the supply chain: an analysis and recent literature. Int J Prod Res 56(1–2):414–430
Dolgui A, Ivanov D, Rozhkov M (2020) Does the ripple effect influence the bullwhip effect? An integrated analysis of structural and operational dynamics in the supply chain. Int J Prod Res 58(5):1285–1301
Dolgui A, Gusikhin O, Ivanov D, Li X, Stecke K (2024) A network-of-networks adaptation for cross-industry manufacturing repurposing. IISE Trans 56(6):666–682
Dolgui A, Ivanov D, Simchi-Levi D (2025) Stress tests for supply chains: towards resilience and viability. Int J Prod Res 63(9):3254–3258
Dubey R, Gunasekaran A, Childe SJ, Papadopoulos A, Blome C, Luo Z (2019) Antecedents of resilient supply chains: an empirical study. IEEE Trans Eng Manag 66(1):8–19

Echefaj K, Charkaoui A, Cherrafi A, Ivanov D (2024) Design of resilient and viable sourcing strategies in intertwined circular supply networks. Ann Oper Res. https://doi.org/10.1007/s10479-024-05873-1

Fahimnia B, Tang CS, Davarzani H, Sarkis J (2015) Quantitative models for managing supply chain risks: a review. Eur J Oper Res 247(1):1–15

Fiksel J (2006) Sustainability and resilience: toward a systems approach. Sustain Sci Pract Policy 2(2):1–8

Garvey MD, Carnovale S (2020) The rippled newsvendor: a new inventory framework for modelling supply chain risk severity in the presence of risk propagation. Int J Prod Econ 228:107752

Garvey MD, Carnovale S, Yeniyurt S (2015) An analytical framework for supply network risk propagation: a Bayesian network approach. Eur J Oper Res 243(2):618–627

Ghadge A, Dani S, Chester M, Kalawsky R (2013) A systems thinking approach for modelling supply chain risk propagation. Supply Chain Manag 18(5):523–538

Ghadge A, Weiß M, Caldwell ND, Wilding R (2019) Managing cyber risk in supply chains: a review and research agenda. Supply Chain Manag 25(2):223–240

Ghadge AEM, Ivanov D, Chaudhuri A (2021) Visualisation of ripple effect in supply chains under long-term, simultaneous disruptions: a system dynamics approach. Int J Prod Res 60(20):6173–6186

Goh M, Meng F (2009) A stochastic model for supply chain risk management using conditional value at risk. In: Wu T, Blackhurst J (eds) Managing supply chain risk and vulnerability. Springer, London

Gupta S, Starr MK, Farahani RZ, Matinrad N (2016) Disaster management from a POM perspective: mapping a new domain. Prod Oper Manag 25:1611–1637

Gupta V, Ivanov D, Choi T-M (2021) Competitive pricing of substitute products under supply disruption. Omega 101:102279

Gurnani H, Mehrotra A, Ray S (2012) Supply chain disruptions: theory and practice of managing risk. Springer, London

Hallikas J, Karvonen I, Pulkkinen U, Virolainen VM, Tuominen M (2004) Risk management processes in supplier networks. Int J Prod Econ 90(1):47–58

Handfield RB, McCormack K (2008) Supply chain risk management: minimizing disruptions in global sourcing. Auerbach Publications, Boca Raton

Haren P, Simchi-Levi D (2020) How coronavirus could impact the global supply chain by mid-march. Harward Business Review, February 28, 2020. https://hbr.org/2020/02/how-coronavirus-could-impact-the-global-supply-chain-by-mid-march?ab=hero-subleft-1. Accessed 2 Jan 2025

Heckmann I (2016) Towards supply chain risk analytics. Springer, Wiesbaden

Heizer J, Render B, Munson C (2022) Operations management: sustainability and supply chain management, 14th edn. Pearson, Harlow

Hendricks KB, Singhal VR (2005) Association between supply chain glitches and operating performance. Manag Sci 51(5):695–711

Ho W, Zheng T, Yildiz H, Talluri S (2015) Supply chain risk management: a literature review. Int J Prod Res 53(16):5031–5069

Hosseini S, Ivanov D (2021) A multi-layer bayesian network method for supply chain disruption modelling in the wake of the COVID-19 pandemic. Int J Prod Res 60(17):5258–5276

Hosseini S, Ivanov D (2022) Resilience assessment of supply networks with disruption propagation considerations: a Bayesian network approach. Ann Oper Res 319:581–607

Hosseini S, Ivanov D, Dolgui A (2019a) Ripple effect modeling of supplier disruption: integrated Markov chain and dynamic Bayesian network approach. Int J Prod Res 58(11):3284–3303

Hosseini S, Ivanov D, Dolgui A (2019b) Review of quantitative methods for supply chain resilience analysis. Transp Res E 125:285–307

Hosseini S, Ivanov D, Blackhurst J (2020) Conceptualization and measurement of supply chain resilience in an open-system context. IEEE Trans Eng Manag. https://doi.org/10.1109/TEM.2020.3026465

Ivanov D (2010) A framework for aligning (re)planning decisions on supply chains strategy, design, tactics, and operations. Int J Prod Res 48(13):3999–4017

Ivanov D (2017) Simulation-based ripple effect modelling in the supply chain. Int J Prod Res 55(7):2083–2101
Ivanov D (2018) Structural dynamics and resilience in supply chain risk management. Springer, New York
Ivanov D (2019) Disruption tails and revival policies: a simulation analysis of supply chain design and production-ordering systems in the recovery and post-disruption periods. Comput Ind Eng 127:558–570
Ivanov D (2020) Predicting the impact of epidemic outbreaks on the global supply chains: a simulation-based analysis on the example of coronavirus (COVID-19/SARS-CoV-2) case. Transp Res E 136:101922. https://doi.org/10.1016/j.tre.2020.101922
Ivanov D (2021) Supply chain viability and the COVID-19 pandemic: a conceptual and formal generalisation of four major adaptation strategies. Int J Prod Res 59(12):3535–3552
Ivanov D (2022a) Lean resilience: AURA (active usage of resilience assets) framework for post-COVID-19 supply chain management. Int J Logist Manag 33(4):1196–1217
Ivanov D (2022b) Viable supply chain model: integrating agility, resilience and sustainability perspectives. Lessons from and thinking beyond the COVID-19 pandemic. Ann Oper Res 319:1411–1431
Ivanov D (2023) Intelligent digital twin (iDT) for supply chain stress-testing, resilience, and viability. Int J Prod Econ 263:108938
Ivanov D (2024a) Cash flow dynamics in the supply chain during and after disruptions. Transp Res E 185:103526
Ivanov D (2024b) Exiting the COVID-19 pandemic: after-shock risks and avoidance of disruption tails in supply chains. Ann Oper Res 335:1627–1644
Ivanov D (2024c) Two views of supply chain resilience. Int J Prod Res 62(11):4031–4045
Ivanov D (2024d) Supply chain resilience: conceptual and formal models drawing from immune system analogy. Omega 127:103081
Ivanov D (2025a) When is the supply chain resilient? Customer and operational perspectives. Int J Prod Res 1–16. https://doi.org/10.1080/00207543.2025.2454331
Ivanov D (2025b) Comparative analysis of product and network supply chain resilience. Int Trans Oper Res. https://doi.org/10.1111/itor.13612
Ivanov D (2025c) Literature review on the stress tests developed and applied in non-food supply chains. Publications Office of the European Union, Luxembourg. https://data.europa.eu/doi/10.2760/9174136, JRC139968
Ivanov D (2025d) Introduction to supply chain resilience and viability, 2nd edn. Springer Nature, Cham
Ivanov D, Dolgui A (2025) Tariff shocks, ripple effect, and deep uncertainty in supply chains: we are entering a turbulence zone, please fasten your seatbelts. Int J Prod Res, https://doi.org/10.1080/00207543.2025.2520598
Ivanov D (2025) Supply chain stress testing for tariff shocks and trade conflicts: methods, models, and reciprocal influence of operations and economics. IEEE Transactions on Engineering Management. https://doi.org/10.1109/TEM.2025.3599711
Ivanov D (2025) No risk, no fun? A bioinspired adaptation-based framework for supply chain resilience in Industry 5.0. Int J Prod Res. https://doi.org/10.1080/00207543.2025.2496962
Ivanov D, Das A (2020) Coronavirus (COVID-19/SARS-CoV-2) and supply chain resilience: a research note. Int J Integr Supply Manag 13(1):90–102
Ivanov D, Dolgui A (2019) Low-certainty-need (LCN) supply chains: a new perspective in managing disruption risks and resilience. Int J Prod Res 57(15–16):5119–5136
Ivanov D, Dolgui A (2020) Viability of intertwined supply networks: extending the supply chain resilience angles towards survivability. A position paper motivated by COVID-19 outbreak. Int J Prod Res 58(10):2904–2915
Ivanov D, Dolgui A (2021) OR-methods for coping with the ripple effect in supply chains during COVID-19 pandemic: managerial insights and research implications. Int J Prod Econ 232:107921

Ivanov D, Dolgui A (2022a) Stress testing supply chains and creating viable ecosystems. Oper Manag Res 15:475–486
Ivanov D, Dolgui A (2022b) The shortage economy and its implications for supply chain and operations management. Int J Prod Res 60(24):7141–7154
Ivanov D, Rozhkov M (2020) Coordination of production and ordering policies under capacity disruption and product write-off risk: an analytical study with real-data based simulations of a fast moving consumer goods company. Ann Oper Res 291(1–2):387–407
Ivanov D, Sokolov B (2010) Adaptive supply chain management. Springer, London
Ivanov D, Sokolov B (2013) Control and system-theoretic identification of the supply chain dynamics domain for planning, analysis, and adaptation of performance under uncertainty. Eur J Oper Res 224(2):313–323
Ivanov D, Sokolov B, Kaeschel J (2010) A multi-structural framework for adaptive supply chain planning and operations with structure dynamics considerations. Eur J Oper Res 200:409–420
Ivanov D, Sokolov B, Dolgui A (2014a) The ripple effect in supply chains: trade-off 'efficiency-flexibility-resilience' in disruption management. Int J Prod Res 52(7):2154–2172
Ivanov D, Sokolov B, Pavlov A (2014b) Optimal distribution (re)planning in a centralized multistage network under conditions of ripple effect and structure dynamics. Eur J Oper Res 237(2):758–770
Ivanov D, Hartl R, Dolgui A, Pavlov A, Sokolov B (2015) Integration of aggregate distribution and dynamic transportation planning in a supply chain with capacity disruption and the ripple effect consideration. Int J Prod Res 53(23):6963–6979
Ivanov D, Dolgui A, Sokolov B, Ivanova M (2017) Literature review on disruption recovery in the supply chain. Int J Prod Res 55(20):6158–6174
Ivanov, D, Dolgui A, Sokolov B (eds) (2025) Handbook of ripple effects in the supply chain. 2nd edn. Springer, New York, ISBN 978-3-031-85507-8
Ivanov D, Dolgui A, Blackhurst J, Choi TM (2023) Toward supply chain viability theory: from lessons learned through COVID-19 pandemic to viable ecosystems. Int J Prod Res 61(8):2402–2415
Jin S, He Y, Ivanov D, Chen W (2024) To act or wait: the impact of disruption duration estimates on manufacturer's response strategies. IISE Trans. https://doi.org/10.1080/24725854.2024.2417377
Jüttner U, Maklan S (2011) Supply chain resilience in the global financial crisis: an empirical study. Supply Chain Manag 16(4):246–259
Kamalahmadi M, Mellat-Parast M (2016) Developing a resilient supply chain through supplier flexibility and reliability assessment. Int J Prod Res 54(1):302–321
Kim Y, Chen YS, Linderman K (2015) Supply network disruption and resilience: a network structural perspective. J Oper Manag 33–34:43–59
Kinra A, Ivanov D, Das A, Dolgui A (2020) Ripple effect quantification by supply risk exposure assessment. Int J Prod Res 58(19):5559–5578
Kleindorfer PR, Saad GH (2005) Managing disruption risks in supply chains. Prod Oper Manag 14(1):53–68
Klibi W, Martel A, Guitouni A (2010) The design of robust value-creating supply chain networks: a critical review. Eur J Oper Res 203(2):283–293
Knight F (1921) Risk, uncertainty and profit: Hart, Schaffner and Marx prize essays 31. Houghton Mifflin, Boston
Kouvelis P, Dong L, Boyabatli O, Li R (2012) Handbook of integrated risk management in global supply chains. Wiley, Hoboken
Kravchenko K, Gruchmann T, Ivanova M, Ivanov D (2024) Responding to the ripple effect from systemic disruptions: empirical evidence from the semiconductor shortage during COVID19. Mod Supply Chain Res Appl 6(4):354–375
Li Y, Zobel CW (2020) Exploring supply chain network resilience in the presence of the ripple effect. Int J Prod Econ 228:107693
Li S, He Y, Huang H, Lin J, Ivanov D (2024) Supply chain hoarding and contingent sourcing strategies in anticipation of Price hikes and product shortages. IISE Trans 56(7):746–761

Liberatore F, Scaparra MP, Daskin MS (2012) Hedging against disruptions with ripple effects in location analysis. Omega 40:21–30
Lissak MR, Letiche H (2002) Complexity, emergence, resilience, and coherence: gaining perspective on organizations and their study. Emergency 4(3):72–94
Macdonald JR, Zobel CW, Melnyk SA, Griffis SE (2018) Supply chain risk and resilience: theory building through structured experiments and simulation. Int J Prod Res 56(12):4337–4355
Manager Magazine (2021) Warum die Chiphersteller die Autoindustrie schmoren lassen (in German). https://www.manager-magazin.de/unternehmen/autoindustrie/autoindustrie-warum-die-versorgung-mit-mikrochips-stockt-und-die-produktion-laehmt-a-970562ab-c6cf-4b23-9097-f089594eb2d5?utm_source=pocket-newtab-global-de-DE. Accessed 2 Jan 2025
March JG, Shapira Z (1987) Managerial perspectives on risk and risk taking. Manag Sci 33:1404–1418
Markowitz H (1952) Portfolio selection. J Financ VII(1):77–91
Melnyk SA, Closs DJ, Griffis SE, Zobel CW, Macdonald JR (2014) Understanding supply chain resilience. Supply Chain Manag Rev 18(1):34–41
Mesarovic MD, Takahara Y (1975) General systems theory: mathematical foundations. Academic Press, New York
Mizgier KJ (2017) Global sensitivity analysis and aggregation of risk in multi-product supply chain networks. Int J Prod Res 55(1):130–144
Mizgier KJ, Jüttner MP, Wagner SM (2013) Bottleneck identification in supply chain networks. Int J Prod Res 51(5):1477–1490
Moetz A, Stylos-Duesmann P, Otto B (2019) Schedule instability in automotive production networks: the development of a network-oriented resequencing method. In: Proceedings of the 9th IFAC conference on manufacturing modelling, management and control (MIM 2019), edited by Dmitry Ivanov, Alexandre Dolgui, Farouk Yalaoui IFAC-PapersOnLine 52 (13):2810–2815
Mosayebi M, Fathi M, Hedayati MK, Ivanov D (2024) Time-to-adapt (TTA). Int J Prod Econ 278:109432
Namdar J, Blackhurst J, Zhao K, Song S (2024) Cascading disruptions: impact of modularity and nexus supplier predictions. J Supply Chain Manag 60(3):18–38
Ojha R, Ghadge A, Tiwari MK, Bititci US (2018) Bayesian network modelling for supply chain risk propagation. Int J Prod Res 56(17):5795–5819
Osadchiy N, Gaur W, Seshadri S (2016) Systematic risk in supply chain networks. Manag Sci 62(6):1755–1777
Paul S, Rahman S (2018) A quantitative and simulation model for managing sudden supply delay with fuzzy demand and safety stock. Int J Prod Res 56(13):4377–4395
Pavlov A, Ivanov D, Dolgui A, Sokolov B (2018) Hybrid fuzzy-probabilistic approach to supply chain resilience assessment. IEEE Trans Eng Manag 65:303. https://doi.org/10.1109/TEM.2017.2773574
Pettit JT, Fiksel J, Croxton LK (2010) Ensuring supply chain resilience: development of a conceptual framework. J Bus Logist 31(1):1–21
Ponomarov S, Holcomb M (2009) Understanding the concept of supply chain resilience. Int J Logist Manag 20(1):124–143
Proselkov Y, Zhang J, Xu L, Hofmann E, Choi T, Rogers D, Brintrup A (2024) Financial ripple effect in complex adaptive supply networks: an agent-based model. Int J Prod Res 62(3):823–845
Quang HT, Hara Y (2017) Risks and performance in supply chain: the push effect. Int J Prod Res 56:1369. https://doi.org/10.1080/00207543.2017.1363429
Queiroz MM, Ivanov D, Dolgui A, Fosso WS (2022) Impacts of epidemic outbreaks on supply chains: mapping a research agenda amid the COVID-19 pandemic through a structured literature review. Ann Oper Res 319:1159–1196
Reuters (2023) Toyota edges past 9.1 mln vehicle output goal, warns chip shortage lingering. Reuters. https://www.reuters.com/business/autos-transportation/toyota-edges-past-91-mln-vehicle-output-goal-warns-chip-shortage-lingering-2023-04-27/. Accessed 6 Jan 2025

Rosenhead J, Elton M, Gupta SK (1972) Robustness and optimality as criteria for strategic decisions. Oper Res Q 23(4):413–431
Rozhkov M, Ivanov D, Blackhurst J, Nair A (2022) Adapting supply chain operations in anticipation of and during the COVID-19 pandemic. Omega 110:102635
Ruel S, El Baz J, Ivanov D, Das A (2024) Supply chain viability: conceptualization, measurement, and nomological validation. Ann Oper Res 335:1107–1136
Sawik T (2016) On the risk-averse optimization of service level in a supply chain under disruption risks. Int J Prod Res 54(1):98–113
Sawik T (2020) Supply chain disruption management, 2nd edn. Springer, New York
Sawik T (2022) Stochastic optimization of supply chain resilience under ripple effect: a covid-19 pandemic related study. Omega 109:102596
Sawik T (2023a) A stochastic optimization approach to maintain supply chain viability un-der the ripple effect. Int J Prod Res 61(8):2452–2469
Sawik T (2023b) Reshore or not reshore: a stochastic programming approach to supply chain optimization. Omega 118:102863
Sawik T, Sawik B (2024) Risk-averse decision-making to maintain supply chain viability under propagated disruptions. Int J Prod Res 62(8):2853–2867
Scheibe KP, Blackhurst J (2018) Supply chain disruption propagation: a systemic risk and normal accident theory perspective. Int J Prod Res 56(1–2):43–59
Schmitt AJ, Singh M (2012) A quantitative analysis of disruption risk in a multi-echelon supply chain. Int J Prod Econ 139(1):23–32
Schmitt TG, Kumar S, Stecke KE, Glover FW, Ehlen MA (2017) Mitigating disruptions in a multiechelon supply chain using adaptive ordering. Omega 68:185–198
Sheffi J (2005) The resilient enterprise. MIT Press, Cambridge, MA
Shih W (2022) Managing supply chain disruptions: lessons from the semiconductor shortage. Harv Bus Rev 15
Simangunsong E, Hendry LC, Stevenson M (2012) Supply-chain uncertainty: a review and theoretical foundation for future research. Int J Prod Res 50(16):4493–4523
Simchi-Levi D (2015) Find the weak link in your supply chain. Harv Bus Rev
Simchi-Levi D, Wei Y (2015) Worst-case analysis of process flexibility designs. Oper Res 63(1):166–185
Simchi-Levi D, Schmidt W, Wei Y (2014) From superstorms to factory fires: managing unpredictable supply chain disruptions. Harv Bus Rev 92(1–2):96
Simchi-Levi D, Schmidt W, Wei Y, Zhang PY, Combs K, Ge Y, Gusikhin O, Sander M, Zhang D (2015) Identifying risks and mitigating disruptions in the automotive supply chain. Interfaces 45(5):375–390
Snyder LV, Atan Z, Peng P, Rong Y, Schmitt AJ, Sinsoysal B (2016) OR/MS models for supply chain disruptions: a review. IIE Trans 48(2):89–109
Sodhi M, Tang CS (2012) Managing supply chain risk. Springer, New York
Sokolov B, Ivanov D, Dolgui A, Pavlov A (2016) Structural quantification of the ripple effect in the supply chain. Int J Prod Res 54(1):152–169
Tang O, Musa SN (2011) Identifying risk issues and research advancements in supply chain risk management. Int J Prod Econ 133:25–34
Tang C, Tomlin B (2008) The power of flexibility for mitigating supply chain risks. Int J Prod Econ 116:12–27
The Guardian (2021). Volkswagen and Toyota face production cuts due to chip shortage. https://www.theguardian.com/business/2021/aug/19/volkswagen-and-toyota-face-production-cuts-due-to-chip-shortage Accessed 6 Jan 2025
Tomlin B (2006) On the value of mitigation and contingency strategies for managing supply chain disruption risks. Manag Sci 52:639–657
Toyota (2021). Toyota Company Background. https://media.toyota.co.uk/wp-content/uploads/sites/5/pdf/210720M-Toyota-Company-Background.pdf Accessed 6 Jan 2025
Toyota (2021b). Toyota Production System. https://global.toyota/en/company/plant-tours/production-system/ Accessed 6 Jan 2025

Toyota (2023). Integrated Report 2023. Toyota Global Site. https://global.toyota/pages/global_toyota/ir/library/annual/2023_001_integrated_en.pdf Accessed 6 Jan 2025

Vicente JJ, Relvas S, Barbosa-Póvoa AP (2018) Effective bullwhip metrics for multi-echelon distribution systems under order batching policies with cyclic demand. Int J Prod Res 56(4):1593–1619

Wagner SM, Bode C (2008) An empirical examination of supply chain performance along several dimensions of risk. J Bus Logist 29(1):307–325

Waters D (2011) Supply chain risk management: vulnerability and resilience in logistics, 2nd edn. Kohan Page, London

Wieland A, Wallenburg CM (2013) The influence of relational competencies on supply chain resilience: a relational view. Int J Phys Distrib Logist Manag 43(4):300–320

Zhao K, Zuo Z, Blackhurst JV (2019) Modelling supply chain adaptation for disruptions: an empirically grounded complex adaptive systems approach. J Oper Manag 65(2):190–212

Digital Supply Chain, Smart Operations, Artificial Intelligence, and Industry 4.0

16

Learning Objectives for This Chapter

- What is Industry 4.0?
- What types of digital technologies can be applied to SCOM?
- How can digital technologies be applied to SCOM?
- How can digital technologies improve decision-making support?
- What is a digital twin in SCOM?
- What are the benefits and challenges of applying digital technologies to SCOM?
- How can artificial intelligence create benefits for SCOM?

16.1 Introductory Case Study: SupplyOn

In this case study, we take you on a journey into the sensor clouds and consider a new perspective on how they could transform working methods within aerospace supplier networks. We will begin with the most important requirement for integrating sensor clouds: Collaboration! Collaboration between all parties involved. Preferably via one central platform that connects all major players within an industry. It was this core idea that a number of aerospace companies envisioned and put

Supplementary Information The online version of this chapter (https://doi.org/10.1007/978-3-030-72331-6_16) contains supplementary material, which is available to authorized users.

Find additional case studies and video streams in the E-Supplement to this book on www.globalsupply-chain-management.de!

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-95859-5_16.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_16

into practice many years ago. In a joint initiative, a central Supply Chain Management hub was launched to effectively connect all relevant players through one single jointly used system.

This SCM hub is called AirSupply and was developed by SupplyOn. Today, more than 2500 customer/supplier relationships are powered by this platform, which has matured into the standard solution in the aerospace industry for supplier collaboration. AirSupply provides the foundation for all main SCM processes such as forecasts, orders, dispatch advices and invoices to be managed in a clearly structured and highly efficient manner.

As a part of this Industry 4.0 initiative, SupplyOn analyzed the existing demand and the incoming goods processes. The demand processes are already comprehensively covered within AirSupply (cf. Chaps. 3 and 6). The challenge came with the incoming goods process: how to efficiently identify supply chain disruptions, detect material damage early on and optimize the goods receipt process to ensure a stable production with the highest possible capacity utilization.

Now let us have a look at the resulting Industry 4.0 Initiative that led up to the solution and then the solution itself. In 2016, an Industry 4.0 working group was set up to collect process optimization suggestions from all departments. The most promising projects would then be piloted and developed within an implementation phase of 90 days, after which a decision would be made on which pilots would go into productive use.

Now let us go online and have a look at the piloted solutions. We started the piloting process by first identifying a specific sensitive part regularly subjected to temperature or humidity impacts during transport. This view shows where the delivery note is generated in AirSupply. All relevant fields are pre-filled with data obtained from the preceding purchase order process.

As AirSupply supports the possibility of "pairing" material to a container and a sensor, the worker can now assign, for example, five packets with ten parts each to a handling unit and then scan the connected sensor number. Now, the supplier only needs to enter the sensor number during the creation of the delivery note. All other processes remain unchanged for the supplier. This was one of the central objectives of the project to ensure the highest possible acceptance level on the supplier side.

From this point on, the parts can be monitored throughout the entirety of the transport cycle. The new SCPM (Supply Chain Performance Management) solution continually displays the position and the condition of the parts in near real-time. The data here is automatically derived from our AirSupply solution. In this particular project, the position, temperature, and humidity of the parts were tracked. The temperature monitoring was important for a temperature-sensitive adhesive and the moisture monitoring was used for metal parts. Tracking the position of the parts was important for two reasons: to monitor the lead time and to allow an automatic "unpairing" of the sensor from the goods upon goods receipt.

The position of the parts or the containers is displayed as well as the distance which has already been covered. The route still to be driven is displayed. The position information in conjunction with the condition information allows a quick analysis on which routes it could become problematic.

SCPM also checks whether the order and the transport are within the agreed lead time. Software evaluates the number of days remaining until the order arrives at the plant and whether or not the lead time will be exceeded. The material planer, therefore, has an exact overview of the current status of an order at the top of the screen.

All the data, such as position and condition, are stored and can be used for reporting and analytics purposes. Different drill-downs are implemented—in this case, the analytics period can also be adjusted. The analysis of the data in this chart allows for the quick identification of all orders where the lead time was exceeded, allowing for the quick identification of transport bottlenecks.

Acknowledgement The introductory case study is based on the materials of SupplyOn and is written with the permission of SupplyOn. We thank Ms. Cornelia Staib and Mr. Arvid Holzwarth (MBA) for their help in preparing this case study.

16.2 Development of Technology in SCOM

16.2.1 Three Industrial Revolutions and Industry 4.0 Formation

The first industrial revolution was the mechanization of the production processes. Steam and water power was used to increase efficiency during the creation of goods. After this, the principles of work separation and the electrification of the production floors took place: this is the second revolution. The combination of the separation of work packages and electrical power helped the industry to produce in larger quantities and use moving production lines. One well-known result of that era was the production of cars, such as the Ford T-Model, on an assembly line. The flow principle and product creation in well-planned work packages led to the final move from craft to mass production.

The era starting with the 1970s was characterized by the increasing usage of computer technology during the value creation processes. This information orientation of production represented the third industrial revolution. Electronics, information technology, and the ambition for automation were dominating objectives that led to, e.g., computer-integrated manufacturing (CIM). During these three industrial revolutions, significant improvements in terms of productivity and efficiency were made. In Fig. 16.1, we summarize the chronological developments toward Industry 4.0 and its formation.

The 3D (three-dimensional) framework comprises management, physical and technological systems, each representing a dimension (Ivanov et al. 2021a). In the 1980s–1990s, transformation from mass production to an increasingly volatile variety and volume market facilitated the development of small lot manufacturing. Flexible manufacturing systems (FMS) and reconfigurable manufacturing systems (RMS) (Stecke 1983; Slack 1987; Bordoloi et al. 1999; Koren et al. 1999) have been established. Production and information technology advances led to the establishment of computer-integrated manufacturing (CIM) followed by enterprise resource planning (ERP) and modular and fractal factories (Warnecke and Braun 1999;

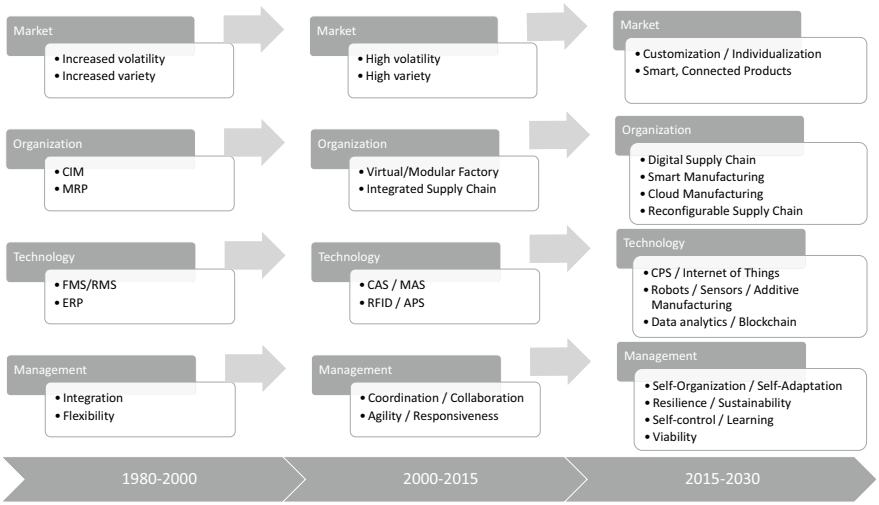


Fig. 16.1 A 3D framework of Industry 4.0 formation (based on Ivanov et al. 2021a)

Wiendahl et al. 2015). Flexibility and integration became the key management and organizational principles of these times (Jordan and Graves 1995; Das et al. 2006).

In the second decade of the twenty-first century, collaborative robots (cobots), sensors, agents, modular factories, and IoT have received broader attention but their relative utility and usage contexts and requirements were not clear (Zühlke 2009). This became possible later, following the rapid progress in data processing and robotics technologies (Choi et al. 2018; Tang and Veelenturf 2019). Digital SC, smart manufacturing, and cloud manufacturing are becoming important outcomes of Industry 4.0 (Ivanov et al. 2021a). Sensors, automated guided vehicles (AGV), blockchain, additive manufacturing, augmented reality, big data analytics, track&trace systems (T&T), and mobile robots are facilitating the formation of cyber-physical systems (CPS) in manufacturing and SCs (Waller and Fawcett 2013; Wamba et al. 2017; Ivanov 2018; Ivanov et al. 2019; Moghaddam and Nof 2018; Dubey et al. 2019a, b, 2020; Fragapane et al. 2022; Panetto et al. 2019; Ivanov and Dolgui 2021).

SC business models started evolving whereas value-adding networks experienced an increasing transformation from rigid physical systems with fixed and static allocation of specific processes to specific firms toward dynamically adaptable and reconfigurable networks, increasingly data-driven in nature. Examples of digitalized SC and operations include logistics and supply chain control with real-time data (Park et al. 2018), dynamic resource allocation in I4.0 customized assembly systems (Ivanov et al. 2016), improving forecasting models using Big Data, combining optimization, machine learning algorithms and agent-based modeling for supply chain resilience (Cavalcantea et al. 2019; Ivanov 2018; Zhao et al. 2019). Industry 4.0 began transforming SCOM (Ivanov et al. 2019; Olsen and Tomlin 2020;

Fragapane et al. 2022; Dolgui et al. 2020a, b; Sokolov et al. 2020; Ivanov and Dolgui 2021).

16.2.2 Fourth Industrial Revolution: Industry 4.0

Improvements did not end with the third revolution; today we are in the midst of the fourth industrial revolution, which is based on digitalization. In other words, it is the decade of Industry 4.0. Industry 4.0 has become a popular term for characterizing the ongoing transformations of manufacturing and supply chain networks. However, its definition depends on the context.

We define Industry 4.0 from the SCOM perspective as follows (Ivanov et al. 2021a):

> Industry 4.0 is an integrity of technologies, organizational concepts and management principles underlying a cost-efficient, responsive, resilient and sustainable network, data-driven and dynamically and structurally adaptable to changes in the demand and supply environment through rapid rearrangement and reallocation of its components and capabilities.

Our definition of Industry 4.0 encompasses technology and management. The main idea of Industry 4.0 is that companies can intensively utilize and connect machines, materials, tools, warehouses, and transportation technology in an intelligent way. This implies the use of cyber-physical approach. Materials, machines, or devices can exchange information mutually, set priorities, trigger action, and help to schedule and sequence the work in process. Value-adding processes will improve themselves, which means continuous improvement would no longer be precisely connected to the human being.

Most new factory concepts share attributes of smart networking (Strozzi et al. 2017). As such, Industry 4.0 represents a smart manufacturing networking concept where machines and products could interact with each other without human control. SCs in such networks have dynamic structures which evolve over time. Industry 4.0 enables companies to produce customized products in small quantities. At the same time, a large amount of variants and throughput can be increased; capacities can be exploited to a higher degree, and stock levels, transports, and emissions can be reduced in parallel. Because it is in complete unison with the classical approach of Lean Thinking, it has also been referred to as "Leandustry 4.0" (Tsipoulanidis 2015, 2017a, b). Industry 4.0 also enables the management of SC risks and resilience (Ivanov 2018, 2024b; Ivanov et al. 2019; Ivanov and Dolgui 2021). These two fundamental aspects of Lean Thinking and risk mitigation offer new opportunities for achieving SCOM Excellence at the times of digitalization (Ivanov 2024c; Rai et al. 2021).

To better explain how digitalization, Industry 4.0, and SCOM fit together, a short hypothetical case is presented. Imagine that you are loading a truck with fresh strawberries. To ensure product quality, the cold chain needs to be permanently monitored in the truck with the cooler (to meet customer expectations). We also

know via GPS where the truck is at each point in time. With Machine-to-Machine (M2M) technology, all collected data can be transmitted in real time, because the M2M technology ensures connectivity and data transmission (it is comparable to a mobile phone with a sim card).

Now imagine that the M2M device in the truck is communicating with a central monitoring center and submits the data that is measured during the journey to a computer in this monitoring facility on a regular basis.

The result is that the machine (truck) is communicating with another machine (computer in the monitoring center). Following that process, the cold chain can be permanently documented (this prevents the shipment of defective products). In other words, we can track and trace the shipped goods on their way. Furthermore, if the truck has a technical problem and due to that the cooling device does not ensure the proper temperature, the driver will be notified (avoidance of insufficient processing).

From a logistical perspective, it does not make sense to continue the journey with perished goods. Based on the GPS data, the next appropriate waste disposal station should be identified and the driver should be instructed to go there. This helps to reduce unnecessary transportation efforts.

Also, the truck will be guided to the destination using the shortest routes: it will be redirected in case there are traffic congestions. Furthermore, the next repair facility can be contacted for the maintenance of the defective truck, where spare parts can be preordered, and the capacity for the repair can be scheduled to ensure the shortest processing time and, thus, the highest usability and availability of the truck. This reduces SC risk, but it also means that over-processing and waiting time are reduced.

In our case, we were assuming the truck was bringing strawberries from the field to a yogurt factory. They will need to be informed early in advance that the truck had a technical defect resulting in rotten fruit: the expected delivery of the fresh strawberries cannot take place as initially planned. The shipment of obsolete inventory is avoided.

Although this is a very unfavorable situation, the producing factory might be able to order strawberries from an alternative supplier or change the production volume to produce an alternative yogurt, maybe with nuts or vanilla flavor. The competitive advantage is generated by the early use of available information that allows higher flexibility over the entire SC and value creation network and better use of existing capacities. Proactive production rescheduling also protects the SC against its vulnerability; even though SC risk cannot be fully eliminated, it can be mitigated.

A lot of the aspects discussed in the short case above are related to the availability of data generated, stored, and analyzed during the various processes. Summarizing, the implementation of Industry 4.0 in SCOM is based on four pillars, i.e., infrastructure, data, engineering, and communication technologies (Fig. 16.2).

In the next sections, we will consider different elements of the technologies shown in Fig. 16.2 in more detail.

Industry 4.0 Technologies			
Infrastructure Technology	**Engineering Technology**	**Data Technology**	**Communication Technology**
a) Internet-of-Things (IoT) b) Cyber-Physical System (CPS)	a) Additive Manufacturing b) Automated Guided Vehicle (AGV) and Drones c) Collaborative Robots (Cobots) d) Augmented Reality	a) Big Data Analytics (BDA) b) Artificial Intelligence (AI) c) Track and Trace (T&T) systems	a) Machine-to-Machine (M2M) b) Cloud Services c) Smart Products d) Blockchain

Fig. 16.2 Industry 4.0 technologies

16.2.3 Internet of Things

Internet of Things (IoT) is a network of interconnected devices, sensors, and appliances, which links physical objects (e.g., machines) to virtual reality. IoT enables the implementation of smart factories and adjustable production and logistics networks enabling smart maintenance or smart logistics.

In IoT, production is fully integrated from planning to execution on the shop-floor and production will react flexibly when changes in the sequence or process become necessary. Production systems are more flexible and more intelligent as a result of new sensor technology, intelligent robotics, and data analysis, so that also lot size 1 and customized products can be created economically. For example, the maintenance of production systems will be optimized through predictive analytics and remote monitoring via mobile technology, augmented reality, and 3D printing of spare parts. With this development, production will move into the cloud and become an on-demand-cloud service.

Data flowing to and from products will allow product use and activities across the value chain to be streamlined in countless new ways. Waste will be cut or eliminated. In other words: "Due to the introduction of Industry 4.0, a whole new level of 'Lean' can be reached because the information and data that has become available now, is necessary to avoid wastes which couldn't be avoided beforehand because of the enhanced information technology, data availability and grown capabilities to analyse generated data" (Tsipoulanidis 2015). A transition from Lean to Leandustry 4.0 is shown in Fig. 16.3.

In sum, "Industry 4.0 will involve the technical integration of Cyber Physical Systems into manufacturing and logistics and the use of the Internet of Things and services in industrial processes. This will have implications for value creation, business models, downstream services and work organisation" (BMBF 2013, p. 14). The principles around the Cyber-Physical Systems are explained subsequently.

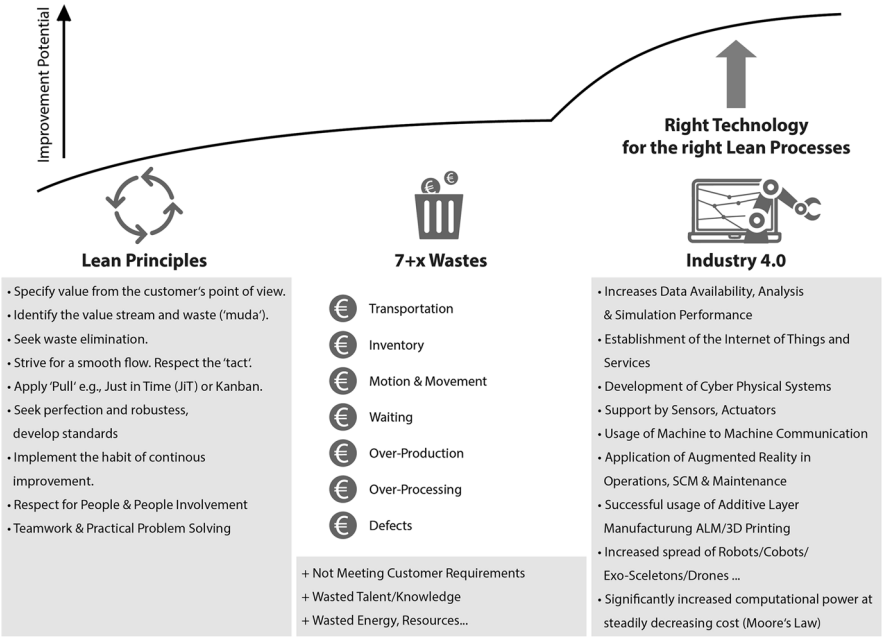


Fig. 16.3 From Lean to Leandustry 4.0 (Tsipoulanidis 2017b)

16.2.4 Cyber-Physical Systems

There are multiple definitions regarding the expression "Cyber-Physical Systems." According to Edward A. Lee, referenced by the VDI (2013), "Cyber-Physical Systems (CPS) are integrations of computation with physical processes. Embedded computers and networks monitor and control the physical processes, usually with feedback loops where physical processes affect computations and vice versa." The fundamental difference between the classical automation processes and CPS is that today an open infrastructure—the Internet—is used for communication and data exchange (VDI 2013, p. 2).

Furthermore, the real and the virtual world are now connected and this is done by converting passive components into active, intelligent systems (Panetto et al. 2019). Due to the interaction of previously passive systems with, e.g., sensors or intelligent communication devices, the components are enabled to transmit data and information from the real into the virtual world, i.e., to other machines, humans for further data analysis.

16.2.5 Smart, Connected Products

To understand the power of digital transformation, the role of smart, connected products has to be clarified. According to Porter and Heppelmann (2015, p. 4), so-called smart, connected products are a combination of three core elements. These

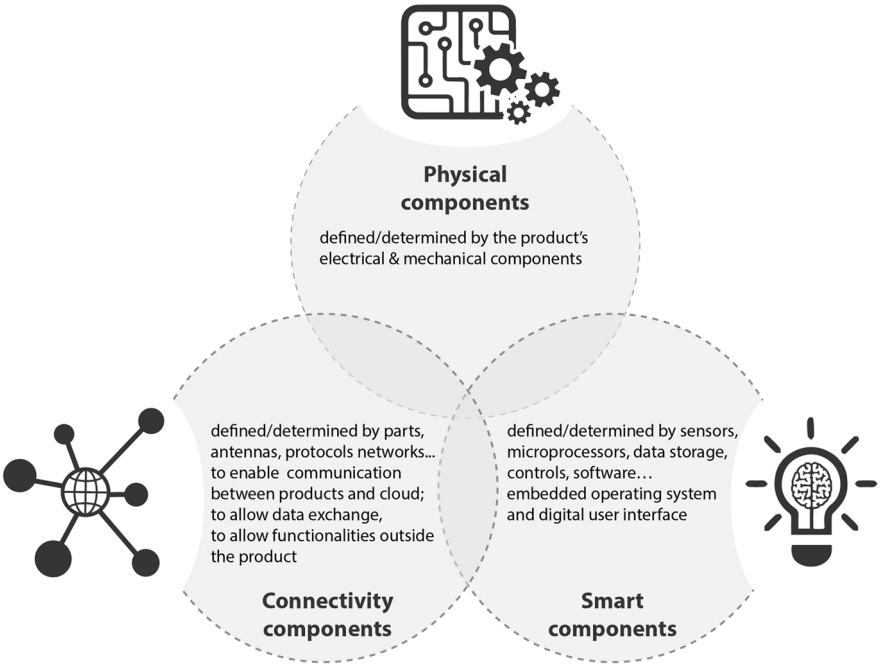


Fig. 16.4 Concept of smart, connected products (based on Porter and Heppelmann 2015)

are physical components, "smart" components, and connectivity components, as shown in the subsequent illustration. Now what does this mean? "Smart components amplify the capabilities and value of the physical components, while connectivity amplifies the capabilities and value of the smart components and enables some of them to exist outside the physical product itself. The result is a virtuous cycle of value improvement" (Porter and Heppelmann 2014, p. 5) (Fig. 16.4).

The physical components are represented by the mechanical and electrical parts or modules, such as the housing or the cooling device of a refrigerator. In addition, a smart component needs to be added to make the physical product intelligent, i.e., "smart." Examples are sensors, control systems, actuators, microprocessors, data storage, software, the embedded operating system, and the user interface (see Porter and Heppelmann 2014). In the example of the refrigerator, a sensor is used to measure temperature to avoid icing or even to scan the items which are stored in the fridge.

To enable the refrigerator to exchange the gathered information with its environment, a connection to the outside world needs to be established. This means the smart product requires a connectivity component. Connectivity takes three forms:

- One-to-one: An individual product connects to the user, the manufacturer, or another product through a port or other interface—for example, when a car is hooked up to a diagnostic machine.

- One-to-many: A central system is continuously or intermittently connected to many products simultaneously—for example, many Tesla automobiles are connected to a single manufacturer system that monitors performance and accomplishes remote service and upgrades.
- Many-to-many: Multiple products connect to many other types of products and often to external data sources—for example, an array of types of farm equipment are connected to one another and geolocation data to coordinate and optimize the farm system. Automated tillers inject nitrogen fertilizer at precise depths and intervals, and seeders follow, placing corn seeds directly in the fertilized soil.

Connectivity serves a dual purpose. First, it allows information to be exchanged between the product and its operating environment, maker, users, and other products and systems. Second, connectivity enables some functions of the product to exist outside the physical device in what is known as the product cloud.

16.2.6 Smart Supply Chains and Smart Value-Adding Networks

A smart product is equipped with sensors, actuators, etc., to exchange data with the environment, and therefore, the third criterion—connectivity—is necessary. How does a smart product influence SCs to become smart value-adding networks? Think about smart products such as a car, an elevator, a train, or an aircraft and their connection to the SC. During their lifetime, these smart products generate data. Actually, a lot of data is generated, which has led to the so-called Big Data.

And this is the difference—smart products generate also data regarding the user behavior, usage conditions, and the use of profiles. This helps us understand why products might fail due to certain usage conditions. This new knowledge supports the improvement of existing products, improved maintenance, and the prediction of failures prior to their potential occurrence.

It also significantly helps to improve product development for the next generation of products. The data generated by the smart product in use also helps in terms of the monitoring of the product during its use (for example, during the operation of a train, the condition of its operating devices is monitored and spare parts can be ordered and the maintenance work scheduled early before something fails). This is important for the preventive maintenance of products (i.e., planned and unplanned repairs, proactive recognition of wearing of components, defects of parts). Unplanned downtimes are avoided or reduced.

Because of monitoring, the right amount of spare parts can be kept, necessary resources for replacement can be scheduled, and the availability of the product increased. This leads to the reduction of unnecessary inventory, resulting in less transportation, and avoidance of overproduction. Thanks to the amount of data generated during the product's lifetime, indications about good or poor supplier quality might also be possible, through the prevention of defective batches. This means that sourcing processes, the process of supplier evaluation and supplier development, and thus the SC are significantly affected.

16.3 Digital SCOM Framework

The technology analysis in Sect. 16.2 allows us to draw the digital SCOM framework as in Fig. 16.5.

BDA, AM, Industry 4.0, and advanced tracking and tracing technologies can be considered in regard to four major SC processes, i.e., plan, source, make, and deliver. Therefore, the areas of digital planning, manufacturing, sourcing, and logistics can be classified. Moreover, IoT, cloud technology, robots, and sensors facilitate the technical implementations of these technologies (MacCarthy and Ivanov 2022a, b).

This classification, in line with the classic SCOR model (cf. Chap. 4), will be further used in Sects. 16.4–16.7 where relevant approaches concerning the usage of digital technology in SCOM will be summarized. However, it is unlikely that the presented approaches and technologies will only be applicable for one of the major processes according to the SCOR Model. The aspect of BDA might be used for sourcing, making, delivering, or returning processes, while the AM might, for example, be used for sourcing or for making processes. Robots could be used in manufacturing, logistical, or retail processes. In essence, multiple applications and, thus, a very broad utilization and process coverage are expected, as the following sections summarize.

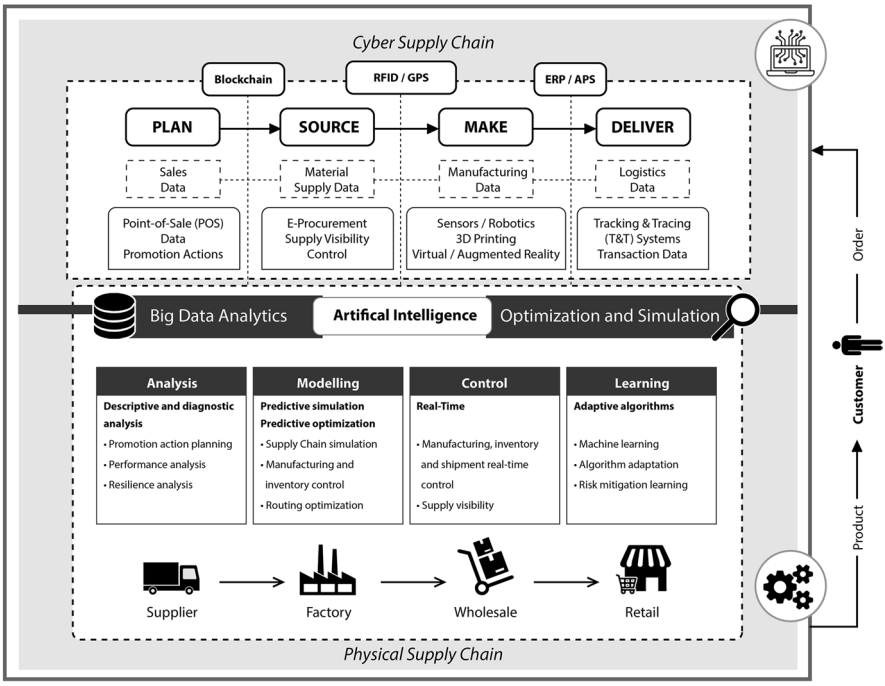


Fig. 16.5 Digital SCOM framework

16.4 Digital Technology in the "Plan" Processes

In this section, a few examples are given of how planning processes can be supported by digital technologies. First, we focus on BDA. Second, the creation of a so-called digital twin in the early beginning of the product life cycle which aids in planning the SC and its operations is given as an example.

16.4.1 Big Data Analytics

Generally, "Big Data" is a widespread expression, which is used to summarize the large amount of unstructured or semi-structured data, which are produced day by day by companies, their devices, machines, or products in use. It requires much effort to structure these data and a lot of work and expertise to analyze the data once it has been uploaded into a relational database. The characteristic aspect of "big" in BDA is related to the sheer amount of data, which is generated, adjusted, or modified day by day.

Big data has been characterized in literature by 5Vs: volume, variety, velocity, veracity, and value (Wamba et al. 2015, 2017). Veracity and value are particularly important since data analysis shows the real value of big data. BDA is based on knowledge extraction from vast amounts of data, facilitating data-driven decision-making.

▶ **Practical Insights** Pharmapacks ships the merchandise that we expect to find in a drug store. Pharmapacks used a pricing software called "Master Mind" to increase their market share and revenue. It updates prices every 45 min and manages inventory and sales/demand forecasts. In another example, to produce cylinder heads at its plant in Untertürkheim, Germany, Mercedes-Benz uses predictive analytics to examine more than 600 parameters that influence quality (Küpper et al. 2016).

BDA has been undoubtedly the most elaborated area of digital technologies applied to SCM over the last decade. Johnson et al. (2016) and Simchi-Levi and Wu (2018) analyzed the application of BDA in retail. Retailers must continuously strive to grow their revenue, margins, and market share. One method for doing this is price optimization models, which calculate the variance of demand as price levels rise or fall, and then combine this information with the relevant cost and inventory data in order to recommend prices that could maximize revenue and profits. Nguyen et al. (2018) showed that optimization is the most popular approach in the prescriptive analytics application to logistics and transportation.

BDA applications to SCM can also be seen in procurement processes, manufacturing shop floors, promotion actions in the omnichannel model, routing optimization, real-time traffic operation monitoring, and proactive safety management. Summarizing, the ability and capacity to analyze the relevant data for a company is a fundamental element for achieving SCOM excellence.

16.4.2 The Digital Twin

In short, a "*digital twin*" is a computerized model that represents the network state for any given moment in time (Ivanov and Dolgui 2021; Ivanov 2024a). A digital twin in SCOM is "a virtual system comprised of (i) a digital visualization of a physical supply chain and its elements (e.g., firms, flows, and products) in a computer model, (ii) digital technologies providing data about the physical object (e.g., sensors, blockchain, clouds), and (iii) descriptive, predictive and prescriptive analytics for decision-making support" (Ivanov 2024a). Figure 16.6 illustrates the digital twin composition.

A digital twin is linked to a real object. Through continuous data exchange, the digital twin always behaves synchronously with its real counterpart, and they influence each other. It mirrors both the state and the performance of the real system and can also send control signals back to the real system. Digital twins can be used for descriptive (e.g., performance visualization), predictive (e.g., demand forecasting), and prescriptive (e.g., supply chain recovery simulation and optimization) decision-making support. Figure 16.7 summarizes the role of digital twins in decision-making in SCOM.

In essence, digital twins are integrators of decision-making support. The principle schema of integrated data-driven and model-based decision-making support in digital twins is composed of the following elements:

- A physical *object* (i.e., an SC or operational process)
- A *subject* (i.e., a decision-maker)
- A *model* (i.e., the replication of the physical object behavior in a controlled digital environment and testing alternatives for the improvement of performance through optimization in the presence of uncertainty and risks
- A digital *object* (i.e., the digital supply chain ecosystem with a regularly updated, sensed footprint of a real object encapsulating data, information, and knowledge about a physical object and its dynamics).

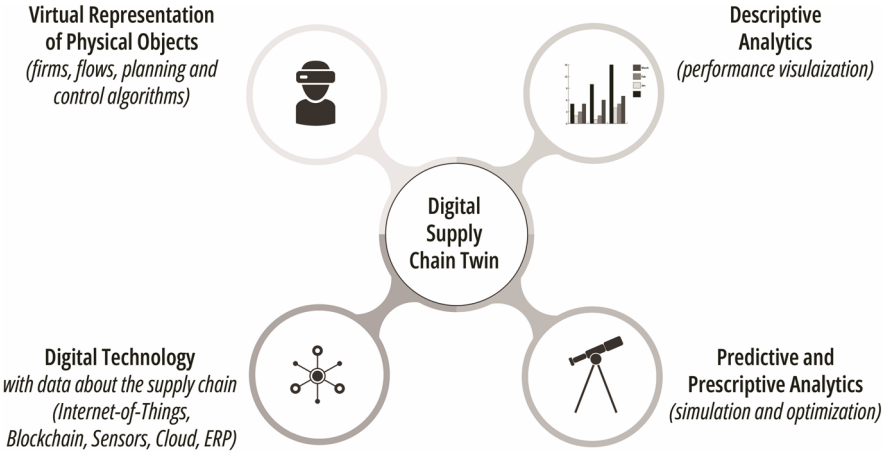


Fig. 16.6 Concept and design of a digital supply chain twin (Ivanov 2024a)

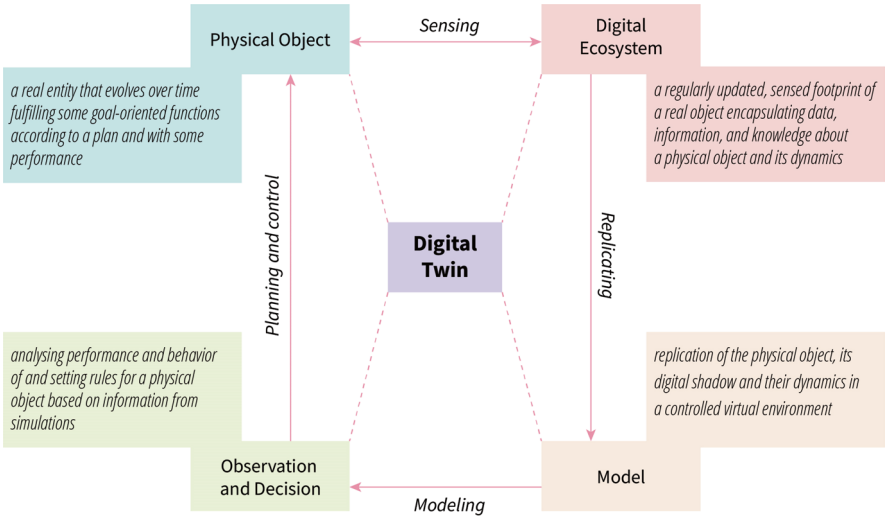


Fig. 16.7 Role of digital twins in decision-making in SCOM

A *subject* (i.e., a decision-maker) observes the *object* (i.e., an SC or operational process) through data and information. Based on data and information combined with expert knowledge, SC *models* are developed. Their objective is to represent the physical process, its dynamics, and decision-making rules. Models are needed to forecast, optimize, and simulate SC and operations dynamics and performance impacts, e.g., for different market development scenarios, disruptions, and changes in production-ordering control policies. However, our data and knowledge about the real object are hardly complete, so the models will be. That is why *environment* is a category that should be considered in order to increase the model's completeness, accuracy, and validity. Environment represents a space of uncertainty, which can be known (e.g., through some probability distribution functions), partially known, or unknown. In the course of object evolution, the subject learns more about the object and its environment, reducing uncertainty and improving model quality. The interactions between the four elements in Fig. 16.7 are driven by sensing, replicating, modeling, and control. Ecosystems are comprised of a set of digital technologies (e.g., Internet-of-Things, Industry 4.0, Blockchain), AI-based knowledge management systems, and data spaces and platforms (e.g., collaborative supplier portals). They are sensing data from the physical object. Building of models and digital twins to enable decision-making support for planning and control is based on physical object and ecosystem observability.

Ivanov and Dolgui (2021) demonstrated how to design and implement digital twins when managing disruption risks in SCs. The combination of model-based and data-driven approaches allows uncovering the interrelations of risk data, disruption modeling, and performance assessment. The SC shocks and adaptations amid the COVID-19 pandemic along with post-pandemic recoveries provide indisputable evidences for the urgent needs of digital twins for mapping supply networks and ensuring visibility (Ivanov 2020, 2022). Figure 16.7 demonstrates the principles of SC disruption risk management by digital twins.

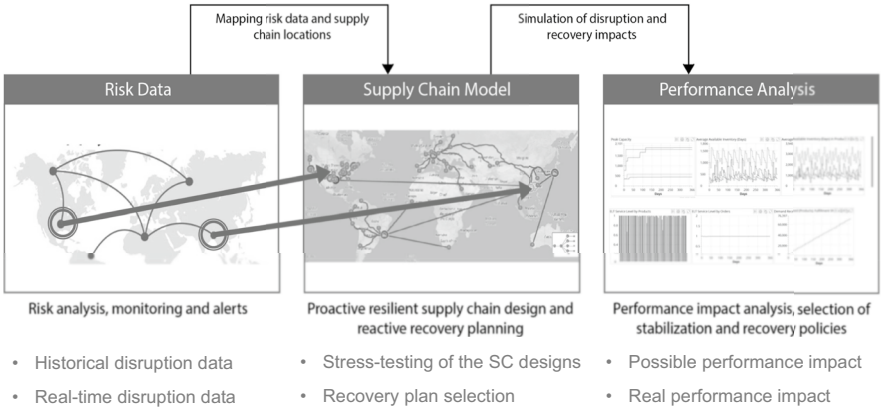


Fig. 16.8 Interrelations between risk data, modeling, and performance analysis (images used are from anyLogistix™ software) (Ivanov and Dolgui 2021)

Figure 16.8 shows the mapping of risk data identification, the locational supply chain model, and performance impact analysis. At the analysis stage, historical risk data from external databases (e.g., natural disaster events in the past and geographical regional risk assessments) and internal sources (e.g., ERP data about supplier reliability performance) are collected. This data can be used to build disruption scenarios for supply chain resilience analysis, which can then be examined and tested in simulation-optimization models. In the real-time mode, the resilience analytics system is used to search for relevant disruption data that might affect supply chain resilience. The simulation can be subsequently run to observe the impact of such a disruption on supply chain performance. Moreover, some recovery policies, such as activation of alternative supply chain designs during the disruption are simulated. Data collected from different sources in real-time are used to update model parameters, such as capacity, inventory, and lead times in terms of production and shipment capacities and inventory availability in the supply chain. Finally, data analytics can be used as data-driven *learning* system to use past experiences to manage future disruptions and so utilizing cyber-physical, AI, and machine learning principles and technologies.

► **Practical Insights** Successful companies are utilizing the advantages of digital technologies to strengthen SC resilience and to benefit from improved efficiencies. For example, Procter & Gamble has not only implemented resilient network design and risk inventory principles but also created a cloud-based platform to provide real-time information on production and external demand to its SC control tower. This strategy helped minimize the fallout when Hurricane Sandy devastated New Jersey in 2012, disrupting production at a factory producing 91% of its perfumes. Insights generated by advanced analytics helped the company make the right decisions quickly, limiting the affected plant's downtime to 2½ days. Based on this experience, P&G used digital tools to prepare

for Hurricane Irma's landfall in Florida in 2017 (Banker 2019). In particular, digital technology helped to predict and identify which suppliers, plants, and distribution centers were most likely to be impacted by the disruption. This foresight allowed the company to prepare contingency plans and relocate its inventory. As a result, the company avoided the disruption and financial loss that other companies suffered. In the future, companies can be similarly well prepared for supply chain disruption by utilizing machine learning and AI for predictive and prescriptive analytics. This can be achieved with the help of technologies such as IoT, cloud computing, 5G, AI, additive manufacturing, and robotics.

16.5 Digital Technology in "Source" Processes

Once the processes have been planned, sourcing activities need to be initiated. Of course, electronic data interchange (EDI) has been a well-known approach for managing the connection and information flow between involved companies for decades. Other means include WebEDI, which connects partners within their supplier–customer relationships, or the usage of Supplier Collaboration Portals (SCPs) for the consideration of online shops for procurement activities.

16.5.1 eProcurement

E-catalogues, e-invoicing, or e-auctions can facilitate sourcing activities. Not new, but still valid is the fact that eProcurement (see also Sect. 3.1.1) is a core process within eSCM since it uses Internet Technology (IT) to facilitate purchasing processes. eProcurement supports companies in improving transparency concerning the supplier base, negotiations, the alignment within the company, or supporting cost-intensive and long-lasting sourcing tasks through a better information exchange. Furthermore, information technology helps to obtain more accurate, fast, and up-to-date information regarding customer needs and business processes. It also provides an overview of market changes and supports better planning, coordination, and controlling. Importantly, information technology better connects companies, suppliers, and customers and, at the same time, provides early warning indications. To sum up, eProcurement helps companies to reduce costs, increase sourcing security, reduce sourcing throughput times, increase flexibility and quality, and improve the relationship between the involved parties.

In addition to the aforementioned factors, Chopra and Meindl (2013, p. 507) saw the following aspects as important trends that will impact IT in SCM:

- The growth in software as a service (SaaS)
- Increased availability of real-time data
- Increased use of mobile technology

16.5.2 Supplier Collaboration Portals

In Sect. 3.1, we introduced eProcurement, Vendor Managed Inventory (VMI), EDI, and Supplier Collaboration Portals (SCPs) to facilitate SC processes with a special focus on VMI. It is the purpose of this section to refer to SCPs as an approach to manage the inbound and outbound flow of materials efficiently, to manage performance, and thus to describe how SCPs can be embedded in a digital SCOM strategy striving for excellence.

SCPs are successful solutions for efficiently connecting suppliers, service providers, and customers via a platform that supports transparent supplier processes. Generally, an SCP supports the management processes via a steering cockpit. This serves the forecasting, *planning,* and monitoring of the value-adding activities and provides transparency and visibility of SCM activities. Furthermore, the project management processes during product development [advanced product quality planning (APQP)] can be managed to secure the smooth start of the project. The fundamental *sourcing* processes [e.g., quotation requests (RFQ)] are supported by the platform to provide the *production* (status of goods received, on-time availability of materials) with the necessary materials. *Delivery* [dispatch and advanced shipment notification (ASN)] of the items is handled via the SCP. For SC managers, the avoidance of SC disruptions, the availability of undamaged components, and the fast usability of the materials from goods receipt are enormously important. SCPs are powerful tools for these critical processes. Even more, the location of parts in certain containers or their status regarding humidity or temperature can also be tracked when the SCP processes the transmitted sensor data.

In parallel, continuous improvement processes can be run, clarity on payment processes provided, and importantly the *return* process, in the sense of managing complaints, can be part of the SCP solution. Often claim management is connected to a practical problem-solving tool that enables companies to handle complaints or defects using the "8D Report" (eight disciplines report). The 8D Report consists of form sheets to document the problem-solving process according to a standard routine of eight steps or disciplines (Schmitt and Pfeifer 2010). To manage SC performance, the related key performance indicators (KPIs) are tracked and allow the SC manager to identify deviations or problems and take corrective actions. The SCP can be seen as a repository of data and information, so that processes can also be analyzed retrospectively. Overall, the SCP helps to connect the involved parties in SCM in real-time and to improve the transparent flow of information and SC visibility and enable collaboration using "what-if" scenarios for better decision-making. The portals also support flexibility and proactivity, leading to improved SCM performance.

As in the example of SupplyOn (see Sect. 16.1) documented, SCPs play a fundamental role in supporting companies to deploy excellently managed digital sourcing processes. In other words, SCPs can be considered a core attribute of digital SCs.

16.5.3 Digital Trends for Excellence in Sourcing

New sourcing technologies to achieve excellence in procurement are considered to have a huge future potential, as digitalization is very likely the future success factor for the efficient sourcing of tasks. In particular, the following digital technologies are considered in sourcing (Decken 2018):

- *Augmented Reality* (e.g., to be used for the visit of suppliers)
- *Blockchain Technology* enables collaborative document management, payment processes, digital signatures, and documentation.
- *Data and Text Mining* helps to analyze texts and data using algorithms.
- *Digital Data Management* recognizes mistakes and errors, sorts, and excludes inactive datasets.
- *Enhanced Procurement Platforms* can be used for the sourcing processes of personnel or technical services.
- *Identification of Sourcing Synergies* using software or AI
- *AI* for the imitation of human interfaces within the sourcing processes (e.g., for the reading, analysis, and evaluation of PDFs or Excel-Tables)
- *SCM Security Rating* as assessed and evaluated by AI
- *Smart-Contracts* to support automated contract adjustments in case of changing parameters

16.5.4 Blockchain

Initially, the Blockchain concept was developed to secure the financial transactions of the cryptocurrency Bitcoin. Recently, the applications of Blockchain technology have begun to revolutionize different aspects of SCOM. The central idea is to increase visibility and efficiency based on record-keeping in the SC. Blockchain applications to SCOM become more and more important to enhance the scale and scope of digital processes along with the creation of information pipeline systems and SC finance applications (Hofmann et al. 2018).

A Blockchain is a decentralized database that exists as copies in a network of computers. It is a chain of blocks, because the data and information stored are captured in blocks. It is also a platform to execute the so-called smart contracts as transactions. The list of all transactions is stored as copies throughout all further evolvements on numerous computers (a network of even hundreds of computers). As a result, if an illegal or unauthorized modification takes place on one of the computers, it can be traced back for verification purposes. The copies are checked regarding their consistency from the perspective of the value (e.g., if the verification value is the same on all but one computer, the manipulation is detected). Thus, the Blockchain is a technology that cannot be manipulated. Participating organizations benefit from high visibility through a secure data sharing mechanism, enabling them to plan with greater certainty. Blockchains store the data about the location of

assets at any point in time, the ownership or the custodianship of assets, and their transactional status.

The Blockchain, a relatively simple technology, is expected to be used extensively in sourcing processes, e.g., for contract management, archiving of customs documents, certificates at deliveries, or for patent management. Contracts in SCs often involve multiparty agreements, with regulatory and logistical constraints. Further complexities may arise from operations in different jurisdictions, as well as dynamic features embedded in the contracts. The flow of information in an SC plays a critical role in the efficiency of the operations. Regulatory processes (e.g., customs) can be expedited using Blockchain by improving confidence in documentations. This, in turn, can result in reductions in wastage, risk, and insurance premiums.

▶ **Practical Insights** In the food industry, it is of critical importance to establish confidence in food safety through accurate and comprehensive documentation of the production process. This includes the location of ingredients' production, as well as methods of processing and distribution. On this frontier, Dole, Driscoll, Tyson, and Nestlé are collaborating with IBM to develop a Blockchain for the food supply system. Among the issues the companies are looking to address is food contamination. IBM and Walmart are also researching how to increase food safety control in the SC using Blockchain technology (IBM 2017).

In SCOM, Blockchain is also being used for SC contracts, which often involve multiparty agreements and regulatory and logistical constraints. Further complexities arise from operations in different jurisdictions, as well as the dynamic features embedded in the contracts. The flow of information in an SC plays a critical role in the efficiency of operations. Regulatory processes (e.g., customs) can be expedited by improving confidence in documentation. This, in turn, reduces waste, risk, and insurance premiums.

Blockchain technology provides a global tamperproof repository for documents and shipping events, and all the data are secure and transparent. Because of this, third-party intermediaries can be eliminated, resulting in efficiency, low costs, less delay, less fraud, and the possibility to have private networks. Additionally, it can also increase customer and partner trust. Blockchain technology can facilitate a truly digital SC.

16.5.5 Robotic Process Automation and Artificial Intelligence in Procurement

Automated processing to support routine sourcing activities such as "purchase-to-pay" is another form of digitalization. These activities include conducting price searches for items or making a purchase requisition. Although it is called "robotic processes automation" (RPA), it is not a physical robot made from steel that acts, but a software application that performs the tasks that are normally done by a

person. It is not necessary that the RPA software does everything, but it supports a person in performing critical or relevant tasks. In other words, the processes are conducted by software robots that run necessary activities using AI. AI software continuously learns and shapes its knowledge using examples. The machine uses experience and feedback, which means that it is not programmed (or coded) to solve one single problem.

Where are potential areas for RPA or AI in procurement? Generally, repetitive or administrative tasks can be automated, especially in purchasing. The Purchase-to-Pay (P2P, see Chap. 5) process often includes manual, repetitive, and error-prone tasks. Operational buyers might validate the incoming material or service requisitions and create purchase orders, which are then sent to the supplier. Through the order confirmation, the supplier commits to the price, quantity, and delivery date of an item. The purchasing team monitors the delivery up to the receipt of the goods. Accounts payable then carries out the last process step: Invoice receipt, i.e., matching the order with the goods receipt and the details regarding the invoice, and finally activating or processing the payments (e.g., for more details on the e-payment processes, see Lysons and Farrington 2020, p. 169).

The key question here is: which process step adds value and which does not? Where can errors occur, where are monotonous and repetitive activities that can be standardized and supported by RPA? This can be answered by the classic approach of process improvement, which aims to identify all aspects of value creation, at the same time eliminating non-value-adding activities and thus increasing efficiency. The philosophy behind this, according to Womack and Jones (1998), is the classic Lean Thinking while simultaneously incorporating the ideas of quality management and the concepts of continuous improvement (Stevenson 2018, p. 610).

▶ **Practical Insights** Let us consider an example based on Knotte et al. (2020). An order confirmation is sent to the buyer's e-mail inbox, it needs to be opened manually, it will be extracted and copied into the corresponding section of the ERP system to confirm a specific delivery date, quantity, and price. Now, what can be robotized? Instead of performing these activities manually, a software robot can take over this task by extracting the contents of the e-mail inbox using intelligent content recognition (ICR). The robot logs into the ERP system, compares the extracted data with the order information. If there is a discrepancy between the order confirmation and the order, the software robot sends an automatic message to the requestor. If there is a match, the robot (the RPA software) updates the new data in the ERP system. The benefits of RPA in this step of the process are not only automation itself but also an improvement in data quality, as the errors in machine verification and data transmission are almost eliminated.

To identify company-specific weaknesses, the first step is to conduct a structured analysis of the current state. This is the basis for deriving potentials and moving toward P2P excellence. Inspired by Lean Thinking, the practical and structured

ESOAR approach is recommended to identify process inefficiencies (see Knotte and Senger [Capgemini invent] and Tsipoulanidis 2020). The ESOAR approach focuses on the following five steps, which are run through in sequence to analyze a process:

1. **E**limination (of irrelevant tasks)
2. **S**tandardization (limiting personalization)
3. **O**ptimization (of ERPs and workflows)
4. **A**utomation (of standardized processes)
5. **R**obotization (of rule-based and repetitive tasks)

According to Capgemini Invent (2020),[1] this approach has already been successfully applied in practice in numerous projects and, in addition to the classic process analysis, includes further elements with a strong focus on digitalization. The following results provide an insight into the optimization potentials that could be uncovered by applying the ESOAR framework:

- Recurring savings in relevant purchasing volume—between 2% and 5%
- Reduction of process costs—between 10% and 40%
- Reduction in paper invoices—up to 80%

A project using the ESOAR method is typically divided into three phases: Actual analysis, target optimization, and development of an action roadmap. For more practical insights on how to achieve Purchase-to-Pay Excellence, see Knotte et al. (2020).

16.6 Digital Technology in "Make" Processes

Many of the digital technologies that belong to Industry 4.0 have a close connection to *make* processes, and are connected via the Internet. This is where the name Internet of Things arose in relation to SCOM. In the following section, core technologies are presented, which mainly focus on physical value-adding processes, but they can also be used in other SC areas, as well as outside the industrial world, e.g., in medical care (3D printing of prosthetics or use of robots to assist staff in hospitals) or in farming.

16.6.1 3D Printing and Additive Manufacturing

Historically, metal subassemblies or components have often been created by casting, milling, drilling, pressing, or turning. Other examples regarding the connection of single loose parts are gluing, riveting, or welding. In this sense, Koether and Rau

[1] https://www.capgemini.com/de-de/resources/operational-excellence-p2p-prozess/

(2012) refer to the use of for example shaping, reshaping, separating, joining, combining, or coating processes. Historically, often material structures took their final shape by material removal (e.g., in the case of milling, drilling, or turning), which is a subtractive technology. However, there are also manufacturing processes in which material is added. In 1990, Fritz and Schulze had already referred to sintering, a process in which (simply explained) metal or ceramic powder is heated and then pressed into the final shape of the component.

According to the International Organization for Standardization (ISO, www.iso.org, 2017), "*Additive manufacturing (AM), also known as 3D printing, is the process of joining materials layer upon layer, as opposed to* '*subtractive manufacturing*' *methods such as machining.*"

AM applications to operations and SCM reach from spare part logistics to redesigning global SC production and sourcing strategies. The core of AM applications to SCM is the use of 3D printers at different stages in the SC to increase manufacturing flexibility, achieve shorter lead times, increase product individualization, and reduce inventory.

▶ **Practical Insights** UPS and SAP developed a joint technology that allows UPS to manufacture items using 3D printing directly at the distribution centers. This contributes to a faster and more efficient SC. Such an integration of production, sourcing, and distribution is also positive for increasing the speed of reactions to possible disruptions in the SC. In another example, Adidas Speedfactory has introduced the innovative technology of 3D printing to its production process, taking the next step toward fully high-tech and automatic manufacturing. The *Speedfactory* was based on an automatic production process to allow manufacturing to move closer to consumer markets while producing faster than ever before.

The advantages of such a project are inspiring. Storage of finished goods becomes less important and even unnecessary since such a factory, which is local and fast, can produce the exact number of shoes that are actually sold. In addition, personalized models or products could be delivered much more easily and quickly to customers because of the short distances. Another economic advantage is an increase in efficiency because of the ongoing work of the machines. It is important to mention that AM allows the creation of structures in ways that were never possible using traditional production techniques. Furthermore, AM allows the simplification of structures to create completely new component geometries, reduce the number of parts, and combine multiple parts into one. AM also is a key technology for reducing the weight of the components through geometry, which also helps to reduce the amount of material needed.

Companies from various industries, such as automotive, aerospace, defense, medicine, consumer goods, or agricultural tractors consider AM in various areas (D'Aveni 2015). In general, they see major potential for AM in the creation of tools, components with functional integration, replacement parts, and the customization of mass products.

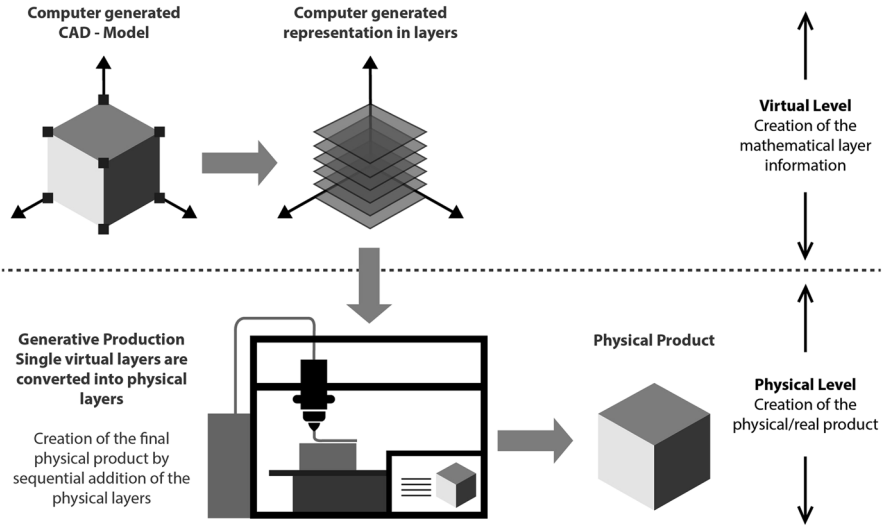


Fig. 16.9 Principles of generative production processes (based on Müller and Töppel 2014, p. 7)

According to Müller and Töppel (2014, p. 7), it is common for the additive or generative manufacturing processes, as they are also called, to follow the sequence shown in Fig. 16.9.

A 3D-CAD volume model is a prerequisite for all generative production processes. This 3D model is sliced into several thin virtual layers using special software. Then, the final product is created in a generative or additive process in which volume elements are added generally in subsequent layers and the final product is created. Materials used for additive production processes include metals, ceramics, or plastics. There is a wide range of various processes that belong to the classification of additive manufacturing (AM). ISO and ASTM International have jointly crafted the Additive Manufacturing Standards Development Structure, a framework that will help meet the needs for new technical standards in this fast-growing field. For SCOM, the core of the AM applications to SCM is the use of 3D printers at different stages in the SC to increase manufacturing flexibility, achieve shorter lead times, increase product individualization, and reduce inventory.

16.6.2 Virtual Reality and Augmented Reality

In the previous section on digital technology in the sourcing process, it was mentioned that augmented reality is a potential tool to prepare and assist procurement team visits to suppliers. In order to understand that striking digital innovation, let us look at the concept of Virtual Reality (VR). VR is computer-generated reality. Like many universities, the University of Sheffield (www.sheffield.ac.uk) operates a VR Laboratory and they summarized that "Virtual Reality is a technology which allows interacting with a 3D computer-simulated environment as if it was real, whether that environment is a simulation of the real world or an imaginary world."

The user enters this virtual environment by either wearing VR glasses or similar devices like a head-mounted display. In the industrial environment and often in the early stages of product development, the user enters a 3D cinema which is a so-called CAVE (Cave Automatic Virtual Reality).

The CAVE is used when products are designed and the engineers, planners, and operators want to assess functionalities, tolerances, and accessibility or even check assembly processes. The CAVE is used, for example, in the automotive, railway, aircraft, or turbine industries. VR glasses are used for training purposes for logistical processes. The operator wears the VR glasses and carries handheld controlling devices. What is shown through the glasses is a completely virtual world, and the user can practice picking and kitting processes. The quality of the virtual representation is fascinating, as practical experience shows. Using VR, it is possible for the automotive industry to share best-practice solutions from one site to another, or train new employees very efficiently.

In contrast to this, it is important to understand the concept of Augmented Reality (AR). AR is a technology that adds a computer-generated image onto the user's view of a real environment. The AR technology converts data and their analyses into pictures and animations, which are then added to our real environment. In this way, AR enhances (in this sense augments) a real view with an unreal additional layer: digital information or visualization are projected on what the user practically sees. This can be compared to information that is written on a window and the user looks through this window into the real environment.

AR is used in production, where operating instructions are displayed in the logical sequence of the assembly processes or to enhance maintenance processes, where a service operator receives instructions about what to replace in which sequence by looking at the defective component. AR is also used within logistical processes, where the pick worker or forklift truck driver receives enhanced information by looking through a device (like an AR glass) which shows which shelf to pick for which item in which quantity. The expression "smart glasses" is used for an AR device that is worn like a pair of eyeglasses. In the medical field, AR is used to educate future medical doctors for surgery, for example. For SCOM, the technology of Augmented Reality surely contributes to the generation of competitive advantage while value-adding transformation processes take place. Both AR and VR are important parts of the Metaverse SCOM (Dolgui and Ivanov 2023; Xiao et al. 2024).

16.6.3 Robotics

The term "robot" was coined by Karel Čapek in 1921 (Brynjolfsson and McAffe 2015, p. 39). The origin is from the word "robota," which means forced labor, i.e., a robot is a device that works for you. A robot is an engineered machine which interacts in the physical world of factories, warehouses, or offices to increase efficiency or perform repetitive, monotonous tasks. In the medical environment, robots can assist nurses, for example, in avoiding heavy activities that are not ergonomic, such as when patients need to be lifted. Robots are also used to perform extremely

dangerous tasks, e.g., when bombs are deactivated (bomb squad) or when unknown territories like buildings need to be entered, e.g., after an earthquake.

According to Robot Institute of America "A robot is a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks", see Gupta and Arora 2013, p. 309).

Brynjolfsson and McAffe (2015, p. 39) referred to the three laws of robotics:

- A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
- A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

Robots commonly have three main characteristics. First, there are sensors that support the gathering of information about the robot's environment. Of course, there are various types of supporting devices employed to guide the robot's behavior and its actions, such as the use of cameras, microphones, thermometers, or accelerometers, to mention a few. Second, are the actuators (e.g., electric motors, hydraulics, pneumatics, etc.) and effectors. These might be the tools at the ends of the robot's arms like pliers, hammers, screwdrivers, spot-welding devices, and motors for manipulation or handling devices, or lights or speakers. In other words, the actuators are the parts that activate the machine to perform the real value-adding work. Third, there is also the control system for each robot, which can be considered the center for AI.

The World Robotics 2024 Report by the International Federation of Robotics (released in Frankfurt, Sep 24, 2024[2]) recorded "4,281,585 units operating robots in factories worldwide - an increase of 10%. Annual installations exceeded half a million units for the third consecutive year. By region, 70% of all newly deployed robots 2023 were installed in Asia, 17% in Europe and 10% in the Americas." In 2016, the number of globally operating robots was just 1.828 million (VDI Nachrichten 2017, p. 3). The main reasons for the increased demand for robots are the need to produce with higher flexibility and to be able to better react to customer needs in shorter cycles. Traditional customers of robots are the automotive, retail, electronics, and logistics industries, who have been using robots for decades. Additionally, small and medium-sized companies are increasingly using robots in their operations.

A typical and classical example is the production and assembly robot. The robots integrate strength and flexibility to assist in production sites for electronics, metal, and machinery, rubber and plastics, food and beverage industries as well as the pharmaceutical and cosmetics industry. Historically, these robots are fixed and

[2] https://ifr.org/ifr-press-releases/news/record-of-4-million-robots-working-in-factories-worldwide (accessed 15.02.2025).

immobile installations, but in the future, moving robots in the shape of snakes or spiders will be developed so they can be very flexibly used, e.g., a crawling robot to conduct welding processes in aircraft production.

Another example is the assisting robot in a car garage. Such a robot would bring wheels, bolts, and tools to the operator. The non-value adding and heavy work of searching, motion and movement, bringing, and lifting is done by the machine, while the value-adding tasks of positioning, fitting, fixing, and checking are performed by the skilled operator who can perform the task under much better and more ergonomic working conditions.

In this context, the expression *Cobot* needs to be mentioned, as human beings and robots increasingly perform tasks in cooperation: they actively collaborate. This is how this term was coined: it is used to describe the collaboration of a robot with a human, thus called "*Cobot*."

A similar scenario is valid for hospitals, where the medical staff is assisted by a robot that lifts the patient when medical treatment is required or for drawing blood from the patient. Mini robots can also be swallowed and then moved externally coordinated through a patient's body for wound detection.

Robots are also used in farming or agriculture. For example, they are used to prune vines in the vineyards (vine-bot) or to assist in pulling lettuce from the fields (lettuce-bots). Robots can also be used for cutting or harvesting crops. In an enlarged scenario, where you have (autonomous) agricultural tractors or other devices that communicate with silos and the monitoring center, the expression "smart farming" is used to describe a system of connected agricultural machines that perform tasks, run diagnosis, provide fertilizers, and are externally monitored. Smart farming goes far beyond the pure use of robots.

For example, the automotive market is shaped by short product life cycles and increased product variety. Using cobot installations in assembly lines enables manufacturers to meet these challenges successfully (Hashemi-Petroodi et al. 2020). Humans still perform a lot of the assembly processes in the automotive industry. However, cobot implementations are becoming more common in the industry. The goal of cobots is not to take over the full workload from humans but to increase productivity and efficiency under unique human supervision. Cobots help improve worker ergonomics and reduce the risk of injury. Furthermore, they can be equipped with 2D or 3D vision sensors and perform quality inspections with high speed and accuracy using computer algorithms. For example, they check the leakage of the doors using a thermographic camera or the quality of car lines using lasers. Cobots are widely used for light duties in car assembly lines without any fences or barriers as opposed to traditional industrial robots. Screw driving, wheel mounting, lifting, and integration of modules to car bodies, placing the valves and other components to the engine, inserting glow plugs into cylinder heads, and assembly of the transmission are the main areas of these cobot implementations.

▶ **Practical Insights** The riveting process was traditionally performed by humans at the BMW Mini plant in Oxford, UK. BMW Mini aimed to shorten the cycle time by using cobots since it was hard to achieve continuous accuracy and repeatability with humans alone. After the

implementation of cobots, the human operator started to put the rivets into place only. Securing the rivets and collecting studs were then performed by cobots, while the operator was free to work on the other side of the car. Furthermore, the cobot could keep track of how many rivets had been fixed. BMW Mini was thus able to reduce the takt time from 64 s to 52 (Reeco Automation 2019). PSA Group, which includes Peugeot, Citroen, Opel, and DS, intended to implement cobots for screw driving applications on body-in-white assembly lines to increase performance and reduce production costs at the Sochaux, France plant. After 1 month of implementation, 200,000 cars were produced with zero failure. The cost was subsequently reduced from 8 to 2 euros per car, resulting in 400,000–1,600,000 euros saved yearly in initial status. In addition, worker ergonomics were also improved by this process (Universal Robots 2019; Robotics and Automation 2018).

On construction sites, automated bulldozers might be guided by the pictures taken by a *drone*, i.e., a *flying robot*, or an "unmanned aerial vehicle" (UAV). During construction work at a site, the quality of the work performed can be monitored and documented. This is very valuable input for progress monitoring under the usage of BIM (Building Information Modeling). How often the construction machine compressed gravel to secure a certain contractually required density or how often certain areas were bulldozed can be recorded and documented.

Of course, drones can be used for more than elevated observation of construction locations during value creation. They can be used to perform replenishment of items or for intralogistics processes. This will be introduced with examples of other machines using digital technology to support delivery processes in the next section.

In retail or warehousing, intelligent robots are also used to assist in cycle counting (also in supermarkets) or in logistics for picking and kitting processes. A very popular example is warehouse robots, which bring items from the shelves to the picking and kitting workers, increasing productivity.

These were some examples of how robots can support "make" processes. Other applications will be discussed in the following chapter on technology in "deliver" processes according to the SCOR model.

16.7 Digital Technology in the "Delivery" Processes

Smart connected products have enabled companies to rethink their entire value chain and have supported them in increasing efficiency in planning, sourcing, making, delivering, and returning products. How is this connected to the subject of smart logistics or the use of digital technology to execute delivery processes? Smart logistics is an approach for connecting transportation devices, vehicles, products, materials, pallets, and load units. Using digital technology and data analytics, tracking and tracing can be performed and companies can optimize routes and loads per shipment. This section is focusing on the investigation of smart logistical devices and their primary application in delivery processes in SCOM.

16.7.1 Drones or Unmanned Aerial Vehicles

A drone is a *flying robot*, or an unmanned airplane or aircraft that can be remotely controlled or reach a destination autonomously using software, known geographic coordinates, embedded sensors, and GPS data. Formally, drones are unmanned aerial vehicles (UAV) or unmanned aircraft systems (UAS).

With regards to SCOM, drones can also be used for the transportation of goods and packages. This might be to deliver goods to areas that are difficult to reach and to ship ordered packages to customers to secure last-mile delivery. In 2017, drones were tested by UPS to carry packages from the delivery truck to the house of the customer (SupplyChain247 2017). Amazon tested the delivery of their goods from a flying warehouse to the customers by drone.

Drones can also assist in conducting cycle counting or checking inventory. For this, the flying robots are equipped with an RFID reading device. As the development of drones, like that of robots, is taking place rapidly, it is only possible to provide an inspiration for drone use within this book.

16.7.2 Smart Driverless Transportation Systems

Driverless transportation systems (DTSs) are mainly used in intralogistics and warehousing. Such DTSs increase safety during the loading and unloading of materials and finished goods. These devices assist operators in every step from goods receipt and storage to order picking and preparation for shipment. In addition to the classical driverless transportation devices that have been used for decades, smart features are related to their precise positioning, guiding, route optimization, machine diagnostics, or real-time condition or loading monitoring abilities. DTSs can operate individually or be coupled and connected. Prominent examples are Automated Guided Vehicles (AGVs). These AGVs are in principle material handling systems or also load carrying devices, which move autonomously through the manufacturing area, the warehouse or a distribution center.

Generally, DTSs provide a higher reliability of operations due to a reduction in human errors, leading to higher productivity in warehousing or intralogistics and cost savings. There are numerous use cases of DTSs and we will take a look at some examples.

One possible DTS application is the flexible connection of two production lines. Such a connection might be necessary for the onward transportation of a vehicle body. Bodies were previously moved from line to line via permanently installed transportation equipment like conveyor belts, for example. Any modifications are, therefore, expensive and time-consuming. With the use of driverless transport vehicles, however, the body is loaded at the end of one line and transported to the start of the other as if by magic. DTS technology is also used for so-called mobile pickers. This form of parts supply, which can replace fixed-route supply trains in production shops, makes it possible to deliver “just in time” or “just in sequence.”

At the Ludwigsfelde Sprinter plant south of Berlin, driverless transport vehicles are already supplying assembly line workers fully automatically with prepared "shopping baskets" from the logistics and picking areas. Take door pre-assembly: where parts containers were once lined up in tight rows along the production line and employees had to find, fetch, and install the parts themselves, you now only see the occasional tool wagon and data terminals. Everything else an employee needs to assemble a side door is brought by a driverless transport vehicle in so-called carsets, which contain the right parts and arrive at the line exactly in sequence with the respective vehicle.

A trailblazing pilot project has also been in place at the Hungarian Mercedes-Benz plant in Kecskemét, where all materials are brought line-side for final assembly in pre-picked carsets by DTS vehicles. Not only does this eliminate walking distances for personnel, but it also means that at the end one look into the "shopping basket" shows whether all parts have been installed—a further simple and effective contribution to quality assurance.

Examples taken from the publication by Daimler AG available at the Global Media Site: "The driverless transport system (DTS): Autonomous mobility around the plant."

These examples above from Daimler show in detail how Operational Excellence can be achieved using digital technology.

16.7.3 Smart Forklifts, Pallet Movers, and Cranes

Using a communication technology, the control system assigns certain pick orders and sends them to the closest forklift, which can then be maneuvered by the operator. In the case of fully automated, or smart, forklifts, the operations are performed completely autonomously, i.e., without an operator.

The Auto Pallet Movers (APM) are examples of such smart forklifts. In principle, they are an automated, reliable, and efficient solution for vertical order picking. According to the producer Jungheinrich, their APM ERC215a has a lifting capacity of 1.5 tons. APMs contribute to eliminating errors in picking and improved safety. Examples are automated signals, speed controls, the use of anti-slip technology, collision detection, and fork speed optimization. Such devices also increase safety in a warehouse by choosing the optimal speed and routes based on the environment. Smart forklifts or Pallet Movers can be equipped with additional technology such as barcode scanners and distance sensors. Furthermore, technology, e.g., freight documentation, proof of delivery, RFID identification, tracking and synchronization of delivery routes, etc. can be installed.

Smart cranes have a broad set of applications, ranging from large warehouses to ports, yards, and the (un-)loading of various heavy freights. Examples for the use of (smart) cranes are related to the movement of steel coils or for paper roll handling. However, they are also used in shipbuilding, train production, and container handling, i.e., in metals processing.

Today, the companies that provide smart cranes also offer smart features or devices for the modernization or retrofitting of existing cranes with smart modules. Smart cranes are used for waste disposal or in container ports to handle containers between ships, trains, and trucks. A well-known example is the so-called smartPORT in Hamburg, which is also called the intelligent port (HPA 2017). The HPA stated that the smartPORT is characterized by 10 key elements:

- Navigation in Real Time
- Green electricity from the land
- Intelligent switching points
- Mobile all-purpose sensors
- Smart maintenance
- Virtual depot
- Port monitor
- E-mobility in the port
- Parking for professionals
- Renewable energies

At the container terminal in the port of Hamburg, automated guided vehicles (AGVs) operate autonomously, and this is done with the help of thousands of transponders that are embedded in the floor. Besides the AGVs, a complementary SmartPORT cloud application was installed and launched in 2015—jointly developed by T-Systems and SAP (Deutsche Telekom AG 2015). It aims to improve the operational traffic and trade flow as well as the overall infrastructure by interconnecting all involved parties. In the cloud, data collected about freight, infrastructure, traffic, and location of vehicles and containers can therefore be consolidated and analyzed. Through the integration of smart logistical devices and an IT system, the automation of the majority of container, truck, and unit load device transportation between the wharf and storage areas could be realized. According to T-Systems (2018), the developed digital transportation management solution has great potential for new value networks because of the following factors:

- Intelligent infrastructures for a faster flow of goods
- Real-time analysis of comprehensive SC data
- Overall platform provides a constantly updated situation picture.
- Better synchronization of dynamic logistics processes
- Unbroken shipment tracking improves distribution planning and delivery reliability.
- Cost transparency by coupling logistics and finance systems

Summarizing, smart logistical devices can positively influence SCOM processes by, for example, supporting processes inside a warehouse with regards to safe (un) loading of goods, more efficient inventory management, or shorter picking cycles. Smart devices can help reduce the number of accidents, increase transport visibility, and support truck drivers. Looking at the delivery processes around the “last mile

delivery," i.e., when the shipment is handed over to the customer, a shorter delivery time could be achieved and the environmental friendliness of the vehicles improved. In combination with tracking and tracing, this means SC visibility is improved.

16.8 Possible Obstacles and Limitations of Digital SCOM

A more connected world no doubt brings many benefits—and some dangers (Table 16.1). Issues of skills, security, and standards are key points to consider. Employees require *new skill sets* that enable them to conduct data analytics, run

Table 16.1 Chances and challenges of digital technology applications to SCOM

Digital technology and enablers	Chances for SCOM	Challenges in SCOM
Predictive analytics, big data	Increase in promotion action quality Better demand forecasts Routing decision optimization supply chain visibility, risk and resilience analysis Better customer experience promotions Reduction in demand risks Reduction in information disruption risks and better quality of contingency plan activation	Data transparency and safety coordination complexity increase in cross-channel logistics Higher time risks because of increases in coordination complexity
Industry 4.0, IoT, smart products, robotics, augmented and virtual reality	Customized production system at the costs of mass production individualized products and higher market flexibility Risk diversification Higher responsiveness Shorter lead times and better capacity utilization Real-time manufacturing monitoring	Radical changes in SC and manufacturing process organization Reduction in the number of SC layers New locations close to the markets Re-qualification of employees, redesign of facility layouts data security Increase in design and control complexity Higher exposure to information security risks Higher supply risks if disruption happens in the upstream SC since no intermediate inventory in between the stages
3D printing, additive manufacturing	Flexibility increase, product variety, Shorter lead time Efficiency increase in MRO inventory control Reduction in demand risks	
Advanced T&T technologies, RFID, sensors, Blockchain	Real-time identification Real-time material flow tracing increase in data quality improved transaction coordination reduction in information disruption risks and better quality of contingency plan activation reductions in supply and time risks due to real-time coordination if activating the contingency policies	Increase investments in ICT data security

prognosis, operate SCPs, collaborate with robots, and so on. Engineers will need to learn new ways of thinking to design parts and products that can be 3D printed. Furthermore, employees on the shop floor and colleagues conducting clerical or administrative tasks might fear they will be replaced by AI. As with all technological developments, there have been jobs lost and new jobs created. In the early days, the workers in warehouses carried goods on their shoulders, and now they use a forklift truck or pallet jack. In the future, there will be staff being responsible for monitoring how the automated systems perform the work. Efficiency and competitiveness are fundamentally important, but companies also have a social responsibility: the right balance must be found. In other words, companies striving for digital SCOM processes need to critically assess cost, benefits, speed, quality, and human involvement during value creation. Ideas to strike this balance include robot taxes or the need for producers to clearly specify the ratio between human and automated value add (like a human labor depth ratio) to limit the associated number of *job losses* due to digitalization.

The *lack of standards* is also frequently mentioned as an obstacle toward digitalization. Especially small and medium-sized companies show hesitance because they do not know exactly if a costly investment in the right new technology will pay off. There is no right answer to this subject. It is more the case that the aspect of standards and standardization needs to be assessed when digital strategies are drafted and respective decisions are going to be taken.

Another crucial subject is related to limited trust in security and cyberattacks. Companies carefully observe *data security* issues, data theft, attacks, dependence on information technology, correctly working algorithms, and so on. Managers raise the question: If all my data is stored in a *cloud* that an external company hosts, what are the possible risks which are connected to that? Again, there is no precise answer. The only answer is that *cybersecurity* or dependence on cloud solution providers is extremely important and companies will need to very carefully assess how safe their data and information are. Thus, it is another item on the list of critical evaluations and decisions when it comes to the implementation of a digital SCOM strategy.

Another result of smart technology in value creation is related to the *redesign of production networks and processes* (Dolgui et al. 2020a, b; Ivanov et al. 2021b). Companies are reassessing their industrial footprint. Companies that were initially under the pressure of cost-efficiency might have shifted their capacities to the best-cost countries in Asia. Now, as salaries and wages increase in Asian production sites, a trend of re-shoring can be diagnosed, which is also supported by sustainability assessments (e.g., climate impact, fair working conditions, long transportation, or dependence on suppliers on other continents for face masks, disinfection or vaccination). There is also a second trend that sees countries in Africa as the new global workbench, as Sun (2017) described. Even Chinese firms consider the shift of production capacity to Africa. European companies might consider the European Balkan countries as an alternative to an Asian factory (here, the main driver besides the cost of production is geographical proximity, so it is not purely related to digitalization).

Finally, *business models* might also be changed in line with the implementation of new smart solutions or new service providers will appear who specialize in the industry of 3D printing on demand.

It is impossible to mention all the obstacles and limitations, but it is important to emphasize that there are two sides to the digital coin. It will be very difficult to avoid digital technology in business and SCOM, but it is important to make a critical assessment and to evaluate what-if-scenarios in the development of a digital SCOM strategy. One very important aspect though is corporate responsibility. This is why the human being should always be at the center of digital SCOM decisions.

16.9 Artificial Intelligence in SCOM

*A*I is a simulation of human decision-making processes by machines and computers. Humans started using computers for decision-making support long time ago (Wiener 1948). Even during its infancy, AI tried to mimic human decision rules and system behaviors, and has been considered important part of decision-making support.

Thanks to rapid technological developments, modern AI approaches such as deep learning, reinforcement learning, federated learning, neural networks, visual and interactive intelligence seem to overcome the limitations of early-stage AI (Kosasih and Brintrup 2022; Rolf et al. 2023). They not only mimic human-defined decision rules (which might be inaccurate due to human's limited ability to describe their decision-making rules) but also create new knowledge through training and frequently outperform humans (Fügener et al. 2022). Most importantly, modern technologies allow to develop human-AI collaboration based on mutual delegating tasks and decision-making authorizations between humans and AI which is referred to as "human-in-the-loop AI."

Figure 16.10 provides an overview of AI-based methods.

Multi-agent systems and *agent-based modeling* represent a system of intelligent entities (i.e., agents), each of which acts based on some rules, interests, and behaviors interacting with other agents. Agent-based simulation modeling is frequently used in SCM to examine competition and collaboration practices with the use of complex adaptive system principles. For example, logistics service providers can be described as competing agents bidding on transportation orders. The agents can not only compete but also collaborate, e.g., in order to maximize the overall profit (Swaminathan et al. 1998, Surana et al. 2005; Nair and Vidal 2010). Agent-based simulation is especially useful for modeling behaviors of firms in supply chains and their individual interests observing how the overall supply chain dynamics evolve through decentralized decisions of agents.

Machine learning methods are based on training principles (Liu 2022). Algorithms that imitate human decision-makers are trained and learned through this training. Machine learning algorithms can be supervised (e.g., labeled datasets are used to train algorithms for accurate data classification or outcome predictions) and unsupervised (e.g., analysis and clustering of unlabeled datasets). One specific area of machine learning is *reinforcement learning* that combines Markov chain and

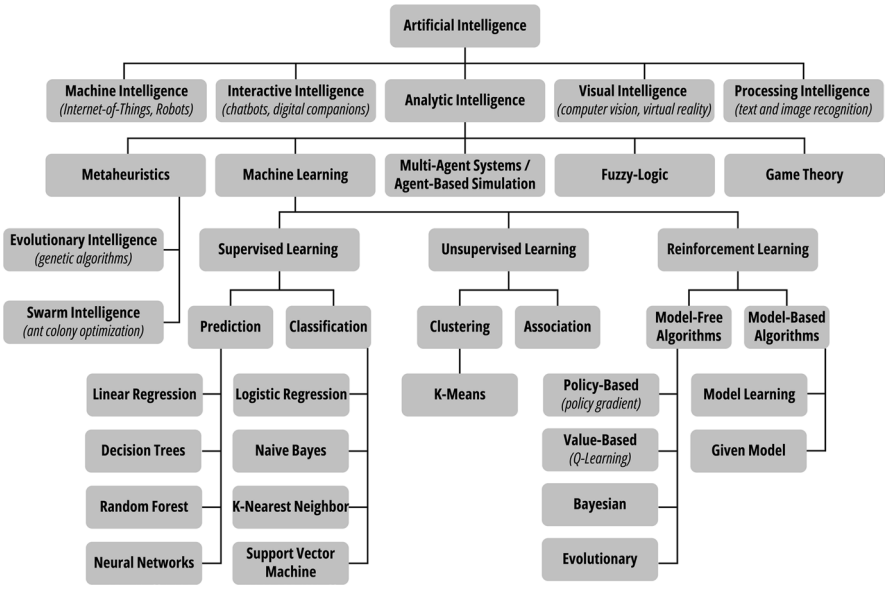


Fig. 16.10 Overview of AI-based methods (Ivanov 2024b)

agent-based simulation elements. Intelligent agents in reinforcement learning algorithms interact with an environment and learn to act within this environment. Supervised machine and reinforcement learning have been applied to SCM in different areas, e.g., inventory management, transportation, and supplier selection (Rolf et al. 2023). For example, an agent can learn how to place optimal orders or select suppliers with highest delivery reliability. Machine learning techniques are frequently combined with discrete-event simulation (Cavalcantea et al. 2019).

Artificial neural networks are inspired by biological neural networks framing mechanisms of animal brains. Neural network models are based on machine learning and training. *Fuzzy theory* is used when some of the modeling elements cannot be quantified and should be described verbally, e.g., high, medium, or low demand. *Evolutionary algorithms* (i.e., metaheuristics such genetic algorithms) are used for solving supply chain optimization problems when exact solutions are impossible due to problem complexity (so called NP-hard problems) (Mirjalili 2018).

Four other groups shown in Fig. 16.10 (i.e., machine intelligence, interactive intelligence, visual intelligence, and processing intelligence) also find numerous applications in SCM. Through machine intelligence, cobots (collaborative robots) help improve manufacturing processes by mutual enhancements of human and robot intelligence (Sheu and Choi 2023). Visual intelligence creates new opportunities for companies to interact with generative AI (e.g., natural language processing such as ChatGPT) allows to simplify simulation model development through automatic coding (Jackson et al. 2024a) and can be used in variety of SCOM problems for learning, perception, prediction, interaction, adaptation, and reasoning (Jackson et al. 2024a, b).

16.10 Case Study "Smart Usage of Big Data Along the Supply Chain: Big Data Analytics Between Companies, Value-Added and the Impact of Risk and Complexity Management"

Learning Goals

- Understand the complexity of big data projects along the supply chain.
- Understand pain points for suppliers, such as data conversion and concerns about sharing machine data.
- Understand how organizational restrictions play a role.
- Understand the challenges of setting the right goals and the need to have a fitting project setup.
- Understand the challenges facing suppliers who need to share some data and keep other data private.

Situation
An Aerospace manufacturing Tier 1 company is in a ramp-up phase for a new Aerospace program (for a cadence increase of a new aircraft model). Therefore, this Tier 1 company is seeking to secure its inbound supply chain with its own key suppliers. First steps are planned to receive, evaluate, and interpret production data from such suppliers. An affected Tier 2 pilot supplier is requested to provide forecast and production progress data to the Tier 1 company.

Target
The CEO of the Tier 1 company wants to secure his supply chain, since he has to manage the ramp-up of a new aircraft program for one major Commercial Aircraft OEM. For this purpose, he requests demand and production data from his strategic A suppliers, which will ideally be available to use for his own company's analytics. Through increased supply chain visibility, the competitiveness of the Tier 1 company in managing the ramp-up will be strengthened. He requests that the Tier 2 supplier listed above acts as the pilot supplier and shares the related production data.

Complication
The Chief Digitalization Officer at the Tier 1 company has just been appointed. As budget and skilled employees are scarce, he seeks to implement a proof-of-concept (PoC) within a few months to show he can perform and reach his objectives even with limited efforts. As the IT department is facing serious bottlenecks, the development of an internal solution is not feasible as long as they are not in the lead for the PoC and receive additional resources.

The Tier 2 supplier does not build upon standard a production system, but for years has built upon dedicated legacy systems with proprietary interfaces, such as a CSV download. The main purpose of the Tier 2 supplier is to stay in business, so he approves joining the proof-of-concept with the Tier 1 company even though the

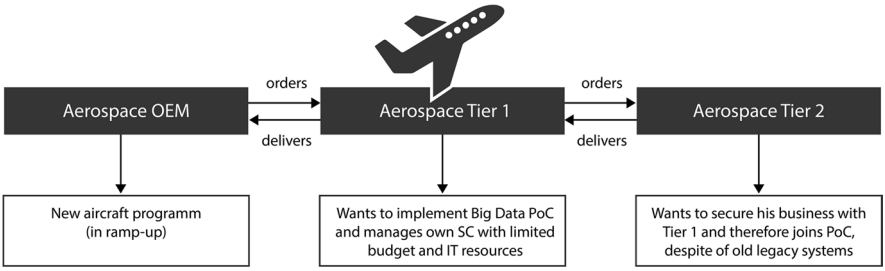


Fig. 16.11 Supply chain structure

value added is questioned by some employees and the workload is already at the limit, because of the program ramp-up for the Aerospace OEM. Furthermore, the Tier 2 supplier wants to keep control of which production data he will share with the Tier 1 company (Fig. 16.11).

Frame Conditions

- The given timeframe for the proof-of-concept is only 3–4 months.
- The budget at the Tier 1 company is limited, but if the proof-of-concept proofs are successful, the board has promised to release a further budget.
- Business processes of the Tier 1 company and Tier 2 supplier (e.g., usage of forecast, before the order process starts) and IT tools (including the ERPs) are different.
- Daily business will need to continue despite the foreseen PoC.

Instructions

Organize yourselves in teams of four people (or at least three). Answer the following five questions and prepare a short (max. 10 min) presentation of the results.

Explore which interest in the Aerospace OEM drives the Tier 1 company to invest in the Big Data PoC (proof of concept). Methodology wise, please fill in the aim grid in Fig. 16.12.

Discuss the pros and cons of a Make-or-Buy decision in terms of the Big Data solution in the Tier 1 company. Consider not only financial factors but also the availability of IT resources, time-to-market and collaboration factors, or if different partners with non-standardized IT tools seek to collaborate.

Discuss in which sequence the following steps (which are randomly listed below) would have to be organized to ensure a successful project. For this purpose, number the steps in the correct sequence in the column on the right (the first step is already entered).

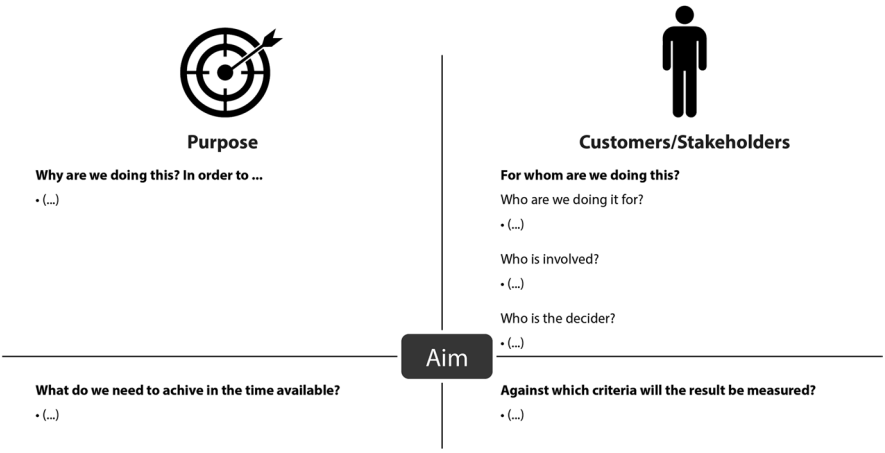


Fig. 16.12 Analysis framework

Project steps (in random order below)	Step number
To-be analysis for future IT ecosystem/landscape	
Data migration	
Deployment of pilot plant	
Pilot productive deployment of suppliers	
Key user training	
Check data sources/data cleansing (ensure good data quality)	
As-is analysis of existing business processes	
Full deployment of all other sites	
As-is analysis of existing IT ecosystem/landscape	
To-be analysis for future needed business processes	
Configuration and setup of to-be solution in test environment	
Quality assurance for each module and related data	
Quality assurance end to end	
Project kickoff/scoping	1
Lessons learned	
Setting up change management	

1 The Tier 1 company has limited available manpower to staff the project. The project lead is a business expert in supplier collaboration but only can invest ½ FTE (1 FTE = 1 full-time equivalent). In addition, the manager of the IT department can only give him limited support of 20 man-days, but only for 3 months.

Explore how the shortage of available internal resources is likely to impact:

(a) Project scope
(b) Project timeline
(c) Probability of developing an in-house solution from scratch vs. involving external expertise

(d) Probability that industry clusters (e.g., cross-company Aerospace organizations, supporting small and medium Aerospace suppliers) are involved
(e) Development of new advanced IT interfaces (e.g., Web Services) vs. integration of in-use file formats (like CSV, Excel), with all their limitations

2 During the course of the project, not all (ideally) required data are available.

(a) Forecast data about future customer-side demands toward the supplier are not always available. Explore which impacts on supply chain security and related stock levels the as-is situation has and which improvements are linked to better supply chain visibility (= more transparency about future material demand, and from the supplier).
(b) Different legacy systems at the Tier 1 company in manufacturing planning and execution only provide limited data export functionalities. Concretely, only CSV files are generated once per night in a so-called batch run. Explain how this bottleneck will impact expectation management toward the sponsoring management for the scope of the project.

Which restrictions apply?

Which benefits (in terms of supply chain predictability and visibility) will still apply? In this context, explore how a high OTD (on-time delivery) performance can contribute to avoid expensive stockouts.

3 Explore why the Tier 2 supplier will seek to maintain control over which production data he shares with the Tier 1 company. In this context, please consider the deviating interests of the Tier 1 company, who wants production-related data from the Tier 2, and of the Tier 2 supplier, who wants to stay in business, even if in his own production, not everything always works perfectly. Then explore how the Tier 2 supplier could still maintain control process- and IT-wise.
4 Explore which benefits the Tier 1 company can expect if the project succeeds. For this purpose, please consider Porter's Competitive Five Forces regarding industry competition.

16.11 Key Points and Discussion Questions

Disruptive innovations such as digitalization and Industry 4.0 influence the development of new paradigms, principles, and models in SCOM. The IoT, Cyber-Physical Systems, and smart, connected products facilitate the development of digital SCs and smart operations. Digital technologies applications to SCOM include BDA, advanced manufacturing technologies with sensors, decentralized agent-driven control, advanced robotics, augmented reality (AR), advanced tracking and tracing technologies, and additive manufacturing (AM) (Fig. 16.13).

While the individual contributors (e.g., robots, sensors, radio frequency identification (RFID), agents, and modular factories) are not really new, they are becoming more practical and companies more receptive to using them to stay competitive. In addition, an attempt to interconnect these local solutions using the progress in data

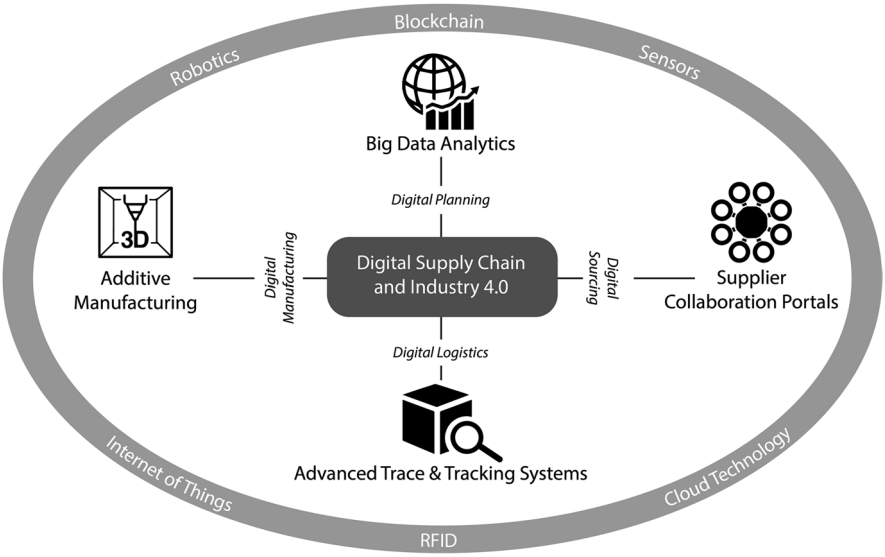


Fig. 16.13 Digital Supply Chain and Industry 4.0

processing technologies can be observed in practice. As such, this trend calls for new principles and models to support the SCM of such future factories.

- Can you summarize the major developments throughout the four industrial revolutions?
- What are the main elements of smart, connected products?
- Where do you see reasons for criticism? Discuss the pros and cons of the digital transformation besides the aspects of the generation of competitive advantage.
- Where do you see the potential to create new business and new jobs and where do you see risk concerning job losses?

On the technology side, ongoing developments, such as Industry 4.0, AM, smart sensors and intelligent materials, open new opportunities, but also create new challenges for ripple effect analysis in the SC. Examples of the impacts of new engineering technologies on the SC are the localization of production, high flexibility, and reduced SC coordination complexity.

The implementation of Industry 4.0 in SCOM consists of the following components:

1 Infrastructure Technology

(a) Internet-of-Things (IoT)
(b) Cyber-Physical System (CPS)

2 Engineering Technology

(a) Additive Manufacturing
(b) Automated Guided Vehicle (AGV)
(c) Collaborative Robots
(d) Augmented Reality

3 Data Technology

(a) Big Data Analytics (BDA)
(b) Artificial Intelligence (AI)
(c) Track and Trace (T&T) systems

4 Communication Technology

(a) Machine-to-Machine (M2M)
(b) Cloud Services
(c) Smart Products
(d) Blockchain

Digital technology drives a rethinking and reinventing of SC design and management from the perspective of reconfigurable and viable networks in the context of three particular transformation drivers, i.e., digital platforms, epidemics/pandemics, and climate change (Ivanov 2021). For the last years, SCs have been experiencing technological transformations on a scale unlike any seen before. Industry 4.0, additive manufacturing, sensors, collaborative robots, and blockchain lead to principally new concepts [such as open manufacturing (Kusiak 2020)] and data utilization opportunities entailing highly flexible and adaptable supply networks with structural variety and multifunctional processes (Blackhurst et al. 2021; Dolgui et al. 2020a, b; Ivanov et al. 2021a; Ruel et al. 2024).

In this setting, *digital platforms* play an increasingly important role in the technological transformations of SCs. Through the digital platforms, SCs are evolving toward technology-driven networks and digital ecosystems operating with the help of procurement (e.g., SupplyOn), manufacturing, sales (e.g., Amazon), and finance (e.g., ChainLink) platforms. The utilization of digital platforms, blockchain, and smart contracting is of high importance and value in SCOM (Queiroz et al. 2019; Cai et al. 2021; Lohmer et al. 2020; Roeck et al. 2020). For example, digital manufacturing platforms can facilitate cloud manufacturing (Frazzon et al. 2021; Sokolov et al. 2020). Digital SC financing with the use of blockchain-driven platforms such as deep tier financing (DTF) create an end-to-end SC visibility and so offer novel context for SCOM. Finally, we take Industry 5.0 is coming (Ivanov 2023b)!

This chapter has shown that there are numerous applications of digital technology in SCOM. Of course, the presented examples only reflect a small portion of the digital possibilities in SCOM and it is important to highlight that the development in this area is taking place at a very high speed.

- Can you give examples for the application of digital technologies and relate them to the processes according to the SCOR Model (plan, source, make, deliver, return)?
- Can you explain the following terms in your own words: Big Data Analytics, Industry 4.0, Blockchain, digital twins, robots, additive layer manufacturing, RFID, and tracking and tracing systems?
- Can you connect these technologies also to the digital SCOM framework and explain how they are successfully connected as a systematic concept?
- Do you recall which suggested framework helps to evaluate a supply chain and to draft a strategy for digital SCOM Excellence?

Immediate benefits of applying digital technology in SCOM are higher transparency, increased capacity usage, cost-saving, and elevated safety as well as reduced SC risk, which are all examples of achieving SCOM Excellence by using technology in the digital transformation.

Finally, we note that this chapter does not pretend to be encyclopedic and rather seeks to highlight the role of digitalization in SCOM with the objective of identifying the perspectives that can be leveraged in the near future in exploring how digital technologies can improve decision-making support and performance in SCOM.

- What are your thoughts regarding the potential which digital SCOM offers to reduce risks and improve resilience and sustainability?

Acknowledgment The case study "Smart usage of Big Data along the Supply Chain" is written by Mr. Arvid Holzwarth (MBA) based on the materials of SupplyOn.

Bibliography

Blackhurst J, Das A, Ivanov D (2021) Supply chain resilience and its interplay with digital technologies: making innovations work in emergency situations. Int J Phys Distrib Logist Manag 51(2):97–103

BMBF (2013) Recommendations for implementing the strategic initiative Industrie 4.0—final report of the Industrie 4.0 working group BMBF

BMWi Bundesministerium für Wirtschaft und Energie (2016) Digitale Strategie 2025, Status March 2016

Bordoloi SK, Cooper WW, Matsuo H (1999) Flexibility, adaptability, and efficiency in manufacturing systems. Prod Oper Manag 8(2):133–149

Bozarth CC, Handfield RB (2019) Introduction to operations and supply chain management. Pearson, Harlow

Brynjolfsson E, McAffe A (2015) The second machine age—Wie die nächste digitale Revolution unser aller Leben verändern wird. Plassen Verlag, Kulmbach

Brynjolfsson E, McAffe A (2017) Von Managern und Maschinen. Harvard Business Manager

Cai Y, Choi TM, Zhang J (2021) Platform supported supply chain operations in the blockchain era: supply contracting and moral hazards. Decis Sci 52(4):866–892

Cavalcantea IM, Frazzon EM, Forcellinia FA, Ivanov D (2019) A supervised machine learning approach to data-driven simulation of resilient supplier selection in digital manufacturing. Int J Inf Manag 49:86–97
Choi TM, Wallace SW, Wang Y (2018) Big data analytics in operations management. Prod Oper Manag 27(10):1868–1883
Chopra S, Meindl P (2013) Supply chain management—strategy, planning and operation. Pearson, Harlow
Chou MC, Chua G, Teo CP, Zheng H (2010) Design for process flexibility: efficiency of the long chain and sparse structure. Oper Res 58(1):43–58
Christopher M (2023) Logistics and supply chain management, 6th edn. FT Press, Upper Saddle River, NJ
D'Aveni R (2015) The 3-D printing revolution. Harv Bus Rev
Das A, Narasimhan R, Talluri D (2006) Supplier integration: finding an optimal configuration. J Oper Manag 24(5):563–582
de Sousa Jabbour ABL, Jabbour CJC, Filho MG, Roubaud D (2018) Industry 4.0 and the circular economy: a proposed research agenda and original roadmap for sustainable operations. Ann. Oper Res 270(1–2):273–286
Decken T (2018) Warum Digitalisierung nicht mehr warten kann, 3rd edn. The Source Magazine
Deutsche Telekom AG (2015) Telemaintenance—remote delivery of first aid. Deutsche Telekom Publications
Dolgui A, Ivanov D (2023) Metaverse supply chain and operations management. Int J Prod Res 61(23):8179–8191
Dolgui A, Ivanov D (2024) Internet of behaviors: conceptual model, practical and theoretical implications for supply chain and operations management. Int J Prod Res. https://doi.org/10.1080/00207543.2024.2372008
Dolgui A, Ivanov D (2025) Internet of behaviors: conceptual model, practical and theoretical implications for supply chain and operations management. Int J Prod Res 63(1):1–8
Dolgui A, Ivanov D, Sokolov B (2018) Ripple effect in the supply chain: an analysis and recent literature. Int J Prod Res 56(1–2):414–430
Dolgui A, Ivanov D, Sethi SP, Sokolov B (2019) Scheduling in production, supply chain and industry 4.0 systems by optimal control. Int J Prod Res 57(2):411–432
Dolgui A, Ivanov D, Sokolov B (2020a) Reconfigurable supply chain: the X-network. Int J Prod Res 58(13):4138–4163
Dolgui A, Ivanov D, Potryasaev S, Sokolov B, Ivanova M, Werner F (2020b) Blockchain-oriented dynamic modelling of smart contract design and execution control in the supply chain. Int J Prod Res 58(7):2184–2199
Dubey R, Gunasekaran A, Childe SJ, Wamba SF, Roubaud D, Foropon C (2019a) Empirical investigation of data analytics capability and organizational flexibility as complements to supply chain resilience. Int J Prod Res 59(1):110–128
Dubey R, Gunasekaran A, Childe SJ, Blome C, Papadopoulos T (2019b) Big data and predictive analytics and manufacturing performance: integrating institutional theory, resource-based view and big data culture. Br J Manag 30(2):341–361
Dubey R, Gunasekaran A, Childe SJ, Bryde DJ, Giannakis M, Foropon C, Roubaud D, Hazen BT (2020) Big data analytics and artificial intelligence pathway to operational performance under the effects of entrepreneurial orientation and environmental dynamism: a study of manufacturing organisations. Int J Prod Econ 226:107599
Fox MS, Barbuceanu M, Teigen R (2000) Agent-oriented supply chain management system. Int J Flex Manuf Syst 12:165–188
Fragapane G, Peron M, Sgarbossa F, Strandhagen JO, Ivanov D (2022) Increasing flexibility and productivity in industry 4.0 production networks with autonomous mobile robots and smart intralogistics. Ann Oper Res 308:125–143
Frank AG, Dalenogare LS, Ayala NF (2019) Industry 4.0 technologies: implementation patterns in manufacturing companies. Int J Prod Econ 210:15–26

Frazzon EM, Freitag M, Ivanov D (2021) Intelligent methods and systems for decision-making support: toward digital supply chain twins. Int J Inf Manag 57:102281
Fritz H, Schulze G (1990) Fertigungstechnik. VDI Verlag, Düsseldorf
Fügener A, Grahl J, Gupta A, Ketter W (2022) Cognitive challenges in human–artificial intelligence collaboration: investigating the path toward productive delegation. Inf Syst Res 33(2):678–696
Ghadge A, Weiß M, Caldwell ND, Wilding R (2019) Managing cyber risk in supply chains: a review and research agenda. Supply Chain Manag 25(2):223–240
Ghadge A, Kara ME, Moradlou H, Goswami M (2020) The impact of industry 4.0 implementation on supply chains. J Manuf Technol Manag 31(4):669–686
Gupta AK, Arora SK (2013) Industrial automation and robotics, 2nd edn. University Science Press, New Delhi
Hashemi-Petroodi SE, Thevenin S, Kovalev S, Dolgui A (2020) Operations management issues in design and control of hybrid human-robot collaborative manufacturing systems: a survey. Annu Rev Control 49:264–276
Hofmann E, Strewe UM, Bosia N (2018) Supply chain finance and blockchain technology. Springer, Cham
Ivanov D (2018) Structural dynamics and resilience in supply chain risk management. Springer, New York
Ivanov D (2020) Predicting the impact of epidemic outbreaks on the global supply chains: a simulation-based analysis on the example of coronavirus (COVID-19/SARS-CoV-2) case. Transp Res E 136:101922
Ivanov D (2021) Introduction to supply chain resilience. Springer Nature, Cham
Ivanov D (2022) Viable supply chain model: integrating agility, resilience and sustainability perspectives. Lessons from and thinking beyond the COVID-19 pandemic. Ann Oper Res 319:411–1431
Ivanov D (2023) Intelligent digital twin (iDT) for supply chain stress-testing, resilience, and viability. Int J Prod Econ 263:108938
Ivanov D (2023b) The industry 5.0 framework: viability-based integration of the resilience, sustainability, and human-centricity perspectives. Int J Prod Res 61(5):1683–1695
Ivanov D (2024a) Conceptualisation of a 7-element digital twin framework in supply chain and operations management. Int J Prod Res 62(6):2220–2232
Ivanov D (2024b) Introduction to supply chain analytics. Springer Nature, Cham. ISBN 978-3-031-51240-7
Ivanov D (2024c) Digital supply chain management and technology to enhance resilience by building and using end-to-end visibility during the COVID-19 pandemic. IEEE Trans Eng Manag 71:10485–10495
Ivanov D, Dolgui A (2021) A digital supply chain twin for managing the disruptions risks and resilience in the era of industry 4.0. Prod Plan Control 32(9):775–788
Ivanov D, Sokolov B (2010) Adaptive supply chain management. Springer, London
Ivanov D, Sokolov B (2012) The inter-disciplinary modelling of supply chains in the context of collaborative multi-structural cyber-physical networks. J Manuf Technol Manag 23(8):976–997
Ivanov D, Sokolov B, Dolgui A, Werner F, Ivanova M (2016) A dynamic model and an algorithm for short-term supply chain scheduling in the smart factory industry 4.0. Int J Prod Res 54(2):386–402
Ivanov D, Dolgui A, Sokolov B (2019) The impact of digital technology and industry 4.0 on the ripple effect and supply chain risk analytics. Int J Prod Res 57(3):829–846
Ivanov D, Tang CS, Dolgui A, Battini D, Das A (2021a) Researchers' perspectives on industry 4.0: multi-disciplinary analysis and opportunities for operations management. Int J Prod Res 59(7):2055–2078
Ivanov D, Sokolov B, Chen W, Dolgui A, Werner F, Potryasaev S (2021b) A control approach to scheduling flexibly configurable jobs with dynamic structural-logical constraints. IISE Trans 53(1):21–38
Ivanov D, Dolgui A, Sokolov B (2022) Cloud supply chain: integrating industry 4.0 and digital platforms in the "supply chain-as-a-service". Transp Res E Logist Transp Rev 160:102676

Ivanov D (2025) Conceptual and formal models for design, adaptation, and control of digital twins in supply chain ecosystems. OMEGA 137:103356
Jackson I, Ivanov D (2023) A beautiful shock? Exploring the impact of pandemic shocks on the accuracy of AI forecasting in the beauty care industry. Transp Res E Logist Transp Rev 180:103360
Jackson I, Ivanov D, Dolgui A, Namdar J (2024a) Generative artificial intelligence in supply chain and operations management: a capability-based framework for analysis and implementation. Int J Prod Res 62(17):6120–6145
Jackson I, Saenz M, Ivanov D (2024b) From natural language to simulations: applying AI to automate simulation modelling of logistics systems. Int J Prod Res 62(4):1434–1457
Jahani H, Jain R, Ivanov D (2023) Data science and big data analytics: a systematic review of methodologies used in the supply chain and logistics research. Ann Oper Res. https://doi.org/10.1007/s10479-023-05390-7
Johnson K, Lee ABH, Simchi-Levi D (2016) Analytics for an online retailer: demand forecasting and price optimization. Manuf Serv Oper Manag 18(1):69–85
Jordan WC, Graves SC (1995) On the principles of the benefits of manufacturing process flexibility. Manag Sci 41:577–594
Khajavi SH, Partanen J, Holmström J (2014) Additive manufacturing in the spare parts supply chain. Comput Ind 65(1):50–63
Knotte D, Senger E, Tsipoulanidis A (2020) Operational Excellence in Ihrem Purchase-to-Pay Prozess – Ein Guide zur Realisierung von Potenzialen im operativen Einkauf, Capgemini Invent
Koether R, Rau W (2012) Fertigungstechnik für Wirtschaftsingenieure. Carl Hanser Verlag, München
Koren Y, Heisel U, Jovane F, Moriwaki T, Pritschow G, Ulsoy G, Van Brussel H (1999) Reconfigurable manufacturing systems. CIRP Ann 48(2):527–540
Kosasih EE, Brintrup A (2022) A machine learning approach for predicting hidden links in supply chain with graph neural networks. Int J Prod Res 60(17):5380–5393
Küpper D, Kuhlmann K, Kocher S, Dauner T, Burggräf P (2016) The factory of the future. https://www.bcg.com/publications/2016/leaning-manufacturing-operations-factory-of-future. Accessed 2 Jan 2025
Kusiak A (2020) Open manufacturing: a design-for-resilience approach. Int J Prod Res 58(15):4647–4658
Li J, Jia G, Cheng Y, Hu Y (2017) Additive manufacturing technology in spare parts supply chain: a comparative study. Int J Prod Res 55(5):1498–1515
Liu KY (2022) Supply chain analytics. Concepts, techniques and applications. Springer Nature, Cham
Lohmer J, Bugert N, Lasch R (2020) Analysis of resilience strategies and ripple effect in blockchain-coordinated supply chains: an agent-based simulation study. Int J Prod Econ 228:107882
Lysons K, Farrington B (2020) Procurement and supply chain management, 10th edn. Person, Harlow
MacCarthy B, Ivanov D (2022a) The digital supply chain—emergence, concepts, definitions, and technologies. In: MacCarthy B, Ivanov D (eds) The digital supply chain. Elsevier, Amsterdam, pp 3–14
MacCarthy B, Ivanov D (eds) (2022b) The digital supply chain. Elsevier, Amsterdam
Mirjalili S (2018) Evolutionary algorithms and neural networks: theory and applications. Springer Nature, Cham
Moghaddam M, Nof SY (2018) Collaborative service-component integration in cloud manufacturing. Int J Prod Res 56(1–2):677–691
Müller B, Töppel T (2014) Technologiescreening Laserstrahlschmelzen und Rapid Prototyping—3D Printing Studie. Fraunhofer-Institut für Werkzeugmaschinen und Umformtechnink IWU
Nair A, Vidal JM (2010) Supply network topology and robustness against disruptions - an investigation using a multi-agent model. Int J Prod Res 49(5):1391–1404
Nayak A, Levalle RR, Lee S, Nof SY (2016) Resource sharing in cyber-physical systems: modelling framework and case studies. Int J Prod Res 54(23):6969–6983

Nguyen T, Zhou L, Spiegler V, Ieromonachou P, Lin Y (2018) Big data analytics in supply chain management: a state-of-the-art literature review. Comput Oper Res 98:254–264
Nof SY (2007) Collaborative control theory for e-work, e-production, and e-service. Annu Rev Control 31(2):281–292
Nof SY, Morel G, Monostori L, Molina A, Filip F (2006) From plant and logistics control to multienterprise collaboration. Annu Rev Control 30(1):55–68
o. V (2017) Roboterabsatz legt deutlich zu. VDI-Nachrichten 40:3
Olsen TL, Tomlin B (2020) Industry 4.0: opportunities and challenges for operations management. Manuf Serv Oper Manag 22(1):113–122
Oztemel E, Gursev S (2020) Literature review of industry 4.0 and related technologies. J Intell Manuf 31:127–182
Panetto H, Iung B, Ivanov D, Weichhart G, Wang X (2019) Challenges for the cyber-physical manufacturing enterprises of the future. Annu Rev Control 47:200–213
Papadopoulos T, Gunasekaran A, Dubey R, Altay N, Childe SJ, Wamba SF (2017) The role of big data in explaining disaster resilience in supply chains for sustainability. J Clean Prod 142(2):1108–1118
Park H, Bellamy MA, Basole RC (2018) Visual analytics for supply network management: system design and evaluation. Decis Support Syst 91:89–102
Porter ME, Heppelmann JE (2014) How smart, connected products are transforming competition. Harv Bus Rev 92(11):64–88
Porter ME, Heppelmann JE (2015) How smart, connected products are transforming companies. Harv Bus Rev 93(10):96–114
Porter ME, Heppelmann JE (2018) Eine Brücke zwischen digitaler und physischer Welt. Harvard Business Manager
Queiroz MM, Telles R, Bonilla SH (2019) Blockchain and supply chain management integration: a systematic review of the literature. Supply Chain Manag 25(2):241–254
Queiroz MM, Ivanov D, Dolgui A, Wamba SF (2022) Impacts of epidemic outbreaks on supply chains: mapping a research agenda amid the COVID-19 pandemic through a structured literature review. Ann Oper Res 319:1159–1196
Rai R, Tiwari MK, Ivanov D, Dolgui A (2021) Machine learning in manufacturing and industry 4.0 applications. Int J Prod Res 59(16):4773–4778
Roeck D, Sternberg H, Hofmann E (2020) Distributed ledger technology in supply chains: a transaction cost perspective. Int J Prod Res 58(7):2124–2141
Rolf B, Jackson I, Müller M, Lang S, Reggelin T, Ivanov D (2023) A review on reinforcement learning algorithms and applications in supply chain management. Int J Prod Res 61(20):7151–7179
Rolston P, Blackhurst J (2020) Industry 4.0 and resilience in the supply chain: a driver of capability enhancement or capability loss? Int J Prof Res 58(16):5006–5019
Rossit DA, Tohmé F, Frutos M (2019) Industry 4.0: smart scheduling. Int J Prod Res 57(12):3802–3813
Ruel S, El Baz J, Ivanov D, Das A (2024) Supply chain viability: conceptualization, measurement, and nomological validation. Ann Oper Res 335:1107–1136
Schmitt R, Pfeifer T (2010) Qualitätsmanagement: Strategien—Methoden—Techniken. Carl Hanser Verlag, München
Sheu JB, Choi T-M (2023) Can we work more safely and healthily with robot partners? A human-friendly robot-human coordinated order Fulfillment scheme. Prod Oper Manag 32(3):794–812
Simchi-Levi D, Wu MX (2018) Powering retailers' digitization through analytics and automation. Int J Prod Res 56(1–2):809–816
Slack N (1987) The flexibility of manufacturing systems. Int J Oper Prod Manag 7(4):35–45
Sokolov B, Ivanov D, Dolgui A (eds) (2020) Scheduling in industry 4.0 and cloud manufacturing. Springer, New York. ISBN 978-3-030-43176-1
Stecke KE (1983) Formulation and solution of nonlinear integer production planning problems for flexible manufacturing systems. Manag Sci 29(3):273–288
Sterman J (2000) Business dynamics: systems thinking and modeling for a complex world. McGraw-Hill, London

Stevenson W (2018) Operations management, 13th edn. McGraw Hill, New York
Strozzi F, Colicchia C, Creazza A, Noè C (2017) Literature review on the 'smart factory' concept using bibliometric tools. Int J Prod Res. https://doi.org/10.1080/00207543.2017.1326643
Sun YI (2017) Die nächste Werkbank der Welt. Harvard Business Manager
Surana A, Kumara S, Greaves M, Raghavan UN (2005) Supply-chain networks: a complex adaptive systems perspective. Int J Prod Res 43(20):4235–4265
Swaminathan JM, Smith SF, Sadeh NM (1998) Modeling supply chain dynamics: a multiagent approach. Decis Sci 29(3):607–632
Tang CS, Veelenturf LP (2019) The strategic role of logistics in the industry 4.0 era. Transp Res E Logist Transp Rev 129:1–11
Tozanli Ö, Saénz ME (2022) Unlocking the potential of digital twins in supply chains. MIT Sloan Manag Rev
Tsipoulanidis A (2015) Von lean thinking zu supply chain & operational excellence durch die digitale transformation. Der Tagesspiegel, 2.11.2015
Tsipoulanidis A (2017a) Leandustry 4.0 – the efficient way from company to customer at times of the digital transformation – how lean thinking is leading to smart supply chain management and operational excellence. ICCMI conference 2017, Thessaloniki, Greece
Tsipoulanidis A (2017b) With lean thinking and industry 4.0 to operational excellence. Supply Chain Management III / 2017:13–19
Tsipoulanidis A (2020) Digital & lean, così la supply chain diventa più efficiente. Logistica Management, no. 308, Ottobre 2020: 86–92
VDI Verein Deutscher Ingenieure e.V (2013) Cyber-physical systems: Chancen und Nutzen aus Sicht der automation. Thesen und Handlungsfelder
Waller MA, Fawcett SE (2013) Data science, predictive analytics, and big data: a revolution that will transform supply chain design and management. J Bus Logist 34(2):77–84
Wamba SF, Chatfield AT (2009) A contingency model for creating value from RFID supply chain network projects in logistics and manufacturing environments. Eur J Inf Syst 18(6):615–636
Wamba SF, Akter S, Edwards A, Chopin G, Gnanzou D (2015) How 'big data' can make big impact: findings from a systematic review and a longitudinal case study. Int J Prod Econ 165:234–246
Wamba SF, Ngai EWT, Riggins F, Akter S (2017) Transforming operations and production management using big data and business analytics: future research directions. Int J Oper Prod Manag 37(1):2–9
Warnecke HJ, Braun J (1999) Vom Fraktal zum Produktionsnetzwerk: Unternehmenskooperation erfolgreich gestalten. Springer, Berlin
Westerman G, Bonnet D, McAfee A (2014) Leading digital—turning technology into business transformation. Harvard Business Review Press, New York
Wiendahl H-P, Reichardt J, Nyhuis P (2015) Handbook factory planning and design. Springer, Berlin
Wiener N (1948) Cybernetics or control and communication in the animal and the machine. Wiley, New York
Womack JP, Jones DT (1998) Lean thinking—banish waste and create wealth in your corporation. Touchstone
Xiao J, Ma S, Wang S, Huang GQ (2024) Meta-inventory management decisions: a theoretical model. Int J Prod Econ 275:109339
Yin Y, Stecke KE, Li D (2018) The evolution of production systems from industry 2.0 through industry 4.0. Int J Prod Res 56(1–2):848–861
Zhang G, MacCarthy B, Ivanov D (2022) The cloud, platforms, and digital twins—enablers of the digital supply chain. In: MacCarthy B, Ivanov D (eds) The digital supply chain. Elsevier, Amsterdam, pp 77–91
Zhao K, Zuo Z, Blackhurst JV (2019) Modelling supply chain adaptation for disruptions: an empirically grounded complex adaptive systems approach. J Oper Manag 65(2):190–212
Zheng G, Ivanov D, Brintrup A (2025) An adaptive federated learning system for information sharing in supply chains. Int J Prod Res. https://doi.org/10.1080/00207543.2024.2432469

Zhong RY, Xu C, Chen C, Huang GQ (2017) Big data analytics for physical internet-based intelligent manufacturing shop floors. Int J Prod Res 55(9):2610–2621
Zühlke D. (2009). SmartFactory – A Vision becomes Reality. IFAC Proceedings Volumes 42(4):31–39

Internet Links

Banker S (2019) Procter & Gamble embraces continuous planning and execution. Forbes, 05 November 2019. [online]. https://www.forbes.com/sites/stevebanker/2019/11/05/procter%2D%2Dgamble-embraces-continuous-planning-and-execution/#69672ecb3ed1. Accessed 30 June 2020
bcg perspectives (Boston Consulting Group) (2015) Industry 4.0—the future of productivity and growth in manufacturing industries. https://www.bcgperspectives.com/content/articles/engineered_products_project_business_industry_40_future_productivity_growth_manufacturing_industries/?chapter=3. Accessed 11 Sep 2015
Capgemini Invent (2020) Operational excellence in ihrem purchase-to-pay prozess. https://www.capgemini.com/de-de/resources/operational-excellence-p2p-prozess/. Accessed 20 Dec 2020
HPA (2017). http://www.supplychain247.com/article/ups_drone_meets_ups_delivery_truck/Drones
IBM (2017). https://www-03.ibm.com/press/us/en/pressrelease/50816.wss. Accessed 20 Nov 2017
IFR. https://ifr.org/ifr-press-releases/news/record-2.7-million-robots-work-in-factories-around-the-globe. Accessed 22 Jan 2021
Pankow G (2019) Prestigeprojekt am Ende: Adidas schließt Speed-Factory. https://www.produktion.de/wirtschaft/prestigeprojekt-am-ende-adidas-schliesst-speed-factory-327.html. Accessed 23 Dec 2020
Reeco Automation (2019) BMW mini robotic riveting. [Online]. https://www.reeco.co.uk/projects/bmw-mini-robotic-riveting/. Accessed 29 Dec 2019
Robotics and Automation (2018). PSA Group modernises its European manufacturing with UR cobots. [Online]. https://roboticsandautomationnews.com/2018/03/22/psa-group-modernises-itseuropean-manufacturing-with-ur-cobots/16552/. Accessed 30 Dec 2019
Selyukh A (2018) Optimized prime: how AI and anticipation power Amazon's 1-hour deliveries. npr: https://www.npr.org/2018/11/21/660168325/optimized-prime-how-ai-and-anticipation-power-amazons-1-hour-deliveries?t=1577276552724
SupplyChain247 (2017). http://www.supplychain247.com/article/ups_drone_meets_ups_delivery_truck/Drones
T-Systems (2018). https://www.t-systems.com/de/en/industries/automotive/scm-solutions/smart-logistics/digital-transport-management-252148
Universal Robots (2019) Group PSA's Sochaux plant in France. [Online]. https://www.universal-robots.com/case-stories/group-psa-s-sochaux-plant-infrance/. Accessed 30 Dec 2019

Pricing and Revenue-Oriented Capacity Allocation

17

Section 17.1 introduces a comprehensive case study from the transportation sector. We discuss methods for the determination of best-selling prices in Sect. 17.2 for situations with sufficient capacity. Suppose the available capacity is scarce and several products and/or customers compete for it. In that case, it is necessary to adopt the price-setting process (Sect. 17.3). Section 17.4 extends the pricing tools to resource networks if more than one potentially scarce resource is involved in the production of a service or product. Prices are set prior to incoming demand, and therefore, pricing requires the involvement of demand estimations. If these estimations are inappropriate, the prices will also be. To overcome this methodological shortcoming in pricing, the concept of "Dynamic Pricing" is proposed, which aims to adjust prices reactively to observed market demand. We will introduce Dynamic Pricing in Sect. 17.5 and summarize the main findings from this chapter along with outlining further issues in RM in Sect. 17.6.

Learning Objectives for This Chapter

Pricing refers to the activity of determining the sales price for a material product or a service. Intuitively, one would derive the price of a product or service from its production costs. However, this is only one pricing strategy: you calculate the corresponding production costs, add a certain profit margin, and get the sales price by summing up the production costs and the profit margin. However, how can you set the sales price if your production costs vary significantly from each unit produced due to non-smooth cost curves? What is an adequate price for a product if you know that different customers come along with different levels of willingness to pay for

Find additional case studies and video streams in the E-Supplement to this book on www.global-supply-chain-management.de!

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_17

it? In these cases, it might become necessary to set an individual price for the next product unit sold and/or the next customer who wants to have this unit.

You might think that pricing belongs to marketing. So why should you as a SCOM manager be educated in pricing? Assume that your company has developed a new product. Its introduction into the market should be made as successfully as possible. Therefore, an offensive advertising strategy accompanies the physical provision of the product. A significantly discounted product price is announced. Consequently, a lot of prospective customers will buy this product and a huge number of units must be transported to the retailers as a result. This is only possible if the responsible transport manager in the supply chain of the new product is informed about the expected quantities to be moved so that he can provide the necessary transport resources. Assume that after a certain period, the market introduction of the new product has been concluded successfully, and the price discount has ended. Consequently, the price increase induces a reduction in the demand from the market so that fewer quantities must be moved into the retail outlets. Again, it is necessary to inform the transport manager about this pricing variation so that he can implement measures to allocate less transport capacity to this product. We have studied the interdependencies among expected or forecasted demand as well as operations planning as sales and operations planning (S&OP) in detail in Chap. 12. Assume now that you use your own capacities to manufacture the product: price variations are now coupled with capacity allocations.

In general, the demand for a product varies if the price is adjusted. A price lift results in a reduction in demand expressed by the number of sold units. A price reduction typically results in a lift in the number of units demanded. In this way, the interaction of price and demand affects supply chain operations planning. As you have read in the other (previous) chapters of this textbook, a huge amount of effort is spent adjusting the capacities of a supply network to fit as tightly as possible to demand. Therefore, a price reduction inducing a demand increase will immediately lead to the scarceness of these capacities. Vice versa, if we increase the supply network capacities to increase the throughput with the goal of lifting the quantity provided to the market, a price reduction for this product will often be necessary. We can observe a correlation between available capacities and pricing decisions. This is the most important reason a SCOM manager should be able to understand pricing decisions.

If your production capacity is scarce, but customers are willing to pay high prices for your products (services), how can you decide which customer(s) you are going to serve at which prices? Revenue Management (RM) is a managerial discipline that tries to formulate answers to questions like these related to pricing decisions and capacity allocations.

RM provides tools that enable us to set every price so that we earn the maximal sum of revenues with the deployed (or available) resources and allocated capacities considering available demand as well as resources. It is unconditionally necessary to manage several challenges to realize this naive idea:

- We have to consider in our pricing scheme the willingness of each individual customer to pay for each product. This is referred to as the customer's perspective in RM, especially in pricing.
- We need to take the available capacities of the requested resources into account if we are setting prices and allocate capacities for customers. In addition, we need to coordinate the interaction of several resources if the provision of a product or service requires the deployment of two or more resources. This is the capacity perspective in pricing and therefore the service provider's perspective in RM.
- We must be able to quantify upcoming customer demand to make the best capacity allocation decisions. This is the forecasting perspective in RM.

After you have carefully worked on the content of this chapter, you:

- Will be familiar with the concept of demand-oriented pricing as a powerful tool to maximize the revenues from a market
- Are able to understand different pricing scenarios and their unique challenges
- Will know that pricing relies on the allocation of capacity to market segments and/or customers and you will be trained to identify adequate capacity allocation decisions
- Will be equipped with tools (models and algorithms) that can support you in your pricing decisions. Especially, you will be able to exploit price–demand and demand–price functions for the determination of profit-maximal prices

17.1 Case Study: FRISIA COASTAL SHIPPING: The Story of Jördis and Tjark

Tjark grew up in the North-Western part of Germany called East Frisia. It is a maritime landscape situated between the North Sea (in the North), the Netherlands (in the West), and the Weser River (in the East). Born in the small harbor village Neuharlingersiel, he spent his childhood as the son of a fisherman who owned a fishing boat. Tjark accompanied his father as often as possible during his daily sailing trips. He liked to be at sea. This was one of the reasons why he decided to become a professional seafarer. After he had finished school, he decided to sign a contract with the Navy to work as a sailor for 2 years. After he had left military service, he started a study program to become a captain. He concluded his training with the certificate "Master Mariner" and a graduation document as a qualified nautical officer. Soon he found a job as the commander of a big container vessel. Over 10 years, he sailed over all the oceans and harbored in all major ports in the world. When he was 38, he married a woman from his home village. Her name is Jördis. She gave birth to twins soon after. Since Tjark wanted to see his two children grow up and to support his wife, he decided to resign from his captain's job. Today, he has just returned from his last trip to his home town Neuharlingersiel and he has a plan to earn enough money for the life of his family in the future.

Since he became the captain of a large container vessel, Tjark also dreamed of following in the footsteps of his father and becoming a captain of a coastal boat sailing along the Frisian Coast. He does not necessarily want to be a fisherman but to spend as much time as possible on a smaller boat in his home waters and be his own boss. From time to time, Tjark studied websites that listed coastal vessels for sale. During his last journey on board the big container vessel cruising through the Pacific Ocean, he found out that the coaster FRISIAN STORM is for sale. The FRISIAN STORM is a very specific vessel. Its full length is around 30 m and it has a 6 m beam. It has been designed to sail through the flat waters around the East Frisian Islands Borkum, Juist, Norderney, Baltrum, Langeoog, Spiekeroog as well as Wangerooge and along the East Frisian Coast (Fig. 17.1). With her very flat ship's hull and a small draught, the FRISIAN STORM can operate in this navigated area independently of the cyclic rise and fall of the sea level (that is regularly around 3.5 m every 6 h). Furthermore, she offers a huge transport capacity of 180 tons and has a maximal speed of approximately 18 knots, which is very fast. In addition, she has a crane on board, so she is independent of landside transshipment equipment. The intended price is fair since the FRISIAN STORM is well in shape and well maintained.

Tjark's business idea is to use a vessel like the FRISIAN STORM to offer a regular, reliable, and quick transport service between selected ports on the Frisian mainland and the aforementioned East Frisian Islands. Such a service is not available at the moment, so Tjark can exploit and close a gap in the supply of the islands. The service can follow a pre-published timetable throughout the year.

To realize his dream of operating a vessel like the FRISIAN STORM, Tjark needs to acquire cargo. Being originally from the East Frisian Coast, Tjark knows the local transport market situation quite well. Every year, 1.4 million tourists travel to the islands and booking approximately 10 million overnight stays. Of course, the peak season is the summer (May until September) when a lot of general cargo (food, beverages, day-to-day necessities) is requested from the islands to provide enough supply to host the tourists. During the winter period, a lot of construction material is needed to perform renovations as well as to construct new buildings on the islands.

Fig. 17.1 The East Frisian Islands and Mainland (c) google.com

Tjark is convinced that his business idea can be realized and he is able to provide the expected transport services. He tells Jördis, his wife, about these plans.

Jördis likes her husband's idea since she can understand his wish to return to the root of the family and become an independent captain with his own vessel sailing through the native coastal area. However, being the mother of two small children she is a little afraid about the economic part of her husband's plan. Therefore, she asks her parents their opinion. Her parents have never been involved in the seafarer's business, but they have been independent business persons for several decades. Together, they run and own a company that sells insurance policies, especially for business customers. Due to this job, both of them know lots of other business persons who are active in East Frisia. When her mother listens to her daughter's sorrows related to her husband's plan to open a small shipping company, she has an idea. She asks Jördis to convince Tjark to wait for a few days until he proceeds developing his new business.

After 2 days, Tjark gets a telephone call from his mother-in-law. She tells him that she has found someone who could support him in starting his new business. Tjark is curious and agrees to meet this person for a business lunch. When Tjark enters the selected restaurant, his mother-in-law introduces Ole. Ole is the project manager of the regional construction company FRISIA REAL ESTATE who is involved regularly in most of the construction projects on the East Frisian Islands. Currently, Ole's company is quite unsatisfied with the existing opportunities to bring the required construction material from the mainland to the islands. With regard to improving the reliability of the material provision, Ole formulates the following offer: during the peak period in winter (October to April) FRISIA REAL ESTATE guarantees to book 60 tons payload every day and during the summer period FRISIA REAL ESTATE will pay for 20 tons payload per day. Furthermore, a 5-year contract will be agreed between Tjark and FRISIA REAL ESTATE in which the freight rate per ton is fixed to 25 € (summer period) and 15 € (winter period).

Tjark has calculated the average monthly operation costs of the FRISIA STORM including staff, bunker oil, administration, insurance rates, and loan interest by 11,000 €. Considering the aforementioned offer, he will have guaranteed monthly revenues from FRISIA REAL ESTATE of 15,000 € (summer) and 27,000 € (winter) for the first 5 years. Consequently, he decides (1) to start his business with the name FRISIA COASTAL SHIPPING and (2) to sign a contract with FRISIA REAL ESTATE. This contract convinces his bank to support him with a loan to buy the FRISIA STORM. Next, he focuses his attention on the development of his business concept for offering maritime transport services between the East Frisian Islands and the East Frisian mainland. He derives the following ideas:

- The contract with FRISIA REAL ESTATE covers all monthly fixed costs.
- The FRISIA STORM's residual payload capacity can be used to serve several other customers and generate additional revenues. These revenues (reduced by a small amount for additional bunker costs and some administrative costs) increase his monthly gain.

- Tjark defines the following transport services. Every day one shuttle between his home base port Neuharlingersiel and one of the seven islands will be established. On Monday, the FRISIA STORM sails from Neuharlingersiel to Borkum and returns the same day. Every Tuesday, Neuharlingersiel is connected with Juist and so on.
- Based on the defined transport services, Tjark wants to offer prospective customers the following products: an "individual transport" of a certain quantity to be moved between Neuharlingersiel and an island (or in the reverse direction) or a "combined transport" that enables a customer to book capacity on two or more services (including a return service) at the same time.

Tjark discusses these pillars of his new business with Jördis. She likes the ideas. However, she asks his husband a very critical question: "Tjark, which price do you want to assign to your prospective customers served in addition to the FRISIA REAL ESTATE CONTRACT?" Currently, Tjark has no answer to this question, but an idea comes up. He remembers a meeting he had with his former classmate Karl, who is now in the maritime cruise business. During this meeting, Karl talked about how cruise shipping companies sell the rooms on the big cruise ships. He decided to contact Karl. A few days later, Tjark calls Karl. Unfortunately, Karl has no explicit knowledge of how the cabin prices are set, but he says that a department called "Revenue Management" takes care of the pricing. From a quick Google query, Tjark learns that revenue management (RM) is a managerial discipline that, among other issues, takes care of the pricing of products and services. A few textbooks are suggested; Tjark selects one and starts to read.

Chopra's textbook (2020) suggests that a balance between the demand and supply associated with a service and/or a product is of critical importance for the creation of value since it prevents costly capacity adaptations over time. In this context, Talluri and Van Ryzin (2004) state that RM follows a strategy that is complementary to SCM, since it tries to level out the fulfilled demand. According to Cachon and Terwiesch (2019, p. 402), "[…] revenue management is also referred to as yield management. Broadly speaking, revenue management is the science of maximizing the revenue earned from a fixed supply." Here, Tjark nods his head. "Yes, this is what I need: a mechanism that helps me to ensure that the FRISIA STORM sails completely filled on each trip at reasonable prices." He continues to read and learn. In particular, Tjark is looking for answers to the following questions related to the pricing of the services of his company:

- What are the interdependencies between the price for a product and the needed resource capacity deployed to produce the service? What is an "optimal" or "best" price for a service and how can it be identified? (Sect. 17.2)
- How can optimal prices for several customers be fixed if these customers compete for a scarce resource, i.e., if there is not enough capacity to allocate enough for all customers? (Sect. 17.3)

- How can he identify the best prices if the services are produced in a network of more than one resource, i.e., if a customer wants to bargain on prices for several different services in parallel? (Sect. 17.4)
- How can he adjust prices during the sales activities if he finds that the originally proposed prices do not work properly, i.e., are too high or too low? (Sect. 17.5)

17.2 Noncompetitive Pricing

Tjark has another former classmate who is now the sales director of a regional retail company called FRISIA-CC. FRISIA-CC is an organization that runs approximately 100 retail outlets in the East Frisian region. Among these outlets, some are on the East Frisian Islands, so FRISIA-CC has to ensure that the island-based outlets are reliably supplied with the complete assortment. This former classmate's name is Susanne, and she is responsible for organizing the supply of the retail outlets. Tjark contacts Susanne and asks to meet. At the beginning of this meeting, Tjark explains his new business. Furthermore, he tries to find out whether Susanne is interested in starting a business relationship between FRISIA-CC and FRISIA COASTAL SHIPPING. Susanne is interested since it was reported to her that the current supply is unreliable and expensive. She decides to give Tjark a chance and starts the following conversation:

Susanne: Tjark, I'm very interested in giving you an order to contribute to the supply of our retail outlets on the island. What kind of service can you offer?

Tjark: Well, I want you to offer a shuttle service between Neuharlingersiel and the seven islands with one weekly return service to each island. For each service, we should fix the capacity to be moved expressed in tons in a contract as well as the price or rate per ton. You can then let me move everything you want to be moved as long as the sum weight does not exceed the agreed capacity. Every week, you pay for the agreed capacity. Would this be an acceptable setup for you?

Susanne: Yes, we should start with this setting. I think the most important issue is the rate per ton to be fixed. My major challenge is the heterogeneity of the retailers on the islands. Depending on the transportation costs, the quantity to be moved (per week) varies. So, after you have suggested your rate, I will be able to give you the tons you can move per week to an island.

Tjark: Ok, I think I got it. The rate determines the demand, right?

Susanne: That's correct! Since we are both interested in setting up efficient and reliable transport services, I have decided to provide you with some information about our customers and their willingness to pay. From recorded data, we can conclude that not more than 50 tons have been demanded from a single island in a week, independently of the transport rate. Furthermore, we know that not a single retailer will use our transport service if the rate per ton is higher than 100 € per ton. We suppose that they are deploying other transport service providers to realize the supply.

Tjark: So, the rate per ton can vary between 0 € per ton as the minimum and 100 € per ton at most?
Susanne: Yes, this is correct!
Tjark: Do you have an idea of how price-sensitive the retailers are? I want to know how many tons we lose if we increase the rate and how many additional tons we can acquire if we reduce the rate per ton.
Susanne: To be honest, we have no explicit knowledge about it. In the past, we documented what happened with the total demanded quantity if we adjusted the transport expenditures after we incorporated several other service providers one after another. Our feeling is that there is a proportional correlation between the announced rate and the demanded quantity.
Tjark: Ok, Susanne, I think that I have enough information now to work out a rate per ton. Thank you very much for your support. I will submit the rate to you tomorrow so that we can start operations as early as possible.

After this conversation, Tjark summarizes what his exercise is. He has to fix a freight rate p^{opt} that will enable him to earn as much revenue with FRISIA-CC as possible. This rate must be set between $p^{\min} = 0$ and $p^{\max} = 100$ €. If he would offer the transport for free, $N^{\max} = 50$ tons payload is possible. If he increases the rate by 1 € he will lose α tons of payload. The payload loss rate is

$$\alpha = \frac{N^{\max} - 0}{0 - p^{\max}} = \frac{50\,\text{tons}}{-100\text{€}} = -\frac{1\,\text{ton}}{2\text{€}}.$$

$$\text{PDF}(p) = N^{\max} + \alpha \cdot p = 50\,\text{tons} - \frac{1}{2}\frac{\text{tons}}{\text{€}} \cdot p \tag{17.1}$$

Now he can write down the acquired payload as a function PDF(p) of the price p per sold unit (ton) as shown in Eq. (17.1). This function is called the *price–demand function*.

This function is plotted in Fig. 17.2 as a think black line. If Tjark sets the rate ("revenue per ton") to$p = 20$ € then he acquires 40 tons payload. For $p = 50$ € he will sell 25 tons and for $p = 70$ € the total payload sold will be 15 tons.

$$\begin{aligned} \text{REV}(p) &= \text{PDF}(p) \cdot p = \left(N^{\max} + \alpha \cdot p\right) \cdot p = N^{\max} \cdot p + \alpha \cdot p^2 \\ &= N^{\max} \cdot p + \frac{N^{\max} - 0}{0 - p^{\max}} \cdot p^2 \end{aligned} \tag{17.2}$$

To calculate the resulting sum of revenues gained in dependence of the rate p, Tjark expresses the revenues by the function REV(p) as the multiplication product between the determined rate p and the induced sales quantity PDF(p) as expressed by Eq. (17.2). For the three aforementioned rates, the revenues are represented by the areas bounded by dotted lines (REV(20) = 20•40 = 800 €), continuous lines (REV (50) = 50 × 25 = 1250 €), as well as dashed lines (REV(70) = 70 × 15 = 1050 €).

$$\text{REV}'(p) = N^{\max} - 2\frac{N^{\max}}{p^{\max}} \cdot p \tag{17.3}$$

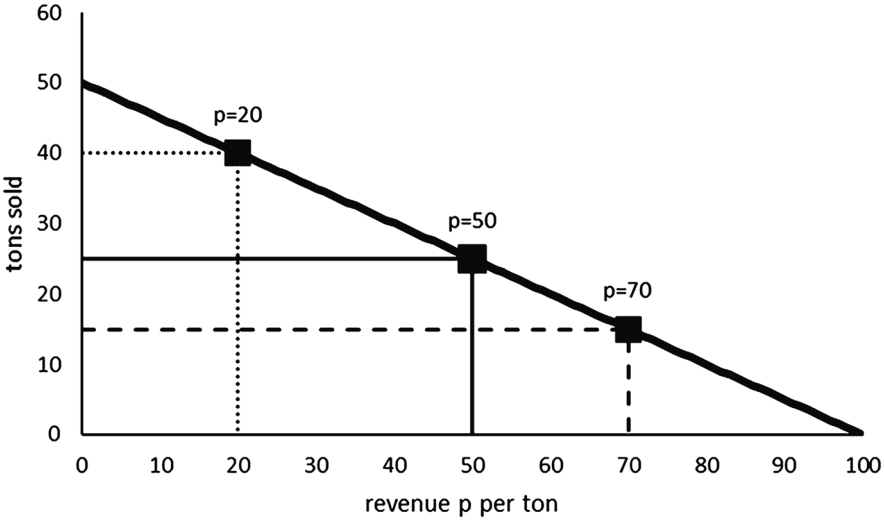


Fig. 17.2 Price–demand function for the customer FRISIA-CC

Tjark wants to know for which rate p he earns the maximal revenue sum. For this, he is going to identify an optimal value p^{opt} that maximizes the function REV. First, Tjark differentiates REV and formulates its first derivative REV′ as given in Eq. (17.3). Next, Tjark is looking for a root of REV′. The only one here is $\frac{p^{max}}{2}$. Finally, since the second derivative function of REV is the constant function $-2\frac{N^{max}}{p^{max}}$ the root $p^{opt} = \frac{p^{max}}{2}$ gives the only maximum of the revenue function REV. In our case, it is p^{opt} = 50 €/ton. If Tjark fixes the rate for FRISIA-CC to 50 €, he maximizes his revenues from this customer. He has to allocate 25 tons for FRISIA-CC. The sum of earnings from FRISIA-CC is 50 €/ton × 25 ton = 1250 €. The fixed rate is called the *individually optimal rate (price) for the customer FRISIA-CC*.

A visualization of the revenue function is provided by Fig. 17.3. It can be seen that the maximal revenue sum is achieved at the rate of 50 €/ton. Table 17.1 compiles the capacity allocation for the shuttle services in the winter period made so far. We see that the agreed rates are different for the two customers.

Tjark finally summarizes the steps for the calculation of the individually optimal rate:

(a) Determine the price–demand function PDF(p) that maps the rate p to the number of sold items.
(b) Determine the revenue function REV(p) = PDF(p)•p to calculate the overall sum of earned revenues if the rate is set to p.
(c) Identify the rate that maximizes REV(p) by calculating the root position(s) of the first derivative function of REV(p).

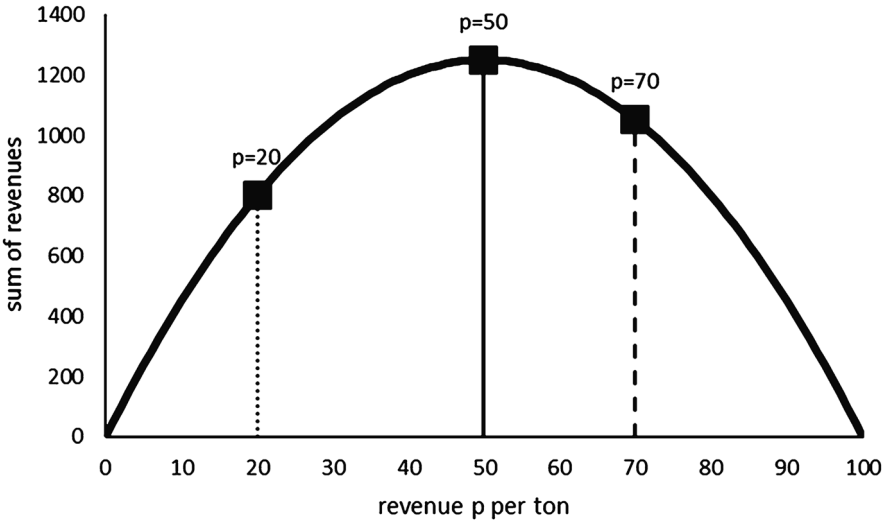


Fig. 17.3 Revenue function for the customer FRISIA-CC

Table 17.1 Capacity allocations for the winter period (agreed rate for the customer)

	Service to						
	Borkum	Juist	Norderney	Baltrum	Langeoog	Spiekeroog	Wangerooge
Customer	Mo.	Tu.	Wed.	Th.	Fr.	Sat.	Sun.
FRISIA REAL-ESTATE	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)
FRISIA-CC	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)
Unused	95 to	95 to	95 to	95 to	95 to	95 to	95 to

This procedure can be applied even if the PDF is nonlinear but differentiable. We demonstrate this by means of another prospective customer that is contacted by Tjark. NORTH-WIND is a maintenance and repair company specializing in MRO-services for off-shore wind generators. This company is responsible for a huge number of wind power stations in the North of the seven East Frisian Islands. They have a combined repair shop and spare part warehouse on each island. NORTH-WIND and FRISIA COASTAL SHIPPING have agreed to start a common business to set up regular supply and disposal of items needed for the NORTH-WIND repair shops.

$$\mathrm{PDF}_{\mathrm{NW}}(p) = \sqrt{\left(N^{\max}\right)^2 - \left(\frac{1}{2}p\right)^2} \tag{17.4}$$

For the determination of the rate to be proposed to NORTH-WIND, Tjark first estimates the $\mathrm{PDF}_{\mathrm{NW}}(p)$ given in Eq. (17.4), which is not an affine linear function of

the rate p, but a strong monotonically decreasing function of the rate p. Tjark decides to use the parameter $N^{\max} = 70$. The resulting curve is shown in Fig. 17.4.

The revenue function $\mathrm{REV}_{\mathrm{NW}}(p)$ stated in Eq. (17.4) now has a maximum rate of around 99 € per ton (cf. Fig. 17.5). This corresponds to an allocation of 49 tons for NORTH-WIND.

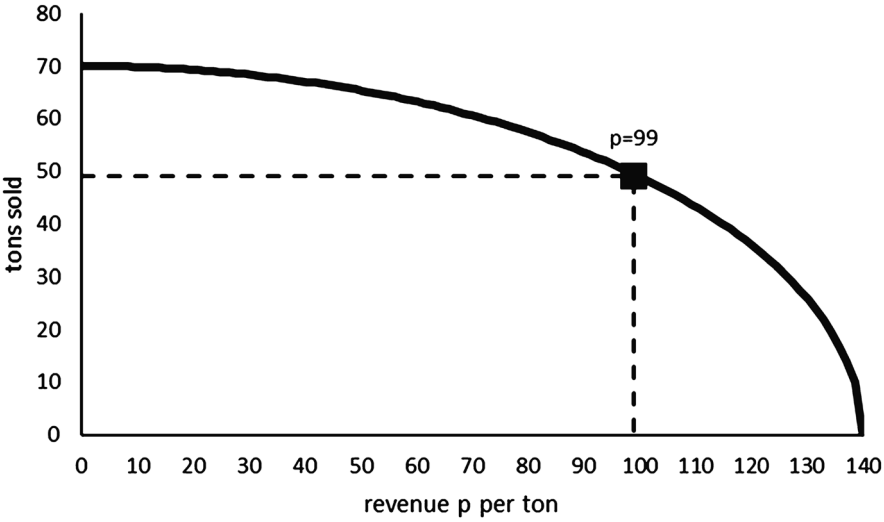


Fig. 17.4 PDF for the customer NORTH-WIND

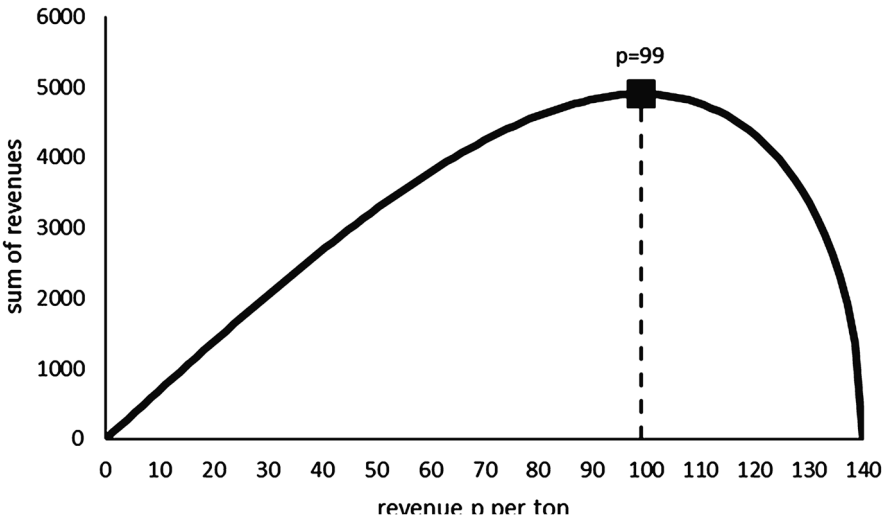


Fig. 17.5 Revenue function for the customer NORTH-WIND

17.3 Pricing with Scarce Capacities

After Tjark has allocated the capacity for the customer NORTH-WIND, he summarizes capacity allocations made so far.

Table 17.2 shows the daily reservations as well as the agreed-upon rates. He discovers that 46 tons are still available to be sold and allocated to customers. This is the reason Tjark decides to put more effort into the acquisition of additional customers and sell the residual capacity. He sets up a website and sends e-mails to potential customers.

Tjark was able to reserve all required capacity to realize the optimal rates, i.e., the resource capacity was not scarce. However, what happens if not enough capacity is available? Such a shortage of capacity appears in two different situations. We will investigate both. We assume that the contract with FRISIA REAL-ESTATE is fixed and can no longer be modified so that we can allocate only 120 tons per day.

Situation 1: Shortage in the Capacity Allocation for Only One Customer

$$\mathrm{PDF}_{\mathrm{X}}\left(p\right) = 250 \cdot \mathrm{e}^{-0.3 \cdot p} \tag{17.5}$$

After Tjark started his marketing activities, he receives a call from another potential customer named Customer-X. This customer offers a maximal demand $N^{\max} = 250$ units (if rate $p = 0$), but the demand decreases if the rate increases according to $\mathrm{PDF}_{\mathrm{X}}(p)$ defined in Eq. (17.5) and plotted in Fig. 17.6.

The plot of the corresponding revenue function $\mathrm{REV}_{\mathrm{X}}(p)$ is shown in Fig. 17.7. The application of the procedure (a)–(c) determines the individually optimal rate $p^{\mathrm{opt}} = 34$ €/ton. This rate can be realized if CUSTOMER-X can be served with approximately 90 tons. Tjark is alarmed since the residual payload capacity on board of the FRISIA STORM is only 46 tons. He decides to increase the rate offered to the customer step by step until $\mathrm{PDF}_{\mathrm{X}}$ indicates a demand not exceeding 46 tons: Tjark has to raise the rate to 57 € per ton.

Table 17.2 Capacity allocations for the winter period (agreed rate for the customer)

	Service to						
	Borkum	Juist	Norderney	Baltrum	Langeoog	Spiekeroog	Wangerooge
Customer	Mo.	Tu.	Wed.	Th.	Fr.	Sa.	Sun.
FRISIA REAL-ESTATE	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)	60 to (15 €)
FRISIA-CC	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)	25 to (50 €)
NORTH-WIND	49 to (99 €)	49 to (99 €)	49 to (99 €)	49 to (99 €)	49 to (99 €)	49 to (99 €)	49 to (99 €)
Unused	46 to	46 to	46 to	46 to	46 to	46 to	46 to

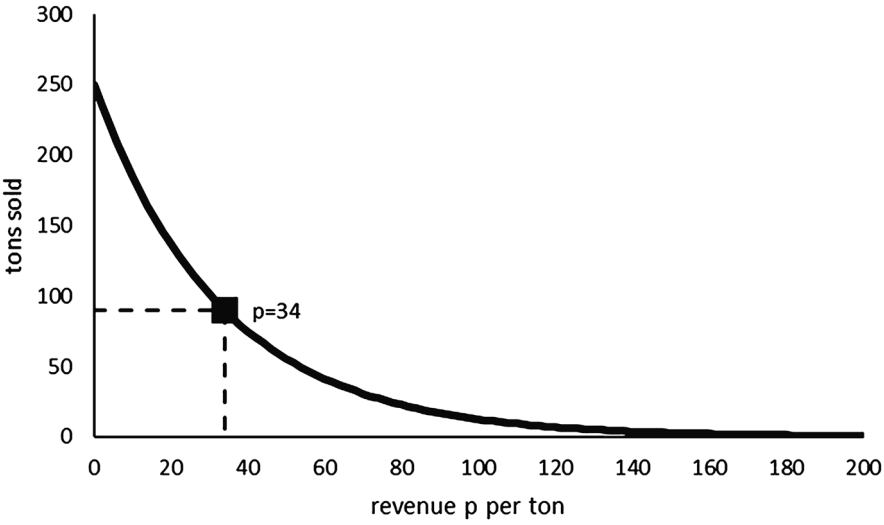


Fig. 17.6 PDF for the CUSTOMER-X

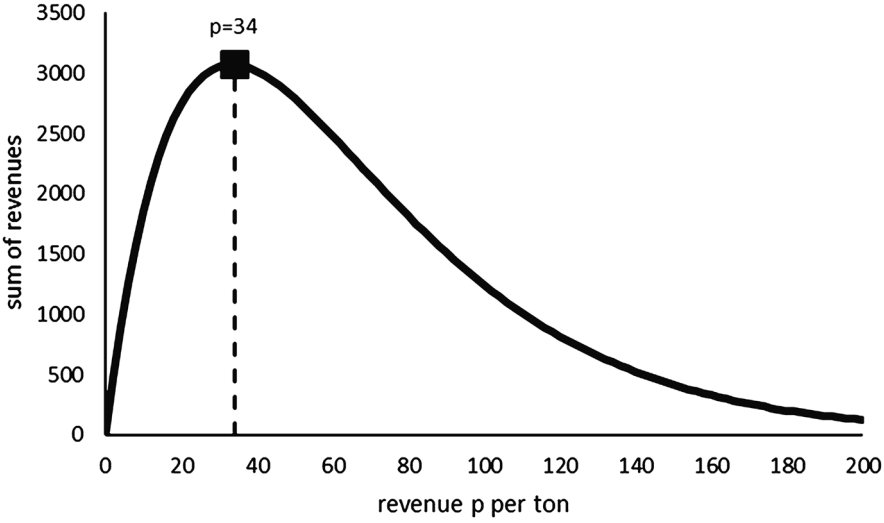


Fig. 17.7 Revenue function REV_X for the CUSTOMER-X

Tjark has solved the capacity shortage problem, but he wants to know the "price" of shortening the allocated capacity for CUSTOMER-X. If he would have sufficient capacity, he could earn 34 €/ton × 90 ton = 3060 €. With the increased rate of 57 € per ton, he is able to sell 46 tons so that the revenue equals 2622 €. If he shortens the capacity allocated to the "last served" customer he cannot realize the 438 € ("lost revenue").

Situation 2: Shortage While the Capacity Allocation for Several Customers
In the previous situation, CUSTOMER-X was the last customer that asked to be served. Immediately before, NORTH-WIND was accepted and before this customer, FRISIA-CC was processed. Tjark has allocated capacity for the three customers in the sequence of their appearance ("First-Come/First-Serve" or FCFS). He wants to understand the impacts of determining rates (prices) sequentially. He understands that for the early customers the "best" pricing decisions are made as long as sufficient capacity is available. As soon as the residual capacity is less than the optimal amount of capacity needed, the previous decisions might become sub-optimal. Here, sub-optimality refers to necessary revisions of previously made pricing and capacity allocation decisions. However, it is impossible to revise a pricing decision after a rate (as well as the accepted quantity) has been announced to the customer.

With respect to the aforementioned findings, Tjark decides to reorganize the pricing process. Instead of immediately processing a customer, Tjark wants to "collect" several pricing requests in order to decide simultaneously about the prices and the capacity allocations for all collected customers. His intention is to improve his pricing decisions by assigning scarce capacity to the most profitable customers (who exhibit the highest willingness to pay) instead of the first arriving customers.

Of course, if n customers are collected, it becomes necessary to set n rates $p_1^{\text{opt}},\ldots,p_n^{\text{opt}}$ and n allocated quantities $\text{PDF}_1\left(p_1^{\text{opt}}\right),\ldots,\text{PDF}_n\left(p_n^{\text{opt}}\right)$ in parallel while taking into account the available scarce resource capacity C^{total}. In addition, these rates must be set so that the available resource capacity generates the maximal possible sum of revenues from all n customers.

The previously described decision problem represents an optimization problem in which the sum of overall revenues received from all n customers is maximized. It is helpful to model this decision task as a mathematical optimization model (cf. Chaps. 7, 8, and 12). To set up such a model, we have (1) to declare the decision variables, (2) to formulate constraints, and (3) to define an objective function.

$$\sum_{i=1}^{n}\text{PDF}_i\left(p_i\right)\le C^{\text{total}} \tag{17.6}$$

$$0\le p_i\le p_i^{\max}\qquad \forall i=1,\ldots,n \tag{17.7}$$

$$\sum_{i=1}^{n}p_i\cdot\text{PDF}_i\left(p_i\right)\to\max \tag{17.8}$$

With respect to (1) we define p_i to be the placeholder for the rate assigned to customer i ($i = 1, \ldots, n$). We use the price–demand function PDF_i for the determination of the required capacity for customer i. The first constraint is that the sum of all allocated capacities for the n customers is bounded by the available resource capacity C^{total}, as shown in Eq. (17.6). Prices must be non-negative and must not exceed the maximal willingness to pay according to Eq. (17.7). The sum of collected revenues from all customers must be maximal subject to Eq. (17.8).

The solving of the optimization model (17.6)–(17.8) requires elaborated procedures. In general, and depending on the type of the price–demand functions, the model is nonlinear. In the following, we demonstrate the general applicability of the model by using the Excel Solver introduced in Chap. 7. Figure 17.8 presents the data (white cells), decision variable cells (light gray), and calculated cells (dark gray cells) used to instruct the Excel Solver Tool. The formulas in column F are as follows: 50 + (50–0)/(0–100)*D4 (cell F4), SQRT(70^2 (0.5*D5)^2) (in cell F5) as well as 250*EXP(−0.03*D6) coded in cell F6. We calculate the revenue sum gained from each customer (cells H4–H6) by multiplying the cell values from columns D and F and the objective function cell H8 contains the sum of the cells H4–H6.

The parameter setting for the Excel Solver is shown in Fig. 17.9. Since neither the constraints nor the objective function are linear expressions, it is necessary to select a general-purpose solver algorithm. Here, we selected the algorithm GRG-nonlinear.

After Tjark has pressed the solve button in the solver's dialog window, the solver tool fetches all data from the spreadsheet and processes all necessary calculations. After this, it returns the prices in the variable cells D4–D6 (Fig. 17.10).

Tjark now compares the revenues gained in FCFS and simultaneous pricing. First, he observes that the revenue sum is larger in the solution for the simultaneous pricing setup. 93.92 € more are earned if Tjark waits until he has collected the relevant data from all three customers. Second, Tjark discovers the reason for the increased revenue sum in the simultaneous pricing setup: CUSTOMER-X who is processed as the last customer in the FCFS setup should be granted access to more capacity than available after FRISIA-CC as well as NORTH-WIND have been processed. Furthermore, the rate agreed with NORTH-WIND should be increased slightly to compensate for the reduced capacity allocated to FRISIA-CC and NORTH-WIND (Table 17.3).

	A	B	C	D	E	F	G	H
1								
2				prices	max. prices	PDF		revenues
3						tons		
4		FRISIA-CC			100.00 €	50.00		- €
5		NORTH-WIND			140.00 €	70.00		- €
6		CUSTOMER-X			300.00 €	250.00		- €
7								
8					required capacity (F4+F5+F6)	370		- €
9					available capacity	120		objective
10								

Fig. 17.8 Spreadsheet with all data needed to solve the optimization model (17.6)–(17.8)

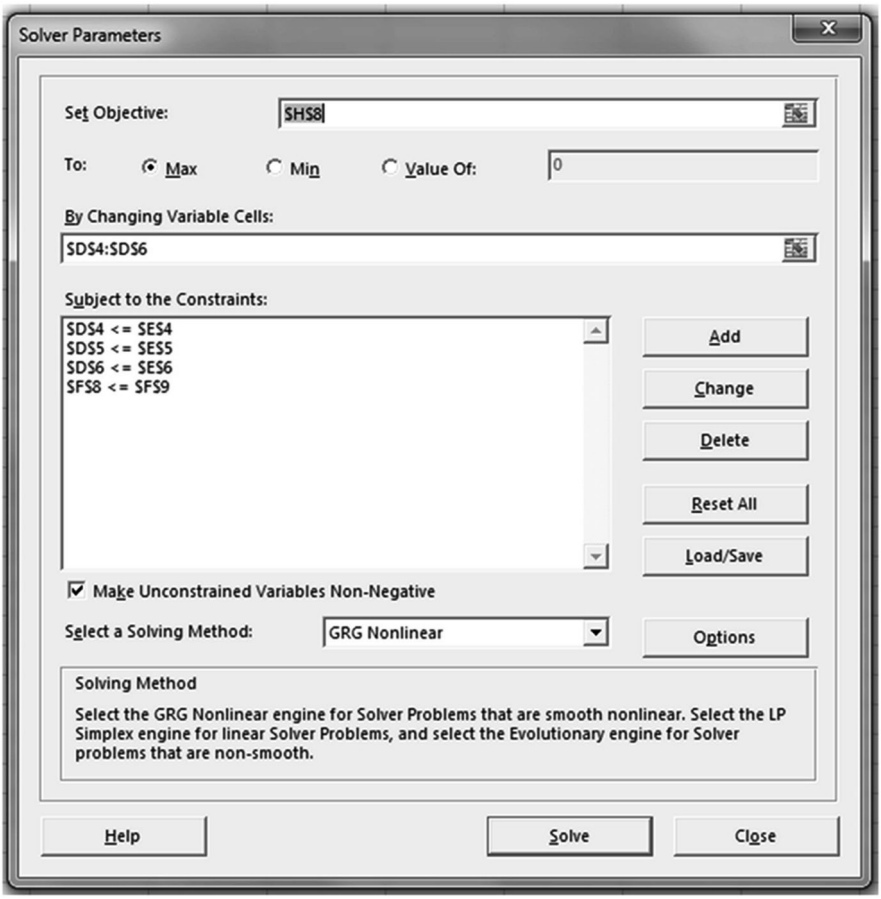


Fig. 17.9 Parameterization of the Excel Solver Tool

	A	B	C	D	E	F	G	H
1								
2				prices	max. prices	PDF		revenues
3						tons		
4		FRISIA-CC		59.30 €	100.00 €	20.35		1.206.79 €
5		NORTH-WIND		103.75 €	140.00 €	47.00		4.876.21 €
6		CUSTOMER-X		51.93 €	300.00 €	52.65		2.733.93 €
7								
8					required capacity (F4+F5+F6)	120		8.816.92 €
9					available capacity	120		objective

Fig. 17.10 Optimal prices returned by the Excel Solver Tool

Table 17.3 Prizing decisions as well as allocated capacity (per day) in the FCFS setup (left) and the simultaneous pricing setup (right)

	Rate/ton	Alloc. tons	Revenue	Rate/ton	Alloc. tons	Revenue
	FCFS pricing			Simultaneous pricing		
FRISIA-CC	50 €	25	1250 €	59.30 €	20.35	1206.79 €
NORTH-WIND	99 €	49	4851 €	103.75 €	47.00	4876.21 €
CUSTOMER-X	57 €	46	2622 €	51.93 €	52.65	2733.93 €
			8723 €			8816.92 €

17.4 Setting Optimal Prices in Resource Networks

Jördis and Tjark have successfully established their business. For several years now their company has sailed with freight on board between Neuharlingersiel and the seven East Frisian islands. All customers are quite satisfied since they have to pay appropriate and acceptable prices for the transport service. Today, Tjark is thinking about a business expansion. He wants to enter into the business of passenger transportation. The current trigger is an offer to buy a second vessel, the SEA DOG. The SEA DOG is in good shape and the price is fair. The maximal capacity is 80 passengers (PAX), but its speed is so great that it can shuttle between up to three islands per day (e.g., three return-trips originating from Neuharlingersiel).

Jördis, the wife of Tjark, has an idea for how this vessel can be deployed. Her business concept for the SEA DOG is based on the observation that a lot of potential PAX need a transport from an island to the mainland and in the reverse direction (or vice versa) while others only book a one-way trip. Together with the regional tourist office and some travel agencies, Jördis wants to offer these customers the opportunity to book capacity for a single trip or a return trip on the SEA DOG. These "service products" are composed of single trips between Neuharlingersiel and an island or between an island and Neuharlingersiel. On each of these trips, the number of seats that can be sold is limited. She, therefore, interprets the trips as the resources.

Table 17.4 contains the available resources. According to long-term contracts with some hotels on the islands, Jördis and Tjark have successfully allocated some seats on the SEA DOG. The residual capacity $\mathrm{CAP}_k^{\mathrm{avail}}$ shown in the right column of Table 17.4 gives the number of SEATS that can be sold directly by Jördis and Tjark.

We call the set of resources R. One-way tickets require one resource, but each return ticket requires the deployment of two resources. It is necessary that both resources offer enough capacity. This is different from the aforementioned setups in which each product corresponds to exactly one resource. Let P denote the set of products. Such a pricing problem is called a network pricing problem (Brotcorne et al. 2011).

The core pricing process should be as follows. All customers are local travel agencies (TAs) that offer foreign tourists trips to and from the island, often with overnight accommodation. These TAs collect the maximal demand $D_{ij}^{\max}$ (expressed as seats on the SEA DOG) in week X and submits it to Jördis and Tjark by Friday 10:00 AM in week X—1. Each TA submits their individually collected maximal

demand. Immediately, Jördis and Tjark consolidate the demand collected from the TAs. They determine the prices and return them to the TAs. All participating TAs are collected in the set C. Tjark likes this concept, both to take the decision to buy the SEA DOG and to extend the service of FRISIA COASTAL SHIPPING to serving the market of infrequent, ad hoc customer requests. They plan to put the SEA DOG in service and start all required preparations together with the TAs.

Table 17.5 lists the submitted demand information from the three participating TAs.

Now Tjark is looking for a procedure to determine the rates for the incoming customer inquiries. He wants to reformulate/adapt the decision model (17.6)–(17.8). He has to set |C|•|P| individual prices in parallel so that the sum of revenues earned in week X is maximized as already done. Now, four restrictions have to be considered during the price determination:

Table 17.4 Definition of the 12 resources offered by the SEA DOG in week X

Resource R_k	Day of week	Start	Destination	Residual capacity CAP_k^{avail}
$k = 1$	Monday	Neuharlingersiel	Norderney	25
$k = 2$	Monday	Norderney	Neuharlingersiel	18
$k = 3$	Monday	Neuharlingersiel	Langeoog	20
$k = 4$	Monday	Langeoog	Neuharlingersiel	22
$k = 5$	Wednesday	Neuharlingersiel	Langeoog	20
$k = 6$	Wednesday	Langeoog	Neuharlingersiel	25
$k = 7$	Wednesday	Neuharlingersiel	Borkum	24
$k = 8$	Wednesday	Borkum	Neuharlingersiel	22
$k = 9$	Saturday	Neuharlingersiel	Norderney	20
$k = 10$	Saturday	Norderney	Neuharlingersiel	18
$k = 11$	Saturday	Neuharlingersiel	Borkum	17
$k = 12$	Saturday	Borkum	Neuharlingersiel	10

Table 17.5 Example booking list for week X

Involved island	Product j	Involved resources	D_{1j}^{max} (PAX)	D_{2j}^{max} (PAX)	D_{3j}^{max} (PAX)	P_{j1}^{max} (€)	P_{j2}^{max} (€)	P_{j3}^{max} (€)
Norderney	$j = 1$	R_1	7	3	5	60	80	70
	$j = 2$	R_1,R_{10}	6	12	6	90	60	90
	$j = 3$	R_2	5	5	4	70	100	60
	$j = 4$	R_{10}	6	5	7	80	70	90
Langeoog	$j = 5$	R_3	12	5	4	60	80	60
	$j = 6$	R_4	3	0	7	90	80	100
	$j = 7$	R_3, R_6	6	7	4	70	60	90
	$j = 8$	R_6	0	2	8	80	90	60
Borkum	$j = 9$	R_7, R_{12}	8	8	5	100	70	120
	$j = 10$	R_7	2	3	0	60	90	70
	$j = 11$	R_{12}	5	4	4	120	60	100

- Each rate must fall into the interval between 0 and the maximal accepted price $P_{ij}^{\max}$.
- The served demand must fall into the interval between 0 and the maximal demand $D_{ij}^{\max}$.
- The residual capacity $\text{CAP}_k^{\text{avail}}$ of a deployed resource must not be exceeded.
- The served demand $\text{PDF}_{ij}()$ must be an integer.

Tjark assumes that all TAs exhibit a simple PDF for each product. In particular, he decides to reuse the affine-linear PDF shown in Eq. (17.1) where the slope $\alpha_{ij} = \dfrac{D_{ij}^{\max} - 0}{0 - R_{ij}^{\max}}$ represents the functional dependency between the fixed rate and the ^{ij} available demand of customer TA_i for product j. He introduces the non-negative decision variable p_{ij} to represent· the rate assigned to TA_i for product j. Furthermore, the PDFs defined as $\text{PDF}_{ij}(p_{ij}) = D_{ij}^{\max} + \dfrac{D_{ij}^{\max} - 0}{0 - p_{ij}^{\max}} \cdot p_{ij}$ are deployed.

$$U_{kj} = \begin{cases} 1, & \text{if and only if product } j \text{ uses resource } k \\ 0, & \text{in all other cases} \end{cases} \tag{17.9}$$

To store the information about whether a product requires the availability of a certain resource, Tjark uses the binary parameter U_{kj} introduced in Eq. (17.9).

$$\sum_{i \in C} \sum_{j \in P} p_{ij} \cdot \text{PDF}_{ij}\left(p_{ij}\right) \rightarrow \max \tag{17.10}$$

$$0 \le p_{ij} \le P_{ij}^{\max} \quad \forall i \in \text{C}, j \in \text{P} \tag{17.11}$$

$$0 \le \text{PDF}_{ij}\left(p_{ij}\right) \le D_{ij}^{\max} \quad \forall i \in \text{C}, j \in \text{P} \tag{17.12}$$

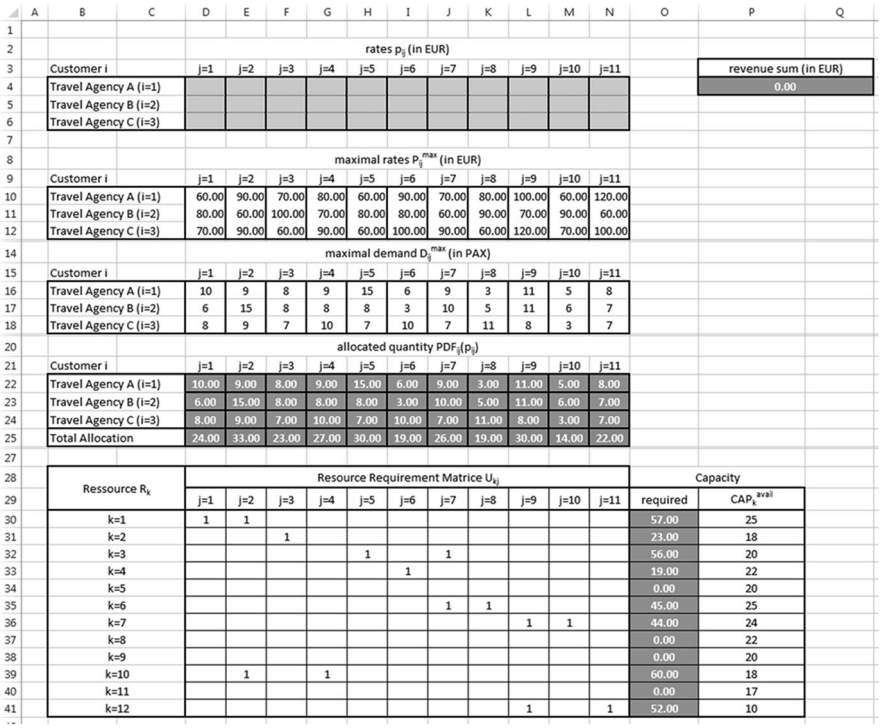
$$\sum_{j \in P} U_{kj} \sum_{i \in C} \text{PDF}_{ij}\left(p_{ij}\right) \le \text{CAP}_k^{\text{avail}} \quad \forall k \in R \tag{17.13}$$

$$\text{PDF}_{ij}\left(p_{ij}\right) \in \left\{0,1,\ldots,D_{ij}^{\max}\right\} \quad \forall i \in \text{C}, j \in \text{P} \tag{17.14}$$

Again, the maximization of the gained revenue sum is addressed (Eq. 17.10). This sum is achieved by adding up the revenues from all customers and all products. The final rate varies between 0 and the maximal possible rate (Eq. 17.11), but the maximal demand cannot be exceeded by the allocated capacity for each customer as well as each product (Eq. 17.12). The network constraint (Eq. 17.13) ensures that the available resource capacities are respected. Only integer values are accepted for the allocated capacity (Eq. 17.14).

Figure 17.11 contains a spreadsheet calculation scheme as the base for an Excel Solver Tool-oriented rate optimization system. The light gray-shaded cells D4:N6 are selected to represent the determined rates.

The configuration of the Solver Tool interacting with the spreadsheet in Fig. 17.11 is presented in Fig. 17.12. The first constraint $D$22:$ N$24 ≤ $D$16:$N$18



rates p_{ij} (in EUR)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)											
Travel Agency B (i=2)											
Travel Agency C (i=3)											

revenue sum (in EUR)
0.00

maximal rates P_{ij}^{max} (in EUR)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	60.00	90.00	70.00	80.00	60.00	90.00	70.00	80.00	100.00	60.00	120.00
Travel Agency B (i=2)	80.00	60.00	100.00	70.00	80.00	80.00	60.00	90.00	70.00	90.00	60.00
Travel Agency C (i=3)	70.00	90.00	60.00	90.00	60.00	100.00	90.00	60.00	120.00	70.00	100.00

maximal demand D_{ij}^{max} (in PAX)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	10	9	8	9	15	6	9	3	11	5	8
Travel Agency B (i=2)	6	15	8	8	8	3	10	5	11	6	7
Travel Agency C (i=3)	8	9	7	10	7	10	7	11	8	3	7

allocated quantity $PDF_{ij}(p_{ij})$

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	10.00	9.00	8.00	9.00	15.00	6.00	9.00	3.00	11.00	5.00	8.00
Travel Agency B (i=2)	6.00	15.00	8.00	8.00	8.00	3.00	10.00	5.00	11.00	6.00	7.00
Travel Agency C (i=3)	8.00	9.00	7.00	10.00	7.00	10.00	7.00	11.00	8.00	3.00	7.00
Total Allocation	24.00	33.00	23.00	27.00	30.00	19.00	26.00	19.00	30.00	14.00	22.00

Ressource R_k	Resource Requirement Matrice U_{kj}											Capacity	
	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11	required	CAP_k^{avail}
k=1	1	1										57.00	25
k=2			1									23.00	18
k=3					1		1					56.00	20
k=4						1						19.00	22
k=5												0.00	20
k=6							1	1				45.00	25
k=7									1	1		44.00	24
k=8												0.00	22
k=9												0.00	20
k=10		1		1								60.00	18
k=11												0.00	17
k=12									1		1	52.00	10

Fig. 17.11 Spreadsheet that contains all information associated with the optimization model (17.10)–(17.14)

realizes (17.11). Constraint (17.12) is represented by the second entry in the list of constraints (Fig. 17.12). Finally, the expression $O$36:$ O$47 $\leq$ $P$36:$P$47 takes care of the consideration of the limited capacities of the involved resources.

The Solver Tool returns the revenue-maximal collection of 33 individual rates (Fig. 17.13). To be able to realize these rates, it is necessary to allocate the number of seats to the three travel agencies given in the cells D22:N24. These values are in general non-integer values. Here, rounding is often not reasonable since the proposed capacity allocation falls exactly into the middle of an interval between two integer values. It is, therefore, necessary to enforce that the Solver Tool uses only integer values for the allocated quantities. Tjark first extends his spreadsheet by the cells D29:N31 that contain the distance of the proposed capacity allocation to the nearest integer value (Fig. 17.14). For example, cell D29 is determined using the formula = ABS(ROUND(D22;0)-D22). Tjark limits the values in the cells D29:N31 to 0.001 each by adding a corresponding additional constraint to the solver configuration (Fig. 17.15). After the Solver Tool re-executes, it now returns integer values

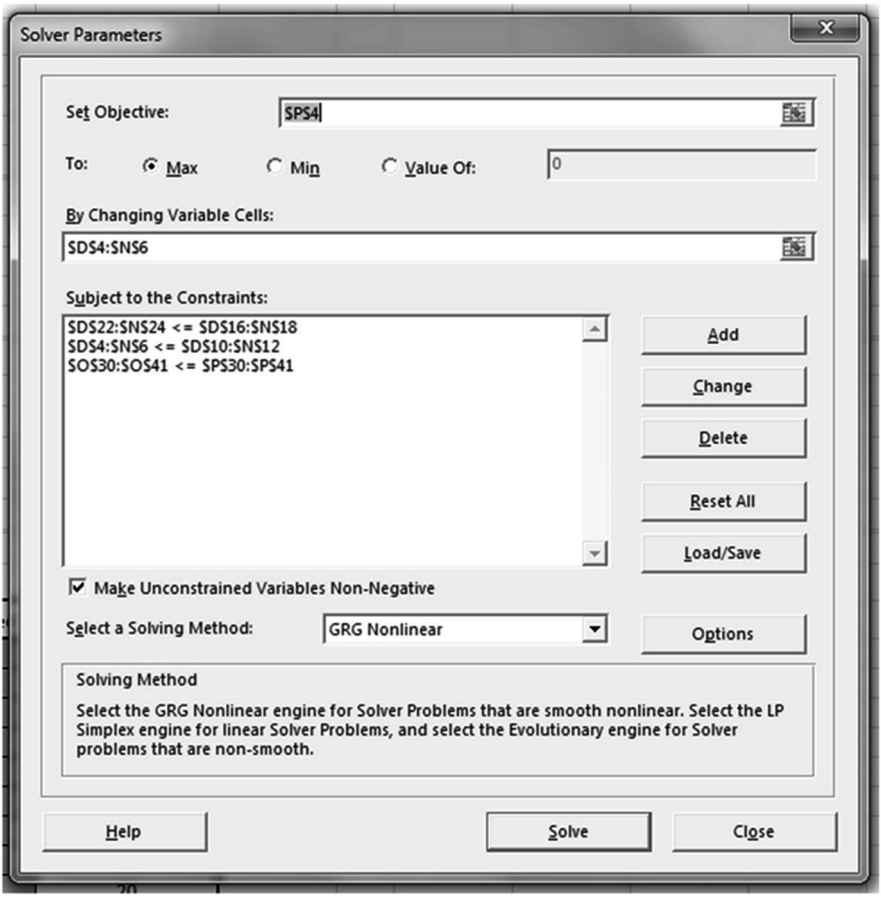


Fig. 17.12 Parameterization of the Solver Tool

(or values that are very close to an integer value) for the allocated capacity (cells D4: N6 in Fig. 17.14).

The Solver Tool has returned 33 prices. Here, we first detect that different prices are again determined for different customers. This phenomenon is called price discrimination (Kimes 2010) when the service provided to all customers is similar. Besides this fact, as we have already observed in the previous setups, we also see that the highest prices are realized if a product requires access to a scarce resource like 3, 10, or 12.

rates p_{ij} (in EUR)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	30.00	60.23	35.00	55.23	39.53	45.00	44.52	40.00	77.25	30.00	87.25
Travel Agency B (i=2)	40.00	45.23	50.00	50.23	49.52	40.00	39.53	44.99	62.25	45.00	57.25
Travel Agency C (i=3)	35.00	60.23	30.00	60.23	39.53	50.00	54.53	30.00	87.25	35.01	77.25

revenue sum (in EUR)
4562.51

maximal rates P_{ij}^{max} (in EUR)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	60.00	90.00	70.00	80.00	60.00	90.00	70.00	80.00	100.00	60.00	120.00
Travel Agency B (i=2)	80.00	60.00	100.00	70.00	80.00	80.00	60.00	90.00	70.00	90.00	60.00
Travel Agency C (i=3)	70.00	90.00	60.00	90.00	60.00	100.00	90.00	60.00	120.00	70.00	100.00

maximal demand D_{ij}^{max} (in PAX)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	10	9	8	9	15	6	9	3	11	5	8
Travel Agency B (i=2)	6	15	8	8	8	3	10	5	11	6	7
Travel Agency C (i=3)	8	9	7	10	7	10	7	11	8	3	7

allocated quantity $PDF_{ij}(p_{ij})$

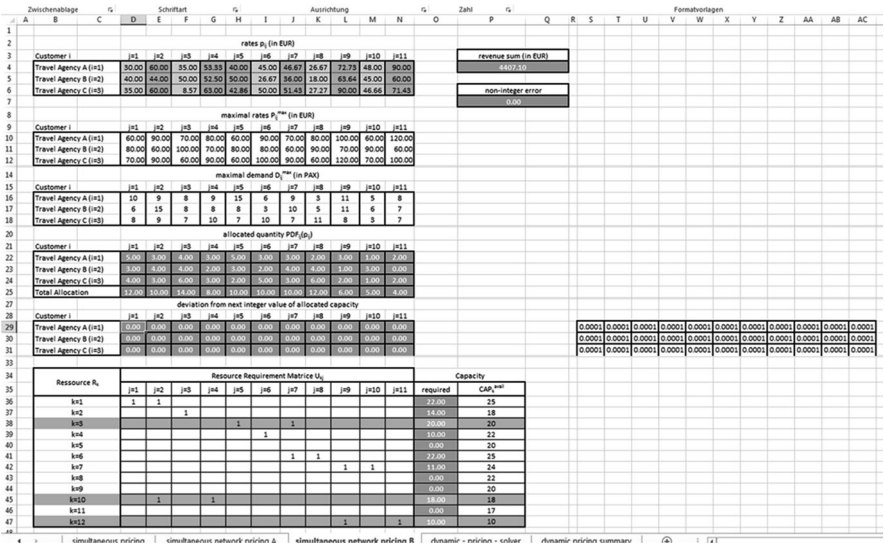
Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	5.00	2.98	4.00	2.79	5.12	3.00	3.28	1.50	2.50	2.50	2.18
Travel Agency B (i=2)	3.00	3.69	4.00	2.26	3.05	1.50	3.41	2.50	1.22	3.00	0.32
Travel Agency C (i=3)	4.00	2.98	3.50	3.31	2.39	5.00	2.76	5.50	2.18	1.50	1.59
Total Allocation	12.00	9.65	11.50	8.35	10.55	9.50	9.45	9.50	5.90	7.00	4.10

Ressource R_k	Resource Requirement Matrice U_{kj}											Capacity	
	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11	required	CAP_k^{avail}
k=1	1	1										21.65	25
k=2			1									11.50	18
k=3					1		1					20.00	20
k=4						1						9.50	22
k=5												0.00	20
k=6							1	1				18.95	25
k=7									1	1		12.90	24
k=8												0.00	22
k=9												0.00	20
k=10		1		1								18.00	18
k=11												0.00	17
k=12									1		1	10.00	10

Fig. 17.13 Returned optimal rates

17.5 Dynamic Pricing: Pricing in Reaction to Observed Market Developments

We have learned in the previous sections that the discrimination of prices for different customers as well as demand-based pricing contribute to maximizing the revenue sum we can earn with a resource of fixed capacity from a heterogeneous market. In all our pricing models, we have exploited the price–demand functions specified for different customers and/or products. If these functions are properly defined then the calculated prices are appropriate. However, if the sales period is long or if the underlying market circumstances change during the sales period, a price–demand function might become inappropriate and, consequently, the derived prices will become inappropriate as well. An idea for handling this situation is to try to adjust the price–demand function and the prices for upcoming customer demand. The next section introduces the reactive and market-driven price update concept, often called Dynamic Pricing (DP).



rates p_i (in EUR)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	30.00	60.00	35.00	53.33	40.00	45.00	46.67	26.67	72.73	48.00	90.00
Travel Agency B (i=2)	40.00	44.00	50.00	52.50	50.00	26.67	36.00	18.00	63.64	45.00	60.00
Travel Agency C (i=3)	35.00	60.00	8.57	63.00	42.86	50.00	51.43	27.27	90.00	46.66	71.43

revenue sum (in EUR)
4407.10

non-integer error
0.00

maximal rates P_i^{max} (in EUR)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	60.00	90.00	70.00	80.00	60.00	90.00	70.00	80.00	100.00	60.00	120.00
Travel Agency B (i=2)	80.00	60.00	100.00	70.00	80.00	80.00	60.00	90.00	70.00	90.00	60.00
Travel Agency C (i=3)	70.00	90.00	60.00	90.00	60.00	100.00	90.00	60.00	120.00	70.00	100.00

maximal demand D_i^{max} (in PAX)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	10	9	8	9	15	6	9	3	11	5	8
Travel Agency B (i=2)	6	15	8	8	8	3	10	5	11	6	7
Travel Agency C (i=3)	8	9	7	10	7	10	7	11	8	3	7

allocated quantity PDF$_i$(p_i)

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	5.00	3.00	4.00	3.00	5.00	3.00	3.00	2.00	3.00	1.00	2.00
Travel Agency B (i=2)	3.00	4.00	4.00	2.00	3.00	2.00	4.00	4.00	1.00	3.00	0.00
Travel Agency C (i=3)	4.00	3.00	6.00	3.00	2.00	5.00	3.00	6.00	2.00	1.00	2.00
Total Allocation	12.00	10.00	14.00	8.00	10.00	10.00	10.00	12.00	6.00	5.00	4.00

deviation from next integer value of allocated capacity

Customer i	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11
Travel Agency A (i=1)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Travel Agency B (i=2)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Travel Agency C (i=3)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001

Ressource R_k	Resource Requirement Matrix U_{kj}											Capacity	
	j=1	j=2	j=3	j=4	j=5	j=6	j=7	j=8	j=9	j=10	j=11	required	CAP_k^{max}
k=1	1	1										17.00	25
k=2			1									14.00	18
k=3					1		1					20.00	20
k=4						1						10.00	22
k=5												3.00	20
k=6							1	1				22.00	25
k=7									1	1		11.00	24
k=8												0.00	22
k=9												0.00	20
k=10		1		1								14.00	18
k=11												0.00	17
k=12									1		1	10.00	10

Fig. 17.14 Returned optimal rates with allocated integer seat numbers

17.5.1 Business Extension by Coastal Tours

FRISIA COASTAL SHIPPING grows further. Jördis and Tjark earn a fair amount of money every season. However, they want to extend and diversify their business in order to protect it. They discuss several ideas and opportunities but do not find an adequate business idea. One evening, they sit together with some old friends who have just returned from their holiday trip to the Caribbean Sea. They report a so-called island hopping tour. This is a round trip by boat starting at their hotel location and visiting five islands in a 5-day trip. Every evening they were on another island where they could do some sightseeing or explore the island on their own. Jokingly, one of the friends says: "Maybe you can offer a similar round trip with one of the FRISIA COASTAL SHIPPING boats. Make some advertising and show the tourist our seven islands within one week."

Back in the office the next morning, Tjark comes back to the friend's remark and heads to Jördis:

Tjark: Jördis, I think our friend had the right idea for a new business when he mentioned the seven-island-round-trip. Some ideas ran through my mind last night. You know that we maintain excellent business connections with most of the hotel owners on the islands. Maybe we can convince them to cooperate together and offer these round trips.

Jördis: The same idea came to my mind during the dinner with our friends and, yes, I think this idea is a realistic one. It is not necessary to offer such a round trip regularly with high frequency. But during the off-peak-seasons, it would be possible if we have a boat available and the hotel owners see

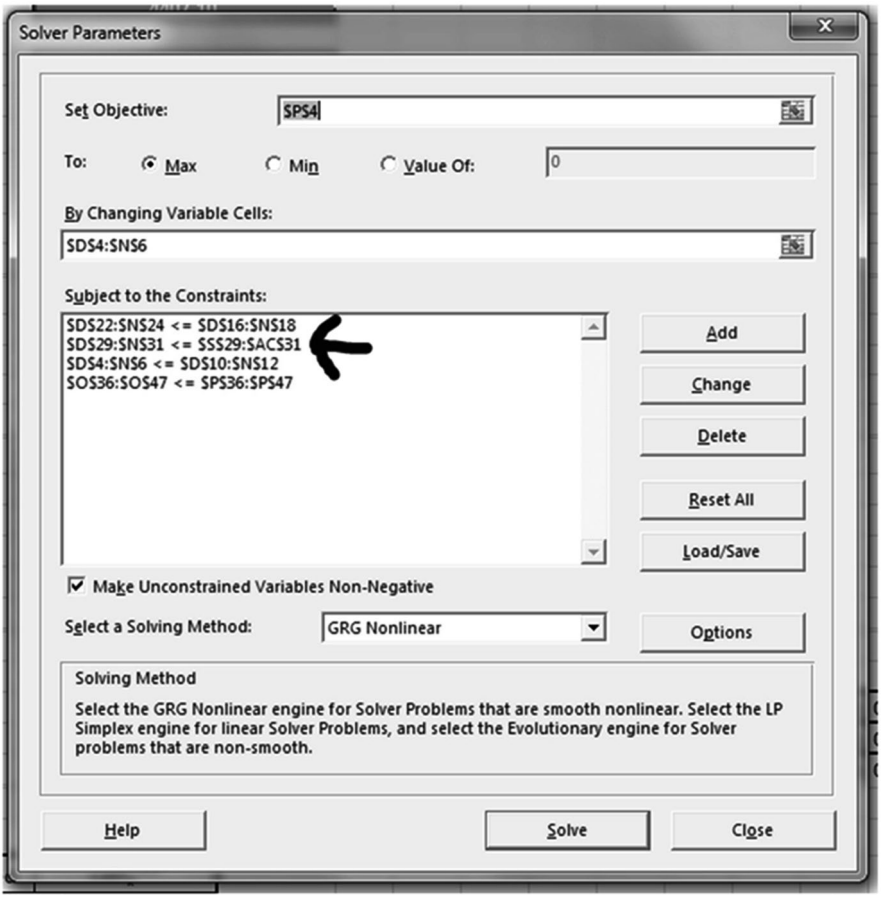


Fig. 17.15 Revised solver parameterization to achieve near-integer capacity allocation values: The labeled additional constraint is now considered while the solver processes the data

the opportunity to sell additional accommodation. The only issue is again the pricing of this new product. I have neither an idea about the willingness to pay for such a quite exhausting seven-day-excursion nor a realistic estimation of which type of tourist will come. We need more information about that, don't we?

Tjark: Of course, I agree with you. Do you remember Christina and Mark?

Jördis: Christina and Mark? Oh, you mean the two young persons we met last year during a tourist convention. They were involved in some tourist projects in Egypt, weren't they?[1]

Tjark: Yes, these two quite polite persons. During a business lunch, they reported a little bit about their projects. If I remember correctly, they were involved

[1] We also know Christina and Mark from the cases discussed in Chap. 14.

> in new project development. Maybe they can help us estimate who the customers might be and/or the market segmentation for the several-day-round-trip offers. I think I will send them an e-mail immediately.

On the same day, just before Tjark finishes his daily work, he receives an e-mail from Christina:

> Hi Tjark, nice to hear from you and Jördis. What a nice idea to think about the "7 islands within 7 days round trip." It is quite special—we have also developed some special desert round trips here in Egypt ☺. If I remember correctly, we got some surprises when we evaluated our first ideas and interviewed some old customers and business partners about the potential willingness to pay. We found that there is no real individual willingness to pay! However, the booking time seemed to influence the possible price. There were three customer groups. We had the "early birds," who booked early (up to 3 weeks before the departure of our bus). All early birds exhibited a similar willingness to pay. They absolutely wanted to participate in such a round trip. The second group was the group of tourists who arrived in the hotels and found out that there is only desert in the surrounding area. They wanted a guided tour to experience life in the desert. We called them "arrived birds." I contrast the early birds who arranged pre-booking often from home to the arrived birds who made on-site bookings in our local travel agencies. Again, all members of the arrived birds exhibited a similar willingness to pay. But this was different from the willingness to pay observed for the "early birds." Finally, the third group was called the "late birds." They could be convinced to take a round trip by progressive on-site advertising to complete a "perfect holiday experience." All late birds have a similar willingness to pay, but it was quite high compared to the willingness to pay observed for the two other groups (I suspect that they are quite relaxed and in a good mood). I hope that my experience helps you to develop your own business. Please let me know about your success. Hope to see you again, Christina.

Jördis also reads the e-mail reply. Both discuss the content and soon find out that the decision situation differs significantly from the decision scenarios before. The pricing scenarios investigated so far report scenarios with the necessity of assigning individual prices to individual customers (or groups of customers). The determined prices were not changed during the complete sales period. Here, it is necessary to assign different prices to different parts of the sales period, but these prices apply to all customers (so they are announced to all customers). In summary, they discovered:

- Early birds book during the first third of the booking period and pay a price of p_1 per person.
- Arrived birds book during the second third of the booking period and accept the price p_2.
- Late birds book during the final third of the booking period and a price of p_3 is possible.

Again, a conversation about the new business opportunity is initiated between Jördis and Tjark.

Tjark: This is funny. There are different prices for the same product, but the price is determined mainly by the time it is sold.

Jördis: That is true, but the booking time is strongly connected to a special group of customers.

Tjark: Yes, you are right. Do you think that we could assume we also have a similar segmentation of potential customers? This is a necessary prerequisite if we want to start the round-trip business.

Jördis: Well, a definitive answer is impossible, but I think, yes, there are some similarities between the two cases in the desert and here at the Frisian coast. Maybe the early birds would be formed by all those tourists who have dreamed about a cruise trip, but do not want or cannot effort to do it on the regular cruise vessels abroad.

Tjark: Oh, that's a good idea. These persons will make a longer plan and book quite early. In another case, we often have very bad weather conditions here in the region. All those customers who have booked a 2-week stay could be convinced to take the island round trip if the weather conditions are bad after they have arrived and they realize that it is absolutely impossible to spend the next days on a beachside. This would be our group of arrived birds.

Jördis: Correct, and here is the third group, the late birds: All tourists who have arrived and who discover that besides the pure nature there is nothing to experience here. For them, such a cruise can be a very interesting experience full of highlights.

Tjark: Haha, now you are exaggerating. But in general, you are right. What do you think: Should we go ahead with the development of our business idea?

Jördis: Yes, I think we should go ahead. However, I see one big challenge: It is again the pricing. How can we decide when to change the price and which prices should we use?

Tjark: We should investigate variable prices. Maybe we can re-apply some of the price-setting tools we have already used.

Jördis and Tjark start learning about the variation of product prices. Customers recognize these price updates as variable prices during the sales period. Here, variable pricing describes the general idea of adjusting the price of a product at certain times during the sales period (McCarthy 2010). There are different implementations of variable prices.

- Seasonal pricing implements price adjustments and the time of the price updates are announced to the customers, so that the customers can anticipate (or even know) when a price will be updated. It is not necessary in seasonal pricing that customers are informed about the time and extent of price adjustments, but it is necessary that the service provider knows exactly when and how much the prices are updated. In seasonal pricing, the prices do not depend on the actual realized demand.

- In Dynamic Pricing (DP), the update of a price is mainly triggered by an exogenous factor like the incoming market demand. The service provider sets the rules for the price update but has only limited (or even no) control of the time and extent of the price adjustment (Maglaras and Meissner 2006).

In the remainder of this section, we accompany Jördis and Tjark and report their findings about DP issues and the application to the decision scenario for the sales of the coastal tour tickets.

17.5.2 Decision Situation Modeling

It is necessary to partition the complete sales period S into T sales subperiods $S_1, \ldots, S_T$. A price p_t ($t = 1, \ldots, T$) has to be fixed for every sales subperiod. If we know the price as well as the quantity q_t sold in period t, then we calculate the revenue gained in period t by $p_t q_t$. Summing up these products over all sales periods determines the sum of gained revenues. This means we are looking for a set of optimal prices that maximizes the revenue sum earned over the complete sales period S. Again, an optimization problem has to be solved. A set of prices must be fixed.

If we assume that the willingness to pay in each subperiod is homogeneous for all customers in a subperiod, and if we assume that the price p_t is a function f_t of the offered quantity described by a known PDF f_t, i.e., $p_t = f_t(q_t)$, then we are able to formulate the decision situation as a mathematical optimization model.

$$\sum_{t=1}^{T} p_t \cdot f_t\left(p_t\right) \rightarrow \max \tag{17.15}$$

$$\sum_{t=1}^{T} f_t\left(p_t\right) \leq \mathrm{CAP} \tag{17.16}$$

The optimization tries to maximize the sum of earned revenues over all subperiods (Eq. 17.15). We can further assume that during the complete sales period only a limited capacity CAP can be allocated for the accepted customer requests. This restriction is expressed by constraint (17.16).

$$0 \leq f_t\left(p_t\right) \leq N_t^{\max} \quad \forall t \in \{1,\ldots,T\} \tag{17.17}$$

$$f_t\left(p_t\right) \in \{0,1,2,\ldots\} \quad \forall t \in \{1,\ldots,T\} \tag{17.18}$$

Since the quantity q_t represents the number of tickets to be sold in period t, this number must be non-negative and is not allowed to exceed the maximal demand (Eq. 17.17). Furthermore, the quantities are restricted to integer numbers (Eq. 17.18).

$$0 \leq p_t \leq p_t^{\max} \quad \forall t \in \{1,\ldots,T\} \tag{17.19}$$

Of course, each period-specific price must be non-negative and must not exceed the period-specific maximal price $P_t^{\max}$ (Eq. 17.19). ^{t}

The mathematical optimization model (17.15)–(17.19) represents the decision task of determining several prices p_t simultaneously for all subperiods. The exploitation of the PDFs $f_1, \ldots, f_T$ also determines the quantity q_t offered in each period t to achieve the sales price p_t. Solving this model means finding the best partition of the available capacity over all subperiods. A certain amount of the available capacity is assigned to each period. Again, this model is very complicated to solve since the objective function (17.15) is, in general, nonlinear like the two constraints (17.16) and (17.17). In Sect. 17.5.3, we will explain how the best capacity distribution over all sales periods is made. We will use the following example that represents the pricing task for the island-trip scenario of Jördis and Tjark.

With the help of some business partners like the regional tourist office, some hotel owners, and a market review, Jördis and Tjark have confirmed that the split-up of the sales period into three subperiods to address the three outlined customer groups is reasonable. Furthermore, they have decided to use PFDs f_1, f_2 as well as f_3 of the form (17.1) with the parameters given in Table 17.6.

Jördis and Tjark decide to plan the deployment of a boat with a capacity of CAP = 100 passengers for the first attempt. They decided to use a booking period comprising $T = 3$ consecutive weeks before the boat's departure. It is expected that the early birds will book in the first week, the arrived birds in the second week, and finally, the late birds in the third and final week. It is necessary to determine three prices p_1, p_2 as well as p_3 for the island round trip. We are now ready to set up the model (17.15)–(17.19) for the island round trip sales scenario.

To ease the following model description, we define $m_i := \dfrac{N_i^{\max} - 0}{0 - p_i^{\max}}$ so that we have $m_1 := \dfrac{100}{-150}, m_2 := \dfrac{60}{-200}, m_3 := \dfrac{80}{-175}$. The PDFs are now $f_1(p) := 100 + \dfrac{100}{-150} p$, $f_2(p) := 60 + \dfrac{60}{-200} p$ and $f_3(p) := 80 + \dfrac{80}{-175} p$. We get the optimization model (17.20)–(17.23) representing the pricing problem for the island round trip sales scenario.

$$p_1 \cdot \left(100 + \frac{100}{-150} p_1\right) + p_2 \cdot \left(60 + \frac{60}{-200} p_2\right) + p_3 \cdot \left(80 + \frac{80}{-175} p_3\right) \tag{17.20}$$

$$\rightarrow \max$$

$$100 + \frac{100}{-150} p_1 + 60 + \frac{60}{-200} p_2 + 80 + \frac{80}{-175} p_3 \leq 100 \tag{17.21}$$

Table 17.6 Parameter of the PDFs in the three periods of the island round trip example

Period i	Customer group	Max. price $P_i^{\max}$	Max. period capacity $N_i^{\max}$ (max. market demand)
1	Early birds	150	100
2	Arrived birds	200	60
3	Late birds	175	80

$$0 \le 100 + \frac{100}{-150} p_1 \le 100$$
$$0 \le 60 + \frac{60}{-200} p_2 \le 60 \tag{17.22}$$
$$0 \le 80 + \frac{80}{-175} p_3 \le 80$$

$$0 \le p_1 \le 150, 0 \le p_2 \le 200, 0 \le p_3 \le 175 \tag{17.23}$$

$$f_1(p_1), f_2(p_2), f_3(p_3) \text{integer} \tag{17.24}$$

17.5.3 Pricing and Capacity Distribution Over the Sales Period

Jördis and Tjark need to find the three optimal prices. The objective function (17.20) is not linear, but quadratic. In addition, 10 restrictions must be considered while evaluating different price vectors $p = (p_1, p_2, p_3)$. Therefore, a manual calculation of the best prices is not a good idea. However, due to the nonlinearity of the objective function, the application of linear programming (Chap. 12) is not possible. To approximate the best price decisions, we can apply the Excel Solver Tool again using a heuristic that applies some rules to determine the prices, but which does not provide any information about the quality of the returned solution.

17.5.3.1 Using the Excel Solver Tool

A spreadsheet that contains all information for the pricing problem is given in Fig. 17.16. White cells contain values that are parameters of the scenario. Gray-shaded cells contain derived or calculated values and the black cells are reserved to host the decided prices. Again, it is necessary to ensure that the determined price p_i

	A	B	C	D	E	F	G	H	I	J
1										
2										
3		Capacity	PAX	CAP	100					
4										
5					Period i					
6				i=1	i=2	i=3				
7		min demand	PAX	0	0	0				
8		max. demand N_i^{max}	PAX	100	60	80		total used capacity (D8+E8+F8)		
9		q_i	PAX	100.00	60.00	80.00		240.00		
10		round(q_i)	PAX	100	60	80				
11		$\lvert q_i\text{-round}(q_i)\rvert$		0.00	0.00	0.00				
12		max. deviation		0.001	0.001	0.001				
13		min. price	EUR	0.00	0.00	0.00				
14		max. price p_i^{max}	EUR	150.00	200.00	175.00				
15		p_i	EUR							
16								total revenue (D16+E16+F16)		
17		revenue	EUR	0.00	0.00	0.00		0.00		

Fig. 17.16 Spreadsheet that contains all information associated with the optimization model (17.20)–(17.24) used by Jördis

(row 15) is associated with an integer value for the associated number of tickets/passengers (PAX) q_i (row 9). We use again the technique to specify a maximal allowed deviation (given in row 12) from an integer value (given in row 10).

The required Solver Tool configuration is presented in Fig. 17.17. Since we do not have a linear optimization model (Chap. 12), we cannot use a simplex-algorithm-based solving method. Therefore, we select the evolutionary algorithm. This is an algorithm to identify feasible solutions with good objective function values. However, the returned solution is not necessarily optimal since the evolutionary algorithm cannot be proved to return optimal solutions. It is a heuristic, which means that it follows some rules to identify a feasible solution without any guarantee that this solution is optimal.

Jördis sets up the aforementioned spreadsheet and configures the Solver Tool. She starts the solver and after a long period a solution is returned into the spreadsheet shown in Fig. 17.18. The complete capacity is allocated. However, the price for the second sales period is set to 0. Consequently, only the first as well as the last period contribute to the total revenue of 3500 €. Jördis is worried about this result. If she manually inserts the tentative price of 33.32 € for the second period then she observes an increase in the total revenue of 5232 € (Fig. 17.19) though only 90 out of 100 tickets are sold.

Tjark also experiments with the spreadsheet tool. Surprisingly, Tjark reported a completely different result, as shown in Fig. 17.20. For the first week, the price is set to 126.00 €, for the second week it is 100.00 €, and for the third week it is

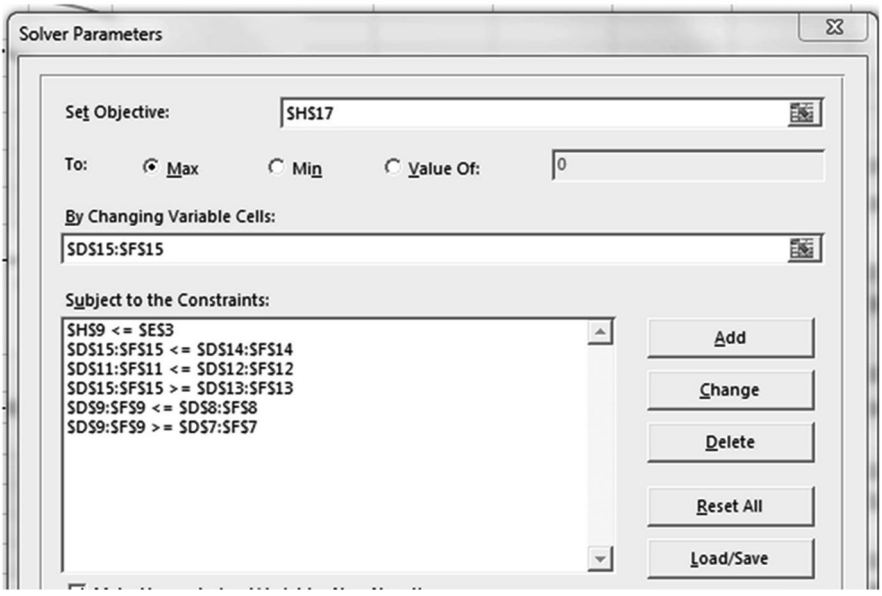


Fig. 17.17 Solver configuration for the optimization model (17.20)–(17.24)

	A	B	C	D	E	F	G	H	I	J
1										
2										
3		Capacity	PAX	CAP	100					
4										
5					Period i					
6				i=1	i=2	i=3				
7		min demand	PAX	0	0	0				
8		max. demand N_i^{max}	PAX	100	60	80		total used capacity (D8+E8+F8)		
9		q_i	PAX	0.00	60.00	40.00		100.00		
10		round(q_i)	PAX	0	60	40				
11		$\lvert q_i$-round(q_i)$\rvert$		0.00	0.00	0.00				
12		max. deviation		0.001	0.001	0.001				
13		min. price	EUR	0.00	0.00	0.00				
14		max. price p_i^{max}	EUR	150.00	200.00	175.00				
15		p_i	EUR	150.00	0.00	87.50				
16								total revenue (D16+E16+F16)		
17		revenue	EUR	0.00	0.00	3500.00		3500.00		
18										

Fig. 17.18 Returned optimal rates p_1, p_2, p_3 for the optimization model (17.20)–(17.24) observed by Jördis

	A	B	C	D	E	F	G	H	I
1									
2									
3		Capacity	PAX	CAP	100				
4									
5					Period i				
6				i=1	i=2	i=3			
7		min demand	PAX	0	0	0			
8		max. demand N_i^{max}	PAX	100	60	80		total used capacity (D8+E8+F8)	
9		q_i	PAX	0.00	50.00	40.00		90.00	
10		round(q_i)	PAX	0	50	40			
11		$\lvert q_i$-round(q_i)$\rvert$		0.00	0.00	0.00			
12		max. deviation		0.001	0.001	0.001			
13		min. price	EUR	0.00	0.00	0.00			
14		max. price p_i^{max}	EUR	150.00	200.00	175.00			
15		p_i	EUR	150.00	33.32	87.50			
16								total revenue (D16+E16+F16)	
17		revenue	EUR	0.00	1666.13	3500.00		5166.13	

Fig. 17.19 Increased revenue sum (cell H17) after manual insertion of the price $p_2 = 33.32$ (cell E15)

87.50 €. The expected numbers of bookings are 16 (week 1), 30 (week 2), and 40 (week 3), so that 86 tickets will be sold in total. The total revenue sum is 8515.95 €.

Jördis wants to know how her husband got these very different price proposals. He explains that he has manually inserted the prices $p_1 = 150$ €, $p_2 = 50$ € as well as $p_3 = 100$ € and, without having cleared the cells D15:F15, he started the solver.

Both Jördis and Tjark discuss their findings. They conclude that the solver approach is not a reliable tool for their purpose. They want to have a more trustful and transparent procedure for price determination over all periods.

	A	B	C	D	E	F	G	H	I	J
1										
2										
3		Capacity	PAX	CAP	100					
4										
5					Period i					
6				i=1	i=2	i=3				
7		min demand	PAX	0	0	0				
8		max. demand N_i^{max}	PAX	100	60	80		total used capacity (D8+E8+F8)		
9		q_i	PAX	16.00	30.00	40.00		86.00		
10		round(q_i)	PAX	16	30	40				
11		$\lvert q_i$-round(q_i)$\rvert$		0.00	0.00	0.00				
12		max. deviation		0.001	0.001	0.001				
13		min. price	EUR	0.00	0.00	0.00				
14		max. price p_i^{max}	EUR	150.00	200.00	175.00				
15		p_i	EUR	126.00	100.00	87.50				
16								total revenue (D16+E16+F16)		
17		revenue	EUR	2016.00	3000.00	3500.00		8516.00		
18										

Fig. 17.20 Returned optimal rates p_1, p_2, p_3 for the optimization model (17.20)–(17.24) observed by Tjark

17.5.3.2 Pricing Heuristic

Jördis and Tjark consult the textbook by Talluri and van Ryzin (2004) to learn about alternative procedures for the determination of the prices for the different periods. Finally, they find a procedure. It is based on the idea of consecutively distributing all capacity units over the available periods based on their marginal profit. The procedure works as follows: the marginal profit of the first ticket is calculated for each of the three periods. The period p^* with the highest marginal profit is identified. If the marginal profit is positive then the first ticket is assigned to period p^*. Otherwise, the procedure stops since no positive profit margin is achieved anymore. The iteration step is complete. In the second iteration, the marginal prices are determined again for all periods. The period with the highest positive marginal profit receives the second ticket and so on. This iteration stops if all tickets are distributed or only negative marginal profits are observed for all periods.

The PDF f_1 maps the price determined for a ticket in period 1 to the number of tickets to be sold in the first period. Jördis and Tjark want to know how much the revenue sum increases if an additional ticket is sold in the first period. It is, therefore, necessary to know which price is appropriate if a desired number of tickets should be sold in the first period. Therefore, they search for an appropriate Demand–Price Function (DPF) g_1 for the first period. This DPF has the property $g_1(f_1(p)) = p$, so that g_1 can be interpreted as the inverse function of the PDF f_1.

In general, an affine linear function $f = a - bx$ has the inverse function $g(y) = \frac{a}{b} - \frac{1}{b}y$, since we have $g\left(f\left(x\right)\right) = \frac{a}{b} - \frac{1}{b}f\left(x\right) = \frac{a}{b} - \frac{1}{b}\left(a - bx\right) = \frac{a}{b} - \frac{a}{b} + x = x$.

$$\overline{R_1}\left(q_1\right) = q_1 \cdot \left(150 - \frac{3}{2}q_1\right) = 150 \cdot q_1 - \frac{3}{2}q_1^2 \tag{17.25}$$

$$\overline{R_2}\left(q_2\right) = q_2 \cdot \left(200 - \frac{10}{3}q_2\right) = 200 \cdot q_2 - \frac{10}{3}q_2^2 \tag{17.26}$$

$$\overline{R_3}(q_3) = q_3 \cdot \left(175 - \frac{35}{16} q_3\right) = 175 \cdot q_3 - \frac{35}{16} q_3^2 \tag{17.27}$$

Table 17.7 contains the resulting DPFs g_1, g_2, g_3 for the three periods. They can be used to determine the required price $g_i(q)$ to sell the quantity q in period i. Jördis and Tjark can use these DPFs to conduct a quantity-based revenue calculation for each period. Therefore, they have to multiply the intended selling quantity q with the associated market price $g_i(q)$ in period i. The function $R_i(q_i) := q_i \cdot g_i(q_i)$ calculates the period-specific revenue sum depending on the quantity q_i assigned to period i. The three functions are defined in (17.25)–(17.27) and printed in Fig. 17.21.

Assume now that Jördis and Tjark have not yet assigned any capacity unit to a period, i.e., $q_1 = 0$, $q_2 = 0$, and $q_3 = 0$. Now, they wanted to earn the maximum possible additional revenue with the first capacity unit (which corresponds to the first ticket). As can be seen in Fig. 17.21, the steepest ascent is achieved in the dashed graph, which belongs to period 2. This means the first ticket earns the maximal revenue sum in period 2. So, the first ticket should be sold (and therefore be assigned) to period 2 and we update $q_2 = q_2 + 1 = 1$.

$$\overline{R_1}'(q_1) = 150 - 3q_1 \tag{17.28}$$

$$\overline{R_2}'(q_2) = 200 - \frac{20}{3} q_2 \tag{17.29}$$

$$\overline{R_3}'(q_3) = 175 - \frac{35}{8} q_3 \tag{17.30}$$

Now the second ticket has to be distributed and assigned to one of the three periods. In order to enable the calculation of the revenue increase caused by the next ticket in the three different periods we can use the first derivative functions $\overline{R_1}'(q_1)$, $\overline{R_2}'(q_2)$ and $\overline{R_3}'(q_3)$ of the period-specific DPFs. These are shown in Eqs. (17.28)–(17.30). Jördis and Tjark have to compare the values $\overline{R_1}'(0) = 150€, \overline{R_2}'(1) = 580/3 \approx 193.30€$ and $\overline{R_3}'(0) = 175€$. Again, the highest revenue increase is

Table 17.7 DPFs g_1, g_2, and g_3 in the three periods of the island round trip example

Period i	f_i	a	b	DPF $g_i(q) = \frac{a}{b} - \frac{1}{b} q$
1	$100 - \frac{100}{150} p$	100	$\frac{100}{150}$	$g_1(q) = 150 - \frac{3}{2} q$
2	$60 + \frac{60}{-200} p$	60	$\frac{60}{200}$	$g_2(q) = 200 - \frac{10}{3} q$
3	$80 + \frac{80}{-175} p$	80	$\frac{80}{175}$	$g_3(q) = 175 - \frac{35}{16} q$

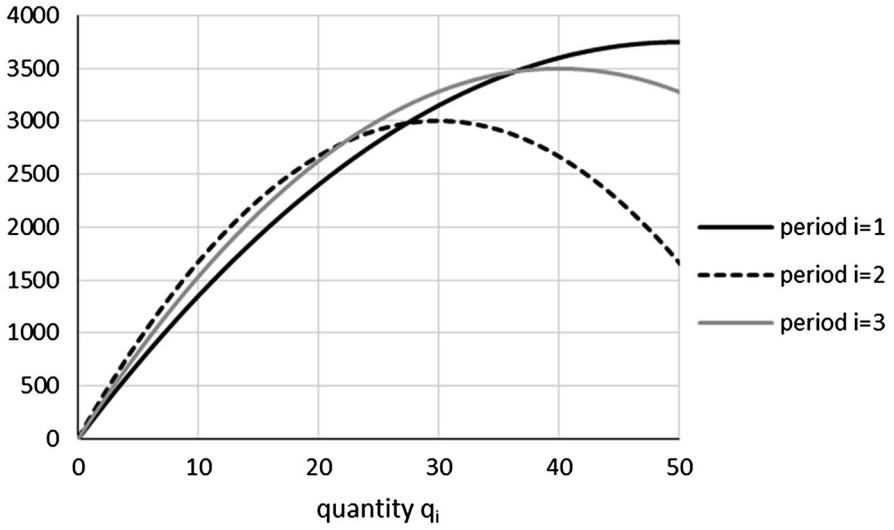


Fig. 17.21 Demand–Price functions $\overline{R_1}'(q), \overline{R_2}'(q)$ as well as $\overline{R_3}'(q)$

observed for period 2 so that also the second ticket goes to period 2 and we update $q_2 = q_2 + 1 = 2$. Next, $\overline{R_1}'(0) = 150€, \overline{R_2}'(2) = {}^{560}/_{3} - \approx 186.67€$ and $\overline{R_3}'(0) = 175€$. The third ticket is assigned to period 2 and it is $q_2 = q_2 + 1 = 3$. For the fourth ticket, Jördis and Tjark continue with the comparison of $\overline{R_1}'(0) = 150€, \overline{R_2}'(3) = 180€$ and $\overline{R_3}'(0) = 175€$ so that also the fourth ticket is sold in period 2. For the fifth ticket, Jördis and Tjark get $\overline{R_1}'(0) = 150€, \overline{R_2}'(4) = {}^{520}/_{3} \approx 173.3€$ and $\overline{R_3}'(0) = 175€$. Now, the third period is the best period and it is $q_3 = q_3 + 1 = 1$. For the sixth ticket, it is necessary to compare $\overline{R_1}'(0) = 150€, \overline{R_2}'(4) = {}^{520}/_{3} \approx 173.30€$ and $\overline{R_3}'(1) = 170.63€$. As a result, the sixth ticket is sold in period 2 and the update $q_2 = q_2 + 1 = 5$ is made, and so on.

Table 17.8 contains the complete distribution procedure over 100 iterations (100 tickets).

The major result of this distribution procedure is that after the complete procedure has terminated we see that $q_1 = 40$ tickets are sold in period 1, $q_2 = 26$ tickets are brought into the market in the second period and the remaining $q_3 = 34$ tickets are sold in period 3. Using the three period-specific DPFs Jördis and Tjark can calculate the appropriate prices p_1, p_2 as well as p_3. They determine $p_1 = g_1(40) = 90$ €, $p_2 = g_2(26) = 113.33$ € and $p_3 = g_3(34) = 100.63$ €. The total revenue is 40×90 € $+ 26 \times 113.33$ € $+ 34 \times 100.63$ € $= 9968$ €.

17.5.3.3 Reactive Price Updating

Before Jördis and Tjark start the first sales period for the island round trips, they summarize what they have done in the round trip pricing process:

Table 17.8 Iterative distribution of the 100 tickets over the three periods

1	A	F	G	H	I	J	K	L	P	Q	R	W	X	Y	Z	AA	AB	AC
2		Quantities q_i			Marginal revenue in period i							Quantities q_i			Marginal revenue in period i			
3	Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period			Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period
4	1	0	0	0	150.00	200.00	175.00	2			52	17	16	18	99.00	93.33	96.25	1
5	2	0	1	0	150.00	193.33	175.00	2			53	18	16	18	96.00	93.33	96.25	3
6	3	0	2	0	150.00	186.67	175.00	2			54	18	16	19	96.00	93.33	91.88	1
7	4	0	3	0	150.00	180.00	175.00	2			55	19	16	19	93.00	93.33	91.88	2
8	5	0	4	0	150.00	173.33	175.00	3			56	19	17	19	93.00	86.67	91.88	1
9	6	0	4	1	150.00	173.33	170.63	2			57	20	17	19	90.00	86.67	91.88	3
10	7	0	5	1	150.00	166.67	170.63	3			58	20	17	20	90.00	86.67	87.50	1
11	8	0	5	2	150.00	166.67	166.25	2			59	21	17	20	87.00	86.67	87.50	3
12	9	0	6	2	150.00	160.00	166.25	3			60	21	17	21	87.00	86.67	83.13	1
13	10	0	6	3	150.00	160.00	161.88	3			61	22	17	21	84.00	86.67	83.13	2
14	11	0	6	4	150.00	160.00	157.50	2			62	22	18	21	84.00	80.00	83.13	1
15	12	0	7	4	150.00	153.33	157.50	3			63	23	18	21	81.00	80.00	83.13	3
16	13	0	7	5	150.00	153.33	153.13	2			64	23	18	22	81.00	80.00	78.75	1
17	14	0	8	5	150.00	146.67	153.13	3			65	24	18	22	78.00	80.00	78.75	2
18	15	0	8	6	150.00	146.67	148.75	1			66	24	19	22	78.00	73.33	78.75	3
19	16	1	8	6	147.00	146.67	148.75	3			67	24	19	23	78.00	73.33	74.38	1
20	17	1	8	7	147.00	146.67	144.38	1			68	25	19	23	75.00	73.33	74.38	1
21	18	2	8	7	144.00	146.67	144.38	2			69	26	19	23	72.00	73.33	74.38	3
22	19	2	9	7	144.00	140.00	144.38	3			70	26	19	24	72.00	73.33	70.00	2
23	20	2	9	8	144.00	140.00	140.00	1			71	26	20	24	72.00	66.67	70.00	1
24	21	3	9	8	141.00	140.00	140.00	1			72	27	20	24	69.00	66.67	70.00	3
25	22	4	9	8	138.00	140.00	140.00	2			73	27	20	25	69.00	66.67	65.63	1
26	23	4	10	8	138.00	133.33	140.00	3			74	28	20	25	66.00	66.67	65.63	2
27	24	4	10	9	138.00	133.33	135.63	1			75	28	21	25	66.00	60.00	65.63	1
28	25	5	10	9	135.00	133.33	135.63	3			76	29	21	25	63.00	60.00	65.63	3

(continued)

Table 17.8 (continued)

1	A	F	G	H	I	J	K	L	P	Q	R	W	X	Y	Z	AA	AB	AC
2		Quantities q_i			Marginal revenue in period i							Quantities q_i			Marginal revenue in period i			
3	Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period			Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period
29	26	5	10	10	135.00	133.33	131.25	1			77	29	21	26	63.00	60.00	61.25	1
30	27	6	10	10	132.00	133.33	131.25	2			78	30	21	26	60.00	60.00	61.25	3
31	28	6	11	10	132.00	126.67	131.25	1			79	30	21	27	60.00	60.00	56.88	1
32	29	7	11	10	129.00	126.67	131.25	3			80	31	21	27	57.00	60.00	56.88	2
33	30	7	11	11	129.00	126.67	126.88	1			81	31	22	27	57.00	53.33	56.88	1
34	31	8	11	11	126.00	126.67	126.88	3			82	32	22	27	54.00	53.33	56.88	3
35	32	8	11	12	126.00	126.67	122.50	2			83	32	22	28	54.00	53.33	52.50	1
36	33	8	12	12	126.00	120.00	122.50	1			84	33	22	28	51.00	53.33	52.50	2
37	34	9	12	12	123.00	120.00	122.50	1			85	33	23	28	51.00	46.67	52.50	3
38	35	10	12	12	120.00	120.00	122.50	3			86	33	23	29	51.00	46.67	48.13	1
39	36	10	12	13	120.00	120.00	118.13	1			87	34	23	29	48.00	46.67	48.13	3
40	37	11	12	13	117.00	120.00	118.13	2			88	34	23	30	48.00	46.67	43.75	1
41	38	11	13	13	117.00	113.33	118.13	3			89	35	23	30	45.00	46.67	43.75	2
42	39	11	13	14	117.00	113.33	113.75	1			90	35	24	30	45.00	40.00	43.75	1
43	40	12	13	14	114.00	113.33	113.75	1			91	36	24	30	42.00	40.00	43.75	3
44	41	13	13	14	111.00	113.33	113.75	3			92	36	24	31	42.00	40.00	39.38	1
45	42	13	13	15	111.00	113.33	109.38	2			93	37	24	31	39.00	40.00	39.38	2
46	43	13	14	15	111.00	106.67	109.38	1			94	37	25	31	39.00	33.33	39.38	3
47	44	14	14	15	108.00	106.67	109.38	3			95	37	25	32	39.00	33.33	35.00	1
48	45	14	14	16	108.00	106.67	105.00	1			96	38	25	32	36.00	33.33	35.00	1
49	46	15	14	16	105.00	106.67	105.00	2			97	39	25	32	33.00	33.33	35.00	3
50	47	15	15	16	105.00	100.00	105.00	1			98	39	25	33	33.00	33.33	30.63	2
51	48	16	15	16	102.00	100.00	105.00	3			99	39	26	33	33.00	26.67	30.63	1
52	49	16	15	17	102.00	100.00	100.63	1			100	40	26	33	30.00	26.67	30.63	3
53	50	17	15	17	99.00	100.00	100.63	3				40	26	34				
54	51	17	15	18	99.00	100.00	96.25	2										

Tjark: Actually, we have assigned a price to each sales period and each of the prices depends on the quantity we plan to sell there. We update the price for the round trip so that we have a variable pricing scheme. So far, I got the clue. But is this really a DYNAMIC pricing approach? The prospective customers do not know when and how we update the price for the round trip, but we do know it. So where is the surprising and a priori unknown element?

Jördis: If we sell exactly 40 tickets in the first period then we do not need to update the prices p_2 and p_3. We can reveal p_2 to the customer at the beginning of the second sales period. However, suppose the capacity (number of tickets) assigned to a period is exhausted before the first period is over. In that case, we have greater than expected demand, meaning less capacity is left for the two remaining periods. Due to this demand underestimation, we have to update the remaining capacity and its distribution over the last two periods. Compared to the initial planning, we have less capacity so that the prices p_2 as well as p_3 should be increased (too much demand, demand under estimation). Since we do not know the actual realized demand in sales period 1 or even a priori if and when 40 tickets will be sold, we do not know in advance when the ticket prices have to be updated and by how much.

Tjark: Ok, I understand. But if at the end of the first period less than 40 tickets are sold, then we have overestimated the demand here. Since we have more capacity left to be sold in the second as well as in the third period we can expect that we have to lower the prices p_2 as well as p_3, don't we?

Jördis: I suppose so. But let's test what happens. We can assume that at the end of period 1, we have sold only 30 tickets instead of 40 tickets. We now have to distribute 70 tickets over the remaining periods 2 and 3 using the iterative procedure introduced in Sect. 17.5.3.2. We can see the results in Table 17.9: Iterative distribution of the 70 tickets over second and third periods (overestimation of demand in the first period). We assign now $q_2 = 30$ tickets to the second period and $q_3 = 40$ tickets to the third period. If we apply the DPFs then we get the updated sales prices $p_2 = g_2(30) = 100$ € as well as $p_3 = g_3(40) = 87.50$ €.

Tjark: In the first period, we have earned 2700.00 €. Without the adaptation of the prices we would have gained 26 × 113.33 € = 2946.58 € in period 2, but 34 × 100.63 € in period 3. Overall, the updated revenue over all three periods is 9068 €. Applying the updated prices in the second and the third period we expect to gain 2700 € + 3000 € + 3500 € = 9200 €. The adaptation of the prices in response to the unexpected low demand in period 1 partially compensates for the revenue loss from the first period in the two remaining periods.

Jördis: That's true. Let's see what happens if we underestimate demand in the first period. If we actually sell 50 tickets instead of 40, then we will have deficits in the two remaining periods or at least in the last period. If we distribute the 50 remaining tickets one after another over the two remain-

ing periods we will assign 22 tickets to period 2 but 28 tickets to period 3 (Table 17.10). The associated ticket prices in the two last periods are $p_2 = g_2(22) = 126.67$ € resp. $p_3 = g_3(28) = 113.75$ €. Is there a financial benefit from the price adaptation?

Tjark: Yes. We have gained a revenue sum of 50×90 € = 4500 € in period 1. Without the adjustments of capacities and prices, we would get a run-out situation in period 3. From period 2 we earn 26×113.33 € = 2946.58 €, but in the third period, we could sell only $24 \times 100.63 = 2415.12$ €. The total sum of revenue without price adaptation would be 4500 € + 2946.58 € + 2415.12 € = 9861.70. After the adaptation, we can earn 4500 € + 22×126.67 € + 28×113.75 € = 10,471.74 €. We were unable to achieve this sum in our initial pricing since the used DPF g_1 is $g_1(50) = 75$. Here, the price update after period 1 enables us to deal better with this unexpected scenario.

The resulting prices obtained in the rolling horizon application of the pricing heuristic during the transition from period 1 to period 3 are summarized in Fig. 17.22. The continuous line shows the proactively determined prices that should be applied if the estimated demand is actually observed after a sales period has ended. If the actual realized demand in period p remains below the forecasted demand, i.e., if the demand arising in this period has been overestimated before period 1, then prices in the subsequent periods must be lowered (dashed line). If the actual demand in the first period exceeds the estimated demand for this period, then we encounter a situation in which the residual capacity is too low to fulfill the quantity assigned to the two remaining capacities. Here, repricing leads to increased prices to reduce the offered capacity (dotted line).

17.5.4 Summary of Dynamic Pricing

We have introduced a multi-period pricing problem in which the overall resource quantity must be distributed over the available sales period. The overall sales period is partitioned into a finite number of subperiods and it must be decided how much will be offered in each subperiod. The application of the demand–price function calculates appropriate prices for the product or service considered in each subperiod. To overcome shortcomings resulting from demand forecast errors, prices can be updated between adjacent subperiods if the end of a subperiod has been reached. As a result, prices for a product or service may be updated unpredictably as well as be demand-driven over time: this is called Dynamic Pricing ("rolling horizon pricing"). The capacity distribution task is represented as a mathematical optimization model. It is nonlinear and so complex that an out-of-the-box-solver is unable to reliably identify appropriate prices. Therefore, we have investigated a heuristic to determine contingents for each subperiod which feed the demand–price functions used to determine period-specific sales prices. This procedure is also applied in a rolling horizon framework to adjust prices to actual demand.

Table 17.9 Iterative distribution of the 70 tickets over second and third period (overestimation of demand in the first period)

1	A	F	G	H	I	J	K	L	P	Q	U	Z	AA	AB	AC	AD	AE	AF
2	Quantities q_i				Marginal revenue in period i							Quantities q_i			Marginal revenue in period i			
3	Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period			Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period
4	1		0	0		200.00	175.00	2			41		18	22		80.00	78.75	2
5	2		1	0		193.33	175.00	2			42		19	22		73.33	78.75	3
6	3		2	0		186.67	175.00	2			43		19	23		73.33	74.38	3
7	4		3	0		180.00	175.00	2			44		19	24		73.33	70.00	2
8	5		4	0		173.33	175.00	3			45		20	24		66.67	70.00	3
9	6		4	1		173.33	170.63	2			46		20	25		66.67	65.63	2
10	7		5	1		166.67	170.63	3			47		21	25		60.00	65.63	3
11	8		5	2		166.67	166.25	2			48		21	26		60.00	61.25	3
12	9		6	2		160.00	166.25	3			49		21	27		60.00	56.88	2
13	10		6	3		160.00	161.88	3			50		22	27		53.33	56.88	3
14	11		6	4		160.00	157.50	2			51		22	28		53.33	52.50	2
15	12		7	4		153.33	157.50	3			52		23	28		46.67	52.50	3
16	13		7	5		153.33	153.13	2			53		23	29		46.67	48.13	3
17	14		8	5		146.67	153.13	3			54		23	30		46.67	43.75	2
18	15		8	6		146.67	148.75	3			55		24	30		40.00	43.75	3
19	16		8	7		146.67	144.38	2			56		24	31		40.00	39.38	2
20	17		9	7		140.00	144.38	3			57		25	31		33.33	39.38	3
21	18		9	8		140.00	140.00	2			58		25	32		33.33	35.00	3
22	19		10	8		133.33	140.00	3			59		25	33		33.33	30.63	2
23	20		10	9		133.33	135.63	3			60		26	33		26.67	30.63	3
24	21		10	10		133.33	131.25	2			61		26	34		26.67	26.25	2

(continued)

Table 17.9 (continued)

1	A	F	G	H	I	J	K	L	P	Q	U	Z	AA	AB	AC	AD	AE	AF
2	Quantities q_i				Marginal revenue in period i							Quantities q_i			Marginal revenue in period i			
3	Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period			Ticket	$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	Selected period
25	22		11	10		126.67	131.25	3			62		27	34		20.00	26.25	3
26	23		11	11		126.67	126.88	3			63		27	35		20.00	21.88	3
27	24		11	12		126.67	122.50	2			64		27	36		20.00	21.50	2
28	25		12	12		120.00	122.50	3			65		28	36		13.33	17.50	3
29	26		12	13		120.00	118.13	2			66		28	37		13.33	13.13	2
30	27		13	13		113.33	118.13	3			67		29	37		6.67	13.13	3
31	28		13	14		113.33	113.75	3			68		29	38		6.67	8.75	3
32	29		13	15		113.33	109.38	2			69		29	39		6.67	4.38	2
33	30		14	15		106.67	109.38	3			70		30	39		0.00	4.38	3
34	31		14	16		106.67	105.00	2					30	40				
35	32		15	16		100.00	105.00	3										
36	33		15	17		100.00	100.63	3										
37	34		15	18		100.00	96.25	2										
38	35		16	18		93.33	96.25	3										
39	36		16	19		93.33	91.88	2										
40	37		17	19		86.67	91.88	3										
41	38		17	20		86.67	87.50	3										
42	39		17	21		86.67	83.13	2										
43.	40		18	21		80.00	83.13	3										

Table 17.10 Iterative distribution of the 50 tickets over second and third period (underestimation of demand in the first period)

1 2 3	Ticket	quantities q_i			marginal revenue in period i			Selected period
		$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	
4	1		0	0		200.00	175.00	2
5	2		1	0		193.33	175.00	2
6	3		2	0		186.67	175.00	2
7	4		3	0		180.00	175.00	2
8	5		4	0		173.33	175.00	3
9	6		4	1		173.33	170.63	2
10	7		5	1		166.67	170.63	3
11	8		5	2		166.67	166.25	2
12	9		6	2		160.00	166.25	3
13	10		6	3		160.00	161.88	3
14	11		6	4		160.00	157.50	2
15	12		7	4		153.33	157.50	3
16	13		7	5		153.33	153.13	2
17	14		8	5		146.67	153.13	3
18	15		8	6		146.67	148.75	3
19	16		8	7		146.67	144.38	2
20	17		9	7		140.00	144.38	3
21	18		9	8		140.00	140.00	2
22	19		10	8		133.33	140.00	3
23	20		10	9		133.33	135.63	3
24	21		10	10		133.33	131.25	2
25	22		11	10		126.67	131.25	3
26	23		11	11		126.67	126.88	3
27	24		11	12		126.67	122.50	2
28	25		12	12		120.00	122.50	3
29	26		12	13		120.00	118.13	2
30	27		13	13		113.33	118.13	3
31	28		13	14		113.33	113.75	3
32	29		13	15		113.33	109.38	2
33	30		14	15		106.67	109.38	3
34	31		14	16		106.67	105.00	2
35	32		15	16		100.00	105.00	3
36	33		15	17		100.00	100.63	3
37	34		15	18		100.00	96.25	2
38	35		16	18		93.33	96.25	3
39	36		16	19		93.33	91.88	2
40	37		17	19		86.67	91.88	3
41	38		17	20		86.67	87.50	3
42	39		17	21		86.67	83.13	2
43	40		18	21		80.00	83.13	3
44	41		18	22		80.00	78.75	2

(continued)

Table 17.10 (continued)

1 2 3	Ticket	quantities q_i			marginal revenue in period i			Selected period
		$i = 1$	$i = 2$	$i = 3$	$i = 1$	$i = 2$	$i = 3$	
45	42		19	22		73.33	78.75	3
46	43		19	23		73.33	74.38	3
47	44		19	24		73.33	70.00	2
48	45		20	24		66.67	70.00	3
49	46		20	25		66.67	65.63	2
50	47		21	25		60.00	65.63	3
51	48		21	26		60.00	61.25	3
52	49		21	27		60.00	56.88	2
53	50		22	27		53.33	56.88	3
54	51		22	28				
55								

17.6 Pricing Lessons Learned and Open Issues

The central component of every demand-oriented pricing approach is the price–demand function (PDF) that maps the price (rate) for a product to the demand for the priced product. Also, its "inverse," the demand–price function, can be used to couple prices with demand. An "optimal" price induces a maximal sum of earned revenues by deploying at least one resource. Suppose this capacity is not scarce and the revenue function is differentiable. In that case, this optimal price can be identified by differential calculus techniques (Sect. 17.2). If the available capacity is scarce, it becomes necessary to identify which customer or product uses the available resource capacity is the most effective way. For this purpose, the pricing decision task has to be represented by a mathematical optimization model in which the prices are the decision variables to be fixed. This model is solved by an adequate model solving software. The returned set of optimal prices represents the best decision for when several customers compete for resource utilization (Sect. 17.3). Network pricing addresses pricing decisions if more than one scarce resource is available and deployed for the realization of products (Sect. 17.4). Again, the determination of optimal prices for the products offered requires the specification and solving of a mathematical decision model. If the price–demand function can only be estimated, then it becomes necessary to adjust prices in response to an observed market behavior. This is called Dynamic Pricing and represents a reactively repeated decision task to be solved (Sect. 17.5).

Although we have seen that the model-based pricing approaches discussed can be used to determine revenue maximal prices, it is necessary to understand under which circumstances these pricing approaches can be applied. Of course, it is necessary that the PDF/DPF is available. However, the market situation also has to be checked. If there is competition in the market considered then impacts of price variations can be boosted or moderated if the other market participants follow their

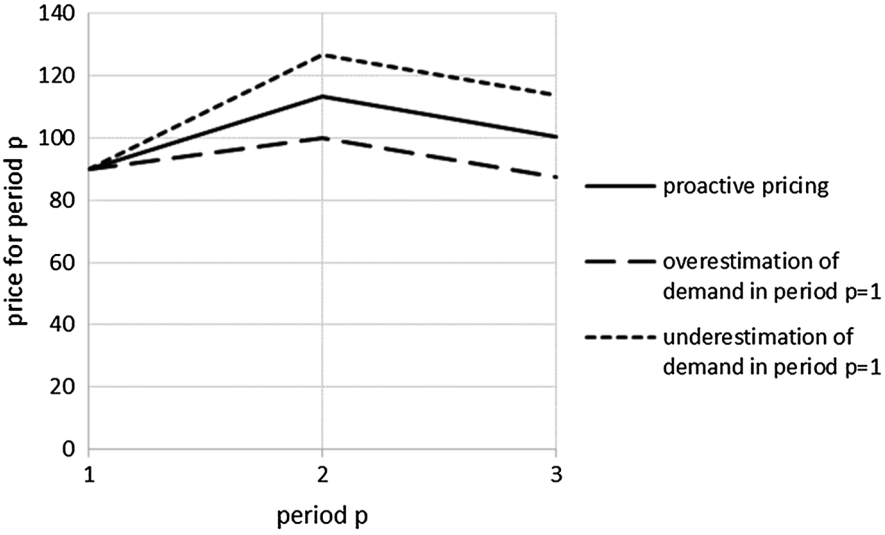


Fig. 17.22 Dynamic pricing results after replanning between period 1 as well as 2

own goal to acquire and attract as much market demand as possible. Therefore, some textbooks and articles assume that the considered service provider/producer has a strong market position, as in a monopolistic market setup.

Pricing is only the first step at the beginning of a sales (sub)period to maximize the revenues by treating different customers individually. RM provides additional techniques for skimming the complete customer surplus. During the sales period, customers with different willingness-to-pay are encountered. In this case, it is necessary that the capacity provider evaluates each customer's request individually. If the willingness to pay is less than the lowest price ("bid price"), the provider must refuse to fulfill this request: it is rejected. This technique is called capacity control (Petrick et al. 2010). Customers whose requests have been accepted and who have already paid sometimes do not use the capacity reserved due to short-term cancelations and no-shows. The unused capacity cannot be filled up 100% again since the sales period is already closed. To avoid a situation where scarce and perishable capacity remains unused, more than 100% of the capacity is sold during the sales period with the goal of ensuring complete capacity usage even if no-shows appear. This RM technique is called overbooking (McGill and van Ryzin 1999).

Bibliography

Alavifard F, Sy M, Ivanov D (2019) Optimal overbooking strategies in the airlines using dynamic programming approach in continuous time. Transp Res E 128:384–399

Brotcorne L, Cirinei F, Marcotte P, Savard G (2011) An exact algorithm for the network pricing problem. Discret Optim 8(2):246–258

Cachon G, Terwiesch C (2019) Matching supply with demand—an introduction to operations management. McGraw Hill, New York

Chopra S (2020) Supply chain management, 7th edn. Pearson Education, Upper Saddle River, NJ

Kimes SE (2010) Strategic pricing through revenue management [electronic version]. Cornell University, School of Hospitality Administration site. http://scholarship.sha.cornell.edu/articles/346. Accessed 05 Mar 2018

Maglaras and Meissner (2006) Dynamic pricing strategies for multiproduct revenue management problems. Manuf Serv Oper Manag 8(2):136–148

McCarthy J (2010) Dynamic and variable pricing [electronic version]. https://www.youtube.com/watch?v=h67sxXVi6oQ. Accessed 05 Mar 2018

McGill JI, van Ryzin G (1999) Revenue management: research overview and prospects. Transp Sci 33(2):233–256

Petrick A, Gönsch J, Steinhardt C, Klein R (2010) Dynamic control mechanisms for revenue management with flexible products. Comput Oper Res 37(11):2027–2039

Talluri K, van Ryzin G (2004) The theory and practice of revenue management. Springer, New York

Yield Management

18

Learning Objectives for This Chapter

This chapter provides a structured introduction to yield management. The following learning objectives serve as a guideline throughout this chapter.

The first objective of this chapter is the illustration of a current and appropriate application context. Section 18.1 introduces the passenger rail service provider Flying On Rail Tracks across Europe (FORTE)-RailServices. This fictive company wants to organize its ticketing in a modern way. This example case supports you in becoming familiar with a typical scenario in which the application of yield management is appropriate and promising. You will also learn that a precise and careful description of the available resources is a valuable initial step for implementing yield management tools.

Section 18.2 aims at working out the core decision task(s) in yield management. You will gain insights into these core decision problems in sales operations, which are dynamic and stochastic at the same time. Decision-relevant information is revealed piece by piece as time progresses, but the requirement to decide immediately about the acceptance or rejection of customer demand represents the major challenge in yield management. Mainly, you will become familiar with situations where you must decide to allocate scarce resources.

Requirements and opportunities associated with the implementation of yield management in sales operations are discussed in Sect. 18.3. You will learn to judge if a given sales situation fulfills those conditions needed for a successful yield management application. This third section also highlights opportunities associated with the integration of yield management techniques in sales operations.

Find additional case studies and video streams in the E-Supplement to this book on www.global-supply-chain-management.de!

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5_18

Section 18.4 teaches you the details of typical challenges associated with the integration of yield management into sales operations. In particular, you will learn which criteria can be used to separate profitable from unprofitable customer requests. Furthermore, this section emphasizes the need to orchestrate access to one resource and an entire resource network. Besides, you will find out that the ability to cope with uncertain future demand is essential. Finally, we emphasize the requirement to handle arriving customer demand (in real-time) one after another.

The core decision task of yield management, the decision about the acceptance or rejection of a particular request, requires optimizing parameters to derive these critical decisions. The discussions' primary objective in Sect. 18.5 is the introduction and application of mathematical optimization models to identify the optimal parameters. It proposes adequate models for both situations, including only one but also two or even more resources.

The aim of Sect. 18.6 comprises the application of the yield management parameters. You will learn how to use these parameters to scan a stream on successively arriving customer demand.

You will accompany Laura through the entire section. FORTERailServices hired her to develop and implement yield management into the train ticketing operations. We eavesdrop on some of the preparatory meetings and see how she discovers and applies quantitative yield management tools. Finally, she demonstrates that yield management's implementation and application are not magic but follow well-structured paths.

18.1 Flying on Rail Tracks Across Europe

FORTE-RailServices is a new company formed by several industrial firms. FORTE stands for *Flying On Rail Tracks across Europe*. Its mission is to establish international high-speed train service operations across Europe. This consortium comprises partners from national railway authorities, the railway industry, and experts who are in charge of specifying and promoting transport operations. FORTE was founded by an initiative of the European Union (EU) in response to the Europeans' inquiry to make transportation more sustainable. The mandate of FORTE contains the innovation of nationally oriented rail services and the development and establishment of new railway technologies. FORTE has quite a high budget, and the consortium partners have agreed to cover every cost for the next years. Therefore, each income from operations is 1:1 transformed into a gain of FORTE.

FORTEs so far have made efforts and are standing on three primary pillars. These pillars are the "enablers" for FORTE's expected success. First, all affected countries have agreed on standards in the control and safety of railway operations. The EU establishes a new European agency to control FORTE's operations to enable reliable and seamless cross-border train operations. Second, rolling stock is improved to enable more efficiency in daily operations. Third, FORTE has compiled innovative rail travel products that attract sufficient demand.

The emphasis on environmental aspects is evident in the promotional efforts undertaken by FORTEs. However, passengers also feel that a precise "deceleration" accompanied by significantly increased travel comfort is needed.

An innovative train composition concept forms the technological backbone of FORTE's daily operations. A mixture of block trains, as well as single wagons, is available as rolling stock. A small auxiliary engine can move individual wagons. It is not necessary to deploy a shunting locomotive. One train conductor can handle the insertion and removal of different wagons between the two-engine end wagons. Therefore, every train reconfiguration takes only a few minutes.

Three different wagon types are available for the train composition. Among them are two different passenger car types and one freight compartment wagon. The latter type of wagon is prepared to carry up to 30 EURO-pallets. A side sliding door on both sides of a wagon enables the loading by a fork lifter on a common nearly same-level platform. An Economy Type Wagon offers 60 passenger seats. The ad-hoc transformation of an economy seat into a reclining chair is possible and makes overnight rides as comfortable as possible. Four shared restrooms offer quite a high standard for personal refreshment. A Business Class Wagon hosts 30 passenger seats in 15 passenger compartments. Each compartment has a private restroom and office equipment (like a desk, a printer, or similar devices). Due to legal issues, a composed train cannot have more than eight wagons between the two engines, but every mixture of wagons is allowed.

After several market surveys and a careful inspection of the available facilities, FORTE decides to construct a star-shaped network. The center of the star is Leipzig in eastern Germany. Enough capacity on the tracks, as well as for a depot, is available there. Five rail routes ("lines") originate from and end here (Fig. 18.1). The

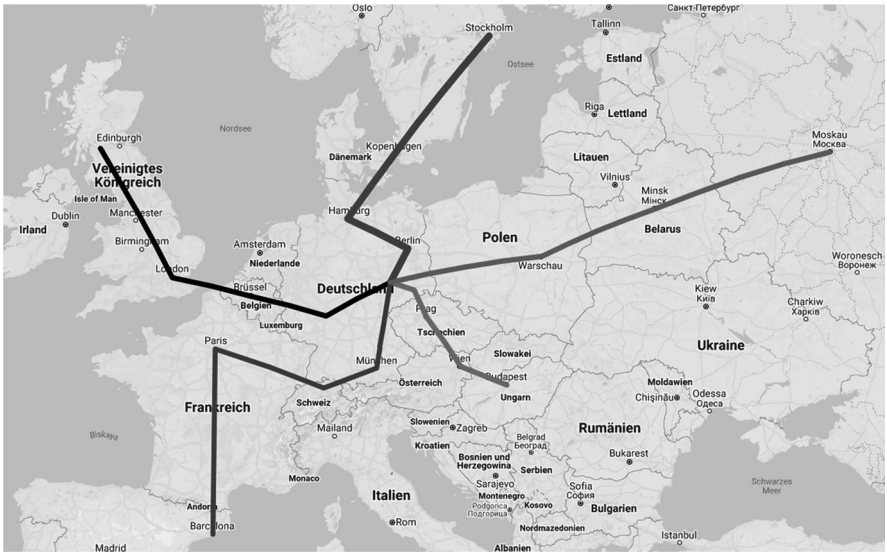


Fig. 18.1 The FORTE Railservice network (© GoogleMaps)

Northwestern Service shuttles daily between Glasgow and Leipzig, calling at London and Brussels and Frankfurt. Daily service in both directions with intermediate stops in Paris, Zurich, and Munich realizes the Southwestern Service. Bookings of the Southeastern Service between Budapest and Leipzig (also serving Vienna, Prague, and Dresden) are possible. The Scandinavian area is connected with Leipzig by the Northern Service, which offers daily shuttles between Stockholm and Leipzig, serving Copenhagen, Hamburg, and Berlin. Finally, the Eastern Service goes from Moscow to Leipzig. Trains on this service are also calling Minsk as well as Warsaw.

The operating concept relies on offering one service on each line in each direction every day. The departure time from one terminus of each service falls into the late morning period. After several technical and organizational improvements become effective, the ride time along a line will be approximately 20 h for each service. A train arrives at its terminus early in the morning. Some hours are available to clear the train for its next ride in the reverse direction (including train recomposition, shunting, cleaning, and replenishment of catering materials).

To ensure that FORTE can deploy the rolling stock with the highest possible efficiency, FORTE has decided to sell tickets and cargo space according to a demand-oriented pricing scheme. The idea is to copy several pricing mechanisms from the airline industry. There, demand-based pricing schemes have been applied successfully for passenger tickets as well as for air cargo. Tickets are sold exclusively online via a website to enable quick and cheap price adjustments.

18.2 Sales Operations by FORTE-RailServices

FORTE implemented a sales team in the headquarters. This team has rolled out some strategic decisions for the organization of the sales processes. First, there is a winter (October until March) and a summer timetable (April until September). The sales phase for train service on such a season starts 1 month before the season opens. Second, the train composition is fixed for each day 6 months before the season starts. The number of carried along wagons is specified for each service before the timetable is published. The total freight capacity, economy seats, and business seats are specified (cf. Chap. 17). Third, FORTE has decided to sell rail freight capacity exclusively based on a dynamic pricing system (cf. Sect. 17.5). Here, it seems impossible to formulate a reliable demand forecast. Fourth, to meet the requirements of as many potential travelers as possible into both ticket classes (ECONOMY and BUSINESS), price differentiation finally results in three ECONOMY class products and two BUSINESS class products (“fare classes”).

- ECONOMY FULLFLEX (one adult, incl. seat reservation(s), one direction, a full refund, free rebooking if capacity is available)
- ECONOMY FAMILY (two persons traveling together along the same path, incl. seat reservation(s), one direction, no refund, free rebooking if capacity is available)

- ECONOMY DEAL (one adult, incl. seat reservation(s), one direction, no refund, no rebooking)
- BUSINESS FULLFLEX (one adult, incl. seat reservation(s), one direction, a full refund, free rebooking if capacity is available)
- BUSINESS SAVER (one adult, incl. seat reservation(s), one direction, no refund, no rebooking)

The sales team conducted a long and intensive discussion about the pricing of the products. Initially, it was planned to apply also a dynamic pricing scheme. However, the sales team manager finally decided that an invariable price is attached to each product to keep the service/products' reliability as high as possible throughout the sales period. However, FORTE sets day- and service-specific prices.

Rob was the responsible senior manager of the sales team of a European Airline. FORTE hired him to organize the network-wide ticket sales for the new train network 18 months before the first FORTE-train departure. After a careful, in-depth inspection of the existing sales department structure, he decides to split up the sales organization. His idea is to establish five so-called line managers. Each line manager will become the responsible sales manager for one of the five services.

Laura is the new line manager for the northwestern service. Today is her first working day after a brainstorming meeting with the other four line managers. During the meeting, they have analyzed and characterized the sales situation for the FORTE services. Laura meets Rob in order to outline the results from the meeting of the line managers.

Rob: "Laura, can you outline the main findings from your meeting. What are the specific needs and requirements as well as prerequisites to set up an adequate ticket-selling system for the rail services?"

Laura: "We found out that we have to separate the sales processes of freight, business class as well as economy class from each other. Here, we have dedicated resources. We cannot mix or interchange seats in economy and business class or freight capacity with passenger seats. Furthermore, we think that we should not operate with upgrades where an economy class passenger is allowed to travel in the business class if all economy seats are sold out."

Rob: "I agree with you. We should protect our high price products. What is, in your opinion, the major challenge for the setup of sales processes for our products?"

Laura: "Let me explain this using the sales data for July first for the Northwestern service, which I summarized in Table 18.1. We have a fixed capacity of seats in the economy class. The numbers vary from day to day according to the selected train configuration. For the service toward Scotland, we have 360 seats available, but we expect to sell only 340 seats with the fixed prizes. However, we expect a total demand of 320 seats for the inbound service toward Leipzig, but we can offer only 300 seats. The availability of some cheap flight connections from Glasgow to the served

stations exerts pressure on the prizes. Thus, we have to reject some of the individual customer demands. However, at the moment, we are unsure how we can separate those requests we should accept from the inquiries we are going to reject."

Rob: "How can you realize the rejection of a request? Customers do not call us, but we sell our tickets only via a website."

Laura: "This is correct. We do not explicitly know that a specific customer is requesting a specific ticket. We can only decide to start or to stop selling products at different times during the sales phase."

Rob: "What are the major obstacles to make these decisions?"

Laura: "First, there is competition among the different products due to low resource capacity. We earn different amounts of money with the different products/ fare classes, and we need to identify the subset of customer demand that maximizes the revenue sum earned with the available resource capacity."

Rob: "So you have a problem similar to the well-known knapsack problems."

Laura: "This is correct. However, we have a stochastic variant to be handled. We have only a forecast of the upcoming demand. We do not know how accurate our forecasts are and how precise they will be. It depends upon our forecast quality how we will handle this problem characteristic."

Rob: "Ok, you are right. We have to get some experience regarding this inherent problem ingredient."

Laura: "Finally, we need to consider the dynamism of the situation. The customers will sequentially try to buy a ticket via our website. We have to decide after each handled request how we update our ticket offer on the website. However, each offer modification impacts the subsequently arriving

Table 18.1 Sales planning data for the NORTHWESTERN services on a given day (July 1st)

	Leipzig → Glasgow	Glasgow → Leipzig
Capacity in economy class	360 seats (6 wagons)	300 seats (5 wagons)
Prices		
ECONOMY FULLFLEX (P_1)	r_1 = 75,- EUR/ticket	r_1 = 70,- EUR/ticket
ECONOMY FAMILY (P_2)	r_2 = 125,- EUR/ticket (62.50 EUR/occupied seat)	r_2 = 125,- EUR/ticket (62.00 EUR/occupied seat)
ECONOMY DEAL (P_3)	r_3 = 45,- EUR/ticket	r_3 = 40,- EUR/ticket
Expected demand		
ECONOMY FULLFLEX (P_1)	D_1 = 40 tickets	D_1 = 20 tickets
ECONOMY FAMILY (P_2)	D_2 = 50 tickets (100 seats)	D_2 = 60 tickets (120 seats)
ECONOMY DEAL (P_3)	D_3 = 200 tickets	D_3 = 180 tickets

requests and, therefore, our opportunities to earn money or to lose money if we cannot serve a customer."

Rob: "Let me summarize your problem description. Our capacity is scarce. If we want to maximize the revenues earned with the available capacity, we have to control access to this capacity carefully. We have to filter profitable requests from a stream of subsequently arriving customer requests. We implement this 'filtering' by offering or not offering products on our website."

Laura: "Exactly. The application of yield management techniques is needed (Belobaba 1987; Donaghy et al. 1995). Yield management provides a toolbox to realize this filtering. Airlines are applying yield management if they vary the offered portfolio of differently prized tickets for a specific flight on the website. We should try to set up a similar yield management system for the FORTE rail service tickets."

Rob: "Ok, I agree with you. Since we already have the defined products, we can establish booking classes to put the products in appropriate quantities. We sell a product as long as its associated booking class is open. If it is closed, we do not sell this product."

Laura: "This statement outlines the required abilities. However, we have to learn and understand how we can decide when we close or open a booking class. This idea seems to be the core decisions to control to access to the scarce resource."

The decision problem mentioned above is a representative example of a central challenge related to revenue management (RM). Suppose we own a resource with an insufficient capacity, then the objective is to maximize the available capacity's benefit. It is often possible to express the benefit by the revenues earned from selling products that access the scarce resource. In contrast to pricing (cf. Chap. 17), we cannot exploit any functional dependency between determining product price and sold quantities. In such a situation, controlling those products that consume an insufficient resource capacity is essential for maximizing resource efficiency. The scientific literature subsumes all tools and efforts in this direction as yield management and classifies it as an integral part of RM. However, RM is a bigger toolbox and contains additional tools to ensure that maximal revenues are earned with the available resources. There are pricing strategies (cf. Chap. 17), price discrimination, forecasting, and overbooking. Another term for yield management is capacity control (Talluri and van Ryzin 2004a, b). Decision problems in yield management are quite complex since they are stochastic and have to incorporate a clear strategy to consider the sequentially arriving requests' dynamics.

18.3 Yield Management: Requirements and Opportunities

18.3.1 Requirements

Laura wants to be sure that yield management is a suitable approach to earn as much revenue as possible. She studies several research papers to check if the FORTE scenario fulfills commonly accepted prerequisites for a successful application. She learns that three conditions must be fulfilled in order to be able to exploit a practical yield management approach.

First, she learns that the *resource capacity under control should be scarce*. Here, scarceness refers to the situation where the total expected demand for the provided capacity units expressed by the potential customers exceeds the available number of capacity units. This scenario is assumed for a large number of days of the season.

Second, the considered *resource is perishable*. It is impossible to postpone the usage of the resource. After a given deadline, the unsold resource capacity cannot earn money at a later time. Laura knows that this is also true for the FORTE application. Every seat that is unsold at the departure time of a train does not earn money.

Third, Laura finds out that it is a necessary assumption that different customers (or customer groups) are *accepting different prices* for the same consumed resource capacity. Higher prices are justified by enriching the core product (allocation of the resource unit(s) like a seat) by extra services that do not consume a significant budget of other resources. Laura inspects the three economy class products. She is sure that the immaterial right to get money back and rebook to another day addresses different customer groups' different needs. Furthermore, these additionally granted opportunities do not produce additional costs. Also, the catering expenditures are negligible. Therefore, Laura feels confident that the FORTE scenario fulfills all necessary preconditions for successfully implementing a yield management system.

18.3.2 Opportunities

Several days later, Laura discusses the idea of booking classes and the required strategies to decide about their opening and closure. The team members feel unsure if and how the application of yield management contributes to earning additional benefit compared to a "first come/first serve" selling strategy. The latter strategy seems more comfortable to implement compared to the implementation of a yield management strategy. Laura recognizes the reluctance of her team. Two members of her team, Paula and Bill, interview Laura. Both will be responsible for the operational setup, monitoring and access control of the Northwestern service trains' booking classes. They intend to learn about the advantages of yield management over a simple first-come-first-serve ticketing strategy.

Bill: "If I got the idea of yield management right, we have to maintain the overview of all planned train services for the future. Furthermore, we need to intervene in the selling process for all these services to decide

upon the opening and the closure of the three different booking classes. Is there an estimation of how much additional revenue we may expect?"

Laura: "This is not easy to work out. It depends on the precise application and scenario. Some business reports estimate the increase in the gained revenues between 2% and 7%."

Paula: "Ok, this is not quite high, but we can realize this additional revenue without the extension of the provided capacity. Therefore, we can save long-term and expensive investments in the rolling stock, which will not pay off in low demand situations. Therefore, I think that the setup and deployment of a yield management system is a good idea. Nevertheless, I want to know whether potential passengers will feel that they benefit from such a ticket selling system."

Bill: "Maybe, we can use the information stored in such a system to offer a prospective passenger available tickets with prices around the originally entered travel date. In case the customer finds out that its favorite ticket is unavailable at the planned journey date, we can automatically present the available tickets one or two days before or later. Such a website helps us to offer this person a 'second-best' purchase option automatically."

Laura: "Finally, we can merge our booking system with our planned customer loyalty program. If we ask passengers to create a user profile and record his/her requests, bookings, and paid ticket prices, we can learn more about this customer. Thus, we can create personalized offers or learn about the individual willingness to pay for our products."

Bill: "In summary, I understand now that setting up a yield management system is far more than implementing just a ticketing system. In contrast, it helps us to understand the wishes of our customers better. We can establish a direct relationship."

18.4 Challenges Associated with Yield Management

Her discussion with Paula and Bill gives Laura the required support for the next steps toward implementing an effective yield management system for the FORTE-ticketing. She returns to her office and searches the Internet to learn about typical challenges associated with the setup of such a sales concept. After several hours of careful reading and studying, she summarizes her findings and reveals four significant challenges to be solved before a yield management system start is reasonable. First, she needs to know how she can operationalize the decisions about booking class openings and closures (Sect. 18.4.1). Second, Laura detects additional challenges if the sold products require the availability of a portfolio of resources instead of only one resource (Sect. 18.4.2). Third, Laura wants to assess the available options to counteract the demand uncertainty (Sect. 18.4.3). Finally, the dynamic of demand, expressed by the dependencies of the subsequently arriving requests, requires a careful update and evaluation of the current booking class state at every time point during the sales phase for a specific train service (Sect. 18.4.4).

18.4.1 Deciding Upon the Opening and Closure of a Booking Class

Laura contacts her old friend Sarah. They have shared a room during their time at the university. Sarah selected a major in pricing and revenue management in the tourism industry and is now responsible for an international hotel group's sales processes.

Sarah: "Hi, Laura, nice to hear from you. What are you currently doing? When you ask me if we can have a phone call, you mentioned a professional issue?"

Laura: "Well, Sarah, I want to talk to you to hear your opinion about an issue of a current project in the train operator company FORTE, where I am currently responsible for the setup of a yield management system."

Sarah: "Perfect, this is our daily routine. When we prepare our hotel room offer, we regularly decide about the type and the number of rooms of different categories for the offer in the different booking channels."

Laura: "This is what we also want to do. Your 'room categories' are equivalent to our 'booking classes'. A booking class represents a combination of a travel path and a specific type of ticket. My current issue is to find out when we open and close a booking class. Tickets associated with an open booking class will be offered on our website, but if a booking class is closed, then the associated tickets will not be offered. I want to formulate rules that help me configure our automatic booking class management system to open or close a booking class based on the already seen and the expected future demand."

Sarah: "Ok, let me try to explain the general concepts we apply. These concepts are not confidential, but you can find them in different textbooks about revenue management and pricing (Akan and Ata 2009; Simpson 1989; Talluri and van Ryzin 2004a, b). Your primary objective is to earn as much money as possible with the available resource capacity independently if these are seats on a train or rooms in a hotel. Therefore, you have to ensure that each resource consumption contributes to this goal. In other words, you should specify a threshold price for the consumption of the resource capacity. If a product's sale leads to revenue below this threshold, you should immediately stop selling this product. In your application, the booking class is closed."

Laura: "I got the clue. Is it easy to determine this threshold?"

Sarah: "No, it depends on different influencing parameters like the residual capacity, the expected upcoming demand, the other products that access this resource, and other factors. There are some basic ideas on how you can find this threshold. Please, have a look into (Talluri and van Ryzin 1998; Klein and Steinhardt 2008) to find a method for the threshold calculation and parameterize it for your case."

Laura: "Thank you very much for this valuable hint. I will inspect the sources you mentioned. It seems that this threshold strategy directly addresses the inspection of the profitability of an individual customer request, doesn't it?"

Sarah: "Yes, for this reason, this approach is sometimes also called a direct yield management criteria. It relies on the idea to calculate the opportunity costs for accepting the request on the table. These opportunity costs can be seen as the lost revenues from future requests if you allocate source capacity for the currently available request. However, this seems to be also the major obstacle of applying direct yield management: You have to calculate the opportunity costs, which is a very challenging task."

Laura: "I can imagine this but suppose that this is hardly possible for FORTE. However, you mentioned a second general acceptance criterion. Can you please also outline the idea behind it?"

Sarah: "Of course, we are applying this so-called indirect yield management to configure our yield management systems. The basic idea is to identify a revenue-maximizing distribution of the available resource capacity among your products. For each product, you determine a quota. This quota gives you the maximal number of, in your case, tickets to put into the booking class at the beginning of the sales phase. Whenever you sell a product, you have to reduce the quota by the number of sold tickets. You close the booking class as soon as the quota falls to 0."

Laura: "Such a quantity-based yield management approach seems to be more suitable for FORTE because it looks easier to implement, isn't it?"

Sarah: "I do not have a clear answer to your question. The calculation of the initial quota is often challenging, and their updating is also not trivial. I want to propose that you inspect (Talluri and van Ryzin 2004a, b). Please, check if you can set up a quantity-based yield management system then."

Laura: "Sarah, thank you very much for your valuable suggestions. Be sure that I am going to inspect and evaluate both approaches to configure rules for the opening and closure of our booking classes."

Laura terminates this call before she summarizes the lessons learned. There are two different approaches to rational decision-making about the availability of a booking class. In direct control, the sale of a product is possible as long as it is profitable (and enough capacity is available). Indirect control specifies sales quantities for booking classes. A booking class remains open as long as these quantities are not exhausted.

18.4.2 Network-Based Products

Laura had come in touch with booking classes when she was looking for flight tickets. She tries to transfer the so far learned issues to the situation when she booked her last business trip flight ticket but struggles with the following observation. When

she booked her ticket for the last business trip from Leipzig to New York, she booked a transfer connection via Munich. However, also connecting flights via Frankfurt as well as Zurich were available. She could combine two flight legs as she liked. Depending on her choice, the total ticket price varied. Today, with her accumulated knowledge about yield management, she assumes that she switches between different booking classes as soon as she combines different flights for the two legs.

She was assigned two seats in different aircrafts for the two trip segments (from Leipzig to Munich and Munich to New York). What was the resource the booking class used to control access? The situation seemed to be different from the introductory situation. When she booked her ticket on the airlines' website, the yield management system checked the seat availability in two resources. Therefore, the system recognized that the booking classes of two different resources were still open.

Laura is curious to know how the booking limit is specified for a product that incorporates two or more resources. It is necessary to control access to several resources instead of controlling access to only one resource. For the booking limit determination, products must be ordered according to their profitability. Is it possible to arrange products similarly if these products require access to two or more resources? What would happen if two resources prefer different products and grant preferred access to different products because their residual capacities are different? Furthermore, different products require access to different resource compositions. Laura is confused because she is convinced now that yield management for a network of resources requires overcoming additional challenges.

18.4.3 Uncertainty of Demand

After she has clarified how to control the booking class availability generally, she turns her attention to the third central challenge: Uncertainty of the demand-to-come compromises every a priori planning of quotas and threshold prices. Therefore, Laura supposes that a central issue with setting up a successful yield management system is her ability to cope effectively with upcoming demand uncertainty.

Both the extent of the demand for the different products are unsure and the time (and therefore the sequence) in which different customers want to purchase one of FORTE's products. Both information is vital to know if the available resource capacity is scarce and requires a careful distribution among the different products in the most profitable way.

If the demand to come is uncertain at planning time, Laura has three options. First, she can ignore the uncertainty (proactive planning) when deciding on quotas and thresholds. Instead, Laura can make assumptions about the time and extent of future demand. After the relevant decisions about the optimal quotas and threshold prices are made, she can wait and see what happens. The advantage of proactive decision-making is that she can exploit detailed data. However, the once made decisions often become suboptimal or even counterproductive if the later realized demand differs significantly from the assumed demand.

Second, the other extreme approach to make the required decisions is to wait that demand occurs and then adjust the booking class control to the updated information. This idea is called the reactive planning approach. Laura feels immensely uncomfortable with the idea to refrain from every proactive planning. She is afraid that the market drives the system. There is the danger of loss of control over the booking system behavior. Besides, she has already read that this so-called online decision-making can result in significantly worse results than proactive planning (Grötschel et al. 2001).

The third way to deal with demand uncertainty is to follow a compromise strategy between a pure proactive and a pure reactive planning approach. Here, Laura thinks about the identification of an acceptance policy. She expects to obtain a set of rules (forming a strategy) for different possible future states instructing her to open or close a booking class after discovering the specific state. For example, she reads the article of Belobaba (1989) about the identification of such a strategy.

18.4.4 Dynamicity: Nesting and Protection

Laura has learned about managing the booking class availability and the general approaches to cope with the inherent demand uncertainty. Now, she wants to understand how ticket-selling websites incorporate booking class management. She slips into a potential flight passenger's role and looks for the cheapest and most comfortable flight tickets for a fictive flight connection. Laura inspects the websites for suitable tickets for travel days far in the future and near travel days.

It takes some time, but she is sure that she has detected some common properties for all the visited websites. First, she recognizes that she can buy the most expensive tickets on all websites of different airlines at any time. Independent from the remaining time until the flight connection starts, high-priced tickets are always available. Therefore, Laura concludes that booking classes of the most valuable tickets is never closed. The only exception she discovered was a situation where a complete flight connection was sold out. Second, Laura is sure that the lowest-priced tickets have the most limited availability. Sometimes, they are available only when the flight departure is far away in the future.

In summary, her findings are that booking classes for the high-priced products stays continuously open. Booking classes for low-priced tickets is closed with a high probability during the ticketing phase. Booking classes of low-price tickets closes earlier than booking classes of high-price products.

Unsure if her observation is correct, Laura invites Rob for a telephone call. She wants to know if her observation describes a generic property of a yield management system. A few minutes later, they talk to each other.

Laura: "I studied the ticketing websites of several airlines in order to understand how they incorporate booking class availability. I discovered that high-prized products are available throughout the sales period if at least one ticket is for sale. On the other hand, low-price booking classes closes

earlier according to my experience. Is this a planned and typical behavior of the yield management system(s)?"

Rob: "Yes, it is a specific and intended property of a booking class management. Every yield management system should exhibit this behavior."

Laura: "Why?"

Rob: "The idea is simple. Throughout the complete sales phase, those customers buying the highest priced tickets must be served. The airline earns the maximum money by selling these tickets. In case that the cheapest products have exhausted their quota, then potential passengers can still select a ticket from a higher-priced booking class. However, a customer who is willing to pay a huge amount of money for the high-priced ticket should never be forced to be diverted to a lower-priced ticket. Such a situation leads to a loss of revenue."

Laura: "Ok, this makes sense. We should have this property implemented into our booking class logic. Nevertheless, to stress this idea to the largest possible extent: Should we grant our economy fullflex ticket access to the full capacity of economy tickets in a train?"

Rob: "Yes, we should do this."

Laura: "But we have three different economy class tickets. I understand that we should grant the highest-priced Fullflex ticket access to all economy seats in service. However, what capacity should we grant to the two other tickets? Is there an 'optimal' limit? In my test with the airline ticketing websites, I discovered that they manage far more than ten booking classes for one flight. So I assume that they follow a systematic review."

Rob: "We should inform us about 'nested booking classes and nested booking limits' (Chapuis 2008). Nesting means allowing high-price booking classes access to the quotas of lower-priced booking classes. The resulting number of total available capacity accessible by a product is called a nested booking limit. On the other hand, quotas for highly prized products are protected against access to lower prized products. We have to set protection levels here."

Laura: "The determination of nested booking limits as well as protection levels seems to be of quite high importance then. I think we should learn more about it."

18.5 Static Decision Tasks in Yield Management

18.5.1 Single Resource Situation

Some time has passed, and FORTE has continued to prepare the first train services. There are still 12 months until the inaugural train service will start. Meanwhile, FORTE receives the definitive date when the rolling stock is ready for passenger services. The FORTE marketing department has developed the idea to offer some "maiden services" before starting the scheduled regular services to introduce

FORTE's new train service network with high PR attraction. Three months before the first scheduled services will start, one train will depart from Glasgow, Stockholm, Moscow, Budapest, and Barcelona toward Leipzig. It is planned that all five trains arrive nearly simultaneously in Leipzig as part of festive events.

The train configurations for these maiden services comprise two Business wagons and six Economy class wagons each. FORTE has sent out invitations to important business partners for the available Business class seats. These VIP guests will stay on board for the entire journey.

To promote the Economy class, FORTE is going to offer exceptional service to the public. Each interested passenger can buy a ticket for short one-stop trips between two adjacent cities according to the planned new services. All ticketing revenues from the maiden services will result in a significant donation to charity projects. Laura wants to use these services to develop and implement a prototypic yield management system into FORTE's website. She feels that the sale products are "easy" to manage since each seat can be sold independently for each trip segment.

Laura collects all required information about the sales scenario for the maiden train services. Table 18.2 shows the collected information for the two trips from Glasgow to London and London to Brussels. She noticed that the pricing department had set the same ticket prices for both trips. The demand for the trip through the Channel Tunnel has a higher demand.

$$160 \cdot x_1 + 300 \cdot x_2 + 70 \cdot x_3 \rightarrow \max \tag{18.1}$$

$$1 \cdot x_1 + 2 \cdot x_2 + 1 \cdot x_3 \leq 360 \tag{18.2}$$

$$x_1 \text{ " } 30 \tag{18.3}$$

Table 18.2 Sales planning data for the maiden Northwestern Services in Economy class

	Glasgow → London	London → Brussels
Capacity in economy class	360 seats (6 wagons)	360 seats (6 wagons)
Prices		
ECONOMY FULLFLEX (P_1)	r_1 = 160,- EUR/ticket	r_1 = 160,- EUR/ticket
ECONOMY FAMILY (P_2)	r_2 = 300,- EUR/ticket (150 EUR/occupied seat)	r_2 = 300,- EUR/ticket (150.00 EUR/occupied seat)
ECONOMY DEAL (P_3)	r_3 = 70,- EUR/ticket	r_3 = 70,- EUR/ticket
Expected demand		
ECONOMY FULLFLEX (P_1)	D_1 = 30 tickets	D_1 = 60 tickets
ECONOMY FAMILY (P_2)	D_2 = 120 tickets (240 seats)	D_2 = 150 tickets (300 seats)
ECONOMY DEAL (P_3)	D_3 = 200 tickets	D_3 = 300 tickets

$$x_2 \text{ " } 120 \tag{18.4}$$

$$x_3 \text{ " } 200 \tag{18.5}$$

$$x_1, x_2, x_3 \geq 0 \tag{18.6}$$

During her studies at the university, she attended an Operations Research course. Now, she remembers some content. The computer use to solve an optimization problem with restrictions requires formalizing the real-world decision scenario. First, she needed to declare the decision variables as placeholders for the required decisions. Here, she has to make three decisions about the three quotas. She uses the decision variables x_1, x_2 as well as x_3. The total revenue maximization is the optimization goal, so that Eq. (18.1) represents the objective function maximization. Constraint (18.2) ensures that the required number of seats (according to the specified quotas) does not exceed the available seats. Constraints (18.3)–(18.5) ensure that the quotas are not set higher than the demand. Finally, constraint (18.6) prevents the specification of negative quotas. Laura recognizes that the mathematical optimization model (18.1)–(18.6) is a linear program. She still knows that she can use the Excel Solver to identify an optimal solution.

Laura sets up a corresponding spreadsheet representation of the model and configures the solver by specifying the cell that hosts the objective function, the variable cells, and the constraints (cf. Chap. 7). After she had started the solver, she got the optimal allocation of capacity to the products. The optimal quota for P_1 (ECONOMY FULLFLEX) is $x_1 = 30$. For P_2 (ECONOMY FAMILY), the optimal quota is $x_2 = 120$, and $x_3 = 90$ is the allocated ticket number for P_3 (ECONOMY DEAL). Laura inspects these decisions. She is not surprised by this solution. The revenue per sold seat is maximal for P_2 (150 EUR), but the revenue per sold seat in P_1 is 80 EUR. Every sold seat of P_3 leads to additional revenue of 70 EUR. Therefore, P_2 is the most profitable product and provides access to the seats with the highest preference. However, the maximal market demand of 120 tickets (240 seats) limits the quota x_2. Next, the product P_1 is selected, but the associated market demand of 30 sets the limit to the growth of the associated quota x_1. Finally, a remaining capacity of 360–240–30 = 90 seats is left and assigned to the most unprofitable product, P_3.

After the solver has identified the optimal quotas, it offers the sensitivity report's inspection (Fig. 18.2). This document reports the shadow price of the resource. This shadow price determines the value of one additional resource capacity unit or, equivalently expressed, the increase of the objective function value if the right-hand side value of constraint (18.2) is increased by one. In the context of yield management, the resource shadow price quantifies the value of the next resource capacity unit that can be allocated. It is the threshold or lower bound of the revenue one can earn with this additional capacity unit. In yield management, the shadow price of a resource is named "bid-price." It can be found in the optimal simplex tableau. A resource bid-price equals 0 if the resource is not scarce but is larger than 0 if it is scarce.

Finally, the resource bid-price might be interpreted as the opportunity costs associated with allocating the last available (or the additional) resource capacity unit. A

Microsoft Excel 15.0 Sensitivity Report
Worksheet: [ym-book-base-model.xlsx]glasgow-london
Report Created: 16.01.2021 11:43:19

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$D$10	xi Products	30	90	160	1E+30	90
$E$10	xi	120	160	300	1E+30	160
$F$10	xi	90	0	70	80	70

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$F$17	required capacity	360	70	360	110	90

Fig. 18.2 Resource shadow price in the sensitivity report

simple example explains this interpretation. Assume that only one seat is left, and Laura knows from the bid-price that she can earn 70 EUR. Now, a passenger who has already booked the seat next to the last free seat asks if he can reserve this seat. This passenger wants to know the price of it. If Laura agrees to assign the last free seat to this passenger, she loses 70 EUR revenue. To avoid any revenue loss, she can offer the reservation to the demanding passenger for a prize of at least 70 EUR or every more considerable amount.

$$\sum_{i=1}^{N} r_i x_i \rightarrow \max \tag{18.7}$$

$$\sum_{i=1}^{N} a_i x_i \leq C \tag{18.8}$$

$$x_i \leq D_i \forall i = 1, \ldots, N \tag{18.9}$$

$$x_i \geq 0 \forall i = 1, \ldots, N \tag{18.10}$$

The general decision situation in yield management requires the determination of optimal quotas $x_1, \ldots, x_N$ for the N products $P_1, \ldots, P_N$. Every sold item of product P_i leads to the revenue r_i, and the objective is to maximize the sum of revenues earned from all products (Eq. 18.7). For each allocated quota unit, product P_i consumes a_i resource units. The total resource consumption must not exceed the available resource capacity of C units (Eq. 18.8). The expected demand D_i of product i

sets an upper bound for P_i's quota (Eq. 18.9). Every quota is non-negative (Eq. 18.10).

Laura summarizes her findings. She can model the primary decision task of yield management as a linear program. Also, it is possible to derive two crucial pieces of information from an optimal model solution. First, she learns the optimal quotas for the different products. Second, she knows the resource bid prices.

18.5.2 Resource Networks

Motivated by her successful handling of the yield management scenario in the unique "maiden service" application, Laura comes back to her original assignment. She has to find a way to identify the optimal quotas for the northwestern train services. Immediately, she discovers a significant structural difference in the two situations. In contrast to the "maiden service" application, there is a wider variety of available products to be sold. Suppose Laura observes the complete service from Glasgow to Leipzig (or in the other direction). In that case, she can offer ten origin-to-destination trips: Glasgow—London ($od = 1$), London—Brussels ($od = 2$), Brussels—Frankfurt ($od = 3$), Frankfurt—Leipzig ($od = 4$), Glasgow—Brussels ($od = 5$), London—Frankfurt ($od = 6$), Brussels—Leipzig ($od = 7$), Glasgow—Frankfurt ($od = 8$), London—Leipzig ($od = 9$), Glasgow—Leipzig ($od = 10$). Laura collects the available ten travel routes in the set $OD := \{1,2, \ldots ,10\}$.

Again, there are the three fare classes ("ticket classes") FULLFLEX, FAMILY, and DEAL. Laura puts them into the set FC. She uses these two sets to describe the products for sale: the ordered pair $(f;od) \in FC \times OD$ defines the product "traveling along the path od after payment of a ticket of the fare class f."

Next, Laura inspects a typical origin-to-destination connection ("od-trip") $od \in OD$. She selects the trip from London to Leipzig. It includes the intermediate stops in Brussels as well as Frankfurt. This trip comprises three trip segments (cf. Fig. 18.3). A journey along this od-trip requires three seats, one for each trip segment. The train that serves this od-trip (on the considered day) provides 300 seats in the Economy class before the first reservation for the NORTHWESTERN service on the considered day is made. Laura defines each trip segment as an individual resource. The number of available seats determines the capacity of this resource. For the NORTHWESTERN service toward Leipzig, there are four resources. The resource portfolio forms a resource network. All resources are needed to offer the portfolio of the $3 \times 10 = 30$ travel products. Laura has to identify the optimal quotas for all 30 products, considering the different products' resource consumptions, the associated revenues, and the available resource capacities. This decision task forms the *basic problem of network yield management*.

Besides the significant increase in the product portfolio content, Laura finds out that the inherent resource capacity check is quite complicated. While all products require access to the only resource under consideration in the initially observed decision task, the required access situation is different in the resource network scenario. First, the simultaneous observation of all resource capacities is needed.

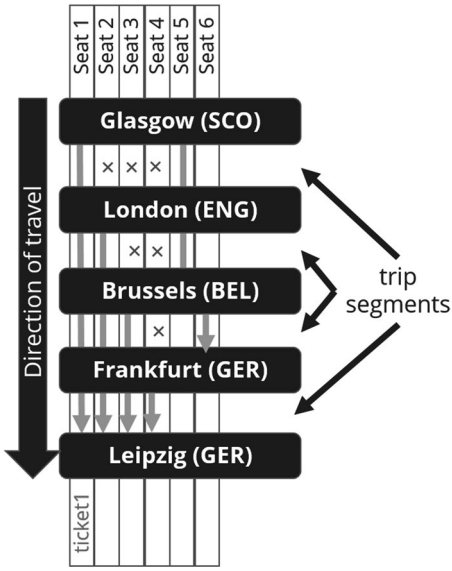


Fig. 18.3 Resource network of the NORTHWESTERN service

Second, each product demands access to an individual subset of the resources in the network.

Laura tries to update the decision models (18.7)–(18.10) of the basic decision task. She declares the non-negative decision variable family $x_{f;od}$ ($(f;od) \in \text{FC} \times \text{OD}$) to host the optimal quotas. The revenue per sold ticket is $r_{f;od}$. Table 18.3 summarizes the defined ticket prizes for the NORTHWESTERN service to Leipzig for July 1st. This date is the example day for Laura to test and inspect the extended decision model and its solution(s).

Furthermore, Laura collects the available resources in set H. For the example decision situation, the set H contains exactly the four resources R_1 (Glasgow to London), R_2 (London to Brussels), R_3 (Brussels to Frankfurt), and R_4 (Frankfurt to Leipzig). Each resource $h \in H$ provides the available capacity of c_h units. Three hundred seats are available for all four resources in the NORTHWESTERN service toward Leipzig on the investigated day. Similar to the primary decision task, the coding of the resource consumption of each product for all resources is required. Laura defines the real-valued parameter $a_{f;od;h,}$ which contains the required consumption of resource $h \in H$ by product $(f;od) \in \text{FC} \times \text{OD}$. In the NORTHWESTERN example, the product $(f;od)$defined by $f =$ "*FAMILY*" and $od \in$OD requires two seats on each involved resource $h,$ but the other two fare classes require only *one* seat.

Finally, it is necessary to integrate a forecast of the expected demand-to-come into calculating optimal quotas. Again, only the expected demand $D_{f;od}$ expressed in future ticket demand for the different products $(f;od)$ is used. Table 18.4 summarizes the demand-to-come for the NORTHWESTERN service toward Leipzig on July 1st.

$$\sum_{f \in \text{FC}} \sum_{od \in \text{OD}} r_{f;od} x_{f;od} \rightarrow \max. \tag{18.11}$$

Table 18.3 Revenue $r_{f;od}$ for the tickets on the NORTHWESTERN service for July 1st (EUR)

f	$od = 1$	$od = 2$	$od = 3$	$od = 4$	$od = 5$	$od = 6$	$od = 7$	$od = 8$	$od = 9$	$od = 10$
Fullflex	100	90	70	50	195	165	125	265	215	315
Family	180	160	120	80	370	310	230	510	420	610
Deal	70	63	49	35	136	115	87	185	150	220

Table 18.4 Expected demand to come $D_{f,od}$ on the NORTHWESTERN service for July 1st

f	$od = 1$	$od = 2$	$od = 3$	$od = 4$	$od = 5$	$od = 6$	$od = 7$	$od = 8$	$od = 9$	$od = 10$
Fullflex	50	60	55	64	40	50	45	30	35	20
Family	30	40	35	44	20	30	25	10	15	5
Deal	70	80	75	85	60	70	65	50	55	40

$$\sum_{f \in \mathrm{FC}} \sum_{od \in \mathrm{OD}} a_{f;od;h} x_{f;od} \leq c_h \forall h \in H \tag{18.12}$$

$$x_{f,od} \leq D_{f,od} \forall f \in \mathrm{FC} \forall od \in \mathrm{OD} \tag{18.13}$$

$$x_{f,od} \geq 0 \forall f \in \mathrm{FC} \forall od \in \mathrm{OD} \tag{18.14}$$

Laura is prepared to formulate the linear program (Eqs. 18.11–18.14). She parameterizes this general formulation and solves the resulting optimization model.

Table 18.5 summarizes the identified optimal quotas for the 30 products. For only 19 products, positive quotas are set. Thus, booking classes for only these products will be open at the beginning of the sales phase. The maximal revenue is 91,540 EUR.

After Laura has inspected the identified quotas, she directs her attention to the resource bid-prices. Since the investigated scenario comprises four resources, she knows that four individual bid-prices, BP_1, BP_2, BP_3, and BP_4, are set. The corresponding values are $BP_1 = 70$ EUR, $BP_2 = 90$ EUR, $BP_3 = 70$, and $BP_4 = 40$.

$$BP_{f;od} = \sum_{h \in H} a_{f;od;h} BP_h \tag{18.15}$$

The resource-specific bid-prices enable the determination of the bid-price of any product (*f*;*od*). It is the sum of the bid-prices of the required resources weighted by the resource consumption (in capacity units) as shown in (Eq. 18.15). In the NORTHWESTERN service toward Leipzig, the bid-price of the product *(Family, Glasgow→Frankfurt)* = $2 \cdot BP_1 + 2 \cdot BP_2 + 2 \cdot BP_3 = 460$,- EUR. The associated booking class is closed since the possible revenue per sold ticket is only 370 EUR and, therefore, less than the product bid price ($x_{Family;5} = 0$).

18.5.3 From Quotas to Booking Limits

Laura has learned before that quotas require processing into nested booking limits (cf. Sect. 18.4.4). The primary reason is to ensure available resource capacity for the most profitable demand as long as resource capacity is left. Two questions occur in this context. First, it is necessary to determine a ranking among the products to identify the most profitable one in the available portfolio. Second, there is the need to calculate the booking limits. The booking limit quantifies the portion of the total available resource capacity to which the yield manager grants access to a product. Nesting means opening the quotas of low-valued products for the access of high-valued products.

18.5.3.1 Single-Resource Scenario

If there is only one resource, then each product requires access to the associated capacity. Different products require different capacity units. However, it is possible to determine the profit associated with product P_i per used resource capacity by calculating the quotient $\frac{r_i}{a_i}$. If Laura sorts the product portfolio by decreasing

Table 18.5 Optimal quotas $x_{f;od}$ the NORTHWESTERN service for July 1st

f	$od = 1$	$od = 2$	$od = 3$	$od = 4$	$od = 5$	$od = 6$	$od = 7$	$od = 8$	$od = 9$	$od = 10$
Fullflex	50	25	10	64	40	50	45	30	35	20
Family	30	–	–	23	20	–	25	10	15	5
Deal	30	–	–	–	–	–	–	–	–	–

quotients, then the highest-ranked product earns the largest revenue per used capacity unit. The efficiency of the next-ranked product is the second-highest efficiency and so on.

Laura applies this approach to the maiden service application outlined in Sect. 18.5.1. According to the ranking mentioned above scheme, the product sequence is P_2, P_1, P_3. The nested booking limits are then defined by b_2:=$x_2 + x_1 + x_3 = 240$ seats, b_1:=$x_1 + x_3 = 210$ seats, and b_3:=90 seats. Here, P_2 can access the total resource capacity, but 240 x_1 x_3 = 30 seats are exclusively reserved for P_1 (this capacity portion is called the protection level).

A general definition of the nested booking limits is derived as follows. Let us assume that N products form the available product portfolio. This portfolio is ordered so that P_1 is the highest-valued product, followed by P_2 and P_N is the least valued (ranked) product. The nested booking limit b_i of Product P_i is defined by b_i:= $x_i + x_{i+1} + \ldots, x_N$. Here, x_i is the optimal quota of product P_i. The definition of the protection level s_i of product P_i is s_i:= C—b_i.

18.5.3.2 Resource-Network Scenario

The evaluation and ranking of products demonstrate an increased complexity if a network of resources requires capacity access control. First, not every product requires access to every resource. Therefore, some products do not compete for scarce resource capacity. However, the consideration of noncompetitive products in a comprehensive ranking might cause undesired preferences among products. Second, the resource consumption of a product varies from resource to resource. Third, individual resources provide different residual capacities. It is, therefore, inappropriate to use the same product preferences for each resource. Therefore, the preferred approach is to determine resource-specific product rankings. These product sequences are exploited to identify resource-specific booking limits.

$$\phi\left(f;od;h\right) := \frac{r_{f;od} - \sum_{l \in H} a_{f;od;l} BP_l}{a_{f;od;h}} \tag{18.16}$$

Laura scans the literature and finds the article of de Boer et al. (2002) that proposes using the net revenue contribution (the difference between the product's revenue and the product's bid price) to sort the products. Klein and Steinhardt (2008) define a specific indicator ϕ (Eq. 18.16) that determines a calculatory surplus per accessed resource unit. Here, the numerator in Eq. (18.16) represents the revenue excess associated with the product (*f;od*). The denominator is the consumption of resource h by product (*f*;*od*). The fraction ϕ(*f*; *od*; *h*) is named the *resource-specific normalized residual surplus* associated with the product (*f*;*od*). This indicator is defined only if (*f*;*od*) requires access to resource h ($a_{f;\,od;\,h} > 0$).

Table 18.6 summarizes the affected products, the resulting ϕ(*f*; *od*; *h*)-values, optimal quotas, and the resulting resource-specific booking limits $b^h_{f;od}$ for all four resources in the NORTHWESTERN service from Glasgow to Leipzig. The number of products with open booking class varies between 8 (resource 4) and 10 (resources 2&3). Although product (1;10) is the highest-valued product in all four resources,

Table 18.6 Resource-specific product rankings and resource-specific nested booking limits

Resource 1				Resource 2				Resource 3				Resource 4			
$(f;od)$	ϕ	$x_{f;od}$	$b^1_{f;od}$	$(f;od)$	ϕ	$x_{f;od}$	$b^2_{f;od}$	$(f;od)$	ϕ	$x_{f;od}$	$b^3_{f;od}$	$(f;od)$	ϕ	$x_{f;od}$	$b^4_{f;od}$
(1;10)	45	20	235	(1;10)	45	20	250	(1;10)	45	20	245	(1;10)	45	20	232
(1;5)	35	40	215	(1;5)	35	40	230	(1;8)	35	30	225	(2;10)	35	5	212
(1;8)	35	30	175	(1;8)	35	30	190	(2;10)	35	5	195	(1;7)	15	45	207
(2;10)	35	5	145	(2;10)	35	5	160	(2;8)	25	10	190	(1;9)	15	35	162
(1;1)	30	50	140	(2;5)	25	20	155	(1;7)	15	45	180	(1;4)	10	64	127
(2;5)	25	20	90	(2;8)	25	10	135	(1;9)	15	35	135	(2;9)	10	15	63
(2;8)	25	10	70	(1;9)	15	35	125	(2;9)	10	15	100	(2;7)	5	25	48
(2;1)	20	30	60	(2;9)	10	15	90	(1;6)	5	50	85	(2;4)	0	23	23
(3;1)	0	30	30	(1;6)	5	50	75	(2;7)	5	25	35				
				(1;2)	0	25	25	(1;3)	0	10	10				

its resource-specific booking limits are different. Resource 4 provides access to up to 232 tickets, but resource 2 is willing to provide sufficient capacity for offering 250 tickets. The number of sold tickets expresses the here reported booking limits. These booking limits are called *product-oriented*. They control the number of sold tickets. It is also possible to define a *capacity-oriented* booking limit. Such a booking limit controls the consumed resource capacity units. The multiplication of the quotas with the resource consumption factors $a_{f;\,od;\,h}$ enables the cumulation of these capacity consumptions to the capacity-oriented booking limits. In the example shown in Table 18.6, the capacity-oriented booking limits are 300, 280, 240, 210, 200, 150, 110, 90, and 30.

After she has compiled Table 18.6, Laura concludes her lessons learned so far. She has specified the products for sale. Also, Laura defined the available resources. Using mathematical programming, she determined optimal quotas for all products. Finally, she concludes the preparation of the yield management system with the determination of nested booking limits. A yield management system uses these limits to decide when to close a booking class. Laura is curious about applying this information during the booking phases when customers consecutively buy the offered products.

18.6 Dynamic Yield Management

18.6.1 Core Procedure for Online Decisions About Acceptance and Rejection

During the booking phase/sales phase, customers contact the service provider because they want to buy a product. These requests occur sequentially, one after another. The service provider has to decide for each request whether it gives the promise to fulfill the customer request ("acceptance") or does not accept the request ("rejection"). Every decision about accepting or rejecting a customer request REQ_A might impact the decision situation for a subsequently arriving customer request REQ_B. If the service provider accepts REQ_A, then some of the available resource capacity is allocated. Sometimes, there is not enough capacity left to fulfill also REQ_B. Besides, bid prices grow after the detection of increased capacity scarceness. The updated bid prices climb above the determined revenue associated with REQ_B. Therefore, yield management requires solving a sequence of interrelated decision situations. A postponement of the solution of a decision problem instance in this sequence is not possible. In the literature (Grötschel et al. 2001), such a decision situation S is called *an online decision problem*. It is a finite sequence $S = (S_0, S_1, \ldots, S_M)$ of M decision problems where the definition of S_{i+1} depends on S_i's solution. Often, customers expect an immediate response or a maximal answer time. Then, S is called a "real-time decision problem."

Different generic approaches are available to solve every decision situation S_i. The first approach is based on identifying the revenue difference between the two,

with and without accepting the currently available request. The decision model (18.11)–(18.14) is solved twice for the two situations. In the first situation, the currently pending request is accepted (leading to reduced remaining capacity but additional revenue). In the second situation, the currently pending request is rejected (causing a revenue loss). If the revenue difference is positive, then the pending request is accepted. Otherwise, it is rejected. This approach is computationally expensive since it comprises the setup and solving of two linear programs. The second approach relies upon the exploitation of the determined bid prices and quotas. These values are used to formulate request acceptance rules. The determination of optimal quotas, booking limits, and bid prices prepares the yield management activities during the booking phase. It is reasonable to understand these three parameters as input into a procedure that automatically filters incoming requests. This procedure restarts iteratively as soon as new customer demand is received.

Table 18.7 presents a pseudo-code representation of a generic yield management procedure. It comprises generic approaches for the solution of the instances of the online decision problem. Step (1) is the determination of quotas (and booking limits) and resource bid-prices before the booking phase starts. Steps (2)–(6) represent an iteration that processes the currently pending request (customer demand). The iteration starts with waiting for the next request (2). Next, the arrived request is analyzed (3). The decision about the acceptance is derived and published (4). If necessary, the quotas and booking limits or bid prices are updated (5). If the booking phase‘s end is reached, then the procedure stops (6); otherwise, it is in an idle state until the next request arrives.

Step (4) represents the derivation of the acceptance decision. In this step, quotas, booking limits, and/or bid prices are exploited to evaluate the pending request.

18.6.2 Bid-Price-Oriented Demand Stream Handling

A bid-price-oriented yield management procedure realizes step (4) by comparing the internally known bid-price with the customer's willingness-to-pay. Additionally, as a secondary criterion, a capacity availability check is required.

Table 18.7 Online decision-making procedure for yield management

(1) Determine quotas x_i as well as bid-prices bp_h using solution of core decision problem model.
(2) Wait for the next customer request.
(3) Analyze the arrived request.
(4) Decide about acceptance.
(5) If necessary, update quotas as well as bid prices using the solution of the core decision problem model with updated parameters.
(6) If the end of sales period has been reached, then stop this procedure; otherwise go to (2).

18.6.2.1 Setup of a Situation with Bid-Price-Based Acceptance Decisions

A necessary prerequisite for installing bid-price-oriented decision-making in step (4) is the knowledge of the customer's price expectation. Thus, every submitted customer demand must be accompanied by the customer's desired price. Such a special yield management setup can be found in customized production scenarios. The steel industry is an example application (Spengler et al. 2007).

18.6.2.2 Bid-Price-Based Acceptance Rules

The representative processing of step (4) starts with the analysis of the required custom-specific demand. The service provider collects all resources required for the provision of the requested product. Besides, the revenue manager determines the required resource capacities. Now, it is possible to determine the bid price for the submitted request using Eq. (18.15). If the submitted willingness to pay exceeds this bid price, then the fulfillment of the customer demand is profitable. Next, the revenue manager checks if all required resources provide sufficient residual capacity. If this is true, then the customer's demand is accepted. In all other situations, the rejection of the demand is the outcome of step (4). After an acceptance decision is made, it is necessary to reserve the involved resources' required capacities.

18.6.2.3 Bid-Price-Update Concepts

As long as the arrived demand corresponds to the demand estimation used to determine the bid prices (solving the model (18.11)–(18.14)), the recent bid prices represent the correct values of the residual capacities. The execution of step (5) is not necessary. However, if an unexpected demand arrives and the yield management system accepts this request, an update of the bid prices becomes necessary. The scientific literature discusses several bid-price-updating strategies. Typical approaches rely on the idea of repeatedly reformulating and solving the updated model (18.11)–(18.14) to recalculate the bid prices. To avoid the computational expensive model resolving, Klein (2007) proposes to use self-adjusting bid-prices.

18.6.3 Indirect Control and Booking Limit Adaptation

Laura decides to desist from incorporating the bid-price-based realization of step (4) in the FORTE yield management system. Customers do not submit their willingness to pay via the ticketing website. Thus, the installation and application of an indirect control system remain the only option. In a meeting of all line managers, they discuss the specification of an example scenario. The intention is to simulate a complete booking phase and evaluate typical booking limit adjustment concepts reported in the scientific literature.

Pierre, who manages the SOUTHWESTERN services, proposes to use the outbound service from Leipzig of this service on May 24th as an example scenario. On this day, there is planned construction and maintenance work between Zurich and Paris, so that the service provides only trips between Leipzig and Zurich. Thus, the

set of *od*-trips is relatively small and comprises the trips Leipzig to Munich ($od = 1$), Munich to Zurich ($od = 2$), and Leipzig to Zurich ($od = 3$). Only one business wagon will be in the train configuration. In business class, only two fare classes are available: Business Fullflex ($fc = 1$) and Business Saver ($fc = 2$). They have to control only six products, and the number of available seats is only 30.

Pierre: "We know how to determine the resource-specific booking limits, but we need a booking limit adaptation strategy to decide when we close a booking class."

Laura: "What exactly is a booking limit adaptation strategy?"

Pierre: "A booking limit adaptation strategy defines how we adjust the booking limits after we have accepted a customer demand. In our application, we need to reduce booking limits after a customer has purchased a ticket to avoid an unwanted, or even worse, undetected capacity runout situation. A booking limit adaptation strategy is part of step (5) of the online decision-making procedure for yield management shown in Table 18.7. This kind of strategy substitutes the model-based re-determination of the quotas. Therefore, they contribute to keeping the computational effort quite low."

Laura: "I remember to read something about these strategies in the contributions (Talluri and van Ryzin 2004a, b). The two authors discuss different strategies. To be honest, I did not understand the details. Maybe you can outline the idea behind such a strategy?"

Pierre: "You are right. There are several strategies to update booking limits throughout the booking phase. Let me explain one of them. First, we should start with determining the nested resource-specific booking limits for our small example scenario."

Laura: "I think we can start with some artificial data. Table 18.8 summarizes the demand-to-come and the revenue for each of the six products."

Pierre: "Next, we need to specify the resource consumption of the different products. We have two resources which are the segments from Leipzig to Munich ($h = 1$) and from Munich to Zurich ($h = 2$). We have $a_{1;1;1} = 1$, $a_{1;1;2} = 0$, $a_{1;2;1} = 0$, $a_{1;2;2} = 1$, $a_{1;3;1} = 1$, $a_{1;3;2} = 1$, $a_{2;1;1} = 1$, $a_{2;1;2} = 0$, $a_{2;2;1} = 0$, $a_{2;2;2} = 1$, $a_{2;3;1} = 1$ and $a_{1;3;2} = 1$."

Laura: "Finally, the available capacity of the resources is 30 seats on both trips, so that we have $c_1 = 30$ and $c_2 = 30$. Let me adjust my example model and re-start the model solver to find the optimal quotas."

Laura: "The maximal possible revenue sum is 3′797 EUR. Table 18.9 presents a proposal for optimal quotas. I let calculate the resource-specific product evaluation indicator values $\phi(\text{fc}; od; h)$ as defined in 18.15. On the first resource, the product ranking is *(1;3)* with $\phi(1; 3; 1) = 28$, *(1;1)* with $\phi(1; 1; h) = 23$, *(2;1)* with $\phi(2; 1; 1) = 7$ and *(2;3)* with $\phi(2; 3; 1) = 0$. The product-oriented nested booking limits on the first resource are therefore $b^1_{1,3} = 8 + 10 + 7 + 5 = 30$, $b^1_{1,1} = 10 + 7 + 5 = 22$, $b^1_{2,1} = 7 + 5 = 12$ and $b^1_{2,3} = 5$. Similarly, we can determine the product-specific evaluation numbers

Table 18.8 Demand-to-come as well as revenues of the scenario of the SOUTHWESTERN service

		$od = 1$ (Leipzig to Munich)	$od = 2$ (Munich to Zurich)	$od = 3$ (Leipzig to Zurich)
Demand	BC FULLFLEX ($fc = 1$)	10	12	18
	BC SAVER ($fc = 2$)	7	15	10
Revenue	BC FULLFLEX ($fc = 1$)	68	72	133
	BC SAVER ($fc = 2$)	52	60	105

Table 18.9 Optimal quotas in the scenario of the SOUTHWESTERN service

$x_{fc;od}$	$od = 1$	$od = 2$	$od = 3$
fc = 1	10	12	8
fc = 2	7	5	5

for the second resource. We get the product ranking *(1,3), (1,2), (2;2)* and *(2;3)* (with the ϕ-values *28,12,0*, and *0*). The product-oriented nested booking limits on the second resource are therefore $b^2_{1,3} = 8 + 12 + 5 + 5 = 30$, $b^2_{1,2} = 12 + 5 + 5 = 22$, $b^2_{2,2} = 5 + 5 = 10$ and $b^2{}_{2,3} = 5$. Pierre, the floor is yours now. Please, show me how we should adjust these booking limits during the booking phase responding to incoming demand."

Pierre: "The most straightforward way to update nested booking limits is called **standard nesting**. In standard nesting, we have two rules.

- Rule I: *If we accept a request for the product of the i-th position in the resource-specific ranking, then we reduce the booking limits by the required capacity units for all products on resource-specific ranking positions 1,2 … , i. We apply these updates to all resources involved in the production of the accepted product.*
- Rule II: *If we observe that the booking limit of a product is larger than the booking limit of a higher ranked product, then we reduce the lower ranked product's booking limit to the value of the booking limit of the higher-ranked product.*"

Laura: "Ok, I understand the idea behind Rule I. If you allocate resource capacity for a product, this capacity is no longer available to higher-ranked products. However, why do we need Rule II?"

Pierre: "We need Rule II to avoid a situation that a product has access to a larger portion of the capacity of a resource than a higher-ranked product. Such a state would compromise the idea of the nested booking limits. Please look into Table 18.10. It contains the development of the booking limits throughout *47* requests. Each line comprises the two resources' booking limits at the beginning of the micro-period and the requested product *(f;od)* appearing in this micro-period. The gray-shaded booking limits in the next line have been updated in response to the acceptance of this request. At first, a demand for the product *(1;3)* appears. The fulfillment of this product requires the availability of enough capacity on both resources. The prod-uct-oriented booking limit for this product on both resources is *30* so that this demand is accepted. Now, these two booking limits are reduced by *1* unit, and the next micro-period starts. Since *(1;3)* is the highest-ranked product on both resources, we made no further booking limit adjustments. The second arriving demand is for the product *(1;1)*. It requires only resource *1*, and its rank is in the second place. We

Table 18.10 Dynamic adjustment of the nested booking limits using "standard nesting"

	Product-oriented nested booking limits														
	Resource 1				Resource 2				Next incoming demand						Accept?
	(1;3)	(1;1)	(2;1)	(2;3)	(1;3)	(1;2)	(2;2)	(2;3)	(1;1)	(1;2)	(1;3)	(2;1)	(2;2)	(2;3)	
1	30	22	12	5	30	22	10	5			1				Yes
2	29	22	12	5	29	22	10	5	1						Yes
3	28	21	12	5	29	22	10	5		1					Yes
4	28	21	12	5	28	21	10	5	1						Yes
5	27	20	12	5	28	21	10	5				1			Yes
6	26	19	11	5	28	21	10	5	1						Yes
7	25	18	11	5	28	21	10	5	1						Yes
8	24	17	11	5	28	21	10	5		1					Yes
9	24	17	11	5	27	20	10	5		1					Yes
10	24	17	11	5	26	19	10	5		1					Yes
11	24	17	11	5	25	18	10	5		1					Yes
12	24	17	11	5	24	17	10	5		1					Yes
13	24	17	11	5	23	16	10	5	1						Yes
14	23	16	11	5	23	16	10	5			1				Yes
15	22	16	11	5	22	16	10	5		1					Yes
16	22	16	11	5	21	15	10	5				1			Yes
17	21	15	10	5	21	15	10	5		1					Yes
18	21	15	10	5	20	14	10	5	1						Yes
19	20	14	10	5	20	14	10	5		1					Yes
20	20	14	10	5	19	13	10	5	1						Yes
21	19	13	10	5	19	13	10	5			1				Yes
22	18	13	10	5	18	13	10	5	1						Yes
23	18	13	10	5	17	12	10	5					1		Yes
24	18	13	10	5	16	11	9	5				1			Yes
25	17	12	9	5	16	11	9	5			1				Yes

26		16	12	9	5		15	11	9	5							1		Yes
27		15	11	8	4		14	10	8	4					1				Yes
28		14	10	7	4		14	10	8	4				1					Yes
29		13	10	7	4		13	10	8	4						1			Yes
30		13	10	7	4		12	9	7	4		1							Yes
31		12	9	7	4		12	9	7	4		1							Yes
32		11	8	7	4		12	9	7	4					1				Yes
33		10	7	6	4		12	9	7	4							1		Yes
34		9	6	5	3		11	8	6	3							1		Yes
35		8	5	4	2		10	7	5	2			1						Yes
36		8	5	4	2		9	6	5	2						1			Yes
37		8	5	4	2		8	7	4	2					1				Yes
38		7	4	3	2		8	7	4	2			1						Yes
39		7	4	3	2		7	6	4	2					1				Yes
40		6	5	2	2		7	6	4	2		1							Yes
41		5	4	2	2		7	6	4	2				1					Yes
42		4	4	2	2		6	6	4	2						1			Yes
43		4	4	2	2		5	5	3	2							1		Yes
44		3	3	1	1		4	4	2	1				1					Yes
45		2	2*	1	1		3	3*	2	1						1			Yes
46		2	2	1	1		2	2	1	1							1		Yes
47		1	1	0	0		1	1	0	0				1					Yes
		0	0*	0	0		0	0*	0	0		10	12	8	7	5	5		

have to check the booking limit of *(1;1)* on resource *1*. This limit is *22*, so that we accept this demand. Now we have to update the booking limits of the product *(1;3)* and the product *(1;1)*."

Laura: "Ok, here you apply Rule I?"

Pierre: "Correct. I have to apply also Rule II in this example but not before the 44th iteration. The 44th demand is for the product *(1;3)*. The booking limits for this product are *3* (resource *1*) and *4* (resource *2*). Therefore, we accept this request. According to Rule I, I have to reduce the booking limit of *(1;3)* on both resources. This update results in the new nested booking limits *(2;3;1;1)*. Now, I have to apply Rule II to ensure that the booking limits are descending from the highest-ranked product to the lowest-ranked product. This update results in the new nested booking limits *(2;2*;1;1)*. I use the * to indicate the application of Rule II. You see, that Rule II also applies in the booking limits adjustment on resource 2."

Laura: "Can you explain to me why we accept all requests? My understanding of yield management is that we also identify requests, which we need to reject?"

Pierre: "The processed sequence of demand complies with the demand forecast you used in the decision model to calculate the optimal quotas. Thus, the realized demand matches the expected demand, so there is no need to reject a single request. In Table 18.11, you can see the processing of an unexpected demand sequence. According to your calculations, the optimal quotas are $x_{1,1} = 10$, $x_{1,2} = 12$, $x_{1,3} = 8$, $x_{2,1} = 7$, $x_{2,2} = 5$, $x_{2,3} = 5$ but the actually arriving first 47 received requests are distributed differently: *9* requests for *(1;1)*, *8* requests for *(1;2)*, *7* requests for *(1;3)*, *7* requests for *(2;1)*, *9* requests, and *7* requests for *(2;3)*. The boldly shaped requests are the differences to the demand stream from Table 18.10. You see that standard nesting results in the rejection of the first request in iteration 36, but this strategy decides to reject four additional requests among the first 47 requests …"

Laura: "… so that there is still capacity available after the first 47 processed requests. This capacity is available for the acceptance of subsequently arriving requests (if additional requests are arriving at all)."

Pierre: "This is precisely the idea of yield management. You try to separate the profitable from the unprofitable requests instead of accepting all requests on the first-come-first-serve strategy."

Laura thanks Pierre for this very fruitful talk. She summarizes her findings concerning the dynamic booking limit adaptation. First, it is necessary to identify the nested booking limits. Here, she can use the quotas derived by solving the basic decision model (18.11)–(18.14). Second, she iteratively processes each request from the incoming demand stream and decides about its acceptance or rejection. If she accepts a request, then she updates the booking limits according to some rules. One candidate rule set is "standard nesting" comprising two specific booking limit adjustment rules.

Table 18.11 Booking limit adaptation using "standard nesting" in the case of a wrong demand estimation

	Product-oriented nested booking limits														
	Resource 1				Resource 2				Next incoming demand						Accept?
	(1;3)	(1;1)	(2;1)	(2;3)	(1;3)	(1;2)	(2;2)	(2;3)	(1;1)	(1;2)	(1;3)	(2;1)	(2;2)	(2;3)	
1	30	22	12	5	30	22	10	5			1				Yes
2	29	22	12	5	29	22	10	5	1						Yes
3	28	21	12	5	29	22	10	5		1					Yes
4	28	21	12	5	28	21	10	5	1						Yes
5	27	20	12	5	28	21	10	5				1			Yes
6	26	19	11	5	28	21	10	5	1						Yes
7	25	18	11	5	28	21	10	5	1						Yes
8	24	17	11	5	28	21	10	5		1					Yes
9	24	17	11	5	27	20	10	5						1	Yes
10	23	16	10	4	26	19	9	4		1					Yes
11	23	16	10	4	25	18	9	4					1		Yes
12	23	16	10	4	24	17	8	4		1					Yes
13	23	16	10	4	23	16	8	4	1						Yes
14	22	15	10	4	23	16	8	4			1				Yes
15	21	15	10	4	22	16	8	4		1					Yes
16	21	15	10	4	21	15	8	4				1			Yes
17	20	14	9	4	21	15	8	4		1					Yes
18	20	14	9	4	20	14	8	4	1						Yes
19	19	13	9	4	20	14	8	4					1		Yes
20	19	13	9	4	19	13	7	4	1						Yes
21	18	12	9	4	19	13	7	4			1				Yes
22	17	12	9	4	18	13	7	4		1					Yes
23	17	12	9	4	17	12	7	4					1		Yes
24	17	12	8	4	16	11	6	4				1			Yes
25	16	11	8	4	16	11	6	4					1		Yes
26	16	11	8	4	15	10	5	4						1	Yes

(continued)

Table 18.11 (continued)

	Product-oriented nested booking limits															
	Resource 1				Resource 2				Next incoming demand							Accept?
	(1;3)	(1;1)	(2;1)	(2;3)	(1;3)	(1;2)	(2;2)	(2;3)	(1;1)	(1;2)	(1;3)	(2;1)	(2;2)	(2;3)		
27	15	10	7	3	14	9	4	3				1				Yes
28	14	9	6	3	14	9	4	3			1					Yes
29	13	9	6	3	13	9	4	3					1			Yes
30	13	9	6	3	12	8	3	3					1			Yes
31	13	9	6	3	11	7	2	2*	1							Yes
32	12	8	6	3	11	7	2	2				1				Yes
33	11	7	5	3	11	7	2	2						1		Yes
34	10	6	4	2	10	6	1	1						1		Yes
35	9	5	3	1	9	5	0	0		1						Yes
36	9	5	3	1	8	4	0	0					1			No
37	9	5	3	1	8	4	0	0				1				Yes
38	8	4	2	1	8	4	0	0						1		No
39	8	4	2	1	8	4	0	0				1				No
40	7	3	1	1	8	4	0	0	1							Yes
41	6	2	1	1	8	4	0	0			1					Yes
42	5	2	1	1	7	4	0	0					1			No
43	5	2	1	1	7	4	0	0						1		No
44	5	2	1	1	7	4	0	0			1					Yes
45	4	2	1	1	6	4	0	0					1			No
46	4	2	1	1	6	4	0	0						1		No
47	4	2	1	1	6	4	0	0			1					Yes
48	3	2	1	1	5	4	0	0	1							Yes
49	2	1	1	1	5	4	0	0		1						Yes
50	2	1	1	1	4	3	0	0				1				Yes
51	1	0	0	0*	4	3	0	0						1		No
52	1	0	0	0	4	3	0	0			1					Yes
53	0	0	0	0	3	3	0	0				1				No
									10	9	8	9	9	8		

18.7 Yield Management Lessons Learned and Open Issues

18.7.1 Summary of Lessons Learned

After you have worked successfully with this article's content, you are familiar with yield management's basic concepts. You have learned typical application scenarios where yield management contributes to preserving scarce resource capacity and maximizes possible revenues (Sect. 18.1). Furthermore, you have acquired knowledge about the importance of a careful and comprehensive specification of resources, capacities, and products.

Section 18.2 taught you about yield management's primary decision task: to decide about the acceptance and rejection of subsequently arriving customer requests. You are aware that this is a crucial ingredient to sales operations, especially in the service industry. The concept of a booking class was presented.

In Sect. 18.3, you learned the foremost prerequisites of a successful yield management application and the opportunities (benefits) of its integration in your sales operations.

Section 18.4 overviewed the significant challenges associated with the conceptualization and implementation of yield management. Now, you are aware that the following issues require a comprehensive analysis. You know that the formulation of criteria for deciding about opening and closing booking classes is crucial. It is also essential to control access to resource networks. Furthermore, the consideration of uncertain future demand requires particular attention. The dynamics of demand require careful decision-making about the acceptance of pending customer demand.

You learned that yield management comprises two fields of action. In the preparatory phase, you parameterize an automatic decision support system. During the active product selling phase, you apply these parameters and update them if it becomes necessary. Section 18.5 introduced you to mathematical programming techniques to optimize the essential yield management parameters: resource bid prices (direct control) and the quotas and nested booking limits (indirect control). Section 18.6 provided insights to applying the parameters above during the product selling phase to process a stream of subsequently arriving requests. Here, the focus was the application of an indirect control procedure exploiting nested booking limits. You learn to apply the technique "standard nesting" to update these decision parameters during the selling phase.

18.7.2 Further Aspects of Yield Management

This article does not discuss the exploitation of known probability distributions of future demand. Such an approximation of the number of expected requests to come and the expected sequencing provides further benefits. For example, it is possible to identify optimal decision policies. Such a policy is a rule set that determines the optimal decisions for (all) future system states (Belobaba 1989).

The here discussed service products require a clearly defined availability of resource capacities. However, service products are often much more complicated. They do not rely on the availability of one specific resource setup, but the resource manager can select from different resource combinations ("flexible products"). Yield management with flexible products is a currently investigated research field (Petrick et al. 2010). Planned upgrades of customers into a higher values product category are also investigated since they promise to increase the earned revenue (Gallego and Stefanescu 2009).

This chapter assumes that we do not serve a customer if we reject his initial demand. However, customers are often willing to buy an alternative product (with a higher revenue) if their initially requested product is unavailable. Therefore, it is necessary to decide about the booking class opening and closure for several similar products but not individually (Strauss et al. 2018; Talluri and van Ryzin 2004a, b).

Bibliography

Akan M, Ata B (2009) Bid-price controls for network revenue management: martingale characterization of optimal bid prices. Math Oper Res 34(4):912–936

Belobaba PP (1987) Survey paper—Airline yield management. An overview of seat inventory control. Transp Sci 21(2):63–73

Belobaba P (1989) OR Practice—Application of a probabilistic decision model to airline seat inventory control. Oper Res 37(2):183–197

Chapuis J (2008) Nesting booking limits in revenue management: the good, the bad and the ugly. SSRN Elect J

de Boer S, Freling R, Piersma N (2002) Mathematical programming for network revenue management revisited. Eur J Oper Res 137:72–92

Donaghy K, McMahon U, McDowell D (1995) Yield management: an overview. Int J Hosp Manag 14(2):139–150

Gallego G, Stefanescu C (2009) Upgrades, upsells and pricing in revenue management. IO Theory

Grötschel M, Krumke S, Rambau J (2001) Online optimization of large scale systems. Springer, Berlin

Klein R (2007) Network capacity control using self-adjusting bid-prices. OR Spect 29:39–60

Klein R, Steinhardt C (2008) Revenue management. Springer, Berlin

Petrick A, Gönsch J, Steinhardt C, Klein R (2010) Dynamic control mechanisms for revenue management with flexible products. Comput Oper Res 37(11):2027–2039

Simpson R (1989) Using network flow techniques to find shadow prices for market demands and seat inventory control. FTL memorandum Hrsg. MIT, Department of Aeronautics and Astronautics, Flight Transportation Laboratory, Boston

Spengler T, Rehkopf S, Volling T (2007) Revenue management in make-to-order manufacturing—an application to the iron and steel industry. OR Spect 29:157–171

Strauss A, Klein R, Steinhardt C (2018) A review of choice-based revenue management: theory and methods. Eur J Oper Res 271(2):375–387

Talluri K, van Ryzin G (1998) An analysis of bid-price controls for network revenue management. Manag Sci 44(1):1577–1593

Talluri KT, van Ryzin GJ (2004a) The theory and practice of revenue management. Springer, s.l.

Talluri K, van Ryzin G (2004b) Revenue management under a general discrete choice model of consumer behavior. Manag Sci 50(1):15–33

Appendix: Case-Study "Re-designing the Material Flow in a Global Manufacturing Network"

With the help of this case-study, materials of many chapters in this book can be applied to practical decision-making. In particular, knowledge on sourcing and production strategies, inventory management, transportation planning, linear and mixed-integer linear programming can be summarized.

Problem Description

In many cases, outsourcing or global sourcing is applied to cost reductions in material flows. At the same time, it can be possible to achieve similar effects by re-designing the material flow within the existing manufacturing network. Especially in industries with deep manufacturing penetration, such as plant engineering, there are many options to extend existing internal customer–supplier relations. In this situation, make-or-buy analysis should be performed for different modules and components at each location. The basis for the comparison of the make-and-buy efficiency is the total cost, comprised of production, logistics, and follow-up costs. In addition, risks should be considered.

Consider an enterprise that produces systems for energy transmission and has two locations: factory A is located in Europe and factory B is located in China. Both factories have deep manufacturing penetration; in other words they are able to produce almost all the components and modules needed for the final product assembly. Both factories can assemble the same final products from the same components, known as shared components (see Fig. A.1).

The final assembly always takes place in the country where the customer builds its energy system. It should be analyzed to see whether the production of the shared components can be distributed within the network so that total network costs are minimized.

For analysis, a module has been selected that is needed for 54% of all the energy system types. This makes the analysis representative and the results scalable. The module is produced according to ATO strategy and built into the final product at both factories. The module is composed of 13 components sourced from seven suppliers. At the first stage, four options for process design have been formulated (see Fig. A.2)

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5

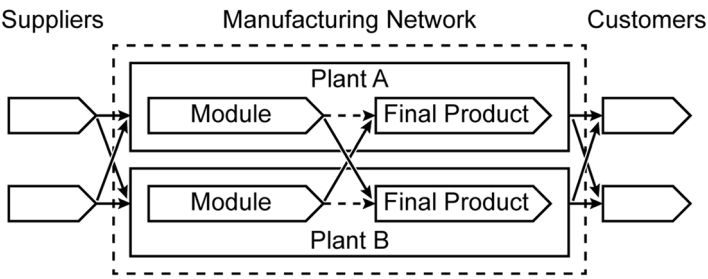


Fig. A.1 Manufacturing network

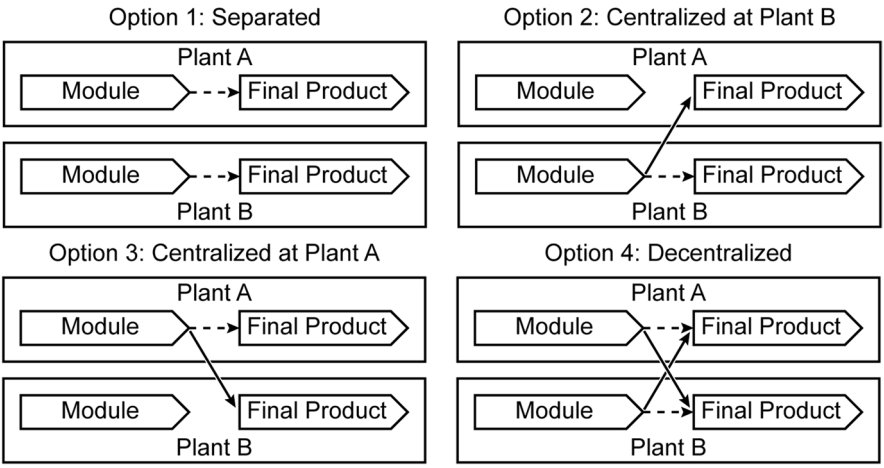


Fig. A.2 Options for process design

At present, option 1 is used.

Questions

1. Formulate a mathematical model for the problem considered above! Select a standard model from Operations Research!
2. The model has now to be filled out with data. Describe your approach to get the necessary data!

Assumptions:

- Material costs at location A are €212, labor costs are €40.50 and overhead costs are €20.78 for one module. Transportation costs for one module are €14.
- Material costs at location B are €143, labor costs are €20.0 and overhead costs are €14 for one module. Transportation costs for one module are €14.

Table A.1 Costs analysis for global sourcing

Costs	GER-GER	GER-CH	CH-GER	CH-CH
Material costs				
Labor costs				
Overhead costs				
Production costs				
Profit (5%)				
Transfer price				
Customs duty (1.7%)				
Transit price				
Transport costs				
Inventory costs				
Total Landed Costs				
Implementation costs				
Coordination costs				
Total Costs				

- Consider inventory costs as composed of cycle and safety inventory costs. Select for calculation appropriate models of inventory management. Use transit price for the calculation.
- The following is true for location A: interest is 10%, fixed costs are €23.2, service level is 95%, standard deviation of lead time is 10 days and there are 250 working days in a year!
- The following is true for location B: interest is 10%, fixed costs are €13.2, service level is 95%, standard deviation of lead time is 10 days and there are 250 working days in a year!
- Consider complexity issues if setting up coordination costs!
- Capacity at each location is 1200 units. Demand for one period is 1000 units at location A and 100 units and location B.

The cost analysis includes the so-called Total Landed Costs and follow-up costs (Table A.1):

3. Solve the model with the help of Excel Solver! Explain the results and link them to one of the four options! Do we have any costs savings if we change from the Option 1? What is your recommendation?

Costs c_{ij}	Location A ($i = 1$)	Location B ($i = 2$)	Capacity a_i
Location A ($j = 1$)			
Location B ($j = 2$)			
Demand b_j			

The objective function can be written as follows: subject to restrictions:

As the basis for comparison, the *as-is* situation can be taken, which is described as follows:

Quantity x_{ij}	Location A ($i = 1$)	Location B ($i = 2$)	Sum
Location A ($j = 1$)			
Location B ($j = 2$)			
Sum			

$Z(x) =$

The optimization model provides the following result:

Quantity x_{ij}	Location A ($i = 1$)	Location B ($i = 2$)	Sum
Location A ($j = 1$)			
Location B ($j = 2$)			
Sum			

$Z(x) =$ €

The network costs are now €. The cost savings are €.

4. Analyze advantages and limitations of each option considering the following criteria:
   (a) Coordination efforts
   (b) Reaction speed
   (c) Supplier management
   (d) Scale effects
   (e) Material and labor costs
   (f) Transportation costs and lead times
   (g) Manufacturing complexity
   (h) Quality
   (i) On-time delivery
   (j) Inventory
   (k) Single sourcing risks

   With the help of scoring analysis, the following results can be indicated (see Table A.2):

5. Consider a third location in India which can be used as a hub in the network. The capacity is 1200 units, and there is no demand in India at present. Sourcing costs from India to Germany are €198 and from India to China - €181.
   (a) Calculate optimal solution!
   (b) What qualitative factors would you consider?
   (c) Perform sensitivity analysis and explain its results!

   Optimal solution is:

Quantity x_{ij}	Location A ($i = 1$)	Location B ($i = 2$)	Location C ($j = 3$)	Sum
Location A ($j = 1$)				
Location B ($j = 2$)				
Location C ($j = 3$)				
Sum				

Table A.2 Scoring analysis

Criterion	Option 1	Option 2	Option 3	Option 4
Coordination efforts				
Reaction speed				
Supplier management				
Scale effects				
Material and labor costs				
Transportation costs and lead times				
Manufacturing complexity				
Quality				
On-time delivery				
Inventory				
Single sourcing risks				
Total				

Z(x) = €

6. Consider five markets (China, Germany, Russia, Egypt, and India) and three factories in China, Germany, and India. The capacity in China can be increased at €3000 (Table A.3).

Formulate and solve a mathematical model for the problem considered above! Select a standard model from Operations Research!

Additional Task

Consider another module for which production capacity of 1000 units in Germany and 600 units in China is available. The corresponding costs are given as follows:

Costs c_{ij}	Location A ($i = 1$)	Location B ($i = 2$)	Capacity a_i
Location A ($j = 1$)	12.276	13.990	1000
Location B ($j = 2$)	9.518	8.167	600
Demand b_j	1000	100	1100/1600

Formulate the mathematical model!
Calculate costs for Option 1!
$Z(x) =$
Calculate optimal solution and costs savings!

Quantity x_{ij}	Location A ($i = 1$)	Location B ($i = 2$)	Sum
Location A ($j = 1$)			
Location B ($j = 2$)			
Sum			

$Z(x) =$
This results in savings of € or % of costs reduction.

Table A.3 Initial data

Supply region	Demand region					Fixed costs	Low capacity	Fixed costs	High capacity
	Production and transportation cost per X units								
	GER	CH	IND	EGT	RUS				
GER	273	303	303	295	292	0	1.200	0	1.200
CH	200	177	182	190	195	0	1.200	3.000	1.600
IND	198	181	170	188	193	0	1.200	0	1.200
Demand	1.000	100	150	50	100				

Additional Discussion Questions

1 Global sourcing is sensible for items with high volumes, low demand fluctuations, and transportation costs as compared with the item value. Which methods of operations management could help you identify such items? Think up a numerical example for each method to describe your approach!
2 In the case-study we considered deterministic demands. Which methods would you use to forecast demand if statistical information is (is not) available?
3 The plant in China was engineered with an excessive capacity. What could be the reasons for that? Which methods could be used to support this decision?
4 We considered one-period analysis. Which methods could you use to lot-size optimization for multi-period problems?
5 Which trade-offs can you see between inventory and transportation costs? Is the ***economic order quantity*** (EOQ) optimal for integrated inventory and transportation costs? If yes, why? If not, which methods could be used to minimize total costs? Think up a simple numerical example to explain your approach!

Index

A
ABC analysis, 111, 403–406, 408, 443
Acceptance rules, 657, 658
Action research, 87
Adaptive planning, 389, 390, 506, 512
Additive manufacturing (AM), 11, 20, 36, 74, 127, 321, 542, 548, 554, 559–561, 577–579
Advanced planning systems (APS), 66, 69–71, 84, 90
Agent-based modeling, 344, 542, 571
Aggregate planning, 376, 378–383, 389, 397
Agile supply chain, 72, 103, 104, 125, 126, 145, 175, 512
Amazon, 35, 49, 87, 147, 316–318, 399–400, 440, 566, 579
Ant colony optimization (ACO), 83, 485
Arc, 253, 340, 450–454, 459, 462, 464, 477
Architecture of information systems (ARIS), 63
Artificial intelligence (AI), 4, 36–38, 49–50, 75, 87, 89, 316–319, 521, 539–579
Artificial neural networks, 572
Assemble-to-order (ATO), 43, 160, 161, 170, 171, 176, 177, 264, 436
Assembly lines, 21, 74, 140, 172, 280, 300–305, 310, 312, 313, 320, 331, 333–335, 400, 541, 564, 565, 567
Augmented reality (AR), 542, 545, 556, 561–562, 577, 578
Automated shipping notification (ASN), 66, 109, 555
Available-to-promise (ATP), 85, 439–440, 444

B
Backlog, 288, 379, 380, 382–384, 398, 419, 480, 481, 483–485, 488
Batch shop, 300, 320
Batch sizes, 169, 313, 331, 402
Bid-prices
  resource-specific, 652
Big data, 67, 73, 88, 89, 542, 548–550, 569, 573–577, 579
Bill-of-materials (BOM), 69, 377, 391–395, 398
Black, 98, 163, 259, 260, 594, 615
Blockchain, 36, 37, 89, 318, 542, 550, 552, 556–557, 569, 578, 579
Booking classes, 637–644, 652, 654, 656, 659, 667, 668
Booking limit
  capacity-oriented, 656
  product-oriented, 656, 659, 661, 662, 665
Booking limit adaptation strategy, 659
Booking phases, 656–659, 661
Bottleneck analysis, 292, 320
Branch-&-bound (b&b), 195
Break-even analysis, 293–296, 320
Bronze, 26
Bullwhip effect, 13, 83, 104–107, 120, 128, 265, 503, 507, 508, 510, 518, 519, 529
Business analytics, 72–76, 87–89
Business cases, 223, 279, 281, 297
Business intelligence (BI), 67, 73, 90
Business process management (BPM), 61–65, 90
Business process modeling, 63–65
Business process reengineering (BPR), 62, 64

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2025

D. Ivanov et al., *Global Supply Chain and Operations Management*, Springer Texts in Business and Economics, https://doi.org/10.1007/978-3-031-95859-5

Business processes, 61, 62, 64, 73, 83, 84, 90, 100, 116, 137, 150, 155, 162, 554, 575, 576
Buyback contracts, 113

C
Capable-to-promise (CTP), 85, 439–440, 444
Capacitated plant location model (CPLM), 85, 200
Capacitated vehicle routing problem (CVRP), 448, 472–478, 487, 490, 491
Capacitated warehouse location problem (CWLP), 186, 200–204, 206, 207, 228
Capacity
  fixed, 608, 635
  freight, 634, 635
Capacity allocation, 237, 251–265, 587–629
Capacity control, 629, 637
Capacity distribution, 614–624
Capacity-oriented booking limit, 656
Capital commitment, 106, 404, 413, 427, 431, 441
Case study research, 87
Causal forecasting, 357, 359
Cell-based layout, 327, 334–336, 345
Center-of-gravity model, 211, 212
Chase strategy, 381, 398
Chocolate, 14, 15, 413, 414, 423, 424
Closed-loop supply chain, 115–118
Cloud computing, 58, 60, 61, 66, 67, 73, 90, 554
Cluster storage, 337, 338
Cobots, 542, 564, 565, 572
Coconut, 11
Coffee, 30, 31, 423
Collaboration, 8, 11, 20, 39, 42, 46, 108, 111, 112, 138, 145, 153, 154, 162, 169, 172–175, 510, 511, 539, 555, 564, 571, 575, 576
Collaborative networks, 74
Collaborative planning forecasting and replenishment (CPFR), 46, 112, 128, 508
Completion time, 85, 288, 481–485, 488
Complexity management, 502, 573–577
Computer integrated manufacturing (CIM), 66, 74, 541
Consolidation of shipments, 238–240, 243
Constraints, 76, 81, 112, 187, 188, 190, 192, 193, 195, 196, 201–205, 222, 226, 246, 278, 282, 284, 292, 293, 335, 343, 376, 377, 382–387, 445, 456, 460, 461, 464, 471, 472, 490, 491, 556, 557, 600, 601, 605, 606, 610, 613, 614, 646
Continuous flow, 305, 320, 489
Continuous improvement, 29, 62, 115, 147, 153, 280, 307–311, 313, 315, 321, 543, 555, 558
Continuous review system, 426
Control variety, 502
Corner point solution method, 386
COVID-19, 43–44, 49, 123, 127, 128, 167, 168, 501, 510, 513, 522
COVID-19 pandemic, 4, 12, 16, 39, 43, 44, 49, 117, 122, 127, 167, 227, 228, 498, 503, 506, 510, 522, 523, 527, 552
Critical ratio (CR), 423
Cross-docking, 21, 253, 264–266, 326, 337
Cyber physical systems (CPS), 74, 86, 124, 317, 542, 545, 546, 577, 578
Cycle inventory, 85, 108, 401, 402, 409–412, 443

D
Data and text mining, 556
Decisions, 4, 41, 98, 137, 163, 183, 236, 279, 352, 377, 401, 447, 499, 540, 588, 631
Decision tree, 198, 296–297, 320
Decision variables, 187, 188, 191–193, 200–202, 204, 208–210, 246, 253, 384, 387, 459, 461, 490, 600, 601, 605, 628, 646, 649
Dedicated storage, 337, 338
Delphi method, 357, 372
Demand planning, 4, 21, 173, 369
Demand-price functions (DPFs), 618–620, 623, 628
Design capacity, 282
Desire, 141
Deviations, 71, 76, 82, 226, 279, 312, 356, 358, 419, 421–423, 425, 430, 437, 499, 500, 507, 508, 513, 529, 555
Digitalization, 8, 11, 20, 21, 315–319, 543, 555, 557, 559, 570, 574, 577, 579
Digital technologies, 4, 8, 39, 41, 67, 73, 123, 125, 317–319, 548–569, 571, 577–579
Digital twin, 36, 75, 86, 90, 549–554, 579
Dijkstra algorithm, 453–456, 490
Direct yield management, 641
Disaster management, 41–43

Disruption management, 40–41, 390, 498, 508, 509, 518
Disruption risks, 122, 147, 225, 226, 318, 499, 506–509, 511, 514, 518–520, 527, 529, 530, 552, 569
Distribute-to-order (DTO), 170, 171, 177
Distribution management, 344
Distribution network design, 224, 225, 233, 238, 248, 263–271
Disturbances, 499–501, 506, 508, 512, 529, 530
Double exponential smoothing, 358, 364–365
Driverless transportation systems (DTSs), 566–567
Drones, 27, 35, 565, 566
Drum-buffer-rope (DBR), 436
Dual sourcing, 6, 120, 122, 144
Due date, 85, 154, 481, 483–487
Dynamic lot-sizing, 83, 402, 431–436
Dynamic pricing, 587, 608–624, 628, 629, 634, 635
Dynamics, 4, 11, 13, 18, 20, 38, 114, 127, 144, 215, 226, 276, 277, 288, 289, 315, 316, 318, 319, 354, 376, 389, 390, 401, 418, 426, 430, 443, 445, 469, 474, 509, 511, 517, 529, 542, 543, 552, 556, 557, 568, 571, 631, 637, 639, 662–664, 667

E
Earliest due date rule (EDD-rule), 484, 485
E-business, 68, 72–75, 90, 401
E-commerce, 67, 73, 268, 316, 337, 501
Economic order quantity (EOQ) model, 409–415, 431, 443
Economic production quantity (EPQ) model, 414–415, 443
Economy of scale, 11, 108, 168, 177, 401, 402, 409, 412, 443
Ecosystems, 122–125, 552, 576, 579
Effective capacity, 282
Effectiveness, 21, 31, 41, 62, 90, 101, 511, 517
Efficiency, 6–12, 26, 31–33, 35, 37, 40, 41, 49, 101, 102, 104, 109, 118, 120, 121, 124, 125, 137, 139, 144, 161, 163, 175, 176, 185, 226–229, 238, 304–306, 308, 312, 315, 318, 319, 321, 355, 389, 437, 440–442, 444, 447, 449, 468, 509, 512, 517, 518, 541, 553, 556–558, 560, 562, 564, 565, 569, 570, 632, 634, 637, 654
Efficiency strategy, 101, 102
Electronic data interchange (EDI), 58, 67, 73, 109, 111, 150, 155, 174, 554
Emergency operation, 42
Engineer-to-order (ETO), 170–172, 175, 178
Enterprise management, 13
Enterprise resource planning (ERP), 58, 59, 66, 68–69, 71, 84, 89, 90, 109, 163, 173, 174, 377, 393, 541, 552, 558
E-procurement, 58–59, 61
Ergonomic workplace, 334
Event-process chains (EPC), 63
Every part every interval (EPEI) lot-size, 436
Evolutionary algorithms, 572, 616
Exact algorithms, 200, 461, 490
Excel solver, 80, 191, 195, 197, 203–205, 228, 247, 461, 601, 602, 605, 615–617, 646
Expected monetary value (EMV), 296

F
Facility location, 6, 79, 80, 102, 206–215, 217–219, 222, 223, 225, 229, 233, 267, 354, 355, 372
Factor-ranking analysis, 215–224
Factor-rating method, 218–220
Factory of future, 321
Factory planning, 4, 16, 275–321
Fare classes, 634, 636, 648, 649, 659
Feasible regions, 386–388
Feasible solutions, 188, 191, 198, 203, 209, 255, 260, 386, 456, 461, 462, 465, 490, 616
Filter, 71, 637, 657
First come/first serve (FCFS), 474, 483–485, 489, 600, 601, 603, 638
First-in-first-out (FIFO), 338
Flexibility, 11, 33, 36, 43, 103, 104, 108, 114, 119, 120, 125, 126, 138, 139, 142, 146, 162, 163, 168, 171, 175, 177, 224, 227, 283, 299, 305, 315, 382, 389, 412, 441, 474, 500, 508, 512–514, 517, 518, 520, 530, 542, 544, 554, 555, 560, 561, 563, 569, 577
Flow shop, 483, 487–488, 490
Flow time, 85, 288, 481–487
Flying robots, 565, 566
Forecasting, 4, 17, 20, 34, 44, 46, 112, 165–167, 283, 317, 351–372, 403, 498, 512, 542, 551, 555, 589, 637
Forecasting process, 354–359, 372
Forecast quality, 359, 636

Forecasts, 4, 5, 46, 49, 107, 109, 112, 128, 164, 177, 283, 316, 352–372, 375–377, 379–381, 389, 397, 406, 419, 444, 503, 540, 550, 552, 569, 574–576, 624, 634, 636, 649, 664
Full truck load (FTL), 241, 243
Fuzzy theory, 78, 572

G
Gantt-chart, 488
Genetic algorithms, 83, 485, 572
Global optimization, 209
Global sourcing, 144–147, 155, 227
Gross requirements, 394

H
Health safety and environment (HSE), 27, 278, 281–282
Heijunka, 436
Heuristics, 77, 82, 83, 85, 200, 226, 302, 343, 431, 434, 435, 444, 458, 461–468, 474, 485, 490, 491, 615, 616, 618–620, 624
Hierarchical planning, 377
Holding costs, 77, 161, 225, 382, 396, 400, 401, 409–411, 413, 415, 417, 430–433, 443, 444, 504
Horizontal coordination, 113–114
Hub-and-spoke network, 266
Humanitarian logistics, 41–43, 127

I
Idle times, 301, 303–305, 380, 488
Indirect yield management, 641
Industry 4.0, 11, 20, 36, 37, 66, 73–75, 88, 90, 123, 318, 321, 489, 540–579
Information technology (IT), 11, 19, 34, 41, 42, 60, 61, 65–69, 71–73, 75, 84, 90, 101, 108, 111, 119, 142, 153–155, 162, 169, 278, 279, 314, 319, 321, 338, 340, 439, 440, 474, 541, 545, 554, 568, 570, 574–576
Innovation strategy, 101
Integrated supply chain, 25–27, 505
Integration definition for function modeling (IDEF), 63
Intelligent, 11, 38, 82, 153, 321, 483, 543, 545, 546, 558, 565, 568, 571, 572, 577
Interest rates, 294, 401, 413, 504
Intermodal transportation, 271
Internet of Things (IoT), 20, 66, 73, 74, 321, 545, 559
Inventory
  control, 7, 16, 17, 88, 162, 314, 403, 406, 412, 418, 425–431, 444, 569
  on hand, 379, 382, 410, 426, 440, 441, 522–524
  management, 11, 13, 18, 21, 27, 36, 68, 85, 98, 110, 162, 355, 365, 372, 399–445, 568, 572
Inventory variance ratio (InvVarR), 518
Iso-profit line, 387, 388
IT projects, 67

J
Jidoka, 312, 498
Job shop, 299–300, 305, 320, 483, 489
Johnson's algorithm, 488, 489
Just-in-time (JIT), 11, 13, 144, 147–148, 155, 166, 227, 312, 315, 355, 400, 406, 498

K
Kaizen, 307, 310, 312
Kanban, 307, 312, 314–315, 400
Knapsack problems, 636

L
Last-in-first-out (LIFO), 338
Layout planning, 280, 325, 326, 330, 337–346
Lead times, 7, 30, 36, 38, 44, 62, 85, 98, 100, 101, 103–105, 112, 120, 139, 142, 145–148, 161, 162, 165, 172, 225, 226, 267, 288, 289, 318, 337, 340, 345, 376, 377, 381, 393–396, 398, 400, 409, 415–422, 424, 425, 427, 430, 431, 439, 441, 444, 469, 481–483, 507, 508, 519, 529, 540, 541, 553, 560, 561, 569
Lean production, 4, 11, 13, 18, 142, 148, 227, 306–315
Lean supply chain, 313–315
Least unit cost heuristics, 431–434
Less than truck load (LTL), 474
Level strategy, 381, 398
Linear programming, 80, 83, 382, 385–386, 615
Linear regression, 359–361
Little's Law, 287–291, 293
Local sourcing, 27, 29, 43, 145, 147
Logistics management, 17

Logistics network, 40, 168, 218, 545

M
Machine learning, 37–38, 67, 88, 89, 316, 355, 365–372, 542, 553, 554, 571, 572
Machine-to-Machine (M2M), 544, 578
Make-or-buy, 137, 139–143, 149, 155, 575
Makespan, 482–485, 488
Make-to-order (MTO), 161–163, 170, 171, 175, 176, 178, 264, 400
Make-to-stock (MTS), 100, 161–163, 166–168, 170, 171, 175–177, 436
Management information systems (MIS), 65–75, 90
Manufacturing execution systems (MES), 36, 66
Manufacturing resource planning (MRP II), 377, 378, 397
Many-to-many network, 237, 238, 251–265
Mass production, 11, 39, 163–164, 171, 177, 301, 308–311, 313, 320, 412, 541, 569
Master production schedule (MPS), 4, 85, 376, 378, 389–390, 393–396, 398
Material requirements planning (MRP), 66, 312, 375–398
Mathematical graph, 447, 451, 452, 490
Maximal revenue, 595, 619, 637, 652
Mean absolute deviation (MAD), 358, 359, 372, 419
Mean absolute percentage error (MAPE), 358, 359, 369, 370, 372
Mean squared error (MSE), 358–364, 372
Meta-heuristics, 485, 572
Miehle algorithm, 207, 211–214, 229
Milk-run, 242–245, 247
Mixed integer linear programming (MILP), 83, 188, 461
Model
  information, 63, 90
Model-based decision approach, 195
Modeling, 6, 36, 38, 63, 78–79, 84, 86–89, 194, 207, 229, 298, 343–345, 365, 458, 459, 490, 502, 552, 553, 565, 571, 572, 613–615
MODI method, 253, 262
Modular factory, 277, 542, 577
Modularization, 163–169
Moving average, 353, 358, 361–362, 372
Muda, 306, 308, 310, 311
Multi-agent systems, 571
Multi-criteria decision-making (MCDM), 77, 153, 226
Multi-objective decision making, 77
Multiple objectives, 62, 76–77
Multiple sourcing, 30, 43, 120, 144–145, 514

N
Nearest neighborhood heuristic, 462
Nested booking classes, 644
Nested booking limits, 644, 652, 654–656, 659, 661–665, 667
Nesting, 643–644, 652
Net requirements, 394–396
Network
  star-shaped, 633
Network yield management, 648
Newsvendor problem, 422–424
Node, 117, 119, 181, 186, 197, 206, 249, 250, 252–255, 258–260, 451–465, 467, 471, 475, 478, 483, 490, 518, 522, 525
Noncompetitive pricing, 593–597

O
Objective function, 187, 188, 191, 192, 195, 197, 208, 210, 211, 225, 226, 343, 383, 385, 387, 461, 464, 465, 490, 600, 601, 614–616, 646
Omnichannel, 238, 550
One-piece single flow, 332, 335
Online decision problem, 656, 657
Operational excellence, 139, 315–316, 567
Operational risks, 117, 225, 226, 504, 506, 510, 518–519, 529
Operations
  management, 3–22, 31, 34, 36, 219, 319, 391, 448
  planning, 5, 85, 354, 375–377, 379–383, 390, 397, 469, 588
  research, 77, 82–83, 88–90, 646
  strategies, 101–104, 128
Opportunity costs, 427, 641, 646
Optimality, 76, 195, 256–260
Optimal prices, 592, 602–607, 613, 615, 628
Optimal rates, 595, 598, 608, 609, 617, 618
Optimizations, 6, 11, 15, 21, 27, 61–63, 69, 71, 77, 82–89, 110, 115, 148, 186, 188, 191–195, 215, 223–226, 228, 233, 246, 252, 253, 298, 308, 316, 340, 346, 384, 402, 431, 434, 439, 443, 444, 449, 452, 456, 461, 464, 465, 468, 469, 490, 542, 550–552, 558, 559, 566, 567, 569, 572, 600,

601, 605, 606, 613–618, 624, 628, 632, 646, 652
Order
management, 139
penetration point, 125, 168–170
pick-up, 338
Order fulfillment, 61, 62, 440
Ordering costs, 401, 409–412, 427, 430, 431, 433–435, 444
Order rate variance ratio (OrdRVarR), 518
Outsourcing, 11, 25, 27, 67, 98, 113, 118, 140–143, 155, 162, 251, 500

P
Pacing, 301
Pareto-optimal, 77
Paths, 150, 243, 261, 449, 452–454, 456, 457, 459, 460, 462–467, 471, 475, 477, 478, 490, 513, 632, 634, 640, 648
Payload, 238–240, 242, 243, 471, 475, 591, 594, 598
Performance, 5, 26, 104, 136, 162, 211, 234, 283, 365, 440, 471, 501, 547
Performance impact, 121–122, 508, 519, 520, 523, 552
Periodic review systems, 426
Perishable
resource, 638
Pick by Light, 338
Pick by Voice, 338
Pick-up list, 338
Planning
proactive, 642, 643
reactive, 643
Portfolio, 4, 30, 31, 147, 148, 234, 280, 300, 637, 639, 648, 652, 654
Postponement, 13, 43, 98, 104, 120, 122, 161, 163–170, 177, 240–241, 245, 247, 248, 272, 506, 512, 656
Predictive analytics, 6, 11, 89, 317, 545, 550, 569
Price
different, 99, 413, 607, 611, 612, 615, 617, 638
threshold, 640, 642
Price-demand function (PDF), 594–597, 599, 605, 613, 618, 628
Price differentiation, 634
Priority rules, 483, 485, 491
Proactive planning, 642, 643
Problems, 5, 28, 98, 136, 164, 188, 246, 277, 353, 376, 402, 452, 498, 544, 599, 631
Process
flow layout, 327–335, 345
optimization, 61–63, 310, 540
Processing time, 46, 302, 338, 461, 481, 484, 486–488, 491
Procurement, 6, 13, 14, 17, 21, 58, 60, 98, 110, 111, 136–139, 142, 148, 162, 163, 234, 235, 376–378, 391–393, 398, 401, 412, 504, 550, 554–559, 561, 579
Product data management, 393
Product flow layout, 327, 331–335, 345
Production floor, 296, 327, 541
Production rate, 39, 105–107, 302–303, 379, 381, 402, 414, 415
Production strategies, 31, 43, 141, 164, 170–177, 263, 400, 436
Product-oriented booking limit, 656, 659, 661, 662, 665, 666
Product-process matrix, 305–306
Protection levels, 644, 654
Pull processes, 168, 170, 176, 177
Purchasing, 4, 5, 16–19, 44, 77, 87, 98, 99, 110, 137–139, 141, 143, 145, 150, 151, 155, 160, 162, 168, 219, 278, 295, 377, 392, 394, 396–398, 402, 406, 530, 554, 557–559
Push/pull, 168, 170, 176, 177
Put-away, 326, 338, 345

Q
Quadratic assignment problem, 340, 342–343, 346
Quality strategy, 101, 102
Quantity flexibility contracts, 113
Queuing theory, 83, 283–287
Quota, 641, 642, 644, 646–649, 652–657, 659, 661, 664, 667

R
Radio frequency identification (RFID), 27, 49, 66, 71–72, 84, 89, 566, 567, 569, 577, 579
Random storage, 337, 338
Ranking, 6, 31, 652, 654, 659, 661
Rapid plant assessment (RPA), 139, 291, 557, 558
Raw materials, 7, 12–14, 27–28, 44, 45, 127, 141, 142, 147, 150, 162, 166, 222, 234, 235, 283, 307, 331, 351, 401, 501, 503
Reactive planning, 643

Real-time control, 66, 71, 89
Real-time (re)planning, 389
Recurrent risks, 508, 529
Redundancy, 10, 120, 498, 499, 508, 510, 512–514, 518, 530
Reengineering, 61–63, 84
Regression analysis, 85, 150, 352, 353, 357, 358, 372
Reinforcement learning, 87, 571, 572
Rejection, 631, 632, 636, 656–658, 664, 667
Relationship (REL)-charts, 340–342, 346
Release time, 394, 481, 483–488
Reorder point (ROP), 409, 415–422, 425–427, 429–431, 443, 444
Replenishment intervals, 425–427, 430, 436, 437
Repurposing, 39, 167–168, 178, 227
Resilience, 6, 36, 145, 226, 267, 315, 441, 498, 542
Resilient SC design, 226, 227
Resilient supply chains, 39–45, 120
Resource capacity, 482, 592, 598, 600, 605, 628, 636–638, 640–642, 646–648, 652, 654, 656, 658, 661, 667, 668
Resource portfolio, 648
Resources
  dedicated, 292, 635
Resource-specific bid-prices, 652
Resource-specific normalized residual surplus, 654
Resource-specific product rankings, 654, 655
Responsiveness, 43, 67, 98, 104, 124, 125, 127, 146, 148, 164, 166, 168, 169, 176, 177, 234, 316, 400, 417, 420, 437, 441, 444, 469, 569
Responsiveness strategy, 102
Retail warehouses, 338, 340, 346
Revenue function, 595–599, 628
Revenue management, 83, 354, 588, 592, 637, 640
Revenue-sharing contracts, 113
Ripple effects, 44, 45, 78, 117–123, 146, 147, 229, 498, 499, 501, 505, 507–511, 514, 518–526, 529, 530, 577
Risk mitigation inventory, 120, 122, 227, 498, 512, 515, 517–518, 520, 530
Risks
  management, 7, 11, 20, 38, 40, 43, 50, 148, 314, 497–530
  objective, 78, 504
  perceived, 78, 105, 504
  pooling, 164
Robotic processes automation (RPA), 557–559
Robotics, 35, 542, 545, 554, 557–559, 562–565, 569, 577
Robustness, 387, 410, 489, 510, 512–514, 517, 520–522, 530
Rolling planning, 376, 389–390

S

Safety inventory, 85, 401, 402, 418, 443, 444
Safety stock, 60, 71, 105, 107, 108, 116, 162, 164, 249, 400, 416–422, 424–425, 436, 437, 444, 445, 498, 500, 506, 512
Sales and operation planning, 382
Sales operations, 631, 632, 634–637, 667
Sales phase, 634, 636, 639, 641, 644, 652, 656
Salvage price, 113, 422
Scarceness
  of resources, 447
SC Event Management (SCEM), 66, 71–72, 90
Schedule, 5, 17, 31, 67, 73, 85, 271, 292, 293, 308, 312, 353, 377–379, 391, 396, 447, 449, 480, 481, 483–489, 514, 543
Scheduled receipts, 393, 394, 440
Scheduling, 16, 17, 66, 69, 83, 85, 88, 153, 344, 376–378, 380, 389, 436, 440, 445, 447–491
Scheduling dilemma, 482
Seasonal inventory, 402
Seasonal pricing, 612
Sensitivity analysis, 153, 220, 229, 389
Sequencing, 279, 280, 317, 345, 447, 448, 472, 482, 483, 490, 667
Service levels, 60, 62, 76, 77, 82, 85, 101, 111, 114, 142, 167, 169, 175, 249, 267, 272, 380, 399, 400, 402, 416–422, 424, 425, 430, 441, 443–445, 468, 474, 509–511, 514, 517, 518, 520, 523
Service operations, 5, 448
Services, 6, 29, 101, 136, 161, 215, 234, 283, 352, 381, 400, 448, 510, 545, 587, 631
Setup costs, 300, 401, 413, 415
Setup times, 280, 300, 481, 482
Shadow prices, 646, 647
Shortest path, 282, 449–458, 490
Shortest-processing time (SPT-rule), 484, 544
Silver-meal heuristic, 431, 433–434
Simple exponential smoothing (SES), 358, 362–365, 367, 370
Simplex method, 386

Simulation, 4, 15, 20, 36, 37, 77, 82, 83, 86–89, 223–226, 297–299, 343–345, 507, 551, 553, 561, 571, 572
Simultaneous planning, 377
Single exponential smoothing, 363
Single period systems, 422–424
Single sourcing, 41, 118, 120, 122, 144, 145, 314
Six sigma, 18, 310, 315, 316
Smart factory, 7, 8, 545
Smart manufacturing, 7, 66, 542, 543
Smart products, 546–548, 569, 578
Society, 4, 10–13, 115, 119, 123–125, 127, 128, 147, 318, 528, 529
Software as a service (SaaS), 58, 60, 73, 90, 554
Sort, systemize, sweep, standardize and sustain (5S), 331
Sourcing, 10, 26, 102, 137, 185, 354, 399, 503, 548
Sourcing strategies, 25–27, 30, 31, 43–44, 120, 135–155, 514, 526, 560
Spend analysis, 149, 150
Spreadsheet approach, 186, 189–195, 203
Standard deviation of demand, 419, 421, 423
Standard nesting, 661–667
Steiner–Weber model, 182, 207, 211
Stochastic, 78, 83, 88, 153, 226, 402, 416–425, 431, 444, 445, 507, 631, 636, 637
Stochastic inventory management, 431
Stockout costs, 19, 577
Storage, 27, 31, 49, 143, 160, 225, 247, 263, 268, 270, 280, 316, 331, 337–338, 340, 345, 351, 355, 381, 384, 434, 436, 440, 444, 546, 560, 566, 568
Strategic fit, 101–104, 128, 142, 184
Structure dynamics, 76
Subcontracting, 278, 379, 381–384
Subsequently arriving requests, 636–637, 639, 664, 667
Supplier analysis, 19, 148–151
Supplier base, 4, 136, 139, 140, 144–147, 149, 154, 227, 228, 314, 554
Supplier collaboration portals (SCPs), 125, 153, 314, 554–555, 570
Supplier development, 136, 140, 148, 153, 154, 403, 548
Supplier integration, 148, 153–154
Supplier relationship management (SRM), 140, 148–154
Supplier reliability, 519, 552
Supplier selection, 136, 148, 149, 151–153, 421, 489, 572
Supply chain (SC)
  coordination, 66, 84, 101, 104–114, 121, 128, 412, 489, 500, 509, 577
  design, 5, 10, 17, 19, 21, 63, 69, 83, 85, 88, 120, 122, 125, 168, 184–205, 224–228, 512, 513, 519, 553, 578
  flexibility, 401, 443, 513, 514, 517, 530
  management, 6, 11, 13, 15, 20, 21, 27–29, 38, 115, 117, 123, 444, 540
  resilience, 11, 12, 40, 49, 114–122, 125, 228, 267, 315, 319, 497–530, 542, 552, 553
  robustness, 513
  segmentation, 147
  strategies, 8, 97–128, 136, 183, 185, 229, 318, 319, 399, 400, 420, 444
  uncertainty factors, 500
Supply chain operations reference (SCOR), 63–65, 138, 319, 548, 565, 579
Supply contract, 113
Sustainability, 4, 6–8, 10–12, 20, 21, 26–30, 34, 37, 38, 48, 62, 65, 100, 101, 114–123, 125, 128, 137, 138, 146, 147, 152, 226, 229, 234, 319, 437, 468, 472, 478, 570, 579
Sustainable SC design, 226
Sustainable supply chains, 7, 10, 11, 45–49
Sweep algorithm, 476–480, 490, 491
Systems, 6, 26, 98, 136, 159, 184, 235, 276, 353, 376, 400, 468, 498, 540, 605, 634

T

Takt time, 292, 301, 303, 304, 315, 333, 345, 565
Tardiness, 481, 482, 484–487
Target inventory levels, 425, 426, 430, 431
Target planning, 278
Task times, 301–303
Technique for Order Performance by Similarity to Ideal Solution (TOPSIS), 153
Theory of constraints, 292, 293, 320
This corresponds to the, 410
3D printing, 36–37, 74, 168, 321, 545, 559–561, 569, 571
Threshold prices, 640, 642
Throughput times, 43, 277, 291, 305, 554
Ticket selling system, 635, 639
Time series analysis, 352–354, 357, 358, 372
Time-to-recovery (TTR), 519–521, 524

Time-to-survive (TTS), 228, 510, 519–522, 524
Total cost of ownership (TCO), 67, 141, 145, 149
Total productive maintenance (TPM), 315, 316, 331
Total quality management (TQM), 11
Toyota production system (TPS), 90, 148, 308–310, 312, 498, 499
Trace and tracking, 66
Tracking and tracing, 71, 548, 565, 569, 577, 579
Trade-off, 62, 77, 120, 175, 176, 226, 396, 399, 402, 413, 417, 431, 437, 443, 444, 482, 498, 517
Train service operations, 632
Transformation processes, 9–12, 14, 17, 150, 562
Transport networks, 40, 233, 235–238, 248, 272
Transportation
  costs, 5, 27, 29, 30, 98, 125, 143, 145, 146, 167, 168, 183, 184, 187, 196, 197, 201, 205, 208, 209, 213, 215, 219, 223, 225, 229, 235, 236, 239–241, 245, 247, 250, 251, 253, 254, 260, 317, 412, 436, 437, 444, 593
  matrix, 254–256, 258, 261–264
  modes, 116, 225, 269–271
  network design, 85, 234–272
  problem, 235, 252–254, 261
Transshipment problem, 253
Transshipments, 102, 146, 234, 243–250, 253, 270, 272, 590
Travelling salesman problem (TSP), 447, 457–465, 471, 472, 474, 480, 481, 483, 487, 490
Triple exponential smoothing, 358, 365

U
Uncertainty
  factors, 78, 499, 500, 529
Unified modeling language (UML), 63
Upgrades, 547, 635, 668
U-shape, 336
Utility value analysis, 220–224, 229

V
Value, 9, 46, 99, 141, 163, 187, 240, 281, 357, 385, 404, 453, 499, 541, 592, 646
Value added, 9, 101, 102, 172, 291, 305, 307, 344, 573–577
Value stream mapping (VSM), 307
Vehicle routing, 16, 340, 467–480
Vendor-managed inventory (VMI), 58–60, 108–111, 128, 162, 163, 431, 445, 554
Vertical integration, 29, 113–114, 167, 504
Viability, 12, 122–128, 226, 526–530
Viable SC designs, 227, 229
Virtual enterprises, 73, 74
Virtual reality (VR), 545, 561–562, 569
Visibility, 6, 38, 39, 50, 58, 59, 65, 88, 110, 112, 114, 116, 163, 172, 228, 314, 318, 510, 552, 555, 556, 568, 569, 574, 576, 577, 579

W
Wagner–Whitin, 434–436, 444
Waiting times, 224, 285, 287–291, 344, 425, 481–483, 544
Warehouse location problem (WLP), 85, 186–205, 228, 252
Warehouse management systems (WMS), 27, 66, 90
Warehouse processes, 337, 341
Weighted moving average, 362
Workforce, 5, 10, 17, 35, 125, 277, 278, 281–283, 309, 314, 338, 351, 379–381
Work-in-process (WIP), 85, 166, 288, 401, 440, 442, 481, 543

X
XYZ analyses, 404–409, 443

Y
Yamazumi chart, 304
Yield management
  direct, 641
  indirect, 641
Yield management systems, 637–645, 656, 658

Z
Zero defects, 147, 315, 336